Prepladder Version X Qbank

Subject: Pediatrics

Contents

Contents (continued)

Growth

1. Arrange the following in the correct sequence of eruption for primary and permanent teeth. Primary lateral incisors Primary central incisors Permanent first premolar Primary second molar Permanent first molar Primary canine

A. 2>3>1>4>5>6

B. 1>3>2>4>6>5

C. 2>1>4>6>5>3

D. 1>2>4>6>5>3

2. At what age does the anterior fontanelle typically close in infants?

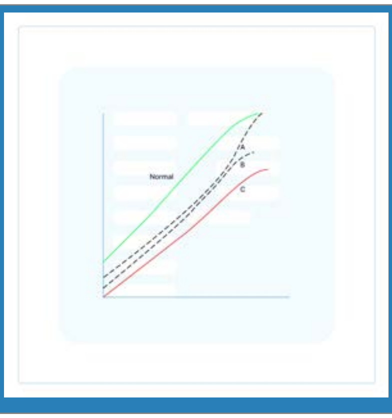
A. 2 months

B. 6 months

C. 12 months

D. 24 months

3. Match the following graphical representations with their respective cause of short stature. i) Familial short stature ii) Constitutional delay iii) Pre-natal onset pathologic short stature

i) Familial short stature	
ii) Constitutional delay	
iii) Pre-natal onset pathologic short stature	

A. i-A, ii-B, iii-C

B. i-B, ii-A, iii-C

C. i-C, ii-B, iii-A

D. i-A, ii-C, iii-B

4. A 2-year-old child presents for evaluation with an abnormal head shape and a distinctive hand deformity as shown in the image. The child's head exhibits early closure of cranial sutures, leading to an unusually shaped skull. Based on these clinical features, which of the following conditions is the most likely diagnosis? {{caption_text}}

A. Crouzon syndrome

B. Apert syndrome

C. Pfeiffer syndrome

D. Carpenter syndrome

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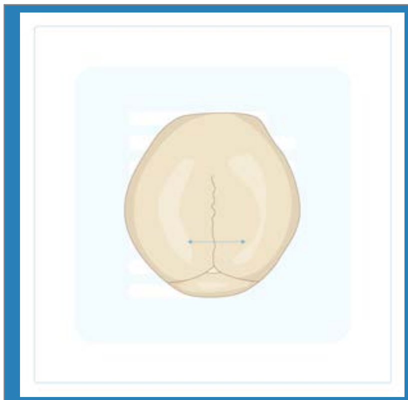
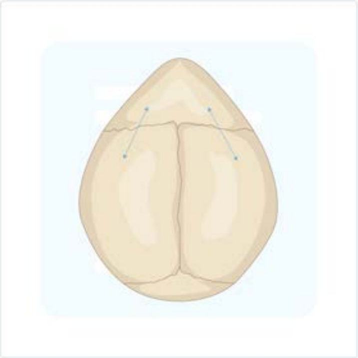
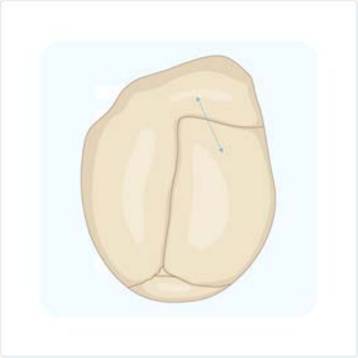
5. Match the following. A. 1. Scaphocephaly B. 2. Plagiocephaly C. 3. Brachycephaly D. 4. Trigonocephaly

 ... (content truncated)	1. Scaphocephaly
 ... (content truncated)	2. Plagiocephaly

 ... (content truncated)	3. Brachycephaly
 ... (content truncated)	4. Trigonocephaly

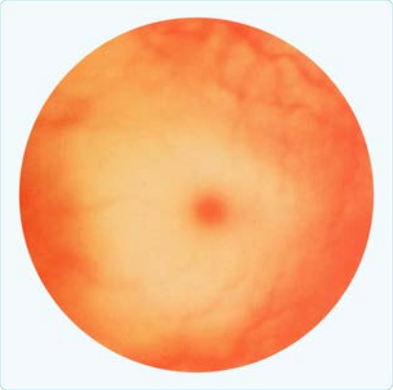
A. A-3, B-2, C-4, D-1

B. A-2, B-3, C-4, D-1

C. A-3, B-1, C-4, D-2

D. A-1, B-4, C-3, D-2

6. A one-year-old child presents with macrocephaly, developmental delay, a significant reduction in motor skills, poor muscle tone, and a low IQ. At birth the child was normal. Fundoscopy reveals the following. What is the most likely diagnosis?



A. Tay-Sachs disease

B. Infantile Hydrocephalus

C. Congenital brain malformation

D. Rett syndrome

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7. Which of the following conditions is not a cause of acquired microcephaly?

A. Seckel Syndrome

B. Down Syndrome

C. Rett Syndrome

D. Angelman Syndrome

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8. A newborn is evaluated and has a head circumference < -2 SD of the mean for age and sex. The baby's head circumference is consistently below this threshold on repeated measurements, No additional findings of developmental delay or neurological impairment. Which of the following is the most appropriate condition for this baby?

A. Microcephaly is probably due to Seckel syndrome.

B. Microcephaly is probably due to the Zika virus in the antenatal period.

C. Normal variation of head size.

D. Microcephaly due to Down’s Syndrome.

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9. A 4-year-old child presents with developmental delays and a recent diagnosis of homocystinuria. Physical examination reveals tall stature, dislocated lens, and a history of recurrent thrombotic events. Genetic testing and biochemical analysis confirm the diagnosis. Which amino acid is most closely linked to the metabolic pathway involved in the development of homocystinuria?

A. Methionine

B. Phenylalanine

C. Tryptophan

D. Valine

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10. Which of the following statements regarding the causes and characteristics of tall stature in children are correct? Tall stature is defined as a height >2SD above the mean for age and gender. Constitutional acceleration of growth (CAG) is characterized by tall stature, normal growth velocity, and advanced bone age, and results in a final height within the mid-parental range. Tall stature is defined as a height >3SD above the mean for age and gender. Obesity-related tall stature is associated with advanced bone age and early onset of puberty, which may limit the increase in final height. Klinefelter syndrome presents with disproportionately tall stature, small testes, and behavioral issues. Marfan syndrome, caused by mutations in the FBN1 gene, features tall stature, lens dislocation, and aortic root dilation.

A. 1,2,4,6.

B. 1,2,3,4.

C. 1,2,4,5,6.

D. 1,4,5,6.

11. A 6-year-old girl is brought to the clinic with a history of frequent bone fractures occurring with minimal trauma. Physical examination reveals a blue sclera, hearing, and dental abnormalities. Which of the following statements best explains the underlying pathophysiology of the patient's condition?

A. Mutation affects collagen type II, leading to bone fractures.

B. Mutation affects collagen type III, resulting in blue sclera.

C. Mutation affects collagen type I, affecting bones and contributing to their fragility.

D. Mutation affects collagen type IV, resulting in dental abnormalities.

12. A 7-year-old boy presents with short stature and disproportionate body proportions. His parents are concerned about his height, which falls well below the 3rd percentile for his age. The child has a normal-sized trunk but exhibits a disproportionately large head and short limbs. X-ray findings show shortening of the long bones and normal bone mineralization. Genetic testing reveals a specific FGFR3 mutation. What is the most likely diagnosis for this child's short stature?

A. Achondroplasia

B. Turner Syndrome

C. Growth hormone deficiency

D. Hypothyroidism

13. A 10-year-old boy presents with short stature and delayed bone age. His growth velocity has been consistently below the 3rd percentile. After thorough evaluation, growth hormone deficiency (GHD) is diagnosed. What is the expected relationship between the Upper and Lower segments of his body?

A. US > LS

B. US = LS

C. US < LS

D. None of the above

14. What will the expected adult height of a male child be if the adult height of the mother is 168 cm and the adult height of the father is 182 cm?

A. 175 cm

B. 176.5 cm

C. 181.5 cm

D. 180 cm

15. A concerned mother brings her 2-year-old child, worried about the child's short stature. On examination, it is found that the child’s bone age matches the chronological age but the height falls more than 2 standard deviations below the average for the child's age. The child has no other developmental anomalies, and both parents are of short stature. What is the most likely cause of the short stature in this child?

A. Familial short stature

B. Growth hormone deficiency

C. Turner syndrome

D. Cushing syndrome

16. Which of the following best defines the time limit used to describe the term " embryo" in human development?

A. 0 to 2 weeks

B. 0 to 8 weeks

C. 9 weeks to birth

D. None of the above

17. Which of the following conditions are most likely to be associated with delayed dentition? Rickets Cleidocranial dysostosis Pierre Robin sequence Down syndrome

A. 1, 3, 4

B. 3, 4

C. 1, 2, 3

D. 1, 2, 4

18. A 2-month-old infant is being evaluated for growth parameters. The pediatrician needs to measure the chest circumference accurately. What is the most appropriate position for this measurement?

A. At the end of the expiration

B. Midway between inspiration and expiration

C. At the end of deep inspiration

D. At the level of nipples, irrespective of the inspiratory or expiratory phase

19. If the head circumference of a newborn is 35 cm, what would be the expected measurement at 1 year of age?

A. 41 cm

B. 44 cm

C. 47 cm

D. 50 cm

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20. A 6-year-old child presents after a road traffic accident, and accurate height measurement using a stadiometer is not possible. Which of the following methods is most appropriate for estimating the child’s height?

A. Using the child's arm span

B. Estimating height based on parental height

C. Head circumference

D. Mid-upper arm circumference

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21. At what age is a child's height typically around half of their predicted adult height?

A. 1 Year

B. 2 Years

C. 9 months

D. 4 Years

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22. A 6-year-old girl is brought to the pediatric clinic for a routine check-up. The pediatrician wants to estimate her weight. What will be the estimated weight of this 6-year-old girl?

A. 12 kg

B. 16 kg

C. 20 kg

D. 24 kg

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23. In the pediatric population, which aspect of growth is most closely correlated with the maturation of the gonads?

A. Skeletal growth

B. Height velocity

C. Weight gain

D. Head circumference

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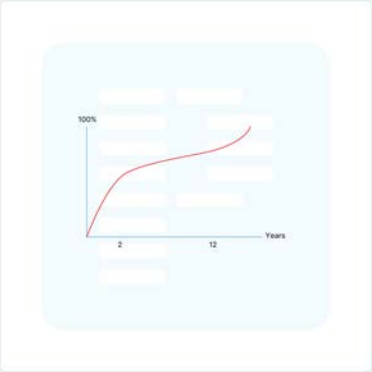
24. A 12-year-old child’s growth chart shows a distinctive pattern in the graph depicting their growth over time as shown in the image below. In this clinical scenario, what does the graph primarily indicate about the child’s growth patterns?



A. Gonadal development

B. Somatic growth

C. Brain growth

D. Lymphoid growth

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25. A 4-year-old child presents for a routine checkup. The pediatrician reviews the child’s health and plots the following parameters on the growth chart. Which of the following parameters is not typically represented on the WHO growth chart?

A. Weight for height

B. Skin fold thickness

C. Mid arm circumference

D. None of the above

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26. A 4-year-old boy is brought to the clinic for a routine check-up. The pediatrician uses the Growth Charts to assess growth patterns. Which of the following charts is used in this patient?

A. IAP Growth charts

B. Centers for Disease Control and Prevention (CDC) Growth Charts

C. National Center for Health Statistics (NCHS) Growth Charts

D. WHO Growth Charts

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27. Arrange the following pubertal changes in females in the correct sequence from the first to the last: Pubarche Thelarche Growth Spurt Menarche

A. 1, 2, 3, 4

B. 2, 1, 3, 4

C. 1, 3, 2, 4

D. 2, 3, 1, 4

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28. A 7-year-old child is brought to the pediatric clinic for a routine check-up. Parents are concerned about her weight gain and overall growth. The pediatrician decides to assess nutritional status and body fat distribution by assessing skin fold thickness to determine if any concerns need to be addressed. Which of the following instruments is best used?

A. Harpenden caliper

B. Shakir Tape

C. Salter scale

D. Infantometer

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29. Which of the following is not assessed by measuring MUAC (Middle upper arm circumference?

A. Body fat reserves

B. Nutritional status

C. Dehydration

D. Growth monitoring

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	2
Question 4	2
Question 5	3
Question 6	1
Question 7	2
Question 8	3
Question 9	1
Question 10	3
Question 11	3
Question 12	1
Question 13	2
Question 14	3
Question 15	1
Question 16	2
Question 17	4
Question 18	2
Question 19	3
Question 20	1
Question 21	2
Question 22	3

Question 23	1
Question 24	2
Question 25	4
Question 26	4
Question 27	2
Question 28	1
Question 29	3

Solution for Question 1:

Correct Answer: C) 2>1>4>6>5>3

Explanation:

The primary teeth appear before permanent teeth in the following order.

Primary Dentition	
Lower central incisor	6 months
Upper central incisor	7 months
Upper lateral incisor	8 months
Lower lateral incisor	9 months
1st Molar	1 year
Canine	1.5 years
2nd Molar	2 years

Permanent teeth.

- First molars – between 6 and 7 years.
- Central incisors – between 6 and 8 years.
- Lateral incisors – between 7 and 8 years.
- Canine teeth – between 9 and 13 years.
- Premolars – between 9 and 13 years.

- Second molars – between 11 and 13 years.
- Third molars (wisdom teeth) – between the ages of 17 and 21 years.

	Primary Dentition	Secondary Dentition
Also known as	Milk or temporary teeth	Permanent teeth
Begin at	6-7 months	6 years
1st tooth to erupt	Lower central incisor	1st molar
Last tooth	2nd molar	3rd molar (or) wisdom tooth
Completion of primary dentition	2½ years-3 years	12 years except the 3rd molar (18-25 years)
Total no. of teeth	20	28-32
Teeth in each quadrant (ICPM)	ICPM: 2102	ICPM: 2123

Solution for Question 2:

Correct Answer: C) 12 months

Explanation:

The anterior fontanelle is a diamond-shaped soft spot on the top of an infant's head, formed by the intersection of several cranial bones.

- This fontanelle plays a crucial role during the early months of life for several reasons:
- Brain Growth: The brain overgrows during the first year, and the fontanelle allows for this expansion without putting pressure on the developing skull.
- Birth Process: The flexibility of the fontanelle helps the baby's head to mold during delivery, making it easier to pass through the birth canal.
- Assessment of Health: Healthcare providers often monitor the fontanelles during check-ups. A fontanelle that is sunken may indicate dehydration, while a bulging fontanelle can be a sign of increased intracranial pressure.

The anterior fontanelle generally closes between 9 to 18 months of age, with 12 months being the average.

- After closure, the bones of the skull begin to fuse, providing a more rigid structure that supports the growing brain.

In contrast, the posterior fontanelle, located at the back of the head, typically closes much earlier—usually between 2 to 3 months.

- Monitoring these closures is essential for tracking an infant's development and ensuring there are no underlying health issues.

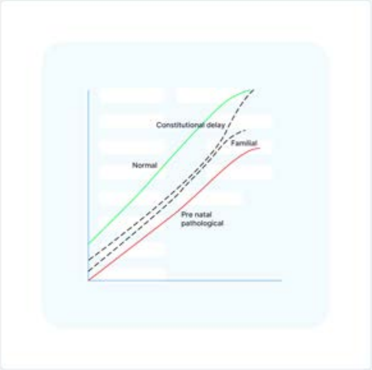
Sphenoidal fontanelle: Closes around 6 months after birth.

Mastoid fontanelle: Closes between 6–18 months after birth.

Solution for Question 3:

Correct Answer: B) i-B, ii-A, iii-C

Explanation:



Difference between familial and constitutional short stature:

Features	Familial short stature	Constitutional delay in growth
Sex	Both equally affected	More common in boys
Length at birth	Normal	Normal (starts falling < 5th centile in 1st 3 years of life)
Family history	Short stature	Delayed puberty
Parents stature	Short (one or both)	Average
Height velocity	Normal	Normal
Puberty	Normal	Delayed

Bone age (BA) and chronological age (CA)	BA = CA > height age	CA > BA = height age
Final height	Short, but normal for target height	Normal

Prenatal onset pathological short stature presents as short stature at birth continuing to short stature at adulthood if untreated.

Solution for Question 4:

Correct Answer: B) Apert Syndrome

Explanation:

- Apert syndrome is a genetic condition that involves both craniosynostosis and a characteristic "mitten hand" deformity.
- Apert syndrome is usually a sporadic condition, although autosomal dominant inheritance can occur.
- It is associated with premature fusion of multiple sutures, including the coronal, sagittal, squamosal, and lambdoid sutures.
- Apert syndrome is characterized by syndactyly of the 2nd, 3rd, and 4th fingers, which may be joined to the thumb and the 5th finger known as mitten hand deformity.
- Similar abnormalities often occur in the feet.
- All patients have progressive calcification and fusion of the bones of the hands, feet, and cervical spine.

Deformities associated with craniosynostosis

Crouzon syndrome (Option A)	Crouzon syndrome is characterized by premature craniosynostosis and is inherited as an autosomal dominant trait. Most often the head is compressed back-to-front diameter or brachycephaly resulting from bilateral closure of the coronal sutures. The orbits are underdeveloped, and ocular proptosis is prominent. Hypoplasia of the maxilla and orbital hypertelorism are typical facial features.
Apert syndrome (Option B)	Apert syndrome is characterized by syndactyly of the 2nd, 3rd, and 4th fingers, which may be joined to the thumb and the 5th finger known as mitten hand deformity. Similar abnormalities often occur in the feet. All patients have progressive calcification and fusion of the bones of the hands, feet, and cervical spine.
Pfeiffer syndrome (Option C)	Pfeiffer syndrome is most often associated with turricephaly. The eyes are prominent and widely spaced, and the thumbs and great toes are short and broad. Partial soft tissue syndactyly may be evident.
Carpenter syndrome (Option D)	Carpenter syndrome is inherited as an autosomal recessive condition, and the many fusions of sutures tend to produce the kleeblattscha■del skull deformity. Soft tissue syndactyly of the hands and feet is always present, and intellectual disability is common.
Chotzen Syndrome	Chotzen syndrome is characterized by asymmetric craniosynostosis and plagiocephaly. It is associated with facial asymmetry, ptosis of the eyelids, shortened fingers, and soft tissue syndactyly of the 2nd and 3rd fingers.

Solution for Question 5:

Correct Answer: C) A-3, B-1, C-4, D-2

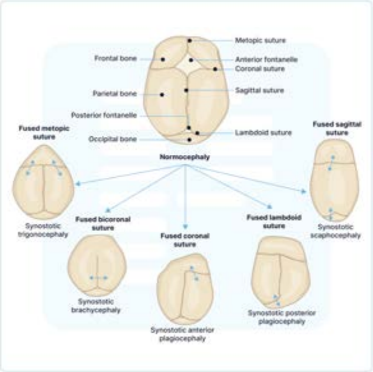
Explanation:

- The above images illustrate the different types of craniosynostosis.
- Craniosynostosis occurs when one or more skull sutures close prematurely, potentially leading to varying impacts on a child's neurodevelopment depending on the number of

sutures affected and the severity of complications.

Posterior fontanelle: This smaller fontanelle at the back of the head usually closes within 2 months of birth.

Anterior fontanelle: This larger fontanelle at the top of the head usually closes between 7 and 18 months.



Scaphocephaly or dolichocephaly (premature fusion of the sagittal suture):

- These patients have a long, narrow-shaped head with a higher anteroposterior diameter.

Anterior plagiocephaly (premature fusion of 1 coronal suture):

- The forehead appears flat on the affected side, Harlequin sign (high supraorbital margins seen in radiographs), frontal bossing on the unaffected side, and nasal deviation towards the normal side.

Posterior plagiocephaly (premature fusion of 1 lambdoid suture):

- Frontal and occipital bossing, ipsilateral ear moved downward, from above the head looks like a trapeze.

Trigonocephaly (premature fusion of the metopic suture):

- The forehead is narrow and pointed; the head has a triangular shape viewed from above, hypotelorism.

Brachycephaly (bi-coronal premature fusion):

- Short skull, forehead, and occiput flattened but the frontal bone is long (vertically) and prominent, hypertelorism, and Harlequin malformation.

Solution for Question 6:

Correct Answer: A) Tay Sachs Disease

Explanation:

- Tay-Sachs disease is a rare, inherited disorder that affects the nervous system. It is caused by a deficiency in an enzyme called Hexosaminidase A (Hex-A).
- This enzyme is crucial for breaking down a type of fat called GM2 ganglioside, which accumulates in the nerve cells of the brain and spinal cord when the enzyme is deficient.
- Macrocephaly is a notable feature in the early stages of Tay-Sachs disease.
- Other common clinical features of Tay-Sachs disease include:

Developmental Delays

- Neurological Decline: Symptoms include irritability, muscle weakness, and a decreased response to stimuli.
- Visual Impairments: The disease can lead to a "cherry-red spot" in the retina, visible upon eye examination.

The onset of Tay-Sachs disease typically occurs between 3 to 6 months of age.

Infantile Hydrocephalus (Option B):

- This condition involves an excess accumulation of cerebrospinal fluid in the brain, which also causes macrocephaly.
- However, hydrocephalus is often accompanied by other signs such as vomiting, irritability, and an abnormal examination of cerebrospinal fluid via imaging studies.

Congenital Brain Malformation (Option C):

- This may present with developmental delays and abnormal head size since birth and is usually identified through imaging studies showing structural brain anomalies.

Rett Syndrome (Option D):

- This is a genetic disorder primarily affecting females, characterized by normal early development followed by a period of regression with loss of motor skills and purposeful hand movements.
- While macrocephaly can be observed in Rett syndrome, it typically presents with a pattern of regression and other specific clinical features like repetitive hand movements and loss of communication skills.

Solution for Question 7:

Correct Answer: B) Down Syndrome

Explanation:

- Down Syndrome is a genetic condition resulting from trisomy 21.
- It is a primary genetic condition that presents with microcephaly among other features.

Acquired (Secondary) causes of Microcephaly

Congenital Infections	Zika Virus	Small size, ocular anomalies.
Cytomegalovirus	Growth retardation, neurological issues.	
Rubella	Growth retardation, heart disease, cataracts.	
Toxoplasmosis	Neurological symptoms, hydrocephalus.	
Drugs and Other Causes	Fetal Alcohol Syndrome	Facial abnormalities, heart disease.
Fetal Hydantoin Syndrome (Phenytoin exposure)	Growth delay, cleft lip/palate, cardiac defects (VSD)	
Radiation Exposure	Severe effects if exposed before 15 weeks gestation.	
Malnutrition and Hypoxia	Variable effects, including neuronal heterotopias and atrophy.	
Metabolic conditions	Phenylketonuria Niemann pick disease Gaucher's disease	Musty odor urine. Cherry red spots are seen in macula and Sphingomyelin deposition. Glucocerebrosidase deficiency, Erlenmeyer flask deformity
Syndromes	Rett syndrome Angelman Syndrome Seckel syndrome	X-linked recessive, Development regression Bird-headed dwarfism, Bead like nose
Clinical Features	Receding forehead, overriding cranial sutures, intellectual disability.	

Diagnosis	Detailed history Regular measurement of HC; is critical for early diagnosis. MRI for brain abnormalities; CT for intracerebral calcification. Screening for infections (TORCH) and metabolic conditions.	

Solution for Question 8:

orrect Answer: C) Normal variation of head size

Explanation:

- A head circumference below -2SD without other abnormal findings can be a normal variation in head size.
- Not all cases of microcephaly are due to pathological conditions; some variations in head size are within the normal range for the population.

Microcephaly

Occipitofrontal circumference <-3 SD for the mean for age, gender, and sex.

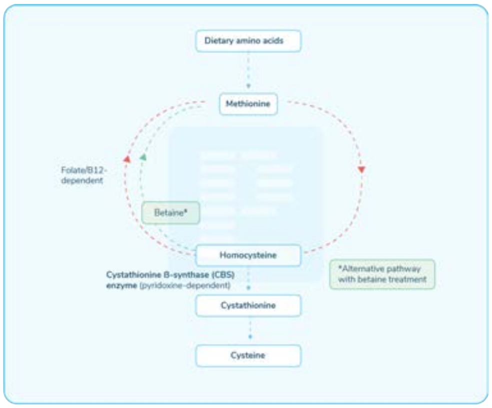
Classification	Causes
Primary (Genetic Causes)	Familial (autosomal recessive) Autosomal dominant Syndromic associations (e.g., Down Syndrome (Option D), Edwards Syndrome, Cri-du-Chat Syndrome).
Secondary (Nongenetic)	Congenital infections (Zika Virus (Option B), Cytomegalovirus, Rubella, Toxoplasmosis) Drugs (Fetal Alcohol Syndrome, Fetal Hydantoin Syndrome), and Environmental factors (Radiation, Malnutrition, Hypoxic-Ischemic Encephalopathy). Metabolic disorders (Phenylketonuria, Lysosomal storage disorders)
Acquired Causes	Postnatal factors (Severe Hypoglycemia, Rett Syndrome, HIV Infection, Seckel Syndrome (Option A)).

Solution for Question 9:

Correct Answer: A) Methionine

Explanation:

Homocystinuria is primarily caused by a deficiency in cystathionine beta-synthase, an enzyme crucial in the metabolism of the amino acid methionine.



Homocystinuria

Etiology	Two variants: a. B6 (pyridoxine)-Responsive b. B6 - Non-responsive homocystinuria.
Pathophysiology	Deficiency of cystathionine beta-synthase (CBS) leading to homocysteine accumulation. Homocysteine accumulation adversely affects the central nervous system, blood vessels, skin, joints, and skeleton.
Clinical Features	Age of Presentation: 3-4 years old. Developmental Delay/Intellectual Disability Ectopia Lentis (dislocated lenses) and severe myopia. Skeletal Abnormalities: Tall stature, marfanoid habitus, pectus carinatum. Thromboembolism: Major cause of early death and morbidity. Additional Symptoms: Seizures, psychiatric issues, extrapyramidal signs, dystonia, hypopigmentation, pancreatitis.
Diagnosis	Plasma & Urine Analysis: Elevated plasma homocysteine (>100 µmol/L), high methionine (>50 µmol/L), increased urine homocysteine. Screening test: Urine Nitroprusside Test. Confirmatory Testing: CBS enzyme activity or molecular testing for CBS gene mutations.

Treatment	Diet: Lifelong methionine-restricted diet with regular metabolic monitoring. Vitamin B6 (Pyridoxine) Oral Folic Acid. Vitamin B12. Oral Betaine.

Phenylalanine (Option B) is an essential amino acid involved in phenylketonuria (PKU) PKU results from a deficiency in phenylalanine hydroxylase, leading to elevated phenylalanine levels.

Tryptophan (Option C) is an essential amino acid related to disorders like Hartnup disorder and pellagra (niacin deficiency).

Valine (Option D) is an essential amino acid involved in maple syrup urine disease (MSUD) when deficient or improperly metabolized.

Solution for Question 10:

Correct answer C) 1,2,4,5,6.

Explanation:

Tall stature

Table rendering failed

Solution for Question 11:

Correct Answer: C) The mutation affects collagen type I, affecting bones-contributing to their fragility.

Explanation:

The COL1A1 gene encodes for collagen type I, which is crucial for bone strength and structure. Mutations in this gene lead to osteogenesis imperfecta, characterized by fragile bones, blue sclera, and dental issues.

Osteogenesis Imperfecta (OI)

Table rendering failed

Mutation affects collagen type II, leading to bone fractures (Option A): Type 2 collagen defects affect cartilage

Mutation affects collagen type III, resulting in the blue sclera (Option B): Type 3 collagen defects affect the blood vessels.

Mutation affects collagen type IV, resulting in dental abnormalities (Option D): Type 4 collagen defects affect the basement membrane.

Solution for Question 12:

Correct answer: A) Achondroplasia

Explanation:

- Disproportionate short limb dwarfism along with normal bone mineralization and specific genetic mutation is likely to be achondroplasia.
- Achondroplasia is characterized by disproportionate short stature, with a normal-sized trunk and shortened long bones.
- The genetic mutation commonly associated with achondroplasia is in the FGFR3 gene.
- Achondroplasia is usually diagnosed through physical examination and confirmed with genetic testing to identify mutations in the FGFR3 gene.
- Management focuses on addressing associated health issues and improving quality of life.
- This includes regular monitoring for complications like spinal problems or ear infections, physical therapy to enhance mobility and strength, and sometimes surgical interventions to address skeletal deformities.
- Growth hormone therapy is not effective for achondroplasia.

Turner Syndrome (Option A) Short stature with additional features like a webbed neck and low-set ears; affects females and involves a missing X chromosome.

Growth Hormone Deficiency (Option C) causes proportionate short stature, not disproportionate; treated with growth hormone therapy, which is ineffective for achondroplasia.

Hypothyroidism (Option D) causes growth delays but with other symptoms like dry skin and fatigue; does not lead to disproportionate body proportions.

Solution for Question 13:

Correct answer: B) US = LS

Explanation:

Growth hormone deficiency is a type of proportionate short stature and hence usually Upper segment = Lower Segment (Proportionately to normal individuals).

- Short stature in growth hormone deficiency (GHD) occurs because growth hormone (GH) is crucial for normal growth and development.
- GH stimulates the liver to produce insulin-like growth factor 1 (IGF-1), which in turn promotes the growth of bones and tissues.
- In GHD, this process is impaired, leading to reduced growth velocity and shorter height compared to peers.
- Children with GHD often present with significantly delayed bone age, meaning their skeletal development lags behind their chronological age.
- Early diagnosis and treatment with recombinant human growth hormone can help normalize growth patterns and improve final height outcomes.

Other causes of proportionate short stature include

- Genetic syndromes: Noonan syndrome, Prader-Willi syndrome, and 3M syndrome,
- Environmental pollutants: lead, cadmium, hexachlorobenzene, and polychlorinated biphenyl
- Family history

US> LS (Option A) is seen in the case of Achondroplasia, Hypothyroidism, and Turner syndrome.

US< LS (Option C) is seen in the case of spondyloepiphyseal dysplasias, and Morquio syndrome.

- US> LS is seen in cases of Achondroplasia, Hypothyroidism, and Turner syndrome.
- US< LS is seen in cases of spondyloepiphyseal dysplasias, and Morquio syndrome.

Solution for Question 14:

Correct answer: C) 181.5 cm

Explanation:

The formula for mid-parental height for males is

MPH = [(Maternal + Paternal height in cm) / 2] + 6.5

For this case scenario, it becomes,

MPH = [(168 + 182) / 2] + 6.5

= 350/2 + 6.5

= 175 + 6.5

= 181.5 cm

In the case of a female child, the formula becomes,

MPH = [(Maternal + Paternal height in cm) / 2] - 6.5

MPH is based on the genetic potential of the child.

Solution for Question 15:

Correct answer: A) Familial short stature

Explanation:

Familial short stature: This is consistent with the child's height being below 2 SD, especially given that her parents are also short, suggesting genetic factors rather than a medical condition.

Familial short stature (FSS) is when a child's height is below average due to genetic factors, specifically the height of their parents.

- Genetics: Short stature runs in the family.
- Growth Pattern: Normal growth rate but consistently below the average height percentile.

- Development: Normal development with no underlying health issues.
- Final Height: Typically reaches a final height consistent with genetic potential but remains lower than average normal height.

Diagnosis: Based on family history and growth patterns, after ruling out other causes of short stature.

Growth hormone deficiency (Option B) & Cushing syndrome (Option D): These typically present with a significantly lower height percentile and delayed growth velocity, not usually seen with normal development and no other abnormalities.

Turner syndrome (Option C) This condition usually presents with specific physical features and typically becomes more apparent with delayed puberty, not commonly diagnosed in a 2-year-old with normal growth and development.

Solution for Question 16:

Correct Answer: B) 0- 8 weeks

Explanation:

In human development, the term "embryo" is used to describe the early stage of development from fertilization from 3rd week until the end of the 8th week of gestation.

Growth phases

Growth Phase	Age Range
Zygote	Product of conception
Embryo	1st 8 weeks
Fetus	9 weeks to birth
Perinatal period	22 weeks to 7 days
Newborn	0 to 28 days
Infant	0-12 months(1st year of life)
Toddler	1-3 years
Preschool	3-6 years
Adolescence	10-19 years

Solution for Question 17:

Correct Answer: D) 1, 2, 4

Explanation:

- Delayed dentition refers to the condition where no teeth erupt by the age of 13 months.
- Typically, primary teeth begin to erupt around 6 months of age.

Causes:

Delayed Dentition (Mnemonic - FRIED Chop)	Natal Teeth (Mnemonic - PEES)
Familial Rickets Idiopathic, Incontinentia pigmenti Endocrine causes Congenital hypopituitarism Congenital hypothyroidism Congenital hypoparathyroidism Down syndrome Cleidocranial dysostosis	Pierre Robin sequence Ellis van Creveld syndrome Epidermolysis bullosa Sotos syndrome

Solution for Question 18:

Correct Answer: B) Midway between inspiration and expiration

Explanation:

The chest circumference is measured at the level of nipples, midway between inspiration and expiration.

- At birth, head circumference (HC) > chest circumference (CC)
- By 9 months to 1 year, HC = CC
- In a normal child beyond 1 year of age, CC > HC

Causes of reduced chest circumference: Skeletal dysplasias, pectus excavatum, malnutrition, Asphyxiating thoracic dystrophy.

Causes of increased chest circumference: Pectus carinatum, obesity, asthma

Solution for Question 19:

Correct Answer: C) 47 cm

Explanation:

The head circumference (HC) at one year is 47 cm.

Period	Rate of increase in Head Circumference
0-3 months	2 cm/month
3-6 months	1 cm/month
6-12 months	0.5 cm/month
1-3 years	0.2 cm/month

Considering the above calculations,

HC at 1 year = 35 cm + (2 cm x 3 months) + (1 cm x 3 months) + (0.5 cm x 6 months)

= 35 + 6 + 3 + 3

= 47 cm

Solution for Question 20:

Correct Answer: A) Using the child's arm span

Explanation:

- In situations where direct height measurement using a stadiometer is challenging, using the arm span is a common and reliable alternative.
- Ideally, height is measured using an infantometer in children <2 years of age, and a stadiometer is used in children >2 years of age.
- In conditions such as scoliosis where a stadiometer cannot be used, alternatives such as arm span and upper segment to lower segment (US:LS) ratio are some of the reliable alternatives.

Arm span

- Measured on arms outstretched at 90 degrees to the body from the tip of the middle finger of one hand to the tip of the middle finger of the other hand.
- Arm span is shorter than the length by 2.5 cm at birth, equals height at 11 years, and after that is slightly (usually, <1 cm) greater than height.

Estimating height based on parental height (Option B) often refers to predicting a child’s health using a ‘mid-parental height’ formula, which is more of a long-term growth prediction rather than an accurate method for determining the current height.

Head circumference (Option C) primarily reflects brain development, particularly in infants and young children. It does not provide an accurate estimate of overall height.

Mid-upper arm circumference (Option D) indicates the nutritional status of a child and is not used in assessing the child’s height.

- Height estimation does not correlate with circumferential measurements like MUAC.

Solution for Question 21:

Correct Answer: B) 2 years

Explanation :

Generally, a child's height at age 2 years is half of their predicted adult height. This milestone is often used as a reference point in pediatrics to estimate the future growth potential of the child.

- Normal length at birth: 50cm.
- In the first year, there is an increase of 25cm, so at 1 year the height is 75cm.
- In the second year there is an increase of 10-15cm, so at 2 years the height is 85-90cm.
- There is a steady increase of 6cm/year till 12 years.

- Height doubles by 4 years
- Height triples by 12 years
- Half of adult height in a normal child (80-85cm) is achieved by 2 years (18-24 months).

AGE	LENGTH
At birth	50cm
By 3 months	60cm
By 9 months (Option C)	70cm
By 1 year (Option A)	75cm
At 2 years	90cm
At 4-4.5 years (Option D)	100cm

Solution for Question 22:

Correct Answer: C) 20 kg

Explanation :

The formula for estimating the weight of a child is given by:

Weight (kg) = 2 × (age in years) + 8

For a 6-year-old girl:

Weight = 2 × 6 + 8 = 12 + 8 = 20 kg

Thus, according to this formula, the estimated weight for this 6-year-old girl is 20 kg.

Weech's formula to estimate the expected weight and height of children using age as a variable.

For <1 year :	x+9 ÷ 2 (Where x is age in months)
For 1-6 years :	2x + 8 (Where x is age in years)
For 7-12 years :	7x - 5 ÷ 2 (Where x is age in years)

Solution for Question 23:

Correct Answer: A) Skeletal growth

Explanation :

The degree of skeletal maturation closely correlates with the degree of sexual maturation.

- During puberty, the hormonal changes associated with gonadal maturation trigger growth spurts and skeletal changes, such as the advancement of bone age and the closure of growth plates.
- Skeletal growth is a continuous process occurring throughout childhood and adolescence.
- It is steady until the pubertal growth spurt when it accelerates and subsequently slows considerably.
- The skeleton matures once the epiphysis or growth plates at the end of long bones fuse to the shaft or diaphysis.
- This occurs by about 18 years in girls and 20-22 years in boys.

Solution for Question 24:

Correct Answer: B) Somatic growth

Explanation :

In this clinical scenario, the sigmoid (S-shaped) curve primarily indicates the pattern of somatic growth, which refers to the child's height growth during childhood and adolescence.

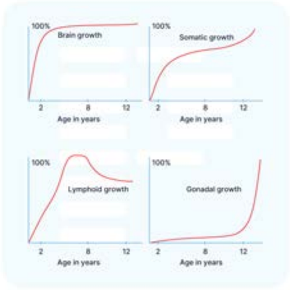
The growth pattern of every individual is unique:

- The order of growth is cephalocaudal and distal to proximal.
- During fetal life, the growth of the head occurs before that of the neck, and arms grow before the legs.
- Distal body parts, such as hands, increase in size before the upper arms.
- In postnatal life, the growth of the head slows down, but limbs continue to grow rapidly.

Different Tissues Grow at Different Rates :

BRAIN GROWTH	The brain enlarges rapidly during the latter months of fetal life and early months of postnatal life.

	At birth, the head size is about 65-70% of the expected head size in adults. It reaches 90% of the adult head size in two years.
GROWTH OF GONADS	Gonadal growth is dormant during childhood and becomes conspicuous during puberty.
GROWTH OF BODY FAT AND MUSCLE MASS	Body tissues can be divided into fat and fat-free (lean) components. The lean body mass includes muscle tissue, internal organs, and skeleton. The growth in lean body mass is primarily due to increased skeletal and muscle mass. Lean body mass correlates closely with stature.
SKELETAL GROWTH	Skeletal growth is a continuous process occurring throughout childhood and adolescence. It is steady until the pubertal growth spurt when it accelerates and subsequently slows considerably. The skeleton matures once the epiphysis or growth plates at the end of long bones fuse to the shaft or diaphysis. This occurs by about 18 years in girls and 20-22 years in boys.
LYMPHOID GROWTH	The growth of lymphoid tissue is most notable during the mid-childhood. During this period, the lymphoid tissue is overgrown, and its mass may appear larger than that of a fully mature adult.



Solution for Question 25:

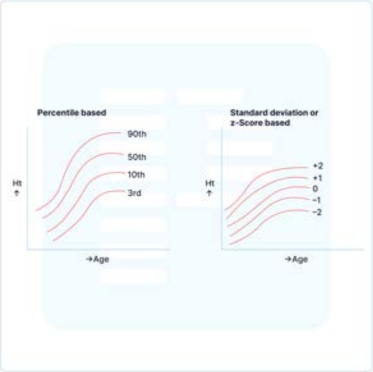
Correct Answer : D) None of the above

Explanation

The WHO growth chart is a standard tool used for monitoring child growth and development. It includes:

- Allowed normal range of variation in observations is conventionally taken as values between the 3rd and 97th percentile curves.
- Growth charts are based on the Multicentre Growth Reference Study (MGRS) conducted by WHO.
- They published new growth charts for infants and children up to 5 years of age in 2006.
- The MGRS was a community-based, multi-country project conducted in Brazil, Ghana, India, Norway, Oman, and the United States.
- Growth standards are not available for children older than five years.

There can be 2 types of growth charts



Solution for Question 26:

Correct answer: D) WHO Growth Charts

Explanation:

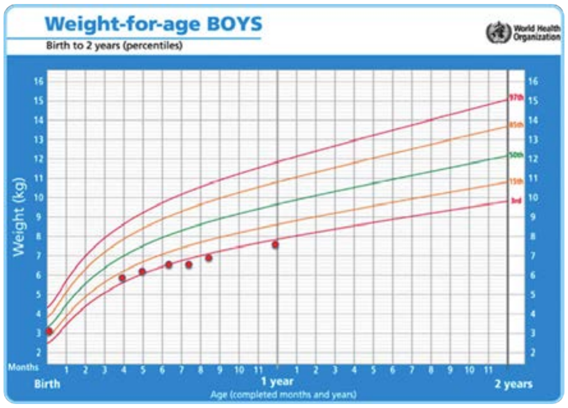
WHO Growth Charts

- Designed specifically for tracking the growth and development of children under 5 years of age globally.

- Growth charts are derived from the Multicentre Growth Reference Study (MGRS) carried out by the WHO.
- This study was a community-based, multi-country initiative that took place in Brazil, Ghana, India, Norway, Oman, and the United States.
- Features: Provide accurate references for weight, length, and head circumference, reflecting normal growth patterns in children.

Percentile Curves: The growth chart displays curves representing percentiles (3rd, 15th, 50th, 85th, and 97th).

- 3rd Percentile: Low end of normal growth.
- 50th Percentile: Median or average growth.
- 97th Percentile: High end of normal growth.



IAP growth charts (Option A) are used most commonly for children aged 5 years and above.

Centers for Disease Control and Prevention (CDC) Growth Charts (Option B) are commonly used in the United States, they are generally more suitable for children aged 2 years and older.

- They do not specifically focus on the under-5 age group as comprehensively.

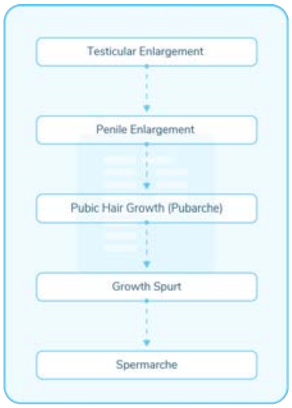
National Center for Health Statistics (NCHS) Growth Charts (Option C) are older and have been replaced by the CDC charts for children aged 2 years and older in the U.S.

- They are less commonly used for under-5.

Solution for Question 27:

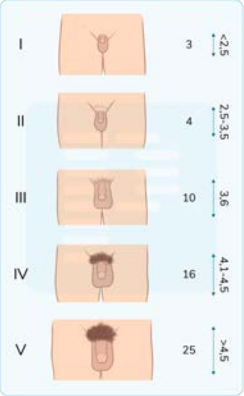
Correct Answer: B) 2,1,3,,4

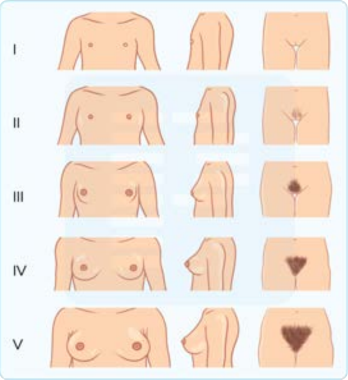
Explanation:

Girls	Boys
Onset of Puberty Age: Around 10 years (range 8-12 years) Duration: 3-5 years	Onset of Puberty Age: Around 11.5 years (range 9-14 years)
Sequence of Pubertal Changes	Sequence of Pubertal Changes
	

Tanner staging

- It is used to assess the stages of physical development during puberty.
- It provides a standardized method to describe and track the progress of pubertal changes





Solution for Question 28:

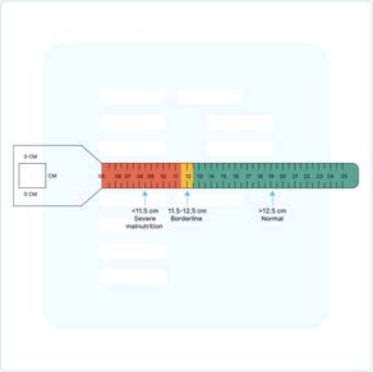
Correct answer: A) Harpenden caliper

Explanation:

The best method used to assess skin fold thickness, to assess the nutritional status of a child, is the Harpenden caliper.

Description	An instrument designed to measure skin fold thickness. Consists of two arms with a calibrated scale to gauge skinfold depth.
Measurement Sites	Commonly used on sites such as the triceps, subscapular, supra iliac, and thigh.
Purpose	Assess Nutritional Status: Measures subcutaneous fat to estimate body fat percentage. Monitor Growth: Helps in tracking growth and nutritional changes over time.
Interpretation	Increased Skin Fold Thickness: Often indicates higher body fat and potential overnutrition. Decreased Skin Fold Thickness: Signify reduced body fat, malnutrition, or undernutrition. Comparison to Norms: Results are compared to standard charts or norms for age and sex to assess body fat levels and nutritional status.

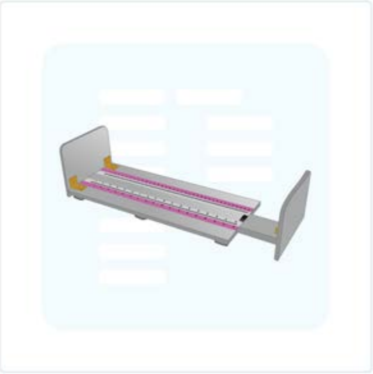
Shakir tape (Option B) is used to measure various body circumferences, such as waist, hip, and chest, to assess growth, nutritional status, and body composition, not skin fold thickness.



Salters scale (Option C) is used to Monitor Growth and assess Nutritional Status by measuring body weight, which can indicate undernutrition, overnutrition, or other health conditions.

- Body fat distribution is not assessed by this method.

Infantometer (Option D) is used to monitor Growth by measuring their length and assess Developmental Milestones by evaluating whether an infant's length is within the expected range for their age.



Solution for Question 29:

Correct Answer: C) Dehydration

Explanation:

MUAC does not assess dehydration; it primarily evaluates body fat reserves, nutritional status, and growth monitoring.

Mid-Upper Arm Circumference (MUAC)

Measurement Location	Measured at the midpoint of the upper arm between the shoulder and the elbow.
Purpose	To assess nutritional status and growth by evaluating body fat and muscle mass, particularly to identify malnutrition or undernutrition in children (6-59 months).
Applications	Useful for monitoring growth, detecting malnutrition, and assessing body fat reserves.
Advantages	Provides a quick, straightforward, and effective indicator of nutritional health without needing specialized equipment.
Interpretation	<11.5cm - Severe acute malnutrition. 11.5 cm -12.5cm - Moderate acute malnutrition.

Previous Year Questions

1. While evaluating a 3-year-old with short stature, height is measured at 90cm and the lower segment is 45 cm. What is the likely diagnosis?

A. Achondroplasia

B. Congenital hypothyroidism

C. Rickets

D. Spondyloepiphyseal dysplasia

---

2. At which of the following ages does maximum brain growth occur?

A. 1 year

B. 2 years

C. 3 years

D. 4 years

---

3. How do you calculate the mid-parental height for a male child with a father who is 183 cm tall and a mother who is 175 cm tall?

A. 185.5

B. 169.5

C. 192.5

D. 195.5

---

4. A 10 year old child presented to the clinic with complaints of recurrent infection. Physical examination shows low height for age and Z line score is <-2SD. The parents also have a history of incorrect feeding method. What is the appropriate term for the above mentioned condition?

A. Stunting

B. Wasting

C. Malnutrition

D. Kwashiorkor

---

5. What is the first primary tooth to erupt?

A. Upper central incisor

B. Lower central incisor

C. Upper canine

D. Lower canine

---

6. Which alternative method can be used for the procedure shown in the image?

A. Head circumference

B. Arm span

C. Crown-rump length

D. Knee height

7. A 10-year-old child is brought to the OPD with a history of repeated fractures for the past 2 years. On examination, blue sclera and dental abnormalities are noted. There is a family history of similar complaints. Which of the following is the diagnosis?

A. Scurvy

B. Rickets

C. Osteogenesis imperfecta

D. Osteomalacia

8. At what age does stranger anxiety appear?

A. 7 months

B. 9 months

C. 12 months

D. 15 months

9. At what age does transfer of one object to another hand happen?

A. 6 months

B. 9 months

C. 7 months

D. 12 months

10. Conjunctival xerosis is caused by which vitamin deficiency?

A. Vitamin A

B. Vitamin K

C. Vitamin C

D. Vitamin D

11. Feature of angular cheilitis, stomatitis, and glossitis is seen in which vitamin deficiency?

A. Niacin

B. Riboflavin

C. Cobalamin

D. Vitamin C

12. Which of the following statements is true regarding the following image? It is called Shelter's tape It is used to assess severe acute malnutrition A reading of 13.5 to 14.5 cm indicates undernourishment It is useful mainly for frontline field workers

A. 1, 3

B. 2, 4

C. 2, 3

D. 1, 4

13. Specific sign of kwashiorkor is:

A. Pitting edema

B. Weight loss

C. Flag sign

D. Muscle wasting

14. From the list below, who is considered a baby at risk? Infant born with a weight of 2.5 kg Infant receiving artificial feeding Infant of a working mother or single parent Infant born 1 week before expected date of delivery Second-born infant

A. 2,3

B. 1,2,3,4

C. 4,5

D. 1,4

15. At what age is it recommended to introduce solid foods to an infant while continuing breastfeeding?

A. 2 months

B. 6 months

C. 12 months

D. 18 months

16. Where would you place the pulse oximeter to measure preductal oxygen saturation in an infant who was born 3 minutes ago?

A. Left upper limb

B. Left lower limb

C. Right upper limb

D. Right lower limb

17. At what age does a child reach half of their adult height?

A. 12- 18 months

B. 20 -24 months

C. 32 - 36 months

D. 40 - 48 months

18. A child is brought to the pediatrician with short stature. On evaluation, growth hormone levels are found to be high and insulin-like growth factor (IGF-1) levels are low. What could be the abnormality here?

A. GH deficiency

B. GHRH deficiency

C. GH receptor resistance

D. IGF-1 deficiency

19. How many primary teeth does a 3-year-old child have?

A. 20

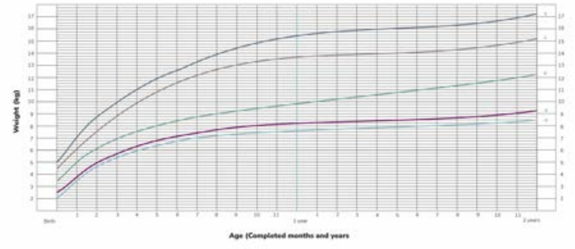
B. 24

C. 16

D. 32

---

20. A 16-month-old boy weighs 8 kg. A WHO growth chart (weight-for-age) is provided below. What is the most appropriate advice for this child?



A. No malnutrition - assure the mother

B. Mild malnutrition - home treatment

C. Severe malnutrition - refer to nutrition rehabilitation center

D. Moderate malnutrition - teach the mother on how to feed

---

21. What is the most reliable indicator of proper growth in an infant weighing 2.8 kg at birth?

A. Increase in length of 25 centimetres in the first year

B. Weight gain of 300 grams per month till 1 year

C. Anterior fontanelle closure by 6 months of age

D. Weight under the 75th percentile and height under the 25th percentile

---

22. A 6-year-old child is brought for evaluation of short stature. His height is comparable with his parents' height and his bone age corresponds to his

chronological age. What is the diagnosis?

A. Familial short stature

B. Constitutional delay

C. Undernutrition

D. Growth hormone deficiency

23. Gonadal growth corresponds with _____.

A. Brain

B. Skeletal system

C. Lymphoid growth

D. Dental growth

24. A mother expresses concern about her child's unwillingness to eat properly. Her 15-month-old son's length falls 2 standard deviations below the median for his age. What is the most probable medical condition?

A. Acute malnutrition

B. Severe wasting

C. Moderately underweight

D. Moderate stunting

25. A 3-year-old child is brought to the OPD for a regular check-up. His growth chart shows a BMI between the 85th and 95th percentile. Which of the following categories would you place him in?

A. Moderate wasting

B. Severe acute malnutrition

C. Overweight

D. Obese

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	1
Question 4	1
Question 5	2
Question 6	2
Question 7	3
Question 8	1
Question 9	3
Question 10	1
Question 11	2
Question 12	2
Question 13	1
Question 14	1
Question 15	2
Question 16	3
Question 17	2
Question 18	3
Question 19	1
Question 20	4
Question 21	1
Question 22	1

Question 23	2
Question 24	4
Question 25	3

Solution for Question 1:

Correct Answer: D) Spondyloepiphyseal dysplasia

Explanation:

In the given scenario, the upper segment: lower segment ratio of the child is 1:1 signifying upper segment short stature which is seen in spondyloepiphyseal dysplasia.

The normal US: LS ratio for a 3-year-old child is 1.3: 1.

Here the height of the child is 90cm with LS = 45cm.

Hence the US = 45cm.

US: LS = 1:1.

This means that the upper segment is shorter than normal.

- Spondyloepiphyseal dysplasia causes short-trunk dwarfism involving the vertebrae and proximal epiphyseal centres.
- The distal epiphyses are spared in this condition.

Proportionate Short Stature	Conditions Causing Short Stature with US > LS	Conditions Causing Short Stature with US < LS
Growth hormone deficiency Genetic syndromes: Noonan syndrome, Prader-Willi syndrome, 3M syndrome Environmental pollutants: Lead, calcium, hexachlorobenzene, polychlorinated biphenyl Family history	Achondroplasia (Option A) Congenital hypothyroidism (Option B) Turner syndrome Diastrophic dysplasia Rickets (Option C)	Spondyliepiphyseal dysplasia (Option D) Morquio syndrome

Solution for Question 2:

Correct Option A - 1 year:

- The first year of life is marked by the most rapid brain growth. By the end of the first year, the brain reaches about 60% of its adult size. This period is critical for neurodevelopment, with significant increases in brain volume and neural connections.

Incorrect options:

Option B - 2 years:

- Brain growth continues, but the rate slows down after the first year. By 2 years of age, the brain is approximately 75% of its adult size, but the most intense phase of growth has already occurred.

Option C - 3 years:

- By 3 years, the brain is about 80-85% of its adult size. Growth is still significant, but it is not as rapid as during the first year.

Option D - 4 years:

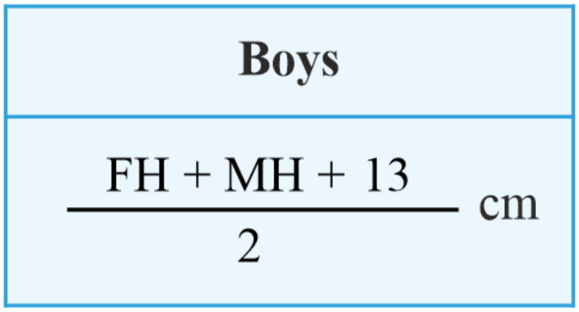
- The brain's growth rate further decreases, reaching around 90% of its adult size by this age. Most brain development in terms of structure is completed, but functional maturation continues.

- - -

Solution for Question 3:

Correct Option A -185.5 cm:

- Mid-parental height for a male child is calculated as follows:



Boys
$\frac{FH + MH + 13}{2}$ cm

- 183+175+13/2 = 185.5 cm

Incorrect Options:

- Options B, C, & D are incorrect, as explained above.

Solution for Question 4:

Correct Option A - Stunting:

- This is the correct answer. Stunting, or chronic malnutrition, is defined as low height-for-age. This condition is a result of long-term nutritional deprivation, often exacerbated by recurrent infections. A Z-score of less than -2 standard deviations (SD) below the median of the World Health Organization's child growth standards is indicative of stunting. The history of incorrect feeding method provided by the parents further supports this diagnosis.

Incorrect Options:

Option B - Wasting: Wasting, or acute malnutrition, is defined as low weight-for-height. It's a sign of recent and severe weight loss, which does not seem to be the case with the child described in the question.

Option C - Malnutrition: While the child is indeed malnourished, this term is broad and doesn't specify the type of malnutrition. In this case, the child's low height-for-age and Z-score of <-2SD indicate stunting, a specific form of malnutrition.

Option D - Kwashiorkor: Kwashiorkor is a form of severe malnutrition characterized by edema (swelling), discolored hair, and dermatosis. It's typically caused by a diet high in carbohydrates but deficient in protein. The question does not provide any information suggesting the child has these symptoms.

Solution for Question 5:

Correct Option B - Lower central incisor:

- The lower central incisor (primary) of the mandibular teeth is the first tooth to erupt, and it erupts between the ages of 6 and 12 months.
- Initial calcification of all of the primary teeth occurs between 14 and 19 weeks in utero.
- All primary incisors are present by 12 months of age. There are 2 lower central incisors in number.
- These teeth then exfoliate at 6 to 7 years of age.

Incorrect Options:

Option A - Upper central incisors: Upper central incisors of the upper teeth (maxillary teeth) erupt at 8 to 12 months and exfoliate at 6 to 7 years.

Option C - Upper canine primary: Upper canine primary teeth canines are also morphologically similar to permanent teeth canines. Canine teeth erupt much later than incisors.

Option D - Lower canine: Lower canine Canine (cuspid) teeth then tend to emerge at 16-23 months.

Solution for Question 6:

Correct Option B - Arm span:

- The image shows the measurement of a child's height using a stadiometer.
- Arm span is an alternative parameter for assessing a child’s height.
- Arm span refers to the distance between the fingertips of one hand and the fingertips of the other hand when the arms are outstretched horizontally.
- Arm span is highly correlated with height, and is a reliable predictor of height.

Incorrect Options:

Option A - Head circumference:

- Head circumference is a measurement commonly used to assess the growth and development of an infant's brain and skull. It is measured in children up to the age of 2 years. It is not a suitable alternative method for measuring the height of a child, as it does not provide an accurate estimation of height.

Option C - Crown-rump length:

- Crown-rump length (CRL) is a measurement used during early pregnancy to estimate gestational age and assess fetal growth. It is the measurement from the top of the head (crown) to the bottom of the buttocks (rump).
- CRL is not applicable for measuring the height of a child, as it is specifically used for prenatal assessments.

Option D - Knee height:

- Knee height is a measurement that can be used to estimate stature or height in older adults or individuals who have difficulty standing upright.
- It is often used in situations where direct height measurement is not possible, such as in bedridden or wheelchair-bound individuals.

Solution for Question 7:

Correct Option C - Osteogenesis imperfecta:

- This child with recurrent fractures with ocular and dental abnormalities most likely has an underlying osteogenesis imperfecta.
- OI is primarily caused by mutations in COL1A1 and COL1A2 genes, which encode type I collagen alpha chains. Collagen is essential for providing strength and flexibility to bones and other connective tissues.
- Clinical features: Increased bone fragility, resulting in frequent fractures, even with minimal trauma or sometimes occurring spontaneously Fractures may be present at birth or occur prenatally, leading to skeletal deformities Blue or grayish tint of the sclera (the white part of the eye) due to the underlying translucency of collagen Dental abnormalities (such as brittle teeth, enamel defects, and increased susceptibility to cavities) Hearing loss (which can be conductive or sensorineural) Ligament laxity
- Increased bone fragility, resulting in frequent fractures, even with minimal trauma or sometimes occurring spontaneously
- Fractures may be present at birth or occur prenatally, leading to skeletal deformities
- Blue or grayish tint of the sclera (the white part of the eye) due to the underlying translucency of collagen
- Dental abnormalities (such as brittle teeth, enamel defects, and increased susceptibility to cavities)

- Hearing loss (which can be conductive or sensorineural)
- Ligament laxity
- Increased bone fragility, resulting in frequent fractures, even with minimal trauma or sometimes occurring spontaneously
- Fractures may be present at birth or occur prenatally, leading to skeletal deformities
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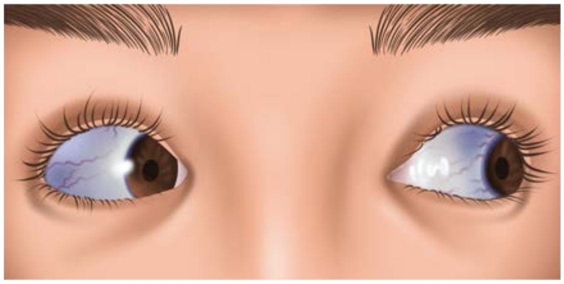


Incorrect Options:

Option A - Scurvy: Scurvy is a vitamin C deficiency disorder. It is characterized by weakness, fatigue, joint pain, easy bruising, gum disease, and poor wound healing.

Option B - Rickets: Rickets leads to impaired bone mineralization, resulting in skeletal deformities. Although rickets can present with fractures, the presence of blue sclera and dental abnormalities are not characteristic.

Option D - Osteomalacia: Osteomalacia is a disorder characterized by softening of the bones due to impaired mineralization. It is caused by vitamin D deficiency in adults.

Solution for Question 8:

Correct Option A - 7 months:

- Stranger anxiety is a social milestone.

- It appears around 6-7 months of age.
- The baby will also be able to recognize strangers.

Incorrect Options:

Option B - 9 months:

- By 9 months the baby waves bye-bye.

Option C - 12 months:

- By 12 months the baby kisses on request and plays a simple ball game.

Option D - 15 months:

- By 15 months the baby points to objects and indicates wet pants.

Solution for Question 9:

Correct Option C - 7 months:

- Transfer of objects from one hand to another is a fine motor skill.
- It appears at 7 months of age.

Incorrect Options:

Option A - 6 months:

- Unidextrous or palmar grasp appears at 6 months.

Option B - 9 months:

- Immature or pincer grasp appears at 9 months.

Option D - 12 months:

- At 12 months, mature or unassisted pincer grasp appears, pulling off caps, mittens, or socks.

Solution for Question 10:

Correct Option A - Vitamin A:

- Conjunctival xerosis (drying of conjunctiva) is the earliest sign of vitamin A deficiency.
- Night blindness is the earliest symptom of vitamin A deficiency.
- Classification of eye lesions due to Vitamin A deficiency.

XN	Night blindness
X1A	Conjuctival Xerosis/Dryness
X1B	Bitot’s spot Bitot’s spot are the triangular spots on the sclera with base of the triangle towards the cornea
X2	Corneal Xerosis
X3	Corneal ulceration/Keratomalacia X3A: Involvement of <1/3rd cornea X3B: Involvement of >1/3rd of cornea
XS	Corneal Scars

Incorrect Options:

Option B - Vitamin K:

- Vitamin K deficiency presents with bruising or mucocutaneous bleeding in older children and hemorrhagic disease in newborns in neonates.

Option C - Vitamin C:

- Vitamin C deficiency presents as scurvy (gum bleeding, scorbutic rosary, painful pseudo paralysis)

Option D - Vitamin D:

- Rickets occur due to Vitamin D deficiency.

Solution for Question 11:

Correct Option B - Riboflavin:

- Riboflavin is Vitamin B2 and its deficiency presents with: Glossitis Cheilosis- inflammation of lips Angular stomatitis- Painful lesions near the angle of the mouth. Seborrheic dermatitis along the nasolabial folds. Photophobia

- Glossitis
- Cheilosis- inflammation of lips
- Angular stomatitis- Painful lesions near the angle of the mouth.
- Seborrheic dermatitis along the nasolabial folds.
- Photophobia
- Glossitis
- Cheilosis- inflammation of lips
- Angular stomatitis- Painful lesions near the angle of the mouth.
- Seborrheic dermatitis along the nasolabial folds.
- Photophobia

Incorrect Options:

Option A - Niacin:

- Niacin deficiency gives rise to Pellagra (4D’s- Diarrhea, Dermatitis, Dementia and Death).

Option C - Cobalamin:

- Cobalamin deficiency manifests with megaloblastic anemia and subacute combined degeneration of the spinal cord.

Option D - Vitamin C:

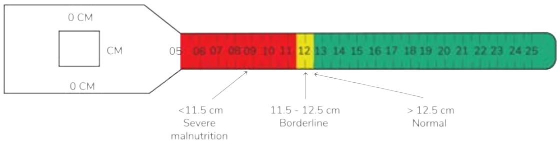
- Vitamin C deficiency presents as scurvy.

Solution for Question 12:

Correct Option B - 2, 4:

- The image shows Shakir's tape, which assesses nutritional status, particularly to screen for severe acute malnutrition in children. It is specifically designed to measure the mid-upper arm circumference (MUAC).
- Tape measurements such as Shakir's tape are designed to be portable, easy to use, and suitable for field settings.
- They are commonly used by frontline field workers, including community health workers, nutritionists, and healthcare professionals, to screen and identify children with severe acute

malnutrition in resource-limited settings.



Incorrect Options:

- Options A, C, and D are incorrect.

Solution for Question 13:

Correct Option A - Pitting edema:

- Kwashiorkor is a disease marked by severe protein malnutrition and bilateral extremity swelling. It usually affects infants and children, most often around the age of weaning through age 5.
- The WHO has a classification system for evaluating malnutrition severity that determines wasting versus kwashiorkor. They use three clinical measures: mid-upper arm circumference (MUAC) weight-for-height/length Z score presence of symmetrical pitting edema
- mid-upper arm circumference (MUAC)
- presence of symmetrical pitting edema
- Children with kwashiorkor have profoundly low levels of albumin.
- Pitting edema is a specific and classic sign of kwashiorkor. It results from a loss of fluid balance between hydrostatic and oncotic pressures across capillary blood vessel walls.
- mid-upper arm circumference (MUAC)
- presence of symmetrical pitting edema

Incorrect Options:

Option B - Weight loss: Weight loss is not a specific sign of this condition.

Option C - Flag sign: It is not a specific sign of kwashiorkor. It can be seen in both kwashiorkor and marasmus

Option D - Muscle wasting: While muscle wasting can occur in severe malnutrition, including kwashiorkor, it is not a specific sign of kwashiorkor alone. Muscle wasting is a more general indicator of malnutrition and can be observed in various forms of protein-energy malnutrition.

Solution for Question 14:

Correct Option A - 2,3:

- Infants who exclusively consume formula may be in danger because they may not get the vital nutrients and antibodies needed for healthy growth and development. Breast milk is thought to be the best food for infants and newborns because it contains all the necessary nutrients and antibodies that fight infections.
- Similarly, children with working mothers or those raised by lone parents may be vulnerable due to a lack of availability of care and assistance.
- For example, working mothers may struggle to nurse their children or make nourishing meals, and single parents may struggle to obtain the money necessary to provide for all of the baby's needs. These elements may cause malnutrition and other health issues.
- Therefore, the babies on artificial feed and working mothers/single parents are at risk.

Incorrect Options:

Option B - 1,2,3,4: Newborns weighing 2.5 kg or less at birth are are not specifically at-risk babies.

Option C - 4,5: A baby that is developing normally and shows no other signs of health issues may not be at risk if born 1 week before the expected date of delivery.

Option D - 1,4: Higher birth order is a risk factor but 2nd in place is not a risk factor.

Solution for Question 15:

Correct Option B - 6 months:

- According to current recommendations by leading health organizations such as the World Health Organization (WHO) and the American Academy of Pediatrics (AAP), solid foods should be introduced to an infant while continuing breastfeeding at around 6 months of age.

Incorrect Options:

Options A, C, and D - 2 months, 12 months and 18 months:

- These ages do not correspond to the age at which solid foods should be initiated.

Solution for Question 16:

Correct Option C - Right upper limb:

- Pre-ductal refers to the circulation before the ductus arteriosus, which is a blood vessel connecting the pulmonary artery to the descending aorta in the fetal circulation.
- In a newborn infant, the ductus arteriosus typically begins to close within minutes to hours after birth.
- To measure preductal oxygen saturation in an infant, the pulse oximeter should be placed on the right upper limb. This is done by attaching the pulse oximeter probe to the infant's right hand or wrist.
- Placing the pulse oximeter on the right upper limb allows for the measurement of oxygen saturation in the right arm before the blood reaches the ductus arteriosus.

Incorrect Options:

Options A, B, and D: These are not the appropriate locations for placing the pulse oximeter to measure preductal oxygen saturation.

Solution for Question 17:

Correct Option B - 20-24 months:

- By the age of 24 months, a child attains the approximate height of half the adult height.

Incorrect Options:

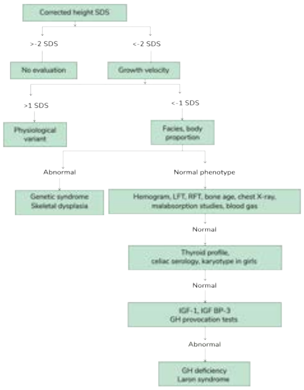
Option A - 12-18 months: By the end of the first birthday, growth velocity slows down compared to the initial velocity, and length is increased by 50% of birth length.

Options C & D - 32-36 months & 40-48 months: There is a rule of 4 for children between the ages of 3 and 4, where an approximately 4-year-old would have a 40-inch height.

Solution for Question 18:

Correct Option C - GH receptor resistance:

- The hormonal features of growth hormone insensitivity (GHI) consist of normal or elevated serum GH levels combined with the inability to generate normal/functional quantities of insulin-like growth factor-1 (IGF-1).
- Circulating concentrations of IGF-1, IGF-II, and IGF-binding protein (BP)-3 are reduced.



Incorrect Options:

- Options A, B, and D are incorrect and can be ruled out based on above explanation.

Solution for Question 19:

Correct Option A - 20:

- An average child develops a full set of 20 primary teeth by age 3.
- The primary teeth begin to shed between the ages of 6 and 7, and permanent teeth appear.

- By age 21, the average person develops 32 teeth, 16 on the upper jaw and 16 on the lower jaw.

	Primary Dentition (milk/temporary teeth)	Secondary Dentition (permanent teeth)
Begin at	6 months	6 years
1 tooth to erupt	Lower central incisor	1st molar
Last tooth	Second molar	3rd molar (or) wisdom tooth
Completes at	2.5-3 years	12 years except for the 3rd molar (18-25 years)
Total no. of teeth	20	28-32
Teeth in each quadrant (ICPM)	I C P M	I C P M
	2 1 0 2	2 1 2 3

Incorrect Options

- Options B, C, and D can be ruled out based on the above explanation.

Solution for Question 20:

Correct Option D - Moderate malnutrition - teach the mother how to feed:

- The child's weight-for-age falls between -2 standard deviations (SD) and -3 SD on the growth chart.
- This child's weight is below the average range for his age, indicating some degree of malnutrition. However, it is not severe enough to warrant immediate referral to a nutrition rehabilitation center.
- Instead, the focus should be on educating the mother or caregiver on proper feeding practices and providing guidance on improving the child's nutritional intake. This may involve promoting a balanced diet, ensuring adequate calorie and nutrient intake, and addressing any feeding difficulties or challenges.

Incorrect Options:

- Options A, B, and C: These options are incorrect based on the above explanation.

Solution for Question 21:

Correct Option A - Increase in length of 25 centimeters in the first year:

- The length at birth is 50 cm (approx), and in the first year, it increases by 25 cm, so at the end of the first year, the height is around 75 cm.
- Length or linear growth is a crucial indicator of overall growth and development. Achieving this milestone suggests that the infant is growing adequately and progressing.

Incorrect Options:

Option B - Weight gain of 300 grams per month till 1 year: While weight gain is an important aspect of growth, it alone may not provide a complete picture of overall growth. Relying solely on weight gain of 300 grams per month may not be sufficient to indicate adequate growth, as it does not account for variations in body composition or length growth.

Option C - Anterior fontanelle closure by 6 months of age: The anterior fontanelle closes between 9 and 18 months.

Option D - Weight under the 75th percentile and height under the 25th percentile: Percentiles are used to compare an infant's growth measurements with a reference population. However, being under the 75th percentile for weight and under the 25th percentile for height does not confirm whether the infant's growth is adequate.

Solution for Question 22:

Correct Option A - Familial short stature:

- In familial short stature, the child follows the family's inherited short stature (shortness).
- The child is short as per definition (height <3rd centile) but is normal according to his own genetic potential determined by the parents' height.
- These children show catch-down growth between birth and 2 years of age, so their height and weight come to lie on their target (mid-parental) centiles by the age of 2.

Incorrect Options:

Option B - Constitutional delay: In constitutional growth delay with delayed adolescence or delayed maturation, the child tends to be shorter than average and enters puberty later than average but is growing at a normal rate. Most of these children tend to eventually grow to approximately the same height as their parents.

Option C - Undernutrition: Undernutrition denotes insufficient energy and nutrient intake to meet a child's needs to maintain good health. It can cause short stature. The height is not comparable with that of the parents and bone age does not correspond to chronological age.

Option D - Growth hormone deficiency: Growth hormone deficiency, also known as dwarfism or pituitary dwarfism, is a condition caused by insufficient amounts of growth hormone in the body. Children with GHD have abnormally short stature with normal body proportions

Solution for Question 23:

Correct Option B - Skeletal system:

- Gonadal growth corresponds with skeletal growth, particularly during puberty.
- Estrogen in females and testosterone in males play a significant role in developing secondary sexual characteristics and contribute to the growth and maturation of the skeleton.
- This hormonal influence leads to changes in bone structure and overall height.

Incorrect Options:

Option A - Brain: Brain undergoes significant development during childhood and adolescence, gonadal growth is not closely associated with the growth of the brain. Various factors, including genetics, nutrition, and environmental factors influence brain development

Option C - Lymphoid growth: Gonadal growth and lymphoid growth are distinct processes. Lymphoid tissues, such as lymph nodes and the spleen, play a role in the immune system but are not directly related to gonadal development.

Option D - Dental growth: Gonadal growth is not unrelated to dental growth. Dental development is a separate process governed by genetics, nutrition, and oral health practices. Teeth develop independently of gonadal growth.

Solution for Question 24:

Correct Option D - Moderate stunting:

- Stunting is defined as a length-for-age more than two standard deviations below the WHO Child Growth Standards median. This indicates that the child has not grown adequately in height and is considered short for age. In this case, the child's length-for-age is ≤ -2 SD below the median, which indicates moderate stunting.
- The reluctance to take feeds appropriately may be one reason for the child's stunted growth. Poor nutrition, frequent infections, and inadequate stimulation can also contribute to stunting.
- Stunting is associated with impaired cognitive development, poor school performance, and reduced economic productivity in adulthood

Incorrect Options:

Option A - Acute malnutrition: Acute malnutrition refers to a recent and severe deficiency of energy, protein, and other nutrients that can lead to wasting and/or edema. The child in this scenario is not showing signs of acute malnutrition.

Option B - Severe wasting: Severe wasting is defined as a weight-for-height greater than three standard deviations below the WHO Child Growth Standards median. The child's weight is not provided in the scenario.

Option C - Moderately underweight: Moderately underweight is defined as a weight-for-age greater than two standard deviations below the WHO Child Growth Standards median. The child's weight is not provided in the scenario.

Solution for Question 25:

Correct Option C - Overweight:

- In children, BMI is calculated using age, sex, height, and weight, and the result is plotted on a growth chart to determine the child’s BMI percentile.
- BMI percentiles in children are classified as follows: Underweight: Less than 5th percentile Healthy weight: 5th percentile to less than 85th percentile Overweight: 85th percentile to less than 95th percentile Obese: Equal to or greater than 95th percentile
- Underweight: Less than 5th percentile
- Healthy weight: 5th percentile to less than 85th percentile
- Overweight: 85th percentile to less than 95th percentile
- Obese: Equal to or greater than 95th percentile
- In this case, the child’s BMI falls between 85th and 95th percentile, which places him in the overweight category. This means that the child’s weight is higher than what is considered healthy for his age, sex, and height.
- Underweight: Less than 5th percentile
- Healthy weight: 5th percentile to less than 85th percentile
- Overweight: 85th percentile to less than 95th percentile
- Obese: Equal to or greater than 95th percentile

Incorrect Options:

- Options A, B and D: The BMI in the given clinical scenario does not fit into any of these categories.

Previous Year Questions

1. When compared to WHO-ORS, low osmolarity ORS has which of the following?

A. Low glucose and high sodium

B. Low glucose and low sodium

C. Low glucose and low potassium

D. Low glucose and high potassium

2. What is the significance of a child transferring an object from one hand to the other?

A. Visual-motor coordination

B. Ability to compare objects

C. Ability to handle small objects

D. Voluntary release

3. According to the expanded New Ballard score, which of the options listed below is considered as a part of the neuromuscular criteria for assessing maturity?

A. Heel-knee test

B. Finger-nose test

C. Scarf sign

D. Cubital angle

4. What is the underlying defect in the congenital malformation shown below?

A. Morulation

B. Neurulation

C. Lateral folding

D. Gastrulation

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5. What is the rate of increase in head circumference in infants less than 3 months of age?

A. 0.5cm/month

B. 1cm/month

C. 2cm/month

D. 4cm/month

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6. What is the likely age of a child who can ride a tricycle, walk up stairs with alternate steps, but cannot hop?

A. 2.5 years

B. 3.5 years

C. 4.5 years

D. 5.5 years

7. During a routine pediatric check-up, a mother brings her 5 month old baby to the clinic. The pediatrician observes that the baby actively looks at his mother’s face and seems to respond with interest as she talks to him, showing recognition and engagement. When is this milestone normally achieved?

A. 2 months

B. 3 months

C. 6 months

D. 9 months

8. Match the following developmental milestones. 1. Transfers objects 2. Social smile 3. Pincer grasp 4. Walks 1-2 steps a. 1-2 months b. 6-7 months c. 12-15 months d. 9-12 months

1. Transfers objects	a. 1-2 months
2. Social smile	b. 6-7 months
3. Pincer grasp	c. 12-15 months
4. Walks 1-2 steps	d. 9-12 months

A. 1-a, 2-b, 3-c, 4-d

B. 1-b, 2-a, 3-d, 4-c

C. 1-c, 2-b, 3-a, 4-d

D. 1-d, 2-c, 3-b, 4-a

9. While you are evaluating a baby, you show him a bright pink teddy bear that he reaches out to with both hands. What is the earliest age by which this milestone is typically achieved?

A. 4 months

B. 3 months

C. 6 months

D. 7 months

10. Which of the following developmental milestones is attained first in infants?

A. Immature pincer grasp

B. Mature pincer grasp

C. Radial palmar grasp

D. Ulnar palmar grasp

11. If a 3-year-old child does not attain one of the following milestones, it is considered developmental delay. Which milestone is being referred to here?

A. Hopping on one leg

B. Drawing a square

C. Feeding by spoon

D. Passing a ball to someone

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	4
Question 5	3
Question 6	2
Question 7	2
Question 8	2
Question 9	1
Question 10	4
Question 11	3

Solution for Question 1:

Correct Option B. Low glucose and low sodium:

- The WHO-ORS formulation, also known as standard ORS, contains a specific ratio of glucose and sodium to promote effective rehydration in individuals with dehydration. The standard ORS has a glucose concentration of 75 mmol/L and a sodium concentration of 75 mmol/L.
- Low osmolarity ORS contains a reduced concentration of both glucose and sodium compared to the standard WHO-ORS.
- The lower glucose levels help prevent osmotic overload, while the lower sodium levels reduce the risk of hypernatremia.

- This formulation is considered safer and more effective in treating dehydration due to diarrhea.

Incorrect options

Option A) Low glucose and high sodium: This option refers to a formulation of ORS that has a reduced glucose concentration and an increased sodium concentration. However, this combination is not consistent with the composition of low osmolarity ORS. High sodium levels in ORS can be detrimental and increase the risk of hypernatremia, especially in vulnerable populations like infants and young children.

Option C) Low glucose and low potassium: This option refers to a formulation of ORS with reduced levels of glucose and potassium. While low glucose is a characteristic of low osmolarity ORS, it still contains a sufficient amount of potassium to address electrolyte imbalances caused by diarrhea. Potassium is an essential electrolyte that helps maintain proper fluid and electrolyte balance in the body.

Option D) Low glucose and high potassium: This option describes a formulation of ORS with reduced glucose levels and increased potassium levels. However, this composition is not consistent with low osmolarity ORS. High potassium levels in ORS can be problematic and potentially harmful, as excessively high potassium intake can lead to hyperkalemia, a condition characterized by elevated levels of potassium in the blood.

Solution for Question 2:

Correct Option: Option B: Ability to compare objects

- When a child transfers an object from one hand to another, it signifies their ability to compare objects

Incorrect Options:

Option A: Visual motor Co-ordination

- Visual motor coordination refers to the ability to coordinate visual input .This action requires the child to use their eyes to guide their hands in a precise manner to successfully transfer the object from one hand to the other.

Option C: Ability to handle small objects

- While handling small objects may be a component of transferring objects, the act of transferring itself is more about coordination and control than the ability to handle small objects.

Option D: Voluntary release

- Voluntary release typically refers to a child's ability to intentionally let go of an object they are holding, which is a different skill from transferring an object between hands. It's related to fine motor skills but not specific to this scenario.

Solution for Question 3:

Correct option is option C: Scarf sign:

The expanded New Ballard score is a method used to assess the maturity of a newborn based on physical and neuromuscular criteria. It includes various assessments to evaluate the neuromuscular development of the baby. One of these assessments is the Scarf sign.

The examiner takes the infant's hand and gently pulls the arm across the front of the chest and around the neck, mimicking the motion of putting on a scarf. With the other hand, gentle pressure is applied to the infant's elbow to help guide the arm around the neck. In mature babies the hand does not cross the shoulder and the elbow does not cross the midline.

Option A: Heel-knee test: This test involves flexing the baby's knee towards the buttocks and assessing the resistance encountered. It is not specifically related to the neuromuscular criteria for maturity according to the expanded New Ballard score.

Option B: Finger-nose test: The finger-nose test is a neurological examination performed in older children and adults to assess coordination and proprioception. It is not part of the neuromuscular criteria for maturity in newborns.

Option D: Cubital angle: The cubital angle is a measurement used to assess the elbow flexion in relation to the forearm. It is not directly related to the neuromuscular criteria for maturity according to the expanded New Ballard score.

Solution for Question 4:

Correct Option D - Gastrulation:

- Gastrulation is a crucial phase during early embryonic development when the blastula (hollow ball of cells) transforms into a trilaminar structure with three germ layers. These germ layers eventually give rise to various organs and tissues. If there is a defect in gastrulation, it can lead to severe congenital malformations due to incorrect formation of these germ layers and their subsequent development. Thus, the correct answer is option D, Gastrulation, as it is the process most directly related to the underlying defect in the congenital malformation mentioned.

Incorrect Options:

Option A - Morulation: Morulation is the process of early embryonic development that involves the division of the zygote into multiple smaller cells called blastomeres. This process occurs during the first few days after fertilization and leads to the formation of the morula, a solid ball of cells. However, morulation is not associated with the development of specific structures like the nervous system, so it is not the underlying defect for the congenital malformation in question.

Option B - Neurulation: Neurulation is the process by which the neural tube is formed, which gives rise to the central nervous system (brain and spinal cord). This process occurs during the third week of embryonic development and involves the folding and fusion of the neural plate to create the neural tube. Neurulation defects can lead to neural tube defects such as spina bifida and anencephaly, but this process is not directly related to the congenital malformation mentioned.

Option C - Lateral Folding: Lateral folding is a process that occurs during gastrulation, which is the early embryonic stage when the three germ layers (ectoderm, mesoderm, and endoderm) form. Lateral folding involves the folding of the embryo's sides towards the midline, allowing the germ layers to develop properly. Defects in lateral folding can lead to malformations, but again, this process is not specifically associated with the malformation mentioned

Solution for Question 5:

Correct Option C - 2cm/month:

At birth, head circumference is around 33 – 35 cm. It increases by approximately 2 cm per month for the first 3 months, 1 cm per month between 3 and 6 months, and 0.5 cm per month for the rest of the first year.

Incorrect options:

Options A, B, and D are incorrect, as explained above.

Solution for Question 6:

Correct Option B - 3.5 years:

- At 3 years, a child goes upstairs with alternating feet & downstairs with 2 per step and rides a tricycle.
- A child hops at 4 years and goes downstairs with alternating feet. Since the child described cannot hop, they are likely younger than 4 years old.
- So, the developmental age of the child is more than 3 years and less than 4 years.

Incorrect Options:

Option A - 2.5 years: At 2.5 years, children usually go upstairs and downstairs with two feet per step but can not ride a tricycle.

Option C - 4.5 years: By 4.5 years, children should be able to hop and go up and down stairs with alternating feet.

Option D - 5.5 years: At 5.5 years, children should be able to skip and stand on one leg for more than 10 seconds.

Solution for Question 7:

Correct Answer: B) 3 months

Explanation:

Key Developmental Milestones				
Age	Gross Motor Milestones	Fine Motor Milestones	Social Milestones	Language Milestones
1 month				Alerts to sounds
2 months			Social smile	
3 months	Neck holding		Recognises mother, anticipates feed (Option B)	Cooing
4 months				Laugh Loud
5 months	Rolls over	Bidextrous reach		
6 months	Sits in tripod position	Unidextrous reach	Recognize strangers, stranger anxiety	Monosyllables
8 months	Sits without support			
9 months	Stands with support	Immature pincer grasp	Waves bye bye	Bisyllables
12 months	Creeps well, stands without support	Mature pincer grasp	Comes when called, plays simple ball game	1 to 2 words with meaning
15 months	Walks alone , creeps upstairs	Imitates scribbling, tower of 2 blocks	Jargon speech	
18 months	Runs, explores drawers	Scribbles, tower of 3 blocks	Copies parents in task	8 to 10 words vocabulary
2 years	Walks up and downstairs (2 feet per step), jumps	Tower of 6 blocks, vertical and circular stroke	Asks for food, drink, toilet, pulls people to show toys	2 to 3 word sentences
3 years	Rides tricycle, alternate feet going upstairs	Tower of 9 blocks, copies circle	Shares toys, knows full name and gender	Asks questions, tells name and gender
4 years	Hops on one foot, alternate feet going downstairs	Copies cross, bridge with blocks	Plays cooperatively in a group, goes to toilet alone	Says song or poem, tells stories
5 years		Copies triangle, gate with blocks	Helps in household tasks, dresses and undresses	Asks meaning of words

Solution for Question 8:

Correct Option B - 1-b, 2-a, 3-d, 4-c:

Transfers objects	6-7 months
Social smile	1-2 months
Pincer grasp	9-12 months
Walks 1-2 steps	12-15 months

Incorrect Options:

- Options A, C and D are incorrect.

Solution for Question 9:

Correct Option A - 4 months:

- Reaching out for objects with both hands is a developmental milestone known as bilateral voluntary grasp. Typically, infants can start to achieve this milestone around 4 months of age.
- At this stage, their hand-eye coordination and motor skills have developed enough to simultaneously reach out and grasp objects with both hands.

Incorrect Options:

- Options B, C, and D: These are not the correct ages at which bilateral voluntary grasp develops.

Solution for Question 10:

Correct Option D - Ulnar palmar grasp:

- Ulnar palmar grasp is the first grasp and first step toward a baby's grip. This happens in the 4th month.
- The baby will hold an object by giving it in the palm while gently wrapping one or two fingers, excluding the thumb, around it.
- At this point, the baby will not have a strong grip on the object, so it may roll away from the hand.

1 Month	Hands kept closed
2 Months	Hands open intermittently
3 Months	Hands kept open Hold an object when placed in the hand “Hand regard” appears (disappears in 20th week) Palmar grasp reflex is lost
4 months	Tries to reach an object, but overshoots
5 Months	Bidextrous grasp
6 Months	Unidextrous or palmar grasp
7 Months	Transfer objects from 1 hand to another
9 Months	Immature / assisted pincer grasp
12 Months	Mature / unassisted pincer grasp Pulls off cap/mittens/socks

Incorrect Options:

Option A - Immature Pincer Grasp: Immature Pincer Grasp is typically developed by 9 months. The child will begin to pick up small objects with only the thumb and index finger.

Option B - Mature pincer grasp: Mature pincer grasp is the ability to hold something between the thumb and first finger. This skill usually develops in babies around 9 to 10 months old and is an important fine-motor milestone.

Option C - Radial palmar grasp: Radial palmar grasp (8 to 10 months) is a radial digital grasp typically developed in which an object is held between the fingers and opposed thumb. The fingers are now strong and coordinated enough to hold an object without exterior stabilization from the palm.

Solution for Question 11:

Correct Option C - Feeding by spoon:

- Feeding by spoon is a self-feeding skill that involves fine motor coordination, hand-eye coordination, and self-help abilities. Most babies can start to use a spoon by themselves at around 10 to 12 months old.
- It is generally expected to be mastered by the age of 2-3 years. If a 3-year-old child is unable to feed themselves with a spoon, it may indicate a developmental delay in fine motor skills or self-care abilities.

Incorrect Options:

Option A - Hopping on one leg: Hopping on one leg is a motor skill milestone that typically develops around 3-4 years of age. Although it is expected by this age, the absence of this milestone alone would not necessarily indicate a developmental delay.

Option B - Drawing a square: Drawing a square involves fine motor skills and visual-spatial coordination. It is a more advanced milestone that is usually achieved around 4-5 years of age. The absence of this milestone in a 3-year-old child would not be considered a developmental delay.

Option D - Passing a ball to someone: Passing a ball to someone involves gross motor skills, hand-eye coordination, and social interaction. It is a developmental milestone that typically develops around 2-3 years of age. The absence of this milestone alone would not necessarily indicate a developmental delay.

Previous Year Questions

1. What is the recommended treatment for a 7-year-old child presenting with bedwetting complaints?

A. Bed alarm

B. Imipramine

C. Desmopressin spray

D. Fludrocortisone

2. Craniopagus is defined as the fusion of:

A. Head and spine

B. Head only

C. Thorax and spine

D. Thorax only

3. Intellectual disability is seen in:

A. Phenylketonuria

B. Alkaptonuria

C. Albinism

D. Von Gierke disease

4. A child with an IQ of 55 comes under:

A. Mild intellectual disability

B. Moderate intellectual disability

C. Severe intellectual disability

D. Profound intellectual disability

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	1

Solution for Question 1:

Correct Option A - Bed alarm:

- A bedwetting alarm system consists of two parts: Moisture sensor Alarm
- Moisture sensor
- Alarm
- When it detects the moisture as the child passes urine, it wakes the child.
- With time, the child gains the awareness to recognize and respond to the sensation of a full bladder, leading to control over bedwetting.
- A bed alarm is most commonly recommended as the first line of treatment for bed wetting.
- Moisture sensor
- Alarm

Incorrect Options:

Option B - Imipramine: Imipramine is recommended when the bed alarm fails to show any result.

Option C - Desmopressin nasal spray: Desmopressin nasal spray is no longer licensed for bedwetting due to an increased incidence of side effects.

Option D - Fludrocortisone: Fludrocortisone is not used to treat bed-wetting.

Solution for Question 2:

Correct Option B - Head only:

- Craniopagus involves the fusion of the heads of two individuals.
- The craniopagus conjoined twins share the skull, meninges, and venous sinuses, and the brains are usually separate, although some cortical fusion can occur.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 3:

Correct Option A - Phenylketonuria:

- Phenylketonuria (PKU) is an autosomal recessive disorder caused by a deficiency of the enzyme phenylalanine hydroxylase, which leads to the accumulation of phenylalanine in the body.
- If left untreated, elevated phenylalanine levels can cause severe intellectual disability and developmental delays.

Incorrect options:

Options B, C, and D do not cause intellectual disability.

Option B -Alkaptonuria:

- Alkaptonuria is a metabolic disorder characterized by the accumulation of homogentisic acid.

Option C - Albinism:

- Albinism is a genetic condition characterized by a lack of melanin production.

Option D - Von Gierke disease:

- Von Gierke disease, also known as glycogen storage disease type I, is a metabolic disorder affecting glycogen metabolism.

Solution for Question 4:

Correct Option A - Mild intellectual disability:

- Intellectual disability: IQ < 70
- IQ = Mental age / Chronological age x 100

	IQ level
Mild intellectual disability	51-70
Moderate intellectual disability	36-50
Severe intellectual disability	21-35
Profound intellectual disability	0-20

Incorrect Options:

Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Neonatal Terminology & Reflexes

1. A newborn is delivered at 36 weeks of gestation with a birth weight of 1,200 grams. How should this newborn be classified based on gestational age and birth weight?

A. Pre-Term and Low Birth Weight (LBW)

B. Term and Very Low Birth Weight (VLBW)

C. Pre-term and Very Low Birth Weight (VLBW)

D. Post-Term and Low Birth Weight (LBW

2. A 38-week gestational-age infant is delivered with a birth weight of 3,200 grams. The neonatal team assesses the infant's weight percentile and notes that it falls between the 10th and 90th percentile for their gestational age. How should this newborn be classified according to gestational age and birth weight?

A. Small for Gestational Age (SGA)

B. Appropriate for Gestational Age (AGA)

C. Large for Gestational Age (LGA)

D. Pre-Term and Low Birth Weight (LBW)

3. A full-term neonate born to a healthy mother presents on the third day of life with a birth weight of 2.9 kg, length of 50 cm, head circumference of 35 cm, heart rate of 130 bpm, respiratory rate of 50 breaths/min, and peripheral cyanosis noted on the extremities. The baby is passing urine but has not passed any stools till now. Which of the following findings in this neonate would be most concerning?

A. Peripheral cyanosis (acrocyanosis) noted on the extremities

B. Passage of the first stool after 48 hours of birth

C. Respiratory rate of 50 breaths/min

D. Head circumference of 35 cm

4. A 43-week-old postterm neonate is delivered vaginally. Which of the following is not a characteristic appearance of a post-term neonate?

A. Overgrown Nails

B. Sparse scalp hair

C. Visible creases on palms and soles

D. Dry, loose, peeling skin

5. A newborn is admitted to the NICU with signs of hypoglycemia and respiratory distress. The mother had multiple gestation and gestational diabetes during her pregnancy. Which of the following is not a pregnancy-related complication that contributes to the high-risk status of this infant?

A. Multiple gestations

B. Gestational diabetes

C. Maternal obesity

D. Preeclampsia

6. A 2-month-old infant is brought to the clinic for a developmental check-up. Which reflex should the clinician expect to see at this stage of development?

A. STNR

B. Moro’s Reflex

C. Parachute Reflex

D. Landau Reflex

7. A 1-day-old newborn shows no startle response to sudden movement or sound. The pediatrician notes an absent Moro reflex, with no asymmetry. The baby is alert but cries weakly. What is the most likely cause?

A. Erb’s palsy

B. Severe asphyxia during birth

C. Hyperekplexia

D. Clavicle fracture

8. A 3-month-old infant is brought for a checkup. The pediatrician notices that when the baby's head is rotated to one side, the arm on the same side extends, and the arm on the opposite side flexes. Which of the following best explains these observations?

A. ATNR

B. STNR

C. Parachute reflex

D. Rooting reflex

9. Match the following images with their distinguishing features 1) a) Benign capillary malformation 2) b) Contains eosinophils 3) c) Contains melanocytes 4) d) Contains keratin

 ... (content truncated)	a) Benign capillary malformation
 ... (content truncated)	b) Contains eosinophils
 3)	c) Contains melanocytes

 ... (content truncated)	d) Contains keratin

A. 1-b, 2-a, 3-d, 4-c

B. 1-c, 2-d, 3-b, 4-a

C. 1-a, 2-b, 3-c, 4-d

D. 1-d, 2-c, 3-a, 4-b

10. Which of the following conditions in newborns typically resolves on its own without treatment?

A. Subconjunctival hemorrhage

B. Acne neonatorum

C. Epstein pearls

D. All of the above

11. Which of the following statements are true regarding the neonatal conditions described? Mastitis neonatorum typically presents with bilateral breast swelling Vaginal bleeding in neonates usually resolves within 10 days. Hymenal tags are usually symptomatic and often require medical treatment. Term neonates can lose up to 10% of their birth weight in the first week of life. Physiologic phimosis usually

resolves by age 3-5 and is generally asymptomatic. Preterm infants regain their birth weight within 7-10 days.

A. 1, 3, 4, 5

B. 3, 4, 6

C. 1, 2, 5, 6

D. 2, 4, 5

12. A 12-hour-old infant on routine examination presents with a well-defined, firm mass on the head. The mass is located over the parietal bones and does not cross the suture lines. The parents mention that the mass became noticeable shortly after birth and has been increasing in size for the past few hours. Which of the following is the most likely diagnosis?

A. Caput Succedaneum

B. Cephalohematoma

C. Subgaleal Hemorrhage

D. Craniotabes

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	2
Question 4	2
Question 5	3
Question 6	2
Question 7	2
Question 8	1
Question 9	4
Question 10	4
Question 11	4
Question 12	2

Solution for Question 1:

Correct Answer: C) Pre-term and Very Low Birth Weight (VLBW)

Explanation:

- The infant was born at 36 weeks gestation, classifying them as Preterm (<37 weeks).
- With a birth weight of 1,200 grams, they are also categorized as Very Low Birth Weight (VLBW) (<1,500 grams).
- Therefore, the newborn is best classified as both Pre-Term and VLBW.

Classification according to gestational age

Pre-Term	Born at < 37 weeks of gestation.
Term (Option B)	Born between 37 completed weeks to <42 weeks of gestation.
Post-Term (Option D)	Born at or beyond 42 weeks of gestation.

Classification according to Birth weight

Low birth weight(LBW) (Option A)	<2500 grams birth weight.
Very low birth weight (VLBW)	<1500 grams Birth Weight.
Extremely low birth weight (ELBW)	<1000 grams birth weight.

Solution for Question 2:

Correct Answer: B) Appropriate for Gestational Age (AGA)

Explanation:

The infant's weight falls within the 10th to 90th percentile for their gestational age, classifying them as Appropriate for Gestational Age (AGA).

Classification according to gestational age and birth weight.

Appropriate for gestational age (AGA)	The baby's weight is appropriate for the gestational age (weight between the 10th and 90th percentile).
Small for gestational age (SGA) (Option A)	Baby's weight is less than expected for the gestational age (weight less than 10th percentile).
Large for gestational age (LGA) (Option C)	The baby's weight is more than expected for the gestational age (weight more than the 90th percentile).

Pre-Term and Low Birth Weight (LBW) (Option D): Pre-Term refers to infants born before 37 weeks of gestation, and LBW is defined as a birth weight of less than 2,500 grams.

- The infant in the scenario was born at 38 weeks with a weight of 3,200 grams, so neither pre-term nor LBW criteria are met.

Solution for Question 3:

Correct Answer: B) Passage of the first stool after 48 hours of birth

Explanation:

Delayed passage of the first stool (beyond 24 hours) may indicate a potential issue like meconium ileus or other gastrointestinal problems.

Characteristics of normal term neonate:

Birth weight of an average Indian baby	2.9 kg
Length	50 cm
US: LS ratio	1.7:1
Head circumference (Option D)	33-37 cm
Heart rate	110-160 bpm
Respiratory rate (Option C)	40 -60 breaths/min
Peripheral cyanosis (Acrocyanosis) normal finding (Option A)	
1st passage of stool	Within 24 hrs.
1st passage of urine	Within 48 hrs.
Murmur	Soft systolic murmur - normal

Solution for Question 4:

Correct Answer: B) Sparse scalp hair

Explanation: Post-term neonates generally have abundant scalp hair, not sparse.

Characteristics of a Post-term Neonate

- Dry, loose, peeling skin (Option D)
- Overgrown nails (Option A)
- Abundant scalp hair
- Visible creases on palms and soles of feet(Option C)

- Minimal fat deposits.
- Green, brown, or yellow skin colouring from meconium staining
- More alert and "wide-eyed" appearance.

Complications of Post-term Neonate

- Increased risk of fetal distress and perinatal asphyxia
- Hypoxic encephalopathy and seizures
- Meconium aspiration syndrome
- Polycythemia
- Hypoglycemia
- PPHN(persistent pulmonary hypertension of the newborn)

Solution for Question 5:

Correct Answer: C) Maternal obesity

Explanation: While maternal obesity can contribute to a high-risk pregnancy, it does not directly contribute to the described complications in the same way as multiple gestations, gestational diabetes, or preeclampsia would.

- Obesity is more of a maternal health condition rather than a pregnancy-related complication.
- The other options—multiple gestations, gestational diabetes, and preeclampsia—are all recognized pregnancy-related complications that can contribute to the infant's high-risk status.

High-risk Infant:

A “high-risk infant” is broadly defined as one who requires more than the standard monitoring and care offered to a healthy-term newborn infant

Factors Associated with High-Risk Infants

1. Maternal Factors:

- Maternal health conditions: High blood pressure Polycystic Ovary Syndrome (PCOS) Diabetes Kidney disease Autoimmune diseases (e.g., Lupus, Multiple Sclerosis) Thyroid disease Obesity HIV/AIDS
- High blood pressure

- Polycystic Ovary Syndrome (PCOS)
- Diabetes
- Kidney disease
- Autoimmune diseases (e.g., Lupus, Multiple Sclerosis)
- Thyroid disease
- Obesity

Maternal health conditions:

- High blood pressure
- Polycystic Ovary Syndrome (PCOS)
- Diabetes
- Kidney disease
- Autoimmune diseases (e.g., Lupus, Multiple Sclerosis)
- Thyroid disease
- Obesity
- Maternal Age Young age (Teen pregnancy) First-time pregnancy after age 35
- Young age (Teen pregnancy)
- First-time pregnancy after age 35

Maternal Age

- Young age (Teen pregnancy)
- First-time pregnancy after age 35
- Lifestyle Factors Smoking/alcohol/drugs
- Smoking/alcohol/drugs

Lifestyle Factors

- Smoking/alcohol/drugs

2 . Pregnancy-Related Conditions

- Multiple gestations (Twins, triplets, etc.) (Option A)
- Gestational diabetes (Option B)
- Preeclampsia and eclampsia (Option D)
- Previous preterm birth
- Congenital anomalies or genetic conditions in the fetus

3. Labour or peripartum

- Pre- or post-term birth

- Requiring resuscitation in the delivery room
- Perinatal asphyxia
- Congenital infections (e.g., syphilis, CMV, toxoplasmosis, herpes simplex)
- Cardiopulmonary distress

Solution for Question 6:

Correct Answer: B) Moro’s Reflex

Explanation: Moro’s Reflex is expected to be present at 2 months as it persists from 28-37 weeks of gestation and disappears around 3-6 months of age.

Reflexes present at birth in a term neonate:

Name of the reflex	Definition	Appears at (weeks of gestation)	Disappears at (Postnatal age)
Rooting Reflex	The rooting reflex is where the mouth turns toward an object. Triggered by lightly stroking the cheek or by bringing an object into the patient's visual field.	32 weeks	1 month
Moro’s Reflex	The Moro reflex is a protective response to sudden loss of balance Elicited by raising the baby’s head slightly, and dropping it suddenly, while the other hand still supports the baby Response: Palm opening; extension, abduction of upper extremities & anterior flexion of body.	28-37 weeks	3-6months
Palmar/Plantar Grasp Reflex	The grasping reflex can be triggered by applying sustained pressure to the palm, causing the patient’s fingers to flex and grasp the object	28 weeks	4 months

	applying the pressure.		
ATNR (Asymmetric Tonic Neck Reflex)	The asymmetric tonic neck reflex is performed by manual rotation of the infant's head to one side The infant will extend its arm to the side of the rotated face and flex the contralateral arm.	35 weeks	3 months

Reflexes present after birth:

Name of the reflex	Definition	Appears at	Disappears at (Postnatal age)
STNR (symmetric tonic neck reflex)(Option A)	Elicited by passive movement of the head, giving the following response: Neck flexion: Upper extremities flex, lower extremities extend. Neck extension: Upper extremities extend, lower extremities flex. Important for activities like sitting, swimming, and playing with a ball.	6-9 months	9-11 months
Parachute reflex (Option C)	The reflex is triggered by turning the child upside down, Response: The arms move forward and the hands spread out in an attempt to catch the fall.	8-9 months	
Landau reflex (Option D)	Elicit by supporting the infant horizontally in a prone position. Positive reaction: The infant raises its head and arches back upward. Subsidiary response: Partial extension of lower limbs at the hips.	3 months	1 year
Neck righting reflex	Elicited by turning the child's head 90 degrees while lying flat. Normal response: the body follows the direction of the head turn due to	Birth	5 years of age

	torsion of the vertebralcolumn.		

Solution for Question 7:

Correct Answer: B) Severe asphyxia during birth

Explanation:

Severe asphyxia can cause brain damage due to oxygen deprivation, which often leads to an absent Moro reflex. In this scenario, the weak cry and the absent reflex strongly suggest a potential neurological injury caused by perinatal hypoxia.

Moro Reflex:

- The Moro reflex or Startle reflex is a primitive neonatal reflex that can be elicited by supporting the baby's head and letting it fall back slightly while still supporting it.
- This triggers a response where the baby first extends and abducts their arms and fingers, then flexes the arms back towards the body, typically accompanied by a cry.
- The reflex begins to develop in utero, with hand opening at 28 weeks, arm extension and abduction at 32 weeks, and full response (including flexion) by 37 weeks.

Normally, the Moro reflex disappears by 3 to 6 months of age.

ABNORMAL MORO REFLEX	CAUSES

Absent Moro Reflex	Birth injuries: Trauma during delivery Intracranial hemorrhage Severe asphyxia: Oxygen deprivation during birth General muscular weakness Cerebral palsy: Especially in spastic forms, the reflex may be absent.
Asymmetric Moro Reflex	Brachial plexus injury e.g., Erb's palsy (Option A) Clavicle fracture (Option D) Cervical spine injury
Exaggerated Moro Reflex	Hyperekplexia: A genetic disorder with an exaggerated startle response, often accompanied by muscle rigidity. (Option C) Severe brain damage: Conditions like microcephaly or hydranencephaly.

Solution for Question 8:

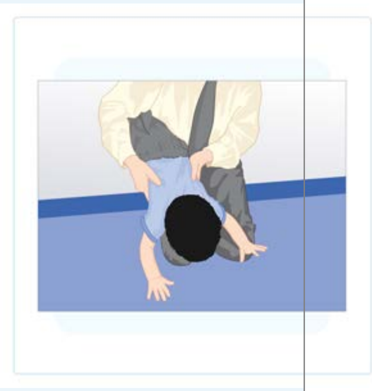
Correct Answer: A) ATNR

Explanation:

- ATNR is typically triggered when the infant's head is rotated to one side, resulting in an extension of the arm on the same side and flexion of the arm on the opposite side.
- This is a normal reflex that helps with hand-eye coordination and usually disappears in 6-7 months.
- The parachute reflex, which appears around 7-8 months and is fully developed by 10-11 months, is a protective reflex where the infant extends their arms when falling to break the fall.

Reflex	Onset	Disappearance	Description

Rooting	32 wk gestation	1 month	 ... (content truncated)
Tonic neck reflexes: 1) Asymmetric Tonic Neck Reflex (ATNR):	35 wk gestation	3 months	ATNR is triggered by rotating the infant's head to one side, which causes extension of the arm on the side the face is turned to and flexion of the arm on the opposite side. This reflex helps in hand-eye coordination development. An obligatory response, where the baby becomes " stuck" in this position, indicates a CNS disorder TONIC NECK REFLEX" data-author="" data-hash="841" data-license="" data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/tYhS5QsAtW85zFBmCwt81741553646.png" />
2) Symmetric Tonic Neck Reflex (ATNR):	6-9 months	9-11 months	STNR involves neck movement impacting arm and leg movement differently—neck flexion leads to arm flexion and leg extension, and neck extension causes arm extension and leg flexion. It helps prepare the baby for crawling. TONIC NECK REFLEX" data-author="" data-hash="842" data-license=""

			data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/oBmMe5qXMaxFsLZ9tNN11741553647.png" />
Parachute	8-9 months postnatal	Remains throughout life	 ... (content truncated)

Solution for Question 9:

Correct Answer: D) 1-d, 2-c, 3-a, 4-b

Explanation:

Condition	Features
 ... (content truncated)	A common benign skin condition characterized by small white papular cystic lesions which are superficial benign keratinous cysts, usually 3 mm in size. Predominantly found on the face and nose.

 ... (content truncated)	Slate-blue, well-defined areas of pigmentation that appear on the buttocks, back, and occasionally other areas of the body. Collection of melanocytes in the deeper layers of the skin. Present in over 50% of Black, Native American, and Asian infants, and sometimes in white infants as well. These benign patches have no known anthropologic significance Usually fades within the first year of life.
 Stork bites/ Salmon Patch/ Angel kiss	The benign capillary malformation can appear as flat red patches on a newborn's forehead and upper eyelids. Do not need treatment because they are harmless and will fade over time.
 ... (content truncated)	Small, white papules on an erythematous base, contain eosinophils and they develop 1-3 days after birth Benign rash, which typically lasts up to a week Usually found on the face, torso, and limbs.

Solution for Question 10:

Correct Answer: D) All of the above

Explanation:

All the conditions listed (subconjunctival haemorrhage, acne neonatorum, and Epstein pearls) are typically self-limiting and do not require medical treatment to resolve.

BENIGN NEONATAL CONDITIONS	
SUBCONJUNCTIVAL HEMORRHAGE (Option A) Subconjunctival haemorrhages are frequently observed due to birth trauma caused by sudden increases in internal or external pressure. Common factors include: Following vaginal delivery Difficult or extended labour Fetal macrosomia Improper use of forceps or vacuum extractors Typically resolve on their own.	
ACNE NEONATORUM (Option B) Acne neonatorum, or neonatal acne, is a common condition in newborns characterized by facial eruptions of inflamed and non-inflamed lesions. It usually appears around 3 weeks of age and lasts about 4 months. Males>females. The condition, linked to sebaceous gland overactivity due to neonatal androgens, typically resolves on its own within 3 months.	
EPSTEIN PEARLS (Option C) Epstein pearls are small, benign cysts that commonly appear in newborns' mouths, particularly on the gums and the roof of the mouth. These white or yellowish bumps, caused by trapped skin cells filled with keratin, can be mistaken for teeth due to their firmness. Present in about 85% of newborns, Epstein pearls typically resolve on their own within a few weeks to within 3 months and do not require treatment. Activities such as breastfeeding or using pacifiers can help them dissolve more quickly.	

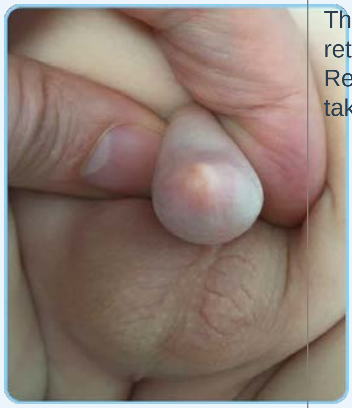
Solution for Question 11:

Correct Answer: D) 2, 4, 5

Explanation:

- Neonatal vaginal bleeding is a normal physiological process caused by decreased maternal estrogen and typically resolves within 10 days. (Statement 2)
- It is normal for term neonates to lose 7-10% of their birth weight in the first week due to loss of extracellular fluid. (Statement 4)
- Physiologic phimosis is a normal condition where the foreskin cannot be retracted and generally resolves by age 3-5 without causing symptoms. (Statement 5)

Condition	Features	Duration	Management
Mastitis Neonatorum	 ... (content truncated)	Resolves within a few weeks	Antibiotics; usually resolve on their own
Vaginal Bleeding	Normal due to decreased maternal estrogen. Symptoms include a small amount of blood-tinged discharge or bleeding.	Resolves within 10 days	Medical evaluation if bleeding persists or if other concerns arise
Hymenal Tags	 Benign remnants of tissue at the vaginal opening. Normal anatomical variations and typically asymptomatic.	Often resolves on its own (Statement 3)	Treatment only if causing discomfort
Physiologic	Expected weight loss in the first week of life.	Term infants regain within 7-10 days; preterm infants	Weight gain of 20-40 grams daily after

Weight Loss	Term neonates lose 7-10% of birth weight; preterm infants lose 10-15%. They then gain about 20 to 40 grams daily. This initial weight loss is primarily due to the loss of extracellular fluid, while subsequent weight gain is attributed to increased intracellular water and solids, supporting cellular growth.	by 10-14 days (Statement6)	regaining birth weight.
Physiologic Phimosis	 ... (content truncated)	The prepuce becomes retractable by 3 yr of age. Resolves by age 3-5; may take longer	Gentle retraction during bathing to aid in loosening.

Solution for Question 12:

Correct Answer: B) Cephalohematoma

Explanation:

Cephalohematoma presents as a well-circumscribed, firm mass over the parietal bones that does not cross suture lines.

- It typically develops in the first few hours after birth, with increasing size for 12-24 hours, and then stabilizes.
- This fits the description provided in the scenario.

Characteristic	Caput Succedaneum (Option A)	Cephalohematoma (Option B)

Incidence	Common	Less common
Location	Subcutaneous plane	Over parietal bones, between the skull and periosteum.
Time of Presentation	Maximum size and firmness at birth.	Develops during the 1st few hours of life. Increasing size for 12-24 hours and then stable.
Time Course	Softens progressively from birth.	Takes 3-6 weeks to resolve, but may resolve within 2-3 days.
Characteristic Findings	Diffuse Circular boggy area of oedema with indistinct borders. Often with overlying ecchymosis Crosses suture line	Well-circumscribed fluid-filled mass. Does not cross the suture line
Association	None	Linear skull fracture (5-25%); hyperbilirubinemia

Subgaleal Hemorrhage (Option C) is a more diffuse, larger haemorrhage that crosses suture lines and is associated with more severe symptoms. It is less likely to present as a firm, well-defined mass and is typically associated with significant blood loss, which is not described in the scenario.

Craniotabes (Option D) refers to soft areas of the skull due to delayed skull bone ossification that may feel soft and dentable, like a ping pong ball, often found in the parietal bones near the sagittal suture, typically in preterm infants or those with uterine compression.

- Craniotabes present as soft areas rather than firm, localized masses and are more common in different contexts, such as preterm birth or specific congenital conditions.

Neonatal Resuscitation

1. A 24-year-old primigravida came for a routine antenatal check at 32 weeks gestation. As the medical officer, which of the following is your responsibility during the antenatal period? a. Antenatal counseling for the mother b. Team briefing and individual responsibility allocation c. Equipment check and preparation d. Proper history taking i.e. prior history of preterm delivery or abortions

A. a & d are correct

B. a correct, b, and c incorrect

C. b & c are incorrect

D. a,b,c,d are correct

2. A 27-year-old mother gave birth to a term male baby through normal vaginal delivery with a birth weight of 3 kg who did not cry immediately after birth. How will you resuscitate the baby?

A. Clear airway if required

B. Position the baby properly to ensure an open airway; the neck should be kept slightly extended and the mouth slightly open.

C. Stimulate the baby i.e. providing physical stimulation by rubbing the back of the baby or flicking the sole of the baby.

D. All of the above

3. During neonatal resuscitation, If the HR is less than 100/min or apnea or gasping is present what measures would you take?

A. Start PPV by self-inflating the Bag and Mask

B. Check SpO2 and monitor ECG

C. Reassess after 30 seconds, if the HR is not less than 100/min then the baby gets post-resuscitation care

D. All of the above

4. During neonatal resuscitation, If the HR is less than 100/min despite PPV, the following steps are followed except:

A. Ensure, whether the positive pressure ventilation is given properly or not

B. Take ventilation corrective step

C. Intubation never done

D. If needed endotracheal intubation should be done.

5. After delivery and resuscitation, a neonate was found to have HR <60 even after effective positive pressure ventilation and no respiratory efforts. What is the line of action?

A. Oxygen by mask

B. Start chest compression

C. Chest compression+adrenaline

D. Adrenaline injection

6. What is the recommended dose of adrenaline for neonatal resuscitation in a preterm baby at NICU with HR persistently below 60/min?

A. 0.002mg/kg/dose

B. 0.02mg/kg/dose

C. 0.2mg/kg/dose

D. 2mg/kg/dose

7. Which among the given statements is wrong regarding resuscitation of newborn? a) The recommended pressure of suction is 80 mmHg b) Correct order of suction: mouth followed by nose. c) The recommended pressure should never exceed >80 mmHg d) Correct order of suction: nose followed by mouth

A. a & b are wrong.

B. b & c are wrong

C. c & d are wrong.

D. a & d are wrong.

8. Which of the following is the wrong statement about PPV?

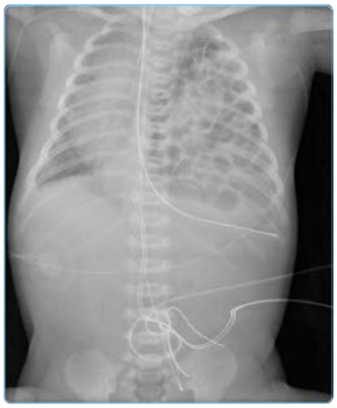
A. The reservoir bag and oxygen tubing may or may not be attached to the self-inflating bag

B. The function of the reservoir is to increase FiO2 delivered to the baby

C. The single most important step in neonatal resuscitation is effective positive pressure ventilation.

D. The rate of PPV should be 30-40/min

9. A 6-hour-old term baby shifted to emergency brought with severe respiratory distress. Chest X-ray is given below. Which of the initial clinical interventions should not be done on this patient?



A. Intubation and insertion of nasogastric tube

B. Bag and mask ventilation

C. Administration of fluids

D. Put the baby in a radiant warmer.

10. All of the following are the key corrective actions to optimize ventilation during neonatal resuscitation except:

A. Ensure the mask is of appropriate size

B. The seal between mask and face should be tight

C. The mask should cover the mouth only

D. The head of the baby should be slightly extended and the mouth of the baby is kept slightly open

11. What is the most reliable method to confirm proper placement of an endotracheal tube immediately after intubation?

A. Auscultation of breath sounds

B. End-tidal carbon dioxide (ETCO2) detection

C. Visualization of the tube passing through the vocal cords

D. Chest X-ray

12. As a part of neonatal resuscitation, you start chest compressions. For maximum effectiveness, what should be the depth of your compressions?

A. ■ of AP diameter of chest

B. ½ of AP diameter of chest

C. ■ of AP diameter of chest

D. ¼ of AP diameter of chest

13. Which of the following is one of the primary benefits of delayed umbilical cord clamping in newborns?

A. It reduces the risk of maternal postpartum hemorrhage

B. It decreases the likelihood of neonatal jaundice

C. It prevents the need for neonatal resuscitation

D. It improves the newborn’s iron stores and reduces the risk of anemia

14. Under which of the following conditions will neonatal resuscitation not be performed?

A. Anencephaly

B. Confirmed case of trisomy 13

C. Gestational age less than 22 weeks

D. All of the above

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	4
Question 4	3
Question 5	2
Question 6	2
Question 7	3
Question 8	4
Question 9	2
Question 10	3
Question 11	2
Question 12	1
Question 13	4
Question 14	4

Solution for Question 1:

Correct Answer: D) a,b,c,d are correct

Explanation:

The components of neonatal resuscitation in the antenatal period include:

- Antenatal counseling, team briefing, and equipment checking

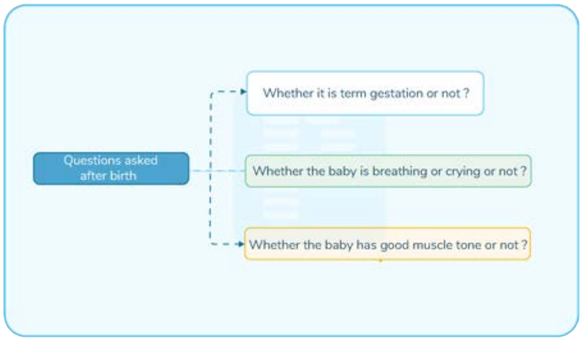
- Antenatal counseling includes proper history taking i.e. prior history of preterm delivery or abortions.
- Team briefing and individual responsibility allocation like receiving the baby, and intubating the baby.
- Equipment check and preparation which ensures the working condition of instruments like suction apparatus, battery of laryngoscope, and radiant warmer.

Solution for Question 2:

Correct Answer: D) All of the above

Explanation:

After birth, as a component of neonatal resuscitation, the following questions are asked:



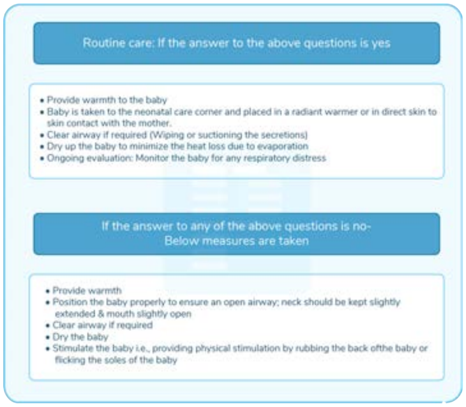
Questions asked after birth

Whether it is term gestation or not ?

Whether the baby is breathing or crying or not ?

Whether the baby has good muscle tone or not ?

The answer to these questions may be yes or no and the next steps would be as follows:



Routine care: If the answer to the above questions is yes

- Provide warmth to the baby
- Baby is taken to the neonatal care corner and placed in a radiant warmer or in direct skin to skin contact with the mother.
- Clear airway if required (Wiping or suctioning the secretions)
- Dry up the baby to minimize the heat loss due to evaporation
- Ongoing evaluation: Monitor the baby for any respiratory distress

If the answer to any of the above questions is no- Below measures are taken

- Provide warmth
- Position the baby properly to ensure an open airway; neck should be kept slightly extended & mouth slightly open
- Clear airway if required
- Dry the baby
- Stimulate the baby i.e., providing physical stimulation by rubbing the back ofthe baby or flicking the soles of the baby

Since in this scenario, the baby is not crying, the aforementioned steps in the table are followed.

Solution for Question 3:

Correct Answer: D) All of the above

Explanation:

If HR <100/min or apnoea or gasping is present, then the following steps are followed:

- Start PPV by self-inflating the Bag and Mask (Option A)
- Check SpO2 and monitor ECG (Option B)
- Reassess after 30 seconds, if the HR is not less than 100/min then the baby gets post-resuscitation care (Option C)
- Baby with labored breathing and persistent cyanosis, improved by CPAP, also needs post-resuscitation care.
- Post resuscitation care: Baby is taken to nursery/NICU, continued monitoring the condition of baby, start IV fluids, oxygen if required, monitor the baby and treat them accordingly i.e by maintaining proper temperature, the glucose of baby.

Solution for Question 4:

Correct Answer: C) Intubation never done

Explanation:

- In case HR is less than 100/min despite PPV, you may intubate the baby or do endotracheal intubation.
- Ensure if the positive pressure ventilation is given properly or not by checking if the chest movement is happening with each breath given or not.(Option A)
- Ventilation corrective steps are used in Neonatal resuscitation to ensure the baby is getting adequate oxygen, clear airway obstruction, correct inadequate breathing efforts, prevent hypoxia, and support circulatory functions. (Option B)

Ventilation corrective steps includes:

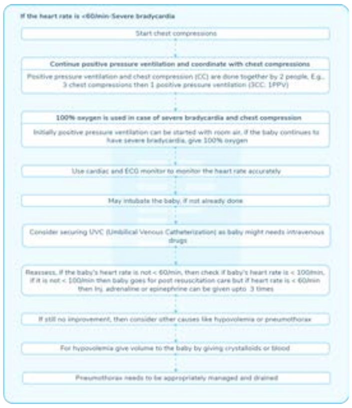
- The mask should cover the mouth and nose should fit well and provide an effective seal
- Ensure the Mask is of appropriate size
- The head of the baby should be slightly extended and the mouth of the baby should be kept slightly open

When HR becomes less than 100/min if needed endotracheal intubation is done (Option D)

Solution for Question 5:

Correct Answer: B) Start chest compression

Explanation:



If the heart rate is <60/min-Severe bradycardia

Start chest compressions

Continue positive pressure ventilation and coordinate with chest compressions

Positive pressure ventilation and chest compression (CC) are done together by 2 people. E.g., 3 chest compressions then 1 positive pressure ventilation (3CC: 1PPV)

100% oxygen is used in case of severe bradycardia and chest compression

Initially positive pressure ventilation can be started with room air, if the baby continues to have severe bradycardia, give 100% oxygen

Use cardiac and ECG monitor to monitor the heart rate accurately

May intubate the baby, if not already done

Consider securing UVC (Umbilical Venous Catheterization) as baby might needs intravenous drugs

Reassess. If the baby's heart rate is not < 60/min, then check if baby's heart rate is < 100/min, if it is not < 100/min then baby goes for post resuscitation care but if heart rate is < 60/min then Inj. adrenaline or epinephrine can be given upto 3 times

If still no improvement, then consider other causes like hypovolemia or pneumothorax

For hypovolemia give volume to the baby by giving crystalloids or blood

Pneumothorax needs to be appropriately managed and drained

- If after 1 min of effective PPV if HR <60/min, chest compression is started
- If the HR is persistently below 60/min, give IV epinephrine.

Solution for Question 6:

Correct Answer: B) 0.02mg/kg/dose

Explanation:

- Dose of Inj.Adrenaline: 0.02mg/kg/dose or 0.2ml/kg/dose given in 1:10000 dilution up to three times in neonatal resuscitation
- The preferred route of administration: IV through umbilical venous catheterization. It can be given intratracheally if IV access is not available.
- Indications: Use of adrenaline is indicated if HR remains below 60/min despite adequate ventilation and chest compression for 30 seconds.

Solution for Question 7:

Correct Answer: C) c & d are wrong

Explanations:

Suction of airways:

- The correct order of suction: is mouth followed by nose.
- When suctioning the nose, vagal stimulation may induce the baby to take a deep breath, potentially resulting in aspiration, if oral secretions are present.
- The recommended pressure of suction is 80 mmHg
- It should never exceed >100 mmHg
- Recommended temperature of delivery room:~25 degrees Celsius
- In the delivery room, the baby should be kept at a warmer temperature of much higher than 25 degrees Celsius.

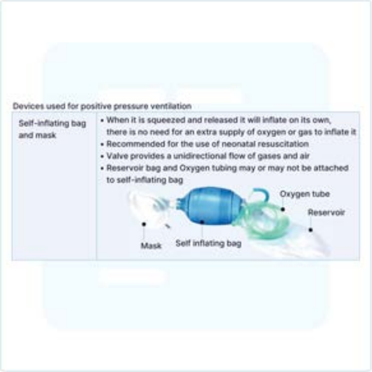
Solution for Question 8:

Correct Answer: D) The rate of PPV should be 30-40/min

Explanation:

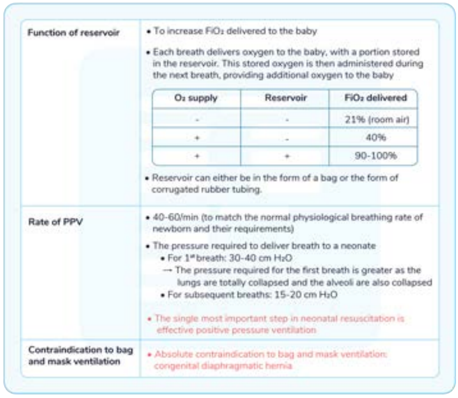
- To match the normal physiological breathing rate of newborns and their requirements the rate of PPV is 40-60/min.

- The reservoir bag and oxygen tubing may or may not be attached to the self-inflating bag. When it is squeezed and released it will inflate on its own, there is no need for an extra supply of oxygen or gas to inflate it. (Option A)
- The function of the reservoir is to increase FiO2 delivered to the baby. Each breath delivers oxygen to the baby, with a portion stored in the reservoir. (Option B)
- The single most important step in neonatal resuscitation is effective positive pressure ventilation. (Option C)
- The pressure required to deliver breath to a neonate for 1st breath: 30-40 cm H20



Devices used for positive pressure ventilation

Self-inflating bag and mask	• When it is squeezed and released it will inflate on its own, there is no need for an extra supply of oxygen or gas to inflate it • Recommended for the use of neonatal resuscitation • Valve provides a unidirectional flow of gases and air • Reservoir bag and Oxygen tubing may or may not be attached to self-inflating bag



<table>
<tr><td>Function of reservoir</td><td>• To increase FiO₂ delivered to the baby
• Each breath delivers oxygen to the baby, with a portion stored in the reservoir. This stored oxygen is then administered during the next breath, providing additional oxygen to the baby
<table>
<tr><th>O₂ supply</th><th>Reservoir</th><th>FiO₂ delivered</th></tr>
<tr><td>-</td><td>-</td><td>21% (room air)</td></tr>
<tr><td>+</td><td>-</td><td>40%</td></tr>
<tr><td>+</td><td>+</td><td>90-100%</td></tr>
</table>
• Reservoir can either be in the form of a bag or the form of corrugated rubber tubing.</td></tr>
<tr><td>Rate of PPV</td><td>• 40-60/min (to match the normal physiological breathing rate of newborn and their requirements)
• The pressure required to deliver breath to a neonate
• For 1st breath: 30-40 cm H₂O
→ The pressure required for the first breath is greater as the lungs are totally collapsed and the alveoli are also collapsed
• For subsequent breaths: 15-20 cm H₂O
• The single most important step in neonatal resuscitation is effective positive pressure ventilation</td></tr>
<tr><td>Contraindication to bag and mask ventilation</td><td>• Absolute contraindication to bag and mask ventilation: congenital diaphragmatic hernia</td></tr>
</table>

Solution for Question 9:

Correct Answer: B) Bag and mask ventilation

Explanation:

The described clinical presentation with a history of sudden noisy breathing and given X-ray is suggestive of congenital diaphragmatic hernia.

- Bag and Mask ventilation is contraindicated in congenital diaphragmatic hernia.
- During the Bag and mask ventilation, the mask envelops both the nose and mouth, directing air through both the trachea and esophagus.
- However, the air passing through the esophagus can result in an extension of bowel loops leading to a mediastinal shift and reduced space for the lungs to expand.
- Immediate steps of management include:

Put the baby in a radiant warmer → Endotracheal intubation→ Placement of a nasogastric tube→ Administration of fluids

X-ray findings in congenital diaphragmatic hernia :

- Mediastinal shift
- Bowel loops in the left hemithorax

Solution for Question 10:

Correct Answer: C) The mask should cover the mouth only

Explanation:

Ventilation corrective steps in Neonatal resuscitation are:

- The mask should cover the mouth and nose, fit well, and provide an effective seal
- Ensure the mask is of appropriate size (Option A)
- The seal between the mask and face should be tight (Option B)
- The head of the baby should be slightly extended and the mouth of the baby should be kept slightly open (Option D)
- MRSOPA- Mask re-apply, head reposition, suction, open airway, increase pressure, alternate airway or ET Tube

Solution for Question 11:

Correct Answer: B) End-tidal carbon dioxide (ETCO2) detection

Explanation:

End-tidal carbon dioxide (ETCO2) detection is the most reliable method for confirming the proper placement of an endotracheal tube immediately after intubation.

Methods to confirm the correct placement of the ET tube:

- On Inspection: Visible B/L chest rise on each breath
- On Auscultation: B/L equal breath sounds
- Absence of breath sounds in epigastrium
- Visible condensation in endotracheal tube
- Large exhaled tidal volumes
- Improvement in vital parameters like spo2, color of baby, peripheral perfusion
- Misting in the endotracheal tube with each breath due to excess water vapor and expired air

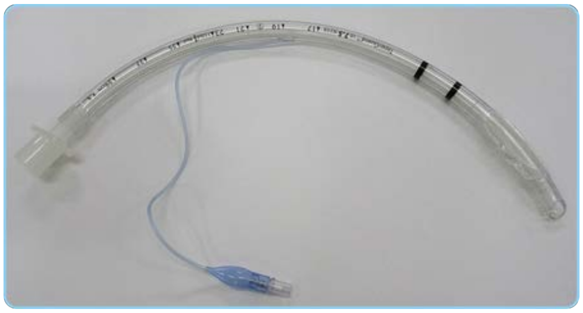
Confirmatory method:

- EtCO2 (End-tidal co2) also known as capnography (Surest confirmatory)
- The tip of ETT should be at the level of the lower border of T2

Different sizes of the Endotracheal tube:

Age	Internal diameter(mm)	Length(cm)
Full term infant	3.5	12
Child	4+ (Age/4)	12+(Age/2){orotracheal} 14+(Age/2){nasotracheal}
Adult female	7.0 - 7.5	20-22
Adult male	7.5 - 9	22-24

- The most preferred type of cuff in ET is a high-volume, low-pressure
- The advantage of a high-volume, low-pressure cuff is the lesser incidence of tracheal mucosal damage
- The cuff pressure should not exceed 25 cm of H20 since excess cuff pressure can lead to tracheal mucosal injury and recurrent laryngeal nerve palsy



- Auscultation of breath sounds (Option A): While auscultation can indicate whether breath sounds are present in both lungs, it is not entirely reliable for confirming placement. It may not detect misplacement into the esophagus or only one lung.
- Visualization of the tube passing through the vocal cords (Option C): While visualizing the tube entering the trachea is a strong indicator of correct placement, it may not be practical in all situations, especially under emergency conditions or if the view is obstructed.
- Chest X-ray (Option D): While a chest X-ray can confirm tube placement, it is not an immediate method and takes time. In emergency situations, relying on a chest X-ray is impractical.

Solution for Question 12:

Correct Answer: A) ■ of AP diameter of chest

Explanation:

For neonatal resuscitation, chest compressions should be delivered at a depth of approximately one-third of the anterior-posterior diameter of the chest.

Site: In the middle, on the lower 1/3rd of the body of the sternum

Just below the line, joining the two nipples.

Depth: At least 1/3rd of the AP diameter of the chest

Technique:

- 2 thumb technique (preferred) encircling the chest generates higher pressure, better perfusion, and lesser rescuers fatigability
- 2 finger technique

Ratio of chest compression: PPV

- 3:1 (for every compression 1 breath)
- In 2 seconds baby should receive 3 compressions and 1 PPV

Solution for Question 13:

Correct Answer: D) It improves the newborn iron stores and reduces the risk of anemia

Explanation:

Umbilical cord clamping must be delayed for 1 to 2 minutes to allow the transfer of the additional amount of blood from the placenta to the infant.

Components of delayed cord clamping include:

- Wait for at least 60 seconds before clamping the cord
- It is recommended for all stable-term as well as preterm neonates
- Advantages:

1) lesser chances of anemia

2) lesser need for blood transfusion

3) lesser risk of shock or hypotension

4) lesser risk of necrotizing enterocolitis

5) lesser risk of intra-ventricular hemorrhage in preterm infants

- Disadvantages: slightly increased risk of neonatal jaundice

Delayed cord clamping provides several benefits for the newborn, but it does not play a direct role in reducing the risk of PPH in the mother. (Option A)

Delayed cord clamping has no benefits in reducing the likelihood of neonatal jaundice. It may cause a slightly increased risk of neonatal jaundice. (Option B)

Delayed cord clamping is not related to preventing the need for neonatal resuscitation. In cases requiring resuscitation early cord clamping is done. (Option C)

Solution for Question 14:

Correct Answer: D) All of the above

Explanation:

Neonatal resuscitation is not performed in the following circumstances:

Anencephaly (Option A) is a severe neural tube defect where major parts of the brain and skull are rudimentary. The prognosis is very poor. Even with resuscitation, the underlying conditions of anencephaly cannot be corrected.

Confirmed case of trisomy 13 (Option B): The resuscitation efforts may not significantly alter the overall outcome or improve the infant's quality of life.

Gestational age less than 22 weeks (Option C) : At 22 weeks of gestation the infant's organs, especially the lungs, heart, and brain, are not sufficiently developed to sustain life.

Neonatal Respiratory Disorders & Perinatal Asphyxia

1. A neonate born after an emergency c-section due to non-reassuring fetal heart tones was noted to have poor muscle tone, a heart rate of 70 bpm, and intermittent gasping respirations. Despite initial tactile stimulation, the infant becomes flaccid and apneic. Which of the following best describes the underlying pathophysiology?

A. Prolonged hypoxia causes progressive neuromuscular depression

B. Severe hypoxia causing pulmonary vasodilation

C. Acute hypoxia leads to metabolic alkalosis

D. Persistent hypocapnia causes central respiratory depression

---

2. A neonate delivered via emergency cesarean section presents with an Apgar score of 2 at 1 minute. Which of the following criteria is most indicative of severe birth asphyxia?

A. Persistent metabolic alkalosis with a high base deficit

B. Apgar score of 5 at 5 minutes and normal blood pH

C. Blood pH <7.0 on umbilical arterial blood gas with a high base deficit

D. Presence of mild hypotonia and a base deficit of 10 mmol/L

---

3. A term newborn is diagnosed with moderate Hypoxic Ischemic Encephalopathy (HIE) after experiencing birth asphyxia. The infant is currently receiving supportive care. Which of the following is the most appropriate next step in the management of this infant?

A. Immediate administration of high-dose corticosteroids

B. Initiation of whole-body cooling therapy

C. Overhydration to prevent hypotension

D. Hyperventilation to reduce intracranial pressure

---

4. A 24-hour-old infant shows lethargy, hypotonia, intermittent seizures, and weak sucking reflex following a complicated delivery. According to the Sarnat classification, which stage of hypoxic-ischemic encephalopathy (HIE) does this infant most likely have?

A. Stage 1

B. Stage 2

C. Stage 3

D. Stage 4

---

5. A multigravida G2P1L1 delivers a baby at 37 weeks of gestation by cesarean section. On assessment, the baby was completely cyanosed having slow and irregular breathing with a heart rate of 76 bpm, and he was showing slight flexion on stimulation along with a slight frown when stimulated by an orogastric tube. Calculate the APGAR score for the baby.

A. 10

B. 4

C. 3

D. 7

---

6. A primigravida delivered a baby at 34 weeks of gestation. The baby was cyanosed and grunting, with a respiratory rate of 72 bpm. Which of the following scores can be used to assess this condition?

A. Downe’s score

B. CRIB score

C. SNAP score

D. Silverman-Andersen score

7. A primigravida with a complicated pregnancy underwent a cesarean section at 32 weeks of gestation. Soon after birth, the baby developed grunting, cyanosis, and a significantly elevated respiratory rate. A chest X-ray was performed immediately. Which of the following findings would not be seen on the given radiograph?

A. Ground glass haziness of lungs

B. Box-shaped heart

C. Presence of air bronchogram

D. Reticulogranular pattern

8. In the lab, amniotic fluid is mixed with 95% ethyl alcohol and then left after shaking it vigorously. After 15 minutes, a complete rim of bubbles is seen. Which of the following tests is described here ?

A. Phosphatidylglycerol test

B. Lamellar body count

C. Shake test

D. Amniotic fluid turbidity test

---

9. Which of the following interventions is NOT typically used in the initial management of neonatal Respiratory Distress Syndrome?

A. Surfactant replacement therapy

B. Continuous positive airway pressure (CPAP)

C. Mechanical ventilation

D. Prophylactic antibiotics

---

10. At what gestational age range is the administration of antenatal corticosteroids typically recommended to reduce the risk of Respiratory Distress Syndrome(RDS) in preterm infants?

A. 20-24 weeks

B. 24-34 weeks

C. 34-37 weeks

D. 37-40 weeks

---

11. A 39-week pregnant woman in active labour experiences spontaneous rupture of membranes, revealing thick, green amniotic fluid. The fetal heart rate tracing shows late decelerations. After delivery, the newborn has poor respiratory effort, cyanosis, and visible meconium staining on the skin. Which of the following is the most appropriate immediate management for this newborn?

A. Immediate intubation and tracheal suctioning

B. Stimulation and provision of positive pressure ventilation

C. Administration of prophylactic antibiotics

D. Immediate administration of surfactant

12. Which of the following is NOT typically associated with Transient Tachypnea of the Newborn (TTN)?

A. More common in infants delivered by cesarean section

B. Usually resolves within 24-72 hours

C. Persistent hypoxemia requiring prolonged mechanical ventilation

D. Delayed clearance of fetal lung fluid

13. A 27-week gestational-age preterm infant in the NICU experiences persistent apnea lasting over 20 seconds despite gentle tactile stimulation and supplemental oxygen. Nasal continuous positive airway pressure (nCPAP) has been used but has not resolved the episodes. What is the most appropriate next step?

A. Continue nasal continuous positive airway pressure (nCPAP)

B. Administer intravenous caffeine citrate

C. Initiate doxapram infusion

D. Start low-dose theophylline

14. A day-old preterm infant is admitted with severe respiratory distress. Physical examination shows a scaphoid abdomen and a chest X-ray image is given below. Despite intensive care, the infant's condition deteriorates. Which of the following statements about the suspected condition is correct?

A. Bochdalek typically occurs on the right side.

B. Morgagni involves the posterior diaphragm.

C. Morgagni is associated with poor prognosis.

D. The condition is mostly sporadic.

15. A 2-day-old newborn diagnosed with congenital diaphragmatic hernia (CDH) presents with severe respiratory distress and is scheduled for urgent surgical intervention. Which of the following management strategies is not recommended in this setting?

A. Bag and mask ventilation

B. Placement of Endotracheal tube

C. Administration of extracorporeal membrane oxygenation

D. Nasogastric tube to decompress the stomach

16. A 3-day-old newborn with congenital diaphragmatic hernia is in NICU showing signs of severe respiratory distress. Despite intensive management, the infant's condition deteriorates and leads to death. What is the most common cause of death in CDH?

A. Perforation of Herniated bowel

B. Pulmonary hypoplasia

C. Pulmonary Embolism

D. All of the above

Correct Answers

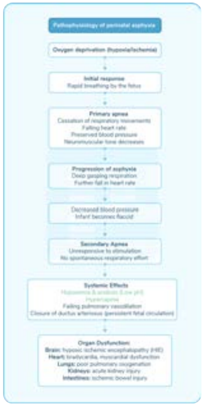
Question	Correct Answer
Question 1	1
Question 2	3
Question 3	2
Question 4	2
Question 5	2
Question 6	4
Question 7	2
Question 8	3
Question 9	4
Question 10	2
Question 11	2
Question 12	3
Question 13	2
Question 14	4
Question 15	1
Question 16	2

Solution for Question 1:

Correct Answer: A) Prolonged hypoxia causes progressive neuromuscular depression

Explanation:

- The given scenario points towards Perinatal asphyxia, where prolonged hypoxia leads to neuromuscular depression, manifesting as poor muscle tone, gasping respirations, and eventually progressing to flaccidity and apnea.
- Birth asphyxia or Perinatal asphyxia is defined as a condition resulting from impaired oxygen delivery to the fetus or neonate that leads to hypoxemia, hypercapnia, and metabolic acidosis, potentially causing significant cellular injury, especially to the brain, and multi-organ dysfunction.
- It is a leading cause of early neonatal mortality.



Solution for Question 2:

Correct Answer: C) Blood pH <7.0 on umbilical arterial blood gas with a high base deficit

Explanation:

Blood pH <7.0 on umbilical arterial blood gas with a high base deficit is a key indicator of severe birth asphyxia, reflecting severe metabolic acidosis and the impact of oxygen deprivation.

Criteria for severe birth asphyxia:

Criterion	Description
Apgar Score	0-3 at >5 minutes
Blood Gas Analysis	pH <7.0 on umbilical arterial blood sample indicates prolonged metabolic or mixed acidemia.
Neurological Manifestations	Includes seizures, coma, hypotonia, or hypoxic-ischemic encephalopathy (HIE) in the immediate neonatal period.

Multiorgan Dysfunction	Evidence of dysfunction in multiple organ systems in the immediate neonatal period.

Persistent metabolic alkalosis with a high base deficit (Option A) is not associated with severe birth asphyxia; severe asphyxia is marked by metabolic acidosis.

Apgar score of 5 at 5 minutes and normal blood pH (Option B) indicates only mild distress, not the severe acidosis typically seen in severe birth asphyxia.

The presence of mild hypotonia and a base deficit of 10 mmol/L (Option D) is not characteristic of severe birth asphyxia; severe cases typically present with more pronounced hypotonia and a higher base deficit.

Solution for Question 3:

Correct Answer: B) Initiation of whole-body cooling therapy

Explanation:

- Whole-body cooling (hypothermia) is the most appropriate next step for infants with moderate to severe HIE.
- It has been shown to reduce brain injury by slowing metabolic processes and is effective in improving neurodevelopmental outcomes in these infants.

Table rendering failed

Immediate administration of high-dose corticosteroids (Option A) is not routinely used in the management of HIE, and they may increase the risk of complications like infection and hyperglycemia.

Overhydration to prevent hypotension (Option C) is avoided as it can worsen cerebral oedema. Fluid management should be carefully balanced to maintain adequate blood flow and cerebral perfusion without overhydrating the infant.

Hyperventilation to reduce intracranial pressure (Option D) is not recommended as a standard treatment in HIE and can lead to complications such as cerebral vasoconstriction, which may worsen the injury.

Solution for Question 4:

Correct Answer: B) Stage 2

Explanation:

In this scenario, the infant's lethargy, hypotonia, intermittent seizures, and weak sucking reflexes align with Stage 2 of the Sarnat classification, indicating moderate HIE.

Table rendering failed

Levene classification system for hypoxic-ischemic encephalopathy (HIE)				
Levene Classification	Level of Consciousness	Muscle Tone	Seizures	Sucking/Respiration
Mild	Irritable	Hypotonia	None	Poor suck
Moderate	Lethargic	Marked Hypotonia	Present	Unable to suck
Severe	Comatose/Unable to Suck	Severe Hypotonia	Present	Unable to sustain spontaneous respiration

Solution for Question 5:

Correct Answer: B) 4

Explanation:

APGAR score

- The APGAR score is an objective method of evaluating the newborn’s condition.
- It is performed at 1 minute and again at 5 minutes after birth.
- Resuscitation must be initiated before the 1 minute score is assigned, therefore, it is not used to guide the resuscitation.
- The components, along with scores, are,

	0	1	2
Appearance	Completely blue or pale	The body is pink, and the extremities are blue	Completely pink
Pulse rate	Absent	<100/min	>100/min
Grimace	No response	Grimaces only	Coughs/sneezes
Activity	Limp and flaccid	Some flexion present	Flexed posture and actively moving limbs
Respiratory effort	None	Slow and irregular	Normal/strong

Interpretation:

- Normal APGAR score is ≥7
- Maximum score = 10; Minimum score = 0
- Score for severe birth asphyxia = 0 - 3

Therefore, the APGAR score for the above case is 4. This indicates moderate distress at birth.

Solution for Question 6:

Correct Answer: D) Silverman- Andersen score

Explanation:

- The above scenario represents respiratory distress in a preterm neonate, in which the Silverman-Andersen score is used.
- Respiratory distress is characterised by tachypnea with lower chest retractions, grunting, and cyanosis.

Silverman-Andersen score (for preterm neonates)			
	0	1	2
Upper chest retractions	Chest and abdomen rise together with respiration	Chest wall lags behind abdomen	Chest wall and abdomen move in opposite directions (see-saw pattern) / paradoxical breathing.

Lower chest retractions	Absent	Minimal	Marked
Xiphisternal retractions	Absent	Minimal	Marked
Nasal flare	Absent	Minimal	Marked
Grunt	None	Audible only with a stethoscope	Audible without a stethoscope
Interpretation: Maximum score is 10; Minimum score is 0 Normal score = 0-3 Severe respiratory distress in a preterm neonate = > 7 score			

Downe's score (for term neonates)			
	0	1	2
Cyanosis	Absent	Present in room air	Present at FiO greater than or equal to 40%
Air- entry	Normal	Decreased	Barely audible
Respiratory rate	<60/min	60-80/min	>80/min
Grunt	Absent	Audible only with a stethoscope	Audible without a stethoscope
Retractions	None	Mild	Severe
Interpretations: Maximum score = 10; Minimum score = 0 Severe respiratory distress in a term baby = >7 score			

CRIB score (Option B) is the Clinical Risk Index for Babies (CRIB), used to predict mortality of neonates by comparing performance of NICUs.

SNAP score (Option C) is the Score for Neonatal Acute Physiology (SNAP) that uses 34 parameters to predict morbidity and mortality for neonates.

Solution for Question 7:

Correct Answer: B) Box-shaped heart

Explanation:

- The most likely diagnosis in this scenario is Respiratory Distress Syndrome (RDS), also known as hyaline membrane disease (HMD), a common condition in preterm infants (<34 weeks of gestation) caused by surfactant deficiency.
- Typical X-ray findings in RDS include: Ground glass haziness of lungs: Represents diffuse alveolar atelectasis, commonly seen in RDS. Presence of air bronchogram: Air-filled bronchi are visible against the hazy, collapsed alveoli, a classic feature of RDS.air bronchogram can be seen in parenchymal pathology of lung such as consolidation, pulmonary alveolar proteiniosis. Reticulogranular pattern: Another term for the ground-glass appearance, indicating the fine, granular pattern throughout the lungs characteristic of RDS.
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-
- A box-shaped heart (Option B) is seen in the Ebstein anomaly; the heart appears box-shaped on X-ray due to the downward displacement of the tricuspid valve, resulting in right atrial enlargement and a distorted heart silhouette.
- Respiratory Distress Syndrome
- Pathophysiology: Surfactant, a lipoprotein produced by type II alveolar cells, lowers surface tension to prevent alveolar collapse at the end of expiration. Its production begins around 20 weeks of gestation and peaks at 35 weeks, so infants born before 35 weeks are at higher risk for RDS due to insufficient surfactant. Without adequate surfactant, alveoli collapse during expiration, impairing gas exchange and causing tissue hypoxia. This results in increased pulmonary pressure, ischemic damage, and the formation of a hyaline membrane from protein transudation.
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- Clinical Features: Respiratory distress within the first 6 hours of life is characterised by tachypnea, retractions, grunting, cyanosis, and decreased air entry.
- Management Neonates with this condition require admission to the neonatal intensive care unit, IV fluids, and appropriate respiratory support. Further management details is depicted in the table below,
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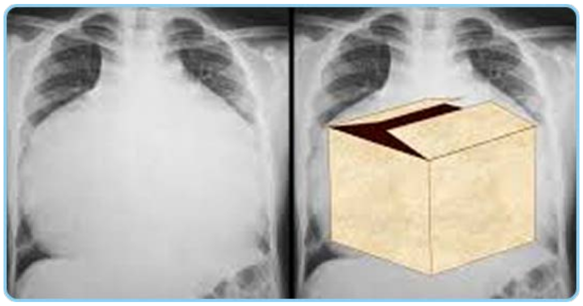

Respiratory Distress Syndrome

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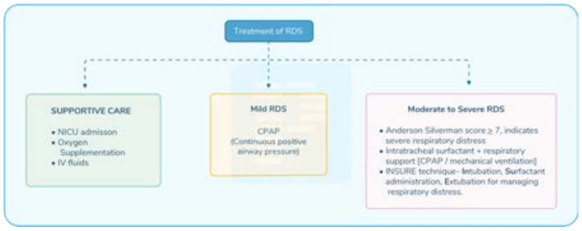
- Its production begins around 20 weeks of gestation and peaks at 35 weeks, so infants born before 35 weeks are at higher risk for RDS due to insufficient surfactant.
- Without adequate surfactant, alveoli collapse during expiration, impairing gas exchange and causing tissue hypoxia. This results in increased pulmonary pressure, ischemic damage, and the formation of a hyaline membrane from protein transudation.
- Neonates with this condition require admission to the neonatal intensive care unit, IV fluids, and appropriate respiratory support.
- Further management details is depicted in the table below,



Solution for Question 8:

Correct Answer: C) Shake test

Explanation:

The procedure is the Foam test or shake test for fetal lung maturity.

Fetal lung maturity tests (detect the adequacy of surfactant in amniotic fluid):

- This is a bedside test, rapidly performed with a fair degree of accuracy.

- Amniotic fluid is mixed with 95% ethyl alcohol. Shake it vigorously for 10-15 minutes and then leave it for another 10-15 minutes.
- A complete ring of bubbles is seen at the meniscus, indicating a positive test for lung maturity.
- Estimation of pulmonary surfactant is done by lecithin/sphingomyelin (L/S) ratio. The ratio at 31-32 weeks is 1 and at 35 weeks it is 2.
- L:S ratio ≥ 2 indicates pulmonary maturity.
- The presence of phosphatidylglycerol (PG) in amniotic fluid reliably indicates lung maturation.
- It is tested by thin-layer chromatography.
- Lamellar body is the storage form of surfactant in Type 2 pneumocytes, which can be detected in amniotic fluid.
- Mature lung: >30,000 / microlitre

Amniotic fluid turbidity test (Option D)

- During the 1st and 2nd trimester, amniotic fluid is yellow and clear.
- At term, it is turbid due to the presence of vernix caseosa, a white, waxy substance that covers and protects the fetal skin and contributes to the fluid's opacity.

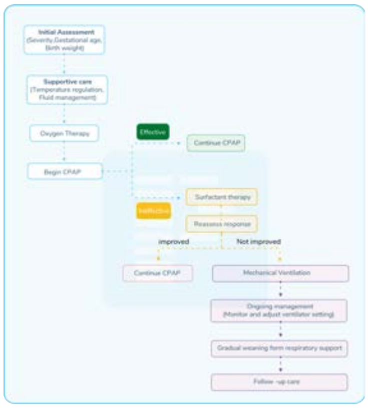
Solution for Question 9:

Correct Answer: D) Prophylactic antibiotics.

Explanation:

Prophylactic antibiotics are not used in the initial management of RDS and are only given if infection, such as sepsis or pneumonia, is suspected. Routine use is not recommended due to the risk of antibiotic resistance.

Management of RDS:



Surfactant replacement therapy (Option A) is crucial for neonatal RDS. It addresses the lack of surfactant in premature infants, reducing alveolar surface tension and improving lung compliance and oxygenation.

Continuous positive airway pressure (CPAP) (Option B) is often used as a first-line treatment for RDS. It helps keep the airways open, improves oxygenation, and can reduce the need for mechanical ventilation.

Mechanical ventilation (Option C): While not always the first choice, mechanical ventilation may be necessary for infants with severe RDS who don't respond adequately to CPAP or surfactant therapy.

Solution for Question 10:

Correct Answer: B) 24-34 weeks

Explanation:

- Antenatal corticosteroids are a crucial intervention in preventing Respiratory Distress Syndrome(RDS) in preterm infants.
- The timing of this intervention is crucial for its effectiveness. Between 24 - 34 weeks, when fetal lungs are rapidly developing and the risk of RDS is highest, corticosteroids can significantly accelerate lung maturation.

Treatment:

Table rendering failed

Steroid of choice: Inj. Betamethasone has a slightly more neuroprotective effect:

20-24 weeks (Option A): This is generally too early for corticosteroid administration. At this stage, the fetus's lungs are not developed enough to benefit significantly from the treatment. Additionally, survival rates for infants born at this gestational age are very low.

34-37 weeks(Option C): While some benefit may still be seen in this late preterm period, the standard recommendation for routine administration typically ends at 34 weeks. After this point, the risk-benefit ratio becomes less favourable as the incidence of RDS naturally decreases.

37-40 weeks(Option D): This is considered full term. By this point, the lungs are usually mature enough that RDS is uncommon, and the potential risks of corticosteroid administration may outweigh the benefits.

Solution for Question 11:

Correct Answer: B) Stimulation and provision of positive pressure ventilation

Explanation:

Current neonatal resuscitation guidelines no longer recommend routine intubation and tracheal suctioning for all meconium-stained infants. Instead, the recommended approach is to provide stimulation and positive pressure ventilation (PPV) if the infant shows poor respiratory effort or has a low heart rate.

Meconium Aspiration Syndrome (MAS):

- It occurs when a newborn inhales meconium-stained amniotic fluid, which can cause airway obstruction, inflammation, and infection.
- Clinical Presentation: Respiratory distress, tachypnea, cyanosis, decreased breath sounds, and meconium staining on the skin or umbilical cord.
- Diagnosis: Based on clinical signs, with chest X-rays showing patchy infiltrates, hyperinflation, flattened diaphragms, increased lung radiolucency, and areas of atelectasis due to airway obstruction by meconium, leading to inflammation and air trapping.
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- Treatment: Supportive respiratory care (oxygen therapy, positive pressure ventilation, mechanical ventilation if needed). Surfactant administration for severe cases and managing

complications.

- Supportive respiratory care (oxygen therapy, positive pressure ventilation, mechanical ventilation if needed).
- Surfactant administration for severe cases and managing complications.
- Complications: Respiratory failure, persistent pulmonary hypertension (PPHN), pneumothorax, and, in severe cases, chronic lung disease.
- Surfactant administration (Option D) may be beneficial in severe cases, but it is not the initial step in management. Routine intubation and tracheal suctioning (Option A) are no longer recommended for all meconium-stained infants. Prophylactic antibiotics (Option C) are not indicated immediately unless there is evidence of infection; they may be considered later if risk factors are present.
- Supportive respiratory care (oxygen therapy, positive pressure ventilation, mechanical ventilation if needed).
- Surfactant administration for severe cases and managing complications.

Surfactant administration (Option D) may be beneficial in severe cases, but it is not the initial step in management.

Routine intubation and tracheal suctioning (Option A) are no longer recommended for all meconium-stained infants.

Prophylactic antibiotics (Option C) are not indicated immediately unless there is evidence of infection; they may be considered later if risk factors are present.

Solution for Question 12:

Correct Answer: C) Persistent hypoxemia requiring prolonged mechanical ventilation

Explanation:

Persistent hypoxemia requiring prolonged mechanical ventilation is not typical of Transient tachypnea of newborns (TTN); it usually self-limits, resolves within days, and rarely needs extended ventilation.

Transient tachypnea of the newborn (TTN):

- A temporary respiratory condition in newborns characterised by rapid breathing (tachypnea), typically occurring shortly after birth.

- Cause: Delayed clearance of fetal lung fluid after birth. This fluid retention in the lungs makes breathing harder for the newborn. (Option D)
- Risk factors: Cesarean delivery (Option A), maternal asthma, male gender, late preterm or early term birth, and macrosomia (large baby).
- Symptoms: Fast breathing (usually >60 breaths per minute), grunting, nasal flaring, mild chest retractions, and cyanosis. The newborn may also exhibit decreased feeding due to respiratory effort.
- Fast breathing (usually >60 breaths per minute), grunting, nasal flaring, mild chest retractions, and cyanosis.
- The newborn may also exhibit decreased feeding due to respiratory effort.
- Onset is usually within a few hours after birth and typically resolves within 24-72 hours.
- The diagnosis is based on the clinical presentation, chest X-ray showing fluid in the lungs, and exclusion of other respiratory conditions.
- Treatment: Primarily supportive care, including oxygen supplementation if needed, monitoring of vital signs, and sometimes continuous positive airway pressure (CPAP). Antibiotics may be started initially but discontinued if infection is ruled out.
- Prognosis: Generally excellent. Most cases resolve within 24-72 hours, though some may take up to 5 days. There are usually no long-term complications. (Option B)
- Fast breathing (usually >60 breaths per minute), grunting, nasal flaring, mild chest retractions, and cyanosis.
- The newborn may also exhibit decreased feeding due to respiratory effort.

Solution for Question 13:

Correct Answer: B) Administer intravenous caffeine citrate

Explanation:

- Caffeine is the treatment of choice for recurrent or persistent apnea of prematurity that does not respond to initial measures like tactile stimulation or nCPAP.
- It increases central respiratory drive, lowers the threshold for response to hypercapnia, enhances diaphragmatic contractility, and prevents diaphragmatic fatigue.

Apnea of Prematurity:

- It is defined as the cessation of breathing for ≥20 seconds or more, or for any duration if accompanied by cyanosis or bradycardia, occurring in an infant born prematurely (<34 weeks GA) due to the immaturity of the central respiratory control mechanisms.
- It occurs in nearly all infants born at <28 weeks (Extreme preterm infants) and decreases in frequency with increasing gestational age.
- Management For mild and intermittent episodes: Gentle tactile stimulation or provision of supplemental oxygen by nasal cannula is often sufficient. If apnea episodes are more frequent or prolonged, nasal continuous positive airway pressure (nCPAP) or heated humidified high-flow nasal cannula (HHHFNC) can be used to maintain airway patency. For recurrent or persistent apnea: Methylxanthines (e.g., caffeine) are preferred because they effectively stimulate respiratory drive and reduce the incidence and severity of apnea. Caffeine aids in extubation and lowers Bronchopulmonary dysplasia (BPD) risk with fewer side effects than theophylline. It is safe for oral or intravenous use with a wide therapeutic window and minimal need for serum monitoring.
- For mild and intermittent episodes: Gentle tactile stimulation or provision of supplemental oxygen by nasal cannula is often sufficient.
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- It is safe for oral or intravenous use with a wide therapeutic window and minimal need for serum monitoring.

Continuing nCPAP (Option A) is beneficial for managing obstructive or mixed apnea by maintaining airway patency but is less effective for central apnea. Since the apnea persists despite nCPAP, additional pharmacological intervention is needed.

Initiate doxapram infusion (Option C): Doxapram is a respiratory stimulant but is less commonly used due to side effects and is typically considered only if methylxanthines are ineffective.

Start low-dose theophylline (Option D): Theophylline is another methylxanthine but is less preferred than caffeine due to a shorter half-life and higher risk of side effects d/t narrow therapeutic range (5-15 microgram), such as tachycardia and feeding intolerance.

Solution for Question 14:

Correct Answer: D) The condition is mostly sporadic.

Explanation:

The given clinical scenario of a preterm infant with severe respiratory distress with a scaphoid abdomen and x-ray showing bowel loops in the thorax points towards Congenital Diaphragmatic Hernia (CDH), which is mostly sporadic.

Congenital Diaphragmatic Hernia:

- Defect in the fetal diaphragm leading to herniation of abdominal contents into the thorax, impairing lung growth.
- Types: Bochdalek Hernia (Defective Development of Pleuroperitoneal Membrane) Morgagni Hernia
- Bochdalek Hernia (Defective Development of Pleuroperitoneal Membrane)
- Morgagni Hernia
- Bochdalek Hernia (Defective Development of Pleuroperitoneal Membrane)
- Morgagni Hernia

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Features	Bochdalek Hernia	Morgagni Hernia
Side	Left-sided (Option A)	Right-sided
Location	Posterolateral	Anteromedial (Option B)
Herniated Organs	Small intestine, stomach	Transverse colon, sometimes liver
Common Symptoms	Scaphoid abdomen Respiratory distress Mediastinal shift	Mostly Asymptomatic and Incidental
Diagnostic Features	Pulmonary hypoplasia Pulmonary hypertension	Less severe respiratory distress
Association	Polyhydramnios, abnormal LHR, and liver position	Rarely detected prenatally; usually diagnosed later
Management	Variable	Better prognosis (Option C)

Diagnosis:

- Antenatal USG (16 - 24 weeks).
- Chest X-ray reveals bowel loops in the thorax, mediastinal shift, and pulmonary hypoplasia.

Solution for Question 15:

Correct Answer: A) Bag and mask ventilation

Explanation:

Bag and Mask Ventilation is contraindicated in CDH patients due to the risk of worsening respiratory distress and increasing intrathoracic pressure, which can further impede lung expansion and exacerbate the hernia.

Management of Congenital Diaphragmatic Hernia:

- Initial Stabilization: Airway Management: Immediate endotracheal intubation (ET) is critical. (Option B) ECMO: If there is no response to ET (Option C) Avoid excessive, mean airway pressure to prevent barotrauma Gastric Decompression: An orogastric tube is inserted to prevent gastric distention that worsens lung compression. (Option D) Bag and Mask Ventilation is Contraindicated in CDH
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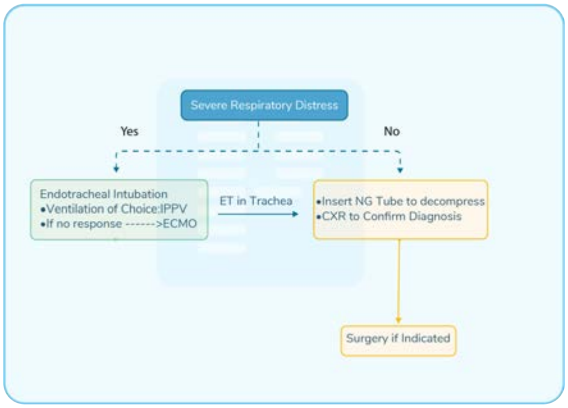
- Bag and Mask Ventilation is Contraindicated in CDH
- Immediate endotracheal intubation (ET) is critical. (Option B)
- ECMO: If there is no response to ET (Option C)



- Surgical Repair is done only after pulmonary stabilisation in CDH (Immediate surgery done in case of Traumatic Hernia)

Solution for Question 16:

Correct Answer: B) Pulmonary hypoplasia

Explanation:

- In congenital diaphragmatic hernia (CDH), a defect in the fetal diaphragm causes abdominal contents to herniate into the thorax, which inhibits lung growth.
- The most common cause of death is pulmonary hypoplasia and its associated pulmonary hypertension.
- Clinically, it manifests with severe respiratory distress, decreased breath sounds on the affected side, and a scaphoid abdomen due to herniated abdominal contents. Cyanosis and mediastinal shift toward the unaffected side are also common.
- Management: Includes surgical repair, extracorporeal membrane oxygenation (ECMO), and supportive care. In-utero interventions.
- Prognosis and Mortality: Good Prognostic Factors: Mild CDH, postnatal repair, and absence of severe pulmonary hypoplasia. Poor Prognostic Factors: Severe pulmonary hypoplasia, severe pulmonary hypertension, need for ECMO, Liver herniation, Lung Heart Ratio<1, Symptoms within first 24 hrs.
- Good Prognostic Factors: Mild CDH, postnatal repair, and absence of severe pulmonary hypoplasia.

- Poor Prognostic Factors: Severe pulmonary hypoplasia, severe pulmonary hypertension, need for ECMO, Liver herniation, Lung Heart Ratio<1, Symptoms within first 24 hrs.
- Good Prognostic Factors: Mild CDH, postnatal repair, and absence of severe pulmonary hypoplasia.
- Poor Prognostic Factors: Severe pulmonary hypoplasia, severe pulmonary hypertension, need for ECMO, Liver herniation, Lung Heart Ratio<1, Symptoms within first 24 hrs.

Perforation of the herniated bowel (Option A) is a severe complication in CDH leading to peritonitis or sepsis but is less common than pulmonary complications as a cause of death.

Pulmonary Embolism (Option C) is a severe condition that can occur due to congenital heart defects or hypercoagulable states, and it is less common in CDH compared to pulmonary hypoplasia.

Neonatology-I

1. A 32-week pregnant woman presents due to concerns about her baby's growth. Her pregnancy has been complicated by chronic hypertension. The ultrasound reveals that the fetal head size is within the normal range, but the body weight and length are significantly below the 10th percentile. What is the most likely etiology of the fetal growth restriction in this case?

A. Genetic disorders

B. Chromosomal disorders

C. Intrauterine infection

D. Placental insufficiency

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2. Which of the following best describes morphological IUGR?

A. Birth weight below the 10th percentile with malnutrition.

B. Birth weight between the 10th and 25th percentiles with malnutrition.

C. Birth weight above the 25th percentile with malnutrition.

D. Birth weight at the 50th percentile with no malnutrition

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3. A 30-year-old pregnant woman is in her second trimester with a history of mild hypertension. Her obstetrician is concerned about the risk of IUGR due to her history . Which of the following strategies should be recommended to prevent IUGR?

A. Low-dose aspirin

B. Increase caffeine intake

C. High-dose antioxidant supplement

D. Avoid all forms of exercise

4. An infant, born at 35 weeks gestation, presents with worsening respiratory distress, irritability, and feeding difficulties. The mother reports that the infant had poor cord care after birth. The infant's temperature is 38°C. He is diagnosed with late-onset sepsis. Which is the most likely timing of the sepsis in this infant?

A. Up to 72 hours after birth

B. After 72 hours but before 7 days after birth

C. Beyond 7 days of life

D. More than 10 days after birth

5. A 2700-g male infant, born at 36 weeks gestation, presents at 6 hours of age with grunting, nasal flaring, and intercostal retractions. He is also experiencing poor feeding. His temperature is 35.8°C, and blood pressure is 42 mm Hg systolic. A CBC shows a WBC count of 2500 cells/mm3 with 80% bands. What is the most likely diagnosis?

A. Neonatal Sepsis

B. Respiratory Distress Syndrome

C. Transient Tachypnea of the Newborn

D. Meconium Aspiration Syndrome

6. A 2-day-old, term neonate presents with lethargy, poor feeding, and mild hypothermia. You suspect neonatal sepsis and order a sepsis screen as part of the diagnostic workup.Which of the following findings would be included in the sepsis screen for this neonate?

A. Total leukocyte count > 5000 /mm3

B. Total leukocyte count < 5000 /mm3

C. Increased absolute neutrophil count

D. Decreased absolute neutrophil count as per Mouzinho's chart.

7. A 3-day-old neonate presents with fever, irritability, and poor feeding. The sepsis screen is equivocal, but a lumbar puncture confirms the diagnosis of bacterial meningitis. Blood culture is pending, but the clinical team decides to initiate treatment immediately. Which of the following is the most appropriate antibiotic regimen and duration of therapy for this neonate?

A. Oral Cefotaxime and Gentamicin for 21 days

B. IV Cefotaxime and Gentamicin for 14 days

C. IV Ampicillin and Gentamicin for 21 days

D. IV Cefotaxime and Gentamicin for 21 days

8. During a routine examination of a newborn weighing 2.5 kg, 15 minutes after birth, you would classify the neonate as having moderate hypothermia if the body temperature is:

A. 36.0–36.4 °C

B. 32.0–35.9 °C

C. 28.0–31.9 °C

D. <28.0 °C

9. Which of the following statements is incorrect about Kangaroo Mother Care (KMC)? I. KMC is only recommended for preterm infants weighing less than 1500 grams. II. KMC must be performed continuously for at least 24 hours a day to be

effective. III. KMC is not necessary once the baby attains 2500 g and a gestational age of 37 weeks. IV. KMC is as good as incubator care in terms of safety and thermal care.

A. I only

B. II only

C. I and II

D. III and IV

---

10. A full-term neonate is born in a resource-limited rural health center. The delivery room temperature is 20°C. Which of the following steps is not part of the warm chain that should be immediately implemented to prevent neonatal hypothermia?

A. Drying the baby immediately after birth.

B. Skin-to-skin contact between the mother and baby.

C. Covering the baby’s head with a warm cap.

D. Bathing the baby in warm water

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11. A 3-day-old preterm neonate is brought to the NICU with an axillary temperature of 31.5°C after being found hypothermic in the maternity ward. The baby is placed under a radiant warmer, and rewarming measures are initiated. After 1 hour, the baby's temperature is rechecked. Which of the following indicates that the rewarming process is proceeding at an appropriate rate for neonatal hypothermia?

A. The baby's temperature rises to 34°C within the first hour.

B. The baby's temperature rises to 36.5°C within 2 hours.

C. The baby's temperature rises by 0.5°C within the first hour.

D. The baby's temperature rises by 0.2°C within the first hour.

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12. A 1-day-old term neonate born to a mother with gestational diabetes is found to have a blood glucose level of 30 mg/dL during routine screening. The baby is asymptomatic and has been breastfeeding moderately well. What is the most appropriate next step in the management of this neonate?

A. Start intravenous glucose infusion immediately.

B. Administer a bolus of 2 mL/kg of 10% dextrose and then start glucose infusion.

C. Provide a trial of oral feeds and recheck blood glucose in 30-45 minutes.

D. Repeat the blood glucose test after 1 hour without feeding and monitor.

13. You are examining an LGA neonate who presents with the physical finding shown in the image below. Given this finding, you suspect the neonate might have an associated congenital heart anomaly. What is the most common congenital heart anomaly you would suspect in this case?

A. Atrial Septal Defect (ASD)

B. Tetralogy of Fallot (TOF)

C. Ventricular Septal Defect (VSD)

D. Patent Ductus Arteriosus (PDA)

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	1
Question 4	3
Question 5	1
Question 6	2
Question 7	4
Question 8	2
Question 9	3
Question 10	4
Question 11	3
Question 12	3
Question 13	3

Solution for Question 1:

Correct Answer: D) Placental insufficiency

Explanation:

The Presence of normal head size with reduced body weight and length, combined with the maternal history of chronic hypertension, points to asymmetric IUGR due to placental insufficiency

	Symmetric IUGR	Asymmetric IUGR

Time of insult	Early pregnancy (1 trimester or early 2 trimester)	Late pregnancy (2 or 3 trimester)
Etiology	Genetic disorders (Option A ruled out) TORCH infections (Option C ruled out) Chromosomal disorders (Option B ruled out)	Maternal undernutrition Maternal Hypertension Anemia
Effect on cells	No. of cells is decreased	Size of cells is mainly affected
Anthropometric parameters	Head circumference, length and weight are equally affected	Head circumference and length are less affected than weight
Ponderal index (P.I)	≥ 2	< 2 Better prognosis

Solution for Question 2:

Correct Answer: B) Birth weight between the 10th and 25th percentiles with malnutrition.

Explanation:

Reflects the specific definition of morphological IUGR where the birth weight is between the 10th and 25th percentiles with accompanying clinical signs of malnutrition.

- Morphological IUGR: Definition: Babies with clinical features of malnutrition. Birth Weight: Between the 10th and 25th percentiles of the expected weight for their gestational age. Classification: These babies are considered IUGR but are not classified as SGA.
- Definition: Babies with clinical features of malnutrition.
- Birth Weight: Between the 10th and 25th percentiles of the expected weight for their gestational age.
- Classification: These babies are considered IUGR but are not classified as SGA.
- Difference Between SGA and IUGR: SGA (Small for Gestational Age): Definition: Birth weight less than the 10th percentile of the expected weight for a given gestational age. Type: Statistical definition based on percentiles. IUGR (Intrauterine Growth Restriction): Definition: Clinical diagnosis based on growth patterns and clinical features. Type: Clinical definition, considering overall health and growth restriction.
- SGA (Small for Gestational Age): Definition: Birth weight less than the 10th percentile of the expected weight for a given gestational age. Type: Statistical definition based on percentiles.

- Definition: Birth weight less than the 10th percentile of the expected weight for a given gestational age.
- Type: Statistical definition based on percentiles.
- IUGR (Intrauterine Growth Restriction): Definition: Clinical diagnosis based on growth patterns and clinical features. Type: Clinical definition, considering overall health and growth restriction.
- Definition: Clinical diagnosis based on growth patterns and clinical features.
- Type: Clinical definition, considering overall health and growth restriction.
- Definition: Babies with clinical features of malnutrition.
- Birth Weight: Between the 10th and 25th percentiles of the expected weight for their gestational age.
- Classification: These babies are considered IUGR but are not classified as SGA.
- SGA (Small for Gestational Age): Definition: Birth weight less than the 10th percentile of the expected weight for a given gestational age. Type: Statistical definition based on percentiles.
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- Definition: Clinical diagnosis based on growth patterns and clinical features.
- Type: Clinical definition, considering overall health and growth restriction.
- Definition: Birth weight less than the 10th percentile of the expected weight for a given gestational age.
- Type: Statistical definition based on percentiles.
- Definition: Clinical diagnosis based on growth patterns and clinical features.
- Type: Clinical definition, considering overall health and growth restriction.

Birth weight below the 10th percentile with malnutrition (Option A ruled out): This describes severe IUGR or SGA, not specifically morphological IUGR.

Birth weight above the 25th percentile with malnutrition (Option C ruled out): Birth weight above the 25th percentile does not fit IUGR criteria.

Birth weight at the 50th percentile with no malnutrition (Option D ruled out): Indicates normal growth, not IUGR.

Solution for Question 3:

Correct Answer: A) Low dose Aspirin

Explanation

Low-dose aspirin (50 mg daily) is recommended for women at risk of IUGR, particularly those with a history of hypertension or other factors that may impair placental blood flow.

- It helps improve placental perfusion and reduce the risk of growth restriction.

Prevention of IUGR:

- Correct malnutrition: With an extra 300 calories per day Balanced energy/protein supplementation is associated with 30% reduction in risk of IUGR.
- With an extra 300 calories per day
- Balanced energy/protein supplementation is associated with 30% reduction in risk of IUGR.
- With an extra 300 calories per day
- Balanced energy/protein supplementation is associated with 30% reduction in risk of IUGR.
- Low-Dose Aspirin: 50 mg daily in very selected cases with a history of thrombotic disease, hypertension, pre-eclampsia, or recurrent FGR is associated with 10% reduction in risk of IUGR.
- 50 mg daily in very selected cases with a history of thrombotic disease, hypertension, pre-eclampsia, or recurrent FGR is associated with 10% reduction in risk of IUGR.
- 50 mg daily in very selected cases with a history of thrombotic disease, hypertension, pre-eclampsia, or recurrent FGR is associated with 10% reduction in risk of IUGR.
- Anti-oxidants (Vit. C & Vit. E): No reduction in risk of IUGR (Option C ruled out)
- Restriction of caffeine intake: is associated with reduction in risk of IUGR (Option B ruled out)

Avoid all forms of exercise (Option D ruled out): Avoiding all forms of exercise is not necessary and could negatively impact overall maternal health.

- Moderate exercise, unless contraindicated, is usually encouraged.

Solution for Question 4:

Correct Answer: C) Beyond 7 days of life

Explanation:

The infant's age of 14 days and the presentation of symptoms, including poor feeding, and respiratory distress fit the pattern of late-onset sepsis, which typically occurs beyond 7 days of life.

Parameter	Early-Onset Sepsis	Late-Onset Sepsis
Timing	Within the first 72 hours of life, but can occur up to day 7 of life (Options A and B ruled out)	Begins beyond the first week of life (Option D ruled out)
Common Organisms	Derived from maternal genital tract: Group B Streptococcus, E. coli, Klebsiella	Community-acquired (e.g., S.aureus), Nosocomial (e.g., Acinetobacter, Klebsiella
Risk factors	Spontaneous onset of preterm labor Prolonged rupture of membranes (>18 hours) Foul-smelling liquor Multiple per vaginal Examinations Maternal fever Prolonged labor	Poor hand hygiene LBW Lack of breastfeeding Poor cord care Superficial infections (pyoderma, umbilical sepsis) Aspiration of feeds Needle pricks Use of IV fluids.
Case fatality rate	High	Low

Solution for Question 5:

Correct Answer: A) Neonatal Sepsis

Explanation:

The combination of respiratory distress(grunting, nasal flaring, and intercostal retractions), hypothermia(35.8 °C), poor perfusion(42 mm Hg systolic), and a low white blood cell count with a high percentage of bands points to Neonatal sepsis.

Clinical Features of Neonatal Sepsis:

- Early Signs: Poor feeding
- Temperature Disturbances: Hypothermia (more common than fever)
- Metabolic Issues: Hypoglycemia metabolic acidosis elevated lactate
- Systemic Manifestations: CNS: Shrill cry, irritability, seizures, abnormal posturing, bulging anterior fontanelle Respiratory: Tachypnea, hypoxia, retractions, grunting GI: Abdominal distension, feed intolerance, recurrent vomiting, necrotizing enterocolitis-like features Severe Sepsis: Septic shock, Multi-organ dysfunction, DIC Sclerema: Generalized non-pitting edema due to severe sepsis or hypothermia
- CNS: Shrill cry, irritability, seizures, abnormal posturing, bulging anterior fontanelle
- Respiratory: Tachypnea, hypoxia, retractions, grunting
- GI: Abdominal distension, feed intolerance, recurrent vomiting, necrotizing enterocolitis-like features
- Severe Sepsis: Septic shock, Multi-organ dysfunction, DIC
- Sclerema: Generalized non-pitting edema due to severe sepsis or hypothermia
- CNS: Shrill cry, irritability, seizures, abnormal posturing, bulging anterior fontanelle
- Respiratory: Tachypnea, hypoxia, retractions, grunting
- GI: Abdominal distension, feed intolerance, recurrent vomiting, necrotizing enterocolitis-like features
- Severe Sepsis: Septic shock, Multi-organ dysfunction, DIC
- Sclerema: Generalized non-pitting edema due to severe sepsis or hypothermia

Respiratory Distress Syndrome (Option B): Presents with respiratory distress but does not commonly cause significant hypothermia or systemic signs such as poor perfusion and low blood pressure.

Transient Tachypnea of the Newborn (Option C): Presents with respiratory distress but usually resolves quickly without the systemic involvement seen in this case.

Meconium Aspiration Syndrome (Option D): MAS typically does not cause hypothermia or the profound leukopenia and bandemia observed here.

Solution for Question 6:

Correct Answer: B) Total leukocyte count < 5000

Explanation:

In neonatal sepsis, a total leukocyte count of less than 5000 is included in a sepsis screen.

Diagnosis of Neonatal Sepsis (NNS):

- Diagnosing sepsis, especially in neonates, can be challenging due to its overlap with other conditions like hypothermia, hyperthermia, and hypoglycemia.
- It's crucial to conduct a thorough clinical examination and relevant investigations to distinguish sepsis from other potential causes and avoid unnecessary antibiotic treatment.

Investigations in NNS:

- For a high probability sepsis case, treatment should start immediately without waiting for investigations.
- Blood cultures are essential for definitive diagnosis and should be collected before initiating antimicrobial therapy.
- A sepsis screen includes:

Components	Abnormal values
Total Leucocyte count (TLC)	<5000/mm3 (Option B)
Absolute Neutrophil Count (ANC)	Low counts as per Manroe chart for term infants Mouzinho's chart for VLBW infants
Immature/Total neutrophil (IT)	>0.2
Micro-ESR	>15mm in 1st hour
C-reactive protein	>1 mg/dl

A positive sepsis screen is indicated by at least two abnormal parameters.

- A lumbar puncture is recommended for all suspected cases of neonatal sepsis (NNS), except for asymptomatic infants being evaluated solely for maternal risk factors.

Increased total leucocyte count (Option A ruled out) and Increased neutrophil count (Option C ruled out) can be ruled out from the above table for sepsis screen.

In screening a term neonate for sepsis, the decreased neutrophil count is evaluated using Manroe's chart rather than Mouzinho's chart (Option D ruled out).

Solution for Question 7:

Correct Answer: D) IV Cefotaxime and Gentamicin for 21 days

Explanation:

In confirmed neonatal meningitis, the recommended antibiotic regimen includes a third-generation cephalosporin like Cefotaxime, often in combination with Gentamicin, for a duration of 21 days to ensure complete eradication of the infection and prevent complications.

Treatment of Neonatal Sepsis:

- Oral antibiotics (Option A ruled out) have no role in the treatment of Neonatal Sepsis.
- Intravenous broad spectrum antibiotics are the treatment of choice.
- In case of suspicion or confirmed meningitis, 3rd generation cephalosporin like cefotaxime (Option D) (Option C ruled out) should be added.
- Duration of antibiotic therapy:

Sepsis screen	Blood culture	Meningitis	Term used	Duration
-	-	-	Clinical sepsis	5-7 days
+	-	-	Screen positive sepsis	7-10 days
+/-	+	-	Culture positive sepsis	14 days
+/-	+/-	+	Meningitis	21 days (Option B ruled out)

- Crucial step in the prevention of NNS: Proper hand washing

Solution for Question 8:

Correct Answer: B) 32.0–35.9 °C

Explanation:

Moderate hypothermia in neonates is classified when the body temperature ranges from 32.0 to 35.9 °C.

Neonatal hypothermia is classified as follows:

Cold stress	36.0 to 36.4 oC (Option A ruled out)

Moderate Hypothermia	32.0–35.9 oC
Severe Hypothermia	<32 oC (Option C and D ruled out)

The clinical thermometer should be kept in the axilla for a minimum of 3 minutes to accurately record the axillary temperature.

Solution for Question 9:

Correct Answer: C) I and II

Explanation:

- KMC is recommended for all low birth weight infants, not just those under 1500 grams.
- KMC does not have to be performed continuously for 24 hours to be effective; shorter, practical periods of KMC are also beneficial.

Kangaroo Mother Care (KMC):

- KMC was first suggested in 1978 by Dr Edgar Rey in Bogota, Colombia, for the prevention of neonatal hypothermia in low birth weight/ premature newborns.
- Kangaroo Mother Care (KMC) can be carried out by the mother or any other caregiver.
- Components: Kangaroo positioning: Skin-to-skin positioning of the newborn on the mother's chest. Kangaroo nutrition: Adequate nutrition through breastfeeding. Early discharge and ambulatory care.
- Kangaroo positioning: Skin-to-skin positioning of the newborn on the mother's chest.
- Kangaroo nutrition: Adequate nutrition through breastfeeding.
- Early discharge and ambulatory care.
- Studies suggest: It is as good as incubators, in terms of safety and thermal protection. (Statement IV) Reduces severe morbidity. Promoted bonding between mother and child.
- It is as good as incubators, in terms of safety and thermal protection. (Statement IV)
- Reduces severe morbidity.
- Promoted bonding between mother and child.
- It is indicated for all hemodynamically stable low birth weight babies. (Statement I).
- Short sessions of KMC (Statement II) can be initiated in babies on medical treatment like IV fluids or oxygen therapy.

- Initiation of KMC can vary with the birth weight of the baby.
- Positioning for KMC: The baby is placed between the mother's breast in an upright position. The hips are abducted and flexed in the frog position and the arms are also flexed. The abdomen of the baby should be at the level of the mother's epigastrium. In this position, the mother's breathing would stimulate the baby and prevent the development of apnea.
- The baby is placed between the mother's breast in an upright position.
- The hips are abducted and flexed in the frog position and the arms are also flexed.
- The abdomen of the baby should be at the level of the mother's epigastrium.
- In this position, the mother's breathing would stimulate the baby and prevent the development of apnea.
- Kangaroo positioning: Skin-to-skin positioning of the newborn on the mother's chest.
- Kangaroo nutrition: Adequate nutrition through breastfeeding.
- Early discharge and ambulatory care.
- It is as good as incubators, in terms of safety and thermal protection. (Statement IV)
- Reduces severe morbidity.
- Promoted bonding between mother and child.
- The baby is placed between the mother's breast in an upright position.
- The hips are abducted and flexed in the frog position and the arms are also flexed.
- The abdomen of the baby should be at the level of the mother's epigastrium.
- In this position, the mother's breathing would stimulate the baby and prevent the development of apnea.

- The duration of KMC should be gradually increased from short sessions of more than one hour to 24 hours a day.
- KMC is not necessary once the baby attains 2500 g of weight or 37 weeks of gestation. (Statement III).

Solution for Question 10:

Correct Answer: D) Bathing the baby in warm water.

Explanation:

- The Warm Chain is a series of 10 interlinked procedures to ensure the thermal protection of newborns.
- These include drying the baby, initiating skin-to-skin contact, and covering the head to prevent heat loss.
- However, bathing the baby in warm water is not part of the Warm Chain.

Components of the warm chain:

Solution for Question 11:

Correct Answer: C) The baby's temperature rises by 0.5°C within the first hour.

Explanation:

A rise in temperature of 0.5°C/hour is considered adequate in the treatment of a neonate with severe hypothermia.

Treatment of Hypothermia in neonates:

Cold stress and Moderate hypothermia (Temperatures up to 32°C)	Severe hypothermia (Temperature < 32°C)

Passive rewarming: Eliminate exposure to cold environments, cold clothing, or wet clothing. Active rewarming: Skin-to-skin contact, warm room, radiant warmer, or incubator. Monitor Temperature Ensure frequent feedings	Rewarming: Initial rapid rewarming up to 34°C followed by slow rewarming. Temperature Monitoring: Every hour for the first 3 hours, every 2 hours until the temperature is normal, then every 3 hours for the next 12 hours. If the rise of temperature has been by 0.5°C per hour then heating is considered adequate (Option C) Additional Support: Provide oxygen. Empirical antibiotics. IV saline if in shock, IV dextrose, and vitamin K. Continuously monitor vital signs

Solution for Question 12:

Correct Answer: C) Provide a trial of oral feeds and recheck blood glucose in 30-45 minutes.

Explanation:

In asymptomatic babies with blood sugar of more than 20 mg/ dL, a trial of oral feeds is given and blood sugar is rechecked after 30-45 minutes.

Management of Hypoglycemia in neonates:

HYPOGLYCEMIA TREATMENT" data-author="" data-hash="835" data-license="" data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/1pxmYeseLDUCYr74Xakf1741552486.png" />

Solution for Question 13:

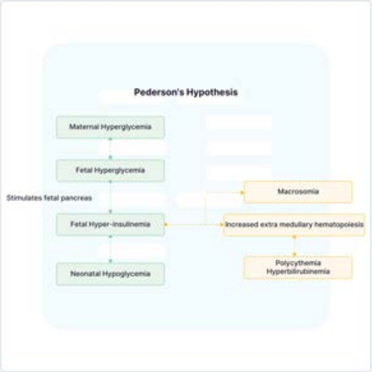
Correct Answer: C) Ventricular Septal Defect (VSD)

Explanation:

A hairy pinna points to the diagnosis of an infant of a diabetic mother, and the most common congenital heart disease in IDM is ventricular septal defect (VSD)

Infant of a diabetic mother:

- Diabetes is one of the most common endocrine disorders affecting women during pregnancy.
- Infants of diabetic mothers (IDM) have a higher incidence of congenital malformations, estimated to be between 2.5 and 12%, with an excess of cardiac problems.
- Pathogenesis:



- Problems in IDM:

Macrosomia/Large for Date Baby	Increased risk of birth trauma if forceps are used in obstructed labor. Increased chances of perinatal asphyxia in prolonged labor.
Metabolic	Hypoglycemia Hypocalcemia Hypomagnesemia Polycythemia Neonatal jaundice
CVS	The most common congenital abnormality in infants of diabetic mothers (IDM) is congenital heart disease (CHD). The most common congenital heart disease in IDM is ventricular septal defect (VSD). (Option C) The most specific congenital heart disease in IDM is transposition of the great arteries (TGA).
Respiratory System	Respiratory distress syndrome or Hyaline membrane disease due to delayed maturation of surfactant.
CNS	Most common congenital neurologic abnormality in IDM: Neural tube defects. Most specific neurologic abnormality in IDM: Sacral agenesis or caudal regression syndrome.
Renal	Renal vein thrombosis Renal agenesis Duplication of ureter Hydroureteronephrosis

GI	Duodenal atresia Lazy/small left colon syndrome
Long Term Problems	Blindness Obesity Nonketotic hypoglycemia Deafness

Neonatology-II

1. A 29-year-old Rh-negative woman, 30 weeks pregnant, presents for a routine follow-up. She has a history of a previous pregnancy complicated by Rh incompatibility but has been receiving appropriate prenatal care. Recent laboratory tests reveal a positive indirect Coombs test with elevated anti-Rh antibodies. The fetal ultrasound shows normal growth and no significant abnormalities. What is the most likely diagnosis for the fetus?

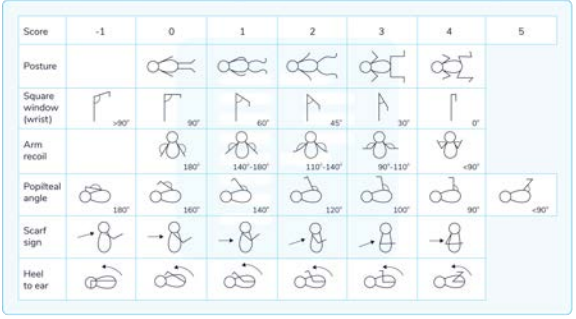
A. Mild Hemolytic Disease of Newborn

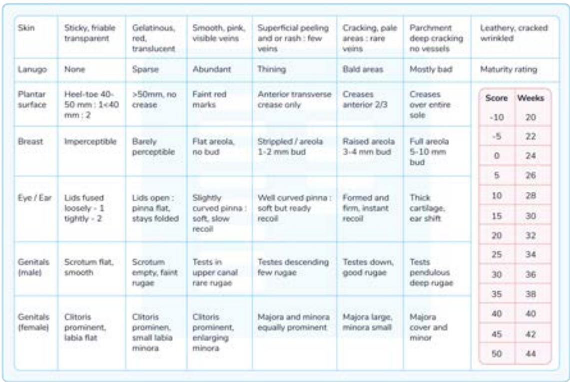
B. Severe Hemolytic Disease of Newborn

C. No Hemolytic Disease of Newborn

D. Severe Hemolytic Disease of Newborn with Normal Growth

2. Which of the following is true about the scoring tool?





Skin	Sticky, friable transparent	Gelatinous, red, translucent	Smooth, pink, visible veins	Superficial peeling and or rash : few veins	Cracking, pale areas : rare veins	Parchment deep cracking no vessels	Leathery, cracked wrinkled
Lanugo	None	Sparse	Abundant	Thining	Bald areas	Mostly bad	Maturity rating
Plantar surface	Heel-toe 40-50 mm : 1<40 mm : 2	>50mm, no crease	Faint red marks	Anterior transverse crease only	Creases anterior 2/3	Creases over entire sole	
Breast	Imperceptible	Barely perceptible	Flat areola, no bud	Strippled / areola 1-2 mm bud	Raised areola 3-4 mm bud	Full areola 5-10 mm bud	
Eye / Ear	Lids fused loosely - 1 tightly - 2	Lids open : pinna flat, stays folded	Slightly curved pinna : soft, slow recoil	Well curved pinna : soft but ready recoil	Formed and firm, instant recoil	Thick cartilage, ear shift	
Genitals (male)	Scrotum flat, smooth	Scrotum empty, faint rugae	Tests in upper canal rare rugae	Testes descending few rugae	Testes down, good rugae	Tests pendulous deep rugae	
Genitals (female)	Clitoris prominent, labia flat	Clitoris prominen, small labia minora	Clitoris prominent, enlarging minora	Majora and minora equally prominent	Majora large, minora small	Majora cover and minor	

Score	Weeks
-10	20
-5	22
0	24
5	26
10	28
15	30
20	32
25	34
30	36
35	38
40	40
45	42
50	44

A. Used solely to assess the neuromuscular maturity

B. Used to assess gestational age based on Physical and Neuromuscular maturity

C. Used only to ascess physical maturity

D. Used exclusively to determine the baby's weight

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3. Refer to the image provided. Which of the following statements is correct regarding the features shown in the image?

A. The system provides an open care environment

B. The system does not include an option for humidification.

C. The system primarily loses heat through radiation.

D. The system is known for being low in cost.

---

4. A preterm neonate is admitted to the neonatal intensive care unit and requires invasive monitoring. The team decides to place an umbilical artery catheter (UAC) and an umbilical venous catheter (UVC).Which of the following statement is correct?

A. UAC is used for drug administration and is associated with sepsis.

B. UVC is used for blood pressure monitoring and blood sampling

C. UAC can cause limb discoloration, while UVC is used for drug administration.

D. Umbilical arterial blood has higher pH and lower pCO2 than umbilical venous blood in the sick.

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5. A baby is born to an HIV-positive mother. The mother has been on antiretroviral therapy (ART) for the past 6 months. What is the appropriate management for this newborn?

A. Daily nevirapine prophylaxis starting at birth for 6 weeks

B. Zidovudine and nevirapine for 6 weeks if the baby is not breastfed

C. Zidovudine and nevirapine for 12 weeks if the baby is breastfed

D. Daily nevirapine prophylaxis starting at birth for 12 weeks

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6. A 28-week preterm baby is found to have recurrent apneas. Which combined approach reduces the need for mechanical ventilation and decreases the risk of bronchopulmonary dysplasia.

A. CPAP and caffeine therapy

B. Intubation and surfactant

C. Surfactant and CPAP

D. High-Flow Nasal Cannula and Caffeine therapy

7. A 28-year-old woman at 32 weeks of gestation undergoes a Doppler ultrasound of the middle cerebral artery (MCA). The results show increased diastolic velocity, and decreased pulsatility index (PI). What do these findings most likely indicate?

A. Normal fetal development.

B. Brain-sparing effect

C. Cerebral edema

D. Cerebral vessel constriction

8. A 32-year-old pregnant woman with blood type: O-negative presents at 28 weeks of gestation with symptoms of significant swelling and difficulty breathing. An ultrasound reveals severe generalized edema, ascites, and hydrothorax in the fetus. The maternal indirect Coombs test is positive. The most likely cause of the hydropic changes in this case can be due to?

A. Rh incompatibility

B. Spherocytosis

C. G6PD deficiency

D. Thalassemia

9. A baby is delivered at 38 weeks gestation to a mother with a history of Rh incompatibility. The delivery was uneventful, but the neonatologist is concerned about the possibility of hemolytic disease due to the mother's sensitization status. The baby appears slightly jaundiced. Which of the following investigations is most appropriate to perform after birth?

A. Direct Coomb's test

B. MCA peak blood flow measurement via USG Dopp

C. Hemoglobin electrophoresis

D. Indirect Coomb's test

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10. A 3-day-old preterm infant, born at 29 weeks of gestation, presents with feeding intolerance, abdominal distension, and bilious vomiting. On examination, the abdomen is tense, and bowel sounds are diminished. An abdominal X-ray reveals pneumatosis intestinalis and portal venous gas. Which of the following anatomical locations is most commonly involved in this condition?

A. Distal part of the jejunum and proximal ileum.

B. Distal part of the ileum and proximal colon.

C. Proximal part of the duodenum and distal jejunum.

D. Sigmoid colon and rectum.

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11. A 29-year-old woman, who is Rh-negative, presents for a routine prenatal visit at 30 weeks of gestation. She has a history of two previous pregnancies, the first was a full-term delivery without complications, and the second ended in a miscarriage at 12 weeks. Her current laboratory tests show a positive indirect Coombs test, and her fetal ultrasound reveals an enlarged liver and spleen. Which of the following mechanisms best explains the pathophysiology of the fetal condition?

A. IgG Antibodies Against Rh Antigen

B. IgM Antibodies Against Rh Antigen

C. IgG Antibodies Against ABO Antigens

D. Fetal Red Blood Cell Destruction Due to Maternal Hyperglycemia

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12. A newborn infant, diagnosed with jaundice, is undergoing phototherapy in the neonatal intensive care unit. Which of the following factors is most critical for maximizing the effectiveness of the phototherapy for this infant?

A. Increasing the distance between the baby and the phototherapy unit.

B. Using a phototherapy unit with a wavelength of 500-550 nm.

C. Ensuring that the baby's exposed surface area is maximized.

D. Using a phototherapy unit with traditional fluorescent lamps instead of LED lamps.

13. A 3-day-old neonate presents with severe jaundice. Despite initial phototherapy, the bilirubin levels remain extremely high. The infant is diagnosed with severe hemolytic disease due to Rh incompatibility. What is the most appropriate next step in management?

A. Continue phototherapy and monitor bilirubin levels.

B. Initiate intravenous immunoglobulin (IVIG) therapy.

C. Perform an exchange transfusion.

D. Prepare for surgical intervention.

14. A neonate presents with jaundice and high bilirubin levels. The condition is associated with various neurological complications. Which of the following is a characteristic feature of kernicterus?

A. Developmental delays

B. Sensorineural hearing loss

C. Disordered executive function

D. Attention deficit/hyperactivity disorder (ADHD)

15. A 2-day-old neonate is assessed for jaundice using Kramer's rule. The neonate's jaundice is noted to extend to the palms and soles. Based on Kramer's scoring system, what is the most appropriate score for this level of jaundice?

A. 1

B. 2

C. 4

D. 5

---

16. A newborn is brought to the clinic on the 7th day of life with mild jaundice. The baby is feeding well, and there are no signs of dehydration. The mother is exclusively breastfeeding, and there are no difficulties with latch or milk production. What is the most appropriate next step in management?

A. Continue exclusive breastfeeding without any intervention

B. Switch to formula feeding temporarily

C. Initiate phototherapy immediately

D. Advise stopping breastfeeding due to risk of worsening jaundice

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17. A newborn presents with yellowing of the skin on the face and sclera at 48 hours of age. The bilirubin levels are elevated but within the range expected for age. The jaundice gradually progresses to the trunk and extremities. Which of the following is the most appropriate management for this patient?

A. Immediate exchange transfusion

B. Administration of intravenous immunoglobulin (IVIG)

C. Initiation of phototherapy

D. Observation and reassurance

---

18. A 10-day-old preterm infant, born at 30 weeks of gestation, presents with sudden abdominal distension and a bluish discoloration of the abdominal wall. The infant appears lethargic and is not feeding well. An abdominal X-ray is performed, which is given below. What Bell's stage of necrotizing enterocolitis (NEC) is most consistent with the radiographic finding?

A. Stage IIa

B. Stage IIIa

C. Stage IIIb

D. Stage IV

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	3
Question 4	3
Question 5	1
Question 6	1
Question 7	2
Question 8	1
Question 9	1
Question 10	2
Question 11	1
Question 12	3
Question 13	3
Question 14	2
Question 15	4
Question 16	1
Question 17	4
Question 18	3

Solution for Question 1:

Correct Answer: A) Mild Hemolytic Disease of Newborn

Explanation:

- The clinical presentation is consistent with mild Hemolytic Disease of Newborn (HDFN)
- Characterized by laboratory evidence of mild hemolysis (positive Coombs test and elevated anti-Rh antibodies) and normal fetal growth without significant symptoms or ultrasound abnormalities.

Clinical features of Erythroblastosis fetalis (HDFN):

Severity and Presentation:

- Mild Hemolysis: Laboratory evidence of hemolysis No significant clinical symptoms
- Laboratory evidence of hemolysis
- No significant clinical symptoms
- Moderate to Severe Hemolysis: Anemia: Pallor Cardiac decompensation (e.g., cardiomegaly, respiratory distress) Massive anasarca (generalized edema) Hydrops Fetalis: (Option B) Excessive abnormal fluid in 2 or more fetal compartments: Skin Pleura Pericardium Placenta Peritoneum Amniotic fluid Frequently leads to death in utero or shortly after birth Associated Symptoms: Severe anemia causing: Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure) Right-sided heart failure (increased pressure, edema, ascites) Pulmonary edema or bilateral pleural effusions (birth asphyxia) Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)
- Anemia: Pallor Cardiac decompensation (e.g., cardiomegaly, respiratory distress) Massive anasarca (generalized edema)
- Pallor
- Cardiac decompensation (e.g., cardiomegaly, respiratory distress)
- Massive anasarca (generalized edema)
- Hydrops Fetalis: (Option B) Excessive abnormal fluid in 2 or more fetal compartments: Skin Pleura Pericardium Placenta Peritoneum Amniotic fluid Frequently leads to death in utero or shortly after birth
- Excessive abnormal fluid in 2 or more fetal compartments: Skin Pleura Pericardium Placenta Peritoneum Amniotic fluid
- Skin
- Pleura
- Pericardium
- Placenta
- Peritoneum
- Amniotic fluid

- Frequently leads to death in utero or shortly after birth
- Associated Symptoms: Severe anemia causing: Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure) Right-sided heart failure (increased pressure, edema, ascites) Pulmonary edema or bilateral pleural effusions (birth asphyxia) Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)
- Severe anemia causing: Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure) Right-sided heart failure (increased pressure, edema, ascites) Pulmonary edema or bilateral pleural effusions (birth asphyxia) Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)
- Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure)
- Right-sided heart failure (increased pressure, edema, ascites)
- Pulmonary edema or bilateral pleural effusions (birth asphyxia)
- Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)
- Laboratory evidence of hemolysis
- No significant clinical symptoms
- Anemia: Pallor Cardiac decompensation (e.g., cardiomegaly, respiratory distress) Massive anasarca (generalized edema)
- Pallor
- Cardiac decompensation (e.g., cardiomegaly, respiratory distress)
- Massive anasarca (generalized edema)
- Hydrops Fetalis: (Option B) Excessive abnormal fluid in 2 or more fetal compartments: Skin Pleura Pericardium Placenta Peritoneum Amniotic fluid Frequently leads to death in utero or shortly after birth
- Excessive abnormal fluid in 2 or more fetal compartments: Skin Pleura Pericardium Placenta Peritoneum Amniotic fluid
- Skin
- Pleura
- Pericardium
- Placenta
- Peritoneum
- Amniotic fluid
- Frequently leads to death in utero or shortly after birth
- Associated Symptoms: Severe anemia causing: Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure) Right-sided heart failure (increased pressure, edema, ascites) Pulmonary edema or bilateral pleural effusions (birth asphyxia) Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)

- Severe anemia causing: Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure) Right-sided heart failure (increased pressure, edema, ascites) Pulmonary edema or bilateral pleural effusions (birth asphyxia) Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)
- Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure)
- Right-sided heart failure (increased pressure, edema, ascites)
- Pulmonary edema or bilateral pleural effusions (birth asphyxia)
- Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)
- Pallor
- Cardiac decompensation (e.g., cardiomegaly, respiratory distress)
- Massive anasarca (generalized edema)
- Excessive abnormal fluid in 2 or more fetal compartments: Skin Pleura Pericardium Placenta Peritoneum Amniotic fluid
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- Peritoneum
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- Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure)
- Right-sided heart failure (increased pressure, edema, ascites)
- Pulmonary edema or bilateral pleural effusions (birth asphyxia)
- Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)
- Hepatic congestion and dysfunction (reduced serum albumin, decreased oncotic pressure)

- Right-sided heart failure (increased pressure, edema, ascites)
- Pulmonary edema or bilateral pleural effusions (birth asphyxia)
- Petechiae, purpura, thrombocytopenia (decreased platelet production or DIC)

Postnatal Manifestations:

- Jaundice: May be absent at birth due to effective placental clearance Severe cases may stain amniotic fluid, cord, and vernix caseosa Pathologic jaundice typically appears within the first 24 hours Risk of kernicterus (bilirubin encephalopathy) due to high levels of unconjugated bilirubin
- May be absent at birth due to effective placental clearance
- Severe cases may stain amniotic fluid, cord, and vernix caseosa
- Pathologic jaundice typically appears within the first 24 hours
- Risk of kernicterus (bilirubin encephalopathy) due to high levels of unconjugated bilirubin
- Other Complications: Hypoglycemia (potentially related to hyperinsulinism and pancreatic islet cell hypertrophy) Risk of severe respiratory distress and asphyxia
- Hypoglycemia (potentially related to hyperinsulinism and pancreatic islet cell hypertrophy)
- Risk of severe respiratory distress and asphyxia
- May be absent at birth due to effective placental clearance
- Severe cases may stain amniotic fluid, cord, and vernix caseosa
- Pathologic jaundice typically appears within the first 24 hours
- Risk of kernicterus (bilirubin encephalopathy) due to high levels of unconjugated bilirubin
- Hypoglycemia (potentially related to hyperinsulinism and pancreatic islet cell hypertrophy)
- Risk of severe respiratory distress and asphyxia

No HDFN (Option C): Less likely because the positive indirect Coombs test and elevated anti-Rh antibodies indicate Rh sensitization

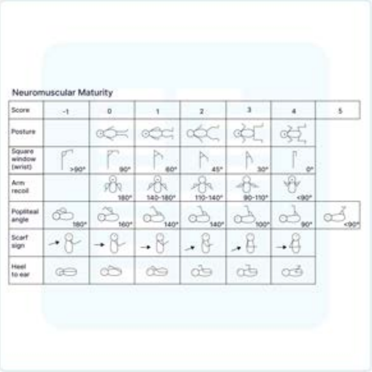
Severe HDFN with Normal Ultrasound (Option D): Severe HDFN would typically present with significant clinical findings and abnormal ultrasound results, not normal findings.

Solution for Question 2:

Correct Answer: B) Helps identify physical maturity and gestational age

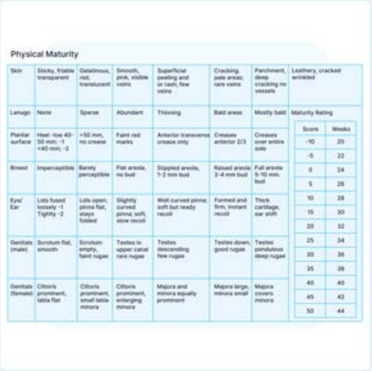
Explanation:

- The given image shows the Expanded New Ballard Score
- The Expanded New Ballard Score assesses Gestational age by evaluating neuromuscular and physical maturity. (Options A, C and D ruled out)
- It assesses various physical characteristics (e.g., skin texture, lanugo, plantar creases).
- It provides an estimate of the gestational age of a preterm baby.
- The score helps in understanding the baby's developmental stage.



Neuromuscular Maturity

Score	-1	0	1	2	3	4	5
Posture							
Square window (wrist)	>90°	90°	60°	45°	30°	0°	
Arm recoil		180°	140-180°	110-140°	90-110°	<90°	
Popliteal angle	180°	160°	140°	140°	100°	90°	<90°
Scarf sign							
Heel to ear							



Physical Maturity

Skin	Sticky, friable transparent	Gelatinous, red, translucent	Smooth, pink, visible veins	Superficial peeling and or rash; few veins	Cracking, pale areas; rare veins	Parchment, deep cracking no vessels	Leathery, cracked wrinkled
Lanugo	None	Sparse	Abundant	Thinning	Bald areas	Mostly bald	
Plantar surface	Heel -toe 40-50 mm; -1 <40 mm; -2	>50 mm, no crease	Faint red marks	Anterior transverse crease only	Creases anterior 2/3	Creases over entire sole	
Breast	Imperceptible	Barely perceptible	Flat areola, no bud	Stippled areola, 1-2 mm bud	Raised areola 3-4 mm bud	Full areola 5-10 mm bud	
Eye/ Ear	Lids fused loosely -1 Tightly -2	Lids open; pinna flat, stays folded	Slightly curved pinna; soft, slow recoil	Well curved pinna; soft but ready recoil	Formed and firm, instant recoil	Thick cartilage, ear stiff	
Genitals (male)	Scrotum flat, smooth	Scrotum empty, faint rugae	Testes in upper canal rare rugae	Testes descending few rugae	Testes down, good rugae	Testes pendulous deep rugae	
Genitals (female)	Clitoris prominent, labia flat	Clitoris prominent, small labia minora	Clitoris prominent, enlarging minora	Majora and minora equally prominent	Majora large, minora small	Majora covers minora	

Maturity Rating

Score	Weeks
-10	20
-5	22
0	24
5	26
10	28
15	30
20	32
25	34
30	36
35	38
40	40
45	42
50	44

Solution for Question 3:

Correct Answer: C) The system primarily loses heat through radiation.

Explanation:

The picture shows it is an incubator. Incubators lose heat primarily through radiation, while radiant warmers primarily lose heat through convection.

Radiant Warmer

- Access to the Baby: Easy
- Insensible Water Loss: Greater (can result in dehydration in small neonates)
- Option to Add Humidification: Not available
- Maintenance and Disinfection: Easy
- Cost: Low
- Most Important Mode of Heating: Radiation
- Heat Loss Mainly Takes Place Through: Convection

Incubator

- Access to the Baby: Restricted(Option A ruled out)
- Insensible Water Loss: Lesser (suitable for neonates below 1500 g)
- Option to Add Humidification: Available in advanced machines(Option B ruled out)
- Maintenance and Disinfection: Difficult to maintain and disinfect
- Cost: High (Option D ruled out)
- Most Important Mode of Heating: Convection
- Heat Loss Mainly Takes Place Through: Radiation

Solution for Question 4:

Correct Answer: C) UAC can cause limb discoloration, while UVC is used for drug administration.

Explanation:

The UAC is associated with complications like limb discoloration due to potential effects on blood flow. The UVC is indeed used for administering drugs and fluids.

- Uses: Invasive blood pressure monitoring (Option B) Repeated blood sampling Exchange transfusion
- Invasive blood pressure monitoring (Option B)
- Repeated blood sampling
- Exchange transfusion
- Most Common Complication: Discoloration of limbs
- Discoloration of limbs
- Invasive blood pressure monitoring (Option B)
- Repeated blood sampling
- Exchange transfusion
- Discoloration of limbs
- Uses: Administer drugs (Option A)
- Administer drugs (Option A)
- Complications: Sepsis
- Sepsis
- Administer drugs (Option A)
- Sepsis
- Umbilical Venous Blood (Option D) Higher pH Lower pCO2 than umbilical arterial blood
- Higher pH
- Lower pCO2 than umbilical arterial blood
- Umbilical Cord Arterial pH: pH < 7 Base deficit > 12 mmol/L Associated with adverse neurological outcomes
- pH < 7
- Base deficit > 12 mmol/L
- Associated with adverse neurological outcomes
- Higher pH
- Lower pCO2 than umbilical arterial blood
- pH < 7
- Base deficit > 12 mmol/L

- Associated with adverse neurological outcomes

Solution for Question 5:

Correct Answer: A) Daily nevirapine prophylaxis starting at birth for 6 weeks.

Explanation:

Since the mother is on ART, the standard prophylaxis for the neonate is daily nevirapine starting at birth for 6 weeks. The other options are relevant for different scenarios or specific high-risk situations.

- If the Mother is on ART: Daily nevirapine prophylaxis starting at birth Duration: 6 weeks (Option D)
- Daily nevirapine prophylaxis starting at birth
- Duration: 6 weeks (Option D)
- For High-Risk Neonates: (Options B and C) High-Risk Factors: Mother not on ART Mother received ART for less than 4 weeks High viral load in the mother (> 1000 copies/ml) Newly identified with HIV within two weeks of delivery Prophylaxis Regimen: Two drugs: zidovudine + nevirapine Duration: If the baby is not breastfed: 6 weeks If the baby is breastfed: 12 weeks
- High-Risk Factors: Mother not on ART Mother received ART for less than 4 weeks High viral load in the mother (> 1000 copies/ml) Newly identified with HIV within two weeks of delivery
- Mother not on ART
- Mother received ART for less than 4 weeks
- High viral load in the mother (> 1000 copies/ml)
- Newly identified with HIV within two weeks of delivery
- Prophylaxis Regimen: Two drugs: zidovudine + nevirapine Duration: If the baby is not breastfed: 6 weeks If the baby is breastfed: 12 weeks
- Two drugs: zidovudine + nevirapine
- Duration: If the baby is not breastfed: 6 weeks If the baby is breastfed: 12 weeks
- If the baby is not breastfed: 6 weeks
- If the baby is breastfed: 12 weeks
- Daily nevirapine prophylaxis starting at birth

- Duration: 6 weeks (Option D)
- High-Risk Factors: Mother not on ART Mother received ART for less than 4 weeks High viral load in the mother (> 1000 copies/ml) Newly identified with HIV within two weeks of delivery
- Mother not on ART
- Mother received ART for less than 4 weeks
- High viral load in the mother (> 1000 copies/ml)
- Newly identified with HIV within two weeks of delivery
- Prophylaxis Regimen: Two drugs: zidovudine + nevirapine Duration: If the baby is not breastfed: 6 weeks If the baby is breastfed: 12 weeks
- Two drugs: zidovudine + nevirapine
- Duration: If the baby is not breastfed: 6 weeks If the baby is breastfed: 12 weeks
- If the baby is not breastfed: 6 weeks
- If the baby is breastfed: 12 weeks
- Mother not on ART
- Mother received ART for less than 4 weeks
- High viral load in the mother (> 1000 copies/ml)
- Newly identified with HIV within two weeks of delivery
- Two drugs: zidovudine + nevirapine
- Duration: If the baby is not breastfed: 6 weeks If the baby is breastfed: 12 weeks
- If the baby is not breastfed: 6 weeks
- If the baby is breastfed: 12 weeks
- If the baby is not breastfed: 6 weeks
- If the baby is breastfed: 12 weeks

Solution for Question 6:

Correct Answer: A) CPAP and caffeine therapy

Explanation:

- CPAP maintains alveolar inflation and reduces the need for invasive ventilation, thereby decreasing the risk of bronchopulmonary dysplasia (BPD).

- Caffeine therapy stimulates respiratory drive and manages apnea of prematurity, reducing the frequency of apnea episodes and the need for additional respiratory support.
- Combined, these approaches effectively lower the risk of BPD.

Delivery Room CPAP

- Continuous Positive Airway Pressure (CPAP): Used in the delivery room for babies born before 32 weeks of gestation.
- Used in the delivery room for babies born before 32 weeks of gestation.
- Benefits of Delivery Room CPAP: Compared to intubation and surfactant administration, CPAP reduces the risk of bronchopulmonary dysplasia or chronic lung disease by 36 weeks of age. (Option B and C ruled out) Results in a decreased need for mechanical ventilation and surfactant therapy.
- Compared to intubation and surfactant administration, CPAP reduces the risk of bronchopulmonary dysplasia or chronic lung disease by 36 weeks of age. (Option B and C ruled out)
- Results in a decreased need for mechanical ventilation and surfactant therapy.
- Used in the delivery room for babies born before 32 weeks of gestation.
- Compared to intubation and surfactant administration, CPAP reduces the risk of bronchopulmonary dysplasia or chronic lung disease by 36 weeks of age. (Option B and C ruled out)
- Results in a decreased need for mechanical ventilation and surfactant therapy.

Caffeine Therapy

Indication:

- Preterm neonate < 32 weeks → Treatment for apnea of prematurity and before extubation from mechanical ventilation.
- < 28 weeks gestation → Use irrespective of the degree of respiratory support required.

Benefits:

- Decreased need for supplemental oxygen and mechanical ventilation.
- Decreased risk of retinopathy of prematurity.
- Decreased risk of bronchopulmonary dysplasia or chronic lung disease.
- Decreased risk of neurodevelopmental impairment → Lesser risk of cerebral palsy.

High-Flow Nasal Cannula (Option D): Useful for some cases but generally less effective than CPAP in preventing BPD and managing severe respiratory distress in extremely preterm infants.

Solution for Question 7:

Correct Answer: B) Brain-sparing effect

Explanation:

- The increased diastolic velocity and decreased PI are characteristic of the brain-sparing effect.
- This phenomenon occurs in response to hypoxemia, where the fetus dilates cerebral vessels to increase blood flow to the brain and protect it from reduced oxygen supply.
- Changes in MCA Doppler: ↑ Diastolic velocity ↓ S/D ratio or Pulsatility Index (PI) Cerebral vessel dilation
- ↑ Diastolic velocity
- ↓ S/D ratio or Pulsatility Index (PI)
- Cerebral vessel dilation
- Clinical Significance: Brain-Sparing Effect: (Option B) Observed in fetal growth restriction (FGR) Due to cerebral vasodilatation in response to hypoxemia (Option D) Abnormal MCA PI: (Option A) Suggests fetal compromise High diastolic flow and decreased PI are indicative of brain-sparing
- Brain-Sparing Effect: (Option B) Observed in fetal growth restriction (FGR) Due to cerebral vasodilatation in response to hypoxemia (Option D)
- Observed in fetal growth restriction (FGR)
- Due to cerebral vasodilatation in response to hypoxemia (Option D)
- Abnormal MCA PI: (Option A) Suggests fetal compromise High diastolic flow and decreased PI are indicative of brain-sparing
- Suggests fetal compromise
- High diastolic flow and decreased PI are indicative of brain-sparing
- Doppler Measurements: Peak Systolic (S) Peak Diastolic (D) Mean (M) S/D Ratio Pulsatility Index (PI): PI = (S–D)/M Resistance Index (RI): RI = (S–D)/S
- Peak Systolic (S)
- Peak Diastolic (D)
- Mean (M)
- S/D Ratio
- Pulsatility Index (PI): PI = (S–D)/M

- Resistance Index (RI): RI = (S–D)/S
- Abnormal MCA Doppler Implications: Brain-Sparing Effect: High diastolic flow, decreased PI Increased PI: Indicates worsening oxygen deficit and possible brain edema (Option C) Reversal in MCA Flow: Suggests cerebral edema
- Brain-Sparing Effect: High diastolic flow, decreased PI
- Increased PI: Indicates worsening oxygen deficit and possible brain edema (Option C)
- Reversal in MCA Flow: Suggests cerebral edema
- ↑ Diastolic velocity
- ↓ S/D ratio or Pulsatility Index (PI)
- Cerebral vessel dilation
- Brain-Sparing Effect: (Option B) Observed in fetal growth restriction (FGR) Due to cerebral vasodilatation in response to hypoxemia (Option D)
- Observed in fetal growth restriction (FGR)
- Due to cerebral vasodilatation in response to hypoxemia (Option D)
- Abnormal MCA PI: (Option A) Suggests fetal compromise High diastolic flow and decreased PI are indicative of brain-sparing
- Suggests fetal compromise
- High diastolic flow and decreased PI are indicative of brain-sparing
- Observed in fetal growth restriction (FGR)
- Due to cerebral vasodilatation in response to hypoxemia (Option D)
- Suggests fetal compromise
- High diastolic flow and decreased PI are indicative of brain-sparing
- Peak Systolic (S)
- Peak Diastolic (D)
- Mean (M)
- S/D Ratio
- Pulsatility Index (PI): PI = (S–D)/M
- Resistance Index (RI): RI = (S–D)/S
- Brain-Sparing Effect: High diastolic flow, decreased PI
- Increased PI: Indicates worsening oxygen deficit and possible brain edema (Option C)
- Reversal in MCA Flow: Suggests cerebral edema

Disappearance of Brain-Sparing Effect or Reversed MCA Flow: Critical for growth-restricted fetus; often precedes fetal death

Solution for Question 8:

Correct Answer: A) Rh incompatibility

Explanation:

Given the maternal blood type (O-negative), the positive indirect Coombs test, and the presence of significant fetal edema and anemia, which are characteristic of immune-mediated hemolysis (Rh incompatibility).

Hydrops Fetalis:

- Description: Most serious form of Rh hemolytic disease (HDFN) Excessive destruction of fetal red cells leads to: Severe anemia Tissue anoxia Metabolic acidosis Adverse effects on: Fetal heart and brain Placenta (hyperplasia to increase oxygen transfer) Progressive diminution of fetal red cells due to hemolysis Resulting damage: Liver damage leading to hypoproteinemia Generalized edema (hydrops fetalis), ascites, and hydrothorax Increased risk for Rh-positive male fetuses (13 times more likely to become hydropic, 3 times more likely to die than female fetuses) Accumulation of extracellular fluid in at least two body compartments Outcomes: Fetal death due to cardiac failure Stillbirth, maceration, or death shortly after birth if born alive
- Most serious form of Rh hemolytic disease (HDFN)
- Excessive destruction of fetal red cells leads to: Severe anemia Tissue anoxia Metabolic acidosis
- Severe anemia
- Tissue anoxia
- Metabolic acidosis
- Adverse effects on: Fetal heart and brain Placenta (hyperplasia to increase oxygen transfer)
- Fetal heart and brain
- Placenta (hyperplasia to increase oxygen transfer)
- Progressive diminution of fetal red cells due to hemolysis
- Resulting damage: Liver damage leading to hypoproteinemia Generalized edema (hydrops fetalis), ascites, and hydrothorax
- Liver damage leading to hypoproteinemia
- Generalized edema (hydrops fetalis), ascites, and hydrothorax

- Increased risk for Rh-positive male fetuses (13 times more likely to become hydropic, 3 times more likely to die than female fetuses)
- Accumulation of extracellular fluid in at least two body compartments
- Outcomes: Fetal death due to cardiac failure Stillbirth, maceration, or death shortly after birth if born alive
- Fetal death due to cardiac failure
- Stillbirth, maceration, or death shortly after birth if born alive
- Types: Immune Hydrops Fetalis: Rh incompatibility ABO incompatibility Other blood group incompatibilities (C, E, Kell, Duffy) Non-Immune Hydrops Fetalis: RBC membrane defects: Spherocytosis (Option B) Elliptocytosis Metabolic disorders: G6PD deficiency (Option C) Systemic diseases (e.g., galactosemia) Hemoglobinopathies: Alpha- and beta-thalassemia syndrome (Option D) Infections: Bacterial Viral (e.g., rubella,parvovirus) Parasitic (e.g., syphilis) Other causes: DIC (Disseminated Intravascular Coagulation) Acute transfusion hemolysis Vitamin E deficiency Drugs (e.g., Vitamin K, nitrofurantoin)
- Immune Hydrops Fetalis: Rh incompatibility ABO incompatibility Other blood group incompatibilities (C, E, Kell, Duffy)
- Rh incompatibility
- ABO incompatibility
- Other blood group incompatibilities (C, E, Kell, Duffy)
- Non-Immune Hydrops Fetalis: RBC membrane defects: Spherocytosis (Option B) Elliptocytosis Metabolic disorders: G6PD deficiency (Option C) Systemic diseases (e.g., galactosemia) Hemoglobinopathies: Alpha- and beta-thalassemia syndrome (Option D) Infections: Bacterial Viral (e.g., rubella,parvovirus) Parasitic (e.g., syphilis) Other causes: DIC (Disseminated Intravascular Coagulation) Acute transfusion hemolysis Vitamin E deficiency Drugs (e.g., Vitamin K, nitrofurantoin)
- RBC membrane defects: Spherocytosis (Option B) Elliptocytosis
- Spherocytosis (Option B)
- Elliptocytosis
- Metabolic disorders: G6PD deficiency (Option C) Systemic diseases (e.g., galactosemia)
- G6PD deficiency (Option C)
- Systemic diseases (e.g., galactosemia)
- Hemoglobinopathies: Alpha- and beta-thalassemia syndrome (Option D)
- Alpha- and beta-thalassemia syndrome (Option D)
- Infections: Bacterial Viral (e.g., rubella,parvovirus) Parasitic (e.g., syphilis)
- Bacterial
- Viral (e.g., rubella,parvovirus)

- Parasitic (e.g., syphilis)
- Other causes: DIC (Disseminated Intravascular Coagulation) Acute transfusion hemolysis Vitamin E deficiency Drugs (e.g., Vitamin K, nitrofurantoin)
- DIC (Disseminated Intravascular Coagulation)
- Acute transfusion hemolysis
- Vitamin E deficiency
- Drugs (e.g., Vitamin K, nitrofurantoin)
- Most serious form of Rh hemolytic disease (HDFN)
- Excessive destruction of fetal red cells leads to: Severe anemia Tissue anoxia Metabolic acidosis
- Severe anemia
- Tissue anoxia
- Metabolic acidosis
- Adverse effects on: Fetal heart and brain Placenta (hyperplasia to increase oxygen transfer)
- Fetal heart and brain
- Placenta (hyperplasia to increase oxygen transfer)
- Progressive diminution of fetal red cells due to hemolysis
- Resulting damage: Liver damage leading to hypoproteinemia Generalized edema (hydrops fetalis), ascites, and hydrothorax
- Liver damage leading to hypoproteinemia
- Generalized edema (hydrops fetalis), ascites, and hydrothorax
- Increased risk for Rh-positive male fetuses (13 times more likely to become hydropic, 3 times more likely to die than female fetuses)
- Accumulation of extracellular fluid in at least two body compartments
- Outcomes: Fetal death due to cardiac failure Stillbirth, maceration, or death shortly after birth if born alive
- Fetal death due to cardiac failure
- Stillbirth, maceration, or death shortly after birth if born alive
- Severe anemia
- Tissue anoxia
- Metabolic acidosis
- Fetal heart and brain
- Placenta (hyperplasia to increase oxygen transfer)
- Liver damage leading to hypoproteinemia

- Generalized edema (hydrops fetalis), ascites, and hydrothorax
- Fetal death due to cardiac failure
- Stillbirth, maceration, or death shortly after birth if born alive
- Immune Hydrops Fetalis: Rh incompatibility ABO incompatibility Other blood group incompatibilities (C, E, Kell, Duffy)
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- RBC membrane defects: Spherocytosis (Option B) Elliptocytosis
- Spherocytosis (Option B)
- Elliptocytosis
- Metabolic disorders: G6PD deficiency (Option C) Systemic diseases (e.g., galactosemia)
- G6PD deficiency (Option C)
- Systemic diseases (e.g., galactosemia)
- Hemoglobinopathies: Alpha- and beta-thalassemia syndrome (Option D)
- Alpha- and beta-thalassemia syndrome (Option D)
- Infections: Bacterial Viral (e.g., rubella,parvovirus) Parasitic (e.g., syphilis)
- Bacterial
- Viral (e.g., rubella,parvovirus)
- Parasitic (e.g., syphilis)
- Other causes: DIC (Disseminated Intravascular Coagulation) Acute transfusion hemolysis Vitamin E deficiency Drugs (e.g., Vitamin K, nitrofurantoin)
- DIC (Disseminated Intravascular Coagulation)
- Acute transfusion hemolysis
- Vitamin E deficiency
- Drugs (e.g., Vitamin K, nitrofurantoin)
- Rh incompatibility
- ABO incompatibility
- Other blood group incompatibilities (C, E, Kell, Duffy)

- RBC membrane defects: Spherocytosis (Option B) Elliptocytosis
- Spherocytosis (Option B)
- Elliptocytosis
- Metabolic disorders: G6PD deficiency (Option C) Systemic diseases (e.g., galactosemia)
- G6PD deficiency (Option C)
- Systemic diseases (e.g., galactosemia)
- Hemoglobinopathies: Alpha- and beta-thalassemia syndrome (Option D)
- Alpha- and beta-thalassemia syndrome (Option D)
- Infections: Bacterial Viral (e.g., rubella,parvovirus) Parasitic (e.g., syphilis)
- Bacterial
- Viral (e.g., rubella,parvovirus)
- Parasitic (e.g., syphilis)
- Other causes: DIC (Disseminated Intravascular Coagulation) Acute transfusion hemolysis Vitamin E deficiency Drugs (e.g., Vitamin K, nitrofurantoin)
- DIC (Disseminated Intravascular Coagulation)
- Acute transfusion hemolysis
- Vitamin E deficiency
- Drugs (e.g., Vitamin K, nitrofurantoin)
- Spherocytosis (Option B)
- Elliptocytosis
- G6PD deficiency (Option C)
- Systemic diseases (e.g., galactosemia)
- Alpha- and beta-thalassemia syndrome (Option D)
- Bacterial
- Viral (e.g., rubella,parvovirus)
- Parasitic (e.g., syphilis)
- DIC (Disseminated Intravascular Coagulation)
- Acute transfusion hemolysis
- Vitamin E deficiency
- Drugs (e.g., Vitamin K, nitrofurantoin)

Solution for Question 9:

Correct Answer: A) Direct Coomb's test

Explanation:

- This test is used to detect antibodies that are bound to the surface of the baby's red blood cells, which is crucial for diagnosing hemolytic disease of the newborn (HDN).
- A positive result indicates that the baby’s RBC are being destroyed by antibodies from the mother, often due to Rh or ABO incompatibility.
- This test is appropriate after birth to assess the newborn’s condition.

Erythroblastosis fetalis

- Determine the blood group (ABO and Rh) of the fetus: Choriovillous sampling Amniocentesis Cordocentesis
- Choriovillous sampling
- Amniocentesis
- Cordocentesis
- Check the Hb levels and hematocrit of the baby
- Non-invasive: USG Doppler (Middle cerebral artery peak blood flow) (Option B)
- USG Doppler (Middle cerebral artery peak blood flow) (Option B)
- Indirect Coomb's test in the mother (to check the sensitization status) (Option D)
- Choriovillous sampling
- Amniocentesis
- Cordocentesis
- USG Doppler (Middle cerebral artery peak blood flow) (Option B)
- Maternal & Neonatal ABO & Rh blood grouping
- Direct Coomb's test (usually positive for the baby)
- Anemia: Hb levels Hematocrit
- Hb levels
- Hematocrit
- Cord blood bilirubin levels (Unconjugated)
- Peripheral smear (to check for evidence of hemolysis)
- Increased reticulocyte count
- Increased LDH
- Hb levels

- Hematocrit

Hemoglobin electrophoresis (Option C): This test is used to diagnose various hemoglobinopathies, such as sickle cell disease or thalassemia,

Solution for Question 10:

Correct Answer: B) Distal part of the ileum and proximal colon

Explanation:

- The above clinical scenario depicts necrotizing enterocolitis (NEC).
- NEC primarily affects the premature infant's gastrointestinal tract, with the most common sites being the distal ileum and proximal colon.
- The reason for this predilection is likely due to the relatively poor blood supply to these regions and their position in the GI tract, where bacterial overgrowth can easily occur.

Solution for Question 11:

Correct Answer: A) IgG Antibodies Against Rh Antigen

Explanation:

- The positive indirect Coombs test indicates that the mother has IgG antibodies against Rh-positive red blood cells from previous sensitizing events (history of previous two pregnancies)
- These antibodies cross the placenta and cause the destruction of the fetal red blood cells, leading to anemia and compensatory splenomegaly in the fetus.

Pathophysiology of Erythroblastosis

IgM Antibodies Against Rh Antigen (Option B): IgM antibodies do not cross the placenta due to their larger size. Therefore, they cannot cause hemolysis of fetal red blood cells

IgG Antibodies Against ABO Antigens (Option C): While IgG antibodies against ABO antigens can cause hemolysis, ABO incompatibility typically causes milder disease compared to Rh incompatibility

Fetal Red Blood Cell Destruction Due to Maternal Hyperglycemia (Option D): Maternal hyperglycemia does not directly cause destruction of fetal red blood cells.

Solution for Question 12:

Correct Answer: C) Ensuring that the baby's exposed surface area is maximized.

Explanation:

A higher exposed surface area of the baby increases the effectiveness of phototherapy.

- 450-460 nm (Option B)
- Lesser distance between the baby and the phototherapy unit increases effectiveness (Option A)
- Higher exposed surface area of the baby increases effectiveness (Option C)
- LED lamps > CFL lamps (Option D)
- Less interruption increases effectiveness
- Effectiveness does not depend on the skin pigmentation of the baby
- Bronze baby syndrome

- Dehydration
- Watery diarrhea
- Hypocalcemia
- Retinal toxicity
- Gonadal toxicity or mutations
- Impaired maternal-child bonding
- Photoisomerization Photoisomer of bilirubin forms excreted via bile Slow & reversible
- Photoisomer of bilirubin forms excreted via bile
- Slow & reversible
- Structural Isomerization Bilirubin → Lumirubin → excreted through kidney without conjugation Faster & irreversible Most important mechanism by which phototherapy acts
- Bilirubin → Lumirubin → excreted through kidney without conjugation
- Faster & irreversible
- Most important mechanism by which phototherapy acts
- Photo-Oxidation Least important
- Least important
- Photoisomer of bilirubin forms excreted via bile
- Slow & reversible
- Bilirubin → Lumirubin → excreted through kidney without conjugation
- Faster & irreversible
- Most important mechanism by which phototherapy acts
- Least important

Solution for Question 13:

Correct Answer: C) Perform an exchange transfusion.

Explanation:

Perform an exchange transfusion: This is the appropriate next step for extremely high bilirubin levels or when phototherapy fails, especially in cases of severe hemolytic disease such as Rh incompatibility. It rapidly decreases bilirubin levels and reduces the risk of complications like kernicterus.

Treatment of Neonatal Jaundice

Treatment Modality	Description	Indication	Effectiveness/Safety
Phototherapy (Option A)	Exposure to specific blue light wavelengths(460-490 nm) to convert bilirubin into a water-soluble form.(aka lumirubin)	First-line treatment for unconjugated hyperbilirubinemia.	Effective for many infants; generally safe with few side effects.
Exchange Transfusion (Option C)	Gradual replacement of the infant's blood with donor blood to rapidly decrease bilirubin levels.	Used for extremely high bilirubin or when phototherapy fails; severe hemolytic disease (e.g., Rh or ABO incompatibility).	Effective for severe cases; reduces risk of kernicterus.
Intravenous Immunoglobulin (IVIG) (Option B)	Reduces antibody production that leads to high bilirubin levels.	For hemolytic jaundice; often used alongside phototherapy.	Enhances effectiveness of phototherapy.
Conjugated Hyperbilirubinemia Management	Addresses specific underlying medical or surgical conditions contributing to elevated bilirubin levels. (Option D)	Complex; dependent on the underlying cause.	Management depends on addressing the root cause.

Solution for Question 14:

Correct Answer: B) Sensorineural hearing loss

Explanation:

Sensorineural hearing loss is a characteristic feature of kernicterus (chronic bilirubin encephalopathy).

Kernicterus (Chronic Bilirubin Encephalopathy, CBE)

- Definition: Severe condition from high bilirubin levels.
- Characteristics: Choreoathetoid cerebral palsy: Involuntary movements and abnormal muscle tone. Impaired upward gaze: Difficulty looking upwards. Sensorineural hearing loss: Hearing impairments. (Option B)
- Choreoathetoid cerebral palsy: Involuntary movements and abnormal muscle tone.

- Impaired upward gaze: Difficulty looking upwards.
- Sensorineural hearing loss: Hearing impairments. (Option B)
- Development: Can arise from acute bilirubin encephalopathy (ABE) with symptoms like lethargy, abnormal behavior, opisthotonus, and seizures.
- Choreoathetoid cerebral palsy: Involuntary movements and abnormal muscle tone.
- Impaired upward gaze: Difficulty looking upwards.
- Sensorineural hearing loss: Hearing impairments. (Option B)

Bilirubin-Induced Neurologic Dysfunction (BIND)

- Definition: Neurological sequelae from milder hyperbilirubinemia.
- Manifestations: Developmental delays: Delayed achievement of milestones. (Option A) Cognitive impairments: Learning and cognitive processing challenges. Disordered executive function: Issues with planning, organization, problem-solving. (Option C) Behavioral and psychiatric disorders: Conditions such as ADHD or autism. (Option D)
- Developmental delays: Delayed achievement of milestones. (Option A)
- Cognitive impairments: Learning and cognitive processing challenges.
- Disordered executive function: Issues with planning, organization, problem-solving. (Option C)
- Behavioral and psychiatric disorders: Conditions such as ADHD or autism. (Option D)
- Complexity: The relationship between bilirubin levels and developmental outcomes is influenced by prematurity, perinatal complications, and hemolytic disease.
- Developmental delays: Delayed achievement of milestones. (Option A)
- Cognitive impairments: Learning and cognitive processing challenges.
- Disordered executive function: Issues with planning, organization, problem-solving. (Option C)
- Behavioral and psychiatric disorders: Conditions such as ADHD or autism. (Option D)

Solution for Question 15:

Correct Answer: D) 5

Explanation:

Jaundice extending to the palms and soles corresponds to a score of 5. This is the highest score on Kramer's scale and indicates severe jaundice.

Overview:

- Visual assessment technique for evaluating neonatal jaundice.
- Scores jaundice based on skin color in different dermal zones.
- Typically used by trained medical professionals.

Scoring System:

Score	Description
0	No jaundice
1	Jaundice at head and neck (Option A)
2	Jaundice at upper trunk (above umbilicus) (Option B)
3	Jaundice at lower trunk and thighs (below umbilicus)
4	Jaundice at arms and lower legs (Option C)
5	Jaundice in palms and soles (Option D)

Assessment Process:

- Performed by trained professionals.
- Blue fluorescent light may be used to enhance visibility.
- Average score from two examiners is recorded.
- Discrepancies resolved with pediatrician's input.

Solution for Question 16:

Correct Answer: A) Continue exclusive breastfeeding without any intervention

Explanation:

The presentation suggests breast milk jaundice, which typically appears after the first 3-5 days of life in an otherwise healthy, well-fed baby. This condition usually requires no intervention, and breastfeeding should continue.

Breastfeeding Jaundice vs Breast Milk Jaundice

Characteristic	Breastfeeding Jaundice	Breast Milk Jaundice
Cause	Insufficient breast milk intake leading to dehydration and decreased bilirubin elimination.	Substances in breast milk interfering with bilirubin breakdown.
Onset	Within the first week of life	After 3-5 days of life
Treatment	Increase breastfeeding frequency, ensure proper latch and milk transfer.	Usually no treatment needed; continue breastfeeding, may use formula supplementation or phototherapy in some cases.
Commonality	More common	Less common
Associated Issue	Breastfeeding difficulties (e.g., latch issues)	Substances(FFA and beta glucouronidase) in breast milk

Switch to formula feeding temporarily (Option B): Switching to formula feeding is not usually necessary unless the bilirubin levels are significantly high and the jaundice persists.

Initiate phototherapy immediately (Option C): Phototherapy is reserved for more severe cases where bilirubin levels are dangerously high, which is not indicated here.

Advise stopping breastfeeding due to risk of worsening jaundice (Option D): Stopping breastfeeding is not recommended in cases of breast milk jaundice, as it does not pose a significant risk of worsening the condition.

Solution for Question 17:

Correct Answer: D) Observation and reassurance

Explanation:

Yellowish discoloration of skin and sclera and elevated bilirubin levels within 48 hours(after 24 hr of birth) of birth indicate physiological jaundice. Observation and reassurance are appropriate for physiological jaundice, which typically resolves within 1-2 week as UDPGT enzyme matures by 1-2 week.

Feature	Physiological Jaundice	Pathological Jaundice

Timing	Appears 24-72 hours after birth; peaks at 3-5 days; resolves by 1-2 weeks	Appears within 24 hours of birth or persists beyond the expected timeframe
Cause	Elevated unconjugated bilirubin due to increased production and immature liver function	Blood group incompatibility, infections, metabolic disorders, genetic conditions
Clinical Presentation	Yellowing of skin and sclera, starting from face, progressing to trunk and extremities	Similar to physiological jaundice, but potentially more severe and may include other symptoms
Treatment	Usually self-resolving; observation and reassurance; phototherapy may be used if needed (Option D)	Depends on the underlying cause; may include phototherapy, exchange transfusion, or specific treatments (Option A, B & C)

Key Distinction:

- Onset timing is crucial. Jaundice appearing within the first 24 hours is always pathological and requires investigation. Or rate of rise of bilirubin >5 mg/dl per day or palm or soles involvement

Solution for Question 18:

Correct Answer: C) Stage IIIb

Explanation:

- The given abdominal radiograph shows pneumoperitoneum.
- Bell's Stage IIIb is characterized by advanced NEC with signs of bowel perforation, such as pneumoperitoneum seen on the X-ray.

Modified Bell’s Staging is used in staging NEC:

Stage	Systemic signs	Abdominal signs	Radiologic signs	Treatment	
I Suspected NEC	IA	Temperature instability Apnea Lethargy	Abdominal distension Occult blood in stools	Normal/ Ileus	NPO IV antibio-tics × 3 days
IB	similar to IA	IA + Gross blood in stools	similar to IA	similar to IA	

II Definite NEC (Option A)	IIA	similar to IA	I + Absent bowel sounds, Abdominal tenderness	Ileus, Pneumatosis intestinalis	NPO IV antibio-tics x 7 to 10 days
IIB	similar to IA + Metabolic acidosis Thrombocytopenia	IIA + Abdominal cellulitis	Portal venous gas shadow	NPO IV antibiotics × 14 days	
III Advanced NEC (Option B)	IIIA	similar to IIB + Hypotension Bradycardia DIC Neutropenia	I + II + Signs of generalized peritonitis, Marked tenderness	IIB + Ascites	NPO IV antibiotics × 14 days Fluid boluses, Inotropes, Mechanical ventilation, Paracentesis
IIIB	similar to IIIA	similar to IIIA	IIIA + Pneumoperitoneum	IIIA + Surgery	

Stage IV (Option D) does not exist in this classification system

Previous Year Questions

1. Which of the following is least likely regarding Rh incompatibility in a newborn?

A. Increased unconjugated bilirubin

B. Increased fecal urobilinogen

C. Hypoalbuminemia

D. Increased stercobilinogen

2. Which of the following is the most common type of congenital diaphragmatic hernia?

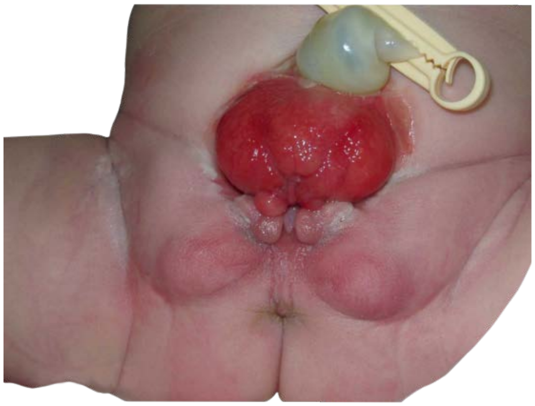
A. Hiatal Hernia

B. Morgagni Hernia

C. Bochdalek Hernia

D. Umbilical Hernia

3. Which neonatal anomaly is depicted in the image?



A. Omphalocele

B. Gastroschisis

C. Patent vitello intestinal duct

D. Exstrophy bladder

---

4. Which of the following are the components of Downe's score? Respiratory rate Feeding Grunting Cyanosis

A. 1,2,3

B. 1,3,4

C. 2,3,4

D. 1,2,4

---

5. What is an indicator of successful neonatal resuscitation?

A. Color change

B. Bilateral Air entry

C. Increased heart rate

D. None of the above

---

6. Banana sign is seen in:

A. Omphalocele

B. NCC

C. Spina bifida

D. Meningitis

---

7. What is the developmental age of a 6-year-old child who exhibits developmental delay, can ride a tricycle, climb stairs with alternate feet, say their name, know their own sex, but cannot narrate a story?

A. 3 years

B. 7 years

C. 5 years

D. 1 year

---

8. An infant in the NICU was observed to have the features as seen in the Image below. In addition to these features, there is an audible grunt. What would the Silverman Anderson score be?

A. 8

B. 6

C. 5

D. 4

---

9. What is a characteristic feature observed in physiological jaundice?

A. Increased indirect bilirubin, increased urobilinogen, absent bilirubin in urine

B. Increased direct bilirubin, increased urobilinogen, absent bilirubin in urine

C. Increased indirect bilirubin, absent urobilinogen, absent bilirubin in urine

D. Increased direct bilirubin, absent urobilinogen, absent bilirubin in urine

10. How much oxygen is given to a term infant with respiratory distress during resuscitation?

A. 21%

B. 50%

C. 70%

D. 100%

11. In accordance with the neonatal resuscitation protocol, at what heart rate do chest compressions commence?

A. 70

B. 60

C. 80

D. 100

12. Which of the following is not a correct match for ABCD according to pediatric advanced life support?

A. Airway

B. Breathing

C. Circulation

D. Dehydration

13. Which of the following is the most reliable indication of successful ventilation effort in neonatal resuscitation?

A. Color change

B. Rise in heart rate

C. Air entry

D. Chest rise

14. Which of the following is not a cause of non-immune hydrops fetalis?

A. ABO incompatibility

B. CCF

C. Parvovirus infection

D. Chromosomal anomalies

15. What is the usual cause of jaundice that occurs either at birth or within 24 hours of birth?

A. Erythroblastosis fetalis

B. Congenital hyperbilirubinemia

C. Biliary atresia

D. Physiological jaundice of the newborn

---

16. A newborn baby presents with irritability and cyanosis, which subsides on crying. On examination, you are unable to pass a nasogastric tube, and no mist formation is noted on the laryngeal mirror test. Which of the following is the best next step in management?

A. Endotracheal intubation

B. Oropharyngeal airway

C. Oxygen inhalation

D. Epinephrine nebulization

---

17. Which of the following primitive reflexes do not disappear?

A. Rooting reflex

B. Parachute reflex

C. Moro’s reflex

D. Palmar grasp reflex

---

18. Identify the condition based on the X-ray of a neonate given in the image.

A. Pneumomediastinum

B. Cystic fibrosis

C. Congenital diaphragmatic hernia

D. Bronchiolitis

19. A child is presenting with palpable blanching purpura, abdominal pain, and pain in the knee joint. Urine examination showed hematuria and proteinuria. What is the most likely diagnosis?

A. Immune thrombocytopenic purpura

B. Dengue

C. Churg-Strauss syndrome

D. Henoch-Schonlein purpura

20. Which of the following are components of kangaroo mother care?

A. Early discharge, skin-to-skin contact, prevent infection

B. Early discharge, skin-to-skin contact, exclusive breastfeeding

C. Skin-to-skin contact, exclusive breastfeeding, prevent infection

D. Early discharge, exclusive breastfeeding, prevent infection

---

21. The following device shown attached to the baby is used in all except:

A. In CPR

B. To know about apnea

C. In neonatal resuscitation

D. Measure central cyanosis

---

22. Which is correct about ABO and Rh incompatibility leading to erythroblastosis fetalis?

A. Cytotoxic

B. Cell-mediated hypersensitivity

C. Antigen-antibody reaction

D. Immune complex

---

23. Which of the following indicates a "infant at risk"?

A. Working mother

B. Mother who has not taken folic acid during pregnancy

C. Infant with history of premature birth <32 weeks

D. Pre-eclampsia

24. The procedure displayed below is:

A. Intubation

B. Orogastric tube insertion

C. Nasogastric tube insertion

D. Oropharyngeal airway insertion

25. A baby assessed 5 minutes after birth is found to be pink, with blue extremities and irregular respiration. The heart rate is 80 beats per minute. There is a grimace and some flexion. What is the APGAR score for this baby?

A. 2

B. 3

C. 4

D. 5

26. A 1-day-old neonate has not passed urine since birth. What is the next step in management?

A. Continue breast feeding and observe

B. Admit to NICU

C. Start artificial feeding

D. Start intravenous fluids

27. A term neonate was born to a G2P1 mother, and initial measures were done by your senior resident, and he asked you a question: A newborn loses maximum heat from?

A. Head

B. Abdomen

C. Palms and soles

D. Neck

28. Phototherapy converts bilirubin to:

A. Stercobilin

B. Lumirubin

C. Biliverdin

D. Urobilin

29. What is the accurate statement concerning the disparities between resuscitation techniques for adults and children?

A. Ventricular dysrhythmias are common in children

B. The no-flow phase precedes the period of cardiac arrest in children

C. More ventilation is to be given compared to chest compressions in children

D. Dysrhythmias are most often due to respiratory insufficiency in children

---

30. Choose the correct order of suctioning during neonatal resuscitation.

A. Mouth-nose

B. Nose-mouth

C. Mouth-nose-oesophagus

D. Trachea-nose-mouth

---

31. When do you consider administering epinephrine to a neonate during resuscitation?

A. Heart rate remains at < 60 beats/minute despite effective compressions and ventilations

B. Heart rate remains at < 100 beats/minute despite effective compressions and ventilations

C. Heart rate does not improve after 30 seconds with bag and mask ventilation

D. Infants with severe respiratory depression fail respond to positive-pressure ventilation via bag and mask

---

32. A baby assessed five minutes after birth is found to have a blue, absent pulse, minimal response to stimulation, flexed arms and legs, and slow irregular movement. What is the APGAR score for this newborn?

A. 2

B. 3

C. 4

D. 5

33. What is the APGAR score for a newborn baby that is assessed 5 minutes after birth and found to be cyanosed with irregular gasping respiration, a heart rate of 60 beats/min, and minimal response to stimulation?

A. 2

B. 5

C. 3

D. 4

34. All of the following are clinical features of necrotizing enterocolitis, except?

A. Vomiting

B. Abdominal mass

C. Erythema of the abdominal wall

D. Metabolic alkalosis

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	4
Question 4	2
Question 5	3
Question 6	3
Question 7	1
Question 8	1
Question 9	1
Question 10	1
Question 11	2
Question 12	4
Question 13	2
Question 14	1
Question 15	1
Question 16	2
Question 17	2
Question 18	3
Question 19	4
Question 20	2
Question 21	4
Question 22	1

Question 23	3
Question 24	3
Question 25	4
Question 26	1
Question 27	1
Question 28	2
Question 29	4
Question 30	3
Question 31	1
Question 32	2
Question 33	4
Question 34	4

Solution for Question 1:

Correct Option D: Increased stercobilinogen: Stercobilinogen is a breakdown product of bilirubin that is further metabolized to stercobilin and excreted in the feces. Rh incompatibility does not directly affect the production or excretion of stercobilinogen. Therefore, increased stercobilinogen is less likely to be seen in Rh incompatibility.

Incorrect Options:

Option A: Increased unconjugated bilirubin: This is a characteristic finding in Rh incompatibility due to the increased breakdown of red blood cells and subsequent release of bilirubin into the bloodstream.

Option B: Increased fecal urobilinogen: Rh incompatibility can lead to increased breakdown of red blood cells, resulting in elevated levels of bilirubin. Bilirubin is eventually converted to urobilinogen, which is primarily excreted in the feces. Therefore, increased fecal urobilinogen can also be seen in Rh incompatibility.

Option C: Hypoalbuminemia

- In Rh incompatibility, Hypoalbuminemia is present.

Solution for Question 2:

Correct Option C - Bochdalek Hernia:

- This is the most common type of congenital diaphragmatic hernia, typically occurring on the left posterolateral side of the diaphragm.
- It results from a failure of the pleuroperitoneal membrane to close during fetal development, allowing abdominal organs to herniate into the chest cavity, which can severely affect lung development.

Incorrect Options:

Option A - Hiatal Hernia:

- This type of hernia occurs when a part of the stomach pushes up through the diaphragm into the chest cavity. While common in adults, it is not a congenital diaphragmatic hernia and is less relevant in neonates.

Option B - Morgagni Hernia:

- This is a rare type of congenital diaphragmatic hernia that occurs through the foramina of Morgagni, located in the anterior part of the diaphragm. It is less common than Bochdalek hernia.

Option D - Umbilical Hernia:

- This is a hernia that occurs when part of the intestine protrudes through the abdominal wall near the navel. It is common in infants but is not considered a congenital diaphragmatic hernia.

Solution for Question 3:

Correct Option D - Exstrophy bladder:

- Exstrophy bladder is characterized by the exposure of the bladder outside the body due to a defect in the lower abdominal wall. This anomaly results in a visible protrusion of the bladder, which is a key feature differentiating it from other conditions such as omphalocele, gastroschisis, and patent vitello intestinal duct.

Incorrect Options:

Option A - Omphalocele:

- This is a congenital defect where the abdominal organs herniate into the base of the umbilical cord due to a defect in the abdominal wall. It is covered by a translucent sac, which is a distinguishing feature.

Option B - Gastroschisis:

- This is a defect in the abdominal wall where the intestines protrude outside the body through a hole next to the umbilicus. Unlike omphalocele, gastroschisis does not have a protective sac covering the exposed organs.

Option C - Patent vitello intestinal duct:

- This anomaly occurs when the duct that connects the fetal intestine to the yolk sac fails to close properly, leading to the formation of a persistent connection between the umbilicus and the intestine, which may result in a small cyst or opening near the umbilicus.

Solution for Question 4:

Correct option B - 1,3,4:

- Downe’s score is used to quantify respiratory distress in a term neonate. Respiratory rate, grunting and cyanosis are components of Downe’s score.

Incorrect options:

Options A, C & D are incorrect as feeding is not a component of Downe's score.

Solution for Question 5:

Correct option is C - Increased heart rate:

During neonatal resuscitation, assessing the effectiveness of the intervention is crucial. One of the signs of effective resuscitation is an increased heart rate. When successful resuscitation measures are implemented, the newborn's heart rate should improve or increase from its initial low or absent state. The heart rate is a vital indicator of the baby's cardiovascular function and response to resuscitation efforts. Therefore, an increased heart rate indicates improved circulation and response to resuscitative measures.

Option A: Colour change: While colour change can be an important sign during neonatal resuscitation, it alone is not considered a definitive indicator of effective resuscitation. Colour change, such as the transition from cyanosis (bluish discoloration) to normal skin colour, may indicate improved oxygenation but does not necessarily reflect the overall success of resuscitation efforts.

Option B: Bilateral air entry: Bilateral air entry is an important goal during neonatal resuscitation to ensure adequate ventilation and oxygenation. However, it is not a specific sign of effective resuscitation.

Option D: None of the above: This option is incorrect as an increased heart rate is indeed a sign of effective neonatal resuscitation, as explained above.

Solution for Question 6:

Correct Option C- Spina bifida:

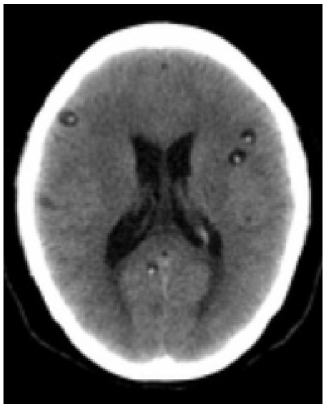
Explanation:

- The "Banana sign" refers to the abnormal shape of the cerebellum seen on imaging, specifically in cases of spina bifida.
- Spina bifida is a congenital neural tube defect that involves incomplete closure of the spinal column during fetal development.
- In spina bifida cases, the cerebellum can be pulled downward and flattened, resembling the shape of a banana on imaging studies such as ultrasound or magnetic resonance imaging (MRI).
- The term is commonly used in the context of prenatal imaging to identify abnormalities and guide appropriate medical management.

Incorrect Options:

Option A- Omphalocele: This option is incorrect The primary symptom of an omphalocele is the protrusion of the abdominal organs through the abdominal muscle wall. It's common for infants born with an omphalocele to have other congenital birth defects as well, such as heart defects, Beckwith-Wiedemann syndrome

Option B- NCC: This option is incorrect as neurocysticercosis on radiology shows the following finding



Option D- Meningitis: This option is incorrect as the banana sign is not visible in patients with meningitis

Solution for Question 7:

Correct Option:

Option A. The developmental age of the child can be determined based on their achieved milestones and abilities compared to typical developmental norms. In this case, the child is 6 years old and demonstrates certain developmental skills and limitations.

The child's ability to ride a tricycle and climb stairs with alternate feet suggests motor skills development beyond the age of 2 years. Additionally, knowing their own name and gender indicates cognitive and self-awareness development.

However, the child's inability to narrate a story suggests a limitation in language and communication skills, which is more typical of a developmental age of around 3 years. Therefore, the most appropriate developmental age for this child would be option A, 3 years.

Incorrect Options:

Option B. 7 years: This option would suggest more advanced language and narrative skills than what the child demonstrates.

Option C. 5 years: This option would imply further advanced cognitive and language abilities, which are not evident based on the given information.

OptionD. 1 year: This option would underestimate the child's developmental age, considering their ability to ride a tricycle, climb stairs, and demonstrate basic self-awareness.

Solution for Question 8:

Correct Option A - 8:

- The Silverman Anderson score assesses five respiratory parameters: intercostal retractions, xiphoid retractions, substernal retractions, nasal flaring, and expiratory grunt. Each parameter is given a score ranging from 0 to 2, with a higher score indicating more severe respiratory distress. Based on the description provided, if the infant in the image exhibits the features associated with respiratory distress, including an audible grunt, it suggests significant respiratory distress. Therefore, the overall Silverman Anderson score would be higher, such as 8, indicating more severe respiratory distress.

Incorrect Options:

Option B - 6: Option B suggests a Silverman Anderson score of 6. This would indicate a moderate level of respiratory distress. However, based on the description provided, which includes features of respiratory distress and an audible grunt, it suggests a more severe respiratory condition. Therefore, a score of 6 would not accurately reflect the severity of the respiratory distress, making it an incorrect option.

Option C - 5: Option C suggests a Silverman Anderson score of 5. This would indicate a moderate level of respiratory distress. However, given the presence of respiratory distress features and an audible grunt, the infant's condition appears to be more severe. A score of 5 would not adequately represent the severity of the respiratory distress observed, making option C incorrect.

Option D - 4: Option D suggests a Silverman Anderson score of 4, which would indicate mild respiratory distress. However, based on the description provided, including the presence of respiratory distress features and an audible grunt, it suggests a more severe respiratory condition. Therefore, a score of 4 would underestimate the severity of the respiratory distress,making option D incorrect.

Solution for Question 9:

Correct option A: Increased indirect bilirubin, increased urobilinogen, absent bilirubin in urine: Physiological jaundice is due to:

- Increased indirect bilirubin: Physiological jaundice is associated with increased levels of indirect (unconjugated) bilirubin in the blood. This occurs because newborns have an immature liver that is still developing its ability to process bilirubin efficiently.
- Physiological jaundice is associated with increased levels of indirect (unconjugated) bilirubin in the blood.
- This occurs because newborns have an immature liver that is still developing its ability to process bilirubin efficiently.
- Increased urobilinogen: Physiological jaundice often leads to increased urobilinogen in the urine. Urobilinogen is a breakdown product of bilirubin and is usually elevated in cases of increased bilirubin production, as seen in jaundice.
- Physiological jaundice often leads to increased urobilinogen in the urine.
- Urobilinogen is a breakdown product of bilirubin and is usually elevated in cases of increased bilirubin production, as seen in jaundice.
- Absent bilirubin in urine: Bilirubin is usually present in the urine in cases of jaundice.
- Physiological jaundice is associated with increased levels of indirect (unconjugated) bilirubin in the blood.
- This occurs because newborns have an immature liver that is still developing its ability to process bilirubin efficiently.
- Physiological jaundice often leads to increased urobilinogen in the urine.
- Urobilinogen is a breakdown product of bilirubin and is usually elevated in cases of increased bilirubin production, as seen in jaundice.

Incorrect options: Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 10:

Correct options A: 21%:

- The amount of oxygen given to a term infant with respiratory distress during resuscitation depends on the severity of the respiratory distress and the infant's oxygen saturation levels.
- The American Academy of Pediatrics recommends starting resuscitation with a 21% to 30% oxygen concentration and adjusting the concentration based on the infant's oxygen saturation levels.
- The goal is to maintain the infant's oxygen saturation between 90% and 95%.

Incorrect options: Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 11:

Correct option B: 60:

- According to the Neonatal Resuscitation Program (NRP) protocol, chest compressions are initiated in newborn infants when their heart rate is less than 60 bpm despite adequate ventilation and oxygenation.
- Before starting chest compressions, it is essential to ensure that the airway is clear and that effective positive pressure ventilation has been established.

Incorrect options: Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 12:

Correct option D: Dehydration:

- Pediatric Advanced Life Support (PALS) uses the ABCD approach to manage pediatric emergencies.
- ABCD stands for: A - Airway B - Breathing C - Circulation D - Disability
- A - Airway
- B - Breathing
- C - Circulation
- D - Disability
- Dehydration is not part of the ABCD approach in PALS, although it can be a concern in pediatric emergencies.
- A - Airway
- B - Breathing
- C - Circulation
- D - Disability

Incorrect options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 13:

Correct option B: Rise in heart rate:

- The rise in heart rate is considered the most effective indicator of successful ventilatory effort in neonatal resuscitation.
- Neonatal resuscitation refers to emergency care for newborn infants who are not breathing or experiencing respiratory distress.
- During this critical procedure, several indicators are monitored to assess the effectiveness of ventilatory efforts and the overall success of the resuscitation.
- The baby's heart rate will typically increase when effective ventilation is provided.
- Oxygenation improves, leading to better perfusion and oxygen delivery to the tissues, increasing heart rate.
- Monitoring the heart rate is crucial, as it provides an objective and immediate assessment of the infant's response to resuscitation efforts.

- If the heart rate fails to rise despite adequate ventilation, further interventions may be required to optimize the resuscitation process.

Incorrect options:

Option A: Colour change:

- Although color change can provide some information about the infant's oxygenation status, it is not the most reliable indicator of successful ventilatory effort.
- Color change alone may not accurately reflect the effectiveness of ventilation, as it can be influenced by various factors such as blood circulation, oxygenation levels, and other systemic conditions.

Option C: Air entry:

- Assessing air entry involves listening for the presence of breath sounds using a stethoscope.
- While it is an important aspect of evaluating ventilation, it may not be the most effective indicator of successful ventilatory effort in neonatal resuscitation.
- It can be difficult to reliably auscultate breath sounds in a chaotic and noisy resuscitation environment, and there could be other reasons for the absence of air entry despite adequate ventilation efforts (e.g., blocked airways or improper positioning).

Option D: Chest rise:

- Chest rise is another crucial component of assessing ventilation in neonatal resuscitation, but alone may not be the most effective indicator of successful ventilatory effort.
- The chest rise can occur even with inadequate ventilation if the airways are partially obstructed or have poor lung compliance.

Solution for Question 14:

Correct Option A - ABO incompatibility:

- Non-immune hydrops fetalis refers to the presence of abnormal fluid accumulation in two or more fetal compartments, such as the subcutaneous tissue, pleural space, pericardial space, or abdominal cavity, without an immune-mediated cause.
- ABO incompatibility is an immune-mediated cause of hydrops fetalis.

- It occurs when there is a mismatch between the blood types of the mother and the fetus, leading to the production of antibodies that can cause hemolysis and subsequent hydrops fetalis.

Incorrect options:

Option B - CCF:

- Congestive cardiac failure refers to the inability of the heart to pump blood effectively, resulting in fluid accumulation in various body compartments.
- It can be a cause of non-immune hydrops fetalis.

Option C - Parvovirus infection:

- Parvovirus B19 infection can lead to severe anemia in the fetus, which can result in hydrops fetalis.
- Parvovirus infection is an example of an infectious cause of non-immune hydrops fetalis.

Option D - Chromosomal anomalies:

- Chromosomal anomalies, such as Turner syndrome and Down syndrome, can be associated with non-immune hydrops fetalis.
- These genetic abnormalities can disrupt normal fetal development and result in the accumulation of fluid in various compartments.

Solution for Question 15:

Correct Option A - Erythroblastosis fetalis:

- Erythroblastosis fetalis, also known as hemolytic disease of the newborn, is a condition that occurs when there is an incompatibility between the blood types of the mother and the fetus.
- This condition can lead to the destruction of fetal red blood cells, resulting in excessive production of bilirubin leading jaundice in the newborn.

Incorrect Options:

Options B, C and D can not cause jaundice at birth or within 24 hours of birth.

Option B - Congenital hyperbilirubinemia:

- Congenital hyperbilirubinemia refers to a group of conditions in which infants have elevated levels of bilirubin in their blood which is caused by various factors, such as genetic disorders affecting bilirubin metabolism or transport, enzyme deficiencies, or liver dysfunction.

Option C - Biliary atresia:

- Biliary atresia is a condition characterized by the absence or blockage of bile ducts, which affects the flow of bile from the liver to the gallbladder and intestine.
- It is a rare condition that typically presents after the first few weeks of life.

Option D - Physiological jaundice of the newborn:

- Physiological jaundice of the newborn is the most common cause of jaundice in newborns and typically appears after the first 24 hours of birth.
- It is a normal and temporary condition caused by the increased breakdown of red blood cells and the immaturity of the newborn's liver in processing bilirubin.

Solution for Question 16:

Correct option B: Oropharyngeal airway:

- The patient's presentation with irritability and cyanosis that subside on crying, along with the inability to pass a nasogastric tube and no mist formation on the laryngeal mirror test, are suggestive of choanal atresia.
- Choanal atresia is a congenital condition characterized by the blockage of the nasal passages due to complete or partial persistence of the buccopharyngeal membrane.
- This obstruction can cause difficulty breathing and feeding, leading to cyanosis and irritability.
- In such cases, facilitation of oral breathing by keeping the mouth open and pulling the tongue forward manually by oral digital manipulation is followed by the placement of a plastic oropharyngeal airway or McGovern open-tip nipple.

Incorrect options: Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 17:

Correct Option B - Parachute reflex:

- Parachute reflex appears at 7-8 months persists throughout life and never disappears.
- It is elicited when a baby is held in a ventral suspension and when dropped suddenly, the baby spreads the upper and lower limbs to avoid falling on the face.

Incorrect Options:

Option A - Rooting reflex:

- Rooting reflex is seen at 32 wks gestation and disappears at 1 month post natal age

Option C - Moro’s reflex:

- Moro’s reflex is seen at 37 wks of gestation and disappears at 5-6 months of age

Option D - Palmar grasp reflex:

- Palmar grasp reflex is seen at 28 wks of gestation and disappears at 3 months of age

Solution for Question 18:

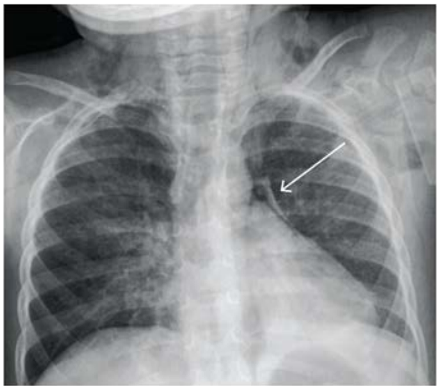
Correct Option C - Congenital Diaphragmatic Hernia:

- The X-ray image shows the presence of air-filled abdominal contents in the thorax. This suggests a diagnosis of a congenital diaphragmatic hernia.
- Congenital diaphragmatic hernia is a condition resulting from the herniation of abdominal contents into the thoracic cavity, resulting in lung hypoplasia and altered pulmonary vascular development.
- Plain radiograph findings:
- Indistinct diaphragm with opacification of part of or all the hemithorax (typically left-sided)
- Scaphoid abdomen
- Deviation of lines: Endotracheal tube Nasogastric tube Umbilical arterial and venous catheters
- Endotracheal tube
- Nasogastric tube
- Umbilical arterial and venous catheters
- Endotracheal tube

- Nasogastric tube
- Umbilical arterial and venous catheters

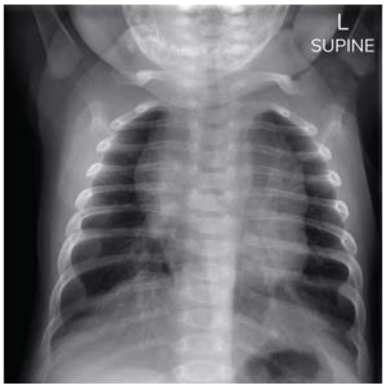
Incorrect Options:

Option A - Pneumomediastinum: Pneumomediastinum is the presence of extraluminal gas within the mediastinum. Small amounts of gas appear as linear or curvilinear lucencies outlining mediastinal contours.



Option B - Cystic fibrosis: Cystic fibrosis predisposes to the development of bronchiectasis. Cylindrical bronchiectasis has tram-track opacities, and cystic bronchiectasis may have air-fluid levels. Bronchovascular markings increase, and bronchi seen end-on may appear as ring shadows. Pulmonary vasculature appears ill-defined.

Option D - Bronchiolitis: Bronchiolitis is a common lung infection in young children and infants. Chest X-ray shows non-specific findings such as ill-defined small or hazy clustered nodules or areas of air trapping characterized by hyperlucency.



Solution for Question 19:

Correct Option D - Henoch-Schonlein purpura:

- Henoch-Schonlein purpura (HSP) is a systemic vasculitis that primarily affects small blood vessels, commonly seen in children.
- The characteristic clinical features include palpable purpura, abdominal pain, joint pain, and renal involvement.
- Hematuria and proteinuria are common findings in urine examination due to kidney involvement.
- The symptoms may be preceded by an upper respiratory tract infection.

Incorrect Options:

Option A - Immune thrombocytopenic purpura:

- While ITP can present with purpura and thrombocytopenia, it does not typically involve joint pain, abdominal pain, or renal manifestations such as hematuria and proteinuria.

Option B - Dengue:

- Dengue is a viral infection that can cause fever, rash, and joint pain, but it does not typically present with palpable purpura or renal involvement.

Option C - Churg-Strauss syndrome:

- Churg-Strauss syndrome, also known as eosinophilic granulomatosis with polyangiitis, is a rare systemic vasculitis that primarily affects medium and small blood vessels.

- It is characterized by asthma, eosinophilia, and systemic vasculitis.

Solution for Question 20:

Correct option B: Early discharge, skin-to-skin contact, exclusive breastfeeding:

- Kangaroo mother care was developed for LBW infants to decrease their mortality.
- There are four components of kangaroo mother care: Skin-to-skin contact Exclusive breastfeeding Early hospital discharge Support to the mother
- Skin-to-skin contact
- Exclusive breastfeeding
- Early hospital discharge
- Support to the mother
- Skin-to-skin contact
- Exclusive breastfeeding
- Early hospital discharge
- Support to the mother

Incorrect options: Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 21:

Correct option is D: Measure central cyanosis:

- Pulse oximeters are not used to directly measure central cyanosis, which refers to the bluish discoloration of the central parts of the body due to decreased oxygenation.
- Pulse oximeters primarily measure peripheral oxygen saturation levels by detecting the amount of oxygen-bound hemoglobin in the blood.

Incorrect options: Pulse oximeters are used in options A, B, and C.

Solution for Question 22:

Correct option A: Cytotoxic:

- ABO and Rh incompatibility leading to erythroblastosis fetalis involves a cytotoxic type of reaction, also known as type II hypersensitivity.
- ABO and Rh incompatibility reactions in erythroblastosis fetalis are characterized by the mother's immune system producing antibodies against the antigens on the fetal red blood cells.
- These antibodies can cause the destruction of the fetal red blood cells by phagocytes in the spleen and liver, leading to hemolysis.
- This is a cytotoxic reaction where antibodies bind to the antigens on the surface of the fetal red blood cells and trigger their destruction.

Incorrect options:

Option B: Cell-mediated hypersensitivity: Cell-mediated hypersensitivity reactions, also known as type IV hypersensitivity, are not involved in the development of erythroblastosis fetalis.

Option C: Antigen-antibody reaction: Erythroblastosis fetalis does involve an antigen-antibody reaction, but the specific mechanism is cytotoxic in nature.

Option D: Immune complex: Immune complex-mediated reactions type III hypersensitivity involves the formation of antigen-antibody complexes that deposit in tissues and trigger inflammation and is not involved in erythroblastosis fetalis.

Solution for Question 23:

Correct option C: Infant with history of premature birth <32 weeks:

- An "infant at risk" refers to a newborn baby who has certain factors or conditions that increase their vulnerability or likelihood of experiencing adverse health outcomes.

- In this case, a history of premature birth before 32 weeks gestation indicates that the infant was born significantly earlier than the normal gestational period.
- Premature infants often require specialized medical care and are at higher risk for various complications due to their underdeveloped organs and physiological systems.

Incorrect options:

Option A: Working mother:

- Being a working mother does not necessarily indicate that the infant is at risk.
- Many mothers successfully balance work and childcare responsibilities without impacting the baby's health.

Option B: Mother who has not taken folic acid during pregnancy: While folic acid supplementation is important for fetal development and reducing the risk of certain birth defects, the absence of folic acid intake alone does not categorize the infant as being at risk.

Option D: Pre-eclampsia:

- Pre-eclampsia is a maternal condition characterized by high blood pressure and organ damage during pregnancy.
- While it can have implications for the mother's health, it does not directly classify the infant as being at risk.
- However, pre-eclampsia may lead to preterm birth, which would then increase the infant's risk due to prematurity.

Solution for Question 24:

Correct option C: Nasogastric tube insertion:

- Nasogastric tube insertion involves the placement of a tube through the nose into the stomach.
- This procedure is commonly performed to administer medications, give feedings, or decompress the stomach.
- It is often used when a patient is unable to take food or medication orally or when there is a need to remove gastric contents.

Incorrect options:

Option A: Intubation: Intubation refers to the placement of an endotracheal tube into the trachea to establish a secure airway for mechanical ventilation.

Option B: Orogastric tube insertion: Orogastric tube insertion involves the placement of a tube through the mouth into the stomach.

Option D: Oropharyngeal airway insertion:

- Oropharyngeal airway insertion involves the placement of a curved device into the mouth to maintain an open airway and facilitate ventilation.
- It is commonly used during resuscitation efforts or in unconscious patients.

Solution for Question 25:

Correct option D: 5:

- The APGAR score is a quick assessment tool used to evaluate the overall condition and well-being of a newborn immediately after birth.
- It consists of five components: Appearance Pulse rate Grimace Activity Respiratory effort
- Appearance
- Pulse rate
- Grimace
- Activity
- Respiratory effort
- Each component is scored from 0 to 2, with a total score ranging from 0 to 10.
- Appearance
- Pulse rate
- Grimace
- Activity
- Respiratory effort

In the given case:

- Appearance: Pink body with blue extremities (score = 1)
- Pulse rate: 80 beats/min (score = 1)
- Grimace: Shows grimace (score = 1)

- Activity: Some flexion (score = 1)
- Respiratory effort: Irregular breathing (score = 1)

Incorrect Options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 26:

Correct option A: Continue breast feeding and observe:

- The clinical scenario where a 1-day-old neonate has not passed urine since birth is within the range of normal findings in neonates.
- Continuing breastfeeding and observing is the next step in management.
- The first urination should occur by 48 hours of age.
- The first passage of meconium is expected by 24 hours of age.
- Delayed urination or passing stool is a cause for concern and must be investigated.

Incorrect options:

Option B: Admit to NICU: If the condition continues, the neonate should be admitted to the NICU.

Option C: Start artificial feeding: When breastfeeding is not possible or is insufficient, such as when a mother is medically unable to breastfeed or when the baby is unable to feed effectively from the breast, artificial feeding may be necessary.

Option D: Start intravenous fluids: It may be required to administer fluids intravenously to a newborn who has not urinated since birth if the infant exhibits signs of dehydration or if there is a suspicion that another medical condition has to be treated.

Solution for Question 27:

Correct option A: Head:

- A newborn loses the maximum amount of heat from the head.
- The head has a large surface area relative to the rest of the body, and the scalp is often uncovered.
- Heat loss from the head can be minimized by providing appropriate head coverings and keeping the baby's head warm.

Incorrect options: Options B, C, and D are incorrect.

Solution for Question 28:

Correct option B: Lumirubin:

- Phototherapy is used to reduce elevated levels of bilirubin in the blood, particularly in cases of neonatal jaundice.
- During phototherapy, the most effective wavelength of light used is 450-460 nm.
- This light interacts with bilirubin, converting it into lumirubin, which is excreted through the kidney without conjugation.

Incorrect options:

Option A: Stercobilin:

- Stercobilin is a breakdown product of bilirubin that gives stool its characteristic brown color.
- It is formed in the intestines after bilirubin is metabolized by gut bacteria.

Option C: Biliverdin: Biliverdin is an intermediate compound in the breakdown of heme, which is eventually converted to bilirubin.

Option D: Urobilin:

- Urobilin is a breakdown product of bilirubin that is excreted in the urine.
- It is formed in the intestines from bilirubin by gut bacteria.

Solution for Question 29:

Correct option D: Dysrhythmias are most often due to respiratory insufficiency in children:

- In children, cardiac dysrhythmias during resuscitation are frequently caused by underlying respiratory insufficiency rather than primary cardiac abnormalities.
- Hypoxia and respiratory failure can lead to bradycardia, asphyxial pulseless electrical activity, or other dysrhythmias.

Incorrect options:

Option A: Ventricular dysrhythmias are common in children:

- Ventricular dysrhythmias, such as ventricular fibrillation or ventricular tachycardia, are less common in children compared to adults.
- Children are more likely to experience bradycardias or pulseless electrical activity during cardiac arrest.

Option B: The no-flow phase precedes the period of cardiac arrest in children:

- The no-flow phase refers to the period when there is no effective circulation, and it occurs during cardiac arrest.
- It does not precede the cardiac arrest.

Option C: More ventilation is to be given compared to chest compressions in children: During pediatric resuscitation, emphasis is placed on high-quality chest compressions rather than excessive ventilation.

Solution for Question 30:

Correct option C: Mouth-nose-oesophagus:

- The correct order of suctioning during neonatal resuscitation is mouth-nose-oesophagus.
- When suctioning the nose, vagal stimulation may induce the baby to take a deep breath, potentially resulting in aspiration if oral secretions are present.
- After suctioning the mouth and nose, the next step is to suction the esophagus, or the back of the throat.
- This step is important because some secretions can accumulate in the esophagus and be aspirated into the airway during resuscitation efforts.

- By suctioning the esophagus, you can prevent the aspiration of any remaining fluids or material into the airway.

Incorrect options: Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 31:

Correct option A: Heart rate remains at < 60 beats/minute despite effective compressions and ventilations:

- During neonatal resuscitation, epinephrine is considered when the heart rate remains below 60 beats per minute despite effective compressions and ventilations.
- In neonatal resuscitation, the primary goal is to establish effective ventilation and oxygenation.
- However, if the heart rate does not improve despite adequate ventilation and chest compressions, the administration of epinephrine may be necessary.

Incorrect options:

Option B: Heart rate remains at < 100 beats/minute despite effective compressions and ventilations: In neonatal resuscitation, a heart rate below 100 beats per minute is still considered within the acceptable range for a newborn and may not require immediate administration of epinephrine.

Option C: Heart rate does not improve after 30 seconds with bag and mask ventilation:

- This statement suggests a lack of response to initial ventilation efforts.
- In this case, it is important to reassess and optimize the ventilation technique, such as by checking the airway, repositioning the mask, or adjusting the ventilation parameters.

Option D: Infants with severe respiratory depression fail respond to positive-pressure ventilation via bag and mask:

- Severe respiratory depression may indicate a need for intervention, but it does not necessarily indicate the immediate administration of epinephrine.
- In such a case, efforts should be focused on improving ventilation and oxygenation before considering the use of additional medications.

Solution for Question 32:

Correct option B: 3:

- The APGAR score is a quick assessment tool used to evaluate the overall condition and well-being of a newborn immediately after birth.
- It consists of five components: Appearance Pulse rate Grimace Activity Respiratory effort
- Appearance
- Pulse rate
- Grimace
- Activity
- Respiratory effort
- Each component is scored from 0 to 2, with a total score ranging from 0 to 10.
- Appearance
- Pulse rate
- Grimace
- Activity
- Respiratory effort

In the given case:

- Appearance: Cyanosed (score = 0)
- Pulse rate: Absent (score = 0)
- Grimace: Shows minimal response to stimulation (score = 1)
- Activity: Some flexion of extremities (score = 1)
- Respiratory effort: Slow and irregular (score = 1)

Incorrect Options: Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 33:

Correct option D: 4:

- The APGAR score is a quick assessment tool used to evaluate the overall condition and well-being of a newborn immediately after birth.
- It consists of five components: Appearance Pulse rate Grimace Activity Respiratory effort
- Appearance
- Pulse rate
- Grimace
- Activity
- Respiratory effort
- Each component is scored from 0 to 2, with a total score ranging from 0 to 10.
- Appearance
- Pulse rate
- Grimace
- Activity
- Respiratory effort

In the given case:

- Appearance: Cyanosed (score = 0)
- Pulse rate: 60 beats/min (score = 1)
- Grimace: Shows minimal response to stimulation (score = 1)
- Activity: Some flexion of extremities (score = 1)
- Respiratory effort: Irregular gasping respiration (score = 1)

Incorrect Options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 34:

Correct option D: Metabolic alkalosis:

- Necrotizing enterocolitis (NEC) is a serious condition that primarily affects premature infants and can lead to inflammation and necrosis of the intestines.
- Metabolic alkalosis is not a clinical feature of necrotizing enterocolitis.
- NEC is associated with metabolic acidosis due to tissue damage and compromised intestinal function.

Incorrect options: Options A, B, and C are clinical features of necrotizing enterocolitis.

Breastmilk & Micronutrients

1. A 10-month-old, exclusively breastfed infant presents with lethargy, poor feeding, restlessness, developmental delay, and abdominal distention. The mother's diet is high in polished rice. Which vitamin deficiency is most likely responsible?

A. Vitamin B12 (Cobalamin)

B. Vitamin B1 (Thiamine)

C. Vitamin B9 (Folate)

D. Vitamin A

2. A 10 y/o child from a rural area with a history of inadequate dietary intake is brought to the clinic for a routine check-up. The child's growth is within normal limits, but the caregiver expresses concerns about ensuring proper nutrition, especially concerning iodine intake. What is the recommended daily intake (RDA) of iodine for a child aged 10 years?

A. 90 µg/day

B. 120 µg/day

C. 150 µg/day

D. 200 µg/day

3. A 4 y/o child is diagnosed with zinc deficiency and is experiencing growth delays, delayed sexual maturation, and poor wound healing. The child's diet consists mostly of cereals and plant-based foods with minimal animal protein. What is the recommended treatment for this child's zinc deficiency?

A. 0.5-1.0 mg elemental zinc/kg/day for several weeks

B. 2-4 mg elemental zinc/kg/day for several weeks

C. 3.5-5.0 mg elemental zinc/day

D. 20-30 mg/kg/day oral zinc therapy

4. A 3-month-old infant is brought to the emergency room with symptoms of irritability, poor feeding, and muscle twitching. An ECG shows a prolonged QTc interval, and the infant exhibits Chvostek and Trousseau signs. What is the most appropriate immediate treatment for this infant's symptoms?

A. Oral calcium supplementation at 40-80 mg/kg/day

B. 2 mL/kg of 10% calcium gluconate IV, administered slowly

C. Calcium chloride 10% IV solution

D. Reassurance and observation.

5. A 6 y/o child presents with irritability, pallor of nail beds, and fatigue. Laboratory tests reveal microcytic, hypochromic red cells, and reduced serum iron levels. What is the most appropriate initial treatment for this child's iron deficiency anemia?

A. Iron-rich dietary counseling.

B. Oral iron supplementation at 3-6 mg/kg/day.

C. Intravenous iron administration.

D. Treatment for intercurrent infections

6. A 5 y/o child with chronic gastrointestinal malabsorption presents with easy bruising and bleeding. Laboratory tests show a prolonged prothrombin time (PT) and elevated levels of PIVKA (protein induced by vitamin K absence). What is the

most appropriate long-term treatment for this child with vitamin K deficiency due to malabsorption?

A. Administer 1 mg of parenteral vitamin K.

B. Administer 2.5-10 mg of parenteral vitamin K.

C. Initiate chronic high-dose oral vitamin K therapy.

D. Administer fresh-frozen plasma (FFP).

7. A 6-month-old premature infant presents with irritability, poor feeding, and unexplained hemolytic anemia. You suspect a vitamin deficiency. Which of the following is the most likely cause of these symptoms?

A. Vitamin A deficiency

B. Vitamin B12 deficiency

C. Vitamin E deficiency

D. Vitamin K deficiency

8. A 3 y/o child is brought with symptoms of nausea, vomiting, and abdominal pain. The child's mother mentions that they recently started a new multivitamin supplement. Laboratory tests reveal elevated serum calcium and 25-hydroxyvitamin D levels. What is the most appropriate management strategy for this child?

A. Immediate hemodialysis.

B. High-dose vitamin D supplementation.

C. Aggressive rehydration with normal saline and discontinuation of vitamin D sources.

D. Administration of glucocorticoids and calcitonin.

9. A 3 y/o child with a diet primarily consisting of ultra-high-temperature (UHT) milk presents with a history of irritability, leg swelling, and gum bleeding. Laboratory findings show anemia and normal prothrombin time. Despite initial treatment with iron supplements, the child's symptoms persist. Which of the following is the most likely underlying cause and the appropriate intervention?

A. Chronic iron deficiency; continue iron therapy and consider a different iron formulation

B. Scurvy due to Vitamin C deficiency; initiate high-dose Vitamin C therapy

C. Vitamin K deficiency due to malabsorption; administer Vitamin K supplementation

D. Excessive calcium intake from fortified milk; reduce calcium and monitor levels

10. A 4-month-old infant is brought with c/o irritability, vomiting, and frequent seizures. The mother reports that the infant's diet consists primarily of formula milk, with minimal introduction of solid foods. O/E, the infant appears listless and has an abnormal EEG pattern. Considering these clinical findings, which of the following deficiencies is most likely responsible for the symptoms described?

A. Riboflavin (Vitamin B2)

B. Vitamin B6 (pyridoxine)

C. Vitamin B12 (cobalamin)

D. Vitamin D

11. A 6-month-old infant is brought in for a routine check-up. The mother has been exclusively breastfeeding and is now concerned about the appropriate time to introduce solid foods. What is the recommended action?

A. Continue exclusive breastfeeding until the baby is 9 months old.

B. Introduce complementary foods along with breastfeeding.

C. Switch entirely to solid foods and stop breastfeeding.

D. Start formula feeding and delay introducing solid foods until 7 months.

12. A 4-year-old child presents to the pediatric clinic with complaints of poor vision during nighttime and dryness of the eyes. On physical examination, you observe Bitot's spots on the conjunctiva. There is concern about vitamin A deficiency. What is the most appropriate WHO grading of vitamin A deficiency and treatment plan for this child?

A. X1B ; 300,000 IU once daily on D1, D2, and D7

B. X3B ; 100,000 IU once daily on D1, D2, and D21

C. X1B ; 200,000 IU once daily on D1, D2, and D28

D. X3B ; 400,000 IU once daily on D1, D2, and D14

13. During a routine visit, the parents of a 2-year-old ask about the best sources of vitamin A and the recommended schedule for preventing deficiency. Which statement about vitamin A prophylaxis is accurate?

A. 100,000 IU of vitamin A at 9 months, 200,000 IU at 16-18 months, and every 6 months up to age 5.

B. 200,000 IU of vitamin A at 9 months, 200,000 IU at 16-18 months, and every 6 months up to age 5.

C. 200,000 IU of vitamin A at 16-18 weeks, 200,000 IU at 9 months, and every 6 months up to age 5.

D. 100,000 IU of vitamin A at 16-18 weeks, 200,000 IU at 9 months, and every 6 months up to age 5.

14. Which of the following statements is FALSE based on the provided information?

A. Fore milk satisfies the thirst of the baby.

B. Hind milk satisfies the hunger of the baby.

C. Preterm breast milk has a higher lactose content than full-term breast milk.

D. Preterm breast milk has higher immunoglobulins and protein content.

15. Match the following: 1. Colostrum a. Higher in fat and sugar content 2. Transitional Milk b. Higher lactose 3. Mature Milk c. High in antibodies, low in lactose

1. Colostrum	a. Higher in fat and sugar content
2. Transitional Milk	b. Higher lactose
3. Mature Milk	c. High in antibodies, low in lactose

A. 1-c, 2-b, 3-a

B. 1-b, 2-c, 3-a

C. 1-b, 2-a, 3-c

D. 1-c, 2-a, 3-b

16. A new mother asks her pediatrician how breast milk's components benefit her baby’s health. Which of the following pairs best describes a protective substance in breast milk and a disease it helps prevent?

A. PABA; Galactosemia

B. Lysozyme; Phenylketonuria

C. Lactoferrin; Maple syrup urine disease

D. Bifidus factor; NEC

17. A preterm infant is being fed breast milk. Which of the following components is likely to be higher in the breast milk of a mother who delivered prematurely?

A. Lactose

B. Fat

C. Protein

D. Vitamin D

18. A healthcare provider is educating a group of expectant mothers about situations where breastfeeding might not be recommended. Which of the following maternal conditions would most likely contraindicate breastfeeding?

A. History of mastitis with successful treatment during pregnancy.

B. Active varicella-zoster infection with lesions on the breast.

C. Use of a low-dose, progestin-only contraceptive pill.

D. Mildly elevated blood pressure controlled with medication.

19. A new mother visits the clinic concerned that her baby might not be latching on correctly during breastfeeding. She describes that feeding sessions are painful, and her baby seems unsettled afterward. Which of the following observations indicates a poor attachment?

A. The baby's mouth is wide open, covering most of the areola.

B. The baby's chin is deeply pressing against the breast, with the lower lip flanged outward.

C. The mother is leaning towards the baby to help them latch onto the breast.

D. The baby’s cheeks are rounded, and swallowing sounds are audible

20. A new mother is struggling with milk production and asks what she can do to increase her breast milk supply. Which of the following strategies would be most effective?

A. Use a pacifier to soothe the baby between feeds.

B. Encourage early and frequent breastfeeding, including night feeds.

C. Skip night feeds to ensure the mother gets more rest.

D. Offer formula supplements to ensure the baby is getting enough nutrition.

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	2
Question 5	2
Question 6	3
Question 7	3
Question 8	3
Question 9	2
Question 10	2
Question 11	2
Question 12	3
Question 13	1
Question 14	3
Question 15	4
Question 16	4
Question 17	3
Question 18	2
Question 19	3
Question 20	2

Solution for Question 1:

Correct Answer: B) Vitamin B1 (Thiamine)

Explanation: Thiamine deficiency in infants, especially in those breastfed by mothers with poor thiamine intake (such as diets high in polished rice), can lead to symptoms such as lethargy, restlessness, developmental delay, and abdominal distention. These symptoms are consistent with infantile beriberi, making thiamine deficiency the most likely cause in this scenario.

Recommended Daily Allowances (RDAs) & Sources for Thiamine, Cobalamin and Folate in Infants:

Vitamin	RDA	Dietary Sources
Thiamine (B1)	0.2-0.3 mg/day	Unpolished rice, oats, legumes, wheat, meat
Cobalamin (B12)	0.3-0.5 µg / day	Milk, meat, eggs (not found in plant sources)
Folate	65-80 µg/day	Green leafy vegetables, citrus fruits, papaya

Vitamin Deficiencies			
	Thiamine Deficiency	Vit B12 Deficiency (Option A)	Vitamin B9 (Folate) Deficiency (Option C)
Cause	Polished rice diet Refined wheat flour diet. Alcoholism Malnutrition Malignancies	Vegan diet. Maternal deficiency during breastfeeding. Malabsorption syndromes (e.g. Celiac disease, Crohn's disease).	Poor dietary intake, Malabsorption (e.g., celiac disease, IBD), Increased demand (e.g., sickle cell disease, malignancies) Inborn errors of folate metabolism, Cerebral folate deficiency
Clinical features:	Early: Lethargy, Irritability, restlessness, depression, drowsiness, poor concentration and GI discomfort. Late: Dry Beriberi (neurologic): Peripheral neuritis, decreased reflexes, loss of vibration sense, muscle	Hematologic manifestations: Megaloblastic anemiaia Neurological manifestations: Developmental delay/regression and involuntary movements (tremors).	Megaloblastic anemia, growth retardation, glossitis, neural tube defects in offspring.

	tenderness, cramping,ataxia, loss of coordination Wet Beriberi (cardiac): Heart failure, tachycardia, edema, cardiomegaly Infantile: Emesis, lethargy, restlessness, ophthalmoplegia, abdominal distention, developmental delay.		

Vitamin A (Option D) deficiency typically presents with visual disturbances, such as night blindness, and may lead to an increased susceptibility to infections. It does not explain the neurological symptoms, lethargy, and abdominal distention seen in this infant. Therefore, it is less likely to be the cause in this case.

Solution for Question 2:

Explanation:

120 µg/day is the correct RDA for iodine for children aged 6-12 years.

- Sources Sea water Seafood Sea plants Soil in mountainous regions (deficient)
- Sea water
- Seafood
- Sea plants
- Soil in mountainous regions (deficient)
- Recommended Daily Intake (RDA) Birth to 5 years: 90 µg/day(Option A). Ages 6-12 years: 120 µg/day. Adolescents and adults: 150 µg/day(Option C). Pregnant and lactating mother: 200µg/day(Option D).
- Birth to 5 years: 90 µg/day(Option A).
- Ages 6-12 years: 120 µg/day.
- Adolescents and adults: 150 µg/day(Option C).
- Pregnant and lactating mother: 200µg/day(Option D).
- Causes of Iodine Deficiency Low iodine content in soil (especially mountainous regions). Lack of iodine in the diet.
- Low iodine content in soil (especially mountainous regions).

- Lack of iodine in the diet.
- Clinical Features of Iodine Deficiency Mental retardation (especially in early life) Goiter Hypothyroidism Lowered IQ Psychoneuromotor and intellectual development issues.
- Mental retardation (especially in early life)
- Goiter
- Hypothyroidism
- Lowered IQ
- Psychoneuromotor and intellectual development issues.
- Prevention of Iodine Deficiency Universal iodization of salt. National Iodine Deficiency Disorders Control Program in India. Target iodine content in salt: 30 ppm at the manufacturing level, and 15 ppm at the distribution level.
- Universal iodization of salt.
- National Iodine Deficiency Disorders Control Program in India.
- Target iodine content in salt: 30 ppm at the manufacturing level, and 15 ppm at the distribution level.
- Sea water
- Seafood
- Sea plants
- Soil in mountainous regions (deficient)
- Birth to 5 years: 90 µg/day(Option A).
- Ages 6-12 years: 120 µg/day.
- Adolescents and adults: 150 µg/day(Option C).
- Pregnant and lactating mother: 200µg/day(Option D).
- Low iodine content in soil (especially mountainous regions).
- Lack of iodine in the diet.
- Mental retardation (especially in early life)
- Goiter
- Hypothyroidism
- Lowered IQ
- Psychoneuromotor and intellectual development issues.
- Universal iodization of salt.
- National Iodine Deficiency Disorders Control Program in India.
- Target iodine content in salt: 30 ppm at the manufacturing level, and 15 ppm at the distribution level.

Solution for Question 3:

Correct Answer: B) 2-4 mg elemental zinc/kg/day for several weeks

Explanation: 2-4 mg elemental zinc/kg/day is recommended for malnourished children and those with significant zinc deficiency, as it accounts for increased needs due to depletion and potential absorption issues.

- Sources: Animal protein: liver, meat, fish, eggs, nuts. Low in cereals, starch, plants, and legumes (high phytate content).
- Animal protein: liver, meat, fish, eggs, nuts.
- Low in cereals, starch, plants, and legumes (high phytate content).
- RDA: Children: 3.5-5.0 mg/day (Option C)
- Causes of Zinc Deficiency: Increased demand in infants, children, adolescents, and pregnant/lactating women. Malnutrition and malabsorption syndromes. Chronic or recurrent diarrhoea. Prolonged IV feeding without adequate zinc.
- Increased demand in infants, children, adolescents, and pregnant/lactating women.
- Malnutrition and malabsorption syndromes.
- Chronic or recurrent diarrhoea.
- Prolonged IV feeding without adequate zinc.
- Clinical Features of Zinc Deficiency: Poor physical growth in children. Delayed sexual maturation (hypogonadism) in adolescents. Anemia, anorexia, diarrhea. Alopecia, dermatitis, impaired immune function. Poor wound healing, and skeletal abnormalities.
- Poor physical growth in children.
- Delayed sexual maturation (hypogonadism) in adolescents.
- Anemia, anorexia, diarrhea.
- Alopecia, dermatitis, impaired immune function.
- Poor wound healing, and skeletal abnormalities.
- Acrodermatitis Enteropathica:
- Animal protein: liver, meat, fish, eggs, nuts.
- Low in cereals, starch, plants, and legumes (high phytate content).
- Increased demand in infants, children, adolescents, and pregnant/lactating women.
- Malnutrition and malabsorption syndromes.

- Chronic or recurrent diarrhoea.
- Prolonged IV feeding without adequate zinc.
- Poor physical growth in children.
- Delayed sexual maturation (hypogonadism) in adolescents.
- Anemia, anorexia, diarrhea.
- Alopecia, dermatitis, impaired immune function.
- Poor wound healing, and skeletal abnormalities.

- Autosomal recessive disorder.
- Severe zinc deficiency due to impaired intestinal absorption (ZIP4 protein defect).
- Early infancy onset: Vesiculobullous, dry, scaly skin lesions (periorificial, buttocks, acral areas).
- Associated with alopecia, eye changes, chronic diarrhea, growth retardation, and stomatitis.
- Treatment: 2-3 mg/kg/day oral zinc therapy.
- Treatment: General deficiency: 0.5-1.0 mg elemental zinc/kg/day for several weeks/months (Option A). Malnourished children: 2-4 mg/kg/day for several weeks.
- General deficiency: 0.5-1.0 mg elemental zinc/kg/day for several weeks/months (Option A).
- Malnourished children: 2-4 mg/kg/day for several weeks.
- General deficiency: 0.5-1.0 mg elemental zinc/kg/day for several weeks/months (Option A).
- Malnourished children: 2-4 mg/kg/day for several weeks.

20-30 mg/kg/day oral zinc therapy (Option D): This dosage is excessively high for treating zinc deficiency. The recommended range is much lower to avoid potential toxicity and side effects.

Solution for Question 4:

Correct Answer: B) 2 mL/kg of 10% calcium gluconate IV, administered slowly

Explanation:

This patient is exhibiting signs and symptoms of severe calcium deficiency.

2 mL/kg of 10% calcium gluconate IV is the immediate treatment for severe hypocalcemia symptoms such as tetany, laryngospasm, and seizures. It provides rapid correction of calcium levels and is administered under cardiac monitoring to prevent complications.

Calcium

Sources: Milk, cheese, yogurt, Green leafy vegetables, legumes, nuts, fruit juice, and some fortified breakfast cereals

RDA:

- Infants- 200- 250mg/day
- Toddlers -700-1000mg/day
- Adolescents - 1300 mg/day

Deficiency of Calcium: Manifestations

- Central nervous system irritability.
- Poor muscular contractility.
- Lethargy, Poor feeding.
- Jitteriness.
- Vomiting, Abdominal distension.
- Seizures.
- Twitching and cramps, Laryngospasm (rare).
- Tetany (muscular twitching, carpopedal spasm, stridor). Chvostek sign: Twitching of orbicularis oculi and mouth upon facial nerve tapping. Trousseau sign: Carpopedal spasm upon arm BP cuff inflation.
- Chvostek sign: Twitching of orbicularis oculi and mouth upon facial nerve tapping.
- Trousseau sign: Carpopedal spasm upon arm BP cuff inflation.
- ECG: Prolonged QTc (>0.45 seconds), Impaired cardiac function.
- Features of rickets (with prolonged hypocalcemia).
- Chvostek sign: Twitching of orbicularis oculi and mouth upon facial nerve tapping.
- Trousseau sign: Carpopedal spasm upon arm BP cuff inflation.

Treatment

- Immediate treatment: Tetany, laryngospasm, seizures: 2 mL/kg of 10% calcium gluconate IV, administered slowly under cardiac monitoring. Repeat IV calcium boluses every 6 hours as needed.
- Tetany, laryngospasm, seizures: 2 mL/kg of 10% calcium gluconate IV, administered slowly under cardiac monitoring.
- Repeat IV calcium boluses every 6 hours as needed.
- Calcium supplementation: IV calcium gluconate: 9.8 mg/mL elemental calcium (0.45 mEq/mL). Calcium chloride 10%: 27 mg/mL elemental calcium (1.4 mEq/mL).
- IV calcium gluconate: 9.8 mg/mL elemental calcium (0.45 mEq/mL).
- Calcium chloride 10%: 27 mg/mL elemental calcium (1.4 mEq/mL).
- Oral calcium therapy: 40-80 mg/kg/day. For asymptomatic patients. Follow-up after IV calcium therapy.
- For asymptomatic patients.
- Follow-up after IV calcium therapy.
- Tetany, laryngospasm, seizures: 2 mL/kg of 10% calcium gluconate IV, administered slowly under cardiac monitoring.
- Repeat IV calcium boluses every 6 hours as needed.
- IV calcium gluconate: 9.8 mg/mL elemental calcium (0.45 mEq/mL).
- Calcium chloride 10%: 27 mg/mL elemental calcium (1.4 mEq/mL).
- For asymptomatic patients.
- Follow-up after IV calcium therapy.

Oral calcium supplementation (Option A) is used for follow-up or asymptomatic patients but is not suitable for acute treatment of severe symptoms.

Calcium chloride 10% (Option C:)While effective, calcium chloride is not as safe- it is more caustic and can cause tissue damage if extravasation from the vein occurs. It is usually reserved for situations where rapid calcium administration is crucial, and central venous access is available. However, calcium gluconate is preferred due to its safer profile.

Reassurance and observation (Option D) would be appropriate for mild cases without serious symptoms. However, immediate intervention is necessary given the severity of the infant's symptoms, including the prolonged QTc interval.

Solution for Question 5:

Correct Answer: B) Oral iron supplementation at 3-6 mg/kg/day

Explanation: The patient's clinical features and lab findings are suggestive of iron deficiency anemia.

Oral iron supplementation at 3-6 mg/kg/day is the first-line treatment for iron deficiency anemia in children. It provides the elemental iron necessary to replenish stores and correct anemia.

Iron

Source:

- Heme sources: meat, poultry, fish.
- Nonheme sources: dairy, eggs, plant-based foods, breads, cereals, breakfast foods.

RDA

- 1-9 yrs: 7-10 mg/day.
- Adolescents:11-18 mg/day

Causes:

- Diminished dietary intake or reduced iron absorption in the proximal small intestine.
- Excessive loss of body iron.
- Intercurrent infections (e.g., hookworm infestation, malaria).
- Dietary inhibitors (e.g., phytates, phosphates, tannates).
- Cow’s milk is a poor source of bioavailable iron and is not advised for children under 1 year old.
- Behavioral: Irritability, anorexia.
- The pallor of Nail beds, oral mucous membranes, and palpebral conjunctivae are reliable indicators
- Physical: Weakness, fatigue, leg cramps, breathlessness, tachycardia.
- Severe cases: Congestive cardiac failure, splenomegaly
- Severe deficiency signs: Angular stomatitis, glossitis, koilonychia, paronychia

Laboratory Findings:

- Peripheral blood smear: Microcytic, hypochromic red cells, anisocytosis, poikilocytosis, increased RDW.
- Reduced MCV and MCHC.
- Decreased serum iron, increased TIBC, transferrin saturation <16%.

- Early reduction in serum ferritin, and high free erythroprotoporphyrin (FEP) before anemia.
- Address Underlying Causes: Identify and correct the cause of anemia, such as treating hookworm infestation.
- Dietary Counseling: Provide guidance on iron-rich foods and avoid inhibitors of iron absorption.
- Iron Supplementation: Oral iron medications (e.g., ferrous sulfate, 20% elemental iron). Dosage: 3-6 mg/kg/day of elemental iron, Administration: On an empty stomach or between meals for best absorption, Duration: Continue for 4-6 months after anemia correction to replenish iron stores, Monitor reticulocyte count: Should increase within 72-96 hours of therapy initiation, Side effects: Gastrointestinal issues (nausea, discomfort, vomiting, constipation, diarrhea)
- Oral iron medications (e.g., ferrous sulfate, 20% elemental iron).
- Dosage: 3-6 mg/kg/day of elemental iron,
- Administration: On an empty stomach or between meals for best absorption,
- Duration: Continue for 4-6 months after anemia correction to replenish iron stores,
- Monitor reticulocyte count: Should increase within 72-96 hours of therapy initiation,
- Side effects: Gastrointestinal issues (nausea, discomfort, vomiting, constipation, diarrhea)

- Oral iron medications (e.g., ferrous sulfate, 20% elemental iron).
- Dosage: 3-6 mg/kg/day of elemental iron,
- Administration: On an empty stomach or between meals for best absorption,
- Duration: Continue for 4-6 months after anemia correction to replenish iron stores,
- Monitor reticulocyte count: Should increase within 72-96 hours of therapy initiation,
- Side effects: Gastrointestinal issues (nausea, discomfort, vomiting, constipation, diarrhea)

- Parenteral iron therapy: eg Injection Iron sucrose, Injection Ferric carboxymaltose.
- Blood transfusion: Packed red cell transfusions are essential in emergencies such as severe bleeding, urgent surgery, or severe anemia with congestive heart failure (administered slowly with monitoring).

Iron-rich dietary counseling (Option A) is important but not sufficient as a standalone treatment for established iron deficiency anemia. It should complement iron supplementation.

Intravenous iron administration(Option C) is generally reserved for cases where oral iron is not effective, not tolerated, or when a rapid response is needed, which is not indicated as the initial treatment for this case.

Treatment for intercurrent infections(Option D) should be addressed if an infection is identified as a contributing factor, but the immediate treatment of iron deficiency anemia requires direct iron supplementation.

Solution for Question 6:

Correct Answer: C) Initiate chronic high-dose oral vitamin K therapy

Explanation: Chronic high-dose oral vitamin K therapy is appropriate for managing vitamin K deficiency due to malabsorption, as it provides a consistent source of vitamin K to counteract ongoing deficiencies.

Sources	Green leafy vegetables. Liver Synthesized by intestinal bacteria.
Recommended Daily Allowance (RDA)	Newborns and infants: 3 to 5 µg/day Older children: 10 to 30 µg/day

- Neonates: Hemorrhagic disease of the newborn (HDN) High levels of protein induced by vitamin K absence (PIVKA)
- Hemorrhagic disease of the newborn (HDN)
- High levels of protein induced by vitamin K absence (PIVKA)
- Children: Bleeding from various sites. Presence of ecchymosis.
- Bleeding from various sites.
- Presence of ecchymosis.
- Hemorrhagic disease of the newborn (HDN)
- High levels of protein induced by vitamin K absence (PIVKA)
- Bleeding from various sites.
- Presence of ecchymosis.
- Prolonged prothrombin time (PT).
- Prolonged activated partial thromboplastin time (aPTT).
- Normal platelet counts.
- Detection of PIVKA.

Infants with VKDB	Administer 1 mg of parenteral vitamin K. Expect PT to decrease within 6 hours and normalize within 24 hours.
For Adolescents	Administer 2.5-10 mg of parenteral vitamin K for rapid correction.

For Severe, Life-Threatening Bleeding	Administer fresh-frozen plasma (FFP) infusion to rapidly correct coagulopathy (along with Vitamin K).
For Vitamin K Deficiency Due to Malabsorption in Children	Chronic administration of high-dose oral vitamin K (2.5 mg twice weekly to 5 mg daily).

1 mg (Option A) or 2.5-10 mg (Option B) of parenteral vitamin K are typically used for acute treatment of vitamin K deficiency, not long-term management in cases of chronic malabsorption.

Fresh-frozen plasma (FFP) (Option D) is used for severe, life-threatening bleeding and coagulopathy, and is not suitable for long-term management of vitamin K deficiency.

Solution for Question 7:

Correct Answer: C) Vitamin E deficiency

Explanation:

Vitamin E deficiency in premature infants can cause hemolytic anemia due to oxidative damage to red blood cells, presenting symptoms such as irritability and poor feeding.

Vitamin E Overview:

Recommended Daily Allowance (RDA):

- No specific RDA.
- Deficiency is rare in the general population but can occur in premature infants and cases of severe malnutrition.

Dietary Sources: Vegetable oils, Seeds, Nuts, Green leafy vegetables, Margarine

Causes of Deficiency:

- Prematurity: Lack of transfer from mother to baby.
- Fat Malabsorption syndromes: Conditions affecting bile acid production (e.g., cholestatic liver disease, cystic fibrosis, coeliac disease, Crohn's disease).
- Abetalipoproteinemia: Autosomal recessive disorder causing fat malabsorption.
- Ataxia with Isolated Vitamin E Deficiency (AVED): Autosomal recessive disorder with mutations in the TTPA gene affecting vitamin E incorporation.

Clinical Manifestations:

- In Premature Infants: Hemolysis, anemia, edema.
- Neurologic Disorder: Cerebellar Disease: Ataxia, dysarthria Posterior Column Dysfunction: Loss of reflexes, decreased proprioception, vibratory sensation Loss of DTR (deep tendon reflexes) Retinal Disease: Ophthalmoplegia, nystagmus, pigmentary retinopathy
- Cerebellar Disease: Ataxia, dysarthria
- Posterior Column Dysfunction: Loss of reflexes, decreased proprioception, vibratory sensation
- Loss of DTR (deep tendon reflexes)
- Retinal Disease: Ophthalmoplegia, nystagmus, pigmentary retinopathy
- Cerebellar Disease: Ataxia, dysarthria
- Posterior Column Dysfunction: Loss of reflexes, decreased proprioception, vibratory sensation
- Loss of DTR (deep tendon reflexes)
- Retinal Disease: Ophthalmoplegia, nystagmus, pigmentary retinopathy

Diagnosis:

- Premature Infants: Serum vitamin E and lipid levels if unexplained hemolytic anemia is present.
- Children with Neurologic Findings: Evaluate vitamin E status
- Unexplained Ataxia: Screen for AVED
- Serum Vitamin E Ratio: <0.8 mg/g (children/adults), <0.6 mg/g (infants) is abnormal.

Treatment:

- Neonates: 25–50 units/day for one week, then adequate dietary intake.
- Malabsorption in children: 1 unit/kg/day, with dose adjustments based on monitoring.
- AVED: High doses of vitamin E, ongoing treatment required.

Vitamin A deficiency (Option A) typically affects vision and skin.

Vitamin B12 deficiency (Option B) often presents with neurological symptoms and macrocytic anemia.

Vitamin K deficiency (Option D) primarily leads to bleeding disorders due to impaired blood clotting.

Solution for Question 8:

Correct Answer: C) Aggressive rehydration with normal saline and discontinuation of vitamin D sources

Explanation:

Hypervitaminosis D leads to hypercalcemia, so the primary treatment involves managing hypercalcemia and discontinuing vitamin D sources.

Aggressive rehydration helps to dilute serum calcium, correct pre-renal azotemia, and increase urinary calcium excretion.

Hypervitaminosis D Overview:

Causes:

- Excessive vitamin D supplementation.
- Accidental over-fortification of milk.
- Contaminated table sugar.
- Inadvertent use of vitamin D as cooking oil.

Upper Intake Limits:

- Children <1 year: 1,000 IU/day.
- Older children and adults: 2,000 IU/day.

Symptoms:

- Gastrointestinal: Nausea, vomiting, constipation, abdominal pain, pancreatitis.
- Cardiac: Hypertension, arrhythmias, decreased QT interval.
- Central Nervous System: Lethargy, hypotonia, confusion, psychosis, hallucinations, coma.
- Renal: Polyuria, dehydration, hypernatremia, acute renal failure, kidney stones, nephrocalcinosis.

Diagnosis:

- Elevated serum calcium and 25-D levels.
- 25-D >100 ng/mL is considered high.
- Normal 1,25-D levels despite high 25-D.

Treatment:

Hemodialysis (Option A) is reserved for severe cases where other treatments are ineffective.

High-dose vitamin D supplementation (Option B) would worsen the condition, not treat it.

Glucocorticoids and calcitonin (Option D) are additional treatments but not the first line in managing acute cases.

Solution for Question 9:

Correct Answer: B) Scurvy due to Vitamin C deficiency; initiate high-dose Vitamin C therapy

Explanation:

The child's symptoms—irritability, leg swelling, gum bleeding, and normal prothrombin time (As the defect is in collagen synthesis and th coagulation cascade is not affected)—align with scurvy, which is caused by Vitamin C deficiency. UHT milk, which lacks Vitamin C, can contribute to this deficiency. High-dose Vitamin C therapy is the appropriate intervention.

Vitamin C Overview:

Recommended Daily Allowance in Infants:

Age Group	RDA (mg/day)
0-6 months	40
7-12 months	50

Vitamin C Source:	Citrus fruits/juices, cauliflower and green leafy vegetables, melons, guava.
Vitamin C Deficiency (Scurvy)	
Causes:	Heat-treated milk or unfortified formula. Diets lacking fruits and vegetables.
Symptoms:	Early: Irritability, loss of appetite, fever, pain, leg tenderness. Later: Swelling, pseudoparalysis, subperiosteal hemorrhages, "rosary" beads at the costochondral junction, gum changes, anemia, bleeding, poor healing.
Treatment:	Vitamin C 100-200 mg/day (oral or parenteral). Improvement within a week; continue for 3 months.

Chronic iron deficiency; continue iron therapy and consider a different iron formulation (Option A): Although iron deficiency can cause anemia, the persistence of symptoms such as gum bleeding and leg swelling, coupled with a diet of UHT milk, strongly suggests a Vitamin C deficiency rather than an ongoing issue with iron therapy.

Vitamin K deficiency due to malabsorption; administer Vitamin K supplementation (Option C): While Vitamin K deficiency can cause bleeding, it typically does not have normal prothrombin time (Elevated Prothrombin time seen). The diet and clinical presentation are more indicative of Vitamin C deficiency.

Excessive calcium intake from fortified milk; reduce calcium and monitor levels (Option D): The symptoms and clinical findings do not match the effects of excessive calcium intake. The dietary history of UHT milk is more consistent with Vitamin C deficiency rather than calcium-related issues.

Solution for Question 10:

Correct Answer: B) Vitamin B6 (pyridoxine)

Explanation:

Vitamin B6 (pyridoxine) deficiency commonly causes irritability, vomiting, seizures, and failure to thrive in infants. It is also associated with EEG abnormalities and can respond to pyridoxine supplementation, which fits the clinical presentation described.

RDAs & sources for Pyridoxine (Vitamin B6) and Riboflavin (Vitamin B2) in Infants:

Vitamin	RDA	Dietary Sources
Pyridoxine (B6)	0.1-0.3 mg/day	Banana, wheat germ, rice, sunflower seeds, fish, poultry.
Riboflavin (B2)	0.3-0.4mg/day	Milk, green leafy vegetables, sprouts, legumes, meat

Vitamin Deficiencies		
	Pyridoxine (B6) deficiency	Riboflavin (B2) deficiency
Cause	High protein intake. Certain medications (eg. Isoniazid) Oral contraceptives. Dialysis.	Malnutrition Malabsorption Certain medications Phototherapy

Clinical features:	Infants: Irritability, seizures, vomiting, failure to thrive. Adults: Peripheral neuritis, cheilosis, glossitis, seborrheic dermatitis, anemia. Dependency Syndromes: Pyridoxine-dependent epilepsy, Vitamin B6-responsive anemia.	Cheilosis, glossitis, keratitis, conjunctivitis, seborrheic dermatitis, anemia, photophobia. (Option A)
Diagnosis:	Low AST/ALT, high urinary xanthurenic acid, EEG response to pyridoxine.	Clinical response to supplementation, EGR (Erythrocyte Glutathione Reductase) activity test, low urinary riboflavin.
Treatment:	Convulsions: 100 mg pyridoxine IM/IV, then oral intake. Dependency: 2-10 mg IM or 10-100 mg orally daily.	Oral riboflavin 3-10 mg/day, balanced diet including dairy products

Vitamin B12 (cobalamin) (Option C): While it can cause developmental delays and irritability, it is less likely to cause seizures at this early age compared to vitamin B6 deficiency. It is mostly associated with a Vegan diet in the mother.

Vitamin D (Option D): Generally leads to rickets and bone pain but does not typically present with seizures or irritability in infants.

Solution for Question 11:

Correct answer: B) Introduce complementary foods along with breastfeeding.

Explanation:

Complementary feeding should begin at 6 months to ensure the infant receives adequate nutrition while continuing to breastfeed.

- It is defined as semi-solid, energy-dense food, given in addition to breast milk.
- It ensures that the infant continues to receive the immunological benefits of breast milk while also obtaining essential nutrients from various solid foods, such as iron-rich and zinc-containing foods.

Continue exclusive breastfeeding until the baby is 9 months old (Option A): While exclusive breastfeeding is highly beneficial during the first six months, continuing it alone can lead to nutritional deficiencies. By 9 months, the baby needs additional nutrients, such as iron, so complementary feeding should begin around 6 months to meet the baby's nutritional needs.

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Switch entirely to solid foods and stop breastfeeding (Option B): While introducing solid foods is necessary at 6 months, breastfeeding should continue to provide the baby with essential antibodies and nutrients that complement the solids. Abruptly stopping breastfeeding may increase the risk of infections and nutritional gaps.

Start formula feeding and delay introducing solid foods until 7 months (Option D): Formula feeding is unnecessary if breastfeeding is adequate, and delaying complementary foods beyond 6 months is not recommended.

Assessment of adequacy of breastfeeding:

- Weight Gain: After an initial weight loss within the first week, a healthy term infant should regain their birth weight by 7 to 10 days of age. Following this, the baby should gain approximately 20 to 40 grams daily.
- Urine Output: A well-nourished baby typically produces 6 to 8 wet diapers daily, indicating adequate hydration.
- Stool Pattern: By the end of the first week, a newborn's stool should change from the initial black-green meconium to a yellow-green colour, reflecting proper digestion and milk intake.
- Infant Behavior: A baby who is content and sleeps well for 2 to 3 hours after feeding is considered to receive enough milk.

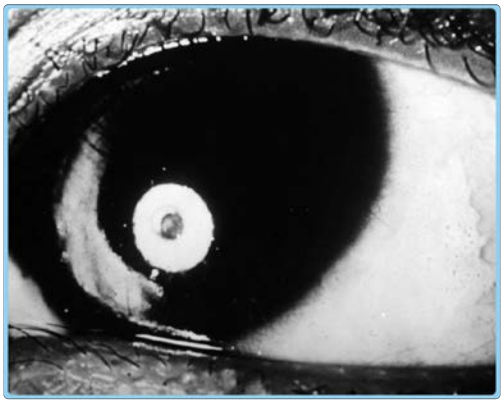
Solution for Question 12:

Correct Answer: C) X1B ; 200,000 IU once daily on D1, D2, and D28

Explanation:

- The presence of Bitot's spots corresponds to the WHO grading X1B for vitamin A deficiency.
- For a child over 1 year old, the recommended treatment is 200,000 IU of vitamin A once daily on D1, D2, and D28 (4 weeks later).

Bitot’s Spot



WHO Grading of eye signs of vitamin A deficiency (VAD) in children	
Grade	Eye Sign
XN	Night blindness
X1A	Conjunctival xerosis
X1B	Bitot's spots
X2	Corneal xerosis
X3A	Corneal ulcer covering less than 1/3 of the cornea
X3B	Corneal ulcer covering at least 1/3 of the cornea (Option B & D ruled out)
XS	Corneal scarring

Age Group	Treatment
Under 6 months	50,000 IU once daily on D1, D2, and 4 weeks later
6 to 12 months	100,000 IU once daily on D1, D2, and 4 weeks later
Over 1 year	200,000 IU once daily on D1, D2, and 4 weeks later (Option A ruled out)

Use in preterm infants:

- Vitamin A supplementation improves respiratory function and reduces the risk of chronic lung disease.
- It reduces death or oxygen requirement, with no impact on neurodevelopmental outcomes.

Solution for Question 13:

Correct Answer: A) 100,000 IU of vitamin A at 9 months, 200,000 IU at 16-18 months, and every 6 months up to age 5.

Explanation:

This child should have received 100,000 IU of vitamin A at 9 months, 200,000 IU at 16-18 months, and 200,000 IU every 6 months up to age 5. Carotenoids in carrots and leafy greens are good dietary sources. The RDA of a 2-year-old child is 400-600 µg.

National Vitamin A Prophylaxis Program by the National Health Mission (NHM) of India		
Age	Dosage	Additional Context
9 months	100,000 IU	Given along with measles immunization.
16-18 months	200,000 IU	Given with a DPT booster.
Every 6 months	200,000 IU	Up to the age of 5 years.
Total Doses: 9 mega doses from 9 months to 5 years.		

Sources of Vitamin A:

- Carotenoids (plant-based): Carrots, leafy greens, squash, oranges, tomatoes; primary carotenoid is β-carotene.
- Preformed Vitamin A (animal-based): Liver, shark/cod liver oil, egg yolk, whole milk, butter.
- Fortified Foods: Many processed foods and infant formulas.

Recommended Daily Allowance (RDA) for Vitamin A:

- Infants- 300-400 µg
- Children- 400-600 µg
- Adolescents- 750 µg
- Pregnant women- 800 µg
- Note: 1 µg retinol equivalent (RE) = 3.3 international units (IU) of vitamin A. Hence, 30 mg retinol = 100,000 IU vitamin A.

Solution for Question 14:

Correct Answer: C) Preterm breast milk has higher lactose content than full-term breast milk.

Explanation:

Preterm breast milk, in fact, has lower lactose content compared to full-term breast milk. It is richer in sodium, immunoglobulins, proteins, iron, and calories.

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Solution for Question 15:

Correct Answer: D) 1-c, 2-a, 3-b

Explanation:

Milk Type	Time of Production	Appearance	Quantity	Key Components
Colostrum	Initial 3-4 days after delivery	Yellow and thick	Small	High in antibodies, immune-competent cells, vitamins A, D, E, and K. Rich in immunoglobulins and proteins. Low in lactose.
Transitional Milk	From 3-4 days until two weeks	Thinner than colostrum	Moderate	Decreased immunoglobulins and proteins. Increased fat and sugar.
Mature Milk	Follows transitional milk	Thinner and watery	Variable	Consistent in nutrient composition Balanced in nutrients essential for growth Higher lactose content.

Solution for Question 16:

Correct Answer: D) Bifidus factor; NEC

Explanation:

Bifidus factor; NEC is the correct answer because it accurately describes a protective substance in breast milk and its associated protective role against NEC.

Anti-Infective Substances in Breast Milk:

- Phagocytic macrophages
- PABA (para-aminobenzoic acid)
- Lactoferrin
- Lysozyme
- Antibodies, especially IgA
- Anti-staphylococcal factor
- Bifidus factor (Option D)
- Bile-stimulated lipase
- Transforming Growth Factor β (TGF-β)

Mnemonic: Teach for PLAB

Benefits of Breast Milk:

- During the Neonatal Period: Helps protect against conditions such as necrotizing enterocolitis (NEC) and neonatal sepsis. (Option D)
- Later in Life: Reduces the risk of obesity(lower total serum cholesterol), hypertension, diabetes, allergies, asthma, dental caries, ulcerative colitis, and Crohn's disease. Reduced risk of post-partum hemorrhage, pre-menopausal breast cancer, and ovarian cancer in the mother.
- Reduced risk of post-partum hemorrhage, pre-menopausal breast cancer, and ovarian cancer in the mother.
- Acute conditions: Diarrhea, pneumonia, ear infection, Haemophilus influenzae infection, meningitis, and UTIs.
- Cognitive Development: Breast milk-fed infants tend to have higher IQs.
- Bonding: Facilitates maternal-child bonding.

- Safety and Accessibility: Breast milk is safe, typically free from contaminants, and readily available even in resource-limited settings.
- Reduced risk of post-partum hemorrhage, pre-menopausal breast cancer, and ovarian cancer in the mother.

Infants who should not receive breast milk

- Galactosemia (Option A ruled out)
- Phenylketonuria (Option B ruled out)
- Maple syrup urine disease (Option C ruled out)

Solution for Question 17:

Correct answer: C) Protein

Explanation:

Preterm breast milk contains higher protein, sodium, iron, immunoglobulins, and calories than full-term milk. The increased protein concentration supports the accelerated growth and development needs of preterm infants, who require more protein for optimal development during their early weeks.

Lactose (Option A): While lactose is present in breast milk, it does not significantly increase in preterm milk compared to full-term milk.

Fat (Option B): The fat content in breast milk varies with the feeding stage but isn't specifically higher in preterm milk compared to full-term milk.

Vitamin D (Option D): Vitamin D, while essential for bone health, is generally present in low amounts in breast milk regardless of whether the infant is preterm or full-term. Supplementation may be required.

Breastmilk Composition	
Colostrum	The initial, thick, yellow milk is produced during the first 3-4 days after birth. Rich in antibodies (particularly secretory IgA), immune cells, and vitamins A, D, E, and K. This early milk provides crucial protection and immunity for the newborn.
Transitional milk	Follows colostrum, appearing from days 3-4 to about two weeks postpartum. A gradual reduction in immunoglobulins and proteins, while fat and sugar

	levels increase. This shift meets the infant's increasingenergy requirements.
Mature milk	Established after the transitional phase. Thinner and more diluted than earlier milk stages, but continues to offer essential nutrients for the baby’s growth.
Carbohydrates - Lactose, Galactose	Lactose: The main sugar in breast milk, which helps with calcium absorption and encourages the growth of beneficial gut bacteria. Galactose: Contributes to the development of brain tissue and nerve cell membranes.
Proteins	Whey proteins (like lactalbumin and lactoglobulin) - easily digestible. Taurine and cysteine- are important for brain development.
Fats	One of the main energy sources in breast milk, with the composition influenced by the mother’s diet. DHA and ARA are essential for the development of the brain and vision.
Vitamins and minerals	Provides necessary vitamins and minerals in forms that are easily absorbed by the infant. Note: Vitamin D may be present in low amounts; supplementation might be recommended.
Water and electrolytes	Comprises around 88% water, which typically eliminates the need for additional fluids. The minimal concentration of minerals reduces stress on the baby’s kidneys.
Immunological components	Secretory IgA offers protection against infections. Includes macrophages, lymphocytes, lactoferrin, and lysozyme, which help fight off pathogens.
Growth factors, enzymes, and hormones	Epidermal Growth Factor (EGF) supports gut health. Enzymes like lipases, aid in fat digestion. Hormones help regulate appetite and metabolism.

Solution for Question 18:

Correct answer: B) Active varicella-zoster infection with lesions on the breast.

Explanation:

Table rendering failed

History of mastitis with successful treatment during pregnancy (Option A): A history of mastitis, once successfully treated, does not contraindicate breastfeeding.

Use of a low-dose, progestin-only contraceptive pill (Option C) is not a contraindication to breastfeeding.

Mildly elevated blood pressure controlled with medication(Option D): Controlled mild hypertension is not a contraindication to breastfeeding, as it does not pose a risk to the infant.

Solution for Question 19:

Correct answer: C) The mother is leaning toward the baby to help them latch onto the breast

Explanation:

Leaning towards the baby to facilitate latching can lead to improper attachment, causing discomfort for the mother and ineffective suckling by the baby. The baby should be brought to the breast, not vice versa, to maintain good attachment.

The baby's mouth is wide open, covering most of the areola (Option A): A wide-open mouth covering most of the areola is a sign of good attachment. This ensures that the baby is effectively compressing the milk sinuses under the areola, facilitating proper milk transfer and reducing discomfort for the mother.

The baby's chin is deeply pressed against the breast, with the lower lip flanged outward (Option B): This helps secure the nipple and areola in the mouth, allowing efficient milk extraction and preventing nipple trauma. It is a sign of good attachment.

The baby’s cheeks are rounded, and swallowing sounds are audible (Option D): Rounded cheeks and audible swallowing sounds are positive signs of effective suckling, indicating that the baby is well-attached and transferring milk successfully.

Solution for Question 20:

Correct answer: B) Encourage early and frequent breastfeeding, including night feeds.

Explanation:

Early initiation and frequent feedings are critical for establishing and maintaining an adequate milk supply. Frequent suckling increases the release of prolactin, which stimulates milk production. Additionally, night feeds are important because prolactin levels are naturally higher at night, further enhancing milk production and helping to sustain a steady milk supply.

Use a pacifier to soothe the baby between feeds (Option A): Using a pacifier can interfere with breastfeeding by causing nipple confusion and reducing the baby’s time at the breast, potentially decreasing milk production.

Skip night feeds to ensure the mother gets more rest (Option C): Skipping night feeds can lead to reduced prolactin levels and milk production. Night feeds are important for sustaining milk output.

Offer formula supplements to ensure the baby is getting enough nutrition (Option D): Offering formula can reduce breastfeeding frequency, leading to less stimulation of milk production and a lower overall milk supply.

Storage of Expressed Breast Milk		
Condition	Safe storage upto	Temperature
Room Temperature	6-8 hours	25 °C
Refrigeration	24 hours	2-8 °C
Freezing	3-6 months	-20 °C
Thawing	Once thawed, used within 24 hours	Under running water

Malnutrition

1. A 2 y/o from a low-income community presents with poor growth. The child's height for age is below -2 SDS on WHO standards, but weight-for-height is normal. What is the most likely diagnosis?

A. Severe Acute Malnutrition (SAM) B

B. Moderate Stunting

C. Moderate Wasting

D. Severe Underweight

---

2. A 2 year old malnourished child is assessed using mid-upper arm circumference (MUAC), and the measurement is 10 cm. What does this indicate?

A. Moderate Stunting

B. Severe Wasting

C. Chronic Malnutrition

D. Obesity

---

3. A 2 y/o child from a rural area presents to the clinic with severe wasting, prominent ribcage, and a "baggy pants" appearance due to loose skin on the buttocks. Despite the child's emaciated state, they remain mentally alert. What is the most likely diagnosis?

A. Kwashiorkor

B. Marasmus

C. Mild Malnutrition

D. Obesity

---

4. A 14-month-old child is brought to the clinic with concerns about poor growth. The child is significantly underweight, and O/E, the mid-upper arm circumference (MUAC) measures 11 cm. There is also noticeable swelling in both feet. What is the most likely diagnosis?

A. Mild Acute Malnutrition

B. Severe Acute Malnutrition

C. Moderate Acute Malnutrition

D. Chronic Malnutrition

---

5. A 2 y/o child is diagnosed with severe acute malnutrition (SAM) and is admitted to the hospital. On admission, the child is found to have hypoglycemia and starts showing signs of hypothermia. Which of the following is the most likely complication of SAM that the child is experiencing?

A. Refeeding Syndrome

B. Electrolyte Imbalance

C. Dehydration

D. Metabolic Disturbances

---

6. A 10-month-old with SAM was treated with F75 and transitioned to F100 and RUTF after the oedema resolved and appetite returned. The child has gained 6 grams/kg/day. Which of the following is NOT an appropriate discharge criterion?

A. The child has shown consistent weight gain of more than 5 grams/kg/day for at least 3 consecutive days.

B. The child's oedema has resolved.

C. The child has been switched from F75 to F100.

D. The child has received all necessary vaccinations.

7. Which of the following is a target of POSHAN Abhiyaan?

A. Increase the average weight of children by 5 kg per year.

B. Achieve a 2% annual reduction in under-nutrition.

C. Eliminate all forms of malnutrition by 2022.

D. Provide free school meals to all children.

8. Which of the following features correctly describes the conditions depicted in images 1 and 2? a)Severe wasting b)Flaky paint dermatitis c)Baggy pants d)Skin and bone appearance e)Nutritional anaemia f)Sugar baby g)Flag sign h)Monkey facies

A. 1- a,d,e,f; 2- b,c,g,h

B. 1- b,d,g,h; 2- a,c,e,f

C. 1- a,b,c,e; 2- d,f,g,h

D. 1- a,c,d,h; 2- b,e,f,g

9. Based on the image provided, which of the following statements is TRUE?

A. Measure from the acromion to the olecranon and mark the midpoint.

B. MUAC assesses the nutritional status of children aged three months to five years.

C. In Shakir's tape, the Yellow zone indicates 11.5 to 13.5 cm.

D. In Shakir's tape, the Green zone indicates >11.5 cm.

10. A 4-month-old infant presents with a 5-week history of scaly lesions on the face, trunk, and limbs as given in the image, along with irritability, dehydration, and generalized oedema. Laboratory tests reveal hypoalbuminemia and hypokalemia. Based on these findings, what is the most likely diagnosis?

A. Marasmus

B. Flaky paint dermatosis

C. Flag sign

D. Acrodermatitis Enteropathica

11. A 2 y/o child is brought to the clinic for evaluation due to concerns about poor growth and changes in hair appearance. The child's hair is sparse, and dull, and shows alternating light and dark bands. Which of the following is NOT a characteristic hair sign in a severely malnourished child?

A. Dyspigmentation

B. Sparse hair

C. Tightening of natural curls

D. Flag sign

12. A 5 y/o male presents with fatigue, glossitis, and tingling in his hands and feet. On examination, the physician notices swollen gums and pinpoint haemorrhages on his legs. All of the following are features of Vitamin B deficiency EXCEPT?

A.

B.

C.

D.

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	2
Question 5	4
Question 6	3
Question 7	2
Question 8	4
Question 9	1
Question 10	2
Question 11	3
Question 12	4

Solution for Question 1:

Correct answer: B) Moderate Stunting

Explanation:

Stunting is indicated by low height-for-age, falling between -2 and -3 SDS, which suggests chronic malnutrition. This child’s growth has been stunted due to long-term inadequate nutrition.

Severe Acute Malnutrition (SAM)(Option A): SAM is characterized by severe wasting, a weight-for-height below -3 SDS, or bilateral oedema. This child does not have these symptoms.

Moderate Wasting (Option C): Wasting refers to low weight-for-height, typically indicating acute malnutrition. The child’s weight for height is normal, so wasting is not present.

Severe Underweight (Option D): Underweight refers to low weight-for-age. Although underweight can be associated with both wasting and stunting, the child's weight-for-height is normal, ruling out underweight as the primary issue.

WHO Classification of Malnutrition	
Moderately underweight	Weight-for-age is between -2 and -3 SDS.
Severely underweight	Weight-for-age is < -3 SDS.
Moderately wasted	Weight-for-length/height is between -2 and -3 SDS.
Severely wasted	Weight-for-length/height is < -3 SDS.
Moderate Acute Malnutrition	Weight-for-length/height or BMI-for-age is between -2 and -3 SDS. Mid-upper arm circumference (MUAC) is between 11.5 cm and 12.5 cm.
Severe Acute Malnutrition:	Weight-for-length/height or BMI-for-age is less than -3SDS. MUAC is less than 11.5 cm. The presence of bilateral pitting oedema.
Moderately Stunted (Moderate Chronic Malnutrition)	Length/height-for-age is between -2 and -3 SDS.
Severely Stunted (Severe Chronic Malnutrition)	Length/height-for-age is < -3 SDS.

Solution for Question 2:

Correct answer: B) Severe Wasting

Explanation:

A MUAC measurement significantly below the recommended threshold (11.5 cm between 6 months to 5 years of age) typically indicates severe wasting. Wasting is a form of acute malnutrition where the child has a skinny appearance due to recent and severe weight loss. Therefore, a low MUAC strongly suggests severe wasting.

- Mid-upper arm circumference (MUAC) is considered an age-independent measure for assessing malnutrition.
- It provides a general indication of acute malnutrition without needing to adjust for the child's age, making it a versatile tool for screening across various age groups within its applicable

range.

Moderate Stunting (Option A): It is assessed using height-for-age measurements, which reflect long-term nutritional status and growth. MUAC is not used to diagnose stunting; thus, a low MUAC measurement would not indicate moderate stunting.

Chronic Malnutrition (Option C): It is associated with long-term nutritional deficiencies and is evaluated through height-for-age measurements that show stunting. MUAC is more suited for detecting acute malnutrition rather than chronic malnutrition.

Obesity (Option D) is identified through high BMI or weight-for-height measurements, not by a low MUAC. Since MUAC is used to assess acute malnutrition, it cannot indicate obesity.

Solution for Question 3:

Correct answer: B) Marasmus

Explanation:

Table rendering failed

Mild Malnutrition (Option C): Mild malnutrition does not present with such severe symptoms as wasting and an emaciated appearance.

Obesity (Option D): Obesity is characterized by excessive fat accumulation, which is the opposite of the symptoms described in the scenario.

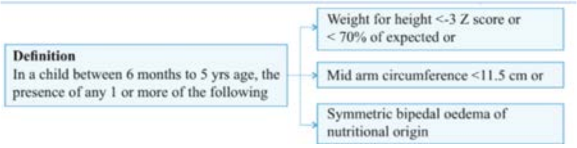
WELLCOME CLASSIFICATION (Protein Calorie Malnutrition)		
Weight (% of standard)	Edema	
Present	Absent	
60-80	Kwashiorkor	Undernourished
<60	Marasmic Kwashiorkor	Marasmus

Solution for Question 4:

Correct answer: B) Severe Acute Malnutrition

Explanation:

Severe Acute Malnutrition is the correct option, characterized by severe wasting (MUAC < 11.5 cm) and the presence of bilateral oedema, both of which are diagnostic criteria for SAM. SAM requires immediate medical attention due to its life-threatening nature.



Definition
In a child between 6 months to 5 yrs age, the presence of any 1 or more of the following

Weight for height <-3 Z score or < 70% of expected or

Mid arm circumference <11.5 cm or

Symmetric bipedal oedema of nutritional origin

Definition of SAM - can have any of the 3 criteria.

Mild Acute Malnutrition (Option A): This refers to a less severe form of malnutrition where weight and MUAC are below normal but not critically low as seen in SAM. The presence of bilateral oedema and a MUAC of 11 cm indicates a more severe condition.

Moderate Acute Malnutrition (Option C) is when a child shows moderate wasting, with a MUAC that is lower than normal but higher than the threshold for severe malnutrition (typically between 11.5 cm and 12.5 cm).

Chronic Malnutrition (Option D): It is characterized by stunting, where a child is shorter for their age due to long-term nutritional deficiency. The key features in this scenario -acute wasting and edema are more consistent with severe acute malnutrition, not chronic malnutrition.

Solution for Question 5:

Correct answer: D) Metabolic Disturbances

Explanation:

SAM is characterized by significant metabolic disturbances, including hypoglycemia (due to impaired gluconeogenesis in the liver) and hypothermia (due to reduced metabolic rate and loss of body fat). The symptoms presented in the scenario align closely with these disturbances, making this the most likely complication the child is experiencing.

Refeeding Syndrome (Option A) is a serious complication that can occur when feeding is reintroduced too rapidly in malnourished patients, leading to dangerous shifts in fluids and electrolytes. Although it is a known risk during the treatment of SAM, the symptoms present as breathlessness, rapid pulse, and watery diarrhoea which are not seen in this patient.

Electrolyte Imbalance (Option B) particularly involving potassium, magnesium, and sodium, is common in children with SAM and can lead to complications like cardiac arrhythmias and muscle weakness. However, the symptoms described (hypoglycemia and hypothermia) are more indicative of metabolic disturbances.

Dehydration (Option C): Dehydration can occur in malnourished children due to inadequate fluid intake, diarrhea, or other factors. However, dehydration typically presents with signs such as dry mucous membranes, sunken eyes, decreased skin turgor, and low blood pressure, none of which are specifically noted in this child. The presentation of hypoglycemia and hypothermia strongly suggests metabolic disturbances related to refeeding rather than dehydration alone.

Solution for Question 6:

Correct Answer: C) The child has been switched from F75 to F100

Explanation:

Switching to F100 is NOT sufficient for discharge. Discharge criteria require resolution of medical issues, sufficient weight gain, and clinical stability.

SAM- Nutritional Rehabilitation Phase and Discharge	

Nutritional Rehabilitation	Entry to Rehabilitation Phase: Indicated by reduced or minimal oedema and a return of appetite. Transition Process: Duration: Over 3 days to prevent refeeding syndrome. Method: Replace F75 with an equal volume of F100 for 2 days. Gradual Increase: Increase feed volume by 10 mL at each feeding until some feed remains uneaten (around 200 mL/kg/day). Post-Transition Feeding: Feeds: Provide unlimited amounts of high-energy, high-protein milk formula like F100 (100 kcal and 3 g protein per 100 mL), ready-to-use therapeutic food (RUTF), or modified family foods with similar energy and protein content. (Option C) Calories and Protein: Offer 150-220 kcal/kg/day and 4-6 g protein/kg/day. Supplements: Continue potassium, magnesium, and micronutrients. Add iron at 3 mg/kg/day. Breastfeeding: Encourage continued breastfeeding if applicable.
RUTF	Standard ready-to-use therapeutic food (RUTF) usually contains milk powder, sugar, peanut butter, vegetable oil, vitamins, and minerals. It has a long shelf life and low moisture content- preventing the growth of pathogens. When using RUTF: Begin with small, frequent meals of RUTF, aiming for 8 meals per day initially and then reducing to 5–6 meals per day. If the child cannot consume the full RUTF portion at each meal, supplement with F-75 to complete the meal until they can eat the full amount of RUTF. If the child is unable to consume at least half of the recommended RUTF amount within 12 hours, discontinue RUTF and provide F-75 instead. Reintroduce RUTF after 1–2 days and continue this approach until the child can consume sufficient RUTF. If the child is still breastfeeding, offer breast milk before each RUTF feeding.
Discharge criteria	All medical issues, including oedema, are resolved for 2 weeks (Option B) They have completed parenteral antibiotic treatment and the infant is clinically stable and alert. The infant is either breastfeeding effectively or feeding well. The weight gain is satisfactory, such as being above the median of WHO growth velocity standards or gaining more than 5g/kg/day for at least 3 consecutive days. (Option A) Before discharge, check and provide any missing vaccinations and routine interventions as needed. Ensure that mothers or caregivers are connected with

	appropriate community follow-up and support. (Option D)

Solution for Question 7:

Correct Answer: B) Achieve a 2% annual reduction in under-nutrition

Explanation:

One of the specific targets of POSHAN Abhiyaan is to achieve a 2% annual reduction in undernutrition among children, women, and adolescent girls.

POSHAN ABHIYAAN

- Prime minister's overarching scheme for holistic nutrition.
- Launch: POSHAN Abhiyaan was launched by Prime Minister Narendra Modi in Jhunjhunu, Rajasthan, in March 2018.
- Objectives: The initiative targets reduction in under-nutrition, stunting, anaemia (in children, women, and adolescent girls), and low birth weight by coordinating various nutrition schemes. Improve nutrition in 1st 1000 days of life (including the antenatal period and 2 years after birth).
- Implementation: The program utilizes existing structural arrangements of line ministries and will be scaled up with support from the World Bank's ISSNIP project to all districts by 2022.
- Annual Targets: Reduce stunting by 2% per year. Reduce Undernutrition by 2% per year (Option B). Reduce Anemia by 3% per year. Reduce low birth weight by 2% per year.
- Reduce stunting by 2% per year.
- Reduce Undernutrition by 2% per year (Option B).
- Reduce Anemia by 3% per year.
- Reduce low birth weight by 2% per year.
- Stunting Reduction Goal: Strives to decrease stunting from 38.4% (NFHS-4) to 25% by 2022, with a target of at least a 2% annual reduction.
- Reduce stunting by 2% per year.
- Reduce Undernutrition by 2% per year (Option B).
- Reduce Anemia by 3% per year.
- Reduce low birth weight by 2% per year.

Increase the average weight of children by 5 kg per year (Option A): POSHAN Abhiyaan does not target increasing the average weight of children by a specific amount annually. Instead, it aims to reduce various forms of malnutrition.

Eliminate all forms of malnutrition by 2022 (Option C): POSHAN Abhiyaan does not aim to eliminate all forms of malnutrition by 2022 but focuses on reducing stunting, under-nutrition, anaemia, and low birth weight.

Provide free school meals to all children (Option D): This refers to the Mid-Day Meal Program also called the PM POSHAN (POshan SHAkti Nirman) Scheme, a separate initiative. The Mid-Day Meal Program aims to provide free meals to children in government primary and upper primary schools, government-aided Anganwadis, Madarsa, and Maqtabs on working days, except for school holidays, primarily to improve nutritional status and encourage school attendance, especially among disadvantaged children.

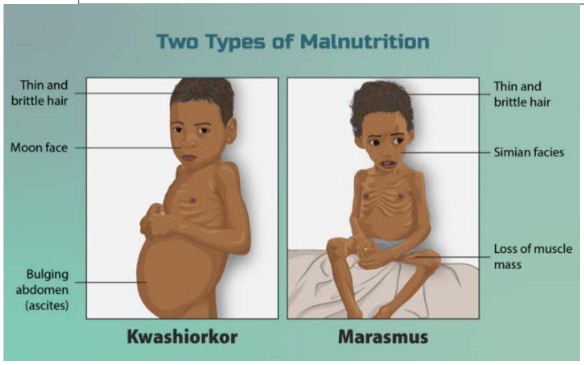
Solution for Question 8:

Correct Answer: D) 1- a,c,d,h; 2- b,e,f,g

Explanation:

MARASMUS	KWASHIORKOR

Severe wasting of fat and muscle. Often due to acute starvation or illness secondary to poor nutritional status. Severe marasmus is a form of severe acute malnutrition (SAM). Skin-and-bones appearance. Monkey facies due to the loss of buccal fat pads. Baggy pants appearance- due to loose skin hanging from the buttocks. Loss of axillary fat pad. Despite the severe wasting, affected children may appear alert. No oedema.	Pitting oedema present. “Well-fed” / “Sugar baby” appearance of the child d/t oedema. Oedema can be mild to severe, representing 5-20% of body weight. Muscle wasting is present. Flaky pain dermatitis present (peeling skin). Mucous membrane lesions- smooth tongue, cheilosis, angular stomatitis. Flag sign in hair- alternating pigmented bands. Hair is sparse. Unhappiness, apathy, irritability, and feeding difficulties. Anorexia, vomiting, abdominal distension, and diarrhoea. Nutritional anaemia is typically present. Cardiovascular issues include cold, pale extremities, prolonged circulation time, bradycardia, and hypotension.



Solution for Question 9:

Correct Answer: A) Measure from the acromion to the olecranon and mark the midpoint

Explanation:

MID UPPER ARM CIRCUMFERENCE (MUAC)	
Definition	MUAC is a vital measurement used primarily to assess the nutritional status of children aged 6 months to 5 years. (Option B) Quick and effective screening tool. In infants under 6 months old, the Mid-Upper Arm Circumference (MUAC) is not utilized as a measurement criterion. (G88)

Procedure	Ensure the child is standing straight with their left arm free of clothing. The arm should be bent at a right angle. Measure the distance between the tip of the shoulder (acromion) and the tip of the elbow (olecranon). Mark the midpoint on the upper arm. (Option A) The child's arm should hang relaxed by their side, with the palm facing the thigh. Use a MUAC tape to wrap around the marked midpoint of the arm. The tape should be snug but not tight enough to pinch the skin. Record the measurement at the zero mark of the tape.
Shakir’s Tape	Shakir's tape is a specific type of MUAC measuring tape designed for use in assessing malnutrition. It is typically three colour-coded to facilitate easy interpretation of nutritional status. The tape is marked with different zones that correspond to various levels of malnutrition: Red Zone: Indicates severe malnutrition (MUAC < 11.5 cm). Yellow Zone: Indicates moderate malnutrition (MUAC between 11.5 cm and 12.5 cm). (Option C) Green Zone: Indicates healthy nutritional status (MUAC > 12.5 cm) (Option D)
SHAKIR’S TAPE	

Solution for Question 10:

Correct Answer: B) Flaky paint dermatosis

Explanation:

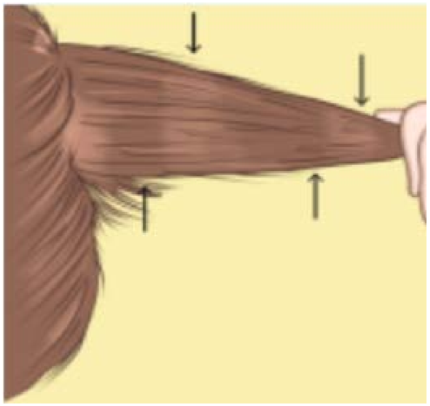
Flaky paint dermatosis, also known as peeling paint dermatosis, is a skin manifestation commonly seen in children with kwashiorkor, a form of severe protein-energy malnutrition. The characteristic skin changes include:

- Dry, desquamative, hyperpigmented, patchy, and macular lesions on the face, trunk, and limbs.
- Fragile skin that peels off irregularly, revealing underlying hypopigmentation.
- Shiny, varnished-looking hyperpigmentation is the most common finding.
- Progression from dryness and atrophy to generalized hyperkeratosis and hyperpigmentation.

The skin changes are attributed to low methionine levels, which impair the sulfation of keratin. Treatment involves improving protein and calorie intake, gradually introducing enteral feeds, and providing supportive care.

Marasmus (Option A) is characterized by severe energy deficiency leading to significant weight loss and muscle wasting. While it presents with malnutrition, it typically does not feature the specific skin manifestations described in the vignette. Instead, a marasmic child presents with a more generalized wasting appearance without the oedema seen in kwashiorkor.

A flag sign (Option C) is characterized by alternating bands of hypopigmented and normally pigmented hair and appears when a child experiences growth spurts. This is seen in a child with kwashiorkor.



Acrodermatitis enteropathica (Option D) is a rare genetic disorder caused by zinc deficiency due to defective zinc absorption in the intestines, often linked to gene mutations that encode the ZIP4 zinc transporter. This condition leads to insufficient dietary zinc absorption, manifesting in distinctive skin, hair, and nail issues. Key symptoms include:

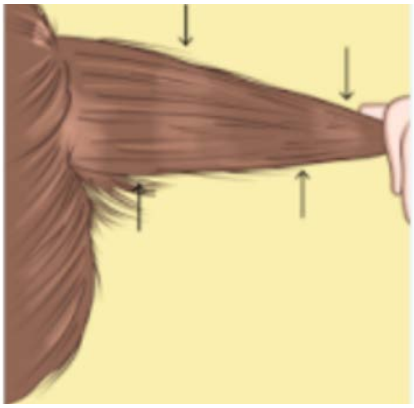
- Periorificial and acral dermatitis: Red, vesicular, and erosive skin lesions around the mouth and extremities.
- Refractory Diarrhoea, Failure to thrive and irritability
- Alopecia: Hair loss.
- Only 20% of cases exhibit all three symptoms of the triad: dermatitis, alopecia, and diarrhoea.

Solution for Question 11:

Correct answer: C) Tightening of natural curls

Explanation:

In malnutrition, the natural curl pattern is usually lost, not tightened. Tightening of curls is not a recognized sign of severe malnutrition.



Flag sign in a severely malnourished child.

- Dyspigmentation: Changes in hair colour. (Option A)
- Loss of characteristic curls: Hair loses its natural curl.
- Sparse hair: Notably over temple and occipital regions. (Option B)
- Dull, easily pluckable hair: Hair lacks lustre and is fragile.
- Flag sign: Alternate bands of hypopigmented and normally pigmented hair, seen during growth spurts. (Option D)

Solution for Question 12:

Correct answer: D)

Explanation:

This image shows swollen and bleeding gums. These are classic signs of scurvy, caused by Vitamin C deficiency, not Vitamin B deficiency.

Clinical features of Vitamin C deficiency

- Increased Capillary Fragility: Petechiae on skin
- Delayed Wound Healing
- Bleeding Gums

Bleeding gums

Scurvy (Severe Vitamin C Deficiency):

- Hemarthrosis (bleeding into joints).
- Painful joint swellings.
- Pseudoparalysis (inability to move limbs due to pain).
- Frog-like posture (semi-flexion at hips and knees)
- Scorbutic rosary (angular swellings at the costochondral junction)- different from rickety rosary seen in vitamin D.

Clinical features of vitamin B deficiencies

- Peripheral neuropathy , Wernicke-Korsakoff syndrome(Mental confusion , ataxia , ocular abnormalities): Caused by thiamine deficiency
- Angular stomatitis, cheilitis(Option A),glossitis(Option B): Caused by riboflavin deficiency

Chelitis	Glossitis
	

- Casal’s necklace: The cutaneous lesions are characterized by a symmetrical, pigmented rash that appears on body parts exposed to sunlight, particularly around the neck or the back. It is caused by a deficiency of Niacin(Option C).

Previous Year Questions

1. In a child presenting with diarrhoea, the following finding is noted. What is the likely micronutrient deficiency?

A. Zinc

B. Copper

C. Vitamin A

D. Vitamin C

2. What is the likely diagnosis for a child who has a swollen abdomen and a pale, unenergetic expression, and is exclusively fed rice water, showing signs of malnourishment with decreased serum protein and albumin levels?

A. Kwashiorkor

B. Marasmus

C. Indian childhood cirrhosis

D. Kawasaki disease

3. A child presented to the clinic with complaints of malnutrition and severe dehydration. What would be the choice of IV fluid for this patient?

A. ORS

B. NS

C. RL in 5% Dextrose

D. DNS

---

4. Which of the following is a reliable sign of some dehydration in severe acute malnutrition?

A. Increased thirst

B. Sunken eyes

C. Skin pinch that goes back slowly

D. Altered sensorium

---

5. What could be the likely diagnosis for a laborer's second child who presents to the outpatient department (OPD) with an enlarged abdomen and a lackluster facial appearance? The child's diet primarily consists of rice water (rice milk). Further investigations reveal low levels of serum protein and albumin.

A. Kwashiorkor

B. Marasmus

C. Indian childhood cirrhosis

D. Kawasaki disease

---

6. Which of the following is not seen in nutritional rickets?

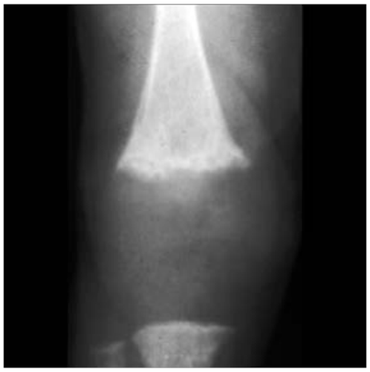
A. Low phosphate

B. Normal calcium

C. Urine Calcium: creatinine > 15mg/kg

D. High PTH

7. A child presents with generalized body aches, poor wound healing, and corkscrew hairs on her arms, and white lines of Frenkel on X-ray as shown below. This clinical presentation is most consistent with which of the following deficiencies?



A. Zinc deficiency

B. Vitamin C deficiency

C. Copper deficiency

D. Iron deficiency

8. Chronic malnutrition is best measured by?

A. Height for age

B. Weight for age

C. Weight for height

D. Body mass index

9. A 4-year-old child with mid-arm circumference of 105mm, upon providing therapeutic food, eagerly completed all of it. This patient is considered under?

A. Uncomplicated SAM

B. Complicated SAM

C. Acute malnutrition

D. Normal nutrition

10. A child presents to the OPD with fatigue, bone pains, and swollen and bleeding gums. Physical examination reveals petechiae on her skin. Which of the following can be given to treat her condition?

A. Vitamin A

B. Vitamin D

C. Vitamin E

D. Vitamin C

11. A child with diarrhea was eager to drink and the skin pinch went back slowly. The child is classified into which of the following categories as per IMNCI?

A. Pink

B. Yellow

C. Green

D. None

---

12. A 10 year old child is got to the OPD with complaints of poor appetite and failure to thrive. Upon further clinical examination, loss of skin turgor, dry mucous membranes are noted. What is the glucose content of the rehydration solution used to treat this condition?

A. 100 mmol/L

B. 125 mmol/L

C. 150 mmol/L

D. 200 mmol/L

---

13. What is the amount of milk given to a term baby in first 24 hours after delivery?

A. 40ml/kg

B. 50ml/kg

C. 60ml/kg

D. 80ml/kg

---

14. A 7-month-old child was brought by her mother with complaints of diarrhoea and mucocutaneous lesions, as shown in the image. The symptoms have been present since weaning was started. What is the diagnosis?

A. Wiskott-Aldrich syndrome

B. Acrodermatitis enteropathica

C. Paediatric epidermolysis bullosa

D. Atopic dermatitis

15. Which of the following is not a criterion for severe acute malnutrition?

A. Weight for height less than -3 SD

B. MUAC < 115 mm

C. Bilateral pedal edema

D. Weight for age less than -3 SD

16. What is the recommended daily dose of iodine for a child?

A. 120-200 µg

B. 90-120 µg

C. 30-60 µg

D. 500 µg

17. In an eye camp, doctors try to make people aware of the causes of blindness. Vitamin A is the leading cause of preventable childhood blindness. A parent brings her 18-month-old child to the doctor for a routine eye check-up. The doctor notices a thinned-out cornea, and from old records, it was seen that the child also has iron-deficiency anemia. If the weight of the child is 10kg, what should be the dose of vitamin A to be given in this case if the doctor marks it as keratomalacia?

A. 50,000 IU

B. 1,00,000 IU

C. 2,00,000 IU

D. 5,00,000 IU

---

18. Pellagra like rash is seen in:

A. Gaucher disease

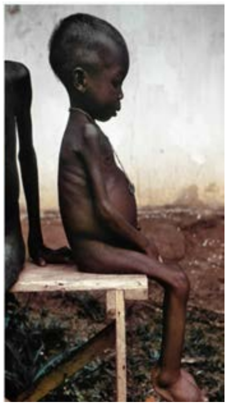
B. Alkaptonuria

C. Hartnup disease

D. Phenylketonuria

---

19. A child comes to your outpatient department exhibiting the findings depicted in the image provided. What is the probable diagnosis?



A. Kwashiorkor

B. Marasmus

C. Nephrotic syndrome

D. Nephritic syndrome

20. A labourer's younger child is brought to the OPD with a swollen belly and dull face. He has mostly fed rice water in his diet. On investigations, the child was found to have low serum protein, low serum protein, and low albumin. What is the probable diagnosis?

A. Kwashiorkor

B. Marasmus

C. Indian childhood cirrhosis

D. Kawasaki disease

21. Which vitamin deficiencies are associated with the presence of cardiac edema and neuropathy?

A. Biotin

B. Thiamine

C. Pyridoxine

D. Riboflavin

22. A child in your unit is diagnosed with severe acute malnutrition (SAM). According to the WHO criteria, SAM is best defined by:

A. Weight for age less than -2 SD

B. Weight for height less than +2 SD

C. Weight for age less than +3 SD

D. Weight for height less than -3 SD

23. Which of the following statements is incorrect about colostrum?

A. It is the breastmilk that is seen in the initial 1-3 days

B. It is yellow in color

C. Contains large number of antibodies

D. Contains high amount of lactose and low amounts of protein

24. Vitamin K supplementation is given to neonates to prevent:

A. Breast milk jaundice

B. Hemorrhagic disease of the newborn

C. Keratomalacia

D. Scurvy

25. Which of the following is an absolute contraindication for breastfeeding?

A. Alcoholic mother

B. Classic galactosemia in the infant

C. A mother with HIV infection

D. Infants born to an HbsAg-positive mother

---

26. What is the mid-arm circumference measurement for severe acute malnutrition?

A. > 14.5 cm

B. 13.5-14.5 cm

C. 11.5-12.5 cm

D. < 11.5 cm

---

27. According to current recommended dietary guidelines for children, energy from saturated fats should be what percentage of total energy intake?

A. < 10%

B. < 5%

C. < 15%

D. < 30%

---

28. All are signs of adequate breastfeeding in a 2-month-old infant, except?

A. Baby sleeps for 2-3 hours after feeding

B. Baby urinates 6-8 times per day

C. Gains weight

D. Good skin tone

---

29. Exclusive breastfeeding should be continued for a minimum of how many months?

A. 6 months

B. 2 months

C. 12 months

D. 5 months

---

30. During the anthropometric evaluation, the height for the age of a child is < -2 SD. This is most likely due to:

A. Chronic malnutrition

B. Acute malnutrition

C. Recent infection

D. No malnutrition

---

31. What can be used as a substitute for breast milk in low-birth-weight (LBW) infants?

A. Cow's milk

B. Preterm formula

C. Term formula

D. Human donor milk

---

32. A child presents with the findings shown in the image below. What is the true statement regarding this condition?

A. Kwashiorkor due to protein malnutrition

B. Kwashiorkor due to calorie malnutrition

C. Marasmus due to protein malnutrition

D. Marasmus due to calorie malnutrition

---

33. What is the fluid of choice for shock in a child with severe acute malnutrition?

A. Normal saline

B. Ringer lactate

C. 10% dextrose

D. Ringer's lactate with 5% dextrose

---

34. Which of the following vitamins is deficient in breast milk?

A. Vitamin A

B. Vitamin C

C. Vitamin E

D. Vitamin K

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	3
Question 4	1
Question 5	1
Question 6	3
Question 7	2
Question 8	1
Question 9	1
Question 10	4
Question 11	2
Question 12	2
Question 13	3
Question 14	2
Question 15	4
Question 16	2
Question 17	3
Question 18	3
Question 19	1
Question 20	1
Question 21	2
Question 22	4

Question 23	4
Question 24	2
Question 25	2
Question 26	4
Question 27	1
Question 28	1
Question 29	1
Question 30	1
Question 31	4
Question 32	1
Question 33	4
Question 34	4

Solution for Question 1:

Correct Answer: A) Zinc

Explanation:

The characteristic rash is acrodermatitis enteropathica, which occurs due to zinc deficiency.

Zinc:

- Sources: Animal protein: liver, meat, fish, eggs, nuts. Low in cereals, starch, plants, and legumes (high phytate content).
- Animal protein: liver, meat, fish, eggs, nuts.
- Low in cereals, starch, plants, and legumes (high phytate content).
- RDA: Children: 3.5-5.0 mg/day
- Causes of Zinc Deficiency: Increased demand in infants, children, adolescents, and pregnant/lactating women. Malnutrition and malabsorption syndromes. Chronic or recurrent diarrhoea. Prolonged IV feeding without adequate zinc.
- Increased demand in infants, children, adolescents, and pregnant/lactating women.

- Malnutrition and malabsorption syndromes.
- Chronic or recurrent diarrhoea.
- Prolonged IV feeding without adequate zinc.
- Clinical Features of Zinc Deficiency: Poor physical growth in children. Delayed sexual maturation (hypogonadism) in adolescents. Anaemia, anorexia, diarrhoea. Alopecia, dermatitis, impaired immune function. Poor wound healing, and skeletal abnormalities.
- Poor physical growth in children.
- Delayed sexual maturation (hypogonadism) in adolescents.
- Anaemia, anorexia, diarrhoea.
- Alopecia, dermatitis, impaired immune function.
- Poor wound healing, and skeletal abnormalities.
- Acrodermatitis Enteropathica:■■■■■ Autosomal recessive disorder. Severe zinc deficiency due to impaired intestinal absorption (ZIP4 protein defect). Early infancy onset: Vesiculobullous, dry, scaly skin lesions (periorificial, buttocks, acral areas). Associated with alopecia, eye changes, chronic diarrhoea, growth retardation, and stomatitis. Treatment: 2-3 mg/kg/day oral zinc therapy.
- Autosomal recessive disorder.
- Severe zinc deficiency due to impaired intestinal absorption (ZIP4 protein defect).
- Early infancy onset: Vesiculobullous, dry, scaly skin lesions (periorificial, buttocks, acral areas).
- Associated with alopecia, eye changes, chronic diarrhoea, growth retardation, and stomatitis.
- Treatment: 2-3 mg/kg/day oral zinc therapy.

Sources:

- Animal protein: liver, meat, fish, eggs, nuts.
- Low in cereals, starch, plants, and legumes (high phytate content).

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Causes of Zinc Deficiency:

- Increased demand in infants, children, adolescents, and pregnant/lactating women.
- Malnutrition and malabsorption syndromes.
- Chronic or recurrent diarrhoea.
- Prolonged IV feeding without adequate zinc.

Increased demand in infants, children, adolescents, and pregnant/lactating women.

Malnutrition and malabsorption syndromes.

Chronic or recurrent diarrhoea.

Prolonged IV feeding without adequate zinc.

Clinical Features of Zinc Deficiency:

- Poor physical growth in children.
- Delayed sexual maturation (hypogonadism) in adolescents.
- Anaemia, anorexia, diarrhoea.
- Alopecia, dermatitis, impaired immune function.
- Poor wound healing, and skeletal abnormalities.

Poor physical growth in children.

Delayed sexual maturation (hypogonadism) in adolescents.

Anaemia, anorexia, diarrhoea.

Alopecia, dermatitis, impaired immune function.

Poor wound healing, and skeletal abnormalities.

Acrodermatitis Enteropathica:

- Autosomal recessive disorder.
- Severe zinc deficiency due to impaired intestinal absorption (ZIP4 protein defect).
- Early infancy onset: Vesiculobullous, dry, scaly skin lesions (periorificial, buttocks, acral areas).
- Associated with alopecia, eye changes, chronic diarrhoea, growth retardation, and stomatitis.
- Treatment: 2-3 mg/kg/day oral zinc therapy.

Autosomal recessive disorder.

Severe zinc deficiency due to impaired intestinal absorption (ZIP4 protein defect).

Early infancy onset: Vesiculobullous, dry, scaly skin lesions (periorificial, buttocks, acral areas).

Associated with alopecia, eye changes, chronic diarrhoea, growth retardation, and stomatitis.

Treatment: 2-3 mg/kg/day oral zinc therapy.

- Treatment: General deficiency: 0.5-1.0 mg elemental zinc/kg/day for several weeks/months Malnourished children: 2-4 mg/kg/day for several weeks.
- General deficiency: 0.5-1.0 mg elemental zinc/kg/day for several weeks/months
- Malnourished children: 2-4 mg/kg/day for several weeks.

Treatment:

- General deficiency: 0.5-1.0 mg elemental zinc/kg/day for several weeks/months
- Malnourished children: 2-4 mg/kg/day for several weeks.

General deficiency: 0.5-1.0 mg elemental zinc/kg/day for several weeks/months

Malnourished children: 2-4 mg/kg/day for several weeks.

Copper (Option B) deficiency presents with anaemia and neutropenia. Acrodermatitis enteropathica is not associated with copper deficiency.

Vitamin A (Option C) deficiency typically presents with visual disturbances, such as night blindness, and may lead to an increased susceptibility to infections. It is not associated with the given rash in the scenario.

Vitamin C (Option D) deficiency in the early stages presents as irritability, loss of appetite, pain, fever, pain, leg tenderness. The later stages are associated with swelling, pseudoparalysis, subperiosteal haemorrhages, "rosary" beads at the costochondral junction, gum changes, anaemia, bleeding, and poor healing. This does not explain the rashes in the given scenario.

Solution for Question 2:

Correct Option A:

- Kwashiorkor is a condition developed due to abrupt weaning in breast milk from a diet that has low protein.
- Kwashiorkor usually affects children aged 1-4 years.
- The general appearance is a Fat sugar baby appearance.
- The babies will have a protuberant abdomen.
- They will have significant pedal edema around Fees and sparse thin.
- They will not have complete alopecia, but the response here is the typical appearance. This type of appearance has been classically described in the literature as a fat sugar baby appearance.

REASONS FOR EDEMA IN KWASHIORKOR

- Hypoalbuminemia: Produces a fall in the oncotic pressure, which causes fluid extravasation into the extravascular space.
- Reduced protein absorption

Incorrect Options:

Option B. Marasmus babies are called Skin and bone babies because it looks like they only have a skeleton, which is covered by a thin layer of skin. There is no fat, and muscles are growing.

Option C. Chronic liver disease of childhood characterised by cirrhosis of the liver due to deposition of copper in the liver.

Option D. Kawasaki disease is an acute febrile mucocutnneous lymph node syndrome mainly affecting infants and young children

Solution for Question 3:

Correct Option C - RL in 5% Dextrose:

- Ringer's Lactate (RL) with 5% Dextrose is often used as an initial intravenous fluid in children who present with severe dehydration and malnutrition. This solution provides hydration and electrolyte replacement, as well as a source of energy from the dextrose.

Incorrect Options:

Option A - ORS: Oral Rehydration Salts (ORS) are typically used for mild to moderate dehydration and not for severe dehydration. ORS is given orally and thus may not be suitable for a severely dehydrated child who might have impaired consciousness or difficulty in swallowing.

Option B - NS: Normal Saline (NS) is an isotonic solution and while it can serve as an intravenous fluid for hydration, it does not contain any source of energy like dextrose. In severe cases of malnutrition, the body requires energy, which can be provided by dextrose.

Option D - DNS: Dextrose Normal Saline (DNS) does contain dextrose, which provides a source of energy, and it also helps in rehydration. However, it lacks the balanced electrolyte content of Ringer's Lactate, which is essential for the proper functioning of body cells and maintaining fluid balance.

Solution for Question 4:

Correct Option A - Increased Thirst:

- Dehydration tends to be over-diagnosed, and its severity is overestimated in severely malnourished children. Increased thirst is a reliable sign of some dehydration in children with severe acute malnutrition.

Dehydration tends to be over-diagnosed, and its severity is overestimated in severely malnourished children. Increased thirst is a reliable sign of some dehydration in children with severe acute malnutrition.

Incorrect Options:

Option B - Sunken eyes:

- Sunken eyes are often observed in children with SAM due to the loss of subcutaneous fat. In these children, the appearance of sunken eyes can be misleading because it is primarily related to malnutrition and reduced fat stores rather than hydration status.

Sunken eyes are often observed in children with SAM due to the loss of subcutaneous fat. In these children, the appearance of sunken eyes can be misleading because it is primarily related to malnutrition and reduced fat stores rather than hydration status.

Option C - Skin pinch that goes back slowly

- In malnourished children, the skin may return slowly to its normal position, even in the absence of dehydration, due to the lack of underlying fat.

In malnourished children, the skin may return slowly to its normal position, even in the absence of dehydration, due to the lack of underlying fat.

Option D - Altered sensorium:

- Altered sensorium is a feature of severe dehydration, not some dehydration.

Altered sensorium is a feature of severe dehydration, not some dehydration.

Solution for Question 5:

Correct Option A:

- Kwashiorkor is a form of severe malnutrition that occurs due to a deficiency in protein intake, particularly essential amino acids. It is commonly seen in children, typically between the ages of 1 and 3 years, who consume a diet lacking in quality protein sources. In the given scenario, the child's symptoms of a swollen belly and dull face, along with low serum protein and low albumin levels, indicate a protein deficiency. These are characteristic features of Kwashiorkor, making it the most probable diagnosis.

Incorrect choices :

Option B. Marasmus: Marasmus is another form of severe malnutrition, but it is characterized by overall energy deficiency (calorie deficiency) rather than a specific protein deficiency. Children with marasmus exhibit severe wasting, loss of muscle mass, and a generally emaciated appearance. Unlike the child in the given scenario, children with marasmus often have little or no edema (swelling).

Option C. Indian childhood cirrhosis: Indian childhood cirrhosis is a liver disease that primarily affects infants and young children in India. It is characterized by liver damage and fibrosis, leading to liver dysfunction. While the child in the scenario does have low serum protein and low albumin levels, Indian childhood cirrhosis is not associated with a swollen belly and dull face as described.

Option D. Kawasaki disease: Kawasaki disease is an acute febrile illness that primarily affects children under the age of 5. It involves inflammation of blood vessels throughout the body, leading to various symptoms such as high fever, rash, red eyes, swollen lymph nodes, and changes in the extremities. However, it is not associated with low serum protein and low albumin levels or a swollen belly.

Solution for Question 6:

Correct Option C - Urine calcium: creatinine > 15mg/kg

- Incorrect: The normal urine calcium: creatinine ratio in children is below 0.2 mg/kg. In nutritional rickets, the urine calcium: creatinine ratio is typically low, not high.

Other Options:

Option A - Low phosphate:

- Correct: Nutritional rickets are associated with low serum phosphate levels.

Option B - Normal calcium:

- Correct: Serum calcium levels in nutritional rickets can be normal or low. Initially, the patient is likely to have hypocalcemia, which is then compensated by increased secretion of PTH.

Option D - High PTH:

- Correct: Elevated PTH (parathyroid hormone) levels are a hallmark of nutritional rickets.

Solution for Question 7:

Correct Option B - Vitamin C deficiency:

- Vitamin C deficiency (scurvy) is known for causing anemia, bleeding gums, and skin changes.
- Collagen is the backbone of blood vessels, in scurvy due to improper collagen synthesis, blood vessels become fragile resulting in bleeding Gum bleeding Painful pseudo paralysis of lower limbs, so there is crying on touching Subperiosteal hemorrhage of long bones. Scorbutic rosary i.e., prominence of the costochondral junction which is sharper, tender, and angulated as compared to rachitic rosary.
- Gum bleeding

- Painful pseudo paralysis of lower limbs, so there is crying on touching Subperiosteal hemorrhage of long bones.
- Scorbutic rosary i.e., prominence of the costochondral junction which is sharper, tender, and angulated as compared to rachitic rosary.
- Xray show "lines of Frenkel" on X-ray.
- Gum bleeding
- Painful pseudo paralysis of lower limbs, so there is crying on touching Subperiosteal hemorrhage of long bones.
- Scorbutic rosary i.e., prominence of the costochondral junction which is sharper, tender, and angulated as compared to rachitic rosary.

Incorrect Options:

Option A - Zinc deficiency:

- Zinc deficiency can cause growth retardation, skin changes, and immune dysfunction. However, the specific feature of "lines of Frenkel" is not typically associated with zinc deficiency.

Option C - Copper deficiency:

- Copper deficiency can lead to anemia, neutropenia, and depigmentation of the hair and skin.

Option D - Iron deficiency:

- Causes pallor, easy fatiguability, iron deficiency anemia, lethargy and irritability.

Solution for Question 8:

Correct Answer: A) Height for age

Explanation:

Chronic malnutrition is best measured by height for age.

WHO Classification of Malnutrition	
Moderately underweight	Weight-for-age is between -2 and -3 SDs.
Severely underweight	Weight-for-age is < -3 SDs.

Moderately wasted	Weight-for-length/height is between -2 and -3 SDs.
Severely wasted	Weight-for-length/height is < -3 SDs.
Moderate Acute Malnutrition	Weight-for-length/height or BMI-for-age is between -2 and -3 SDs. Mid-upper arm circumference (MUAC) is between 11.5 cm and 12.5 cm.
Severe Acute Malnutrition:	Weight-for-length/height or BMI-for-age is less than -3SDs. MUAC is less than 11.5 cm. The presence of bilateral pitting oedema.
Moderately Stunted (Moderate Chronic Malnutrition)	Length/height-for-age is between -2 and -3 SDs.
Severely Stunted (Severe Chronic Malnutrition)	Length/height-for-age is < -3 SDs.

Weight for age (Option B) is used to determine whether a child is moderately or severely underweight.

Weight for height (Option C) is used to determine the severity of acute malnutrition and wasting.

Body mass index (Option D) is used to determine the severity of acute malnutrition.

Solution for Question 9:

Correct Answer: A) Uncomplicated SAM

Explanation:

- In this case, the 4-year-old child has a MUAC of 105 mm (10.5 cm), which falls below the Severe Acute Malnutrition (SAM) threshold of 115 mm (11.5 cm), indicating severe malnutrition.
- The child eagerly consumed the therapeutic food, successfully passing the appetite test.
- There is no severe edema or medical complications requiring inpatient care, and no immediate life-threatening conditions.

Based on these findings, the child is classified as having Uncomplicated SAM and can be managed with supervised home-based therapeutic feeding.

Severe Acute Malnutrition (SAM) in Children (6 to 59 months of age):

Criteria	Details
Diagnostic Criteria for SAM	
Assessment for Complications	Check for the following complications: Severe edema (+++) Low appetite (fails appetite test) Medical complications Presence of one or more danger signs (as per IMNCI guidelines)
Management Approach	Based on the presence or absence of complications: If NO complications → Uncomplicated SAM → Supervised home management If YES (complications present) → Complicated SAM → Inpatient management in a healthcare facility

Solution for Question 10:

Correct Option D - Vitamin C:

- The above-given clinical history is diagnostic of scurvy caused by a deficiency of vitamin C.
- This occurs due to improper collagen synthesis leading to fragile blood vessels resulting in bleeding.
- Management includes vitamin C supplementation of 100-200 mg daily.

Incorrect Options:

Option A - Vitamin A :

- Vitamin A deficiency presents with night blindness, and dry and scaly hyperkeratotic patches on arms, shoulders, legs, and buttocks.

Option B - Vitamin D:

- Vitamin D deficiency presents with features of rickets.

Option C - Vitamin E:

- Vitamin E deficiency presents with cerebellar and retinal disease.

Solution for Question 11:

Correct Option B - Yellow:

- In the above clinical scenario, the child with diarrhea is eager to drink and the skin pinch goes back slowly indicating some dehydration.
- According to IMNCI, some dehydration is categorized under yellow.

Assessment of Dehydration in a Child with Diarrhea

Parameters	No Dehydration	Some Dehydration	Severe Dehydration
Sensorium	Well alert	Restless, irritable	Lethargic, floppy
Eyes	Normal	Sunken	Very sunken
Tears	Present	Absent	Absent
Mucosa	Moist	Dry	Very dry
Thirst	Drinks normally, not thirsty	Thirsty, drinks eagerly	Drinks poorly/not able to drink
Skin pinch	Goes back quickly	Goes back slowly (<2 sec)	Goes back very slowly (>2 sec)

Incorrect Options:

Option A - Pink:

- Severe dehydration is categorized as pink.

Option C - Green:

- No dehydration is categorized as green

Solution for Question 12:

Correct Option B - 125 mmol/L:

- This child presented with symptoms of malnutrition. ReSoMal is the rehydration solution given for dehydrated children with malnutrition. It is given orally. ReSoMal has a glucose concentration of 125 mmol/L

Incorrect Options:

Options A, C and D: ReSoMal's glucose concentration is neither of the above values.

Solution for Question 13:

Correct option C: 60ml/kg:

- In the first 24 hours after delivery, a term baby is typically given around 60ml of milk per kilogram of body weight.
- This amount accounts for the newborn's small stomach size and the gradual increase in milk intake as the baby's digestive system continues to develop.

Incorrect options: Options A, B, and D are incorrect.

Solution for Question 14:

Correct option B: Acrodermatitis enteropathica:

- The patient's presentation with diarrhoea and mucocutaneous lesions, which started since weaning, is suggestive of acrodermatitis enteropathica.
- Acrodermatitis enteropathica is a rare autosomal recessive disorder characterized by impaired zinc absorption, leading to zinc deficiency.
- It typically manifests when breast milk or formula is replaced with a zinc-deficient diet, such as solid foods.
- The condition is characterized by diarrhea, mucocutaneous lesions, and a failure to thrive.

Incorrect options:

Option A: Wiskott-Aldrich syndrome: Wiskott-Aldrich syndrome is a primary immunodeficiency disorder characterized by recurrent infections, eczema, thrombocytopenia, and a small platelet size.

Option C: Pediatric epidermolysis bullosa: Pediatric epidermolysis bullosa is a group of inherited disorders characterized by fragile skin and mucous membranes, leading to blistering and erosions with minimal trauma.

Option D: Atopic dermatitis:

- Atopic dermatitis is a chronic, inflammatory skin condition characterized by dry, itchy skin and a rash.
- It often starts in infancy or early childhood and can be associated with a personal or family history of allergies.

Solution for Question 15:

Correct option D: Weight for age less than -3 SD:

- Weight for age less than -3 SD is not a criteria for severe acute malnutrition.
- Severe acute malnutrition: In a child between 6 months to 5 years of age, the presence of any one or more of the following: Weight for height < -3 Z-score or < 70% of expected or Mid arm circumference < 11.5 cm or Symmetric bipedal oedema of nutritional origin
- Weight for height < -3 Z-score or < 70% of expected or
- Mid arm circumference < 11.5 cm or
- Symmetric bipedal oedema of nutritional origin
- Weight for height < -3 Z-score or < 70% of expected or
- Mid arm circumference < 11.5 cm or
- Symmetric bipedal oedema of nutritional origin

Incorrect options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 16:

Correct option B: 90-120 µg:

- The recommended daily dose of iodine in a child is 90-120 µg.
- Iodine is an essential nutrient required for the synthesis of thyroid hormones.
- Adequate iodine intake is important for normal thyroid function and proper growth and development.

Incorrect options: Options A, C, and D are incorrect.

Solution for Question 17:

Correct option C: 2,00,000 IU:

- Keratomalacia is a severe condition of the cornea caused by vitamin A deficiency.
- In cases of keratomalacia, a high dose of vitamin A is required to correct the deficiency and prevent further damage.
- The recommended dose of vitamin A for the treatment of keratomalacia in children is 2,00,000 IU.
- According to WHO guidelines, children aged 12-59 months should receive 200,000 IU of vitamin A immediately, with additional doses on subsequent days as necessary.

Incorrect options: Options A, B, and D are incorrect.

Solution for Question 18:

Correct option C: Hartnup disease:

- Hartnup disease is a metabolic disorder characterized by impaired transport of certain amino acids, including tryptophan.

- Tryptophan is a precursor for niacin (vitamin B3) synthesis.
- A deficiency of niacin can lead to pellagra-like symptoms, including a rash, gastrointestinal disturbances, and neurological abnormalities.

Incorrect options: Options A, B, and D do not cause a pellagra-like rash.

Option A: Gaucher disease: Gaucher disease is a lysosomal storage disorder characterized by the accumulation of glucocerebroside in various organs.

Option B: Alkaptonuria: Alkaptonuria is a disorder of phenylalanine and tyrosine metabolism, leading to the accumulation of homogentisic acid.

Option D: Phenylketonuria: Phenylketonuria is a genetic disorder characterized by the inability to metabolize phenylalanine properly.

Solution for Question 19:

Correct option A: Kwashiorkor:

- The image showing a swollen abdomen and edema in the feet is suggestive of Kwashiorkor.
- Kwashiorkor is a type of protein-energy malnutrition that can occur in children with inadequate protein intake.
- The disease is characterized by a swollen belly, a dull face, and edema in the limbs.
- Low serum protein and albumin levels can lead to edema in Kwashiorkor.

Incorrect options:

Option B: Marasmus:

- Marasmus is caused by a deficiency in calories and energy.
- The disease is characterized by severe wasting.
- The child appears thin and has no fat.

Option C: Nephrotic syndrome: Nephrotic syndrome is charaterized by massive proteinuria, hypoalbuminemia with hyperlipidemia, and edema.

Option D: Nephritic syndrome: The nephritic syndrome is a clinical syndrome presenting as hematuria, hypertension, oliguria, and edema.

Solution for Question 20:

Correct option A: Kwashiorkor:

- The patient's presentation with a swollen abdomen and a tired-looking face with low levels of serum protein and albumin on investigation and a history of consumption of rice water as a major part of the diet are suggestive of Kwashiorkor.
- Kwashiorkor is a type of protein-energy malnutrition that can occur in children with inadequate protein intake.
- The disease is characterized by a swollen belly, a dull face, and edema in the limbs.
- Low serum protein and albumin levels can lead to edema in Kwashiorkor.

Incorrect options:

Option B: Marasmus: Marasmus is a type of protein-energy malnutrition characterized by severe calorie deficiency, which leads to muscle wasting and weight loss.

Option C: Indian childhood cirrhosis:

- Indian childhood cirrhosis is a chronic liver disease of childhood characterised by cirrhosis of the liver associated with the deposition of copper in the liver.
- It primarily affects children of 1–3 years of age and has a genetic predisposition.

Option D: Kawasaki disease: Kawasaki disease is an acute febrile illness that affects young children and causes inflammation in blood vessels.

Solution for Question 21:

Correct option B: Thiamine:

- Thiamine deficiency can lead to a condition called beriberi, which can manifest as two different types: Wet beriberi Dry beriberi
- Wet beriberi
- Dry beriberi

- Wet beriberi is characterized by cardiovascular symptoms, including cardiac edema due to heart failure.
- The cardiac edema results from the impaired pumping function of the heart.
- Thiamine deficiency can also cause neuropathy.
- Wet beriberi
- Dry beriberi

Incorrect options:

Option A: Biotin:

- A biotin deficiency is rare.
- Biotin is involved in various metabolic processes and is necessary for the metabolism of fats, carbohydrates, and proteins.

Option C: Pyridoxine:

- Pyridoxine, also known as vitamin B6, is important for the metabolism of amino acids and the synthesis of neurotransmitters.
- While severe pyridoxine deficiency can lead to neurological symptoms, it is not primarily associated with cardiac edema.

Option D: Riboflavin:

- Riboflavin, also known as vitamin B2, plays a role in energy production and cellular growth.
- Riboflavin deficiency can lead to a condition called ariboflavinosis, which primarily affects the mucous membranes and skin, causing symptoms like angular cheilitis (cracked corners of the mouth) and dermatitis.

Solution for Question 22:

Correct option D: Weight for height less than -3 SD:

- When assessing and classifying malnutrition in children, the World Health Organization (WHO) uses two primary indicators: Weight for age Weight for height
- Weight for age
- Weight for height

- These indicators are expressed as standard deviation (SD) scores, also known as Z-scores, which compare a child's measurements to the reference population.
- In the case of severe acute malnutrition (SAM), the WHO focuses on the weight for height indicator.
- A weight for height Z-score of less than -3 SD is used to define severe acute malnutrition.
- Weight for age
- Weight for height

Incorrect options: Options A, B, and C are incorrect.

Solution for Question 23:

Correct option D: Contains high amount of lactose and low amounts of protein: Colostrum is rich in proteins and low in lactose.

Colostrum:

- Colostrum is produced during the first 72 hours after the birth of the baby.
- It is a thick, yellowish-colored milk produced in small quantities that is rich in immunoglobulins, macrophages, and proteins and is low in lactose.
- It is also known as the first immunization of the baby and must be fed to all babies.

Incorrect options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 24:

Correct option B: Hemorrhagic disease of the newborn:

- Hemorrhagic disease is a bleeding problem that occurs in a baby during the first few days of life.
- Babies are normally born with low levels of vitamin K, an essential factor in blood clotting.

- A deficiency in vitamin K is the main cause of hemorrhagic disease in newborn babies.
- All neonates should receive a single dose of vitamin K (1 mg) intramuscularly at birth to prevent hemorrhagic disease in newborns.

Incorrect options:

Option A: Breast milk jaundice: Breast milk jaundice occurs due to substances in breast milk that decrease bilirubin conjugation or glucuronidase action.

Option C: Keratomalacia: Keratomalacia is an eye disorder that involves drying and clouding of the cornea due to vitamin A deficiency.

Option D: Scurvy: Scurvy is due to vitamin C deficiency.

Solution for Question 25:

Correct option B: Classic galactosemia in the infant: Absolute contraindications for breastfeeding are seen:

- If the mother is on radiotherapy or chemotherapy
- If the child suffers from galactosemia or lactose intolerance

Incorrect options: Options A, C, and D are not an absolute contraindication for breastfeeding.

Solution for Question 26:

Correct option D: < 11.5 cm:

- Mid Arm Circumference (MAC) / Mid Upper Arm Circumference (MUAC) is the circumference of the right upper arm measured at the midpoint between the tip of the shoulder and the tip of the elbow (olecranon process and the acromium). Red: < 11.5 cm = Severe acute malnutrition Yellow: 11.5-12.5 cm = Under nourished Green: >12.5 cm = Normal
- Red: < 11.5 cm = Severe acute malnutrition

- Yellow: 11.5-12.5 cm = Under nourished
- Green: >12.5 cm = Normal
- Red: < 11.5 cm = Severe acute malnutrition
- Yellow: 11.5-12.5 cm = Under nourished
- Green: >12.5 cm = Normal

Incorrect options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 27:

Correct option A: < 10%:

- According to the current recommended dietary guidelines for children, the intake of saturated fats should be less than 10% of total energy intake.
- This guideline is in place to promote a diet that is low in saturated fats and encourages the consumption of healthier sources of fat, such as unsaturated fats.

Incorrect options: Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 28:

Correct option A: Baby sleeps for 2-3 hours after feeding: Baby sleeping for 1-2 hours after feeding is a sign of adequate breastfeeding.

Incorrect options: Options B, C, and D are signs of adequate breastfeeding.

Solution for Question 29:

Correct option A: 6 months:

- Breastfeeding is exclusively recommended for the first 6 months of life.
- It ensures child health and higher survival rates.

Incorrect options: Options B, C, and D are incorrect.

Solution for Question 30:

Correct option A: Chronic malnutrition:

- The child's height for age being below -2SD suggests chronic malnutrition.
- The best indicator of chronic malnutrition is a decrease in height for age, i.e., stunting.
- It can be caused by factors such as insufficient food availability, poor dietary diversity, inadequate breastfeeding, or repeated infections that affect nutrient absorption.

Incorrect options:

Option B: Acute malnutrition:

- Acute malnutrition refers to a severe and sudden lack of adequate nutrition, often resulting from a recent significant decrease in food intake or a severe illness.
- The best indicator of acute malnutrition is a decrease in weight for height, i.e., wasting.

Option C: Recent infection: While infections can temporarily affect a child's growth and nutritional status, a recent infection alone is less likely to cause a significant and sustained decrease in height for age.

Option D: No malnutrition:

- The child's height for age is below -2 SD, which indicates a deviation from the average height of healthy children.
- This suggests a form of malnutrition or an underlying health condition affecting growth.

Solution for Question 31:

Correct option D: Human donor milk:

AIIMS protocols - Feeding of LBW infants: All LBW infants, irrespective of their initial feeding method, should receive ONLY breast milk. This can be ensured even in infants who are fed by paladai or gastric tube by giving expressed breast milk (mother's own milk or human donor milk)

Incorrect options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 32:

Correct option A: Kwashiorkor due to protein malnutrition:

- The image showing a swollen abdomen and a tired-looking face is suggestive of Kwashiorkor.
- Kwashiorkor is a type of protein-energy malnutrition that can occur in children with inadequate protein intake.
- The disease is characterized by a swollen belly, a dull face, and edema in the limbs.
- Low serum protein and albumin levels can lead to edema in Kwashiorkor.

Incorrect options: Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

- Marasmus is caused by a deficiency in calories and energy.
- The disease is characterized by severe wasting.
- The child appears thin and has no fat.

Solution for Question 33:

Correct option D: Ringer's lactate with 5% dextrose: WHO recommends ringer lactate with 5% dextrose as preferred for treating shock in a child suffering from severe acute malnutrition.

Incorrect options: Options A, B, and C are not incorrect.

Solution for Question 34:

Correct option D: Vitamin K:

- Breast milk is relatively low in vitamin K, which is essential for blood clotting.
- Newborns are particularly susceptible to vitamin K deficiency, which can lead to bleeding disorders such as hemorrhagic disease of the newborn.
- It is common practice to administer a vitamin K injection to newborns shortly after birth to prevent deficiency.
- Breast milk contains adequate amounts of all vitamins except vitamin D, vitamin K, and vitamin B12.

Incorrect options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Normal Development, Developmental & Behavioral Disorders

1. A child has an IQ of 65 and is able to perform basic self-care tasks with some supervision.The boy is struggling academically and has limited vocabulary. Which level of intellectual disability is most likely?

A. Mild

B. Moderate

C. Severe

D. Profound

2. A 7-year-old child has been experiencing involuntary urination in clothes at night, despite having been toilet trained for over a year. The episodes occur only at night and have been persistent for the past six months. Which of the following is the most appropriate initial management approach for the child?

A. Initiate treatment with desmopressin as the first-line treatment.

B. Comprehensive evaluation including urine culture, renal ultrasound, and voiding studies.

C. Behavioral therapy with positive reinforcement and using a bladder alarm if needed.

D. Refer to a specialist for a psychological assessment.

3. A 10-year-old child has been diagnosed with Attention Deficit Hyperactivity Disorder (ADHD). The primary approach for managing ADHD involves several strategies. Which of the following treatment options is the most appropriate first-line approach for managing ADHD in children?

A. Immediate initiation of treatment with methylphenidate.

B. Intensive cognitive-behavioral therapy (CBT) as the sole treatment option.

C. Implementation of behavioral interventions and psychotherapy tailored to the individual, with medication if necessary.

D. Referral to a specialist for neurological evaluation before starting behavioral therapy.

4. A 7-year-old is brought to the clinic with complaints of low academic performance. He has difficulty listening in class, often makes careless mistakes, frequently interrupts others, is restless and is often fidgeting during lessons. He has had symptoms for the past year.Based on the above scenario which diagnosis is most likely ?

A. Autism Spectrum Disorder (ASD)

B. Generalized Anxiety Disorder (GAD)

C. Specific Learning Disability (SLD)

D. Attention Deficit Hyperactivity Disorder (ADHD)

5. Which of the following statements is true regarding PICA in children?

A. PICA is observed only in children who have experienced severe emotional trauma or neglect.

B. PICA is characterized by the persistent ingestion of non-nutritive substances for at least 1 month..

C. Typically occurs in children between 5-10 years.

D. Low socioeconomic status is not a risk factor.

6. A 2-year-old boy has had several episodes- of turning pale or bluish and losing consciousness for a brief period. These episodes occur when he is upset or frustrated, no other significant symptoms and development is normal. During these

events, he is unresponsive for about 30 seconds and then recovers spontaneously. What is the most appropriate initial management?

A. ECG to rule out underlying cardiac issues and reassure about the disease.

B. Reassure and educate about the disease.

C. Administration of oral iron supplements to address possible iron deficiency anemia.

D. Initiation of anticonvulsant medication to prevent further episodes.

7. A 4-year-old girl is brought to the OPD by her parents reporting that she initially met her developmental milestones but has recently begun to lose previously acquired skills. They also mention that she has started to exhibit repetitive hand movements and difficulty with motor control. On examination, you observe that she has a characteristic hand-wringing movement and a loss of purposeful hand movements. Which of the following is the most likely diagnosis for this child?

A. Asperger syndrome

B. Rett Syndrome

C. Cerebral Palsy

D. ADHD

8. Above what age should the absence of a social smile be considered a developmental red flag?

A. 1 month

B. 2 months

C. 4 months

D. 6 months

9. A 6-month-old infant presents with hypotonia, poor feeding, constipation and developmental delays. The child was born at term with a low birth weight. Which of the following conditions is most likely to be the underlying cause?

A. Down syndrome

B. Congenital Hypothyroidism

C. Cerebral palsy

D. Fragile X syndrome

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10. A child who is 4 years old is capable of using only 2 word sentences and pronouns.What is the child's Developmental Quotient (DQ) of the child in the language domain?

A. 25

B. 50

C. 75

D. 100

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11. Which among the following developmental milestones is not developed by the age of 9 months?

A. Mature pincer grasp

B. Bisyllables

C. Unidextrous reach

D. Waves bye bye

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12. A 9-month-old watches as her mother hides her favorite toy bunny under a blanket. Instead of forgetting about it, Emma lifts the blanket to find the bunny.

What does it imply?

A. Object permanence

B. Transferring object

C. Visuomotor coordination

D. Object comparison

13. Match the following months with the milestones achieved at that age 1. 5 months a. Rolling over 2. 7 months b. Transfers objects 3. 9 months c. Waves bye-bye 4. 12 months d. 1-2 words with meaning

1. 5 months	a. Rolling over
2. 7 months	b. Transfers objects
3. 9 months	c. Waves bye-bye
4. 12 months	d. 1-2 words with meaning

A. 1-a, 2-b, 3-c, 4-d

B. 1-d, 2-c, 3-b, 4-a

C. 1-b, 2-a, 3-d, 4-c

D. 1-a, 2-c, 3-b, 4-d

14. A 3-year-old girl can jump, copy circles, share toys, and ask questions. Which of these milestones would she have achieved before 3 years?

A. Jumping

B. Copying circles

C. Sharing toys

D. Asking questions

15. A mother found her healthy baby boy sitting with the support of his hands for the first time. How old is he likely to be based on this finding ?

A. 1 year

B. 9 months

C. 6 months

D. 15 months

16. A 15-month-old baby girl can walk alone, imitate scribbling, creep upstairs, and is heard saying 1-2 words other than mama and dada. Which among these is not a developmental milestone she achieved at 15 months?

A. Walking alone

B. Imitates scribbling

C. 2 words sentences

D. Creeps upstairs

17. By which age the milestone given in the image below is achieved by the child?

A. 1 month

B. 2 months

C. 3 months

D. None of the above.

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18. Which among the following statements is false regarding the rules of development of a child?

A. Sequence of attainment of milestone is same in all children

B. Disorganized mass activity is replaced by specific actions

C. Primitive reflexes have to be lost before relevant milestone are attained

D. The process of development progresses from toe to head.

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19. During a routine pediatric check-up, a mother brings her 5 month old baby to the clinic. The pediatrician observes that the baby actively looks at his mother’s face and seems to respond with interest as she talks to him, showing recognition and engagement. When is this milestone normally achieved?

A. 2 months

B. 3 months

C. 6 months

D. 9 months

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20. During a developmental check-up, a pediatrician observes a group of toddlers at a local playgroup. The children are engaged in playing with toys independently but are situated close to one another. Each child seems focused on their own activity, occasionally glancing at the others but not interacting directly.At what age

is this type of play most commonly observed?

A. 6 months

B. 1 years

C. 2 years

D. 4 years

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	3
Question 4	4
Question 5	2
Question 6	2
Question 7	2
Question 8	4
Question 9	2
Question 10	2
Question 11	1
Question 12	1
Question 13	1
Question 14	1
Question 15	3
Question 16	3
Question 17	3
Question 18	4
Question 19	2
Question 20	3

Solution for Question 1:

Correct Answer: A) Mild

Explanation:

- IQ of 65 with ability to perform basic self care tasks with supervision is of Mild ID.

Intellectual disability (ID) is a condition characterized by significant limitations in both intellectual functioning and adaptive behavior, which covers a range of everyday social and practical skills.

- Intellectual Disability levels is assessed by Intelligence Quotient(IQ)
- IQ = Mental age/Chronological age *100

Intellectual Disability levels:

IQ Score	Degree	Characteristics
51-70	Mild ID	Can learn practical life skills- May live independently with support- Can work in jobs with minimal supervision.
36-50	Moderate ID	May develop basic communication skills- Can learn simple daily living activities- Often requires supervised living.
21-35	Severe ID	Limited communication skills- Requires extensive supervision- Little understanding of written language.
0-20	Profound ID	Very limited communication- Dependent on others for all aspects of daily care- Significant physical and sensory impairments.

Solution for Question 2:

Correct Answer: C) Behavioral therapy with positive reinforcement and using a bladder alarm if needed.

Explanation:

Nocturnal enuresis

Definition	Involuntary urination during sleep past the age of 5 years, with near-complete bladder emptying occurring at least twice a month for 3 months.
Prevalence	More common in males than females.
Types	Primary Monosymptomatic Nocturnal Enuresis: Present since birth without additional symptoms. Secondary Nocturnal Enuresis: Occurs after being dry at night for at least 6 months; may be due to UTIs, diabetes, stress, or bowel-bladder dysfunction. Polysymptomatic Enuresis: Includes lower urinary tract symptoms like hesitancy, urgency, or dribbling.
Causes	Delayed Bladder Maturation: Immaturity in nighttime bladder control. Reduced ADH Secretion: Decreased antidiuretic hormone production at night. Genetic Factors: Family history increases risk. Organic Factors: UTIs, constipation, structural urinary tract abnormalities. Sleep Disorders: Includes obstructive sleep apnea. Stress and Emotional Factors: Stressful events or emotional issues.
Management	Behavioral Modifications: Reassurance. Encourage voiding before bed. Restrict evening fluid intake (40% in the morning, 40% at noon, 20% after 6 PM). Avoid caffeinated drinks. Motivational Therapy: Use verbal praise and rewards for dry nights. Alarm Therapy: Use alarms that alert when the child starts to urinate, to condition recognition of a full bladder. Medication: Short-Term: Oral desmopressin for immediate relief. Long-Term: Oxybutynin or tolterodine to reduce bladder contractions. Imipramine is avoided due to side effects.

Initiate treatment with desmopressin as the first-line treatment (Option A).

- Desmopressin is typically reserved for cases resistant to initial behavioral interventions.

Comprehensive evaluation including urine culture, renal ultrasound, and voiding studies (Option B).

- In cases of primary monosymptomatic nocturnal enuresis, detailed investigations such as urine cultures and renal ultrasounds are usually not necessary unless there are additional signs or symptoms suggesting a possible underlying condition.

Refer to a specialist for a psychological assessment (Option D).

- Psychological stressors can contribute to enuresis, primary monosymptomatic nocturnal enuresis is not usually caused by psychological issues alone.

Solution for Question 3:

Correct Answer: C) Implementation of behavioral interventions and psychotherapy tailored to the individual, with medication if necessary.

Explanation:

ADHD management

Management Approach	Psychotherapy: The primary approach involves tailored psychotherapy for individuals and their families. Medication: Considered if psychotherapy alone is insufficient.
Behavioral Interventions	Behavioral Parent Training: Training for parents to manage ADHD-related behaviors effectively. Combined Training Interventions: Integrates multiple behavioral strategies. Behavioral Classroom Management: Techniques to manage behavior in the classroom setting. Behavioral Peer Interventions: Strategies to improve interactions with peers. Organization Training: Focuses on improving executive functions such as organization and planning.

Medication Categories	Stimulants: Boost the central nervous system, enhancing neurotransmitters like dopamine and norepinephrine. Examples: Methylphenidate (Option A), Amphetamine, Dexmethylphenidate . Alpha-2 Agonists: Used alone or with stimulants to regulate norepinephrine for managing symptoms. Examples: Clonidine,Guanfacine. Selective Norepinephrine Reuptake Inhibitors (SNRIs): Primarily for depression but also used for ADHD by increasing norepinephrine levels. Example: Atomoxetine Atypical Antidepressants: Considered if ADHD co-occurs with other conditions like depression or anxiety. Examples: Bupropion, Mirtazapine. Tricyclic Antidepressants (TCAs): Less commonly used , but may be an option if other treatments fail. Example: Imipramine
Monitoring and Adjustments	Regular medical monitoring is crucial to manage side effects and adjust dosages.

Intensive cognitive-behavioral therapy (CBT) as the sole treatment option (Option B).

- CBT can be an effective component of ADHD treatment, it is not usually the sole approach.

Referral to a specialist for neurological evaluation before starting treatment (Option D).

- Comprehensive neurological evaluation is not the standard first step in ADHD management.

Solution for Question 4:

Correct Answer: D) Attention Deficit Hyperactivity Disorder (ADHD)

Explanation:

ADHD

Prevalence	Most common neurobehavioral disorder of childhood; prevalence in India is approximately 1.3 per 1000.
Definition	Children aged 4 to 18 years with academic or behavioral problems and symptoms of inattention, hyperactivity, and impulsivity.

Diagnosis	Based on DSM-5 criteria: Onset up to age 12; symptoms must occur in at least two settings and interfere with functioning. Symptoms: Inattention Hyperactivity impulsivity
Management	Psychotherapy tailored to the individual and family. If psychotherapy is insufficient, medications like methylphenidate and atomoxetine may be used.

Autism Spectrum Disorder (ASD) (Option A):

- ASD involves impaired communication and social reciprocity, it is usually characterized by symptoms present before the age of 3, and includes stereotypic and restrictive behaviors.

Generalized Anxiety Disorder (GAD) (Option B):

- GAD primarily involves excessive anxiety and worry about various aspects of life, including academic performance

Specific Learning Disability (SLD) (Option C):

- SLD involves difficulties with academic skills such as reading (dyslexia), writing (dysgraphia), or math (dyscalculia); it does not typically include symptoms of hyperactivity and impulsivity.

Solution for Question 5:

orrect Answer: B) PICA is characterized by the persistent ingestion of non-nutritive substances for at least 1 month.

Explanation:

PICA DISORDER

Condition	Consumption of inedible or non-nutritive substances (e.g., chalk, mud, paint) for at least 1 month
Age	Typically occurs in children under 5 years (Option C)
Inappropriate	Behavior is not aligned with developmental stages or cultural practices.

Risk Factors	Malnutrition Iron deficiency anemia Low socioeconomic status (Option D) Psychosocial stress (e.g., parental neglect, maternal deprivation) Developmental delays (e.g., cerebral palsy)
Complications	Lead poisoning (from paint) Parasitic infestations (from mud)
Management	Behavioral therapy Deworming (one dose of Albendazole) Low-dose iron supplements (0.5 to 1 mg/kg/day)

PICA is most commonly observed in children who have experienced severe emotional trauma or neglect (Option A).

- Emotional trauma or neglect may contribute to PICA, but it is not the most common or primary association.
- PICA is more frequently linked to developmental disorders (e.g., autism spectrum disorder) and nutritional deficiencies (e.g., iron deficiency).

Solution for Question 6:

Correct answer: B) Reassure and educate about the disease.

Explanation:

Breath holding spells

Age and onset	Reflex event, lasts usually for about 2 minutes Begins around 6 months of age Peaks by 2 years; often resolves by 5 years
Sequence of Events	Triggered by anger, frustration, or pain ↓ Leads to crying ↓ Breath holding follows a full expiration

Types	1. Cyanotic Type Most common Increased sympathetic activity Child turns cyanotic (bluish) May develop tonic-clonic movements Recovers in a few seconds 2. Pallid Type Increased parasympathetic activity. Child becomes pale, mimicking syncope. Lasts for a few seconds. If prolonged, ECG may be done to rule out long QT syndrome.
Management	Reassure about the benign nature of the disease. During a seizure like episode: place the child laterally to avoid aspiration. Do Not pick up a child during an episode-can lead to spontaneous reduced cerebral flow.

Main differences between Breath holding and Cardiac involvement for a pallid type

	Breath-Holding Spells	Cardiac Involvement
Typical Duration	Generally lasts less than 2 minutes.	May last longer than 2 minutes (Option A)
Symptoms	Brief loss of consciousness, pallor, or cyanosis.	May include chest pain, irregular heartbeats, or prolonged symptoms.
Resolution	Resolves spontaneously as the child starts breathing again.	May require medical intervention to restore heart function.
When to Investigate	If the spell exceeds 2 minutes or has atypical symptoms, further evaluation is necessary.	Persistent or atypical symptoms should prompt investigation.
Management	Reassure and educate about the benign nature of the disease.	Manage according to the diagnosis.

Administration of oral iron supplements to address possible iron deficiency anemia (Option C).

- Breath-holding spells are not typically related to iron deficiency anemia.
- The primary management for breath-holding spells focuses on reassurance and education.

Initiation of anticonvulsant medication to prevent further episodes (Option D), anticonvulsant medications are not typically used to treat breath-holding spells.

Solution for Question 7:

Correct Answer: B) Rett Syndrome

Explanation:

The above features are suggestive of Rett syndrome.

Rett syndrome:

- Caused by mutations in the MECP2 gene with X-linked dominant inherit
- More common in girls and begins between 6-18 months of age after a period of normal development.
- Loss of previously acquired skills, such as purposeful hand movements and speech.
- Motor Abnormalities: Includes repetitive hand movements (e.g hand-wringing), gait disturbances, and motor impairments.
- Cognitive Impairment: Progressive cognitive decline, leading to severe intellectual disability.
- Communication Issues: Significant impairment in communication abilities.
- Autonomic Dysregulation: Breathing abnormalities (e.g hyperventilation or hypoventilation)
- Behavioral Symptoms: Social withdrawal and anxiety.
- Neurological Findings: Seizures are common and have acquired microcephaly.

Asperger syndrome (Option A): Asperger Syndrome typically have average to above-average intelligence and may have strong verbal skills, but they often experience challenges with social interactions and exhibit restricted or repetitive behaviors and interest.

Cerebral Palsy (Option C): Results in motor impairment and does not involve a period of normal development followed by skill loss.

ADHD (Option D): Characterized by inattention, hyperactivity, and impulsivity but does not feature the loss of acquired skills or the repetitive hand movements seen in Rett syndrome.

Solution for Question 8:

Correct Answer: D) 6 months

Explanation:

- The social smile typically emerges between 6-8 weeks of age. By 2 months, most infants should be smiling in response to social interactions, such as when a caregiver talks or smiles at them.
- The absence of a social smile above 6 months is considered a developmental red flag and may indicate delays in social-emotional development.

Developmental Red Flags:

Gross motor	Upper limit	Usual time
Sitting with support	9 months	6 months
Standing with support	12 months	9 months
Walking with support	15 months	12 months
Social	Upper limit	Usual time
Social smile	6 months	2-3 months
Waving bye-bye	12 months	9 months
Language	Upper limit	Usual time
Babbling	12 months	6 months
Single words	15-16 months	1 months
Fine motor	Upper limit	Usual time
Immature pincer grasp	12 months	9 months
Scribbling	24 months	15 months

- Beyond the upper limit is considered an abnormality.

Solution for Question 9:

orrect Answer: B) Congenital Hypothyroidism

Explanation:

The presenting features along with low birth weight suggests Congenital Hypothyroidism.

Congenital Hypothyroidism presents with features like:

- Hypotonia: Low muscle tone due to inadequate thyroid hormone levels.
- Poor Feeding: Infants with congenital hypothyroidism often have difficulty feeding, which can result in poor weight gain.
- Developmental Delays: Delay in reaching developmental milestones is common, including motor and cognitive delays.
- Congenital hypothyroidism can be associated with intrauterine growth restriction (IUGR), leading to a low birth weight.

Down Syndrome (Option A): While Down syndrome can cause developmental delays, it usually presents with characteristic features such as hypotonia and distinct facial dysmorphisms, and it is not specifically associated with low birth weight.

Cerebral Palsy (Option C): This condition often results from brain injury and might present with hypotonia, but it typically becomes apparent later and is less likely to be the cause of poor feeding and low birth weight at 6 months.

Fragile X Syndrome (Option D): This genetic disorder leads to developmental delays and behavioral issues but does not usually cause hypotonia or poor feeding in the early months of life.

Causes of developmental delay:

Antenatal Causes	Fragile X syndrome Rett syndrome Down syndrome Cortical malformations TORCH group infections Inborn errors of metabolism William syndrome Noonan syndrome
Perinatal Causes	Hypoxic ischemic encephalopathy Kernicterus Encephalitis Hypoglycemic brain injury Hypothyroidism
Postnatal Causes	B12,B1 and Iodine deficiency Lead toxicity

Solution for Question 10:

Correct Answer: B) 50

Explanation:

Developmental quotient (DQ) is a measure used in developmental psychology and pediatrics to assess a child's developmental progress compared to age-related norms.

DQ = Developmental age/Chronological age*100

- Capable of using only 2-3 word sentences and pronounce like I, You, Me is a marker of language milestone of 2 years age which is the developmental age.
- Developmental age= 2 years
- Chronological age= 4 years
- DQ= 2/4*100 = 50

Solution for Question 11:

Correct Answer: A) mature pincer grasp

Explanation:

Key Developmental Milestones				
Age	Gross Motor Milestones	Fine Motor Milestones	Social Milestones	Language Milestones
1 month				Alerts to sounds
2 months			Social smile	
3 months	Neck holding		Recognises mother, anticipates feed	Cooing
4 months				Laugh Aloud

5 months	Rolls over	Bidextrous reach		
6 months	Sits in tripod position	Unidextrous reach (Option C)	stranger anxiety(7 months)	Monosyllables
8 months	Sits without support			
9 months	Stands with support	Immature pincer grasp	Waves bye bye (Option D)	Bisyllables (Option B)
12 months	Creeps well, stands without support	Mature pincer grasp	Comes when called, plays simple ball game	1 to 2 words with meaning
15 months	Walks alone (Option A), creeps upstairs	Imitates scribbling, tower of 2 blocks	Jargon speech	
18 months	Runs, explores drawers	Scribbles, tower of 3 blocks	Copies parents in task	8 to 10 words vocabulary
2 years	Walks up and downstairs (2 feet per step), jumps	Tower of 6 blocks, copies straight line	Asks for food, drink, toilet, pulls people to show toys	2 to 3 word sentences

Solution for Question 12:

Correct Answer: A) Object permanence

Explanation:

- Object permanence is typically achieved by 9 months of age. Infants understand that objects continue to exist even when out of sight.
- They persist in searching for hidden objects.
- Younger infants look down for a dropped object and quickly give up searching if the object is not visible.

Solution for Question 13:

Correct Answer: A) 1-a, 2-b, 3-c, 4-d

Explanation:

Month	Milestone
5	Gross motor - Rolls over
6	Gross motor - Tripod sitting Fine motor - Unidextrous reach, object transfer(7 months) Social - Stranger anxiety(7 months) Language - Monosyllable
9	Gross motor - Stands with support Fine motor - Immature pincer grasp, probes with forefinger Social - Waves bye-bye Language - Bisyllables
12	Gross motor - Creeps well, walks but falls, stands without support Fine motor - Mature pincer grasp Social - Comes when called, plays simple ball game Language - 1-2 words with meaning

Solution for Question 14:

Correct Answer: A) Jumping

Explanation:

Jumping is a milestone she would have achieved at 2 years of age. Rest all are milestones of 3 years

Milestones achieved at 3 years of age:

- Gross motor - Rides tricycle, alternate feet going upstairs
- Fine motor - Tower of 9 blocks, copies circles
- Social - Shares toys, knows full name and gender
- Language - Asks questions

Solution for Question 15:

Correct Answer: C) 6 months

Explanation:

Baby sits in tripod fashion (with the supports of his own hands) at 6 months

Gross motor development milestones:

3 months	Neck holding
5 months	Rolls over
6 months	Sits in tripod fashion (sitting with own support)
8 months	Sits without support
9 months	Stands holding on (with support)
12 months	Creeps well , walks but falls; stands without support
15 months	Walks alone; creeps upstairs
18 months	Runs; explores drawers
2 years	Walks up and downstairs (2 feet/step); jumps
3 years	Rides tricycle; alternates feet going upstairs
4 years	Hops on one foot; alternates feet going downstairs

Solution for Question 16:

Correct Answer: C) 2 word sentences

Explanation:

Saying 1-2 words with meaning is achieved at 12 months, and 2 word sentences , at 2 years

Milestones achieved at 15 months:

- Gross motor - Walks alone, creeps upstairs
- Fine motor - Imitates scribbling, tower of 2 blocks
- Social - Jargon

Age	Gross Motor Domain Milestones	Fine Motor Domain Milestones	Language Domain Milestones	Social Domain Milestones
1 month	-	-	Alerts to sound	-
2 months	-	-	-	Social smile
3 months	Neck holding	-	Coos (musical vowel sounds)	Recognizes mother; anticipates feeds
4 months	-	Bidextrous reach	Laughs loud	-
5 months	Rolls over	-	-	-
6 months	Sits in tripod fashion (sitting with own support)	Unidextrous reach; transfers objects	Monosyllables (ba, da, pa), ah-goo sounds	Recognizes strangers, stranger anxiety
8 months	Sitting without support	-	-	-
9 months	Stands holding on (with support)	Immature pincer grasp; probes with forefinger	Bisyllables (mama, baba, dada)	Waves “bye-bye”
12 months	Creeps well; walks but falls; stands without support	Pincer grasp mature	1-2 words with meaning	Comes when called; plays simple ball game
15 months	Walks alone; creeps upstairs	Imitates scribbling; tower of 2 blocks	Jargon	-
18 months	Runs; explores drawers	Scribbles; tower of 3 blocks	8-10 word vocabulary	Copies parents in task (e.g., sweeping)
2 years	Walks up and downstairs (2 feet/step); jumps	Tower of 6 blocks; vertical and circular stroke	2-3 word sentences, use pronouns “I”, “me”, “you”	Asks for food, drink, toilet; pulls people to show toys
3 years	Rides tricycle; alternate feet going upstairs	Tower of 9 blocks; copies circle	Asks questions; knows full name and gender	Shares toys; knows full name and gender
4 years	Hops on one foot; alternate feet going	Copies cross; bridge with blocks	Says song or poem; tells stories	Plays cooperatively in a group; goes to

	downstairs			toilet alone
5 years	Skips	Copies triangle; gate with blocks	Asks meaning of words	Helps in household tasks, dresses, and undresses

Solution for Question 17:

Correct Answer: C) 3 months

Explanation:

The image depicts ventral suspension of the child (this co relates with neck holding milestone which also develops by 3 months)

- Method: The child is held in a prone position and then lifted from the couch, with the examiner supporting the chest and abdomen of the child with the palm of his hand.
- Different stages:- Till 4 weeks (1 month): Head flops down 6 - 8 weeks: Child momentarily holds the head in the horizontal plane 8 weeks: He can maintain the position well. 12 weeks: Lift the head above the horizontal plane
- Till 4 weeks (1 month): Head flops down
- 6 - 8 weeks: Child momentarily holds the head in the horizontal plane
- 8 weeks: He can maintain the position well.
- 12 weeks: Lift the head above the horizontal plane
- Till 4 weeks (1 month): Head flops down
- 6 - 8 weeks: Child momentarily holds the head in the horizontal plane
- 8 weeks: He can maintain the position well.
- 12 weeks: Lift the head above the horizontal plane

Solution for Question 18:

Correct Answer: D) The process of development progresses from toe to head.

Explanation:

- Development is a continuous process, starting in utero and progressing in an orderly manner until maturity.
- Development depends on the functional maturation of the nervous system.
- Maturity of the Nervous system is essential for a child to learn a particular milestone or skill.
- The sequence of attainment of milestones is the same in all children. Variation may exist in the time and manner of their attainment.
- Variation may exist in the time and manner of their attainment.
- The process of development progresses in a cephalocaudal direction. Head control proceeds trunk control.
- Head control proceeds trunk control.
- Certain primitive reflexes have to be lost before relevant milestones are attained. Eg. Asymmetric tonic neck reflex has to disappear to allow the child to turnover.
- Eg. Asymmetric tonic neck reflex has to disappear to allow the child to turnover.
- The initial disorganized mass activity is gradually replaced by specific actions. Eg. When a bright toy is shown to a 3 to 4 month old child, the child sequels loudly and moves all limbs in excitement, whereas a 3 to 4 year old may just smile and ask for it.
- Eg. When a bright toy is shown to a 3 to 4 month old child, the child sequels loudly and moves all limbs in excitement, whereas a 3 to 4 year old may just smile and ask for it.
- Variation may exist in the time and manner of their attainment.
- Head control proceeds trunk control.
- Eg. Asymmetric tonic neck reflex has to disappear to allow the child to turnover.
- Eg. When a bright toy is shown to a 3 to 4 month old child, the child sequels loudly and moves all limbs in excitement, whereas a 3 to 4 year old may just smile and ask for it.

Solution for Question 19:

Correct Answer: B) 3 months

Explanation:

Key Developmental Milestones				

Age	Gross Motor Milestones	Fine Motor Milestones	Social Milestones	Language Milestones
1 month				Alerts to sounds
2 months			Social smile	
3 months	Neck holding		Recognises mother, anticipates feed (Option B)	Cooing
4 months				Laugh Loud
5 months	Rolls over	Bidextrous reach		
6 months	Sits in tripod position	Unidextrous reach	Recognize strangers, stranger anxiety	Monosyllables
8 months	Sits without support			
9 months	Stands with support	Immature pincer grasp	Waves bye bye	Bisyllables
12 months	Creeps well, stands without support	Mature pincer grasp	Comes when called, plays simple ball game	1 to 2 words with meaning
15 months	Walks alone , creeps upstairs	Imitates scribbling, tower of 2 blocks	Jargon speech	
18 months	Runs, explores drawers	Scribbles, tower of 3 blocks	Copies parents in task	8 to 10 words vocabulary
2 years	Walks up and downstairs (2 feet per step), jumps	Tower of 6 blocks, vertical and circular stroke	Asks for food, drink, toilet, pulls people to show toys	2 to 3 word sentences
3 years	Rides tricycle, alternate feet going upstairs	Tower of 9 blocks, copies circle	Shares toys, knows full name and gender	Asks questions, tells name and gender
4 years	Hops on one foot, alternate feet going downstairs	Copies cross, bridge with blocks	Plays cooperatively in a group, goes to toilet alone	Says song or poem, tells stories
5 years		Copies triangle, gate with blocks	Helps in household tasks, dresses and undresses	Asks meaning of words

Solution for Question 20:

Correct Answer: C) 2 years

Explanation:

- Parallel play occurs in children aged 2 years, where they play side by side with similar toys but without direct interaction.
- They enjoy the presence of others but do not engage in shared activities or communication.
- This stage is important for developing social skills, as children begin to recognize others' feelings and thoughts. For example, if one child gets hurt, another may seek help, showing early signs of empathy.
- As they grow, children move from parallel to cooperative play, where they start sharing, collaborating, and learning essential social skills like empathy, negotiation, and turn-taking.

Acid-Base Balance, Fluid and Electrolyte Disorders, and IVF

1. A 5-year-old boy is brought to the emergency department with a history of profuse watery diarrhea and multiple episodes of vomiting for the past 3 days. On examination, he appears lethargic, has poor oral acceptance, and has dry mucous membranes. Laboratory tests reveal serum potassium of 2.8 mEq/L. Which of the following is the most appropriate initial treatment for this patient?

A. Oral potassium chloride solution

B. Intravenous potassium chloride at 0.5 mEq/kg/hour

C. Intravenous potassium chloride at 1.5 mEq/kg/hour

D. Potassium-sparing diuretic

2. A 10-kg child with severe dehydration requires rapid fluid resuscitation. Which of the following IV fluids is most appropriate for initial management?

A. 5% Dextrose in 0.45% Normal Saline

B. Normal Saline (0.9% NaCl)

C. Ringer's Lactate

D. Isolyte P Solution

3. Which of the following is the most appropriate cannula size for peripheral IV access in a 38-week gestational age neonate on day-3 of life?

A. 22 gauge

B. 24 gauge

C. 20 gauge

D. 18 gauge

---

4. Before performing radial artery cannulation on a 7-year-old child for continuous blood pressure monitoring during major surgery, the anesthesiologist compresses both the radial and ulnar arteries, asks the child to make a fist, then releases the ulnar artery while keeping the radial artery compressed. What is the primary purpose of this maneuver?

A. To assess the elasticity of the arterial walls

B. To evaluate collateral circulation in the hand

C. To determine the size of the radial artery

D. To check for arteriovenous malformations

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5. A 5-year-old boy weighing 16 kg presents to the ER with partial-thickness burns on his entire face, the front of his right thigh, and the front of his right leg. Estimate the volume of fluid to be replaced over the next 24 hours for initial resuscitation.

A. 1500 ml

B. 1300 ml

C. 850 ml RL

D. 850 ml NS

---

6. A 3-year-old boy is brought to the emergency department with lethargy, tachycardia, and cool extremities. His blood pressure is 70/40 mmHg, heart rate is 160 bpm, and capillary refill time is 4 seconds. Which of the following is the most appropriate initial management?

A. Administer 20 mL/kg of normal saline over 1 hour

B. Start dopamine infusion at 5 mcg/kg/min

C. Give 20 mL/kg of normal saline as a rapid bolus

D. Administer 1 mg/kg of hydrocortisone IV

---

7. A preterm newborn weighing 1200 g is admitted to the neonatal intensive care unit. What is the recommended fluid requirement for this infant on day 2 of life?

A. 80 mL/kg/day

B. 95 mL/kg/day

C. 110 mL/kg/day

D. 120 mL/kg/day

---

8. A 4-year-old boy weighing 18 kg is admitted for minor elective surgery. According to the Holliday-Segar method, what is his estimated daily fluid requirement?

A. 1300 mL

B. 1400 mL

C. 1700 mL

D. 1900 mL

---

9. A 10-year-old boy is brought to the emergency department with severe vomiting. His serum potassium level is 2.7 mEq/L. Which of the following ECG changes is most likely to be seen in this patient?

A. Peaked T waves

B. Prolonged PR interval

C. U waves

D. ST segment elevation

---

10. A 3-year-old girl with a history of recurrent urinary tract infections presents with lethargy and muscle weakness. Laboratory tests reveal a serum potassium level of 6.8 mEq/L. Which of the following is the most likely etiology of her condition?

A. Congenital adrenal hyperplasia

B. Obstructive uropathy

C. Excessive potassium intake

D. Rhabdomyolysis

---

11. Which of the following statements is false?

A. TBW constitutes 75% of body weight at the time of birth

B. Preterm neonates have lower TBW than the term neonates

C. After birth, ECF decreases and ICF increases

D. After puberty, males have slightly more total body water than females.

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12. A 6-year-old boy is brought to the emergency department with a 2-day history of headache, nausea, and confusion. He was recently treated for bacterial meningitis. Physical examination reveals no signs of dehydration or edema. Laboratory studies show: Serum sodium: 128 mEq/L Serum osmolality: 265 mOsm/kg Urine sodium: 55 mEq/L Urine osmolality: 450 mOsm/kg Blood pressure: 100/60 mmHg According to the Schwartz and Bartter Clinical Criterion, which of the following is NOT a diagnostic criterion for Syndrome of Inappropriate Antidiuretic Hormone Secretion (SIADH)?

A. Serum sodium less than 135 mEq/L

B. Urine osmolality greater than 100 mOsm/kg

C. Urine sodium more than 40 mEq/L

D. Low blood pressure

13. A 7-year-old boy presents to the emergency department with confusion and lethargy. His medical history includes cirrhosis. Physical examination reveals peripheral edema and ascites. Laboratory tests show: Serum sodium: 128 mEq/L Serum creatinine: 1.4 mg/dL Total protein: 5.5 g/dL (low) Albumin: 2.8 g/dL (low) Which of the following best describes the type of this patient's condition?

A. Hypovolemic hyponatremia

B. Euvolemic hyponatremia

C. Hypervolemic hyponatremia

D. None of the above

14. Which of the following is not likely the cause of hypernatremia in the pediatric age group?

A. Nephrogenic diabetes insipidus

B. Acute tubular necrosis

C. Inadequate breastfeeding

D. Hypothyroidism

15. A 4-year-old boy has been brought to the emergency department with severe vomiting for the past 2 days. He appears dehydrated and lethargic. Laboratory studies reveal: pH: 7.52 pCO2: 38 mmHg HCO3-: 32 mEq/L Na+: 138 mEq/L Cl-: 90

mEq/L Urinary Chloride: 10 mEq/L Which of the following is the MOST appropriate initial treatment for this patient's acid-base disorder?

A. Administer sodium bicarbonate

B. Provide oxygen therapy

C. Initiate normal saline infusion

D. Start acetazolamide

16. A 5-year-old patient presents with resistant hypertension that has not improved despite various antihypertensive treatments. Laboratory tests reveal normal levels of potassium and no significant kidney damage. The patient's family history shows a pattern of early-onset hypertension in multiple relatives. Which of the following is most likely to be the underlying cause of the patient’s hypertension?

A. An activating mutation of sodium channels in the distal nephron

B. Hyperaldosteronism due to adrenal hyperplasia

C. A loss-of-function mutation in the renin-angiotensin-aldosterone system

D. An activating mutation of sodium channels in the loop

17. Which of the following is not a feature of Chloride Responsive Metabolic Alkalosis in children?

A. It responds to fluid replacement with normal saline

B. Urinary chloride >20 meq/L

C. Decrease in ECF volume

D. Can occur in cystic fibrosis.

18. Match the following: Etiology Diagnosis a) Tissue hypoxia (Shock) i) Respiratory alkalosis b) Renal tubular acidosis ii) Metabolic Alkalosis c) Paralysis of respiratory muscles iii) Respiratory Acidosis d) Excessive use of loop diuretics iv) High anion gap metabolic acidosis e) Over Ventilation of intubated child v) Normal anion gap metabolic acidosis

Etiology	Diagnosis
a) Tissue hypoxia (Shock)	i) Respiratory alkalosis
b) Renal tubular acidosis	ii) Metabolic Alkalosis
c) Paralysis of respiratory muscles	iii) Respiratory Acidosis
d) Excessive use of loop diuretics	iv) High anion gap metabolic acidosis
e) Over Ventilation of intubated child	v) Normal anion gap metabolic acidosis

A. a-i, b-iii, c-ii, d-iv, e-v

B. a-v, b-iii, c-iv, d-ii, e-i

C. a-iv, b-v, c-ii, d-iii, e-i

D. a-iv, b-v, c-iii, d-ii, e-i

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19. Match the following: Plasma Osmolality Urine Osmolality Diagnosis 1) Increased a) Increased i) Dehydration/water deprivation 2) Decreased b) Decrease ii) Diabetic insipidus iii) Syndrome of inappropriate antidiuretic hormone secretion (SIADH)

Plasma Osmolality	Urine Osmolality	Diagnosis
1) Increased	a) Increased	i) Dehydration/water deprivation
2) Decreased	b) Decrease	ii) Diabetic insipidus
		iii) Syndrome of inappropriate antidiuretic hormone secretion (SIADH)

A. 1-a-iii

B. 2-a-ii

C. 2-b-i

D. 2-a-iii

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	2
Question 5	3
Question 6	3
Question 7	2
Question 8	2
Question 9	3
Question 10	2
Question 11	2
Question 12	4
Question 13	3
Question 14	4
Question 15	3
Question 16	1
Question 17	2
Question 18	4
Question 19	4

Solution for Question 1:

Correct Answer: B) Intravenous potassium chloride at 0.5 mEq/kg/hour

Explanation:

- Normal serum potassium level: 3.5-5.5 meq/L.
- Hypokalemia is defined as a serum potassium level below 3.5 mEq/L. Mild hypokalemia: 3 - 3.5 mEq/L Moderate hypokalemia: 2.5 - 3.0 mEq/L Severe hypokalemia: <2.5 mEq/L
- Mild hypokalemia: 3 - 3.5 mEq/L
- Moderate hypokalemia: 2.5 - 3.0 mEq/L
- Severe hypokalemia: <2.5 mEq/L
- In this case, the patient has moderate hypokalemia with symptoms of lethargy and dehydration.
- Mild hypokalemia: 3 - 3.5 mEq/L
- Moderate hypokalemia: 2.5 - 3.0 mEq/L
- Severe hypokalemia: <2.5 mEq/L

Etiology:

- Common causes of hypokalemia in children include: Gastrointestinal losses (diarrhea, vomiting) Renal losses (diuretics, renal tubular disorders) Inadequate intake Transcellular shifts (e.g., insulin administration, beta-agonists)
- Gastrointestinal losses (diarrhea, vomiting)
- Renal losses (diuretics, renal tubular disorders)
- Inadequate intake
- Transcellular shifts (e.g., insulin administration, beta-agonists)
- Gastrointestinal losses (diarrhea, vomiting)
- Renal losses (diuretics, renal tubular disorders)
- Inadequate intake
- Transcellular shifts (e.g., insulin administration, beta-agonists)

Clinical features:

- Muscle weakness
- Fatigue
- Constipation
- Cardiac arrhythmias (in severe cases)
- Polyuria and polydipsia

Treatment:

- The treatment of hypokalemia depends on its severity and the presence of symptoms.
- In this case, with moderate hypokalemia and symptoms, intravenous replacement is appropriate.

Potassium Supplementation:

Indications for IV Potassium Supplementation:	Symptomatic patients Severe hypokalemia (<2.5 mEq/L) Presence of ECG abnormalities Unable to tolerate oral feeds.
IV Potassium Chloride Supplementation:	Dosage: 0.5-1 mEq/kg per dose Administration: IV infusion over 1-2 hours Infusion Rate: Do not exceed 1 mEq/kg per hour Potassium Concentration: Maximum of 60 mEq/L for peripheral lines Maximum of 80 mEq/L for central lines Avoid using glucose-containing fluids to prevent insulin-mediated intracellular potassium shift.
Oral Potassium Supplementation:	Dosage: 2-4 mEq/kg per day, divided into 3-4 doses Common Preparations: Potassium chloride (10%; 20 mEq/15 mL) Potassium citrate (especially for renal tubular acidosis) Administration Tip: Liquid preparations can be bitter; dilute with juice or water to improve taste.

Additional Management Considerations:

- Replace ongoing potassium losses
- Ensure volume resuscitation with normal saline
- Correct any associated hypomagnesemia
- Treat underlying conditions (e.g., monogenic hypertension, Bartter syndrome, Gitelman syndrome)

Oral potassium chloride solution (Option A):

- While oral replacement is preferred for mild, asymptomatic hypokalemia, this patient has moderate hypokalemia with symptoms and poor oral intake due to diarrhea.
- Intravenous replacement is more appropriate in this situation.

Intravenous potassium chloride at 1.5 mEq/kg/hour (Option C):

- This rate exceeds the maximum recommended rate and could lead to complications such as cardiac arrhythmias.

Potassium-sparing diuretic (Option D):

- This is not an appropriate initial treatment for acute hypokalemia.

- These medications are used in specific conditions causing chronic potassium loss, such as Bartter syndrome or Gitelman syndrome.

Solution for Question 2:

Correct Answer: B) Normal Saline (0.9% NaCl)

Explanation:

- Normal Saline (0.9% NaCl) is the fluid of choice for initial rapid fluid resuscitation in children with severe dehydration or shock.
- It has the following characteristics: Isotonic solution with osmolality of 308 mOsm/L Contains 154 mEq/L of both sodium and chloride Does not contain glucose, potassium, or other electrolytes
- Isotonic solution with osmolality of 308 mOsm/L
- Contains 154 mEq/L of both sodium and chloride
- Does not contain glucose, potassium, or other electrolytes
- For a 10-kg child, the initial bolus would typically be 20 mL/kg (200 mL) given rapidly over 15-20 minutes.
- Infusion of large volumes of normal saline may be associated with hyperchloremic metabolic acidosis and associated adverse effects.
- Balanced salt solutions(RL) are physiologically similar to human plasma and contain less chloride.
- Ringer's Lactate is also an acceptable alternative, especially in cases of prolonged shock or diarrhoea, as it is a more balanced solution.
- Isotonic solution with osmolality of 308 mOsm/L
- Contains 154 mEq/L of both sodium and chloride
- Does not contain glucose, potassium, or other electrolytes

IV fluids used in pediatrics include:

1. Normal Saline (0.9% NaCl):
    - Used for initial fluid resuscitation in shock or severe dehydration
    - Also used for the correction of hypernatremia
2. Lactated Ringer's Solution:

- An alternative to Normal Saline for fluid resuscitation
- Contains electrolytes closer to plasma composition

3. 5% Dextrose in Water (D5W):
   - Used when free water is needed without additional electrolytes
   - Caution is needed as it can cause hyponatremia if overused
4. 3% Saline:
   - Used in the treatment of severe symptomatic hyponatremia
   - Requires careful monitoring due to the risk of rapid sodium correction
5. 10% Dextrose:
   - Used for treatment of hypoglycemia
6. Plasma-Lyte:
   - A balanced electrolyte solution
   - May be used for fluid resuscitation or as a maintenance fluid in some cases
7. Albumin Solutions:
   - 5% or 25% albumin may be used in specific situations like burn management or certain types of shock
8. 5% Dextrose in 0.45% Normal Saline.
   - Appropriate free water
   - Glucose to prevent hypoglycemia and ketosis
   - Sodium and chloride to maintain electrolyte balance
   - 5% Dextrose in 0.45% Normal Saline is hypotonic and not suitable for rapid fluid resuscitation (Option A)
   - While Ringer's Lactate is acceptable, Normal Saline is preferred for initial rapid resuscitation (Option C)
   - Isolyte P is hypotonic (Na 23 mEq/L) and not appropriate for rapid fluid resuscitation (Option D)

Solution for Question 3:

Correct Answer: B) 24 gauge

Explanation:

- The 24 gauge cannula is the most appropriate for a 38-week gestational age neonate as it provides a good balance between ease of insertion and adequate flow rate for most purposes in this population.

The choice of cannula size in pediatrics depends on the patient's age, size, and the purpose of the IV access.

- 26-gauge and 24-gauge cannulas are suitable for neonates, including preterm infants.
- 22-gauge cannulas are generally used for older infants. (Options A & D)
- 20-gauge cannulas are typically reserved for children. (Option C & D)

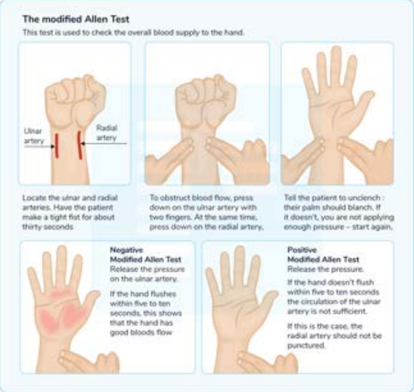
Solution for Question 4:

Correct Answer: B) To evaluate collateral circulation in the hand

Explanation:

- This maneuver, known as the Modified Allen's test, is a non-invasive procedure used primarily to evaluate the collateral circulation in the hand before radial artery cannulation. Its main purposes are: To assess the patency and adequacy of the ulnar artery To evaluate the completeness of the palmar arch
- To assess the patency and adequacy of the ulnar artery
- To evaluate the completeness of the palmar arch
- The test is performed as follows: Both ulnar and radial arteries are compressed. The child is asked to clench their fist several times to expel blood from the hand. The hand is then opened, appearing pale. Pressure on the ulnar artery is released while maintaining compression on the radial artery. If the hand flushes within 5-7 seconds, it indicates adequate collateral circulation through the ulnar artery.
- Both ulnar and radial arteries are compressed.
- The child is asked to clench their fist several times to expel blood from the hand.
- The hand is then opened, appearing pale.
- Pressure on the ulnar artery is released while maintaining compression on the radial artery.

- If the hand flushes within 5-7 seconds, it indicates adequate collateral circulation through the ulnar artery.
- To assess the patency and adequacy of the ulnar artery
- To evaluate the completeness of the palmar arch
- Both ulnar and radial arteries are compressed.
- The child is asked to clench their fist several times to expel blood from the hand.
- The hand is then opened, appearing pale.
- Pressure on the ulnar artery is released while maintaining compression on the radial artery.
- If the hand flushes within 5-7 seconds, it indicates adequate collateral circulation through the ulnar artery.



- A positive test (rapid flushing) suggests that it's safe to proceed with radial artery catheterization, as the ulnar artery can adequately perfuse the hand if the radial artery is compromised during or after the procedure.
- A negative test, where the hand fails to flush within 5-15 seconds, suggests inadequate or absent ulnar circulation.
- In this case, avoid puncturing the radial artery, as it is the sole source of arterial blood to that hand.
- The test does not assess the elasticity of arterial walls.(Option A)
- This maneuver does not determine the size of the radial artery (Option C)
- The test is not designed to check for arteriovenous malformations (Option D)

Solution for Question 5:

Correct Answer: C) 850 ml RL

Explanation:

Fluid resuscitation: Parkland formula: Ringer Lactate Volume (mL) = 4 mL × weight (kg) × % TBSA burn

- The Modified Lund & Browder chart is essential for estimating the total body surface area (TBSA) affected by burns in children.
- According to this chart:

- For a 5-year-old child: Entire face (front half of head) = 6½% Front of thigh (half of thigh) = 4% Front of the leg (half of the leg) = 2¾%
- Entire face (front half of head) = 6½%
- Front of thigh (half of thigh) = 4%
- Front of the leg (half of the leg) = 2¾%
- Entire face (front half of head) = 6½%
- Front of thigh (half of thigh) = 4%
- Front of the leg (half of the leg) = 2¾%

Therefore, the total affected area is: 6½% (face) + 4% (front of right thigh) + 2¾% (front of right leg) = 13.25% TBSA

Treatment of Burns:

Indications for inpatient care	Third-degree burns at any age Second-degree burns >10% TBSA Burns involving the face, hands, or genitals Electrical, chemical, or inhalational injuries Burn patients with concomitant trauma
Early management goals	Adequate fluid replacement Correction of hypoxia and ventilatory disturbances Prevention of hypothermia Pain and anxiety control Wound care Nutrition Supportive care
Fluid resuscitation	Parkland formula: Ringer Lactate Volume (mL) = 4 mL × weight (kg) × % TBSA burn 4 ml x 16 x 13.25% = 848 ml of RL. Half given in the first 8 hours, the remainder in the next 16 hours Monitor urine output (target >1 mL/kg/hr in infants and young children) resuscitation in burns" data-author="" data-hash="910" data-license="" data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/8xJcBRMlgrRv

	Ejv0GsJW1742028009.png" />
Wound care and topical therapy	Common agents: 0.5% silver sulfadiazine, 0.5% silver nitrate, mafenide acetate Daily dressing changes
Nutrition	High calorie and protein intake required Preferably enteral feeding when possible Parenteral nutrition for extensive burns or prolonged ileus
Pain management	Opioids commonly prescribed for adequate pain control

Infusion of large volumes of normal saline may be associated with hyperchloremic metabolic acidosis and associated adverse effects. Balanced salt solutions(RL) are physiologically similar to human plasma and contain less chloride making RL the fluid of choice for burns resuscitation (Option D).

Solution for Question 6:

Correct Answer: C) Give 20 mL/kg of normal saline as a rapid bolus

Explanation:

- Shock is a state of inadequate tissue perfusion that can result from various etiologies, including hypovolemic, cardiogenic, distributive, or obstructive causes.
- The clinical features of shock in children include tachycardia, weak peripheral pulses, prolonged capillary refill time, cool extremities, altered mental status, and hypotension (a late sign in pediatric shock).

Treatment of shock in children follows a stepwise approach:

Initial Resuscitation	Rapid administration of 10-20 mL/kg isotonic crystalloid (usually normal saline or Ringer's lactate) over 5-10 minutes. This can be repeated up to 60 mL/kg within the first hour if needed. Reassessment: After each fluid bolus, reassess for signs of improved perfusion and fluid overload.
Vasoactive Medications	If shock persists after adequate fluid resuscitation, initiate vasoactive drugs. Dopamine (5-20 mcg/kg/min) or epinephrine (0.05-0.3 mcg/kg/min) are commonly used first-line agents.

Corticosteroids	Consider hydrocortisone (2 mg/kg) in catecholamine-resistant shock or if adrenal insufficiency is suspected.
Specific Therapies	Antibiotics for suspected septic shock Blood products for hemorrhagic shock Antidotes for toxic shock
Supportive Care	Ensure adequate oxygenation and ventilation Correct electrolyte imbalances and hypoglycemia Maintain normothermia Continuous Monitoring: Vital signs, urine output, mental status, and perfusion markers should be closely monitored.

- Administering 20 mL/kg of normal saline over 1 hour is too slow for a child in shock.
- Rapid fluid resuscitation is crucial in the initial management (Option A)
- Starting dopamine infusion at this stage is premature.
- Vasoactive medications are considered after adequate fluid resuscitation and still the patient remains in shock (Option B)
- While hydrocortisone may be beneficial in certain types of shock (e.g., septic shock with suspected adrenal insufficiency), it is not the first-line treatment.
- Fluid resuscitation takes precedence (Option D)

Solution for Question 7:

Correct Answer: B) 95 mL/kg/day

Explanation:

95 mL/kg/day is the correct answer for a preterm infant weighing 1200 g on day 2 of life.

- Fluid management in neonates is crucial and varies based on factors such as gestational age, birth weight, and postnatal age.

Fluid requirements for neonates:

For neonates weighing <1500 g	For neonates weighing ≥1500 g

Day 1: 80 mL/kg/day Day 2: 95 mL/kg/day Day 3: 110 mL/kg/day Day 4: 120 mL/kg/day Day 5: 130 mL/kg/day Day 6: 140 mL/kg/day Day 7 and onwards: 150 mL/kg/day	Day 1: 60 mL/kg/day Day 2: 75 mL/kg/day Day 3: 90 mL/kg/day Day 4: 105 mL/kg/day Day 5: 120 mL/kg/day Day 6: 135 mL/kg/day Day 7 and onwards: 150 mL/kg/day

- Preterm neonates have higher total body water and may lose up to 10-15% of birth weight during the first week of life.
- Insensible water loss is higher in preterm infants due to their thin skin.

80 mL/kg/day (Option A): This is the recommended fluid requirement for day 1, not day 2.

110 mL/kg/day (Option C): This is the recommended fluid requirement for day 3.

120 mL/kg/day (Option D): This is the recommended fluid requirement for day 4.

Solution for Question 8:

Correct Answer: B) 1400 mL

Explanation:

The Holliday-Segar method is a widely used approach for calculating maintenance fluid requirements in children.

- It's based on the child's weight and provides an estimate of daily fluid needs.
- It estimates a child's energy expenditure in kilocalories and aligns it with their fluid needs.
- For every 100 kilocalories burned during metabolism, approximately 100 mL of fluid is required to replace the loss.
- The method assumes that each kilocalorie lost necessitates 1 mL of fluid for adequate replacement.

The method uses the following formula:

- 100 mL/kg for the first 10 kg
- 50 mL/kg for the next 10 kg (11 - 20 kg)
- 20 mL/kg for each kg above 20 kg

For this 18 kg child:

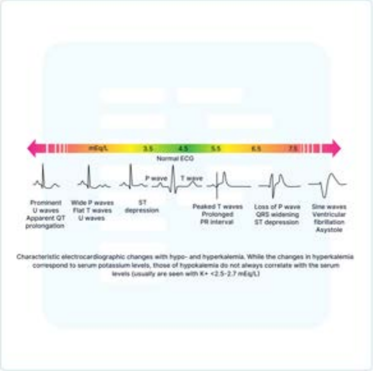
- First 10 kg: 10 x 100 mL = 1000 mL
- Next 8 kg: 8 x 50 mL = 400 mL
- Total: 1000 + 400 = 1400 mL

Solution for Question 9:

Correct Answer: C) U waves

Explanation:

- In hypokalemia, U waves become prominent and are typically best seen in leads V2-V4.
- They appear as a small positive deflection after the T wave.
- ECG changes are important indicators of electrolyte imbalances, particularly potassium disturbances.



Characteristic electrocardiographic changes with hypo- and hyperkalemia. While the changes in hyperkalemia correspond to serum potassium levels, those of hypokalemia do not always correlate with the serum levels (usually are seen with K+ <2.5-2.7 mEq/L)

Hypokalemia	Hyperkalemia
Flattened T waves ST segment depression Prominent U waves Prolonged QT interval Wide P waves	Peaked T waves (earliest and most sensitive sign) Prolonged PR interval Widened QRS complex Sine wave pattern (in severe cases) Loss of P wave

Peaked T waves (Option A):

- This is a characteristic ECG finding in hyperkalemia, not hypokalemia.
- Peaked T waves are typically narrow-based and symmetrical.

Prolonged PR interval (Option B):

- This is also a feature of hyperkalemia, occurring due to delayed conduction through the AV node.

ST-segment elevation (Option D):

- This is not typically associated with hypokalemia or hyperkalemia.
- ST-segment elevation is more commonly seen in conditions like myocardial infarction or pericarditis.

Solution for Question 10:

Correct Answer: B) Obstructive uropathy

Explanation:

- The history of recurrent urinary tract infections suggests the possibility of an underlying urological abnormality.
- Obstructive uropathy can lead to impaired renal potassium excretion, resulting in hyperkalemia.
- Normal serum potassium level: 3.5-5.5 meq/L.
- Hyperkalemia is defined as a serum potassium level above 5.5 mEq/L.

HYPERKALEMIA

Etiology:

- Decreased renal excretion (acute kidney injury, chronic kidney disease, obstructive uropathy)
- Transcellular shifts (acidosis, cell lysis, rhabdomyolysis)
- Excessive intake (rare in children with normal renal function)
- Pseudohyperkalemia (hemolysis during blood draw)
- Endocrine disorders (adrenal insufficiency, hypoaldosteronism)

Clinical features:

- Muscle weakness
- Fatigue
- Paralysis (in severe cases)
- Cardiac arrhythmias

- Often asymptomatic in mild cases

Treatment:

Prevent further progression of hyperkalemia	Discontinue Potassium-Containing Substances
Stabilize the Myocardial Membrane	To prevent cardiac arrhythmias, administer IV calcium gluconate (10% solution) at a dose of 0.5-1 mL/kg over 5-10 minutes, with continuous cardiac monitoring. If bradycardia occurs, discontinue the infusion.
Enhance Potassium Uptake into Cells:	Regular Insulin and Glucose: Administer 0.3 units of regular insulin per gram of glucose IV over 2 hours. Sodium Bicarbonate: Infuse 1-2 mEq/kg body weight of sodium bicarbonate IV over 20-30 minutes. Beta-Adrenergic Agonists: Use nebulized or IV beta-agonists (e.g., salbutamol, terbutaline) to drive potassium into cells.
Eliminate Potassium from the Body:	Sodium Polystyrene Sulfonate (Kayexalate): Administer 1 g/kg orally or via rectal enema, with a maximum dose of 15 g, mixed in 20-30% sorbitol. Diuretics: Consider loop or thiazide diuretics if renal function is adequate. Renal Replacement Therapy: For severe hyperkalemia, especially in cases of renal impairment or tumor lysis syndrome, hemodialysis or peritoneal dialysis may be required. For Hypoaldosteronism: Administer maintenance doses of steroids and fludrocortisone. For severe hyperkalemia (K+ >7.0 mEq/L) or when ECG changes (e.g., tall peaked T waves) are present, immediate treatment is necessary, starting with IV calcium gluconate, followed by sodium bicarbonate, insulin-glucose infusion, and/or nebulized beta-agonists. In refractory cases, hemodialysis may be needed. Mild hyperkalemia (5.5-6.5 mEq/L) should be managed by stopping potassium intake, discontinuing potassium-sparing drugs, and addressing the underlying cause.

Congenital adrenal hyperplasia (Option A):

- While this condition can cause electrolyte imbalances, it typically results in hypokalemia due to aldosterone deficiency.
- It's unlikely to cause hyperkalemia.

Excessive potassium intake (Option C):

- This is rarely a cause of hyperkalemia in children with normal renal function.
- The history doesn't suggest excessive intake.

Rhabdomyolysis (Option D):

- While this can cause hyperkalemia due to the release of intracellular potassium, it's typically associated with a history of trauma, severe exercise, or certain medications.
- The clinical presentation doesn't suggest rhabdomyolysis.

Solution for Question 11:

Correct Answer: B) Preterm neonates have lower TBW than the term neonates

Explanation:

Preterm neonates have higher TBW than the term neonates.

- Extremely preterm infants (<28 weeks gestation) are at particularly high risk of fluid and electrolyte imbalance due to their high TBW content and immaturity of renal function and skin barriers.
- Neonates undergo volume contraction because of high renal function at birth.

Total Body Water (TBW) Composition:

- 90% of body weight in early fetal life.
- 75% of body weight at birth. (Option A)
- 60% of body weight by the end of the first year, remaining consistent until puberty.
- After puberty, males have slightly more TBW compared to females. (Option D)

Preterm vs. Term Neonates:

- Preterm neonates have a higher TBW compared to term neonates.

TBW Distribution:

- TBW is divided into: Extracellular Fluid (ECF) Intracellular Fluid (ICF)
- Extracellular Fluid (ECF)
- Intracellular Fluid (ICF)
- The equation is: TBW = ECF + ICF ECF = plasma + interstitial fluid
- ECF = plasma + interstitial fluid
- Extracellular Fluid (ECF)
- Intracellular Fluid (ICF)

- ECF = plasma + interstitial fluid

ECF and ICF Changes:

- In fetus and newborns, ECF > ICF volume.
- After birth, ECF volume decreases and ICF volume increases. (Option C)

Solution for Question 12:

Correct Answer: D) Low blood pressure

Explanation:

- According to Schwartz and Bartter Clinical Criterion, blood pressure is not considered amongst the diagnostic criteria.

The Schwartz and Bartter Clinical Criterion for SIADH diagnosis includes:

Solution for Question 13:

Correct Answer: C) Hypervolemic hyponatremia

Explanation: The patient has signs of fluid overload (edema, ascites), indicating hypervolemic hyponatremia. Cirrhosis is one of the causes of hypervolemic hyponatremia.

Causes of Hyponatremia	
Hypovolemic (Option A)	GI/skin losses Renal losses: Intake of diuretics Underlying renal conditions: Nephronophthisis Autosomal recessive polycystic kidney disease Cerebral salt wasting syndrome: Sodium loss through urine

Euvolemic (Option B)	SIADH (syndrome of inappropriate antidiuretic hormone) Glucocorticoid deficiency Hypothyroidism
Hypervolemic	Conditions with lots of third spacing or edema Total fluid in body is increasing but fluid in intravascular compartment may not be much Specific conditions: Cardiac failure Cirrhosis of liver Nephrotic syndrome Hypoalbuminemia Renal failure

Solution for Question 14:

Correct Answer: D) Hypothyroidism

Explanation: Hypothyroidism can lead to hyponatremia due to impaired free water excretion. It does not result in hypernatremia.

Causes of Hypernatremia	
Excess Sodium Intake	Salt-rich diet IV 3% NaCl (Hypertonic saline) Sodium bicarbonate (oral or IV) Excess sodium intake Hypertonic feeds, boiled skimmed milk Ingestion of seawater
Water Deficit	Diabetes insipidus (central or nephrogenic) (Option A) Increased insensible losses Inadequate fluid intake Inadequate breastfeeding (Option C)
Sodium and Water Loss (water loss > Sodium loss)	GI losses: Vomiting or diarrhea Cutaneous losses: Burns involving large body surface area Renal losses: Polyuric phase of acute tubular necrosis (Option B) Osmotic diuresis: Free water loss Loop, osmotic diuretics; post obstructive diuresis
Hormonal	Hyperaldosteronism: More and more sodium retention in body Primary hyperaldosteronism, Cushing syndrome

Solution for Question 15:

Correct Answer: C) Initiate normal saline infusion

Explanation:

- The patient presents with metabolic alkalosis (pH > 7.45, increased HCO3-) likely due to severe vomiting.
- Also, the low urinary chloride (< 15 mEq/L) indicates a chloride-responsive metabolic alkalosis.
- The most appropriate treatment is to initiate normal saline infusion.

Acid-Base Disorder	Primary Cause	Initial Treatment	Additional Considerations
Metabolic Acidosis	Excess HCO3- loss or H+ retention	Treat underlying cause Fluid resuscitation (e.g., normal saline) Start sodium bicarbonate (Option A)	Sodium bicarbonate rarely used, only if pH < 7.1 or life-threatening Address electrolyte imbalances after initial resuscitation
Respiratory Acidosis	CO2 retention due to respiratory compromise	Assisted ventilation (NIV/IMV) Treat underlying cause	For every 10 mmHg increase in pCO2, there is 4 mEq/L rise in HCO3- (metabolic compensation)
Metabolic Alkalosis (Chloride-Responsive)	HCl loss (e.g., vomiting) Urinary chloride < 15 mEq/L	Volume repletion with normal saline	Most common type Responds to fluid and chloride replacement
Metabolic Alkalosis (Chloride-Resistant)	Various endocrine disorders Urinary chloride > 20 mEq/L	Treat underlying cause May require specific treatments based on etiology (e.g., spironolactone for Liddle syndrome) Acetazolamide	Does not respond to volume repletion alone Requires targeted therapy based on specific diagnosis
Respiratory Alkalosis	CO2 washout due to hyperventilation	Treat underlying cause (e.g., fever, sepsis, asthma) Adjust ventilator settings if over ventilated	Monitor for hypocalcemia (may cause tetany, seizures) Metabolic compensation occurs over days

Provide oxygen therapy (Option B):

- Oxygen therapy is not indicated in the treatment of metabolic alkalosis.
- The patient's primary issue is related to fluid and electrolyte imbalances due to vomiting, not a respiratory or oxygenation problem.

Start acetazolamide (Option D):

- Acetazolamide is a carbonic anhydrase inhibitor that can be used to correct metabolic alkalosis by increasing renal bicarbonate excretion.
- However, the immediate need is to address the underlying cause of the alkalosis, which in this case is due to chloride loss from vomiting.
- Normal saline is usually the initial treatment of choice.

Solution for Question 16:

Correct Answer: A) An activating mutation of sodium channels in the distal nephron

Explanation:

- The patient experienced resistant hypertension, normal potassium levels and no significant kidney damage with family history of early-onset hypertension suggesting a genetic component.
- Thus, the patient is most likely suffering from Liddle syndrome.

Liddle Syndrome:

- An autosomal dominant disorder
- Caused by an activating mutation in sodium channels in the collecting duct (distal part of nephron) (Option D)
- These sodium channels remain perpetually open
- This leads to sodium retention and resistant hypertension

Differential diagnosis of Liddle Syndrome: (Common clinical feature: Hypertension)

- An autosomal dominant disorder
- It is characterized by overproduction of aldosterone thus causing hypertension
- Treatment involves glucocorticoids, which suppress ACTH production from the pituitary
- Reduced ACTH lowers aldosterone synthase production, thereby decreasing hypertension

- Caused by a deficiency in the 11 beta-hydroxysteroid dehydrogenase enzyme
- Results in elevated cortisol levels and reduced cortisone levels in the kidney
- Elevated cortisol exhibits mineralocorticoid activity, leading to hypertension

Hyperaldosteronism from adrenal hyperplasia (Option B) would generally cause hypokalemia due to excessive potassium excretion.

A loss-of-function mutation in RAAS system (Option C) would usually result in reduced aldosterone production and potentially lower blood pressure, not resistant hypertension.

Solution for Question 17:

Correct Answer: B) Urinary chloride >20 meq/L

Explanation:

Chloride-Responsive Metabolic Alkalosis:

- Decrease in extracellular fluid (ECF) volume (Option C)
- Urinary chloride levels < 15 meq/L
- Responsive to volume replenishment with normal saline (Option A)
- Causes include: (chloride or fluid loss from the gastrointestinal tract, kidneys, or skin) Gastrointestinal losses: persistent vomiting, continuous nasogastric drainage, congenital chloride diarrhoea Renal losses: loop diuretics (e.g., furosemide), thiazide diuretics Skin losses: cystic fibrosis (Option D)
- Gastrointestinal losses: persistent vomiting, continuous nasogastric drainage, congenital chloride diarrhoea
- Renal losses: loop diuretics (e.g., furosemide), thiazide diuretics
- Skin losses: cystic fibrosis (Option D)
- Gastrointestinal losses: persistent vomiting, continuous nasogastric drainage, congenital chloride diarrhoea
- Renal losses: loop diuretics (e.g., furosemide), thiazide diuretics
- Skin losses: cystic fibrosis (Option D)

Chloride-Resistant Metabolic Alkalosis:

- Urinary chloride levels exceed 20 meq/L

- Does not respond to volume replenishment with normal saline

Chloride Resistant Metabolic Alkalosis
Normal BP Barter syndrome Gitelman syndrome
High BP Mnemonic: CLGADA C-Congenital adrenal hyperplasia(11 beta hydroxylase/17 alpha hydroxylase deficiency) L-Liddle syndrome G-GRA (Glucocorticoid remediable Aldosteronism) A-AME (Apparent mincralocorticoid exccss) D-Disorders of sexual development A-Adrenal Adenoma/carcinoma

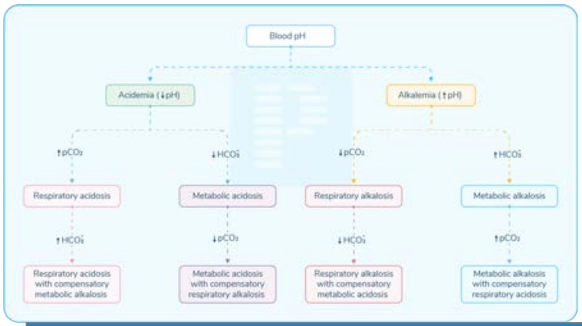
Solution for Question 18:

Correct Answer: D) a-iv, b-v, c-iii, d-ii, e-i

Explanation:

Acidosis:

- Defined as a pH level below 7.35
- Causes include: Metabolic: Resulting from an excess loss of bicarbonate (HCO■) or retention of hydrogen ions (H■) Respiratory: Due to accumulation of carbon dioxide (CO■)
- Metabolic: Resulting from an excess loss of bicarbonate (HCO■) or retention of hydrogen ions (H■)
- Respiratory: Due to accumulation of carbon dioxide (CO■)
- Metabolic: Resulting from an excess loss of bicarbonate (HCO■) or retention of hydrogen ions (H■)
- Respiratory: Due to accumulation of carbon dioxide (CO■)



Causes of Metabolic Acidosis	
Metabolic Acidosis with Normal Anion Gap (NAGMA)	Metabolic Acidosis with Increased Anion Gap (HAGMA)
Post hypocapnia Urinary tract diversions Renal tubular acidosis Diarrhea Ammonium chloride ingestion	Ketoacidosis (DKA) Kidney failure Lactic acidosis Liver failure Metformin Tissue hypoxia (shock) Poisoning by Ethylene glycol

It arises from respiratory compromise that leads to carbon dioxide (CO■) retention due to factors such as:

- Reduced central respiratory drive
- Paralysis of respiratory muscles
- Diseases affecting lung tissue or airways
- Progressive neuromuscular disorders
- Scoliosis

Alkalosis is defined as a pH > 7.45

Causes of Metabolic Alkalosis	
Chloride Responsive Metabolic Alkalosis	Chloride Resistant Metabolic Alkalosis
Gastrointestinal losses: ongoing vomiting, continuous nasogastric (NG) drainage, or congenital chloride diarrhea Renal losses: use of loop diuretics (such as furosemide) or thiazide diuretics Cutaneous losses: cystic fibrosis	Normal BP Barter syndrome Gitelman syndrome

High BP Mnemonic: CLGADA C-Congenital adrenal hyperplasia (11 beta hydroxylase/17 alpha hydroxylase deficiency) L-Liddle syndrome G-GRA (Glucocorticoid remediable Aldosteronism) A-AME (Apparent mincralocorticoid exccss) D-Disorders of sexual development A-Adrenal Adenoma/carcinoma	

IV. Causes of Respiratory Alkalosis:

- Primary Cause: Excessive loss of CO2 due to hyperventilation Elevated body temperature (high fever) Sepsis Bronchial asthma Central nervous system (CNS) disorders Overventilation in an intubated child
- Elevated body temperature (high fever)
- Sepsis
- Bronchial asthma
- Central nervous system (CNS) disorders
- Overventilation in an intubated child
- Elevated body temperature (high fever)
- Sepsis
- Bronchial asthma
- Central nervous system (CNS) disorders
- Overventilation in an intubated child

Solution for Question 19:

Correct Answer: D) 2-a-iii

Explanation:

Plasma Osmolality	Urine Osmolality	Diagnosis
1. Increased	Increased	Dehydration/water deprivation
2. Increased	Decreased	Diabetic insipidus
3. Decreased	Increased	Syndrome of inappropriate antidiuretic hormone secretion

		(SIADH)

- Osmolality refers to the concentration of solute per unit weight of solvent and is measured in mOsm/kg.

Normal Values:

- Plasma Osmolality: Ranges from 285 to 295 mOsm/kg.
- Urine Osmolality: Can reach up to 1200-1400 mOsm/kg, typically achieved by or after 1 year of age.
- Calculation of plasma osmolality = 2 [sodium] + Glucose (mg/dl)/18 + BUN/2.8

Previous Year Questions

1. A 7-year-old boy weighing 30 kg has been posted for surgical resection of a tumor. What is the daily maintenance fluid required for this child while he's NPO?

A. 1000 mL

B. 1700 mL

C. 2500 mL

D. 3000 mL

---

2. A 5-year-old child presents with symptoms of mild to moderate dehydration. You are evaluating the clinical signs to confirm the diagnosis. Which of the following clinical signs are consistent with some dehydration in a child? Decreased urine output Lethargy Delayed skin pinch Sunken eyeballs Which of the following combinations is true for dehydration in this child?

A. 1, 2 and 3

B. 4,3 and 2

C. 1,3 and 4

D. 1,2 and 4

---

3. Which of the following fluid solutions is preferred as the maintenance fluid for pediatric patients?

A. N/5 5% D

B. N 5% D

C. N/2 10% D

D. N/4 5% Dextrose

4. What is the daily fluid requirement for a 3-day-old baby with a birth weight of 1500 grams?

A. 100-110 ml/kg/day

B. 80-90 ml/kg/day

C. 120-130 ml/kg/day

D. 130-150 ml/kg/day

5. What is the most common electrolyte imbalance observed in pyloric stenosis?

A. Hyponatremia

B. Hypokalemia

C. Hypochloremia

D. Hypophosphatemia

6. A child was brought to the casualty with complaints of vomiting and loose stools with a history of laxative use. On examination, arrhythmia is present. What will be the abnormality present?

A. Hyponatremia

B. Hypocalcemia

C. Hypokalemia

D. Hyperkalemia

7. What is the most appropriate next step in managing the condition of a 7-year-old boy who is experiencing abdominal pain, vomiting, oliguria, and periorbital puffiness after undergoing chemotherapy? The investigations have shown hyperuricemia, elevated creatinine levels, and hyperkalemia.

A. Hydration

B. Probenecid

C. Allopurinol

D. Rasburicase

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8. A 10-year-old child weighing 30 kg presents with a history of loose stools for 2 days. On examination, there is severe dehydration. Laboratory investigations are as follows. What is the initial management as per ISPAD guidelines? RBS 550 mg/dL pH 7.01 Na+ 158 mEq/L Urine glucose 3+

RBS	550 mg/dL
pH	7.01
Na+	158 mEq/L
Urine glucose	3+

A. Manage ABC, NS 20 mL/kg and start insulin after 1 hour

B. Manage ABC, NS 20 mL/kg along with insulin 0.1 IU/kg/hr

C. Manage ABC, NS 10 mL/kg along with insulin 0.1 IU/kg/hr

D. Manage ABC, NS 10 mL/kg and start insulin after 1 hour

---

9. In a child presenting with infantile hypertrophic pyloric stenosis, what metabolic abnormality is commonly observed given the presence of recurrent episodes of non-bilious vomiting?

A. Hypochloremic, hypokalemic acidosis

B. Hypochloremic, hypokalemic alkalosis

C. Hypokalemic hyperchloremic alkalosis

D. Hyperchloremic hypokalemic acidosis

---

10. What is the probable reason for the ineffectiveness of the treatment in a 10-month-old child with diarrhea and low potassium levels who also exhibits muscular weakness during examination?

A. Hypomagnesemia

B. Hyponatremia

C. Hypermagnesemia

D. Hypernatremia

---

11. What is the most appropriate course of action for the treatment of a 3-year-old child weighing 12 kg who is experiencing diarrhea, exhibiting signs of thirst, drinking eagerly, lacking tears, and demonstrating slow skin pinch recovery?

A. 1200 ml RL over 12 hours

B. 600 ml RL over 6 hours

C. 900 ml ORS over 4 hours

D. 300 ml ORS per episode of loose stool

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	2
Question 4	2
Question 5	2
Question 6	3
Question 7	1
Question 8	1
Question 9	2
Question 10	1
Question 11	3

Solution for Question 1:

Correct Option B: 1700 mL:

Based on the Holliday-Segar method, the daily maintenance fluid requirement for a 7-year-old boy who weighs 30 kg can be estimated as follows:

For the first 10 kg of body weight: 100 mL/kg/day

= 10 kg x 100 mL/kg/day

= 1000 mL/day

For the next 10 kg (from 11 to 20 kg): an additional 50 mL/kg/day

= 10 kg x 50 mL/kg/day

= 500 mL/day

For the remaining 10 kg (from 21 to 30 kg): an additional 20 mL/kg/day

= 10 kg x 20 mL/kg/day

= 200 mL/day

Adding the results from the above calculations:

1000 mL/day + 500 mL/day + 200 mL/day = 1700 mL/day

Therefore, based on the Holliday-Segar method, the estimated daily maintenance fluid requirement for a 7-year-old boy weighing 30 kg would be approximately 1700 mL/day.

Incorrect Options:

Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 2:

Correct Option C - 1,3 and 4:

- Signs of some dehydration include increased heart rate and decreased urine output, indicating a depletion in intravascular space. Additional symptoms may include irritability, restlessness, sunken eyes, no tear production, dry mucous membranes, and slightly delayed skin pinch test (<2 sec).

Incorrect options:

2 - Lethargy: Lethargy is not seen in some dehydration; rather, it is a feature of severe dehydration.

Solution for Question 3:

Correct Option B - N 5% D:

- The preferred fluid for fluid correction in pediatric patients is Dextrose Normal Saline (DNS), comprising 5% dextrose and normal saline. Moreover, a maintenance dose of potassium, with 1ml of KCl per 100ml, is also added to the solution.
- DNS (Dextrose Normal Saline) is used in children as a maintenance fluid because: Prevents Hypoglycemia: Provides glucose for energy, crucial for children with higher metabolic rates. Maintains Fluid and Electrolyte Balance: Sodium chloride in DNS preserves extracellular fluid volume and prevents dehydration. Isotonic Solution: Stabilizes intravascular volume without causing fluid shifts.
- Prevents Hypoglycemia: Provides glucose for energy, crucial for children with higher metabolic rates.
- Maintains Fluid and Electrolyte Balance: Sodium chloride in DNS preserves extracellular fluid volume and prevents dehydration.
- Isotonic Solution: Stabilizes intravascular volume without causing fluid shifts.
- Prevents Hypoglycemia: Provides glucose for energy, crucial for children with higher metabolic rates.
- Maintains Fluid and Electrolyte Balance: Sodium chloride in DNS preserves extracellular fluid volume and prevents dehydration.
- Isotonic Solution: Stabilizes intravascular volume without causing fluid shifts.

Incorrect options:

Options A, C, and D: These hypotonic solutions are not preferred for maintenance therapy in children.

Solution for Question 4:

Correct Option B - 80-90 ml/kg/day:

- The fluid requirement of a baby weighing 1500gm or more is: Day 1-60ml/kg/day Day 2(increase it by 10-15 ml/kg/day) - 70-75 ml/kg/day Day 2(increase it by 10-15 ml/kg/day) - 80-90 ml/kg/day
- Day 1-60ml/kg/day
- Day 2(increase it by 10-15 ml/kg/day) - 70-75 ml/kg/day
- Day 2(increase it by 10-15 ml/kg/day) - 80-90 ml/kg/day
- Day 1-60ml/kg/day

- Day 2(increase it by 10-15 ml/kg/day) - 70-75 ml/kg/day
- Day 2(increase it by 10-15 ml/kg/day) - 80-90 ml/kg/day

Incorrect Options:

- Options A, C & D: These are not the daily fluid requirements for a three-day-old baby with a birth weight of 1500 grams.

Solution for Question 5:

Correct Option B - Hypokalemia:

- Hypokalemia is the most common metabolic disorder observed in pyloric stenosis.

Incorrect Options:

- Options A, C & D: Hyponatremia, Hypochloremia, and Hypophosphatemia are not the most common metabolic disorders associated with Pyloric stenosis

Solution for Question 6:

Correct Answer: C) Hypokalemia

Explanation

Vomiting and loose stools, along with cardiac arrhythmia, point to the diagnosis of hypokalemia.

- Normal serum potassium level: 3.5-5.5 meq/L.
- Hypokalemia is defined as a serum potassium level below 3.5 mEq/L. Mild hypokalemia: 3 - 3.5 mEq/L Moderate hypokalemia: 2.5 - 3.0 mEq/L Severe hypokalemia: <2.5 mEq/L
- Mild hypokalemia: 3 - 3.5 mEq/L
- Moderate hypokalemia: 2.5 - 3.0 mEq/L
- Severe hypokalemia: <2.5 mEq/L
- Mild hypokalemia: 3 - 3.5 mEq/L

- Moderate hypokalemia: 2.5 - 3.0 mEq/L
- Severe hypokalemia: <2.5 mEq/L

Etiology:

- Common causes of hypokalemia in children include: Gastrointestinal losses (diarrhoea, vomiting) Renal losses (diuretics, renal tubular disorders) Inadequate intake Transcellular shifts (e.g., insulin administration, beta-agonists)
- Gastrointestinal losses (diarrhoea, vomiting)
- Renal losses (diuretics, renal tubular disorders)
- Inadequate intake
- Transcellular shifts (e.g., insulin administration, beta-agonists)
- Gastrointestinal losses (diarrhoea, vomiting)
- Renal losses (diuretics, renal tubular disorders)
- Inadequate intake
- Transcellular shifts (e.g., insulin administration, beta-agonists)

Clinical features:

- Muscle weakness
- Fatigue
- Constipation
- Cardiac arrhythmias (in severe cases)
- Polyuria and polydipsia

Treatment:

- The treatment of hypokalemia depends on its severity and the presence of symptoms. In this case, with moderate hypokalemia and symptoms, intravenous replacement is appropriate
- Potassium Supplementation:

Indications for IV Potassium Supplementation:	Symptomatic patients Severe hypokalemia (<2.5 mEq/L) Presence of ECG abnormalities Unable to tolerate oral feeds.
IV Potassium Chloride Supplementation:	Dosage: 0.5-1 mEq/kg per dose Administration: IV infusion over 1-2 hours Infusion Rate: Do not exceed 1 mEq/kg per hour Potassium Concentration: Maximum of 60 mEq/L for peripheral lines Maximum of 80 mEq/L for central lines Avoid using glucose-containing fluids to prevent insulin-mediated intracellular potassium shift.

Oral Potassium Supplementation:	Dosage: 2-4 mEq/kg per day, divided into 3-4 doses Common Preparations: Potassium chloride (10%; 20 mEq/15 mL) Potassium citrate (especially for renal tubular acidosis) Administration Tip: Liquid preparations can be bitter; dilute with juice or water to improve taste.

- Additional Management Considerations: Replace ongoing potassium losses Ensure volume resuscitation with normal saline Correct any associated hypomagnesemia Treat underlying conditions (e.g., monogenic hypertension, Bartter syndrome, Gitelman syndrome)
- Replace ongoing potassium losses
- Ensure volume resuscitation with normal saline
- Correct any associated hypomagnesemia
- Treat underlying conditions (e.g., monogenic hypertension, Bartter syndrome, Gitelman syndrome)
- Replace ongoing potassium losses
- Ensure volume resuscitation with normal saline
- Correct any associated hypomagnesemia
- Treat underlying conditions (e.g., monogenic hypertension, Bartter syndrome, Gitelman syndrome)

Hyponatremia (Option A) can be caused due to GI losses but is not associated with cardiac arrhythmias.

Hypocalcemia (Option B) is not associated with arrhythmias but rather presents with prolonged QT intervals and congestive cardiac failure.

Hyperkalemia (Option D) is associated with cardiac arrhythmias but is not caused due to vomiting and diarrhoea.

Solution for Question 7:

Correct Option A - Hydration:

- The patient’s presentation with abdominal pain, vomiting, oliguria, and periorbital puffiness after undergoing chemotherapy, with investigations showing hyperuricemia, elevated creatinine levels, and hyperkalemia, is suggestive of tumor lysis syndrome.

- Initial hydration and management of electrolyte imbalances are the mainstays of tumor lysis syndrome treatment.
- Hydration increases urine production, which protects the kidneys and facilitates the removal of excess uric acid and other metabolites.

Incorrect Options:

Options B, C, and D: They can be used in the management of tumor lysis syndrome but are not the first-line treatment.

Solution for Question 8:

Correct Option A - Manage ABC, NS 20 mL/kg and start insulin after 1 hour:

The initial management for the 10-year-old child with severe dehydration with diabetic ketoacidosis, as per the International Society for Pediatric and Adolescent Diabetes (ISPAD) guidelines, is:

- Manage ABC (Airway, Breathing, Circulation): This involves ensuring a patent airway, providing oxygen if necessary, assessing and maintaining adequate breathing, and restoring circulation.
- NS 20 mL/kg: The initial fluid resuscitation in cases of severe dehydration is NS at a rate of 20 mL/kg.
- Start insulin after 1 hour: An IV insulin bolus should not be used at the start of therapy as it can precipitate shock by rapidly decreasing osmotic pressure and can exacerbate hypokalemia.

Incorrect Options:

Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 9:

Correct Option B - Hypochloremic, hypokalemic alkalosis:

- In infantile hypertrophic pyloric stenosis (IHPS), there is a thickening of the pyloric muscle, leading to the obstruction of the passage between the stomach and the small intestine.
- This obstruction results in persistent non-bilious vomiting.
- The vomiting in IHPS leads to the loss of gastric acid (HCl) causing a decrease in both chloride and hydrogen ions.
- This leads to hypochloremic metabolic alkalosis.
- Additionally, the loss of gastric fluid also leads to the loss of potassium (K+), resulting in hypokalemia.

Incorrect Options:

Options A, C, and D are incorrect. Refer to the explanation of the correct

Solution for Question 10:

Correct Option A - Hypomagnesemia:

- Concomitant magnesium deficiency leads to aggravated hypokalemia.
- Hypokalemia associated with magnesium deficiency is often refractory to treatment with K+.
- Hypomagnesemia develops from decreased intake or more commonly increased losses which could be gastrointestinal (diarrhea and vomiting) or renal.
- Symptomatic magnesium depletion (serum levels below 1.2 mg/dL) is often associated with hypokalemia, hypocalcemia and metabolic acidosis.

Incorrect Options:

Options B, C, and D do not cause refractoriness to treatment with K+ in patients with hypokalemia.

Solution for Question 11:

Correct Option C: 900 ml ORS over 4 hours:

- The patient’s presentation with signs of thirst, drinking eagerly, lacking tears, and demonstrating slow skin pinch recovery is suggestive of some dehydration.
- The treatment for patients with some dehydration is 75 ml/kg over 4 hours orally; if they cannot accept it orally, they should be given IV fluids.
- The patient’s weight is 12 kg; hence, the quantity of ORS that will be given to the patient is 12kg x 75 ml/kg, i.e.900 ml.

Incorrect Options:

Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Genetic Disorders and Syndromes

1. A 10-year-old boy presents with recurrent weakness and confusion resembling stroke-like symptoms. His mother reports episodes of fatigue and lactic acidosis. Family history shows that his maternal grandmother had similar symptoms, while his grandfather was unaffected. Which of the following statements is TRUE about the inheritance pattern of this condition?

A. The patient's siblings have a risk of inheriting the disease from their mother.

B. Mitochondrial disorders can only exhibit homoplasmy.

C. The threshold effect is related to disorders associated with homoplasmy.

D. Kearns-Sayre syndrome and Pearson syndrome do not follow this type of inheritance.

2. The image shows a child with distinctive facial features. Which of the following are the features associated with this syndrome? {{caption_text}}

A. Short stature and triangular face

B. Hypoplastic mandible cheek and ear abnormalities

C. Facial asymmetry and triangular face

D. Retracted mandible

3. A 33-year-old pregnant woman was found to have an abnormal NT scan. An amniocentesis revealed a chromosomal karyotype given below. This result most commonly occurs due to which of the following?

A. Non-disjunction in maternal meiosis

B. Non-disjunction in paternal meiosis

C. Robertsonian translocation

D. Mosaic pattern

4. A 15-year-old male presents with tall stature, long limbs, and a history of frequent joint dislocation. O/E he has a high-arched palate and aortic regurgitation is noted on auscultation. Which of the following is the common genetic abnormality associated with this condition?

A. Mutation in BRCA1 gene

B. Mutation in the FBN1 gene

C. Mutation in the PTEN gene

D. Deletion of chromosome 22

5. A 12-year-old boy presents with short stature, webbed neck, broad chest, and undescended testis. What is the likely diagnosis?

A. Turner syndrome

B. Noonan syndrome

C. Klinefelters syndrome

D. Edward syndrome

6. A newborn presents with microcephaly, cleft lip and palate, and polydactyly. Cardiac evaluation reveals a ventricular septal defect. Which of the following is most likely to be associated with this clinical presentation?

A. Trisomy 21 (Down syndrome)

B. Trisomy 18 (Edwards syndrome)

C. Trisomy 13 (Patau syndrome)

D. Turner syndrome (45, X)

7. A 14-year-old girl presents with short stature and delayed puberty. On physical examination, she has a webbed neck, wide-set nipples, and cubitus valgus. Which of the following is not typically associated with this patient's likely diagnosis?

A. Coarctation of the aorta

B. Streak gonads

C. Intellectual disability

D. Horseshoe kidney

8. A 32-year-old primigravida at 12 weeks of gestation undergoes first-trimester screening for Down syndrome. Which of the following is not typically included in this screening?

A. Maternal serum β-hCG level

B. Nuchal translucency measurement

C. Maternal serum PAPP-A level

D. Maternal serum alpha-fetoprotein (AFP) level

9. A 4-year-old child presents with global developmental delay, hypotonia, and characteristic dysmorphic facial features. Karyotype analysis confirms trisomy 21. Given the genetic basis of the condition, which of the following additional findings is least likely to be directly associated with this disorder?

A. Increased risk of early-onset Alzheimer's disease

B. Increased risk of acute lymphoblastic leukemia (ALL)

C. Increased risk of ataxia-telangiectasia

D. Increased risk of congenital heart defects

10. A 3-year-old girl presents with developmental delay, ataxic gait, and frequent inappropriate laughter. Genetic testing reveals a deletion in the 15q11-q13 region of the maternal chromosome. Which of the following conditions is most likely?

A. Disorder with maternal uniparental disomy

B. Disorder with paternal uniparental disomy

C. Beckwith-Wiedemann syndrome

D. Silver-Russell syndrome

11. Which of the following disorders manifest when only one allele on a somatic chromosome is affected?

A. Achondroplasia

B. Galactosemia

C. Mucopolysaccharidosis type II

D. Duchenne muscular dystrophy

---

12. A 13-year-old adolescent boy presents with intellectual disability, a long face, large ears, and macroorchidism. Genetic testing reveals more than 200 CGG repeats in the X-chromosome. Which of the following statements about this condition is incorrect?

A. The gene involved is the FMR-1 gene on the X-chromosome

B. Normal population carries 55-200 CGG repeats

C. Carriers may experience premature ovarian failure

D. Hyperextensible joints and mitral valve prolapse are associated features

---

13. A geneticist is reviewing various trinucleotide repeat disorders. Match the disorders with their correct trinucleotide repeats a. Huntington's disease 1. GAA b. Fragile X syndrome 2. CTG c. Friedreich's ataxia 3. CAG d. Myotonic dystrophy type 1 4. CGG

a. Huntington's disease	1. GAA
b. Fragile X syndrome	2. CTG
c. Friedreich's ataxia	3. CAG
d. Myotonic dystrophy type 1	4. CGG

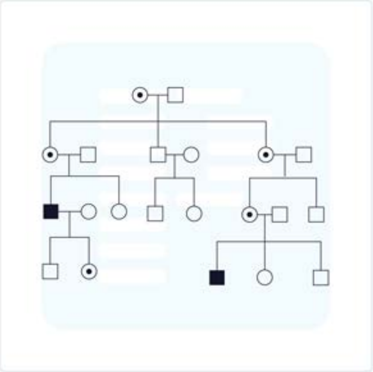
A. a-1, b-4, c-3, d-2

B. a-1, b-2, c-3, d-4

C. a-3, b-4, c-1, d-2

D. a-3, b-2, c-1, d-4

14. A genetic counselor is analyzing a family pedigree for a rare condition. The pedigree shows the following characteristics: Which of the following best describes the most likely mode of inheritance for this condition?



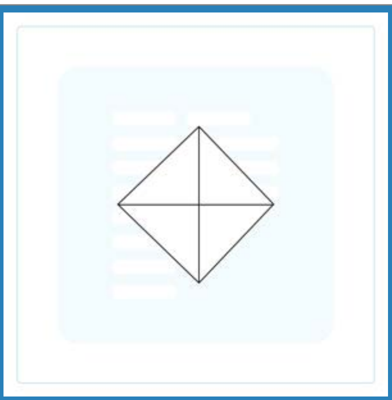
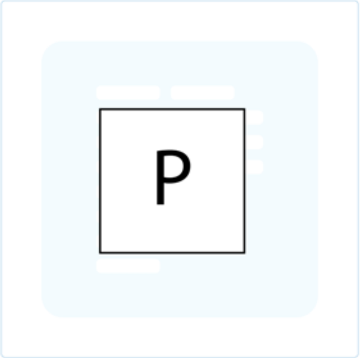
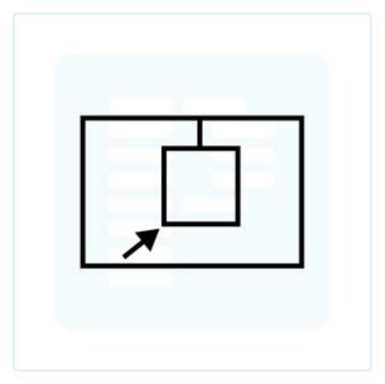
A. Autosomal recessive

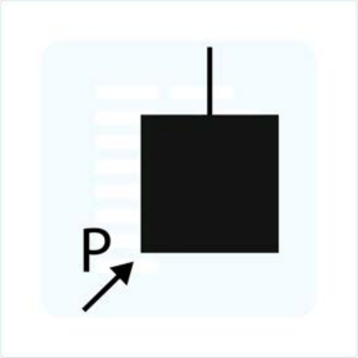
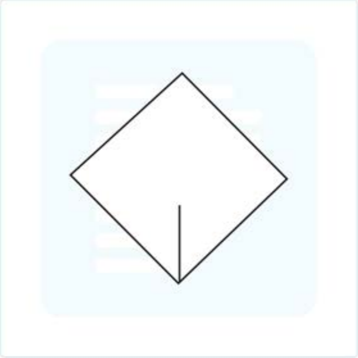
B. X-linked dominant

C. X-linked recessive

D. Mitochondrial inheritance

15. Match the symbols with their correct representation a. Proband (male) 1. b. Consultand (male) 2. c. Pregnancy (male) 3. d. Miscarriage 4. e. Stillbirth 5.

a. Proband (male)	 ... (content truncated)
b. Consultand (male)	 ... (content truncated)
c. Pregnancy (male)	 ... (content truncated)

d. Miscarriage	 ... (content truncated)
e. Stillbirth	 ... (content truncated)

A. a-2, b-3, c-4, d-1, e-5

B. a-4, b-3, c-2, d-5, e-1

C. a-2, b-3, c-4, d-5, e-1

D. a-4, b-3, c-2, d-1, e-5

16. Which of the following disorders are manifested when one allele on the X chromosome is affected?

A. Hypophosphatemic vitamin D-resistant rickets

B. Rett syndrome

C. Incontinentia pigmenti

D. All of the above

---

17. Which of the following statements is true?

A. All sons of affected males are affected by X-linked dominant disorders

B. Lyonization can cause the expression of X-linked recessive disorder in females

C. All offspring of an affected female are affected in an autosomal dominant disorder

D. None of the above

---

18. Which of the following disorders are manifested when both alleles on the X chromosome are affected?

A. Hemophilia

B. Leigh disease

C. Incontinentia pigmenti

D. MELAS syndrome

---

19. Which of the following disorders are manifested when both alleles on a somatic chromosome are affected?

A. Spinal muscular atrophy

B. Marfan syndrome

C. Neurofibromatosis

D. Colour blindness

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	2
Question 5	2
Question 6	3
Question 7	3
Question 8	4
Question 9	3
Question 10	2
Question 11	1
Question 12	2
Question 13	3
Question 14	3
Question 15	2
Question 16	4
Question 17	2
Question 18	1
Question 19	1

Solution for Question 1:

- The clinical scenario is indicative of MELAS (Mitochondrial Encephalo-myopathy, Lactic acidosis, Stroke-like episodes), which is due to mitochondrial inheritance.
- Diseases related to mitochondrial inheritance can show either homoplasmy or heteroplasmy. (Option B)
- Homoplasmy implies all mtDNA are identical which could be all wild type or all mutated. Heteroplasmy is a mixture of mutated and wild-type mtDNA.
- In the presence of heteroplasmy there is a threshold effect and clinical expression can vary between different tissues due to variation in mtDNA mutations. (Option C)

Mitochondrial Inheritance Disorders:

Inheritance Pattern	Maternal inheritance: All children of an affected female inherit the disease. Homoplasmy: Presence of mtDNA which is either all wild type or all mutated. Heteroplasmy: Presence of both wild type (normal) and mutated mtDNA in the same individual. Threshold effect: Minimum percentage of mutated mtDNA required for disease manifestation. In the presence of heteroplasmy, the threshold effect leads to variability in clinical expression across different tissues and mtDNA mutations.
Examples (Mnemonic: KLMNOP)	Kearns-Sayre syndrome- Pigmentary retinopathy, cerebellar ataxia, heart block (Option D) Leber's hereditary optic neuropathy- Subacute painless bilateral visual failure MELAS- Mitochondrial Encephalomyopathy, Lactic acidosis, Stroke-like episodes MERRF- Myoclonic epilepsy, Seizures, Ataxia NARP- Neuropathy, Ataxia, Retinitis pigmentosa CPEO- Chronic progressive external ophthalmoplegia, Bilateral Ptosis Pearson syndrome- Sideroblastic anemia of childhood, pancytopenia, pancreatic involvement (Option D)

Solution for Question 2:

The given image is suggestive of Treacher-Collins syndrome.

Treacher- collins syndrome

- Autosomal dominant
- Facial appearance: downward sloping palpebral fissures, colobomas and drooping of the lower eyelids, sunken cheekbones, deformed pinnae, atypical hair growth extending towards cheeks.
- Hypoplastic mandible
- Facial clefts, malformed ears, deafness

Pierre-Robin syndrome:

- Autosomal dominant
- Clinical features: Micrognathia accompanied by a high-arched or cleft palate Retracted mandible (Option D) Normal sized tongue Retraction of tongue (Glossoptosis)
- Micrognathia accompanied by a high-arched or cleft palate
- Retracted mandible (Option D)
- Normal sized tongue
- Retraction of tongue (Glossoptosis)
- Micrognathia accompanied by a high-arched or cleft palate
- Retracted mandible (Option D)
- Normal sized tongue
- Retraction of tongue (Glossoptosis)

Russell-Silver syndrome:

- Clinical features: Short stature, triangular face, microcephaly, limb and facial asymmetry, 5th finger clinodactyly (Options A and C).

Solution for Question 3:

- The chromosomal karyotype of 47, XX, +21 is suggestive of Down's syndrome which occurs due to Non-disjunction in maternal meiosis.
- Non-disjunction points to the failure of the homologous chromosome to separate during meiotic division.

Mechanism:

- Trisomy 21 (meiotic material nondisjunction 95%)
- obertsonian translocation (4%) - involves 2 acrocentric chromosomes (chromosomes 13, 14, 15, 21, and 22) that fuse near the centromeric region, with a subsequent loss of the short arms. (Option C)
- Mosaic pattern (1%) (Option D)

Solution for Question 4:

Marfan syndrome is commonly associated with mutations in the FBN1 gene, which encodes the connective protein fibrillin.

- Autosomal dominant inheritance.

- Multisystem disorder with involvement of the connective tissues.

Clinical features:

Skeletal abnormalities:	Tall stature Arachnodactyly Chest wall abnormalities like pectus excavatum/Pectus carinatum Thumb sign(On folding thumb, it will protrude beyond the ulnar border of palm) or wrist sign is positive Increased joint laxity
CVS abnormalities:	Aortic root dilation Mitral insufficiencies Arrhythmias
Eye abnormalities:	Ectopia lentis or subluxation of the lens of the eye Blindness
Pulmonary involvement:	Increased risk of pneumothorax

Diagnosis:

- Marfan syndrome is clinically diagnosed using Ghent criteria

Mutation in BRCA1 gene (Option A) is associated with an increased risk of breast and ovarian cancer, not with the symptoms described in this case.

Mutation in the PTEN gene (Option C) is linked to PTEN hamartoma tumor syndrome, which includes Cowden syndrome, characterized by multiple hamartomas and an increased risk of certain cancers, not the features presented here.

Deletion of chromosome 22 (Option D) is associated with DiGeorge syndrome, which typically presents with congenital heart defects, cleft palate, and immune deficiencies, but not with the specific physical features described in this scenario.

Solution for Question 5:

The adolescent presenting with short stature, webbed neck, broad chest, and undescended testis suggests Noonan syndrome.

Noonan syndrome:

- Autosomal dominant inheritance

- It is one of the disorders of the group called RASopathies; the RAS MAP kinase pathway is involved.
- The most common gene involved: is the PTPN 11 gene
- There is microdeletion in the PTPN 11 gene; the deletion is so small that it cannot be seen in the karyotype.
- So, karyotype is usually normal in these individuals.

Turner syndrome:

- Basic defect: 45, XO
- (Option A) Always seen in females
- Most common congenital heart disease: Bicuspid aortic value> Coarctation of the aorta
- In Turner's syndrome, the Barr body is absent

Similarities of Noonan syndrome and Turner syndrome:

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Differences between Noonan syndrome and Turner syndrome:

Noonan Syndrome	Turner Syndrome
Autosomal dominant inheritance Karyotype- normal(Microscopic deletion) Seen in both boys and girls Intellectual disability present Delayed puberty but fertility preserved	Do not follow any inheritance pattern Karyotype-45,XO Seen only in girls Normal intelligence Infertile(Streak ovaries and rudimentary uterus)

Noonan syndrome and Turner syndrome" data-author="" data-hash="928" data-license="" data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/BNCWoaGPmLC5CENPrZki1742031037.png" />

Edward’s syndrome (Option C):

- Trisomy 18
- Clinical features include Facial dysmorphism, index finger overlapping other fingers, and rocker bottom feet

Klinefelter’s syndrome (Option D):

- Results from meiotic nondisjunction of sex chromosomes.
- Karyotype: 47, XXY
- Clinical features: Disproportionately long arms and legs Gynecomastia Testicular dysgenesis Infertility/azoospermia

- Disproportionately long arms and legs
- Gynecomastia
- Testicular dysgenesis

- Disproportionately long arms and legs
- Gynecomastia
- Testicular dysgenesis

Solution for Question 6:

The clinical presentation described in the question is most consistent with Trisomy 13, also known as Patau syndrome.

Clinical features of Trisomy 13 and Trisomy 18:

Feature	Trisomy 18 (Edwards Syndrome)	Trisomy 13 (Patau Syndrome)
Prenatal Findings	Intrauterine growth restriction (IUGR), polyhydramnios, single umbilical artery	IUGR, holoprosencephaly, omphalocele, polyhydramnios
Craniofacial Features	Micrognathia, low-set ears, prominent occiput	Cleft lip/palate, microphthalmia, holoprosencephaly, scalp defects
Extremities	Clenched hands with overlapping fingers, rocker-bottom feet	Polydactyly, clenched hands, rocker-bottom feet
Cardiac Anomalies	Ventricular septal defect (VSD), patent ductus arteriosus (PDA), polyvalvular dysplasia	VSD, atrial septal defect (ASD), PDA, dextrocardia
Gastrointestinal Abnormalities	Omphalocele, malrotation	Omphalocele, umbilical hernia, imperforate anus
Genitourinary Anomalies	Horseshoe kidney, hydronephrosis	Polycystic kidneys, cryptorchidism, abnormal genitalia
Neurological Abnormalities	Severe intellectual disability, seizures, microcephaly	Severe intellectual disability, seizures, microcephaly, holoprosencephaly
Survival	Median survival is 3-14 days; <10% survive beyond the first year	Median survival is 2.5 days; <10% survive beyond the first year

Trisomy 21 (Down syndrome) (Option A):

- While Down syndrome can present with cardiac defects, it typically doesn't include microcephaly, cleft lip/palate, or polydactyly.
- Common features include upslanting palpebral fissures, flat facial profile, and hypotonia.

Trisomy 18 (Edwards syndrome) (Option B):

- Trisomy 18 can present with similar cardiac defects and occasionally cleft lip/palate, but microcephaly and polydactyly are more characteristic of Trisomy 13.

Turner syndrome (45, X) (Option D)

- Turner syndrome affects females and is characterized by short stature, webbed neck, and gonadal dysgenesis.
- It doesn't typically present with the features mentioned in the question.

Solution for Question 7:

- The clinical presentation described in the question is highly suggestive of Turner syndrome, a chromosomal disorder characterized by the complete or partial absence of one X chromosome in females (45, X, or mosaic karyotypes).
- Intellectual disability is NOT typically associated with Turner syndrome, but it is associated with Noonan syndrome. While some individuals may have specific learning difficulties (particularly in spatial-temporal processing and mathematics), general intellectual ability is usually normal.

Clinical Features of Turner Syndrome:

- Short Stature: A prominent feature, often becoming noticeable early in childhood.
- Skeletal Abnormalities: Short Fourth Metacarpal: A shortened bone in the hand. Cubitus Valgus: Increased carrying angle of the arms.
- Short Fourth Metacarpal: A shortened bone in the hand.
- Cubitus Valgus: Increased carrying angle of the arms.
- Auditory Issues: Sensorineural Hearing Loss.
- Sensorineural Hearing Loss.
- Gonadal Dysgenesis: Primary Amenorrhea: Due to underdeveloped (streak) ovaries and a rudimentary uterus, leading to infertility. (Option B) Absent Barr Body: Seen in cytogenetic analysis, indicative of Turner syndrome.

- Primary Amenorrhea: Due to underdeveloped (streak) ovaries and a rudimentary uterus, leading to infertility. (Option B)
- Absent Barr Body: Seen in cytogenetic analysis, indicative of Turner syndrome.
- Cardiovascular Abnormalities: Coarctation of the Aorta (Option A), bicuspid aortic valve, hypertension - May develop over time due to cardiovascular abnormalities.
- Coarctation of the Aorta (Option A), bicuspid aortic valve, hypertension - May develop over time due to cardiovascular abnormalities.
- Lymphatic System Abnormalities: Cystic Hygroma: Present at birth, leading to a characteristic webbed neck. Lymphedema: Swelling of the hands and feet, especially noticeable in infancy.
- Cystic Hygroma: Present at birth, leading to a characteristic webbed neck.
- Lymphedema: Swelling of the hands and feet, especially noticeable in infancy.
- Distinctive Physical Features: Webbed Neck Low Posterior Hairline Shield-Shaped Chest High-Arched Palate Micrognathia Epicanthal Folds
- Webbed Neck
- Low Posterior Hairline
- Shield-Shaped Chest
- High-Arched Palate
- Micrognathia
- Epicanthal Folds
- Endocrine Disorders: Hypothyroidism: Common, requiring regular monitoring and treatment. Diabetes Mellitus: Increased risk for type 2 diabetes. Osteoporosis: Increased risk due to estrogen deficiency.
- Hypothyroidism: Common, requiring regular monitoring and treatment.
- Diabetes Mellitus: Increased risk for type 2 diabetes.
- Osteoporosis: Increased risk due to estrogen deficiency.
- Renal Abnormalities: Horseshoe Kidney: A fusion of the kidneys across the midline of the body. (Option D) Congenital Malformations: Various other structural abnormalities may be present.
- Horseshoe Kidney: A fusion of the kidneys across the midline of the body. (Option D)
- Congenital Malformations: Various other structural abnormalities may be present.
- Gastrointestinal Issues: Increased Risk of Inflammatory Bowel Disease (IBD).
- Increased Risk of Inflammatory Bowel Disease (IBD).
- Vision and Ocular Issues: Strabismus, ptosis, refractive errors.
- Strabismus, ptosis, refractive errors.

- Developmental and Cognitive Features: Normal Intelligence: Cognitive abilities are generally within the normal range, in contrast to the Noonan syndrome, though some may have specific learning difficulties, particularly in spatial reasoning and mathematics. (Option C) Social Challenges: Some may experience difficulties with social interactions.
- Normal Intelligence: Cognitive abilities are generally within the normal range, in contrast to the Noonan syndrome, though some may have specific learning difficulties, particularly in spatial reasoning and mathematics. (Option C)
- Social Challenges: Some may experience difficulties with social interactions.
- Short Fourth Metacarpal: A shortened bone in the hand.
- Cubitus Valgus: Increased carrying angle of the arms.
- Sensorineural Hearing Loss.
- Primary Amenorrhea: Due to underdeveloped (streak) ovaries and a rudimentary uterus, leading to infertility. (Option B)
- Absent Barr Body: Seen in cytogenetic analysis, indicative of Turner syndrome.
- Coarctation of the Aorta (Option A), bicuspid aortic valve, hypertension - May develop over time due to cardiovascular abnormalities.
- Cystic Hygroma: Present at birth, leading to a characteristic webbed neck.
- Lymphedema: Swelling of the hands and feet, especially noticeable in infancy.
- Webbed Neck
- Low Posterior Hairline
- Shield-Shaped Chest
- High-Arched Palate
- Micrognathia
- Epicanthal Folds
- Hypothyroidism: Common, requiring regular monitoring and treatment.
- Diabetes Mellitus: Increased risk for type 2 diabetes.
- Osteoporosis: Increased risk due to estrogen deficiency.
- Horseshoe Kidney: A fusion of the kidneys across the midline of the body. (Option D)
- Congenital Malformations: Various other structural abnormalities may be present.
- Increased Risk of Inflammatory Bowel Disease (IBD).
- Strabismus, ptosis, refractive errors.
- Normal Intelligence: Cognitive abilities are generally within the normal range, in contrast to the Noonan syndrome, though some may have specific learning difficulties, particularly in spatial reasoning and mathematics. (Option C)
- Social Challenges: Some may experience difficulties with social interactions.

Differences between Noonan syndrome and Turner syndrome:

Noonan Syndrome	Turner Syndrome
Autosomal dominant inheritance	Do not follow any inheritance pattern
Karyotype- normal(Microscopic deletion)	Karyotype-45,XO
Seen in both boys and girls	Seen only in girls
Intellectual disability present	Normal intelligence
Delayed puberty but fertility preserved	Infertile(Streak ovaries and rudimentary uterus)

Solution for Question 8:

- Maternal serum alpha-fetoprotein (AFP) level is not typically included in first-trimester screening.
- AFP is part of the second-trimester quad screen.

Diagnosis of Down syndrome:

Definitive diagnosis requires chromosomal analysis, which can be done prenatally or postnatally.

1. Prenatal diagnostic tests:
    - Chorionic Villus Sampling (CVS): Usually performed between 10-13 weeks
    - Amniocentesis: Typically performed after 15 weeks
2. Postnatal diagnosis:
    - Physical examination showing characteristic features
    - Confirmed by karyotype analysis

Antenatal Screening and Diagnosis:

1. Confirmatory Test:
    - Fetal Karyotype Analysis is the gold standard for the prenatal diagnosis of Down syndrome, confirming the presence of an extra copy of chromosome 21. Chorionic Villous Sampling (CVS), Amniocentesis, Cordocentesis (Percutaneous Umbilical Blood Sampling).
    - Chorionic Villous Sampling (CVS), Amniocentesis, Cordocentesis (Percutaneous Umbilical Blood Sampling).
    - Chorionic Villous Sampling (CVS), Amniocentesis, Cordocentesis (Percutaneous Umbilical Blood Sampling).
2. Radiological Markers:

  - First Trimester: Increased Nuchal Translucency (NT): An NT thickness greater than 3 mm, measured between 11-14 weeks, is a significant marker for Down syndrome. (Option B)
  - Increased Nuchal Translucency (NT): An NT thickness greater than 3 mm, measured between 11-14 weeks, is a significant marker for Down syndrome. (Option B)
  - Second Trimester: Soft Markers: Include shortened femur length, hypoplastic nasal bones, cardiac anomalies, and duodenal atresia.
  - Soft Markers: Include shortened femur length, hypoplastic nasal bones, cardiac anomalies, and duodenal atresia.
  - Increased Nuchal Translucency (NT): An NT thickness greater than 3 mm, measured between 11-14 weeks, is a significant marker for Down syndrome. (Option B)
  - Soft Markers: Include shortened femur length, hypoplastic nasal bones, cardiac anomalies, and duodenal atresia.

3. Biochemical Markers:
   - First Trimester (T1): β-HCG & PAPP-A: Elevated β-HCG and low PAPP-A levels are suggestive of Down syndrome. (Option A and C)
   - β-HCG & PAPP-A: Elevated β-HCG and low PAPP-A levels are suggestive of Down syndrome. (Option A and C)
   - Second Trimester (T2): Triple Test: Measures β-HCG, unconjugated estriol, and α-fetoprotein. Quadruple Test: Adds Inhibin-A to the triple test markers. Elevated Markers in Down Syndrome: Increased β-HCG and Inhibin-A levels are characteristic.
   - Triple Test: Measures β-HCG, unconjugated estriol, and α-fetoprotein.
   - Quadruple Test: Adds Inhibin-A to the triple test markers.
   - Elevated Markers in Down Syndrome: Increased β-HCG and Inhibin-A levels are characteristic.
   - β-HCG & PAPP-A: Elevated β-HCG and low PAPP-A levels are suggestive of Down syndrome. (Option A and C)
   - Triple Test: Measures β-HCG, unconjugated estriol, and α-fetoprotein.
   - Quadruple Test: Adds Inhibin-A to the triple test markers.
   - Elevated Markers in Down Syndrome: Increased β-HCG and Inhibin-A levels are characteristic.
4. Integrated Test:
   - Combines maternal age, T1 (NT thickness, PAPP-A), and T2 (quadruple test) to calculate a composite risk for Down syndrome.
5. Non-Invasive Prenatal Screening/Test (NIPS/NIPT):

- Cell-Free Fetal DNA Testing: Analyzes fetal DNA circulating in maternal blood to detect chromosomal abnormalities like Down syndrome.

Solution for Question 9:

Down's syndrome, caused by trisomy 21, is associated with multiple systemic manifestations due to the presence of an extra copy of chromosome 21.

- This is due to meiotic nondisjunction in either maternal or paternal gamete.
- Occurs at a frequency of 1:800 to 1:1000 newborns.
- The overexpression of genes located on chromosome 21 contributes to the clinical features.

Clinical Features and Diagnosis:

General Features:

- Intellectual disability
- Flat facial profile, upward-slanting eyes with epicanthic folds, oblique palpebral fissure (visible when eyes are open).
- Small nose with flat nasal bridge
- Narrow palate, small teeth, and furrowed, protruding tongue
- Significant hypotonia
- Small, brachycephalic skull with flat occiput, and small, dysplastic ears
- Short, broad hands with clinodactyly (hypoplasia of the middle phalanx of the fifth finger)
- Simian crease on the palm
- A wide gap between the first and second toes (sandal gap)

Comorbidities

- Congenital Heart Disease (Option D): 40% of children have congenital heart defects, primarily endocardial cushion defects (40-60% of cases) - AVSD, VSD, ASD, PDA, TOF. Early cardiac evaluation is crucial.
- Early cardiac evaluation is crucial.
- Gastrointestinal Malformations: 12% have atresias, especially duodenal atresia. There is an increased risk of annular pancreas and Hirschsprung disease.
- There is an increased risk of annular pancreas and Hirschsprung disease.

- Ophthalmic Problems: Increased risk of cataracts, nystagmus, strabismus, and visual acuity issues. Regular eye evaluations are advised from infancy onward.
- Regular eye evaluations are advised from infancy onward.
- Hearing Defects: 40-60% experience conductive hearing loss, often due to serous otitis media in the first year. Routine hearing assessments are recommended before 6 months of age and annually.
- Routine hearing assessments are recommended before 6 months of age and annually.
- Thyroid Dysfunction: Hypothyroidism occurs in 13-54% of patients.
- Atlanto-occipital Subluxation: Incidence ranges from 10-30%. A lateral neck radiograph is recommended.
- Physical Growth: Linear growth is typically slower, and children may become obese with age. Muscle tone often improves, but regular monitoring of height and weight is necessary.
- Muscle tone often improves, but regular monitoring of height and weight is necessary.
- Malignancies: Increased risk of lymphoproliferative disorders, such as acute lymphoblastic leukemia (Option B), acute myeloid leukemia, myelodysplasia, and transient lymphoproliferative syndrome.
- Early-Onset Alzheimer's Disease (Option A): The APP gene, located on chromosome 21, plays a significant role in amyloid plaque formation, a hallmark of Alzheimer's disease.
- Early cardiac evaluation is crucial.
- There is an increased risk of annular pancreas and Hirschsprung disease.
- Regular eye evaluations are advised from infancy onward.
- Routine hearing assessments are recommended before 6 months of age and annually.
- Muscle tone often improves, but regular monitoring of height and weight is necessary.

Ataxia-telangiectasia is a rare autosomal recessive disorder caused by mutations in the ATM gene on chromosome 11, unrelated to trisomy 21. This condition involves progressive cerebellar ataxia, telangiectasias, and immunodeficiency.

Solution for Question 10:

- Angelman disorder is a disorder with paternal uniparental disomy.
- Genomic imprinting is an epigenetic phenomenon (alteration in DNA without any change in nucleotide sequence) where certain genes are expressed in a parent-of-origin-specific

manner.

- The maternal or paternal allele is silenced, while the other is expressed.
- This gene silencing is commonly linked to DNA methylation.

Angelman syndrome (AS) is caused by the loss of function of the maternally inherited UBE3A gene in the 15q11-q13 region.

- This can occur through various mechanisms, including deletions, mutations, or imprinting defects.
- The key features of AS include developmental delay, ataxic gait, and inappropriate laughter, all of which are present in this patient.

Feature	Angelman Syndrome	Prader-Willi Syndrome
Genetic Cause	Deletion or mutation of the maternal UBE3A gene on chromosome 15q11-q13.	Deletion or mutation of the paternal region on chromosome 15q11-q13, or maternal uniparental disomy.
Developmental Milestones	Severe developmental delay; speech is often minimal or absent.	Mild to moderate developmental delay; speech and motor skills are delayed.
Physical Features	Microcephaly, frequent smiling/laughter, ataxia, seizures.	Hypotonia, obesity, short stature, small hands and feet.
Behavioral Characteristics	Happy demeanor, hyperactivity, fascination with water.	Food-seeking behavior, temper tantrums, compulsive behaviors.
Intellectual Disability	Severe to profound intellectual disability.	Mild to moderate intellectual disability.
Seizures	Common and often severe.	Rare, typically not a prominent feature.
Growth Patterns	Normal or mildly affected growth; often underweight.	Hyperphagia leading to obesity if not managed.
Sleep Disturbances	Frequent, often severe sleep disturbances.	Common, often associated with excessive daytime sleepiness.

Disorder with maternal uniparental disomy(Option A):

- While PWS also involves the 15q11-q13 region, it results from the loss of paternally expressed genes in this region with maternal disomy.
- PWS presents with hypotonia, obesity, and cognitive impairment, which don't match the given clinical picture.

Beckwith-Wiedemann syndrome (BWS) (Option C):

- BWS is an overgrowth disorder caused by dysregulation of imprinted genes on chromosome 11p15.5.
- It typically presents with macrosomia, macroglossia, and an increased risk of embryonal tumors, which are not described in this case.

Silver-Russell syndrome (SRS) (Option D):

- SRS is a growth restriction disorder associated with imprinting defects on chromosomes 7 and 11.
- It is characterized by intrauterine growth restriction, postnatal growth failure, and body asymmetry, which don't align with the presented symptoms.

Solution for Question 11:

Achondroplasia is inherited in an autosomal dominant manner.

Autosomal Dominant disorders

- These disorders appear when just one allele is affected, unlike recessive disorders which require both alleles to be affected.
- In cases of dominant disorders, at least one parent will have the same condition.

Examples:

- Hereditary Spherocytosis, Hypercholesterolemia
- Ehlers-Danlos Syndrome
- Achondroplasia
- Von Willebrand Disease
- Pseudohypoparathyroidism
- Myotonic Dystrophy
- Osteogenesis Imperfecta (most types are autosomal dominant)
- Marfan Syndrome
- Intermittent Porphyria
- Noonan Syndrome
- Adenomatous Polyposis Coli (a precancerous condition for colorectal carcinoma)
- Neurocutaneous Conditions - Neurofibromatosis and Tuberous Sclerosis (Hereditary)

Mucopolysaccharidosis type II and Duchenne muscular dystrophy (Options C and D) are X-linked recessive diseases.

Galactosemia (Option B) is an autosomal recessive disease.

Solution for Question 12:

The patient’s clinical presentation is indicative of Fragile-X Syndrome.

- More than 200 CGG repeats is diagnostic of Fragile-X syndrome.
- However, repeat expansion of CGG in the range of 5-55 repeats is seen in the normal population.

Fragile X Syndrome:

Genetic Basis	CGG repeats on FMR-1 (Familial mental retardation) gene on X chromosome (Option A)
CGG Repeat Numbers	Normal population: 5-55 repeats Carriers (premutation): 55-200 repeats Fragile X Syndrome: >200 repeats
Gender Prevalence	More common in males Females are mostly carriers
Clinical Features	Long face & large ears Large mandible/Prominent jaw High-arched palate Hyperextensible joints (Option D) Mitral valve prolapse Macroorchidism/large testes (seen in adolescent age group) Intellectual disability
Clinical features of female carriers	May be associated with premature ovarian failure (Option C) 10% may have an intellectual disability/low IQ

Solution for Question 13:

- Trinucleotide repeat disorders are caused by the expansion of specific three-nucleotide sequences in genes.
- Threshold effect: There's a threshold effect, where symptoms appear only above a certain number of repeats.
- Anticipation: Many, but not all trinucleotide repeat disorders exhibit anticipation, where the disease appears earlier and with increased severity in successive generations.
- Dynamic mutations: Unlike traditional mutations, the number of repeats can change (usually increase) from one generation to the next or even within an individual's lifetime.
- This instability is a hallmark of these disorders.

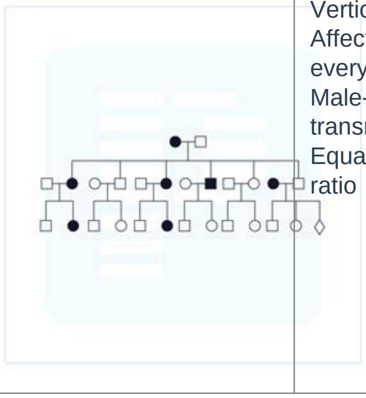
Examples of Trinucleotide repeat disorders include:

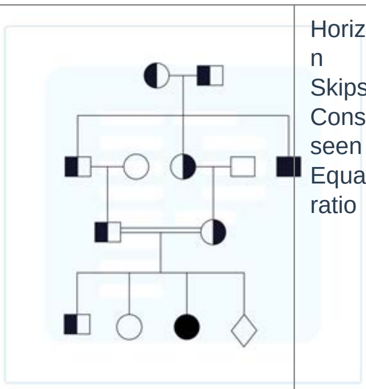
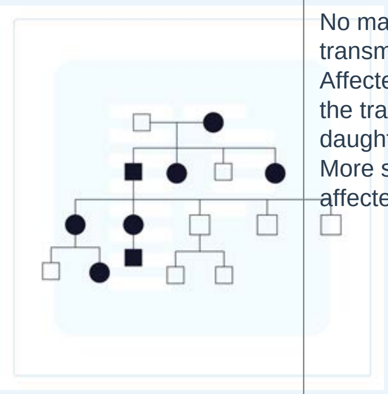
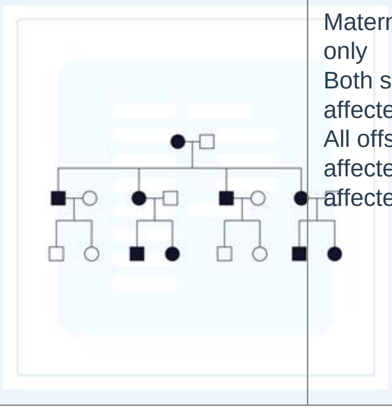
Disease	Trinucleotide Repeat
Huntington's disease	CAG
Fragile X syndrome	CGG
Friedreich's ataxia	GAA
Myotonic dystrophy type 1	CTG

Solution for Question 14:

- The pedigree indicates an X-linked recessive condition, as the expression of disease occurs only in males. In males, since there is only one X chromosome, X-linked recessive disorders manifest.
- In females, the disorder does not manifest (usually) since the mutant gene is compensated for by the normal allele on the other X chromosome.
- Females thus act as carriers of the mutant allele.

The table given below explains various modes of inheritance:

Mode of Inheritance	Pedigree	Explanation	Diseases
Autosomal Dominant (Option A)		Vertical transmission Affected individuals in every generation Male-to-male transmission Equal male: female ratio	Huntington's disease Marfan syndrome Neurofibromatosis type 1

Autosomal Recessive		Horizontal transmission Skips generations Consanguinity often seen Equal male: female ratio	Cystic fibrosis Sickle cell anemia Beta-thalassemia
X-Linked Dominant (Option B)		No male-to-male transmission Affected fathers pass the trait to all daughters More severely affected males	Orofaciodigital syndrome Incontinentia Pigmenti Vitamin D-resistant rickets
X-Linked Recessive	recessive " data-author="" data-hash="917" data-license="" data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/Y7guwGOcb2ev0HfqXuov1742029792.png" />	Primarily affects males No male-to-male transmission Unaffected carrier females	Hemophilia A Duchenne muscular dystrophy Hunter Syndrome
Mitochondrial (Option D)		Maternal transmission only Both sexes can be affected All offspring of the affected mother are affected	Leigh Disease MELAS syndrome

Solution for Question 15:

Common pedigree symbols include:

pedigree symbols" data-author="" data-hash="916" data-license="" data-source="" data-tags="March2025" src="https://image.prepladder.com/notes/iEHXnJTvnKBygz4wj3Vj1742029648.png" />

Solution for Question 16:

Examples of X-linked dominant Disorders:

- Hypophosphatemic vitamin D-resistant rickets (Option A)
- Orofaciodigital syndrome
- Incontinentia pigmenti (Option C)
- Charcot Marie tooth disease
- Rett syndrome(Option B)

X-Linked Dominant Conditions:

- These conditions are rare.
- Both heterozygous females and hemizygous males are affected.

Inheritance Patterns:

- Affected Males: All sons of affected males are normal. All daughters of affected males are affected.
- All sons of affected males are normal.
- All daughters of affected males are affected.
- Affected Females: Transmit the condition to 50% of their sons and 50% of their daughters.
- Transmit the condition to 50% of their sons and 50% of their daughters.
- All sons of affected males are normal.
- All daughters of affected males are affected.

- Transmit the condition to 50% of their sons and 50% of their daughters.

Severe Cases:

- In some instances, the mutant gene causes severe developmental effects, resulting in affected males rarely being born alive.

Solution for Question 17:

- X-inactivation (Lyonization) is a random process that typically involves the equal inactivation of one of the two X chromosomes in a heterozygous female.
- However, it is possible for skewed X-inactivation to occur where the X chromosome with the mutant allele is predominantly active in most cells.
- This can result in the expression of the disorder if the majority of cells have the X chromosome carrying the mutant allele.

Other scenarios where females manifest X-linked recessive disorders:

- The female has only one X chromosome: In Turner syndrome (45,X), females only have one X chromosome. Therefore, if they inherit a mutated X chromosome, they will manifest the X-linked recessive disorder.
- In Turner syndrome (45,X), females only have one X chromosome.
- Therefore, if they inherit a mutated X chromosome, they will manifest the X-linked recessive disorder.
- In Turner syndrome (45,X), females only have one X chromosome.
- Therefore, if they inherit a mutated X chromosome, they will manifest the X-linked recessive disorder.

All sons of affected males are affected by X-linked dominant disorders (Option A): All sons of affected males are normal in X-linked dominant disorder.

All offspring of an affected female are affected in an autosomal dominant disorder (Option C): All offspring of an affected female are affected by a mitochondrial inheritance

Solution for Question 18:

Hemophilia is an X-linked recessive disorder.

X-linked Recessive Disorders in Males:

- Manifests in males because they have only one X chromosome. If the X chromosome carries the recessive mutant allele, the disorder will be expressed.

X-linked Recessive Disorders in Females:

- Typically do not manifest clinically because the normal allele on the other X chromosome compensates for the mutant allele.
- Females with one mutant allele are carriers of the disorder.

Inheritance Pattern:

- Carrier females pass the mutant allele to half of their male children, who will then be affected by the disorder.

Examples of X-linked Recessive Disorders:

- G6PD (Glucose-6-Phosphate Dehydrogenase) Deficiency
- Duchenne Muscular Dystrophy (DMD)
- Color blindness
- Fabry Disease
- Chronic Granulomatous Disease (CGD)
- Hemophilia A and B
- Hunter disease
- Agammaglobulinemia (Bruton's Disease)
- Wiskott-Aldrich Syndrome
- Albinism
- Lesch-Nyhan Syndrome

Leigh disease (Option B) and MELAS syndrome (Option D) are mitochondrial disorders.

Incontinentia pigmenti (Option C) is an X-linked dominant disorder can affect females with one mutated allele; both alleles don’t need to be affected for the disorder to manifest.

Solution for Question 19:

Spinal muscular atrophy is an autosomal recessive disorder.

Autosomal Recessive Disorders:

- These disorders appear only when both copies (alleles) of a gene are affected.
- If both parents are carriers of the gene, the disorder will manifest only in their children who inherit both affected alleles.

Examples of Autosomal Recessive disorders include:

- Albinism, Alkaptonuria
- β-Thalassemia
- Cystic Fibrosis, Congenital Adrenal Hyperplasia
- Congenital Sensorineural Deafness
- Emphysema due to α1-Antitrypsin Deficiency
- Gaucher Disease, Galactosemia

Marfan syndrome and Neurofibromatosis (Option B and C) are autosomal dominant diseases.

Colour blindness (Option D) is an X-linked recessive disorder.

Previous Year Questions

1. A child with fused carpal bones comes to your clinic. What is the gene associated with hand-feet syndrome?

A. HOXA13

B. BCL6

C. FGFR

D. COl11

2. At what age is surgery typically performed for cleft lip?

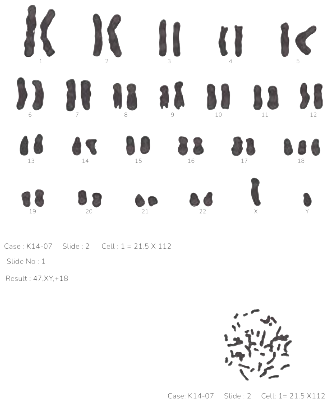
A. 3 months

B. 6 months

C. 1 year

D. 18 months

3. Which trisomy is suggested by the karyotyping results showing chromosomal defects?



A. Trisomy 18

B. Trisomy 13

C. Trisomy 21

D. Trisomy 11

4. What is the diagnosis of the patient having the following type of karyotype as shown in the image?

A. Turner Syndrome

B. Down Syndrome

C. Klinefelter syndrome

D. Edward syndrome

---

5. A child was diagnosed with Turner syndrome. Which of the following are features of Turner syndrome?

A. Webbed neck

B. Normal height

C. Decreased carrying angle

D. Gynecomastia

---

6. A teenage female was brought to the OPD with delayed puberty and was found to have short stature and widely spaced nipples. What is the probable karyotype?

A. 46, XO

B. 45, XO

C. 47, XXX

D. 46, XY

---

7. In which trimester is a quadruple test done for antenatal screening of Down syndrome?

A. 1st trimester

B. 3rd trimester

C. 2nd trimester

D. 4th trimester

8. Match the following: 1. Trisomy 13 a. GLU-VAL replacement 2. Trisomy 18 b. Edward syndrome 3. Trisomy 21 c. Down syndrome 4. Sickle cell anemia d. Patau syndrome

1. Trisomy 13	a. GLU-VAL replacement
2. Trisomy 18	b. Edward syndrome
3. Trisomy 21	c. Down syndrome
4. Sickle cell anemia	d. Patau syndrome

A. 1-d, 2-b, 3-c, 4-a

B. 1-b, 2-c, 3-a, 4-d

C. 1-c, 2-b, 3-a, 4-d

D. 1-a, 2-c, 3-a, 4-d

9. Which chromosome is affected in Hunter disease, leading to an abnormality?

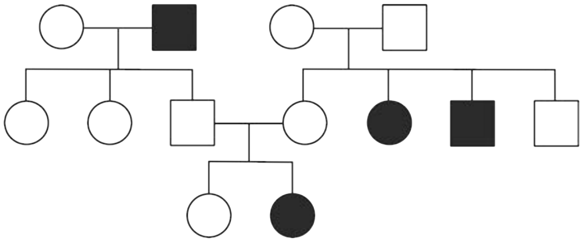
A. X chromosome

B. Y chromosome

C. Chromosome 21

D. Chromosome 22

10. What type of genetic inheritance is indicated by the pedigree chart?



A. Autosomal-dominant disorder

B. X- linked dominant disorder

C. Autosomal-recessive disorder

D. X- linked recessive disorder

---

11. Choose the most appropriate answer. Assertion: In phenylketonuria, the urine has a mousy odor. Reason: The serum tyrosine levels are decreased in phenylketonuria.

A. Both assertion and reason are independently false statements.

B. The assertion is a true statement, but the reason is a false statement.

C. Both assertion and reason are independently true statements, but the reason is not the correct explanation for the assertion.

D. Both assertion and reason are independently true statements, and the reason is the correct explanation for the assertion.

---

12. What are the accurate characteristics of ovotesticular disorders of sex development?

A. Karyotype is 46 XY.

B. Ovotesticular disorder of sex development can be identified in the gonadal biopsy

C. The response of testosterone to hCG is increased.

D. Uterus is usually absent.

---

13. Which of the subsequent conditions is characteristic of an X-linked disorder?

A. Color blindness

B. Thalassemia

C. Sickle cell anemia

D. Cystic fibrosis

---

14. Which of the following statements about ataxia telangiectasia is false?

A. Mutations in 11q gene is implicated

B. Follows autosomal dominant mode of inheritance

C. Humoral and cellular immunodeficiency

D. Linked to adenocarcinomas

---

15. What is the most common presentation of congenital adrenal hyperplasia?

A. Male pseudohermaphroditism

B. Female pseudohermaphroditism

C. True hermaphroditism

D. 46, XY intersex

---

16. What could be a potential cause for a child with short stature who is brought to the pediatrician and has high growth hormone levels and low insulin-like growth factor (IGF-1) levels?

A. Turner syndrome

B. Pituitary dwarfism

C. Laron dwarfism

D. Maternal deprivation dwarfism

17. What is the diagnosis of a 10-year-old boy who was brought for psychiatric evaluation due to intellectual disability and exhibits an IQ of 45? The boy also presents with physical characteristics such as a long face, large ears, and large testes, along with a cardiac murmur. Additionally, there is a history of poor intellectual function in the paternal family.

A. Cushing’s syndrome

B. Rett syndrome

C. Fragile X syndrome

D. William syndrome

18. Which statement correctly describes X-linked hypophosphatemic rickets?

A. FGF-23 levels decreased

B. PHEX gene is affected

C. It is the most common cause of rickets

D. Excretion of phosphorus is decreased

19. What is the most likely karyotype for a 16-year-old girl who presents with short stature, primary amenorrhea, and widely spaced nipples?

A. 45, XO

B. 46, XO

C. 47, XXX

D. 46, XY

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	2
Question 5	1
Question 6	2
Question 7	3
Question 8	1
Question 9	1
Question 10	3
Question 11	3
Question 12	2
Question 13	1
Question 14	2
Question 15	2
Question 16	3
Question 17	3
Question 18	2
Question 19	1

Solution for Question 1:

Correct Answer: A) HOXA13

Explanation:

Hand-Foot-Genital Syndrome (HFGS):

- Rare, dominantly inherited condition affecting distal limbs and genitourinary tract.
- Genetic cause: Identified mutation in the HOXA13 homeobox gene. HFGS is the second human syndrome linked to a HOX gene mutation.
- Identified mutation in the HOXA13 homeobox gene.
- HFGS is the second human syndrome linked to a HOX gene mutation.
- Clinical Characteristics: Limb Malformations: Bilateral shortening of thumbs and great toes, typically involving distal phalanx or first metacarpal/metatarsal. Impaired dexterity or thumb apposition due to the malformations. Urogenital Defects: In females: Abnormalities in the ureters and urethra, incomplete müllerian fusion. In males: Hypospadias with or without chordee. Associated conditions: Vesicoureteral reflux, recurrent urinary tract infections, and chronic pyelonephritis. Normal fertility.
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- In females: Abnormalities in the ureters and urethra, incomplete müllerian fusion.
- In males: Hypospadias with or without chordee.
- Associated conditions: Vesicoureteral reflux, recurrent urinary tract infections, and chronic pyelonephritis.
- Normal fertility.
- Diagnosis and Testing: Clinical Diagnosis: Physical examination, radiographs of hands/feet, and imaging of kidneys, bladder, and female reproductive tract. Genetic Diagnosis: Identification of a heterozygous HOXA13 pathogenic variant confirms diagnosis if clinical/radiographic features are unclear.
- Clinical Diagnosis: Physical examination, radiographs of hands/feet, and imaging of kidneys, bladder, and female reproductive tract.

- Physical examination, radiographs of hands/feet, and imaging of kidneys, bladder, and female reproductive tract.
- Genetic Diagnosis: Identification of a heterozygous HOXA13 pathogenic variant confirms diagnosis if clinical/radiographic features are unclear.
- Identification of a heterozygous HOXA13 pathogenic variant confirms diagnosis if clinical/radiographic features are unclear.
- Management: Treatment of Manifestations: Surgery for hand/foot malformations is usually unnecessary. Ureteric reimplantation and bladder outlet abnormality correction often needed. Rare surgical correction for vaginal septum or uterine abnormalities, unless recurrent pregnancy loss occurs. Prevention of Secondary Complications: Prophylactic antibiotics or surgery for urinary tract issues. Gynecological exams before menstruation for hymenal constriction. Pre-pregnancy evaluation for uterine and vaginal anatomy due to risks in pregnancy.
- Treatment of Manifestations: Surgery for hand/foot malformations is usually unnecessary. Ureteric reimplantation and bladder outlet abnormality correction often needed. Rare surgical correction for vaginal septum or uterine abnormalities, unless recurrent pregnancy loss occurs.
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- Gynecological exams before menstruation for hymenal constriction.
- Pre-pregnancy evaluation for uterine and vaginal anatomy due to risks in pregnancy.
- Identified mutation in the HOXA13 homeobox gene.
- HFGS is the second human syndrome linked to a HOX gene mutation.
- Limb Malformations: Bilateral shortening of thumbs and great toes, typically involving distal phalanx or first metacarpal/metatarsal. Impaired dexterity or thumb apposition due to the malformations.
- Bilateral shortening of thumbs and great toes, typically involving distal phalanx or first metacarpal/metatarsal.
- Impaired dexterity or thumb apposition due to the malformations.
- Urogenital Defects: In females: Abnormalities in the ureters and urethra, incomplete müllerian fusion. In males: Hypospadias with or without chordee. Associated conditions:

Vesicoureteral reflux, recurrent urinary tract infections, and chronic pyelonephritis. Normalfertility.

- In females: Abnormalities in the ureters and urethra, incomplete müllerian fusion.
- In males: Hypospadias with or without chordee.
- Associated conditions: Vesicoureteral reflux, recurrent urinary tract infections, and chronic pyelonephritis.
- Normal fertility.
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- In males: Hypospadias with or without chordee.
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- Normal fertility.
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- Identification of a heterozygous HOXA13 pathogenic variant confirms diagnosis if clinical/radiographic features are unclear.
- Treatment of Manifestations: Surgery for hand/foot malformations is usually unnecessary. Ureteric reimplantation and bladder outlet abnormality correction often needed. Rare surgical correction for vaginal septum or uterine abnormalities, unless recurrent pregnancy loss occurs.
- Surgery for hand/foot malformations is usually unnecessary.
- Ureteric reimplantation and bladder outlet abnormality correction often needed.
- Rare surgical correction for vaginal septum or uterine abnormalities, unless recurrent pregnancy loss occurs.

- Prevention of Secondary Complications: Prophylactic antibiotics or surgery for urinary tract issues. Gynecological exams before menstruation for hymenal constriction. Pre-pregnancy evaluation for uterine and vaginal anatomy due to risks in pregnancy.
- Prophylactic antibiotics or surgery for urinary tract issues.
- Gynecological exams before menstruation for hymenal constriction.
- Pre-pregnancy evaluation for uterine and vaginal anatomy due to risks in pregnancy.
- Surgery for hand/foot malformations is usually unnecessary.
- Ureteric reimplantation and bladder outlet abnormality correction often needed.
- Rare surgical correction for vaginal septum or uterine abnormalities, unless recurrent pregnancy loss occurs.
- Prophylactic antibiotics or surgery for urinary tract issues.
- Gynecological exams before menstruation for hymenal constriction.
- Pre-pregnancy evaluation for uterine and vaginal anatomy due to risks in pregnancy.

BCL6 gene (Option B) mutations are associated with B cell lymphomas and other cancers.

FGFR gene (Option C) mutation causes skeletal disorders, mainly dwarfism and craniofacial malformations.

COI 11 (Option D) mutation is associated with hearing loss.

Solution for Question 2:

Correct Option A - 3 months:

- Surgery for cleft lip is typically performed around 3 months of age. This timing follows the " Rule of 10s" guideline, which suggests the baby should be at least 10 weeks old, weigh at least 10 pounds, and have a hemoglobin level of at least 10 g/dL. Early repair at this age helps with feeding, speech development, and social integration.
- Surgery at 3 months allows for timely intervention that promotes better outcomes for speech, feeding, and overall development, making it the preferred age for cleft lip repair.

Incorrect Options:

Option B - 6 months:

- By 6 months, cleft palate repair may be considered, but cleft lip surgery is ideally done earlier to aid in normal development. Waiting until 6 months for cleft lip surgery could delay essential early interventions.

Option C - 1 year:

- Cleft palate repair is more common around this time, but cleft lip repair is usually already completed to support early developmental milestones, particularly speech and feeding.

Option D - 18 months:

This is later than recommended for cleft lip surgery. By 18 months, speech development is in progress, and delaying surgery could impact speech and social development.

Solution for Question 3:

Correct Option A - Trisomy 18:

- The given pedigree depicts Trisomy 18 or Edward syndrome.
- The chromosome 18 has three copies in the above-given pedigree.

Incorrect Options:

- Options B, C, & D: Trisomy 13,21 and 11 are not depicted in the given pedigree.

Solution for Question 4:

Correct Option B - Down Syndrome:

- The above image shows three copies of chromosome 21 and is suggestive of Down’s syndrome.
- Down syndrome, also known as Trisomy 21, is caused by the presence of an extra copy of chromosome 21.

Incorrect Options:

Option A: Turner Syndrome:

- Turner syndrome, also known as Monosomy X, occurs in females.
- It is characterized by the absence or partial deletion of one of the X chromosomes, resulting in a karyotype of 45, XO.

Option C: Klinefelter Syndrome: Klinefelter syndrome characterized by the presence of an extra X chromosome, resulting in a karyotype of 47, XXY.

Option D: Edwards Syndrome:

- Edwards syndrome, also known as Trisomy 18, is caused by the presence of an extra copy of chromosome 18.
- The karyotype of individuals with Edwards syndrome is usually 47, XX+18, or 47, XY+18.

Solution for Question 5:

Correct Option A - Webbed neck:

- The above image shows a webbed neck which is a feature of Turner’s syndrome.
- Clinical features of turner's syndrome.

S	Short stature Short 4th metacarpal Sensorineural hearing loss
A	Amenorrhea (Primary)
B	Barr body absent
C	Cardiac abnormalities Cystic hygroma Cubitus valgus
L	Low posterior hairline Lymphedema of hands and feet Low thyroid (Hypothyroidism)
O	Ovaries streak-shaped/underdeveloped Rudimentary uterus
W	Webbed neck
N	Nipples are widely spaced on a shield-shaped chest Normal intelligence

Incorrect Options:

Option B - Normal height:

- Turner’s syndrome patients are short statured.

Option C - Decreased carrying angle:

- Carrying angle is increased in patients with Turner’s syndrome.

Option D - Gynecomastia:

- Gynecomastia is not a feature of Turner’s syndrome

Solution for Question 6:

Correct Option B - 45, XO:

- The above clinical features are diagnostic of Turner's syndrome.
- The karyotype is 45 XO as there is only a single X chromosome and no Y chromosome.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 7:

Correct Option C - 2nd trimester:

- The quadruple screening test is offered to pregnant women in the second trimester (from 14 to 20 weeks gestation).
- Four biochemical markers are analysed and combined with maternal age to generate a quadruple screening test result.
- These comprise: α-fetoprotein (AFP) Human chorionic gonadotropin (hCG) or free β-hCG Inhibin A Unconjugated oestriol
- α-fetoprotein (AFP)

- Human chorionic gonadotropin (hCG) or free β-hCG
- Inhibin A
- Unconjugated oestriol
- α-fetoprotein (AFP)
- Human chorionic gonadotropin (hCG) or free β-hCG
- Inhibin A
- Unconjugated oestriol

Incorrect Options: Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 8:

Correct Option A - 1-d, 2-b, 3-c, 4-a:

Trisomy 13	Patau syndrome
Trisomy 18	Edward syndrome
Trisomy 21	Down syndrome
Sickle cell anemia	GLU-VAL replacement

Incorrect Options: Options B, C, and D are incorrect.

Solution for Question 9:

Correct Option A - X chromosome:

- Hunter syndrome, also known as mucopolysaccharidosis II, is an X-linked recessive disorder caused by a deficiency of the enzyme iduronate sulfatase.
- The defective gene responsible for the condition is located on the X chromosome.

Incorrect Options -B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 10:

Correct Option C - Autosomal-recessive disorder:

- In an autosomal-recessive disorder, the affected individual must inherit two copies of the mutated gene, one from each parent, in order to express the disorder.
- Carriers (heterozygous individuals) do not show symptoms but can pass on the mutated gene to their offspring.
- The disorder can skip generations and is typically seen in individuals who have two affected parents or when both parents are carriers.
- Examples of autosomal-recessive disorders include cystic fibrosis and sickle cell anemia.

Incorrect Options:

Option A: Autosomal-dominant disorder:

- In an autosomal-dominant disorder, the affected individual has a 50% chance of passing on the mutated gene to each offspring.
- The disorder typically appears in every generation, and both males and females can be affected.
- Unaffected individuals do not transmit the disorder to their offspring.
- Examples of autosomal-dominant disorders include Huntington's disease and Marfan syndrome.

Option B: X-linked dominant disorder:

- In an X-linked dominant disorder, an affected male passes the mutated gene to all of his daughters but none of his sons.
- Affected females have a 50% chance of passing on the mutated gene to each offspring, regardless of gender.
- The disorder can appear in every generation, but it is more commonly observed in females.
- Examples of X-linked dominant disorders include Rett syndrome and Incontinentia Pigmenti.

Option D: X-linked recessive disorder:

- In an X-linked recessive disorder, affected males pass the mutated gene to all of their daughters but none of their sons.
- Carrier females have a 50% chance of passing on the mutated gene to each offspring.
- The disorder is more commonly observed in males.
- Examples of X-linked recessive disorders include hemophilia and Duchenne muscular dystrophy.

Solution for Question 11:

Correct Option C - Both assertion and reason are independently true statements, but the reason is not the correct explanation for the assertion:

- Phenylketonuria is a genetic disorder caused by a deficiency of the enzyme phenylalanine hydroxylase, which is responsible for converting the amino acid phenylalanine into tyrosine.
- One of the symptoms associated with phenylketonuria is a distinct odor of the urine, often described as "mousy" or "musty."
- This odor is caused by the accumulation of phenylalanine and its breakdown products, such as phenyl pyruvic acid, in the urine.
- Since the enzyme phenylalanine hydroxylase is deficient in phenylketonuria, phenylalanine cannot be converted into tyrosine effectively.
- As a result, tyrosine levels are decreased in the blood.

Incorrect Options - A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 12:

Correct Option B - Ovotesticular disorder of sex development can be identified in the gonadal biopsy:

- Ovotestis, which contains both ovarian and testicular tissue, can be identified in the gonadal biopsy of individuals with ovotesticular disorders of sex development.

Incorrect Options:

Option A: The karyotype is 46 XY: Ovotesticular disorders of sex development can have various karyotypes, including 46 XX, 46 XY, or mosaic patterns.

Option C: The response of testosterone to hCG is increased: The response of testosterone to human chorionic gonadotropin (hCG) stimulation is typically decreased in individuals with ovotesticular disorders of sex development.

Option D: Uterus is usually absent: While the presence or absence of a uterus can vary in ovotesticular disorders of sex development, it is possible for individuals with this condition to have a uterus.

Solution for Question 13:

Correct Option A - Color blindness:

- Color blindness is a X-linked disorder.
- It is caused by mutations in the genes responsible for the production of photopigments in the cone cells of the retina.
- These mutations lead to a deficiency or absence of the photopigments, resulting in an inability to distinguish certain colors, particularly shades of red and green.

Incorrect Options - Options B, C, and D are autosomal recessive disorders.

Solution for Question 14:

Correct Option B - Follows autosomal dominant mode of inheritance:

- Ataxia telangiectasia is an autosomal recessive disorder.

Incorrect Options:

Option A: Mutations in 11q gene is implicated: Ataxia telangiectasia is caused by mutations in the ATM gene located on chromosome 11q.

Option C: Humoral and cellular immunodeficiency: Ataxia telangiectasia is associated with both humoral (antibody-mediated) and cellular (T-cell-mediated) immunodeficiency.

Option D: Linked to adenocarcinomas: Individuals with ataxia telangiectasia have an increased risk of developing certain types of cancer, including lymphomas, leukemia, and certain solid tumors such as breast and gastric adenocarcinomas.

Solution for Question 15:

Correct Option B - Female pseudohermaphroditism:

- In congenital adrenal hyperplasia, female pseudohermaphroditism occurs when there is prenatal exposure to excess androgens due to the enzyme deficiency.
- It leads to virilization of the external female genitalia, causing them to appear more masculine.

Incorrect Options: Options A, C, and D are not seen in congenital adrenal hyperplasia.

Solution for Question 16:

Correct Option C - Laron dwarfism:

- The patient’s presentation with short stature, high growth hormone levels and low insulin-like growth factor levels is suggestive of Laron dwarfism.
- Laron dwarfism, also known as growth hormone insensitivity syndrome, is a rare genetic disorder characterized by an inability of the body to respond to growth hormone.
- It is typically caused by mutations in the growth hormone receptor gene or post-receptor signaling pathway genes.
- In Laron dwarfism, despite high levels of growth hormone, the body fails to produce adequate levels of IGF-1, which is the main mediator of growth hormone's effects on growth.

Incorrect options:

Option A- Turner syndrome:

- In Turner syndrome, growth hormone levels are usually reduced rather than elevated.

Option B- Pituitary dwarfism:

- Pituitary dwarfism, also known as growth hormone deficiency, is a condition where there is insufficient production or release of growth hormone by the pituitary gland.
- In pituitary dwarfism, both growth hormone levels and IGF-1 levels are typically low.

Option D- Maternal deprivation dwarfism:

- Maternal deprivation dwarfism, also known as psychosocial dwarfism, is a condition that can occur in children who experience severe emotional deprivation or neglect during early childhood.
- It is characterized by growth failure and short stature, but it is not associated with high growth hormone levels.

Solution for Question 17:

Correct Option C - Fragile X Syndrome:

- The patient’s presentation with a long face, large ears, large testes, cardiac murmur, intellectual disability, and a history of poor intellectual function in the paternal family is suggestive of fragile X syndrome.
- Fragile X syndrome is a genetic disorder caused by the expansion of a CGG trinucleotide repeat in the FMR1 gene on the X chromosome.
- It is the most common inherited cause of intellectual disability and affects males more severely than females.
- Clinical features of fragile X syndrome: Long face Large ears Large mandible/Prominent jaw High arched palate Hyperextensible joints Mitral valve prolapse Macroorchidism/Large testes: Seen in the adolescent age group Intellectual disability (developmental delay, mental retardation)
- Long face
- Large ears
- Large mandible/Prominent jaw
- High arched palate

- Hyperextensible joints
- Mitral valve prolapse
- Intellectual disability (developmental delay, mental retardation)
- Long face
- Large ears
- Large mandible/Prominent jaw
- High arched palate
- Hyperextensible joints
- Mitral valve prolapse
- Intellectual disability (developmental delay, mental retardation)

Incorrect Options -A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 18:

Correct Option B - PHEX gene is affected:

- X-linked hypophosphatemic rickets is caused by mutations in the PHEX gene, which encodes for the phosphate-regulating endopeptidase homolog X-linked (PHEX) protein.
- Mutations in this gene lead to impaired phosphate reabsorption in the renal tubules and subsequent hypophosphatemia.

Incorrect Options:

Option A: FGF23 levels decreased:

- In X-linked hypophosphatemic rickets (XLH), FGF-23 levels are increased rather than decreased.
- FGF-23 is a hormone that regulates phosphate metabolism, and its increased levels in XLH result in increased renal phosphate wasting and decreased levels of active vitamin D.

Option C: It is the most common cause of rickets:

- XLH is the most common inherited form of rickets.
- Nutritional deficiencies (such as vitamin D deficiency) are the most common overall cause of rickets.

Option D: Excretion of phosphorus is decreased: In XLH, there is increased renal phosphate wasting, resulting in increased excretion of phosphorus in the urine.

Solution for Question 19:

Correct option A - 45, XO

- The patient’s presentation with short stature, primary amenorrhea, and widely spaced nipples is suggestive of Turner syndrome.
- The karyotype of this patient is 45, XO.

Incorrect Options - B, C, and D are incorrect. Refer to the explanation of the correct answer.

Disorders of Carbohydrate and Amino Acid Metabolism

1. A 10-year-old boy is brought to the pediatrician with complaints of chronic back pain and stiffness that have worsened over the past year. On examination, the physician notes bluish-black discoloration in the sclerae and dark pigmentation in the cartilage of the ears. The mother mentions that the child's urine turns dark when left standing for a few hours. Based on these findings, what is the most likely diagnosis?

A. Ehlers-Danlos Syndrome

B. Alkaptonuria

C. Marfan Syndrome

D. Osteogenesis Imperfecta

2. A 4-year-old boy is brought to the clinic due to developmental delay and vision problems. Physical examination reveals a tall, slender child with elongated limbs, pectus excavatum, and blue eyes. Ophthalmologic evaluation shows bilateral lens dislocation. Laboratory tests reveal elevated levels of homocystine and methionine in the blood and urine. Which of the following is the most likely underlying enzyme deficiency?

A. Phenylalanine hydroxylase

B. Cystathionine β-synthase

C. Methylenetetrahydrofolate reductase

D. Methionine synthase

3. A 6-year-old boy is brought to the clinic. His parents mention occasional episodes of unsteady gait and irritability. Physical examination reveals a

pellagra-like rash on sun-exposed areas. Laboratory tests show increased urinary excretion of neutral amino acids. Which of the following best describes the underlying pathophysiology of this condition?

A. Defective metabolism of tryptophan

B. Impaired transport of monoamine monocarboxylic amino acids (neutral amino acids)

C. Deficiency of enzyme tyrosine hydroxylase

D. Deficiency of biotinidase enzyme

4. A 2-day-old male neonate, born to consanguineous parents, presents with poor feeding, vomiting, and lethargy. Laboratory investigations reveal severe hyperammonemia (plasma ammonia 800 µmol/L) and respiratory alkalosis. Which of the following is the most appropriate immediate management step?

A. Start a protein-based diet

B. Administer intravenous sodium ammonium benzoate and arginine

C. Perform exchange transfusion

D. Initiate valproate for seizure prophylaxis

5. A 3-day-old neonate presents with lethargy, poor feeding, and hyperammonemia. Plasma amino acid analysis shows elevated citrulline levels. Which of the following enzyme deficiencies is most likely responsible for this presentation?

A. Ornithine transcarbamylase

B. Argininosuccinate synthetase

C. Carbamoyl phosphate synthetase I

D. Arginase

6. A 5-day-old neonate presents with poor feeding, lethargy, and a sweet, maple syrup-like odor in the urine. Initial laboratory tests show metabolic acidosis and ketonuria. Which of the following is the most appropriate next step in management?

A. Initiate a keto diet

B. Administer phenobarbital for seizure prophylaxis

C. Begin immediate hemodialysis

D. Start a leucine-restricted diet and provide thiamine supplementation

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7. Match the following Inborn Errors of Metabolism (IEMs) with their characteristic odors: Column A (IEMs) Column B (Odors) a. Maple Syrup Urine Disease 1. Tom cat urine b. Phenylketonuria 2. Cabbage-like c. Multiple Carboxylase Deficiency 3. Mousy or musty d. Tyrosinemia 4. Burnt sugar

Column A (IEMs)	Column B (Odors)
a. Maple Syrup Urine Disease	1. Tom cat urine
b. Phenylketonuria	2. Cabbage-like
c. Multiple Carboxylase Deficiency	3. Mousy or musty
d. Tyrosinemia	4. Burnt sugar

A. a-4, b-3, c-1, d-2

B. a-3, b-4, c-2, d-1

C. a-1, b-2, c-3, d-4

D. a-2, b-1, c-4, d-3

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8. Which of the following is not a common clinical feature of organic acidemias in neonates?

A. Lethargy

B. Vomiting

C. Respiratory acidosis

D. Seizures

9. A 2-week-old infant presents with prolonged jaundice, vomiting, and failure to thrive with a history of hepatorenal failure. Preliminary blood tests show elevated liver enzymes and a low platelet count. The infant's urine has a distinctive " cabbage-like" odour. Which of the following investigations is most likely to confirm the suspected diagnosis?

A. Urine culture and sensitivity

B. Serum ammonia levels

C. Urine succinylacetone test

D. Urine glucose levels

10. Match the following types of Glycogen storage disorders and their respective enzyme deficiencies: a. McArdle disease 1. Glucose 6 Phosphatase b. Her's disease 2. Muscle phosphorylase c. Von Gierke disease 3. Branching enzyme d. Anderson disease 4. Hepatic phosphorylase

a. McArdle disease	1. Glucose 6 Phosphatase
b. Her's disease	2. Muscle phosphorylase
c. Von Gierke disease	3. Branching enzyme
d. Anderson disease	4. Hepatic phosphorylase

A. a-3, b-1, c-4, d-2

B. a-3, b-4, c-1, d-2

C. a-2, b-3, c-1, d-4

D. a-2, b-4, c-1, d-3

11. A 6-month-old infant is brought to the clinic by their parents reporting that their infant is not meeting developmental milestones, such as rolling over or sitting up. On examination, the infant has eczema and a noticeable musty odor. Hair and skin pigmentation appear lighter. Which of the following enzymes is deficient in this infant?

A. Phenylalanine hydroxylase

B. Tyrosinase

C. Tyrosine aminotransferase

D. 4 HPPD (Hydroxy Phenyl pyruvate dioxygenase)

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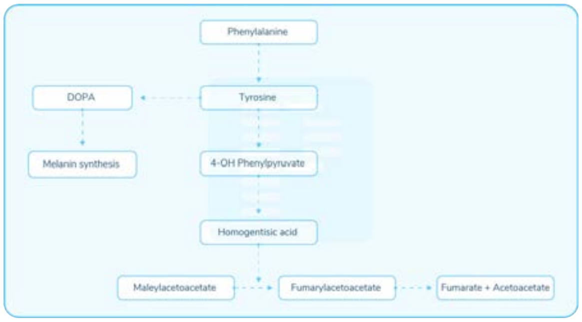
12. Which enzyme converts tyrosine to 4-OH Phenylpyruvate?



A. Tyrosinase

B. Tyrosine aminotransferase

C. 4 HPPD (Hydroxy Phenyl pyruvate dioxygenase)

D. None of the above

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13. A 7-month-old infant is brought to the clinic for evaluation of vomiting and poor weight gain. The infant started having these symptoms after adding sugar to the

diet. The physical exam is notable for hepatomegaly (enlarged liver). Which of the following is the most likely diagnosis?

A. Phenylketonuria

B. Hereditary fructose intolerance

C. Galactosemia

D. Glycogen storage disease type I

14. A 5-day-old newborn presents with poor feeding, vomiting, and jaundice. Physical examination reveals hepatomegaly and cataracts. Laboratory tests show elevated liver enzymes, hypoglycemia, and a positive reducing substance in the urine. Which of the following is the most appropriate initial management step?

A. Start phototherapy for jaundice

B. Administer intravenous glucose

C. Discontinue all lactose-containing foods

D. Begin enzyme replacement therapy

15. A 3-month-old infant presents with hypotonia, feeding difficulties, and heart failure. Laboratory tests reveal elevated serum creatine kinase and liver enzymes. Which of the following is not a characteristic of this condition?

A. It is caused by a deficiency of acid α-1,4-glucosidase

B. Glycogen accumulates primarily in the cytoplasm of affected cells

C. Enzyme replacement therapy with alglucosidase alfa is available

D. Cardiac involvement can include cardiomyopathy

16. Which of the following statements is incorrect with respect to Type 1 (Von Gierke Disease) and Type 3 (Cori Disease) Glycogen Storage Disease?

A. Type 1 GSD is characterized by enlarged kidneys, while Type 3 GSD typically has normal kidney size

B. Muscle involvement is seen in Type 3 GSD but not in Type 1 GSD

C. Lactate levels are typically normal in Type 3 GSD but elevated in Type 1 GSD

D. Liver biopsy in Type 1 GSD shows fibrosis, while in Type 3 GSD it shows distension of hepatocytes by glycogen and fat deposition

17. A 4-month-old infant presents with hepatomegaly, hypoglycemic seizures, and a doll-like face with fat cheeks. Laboratory tests reveal hypoglycemia, lactic acidosis, hyperuricemia, and hyperlipidemia. Which of the following is not a correct statement about this condition?

A. It is caused by a deficiency in glucose-6-phosphatase or its translocase

B. The condition is inherited in an autosomal recessive pattern

C. Glucagon administration typically leads to a significant increase in blood glucose levels

D. Treatment involves maintaining normal blood glucose levels through multiple feeds at regular intervals.

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	2
Question 5	2
Question 6	4
Question 7	1
Question 8	3
Question 9	3
Question 10	4
Question 11	1
Question 12	2
Question 13	2
Question 14	3
Question 15	2
Question 16	4
Question 17	3

Solution for Question 1:

Correct Answer: B) Alkaptonuria

Explanation:

The patient's finding of urine darkening upon standing aligns with a hallmark symptom of alkaptonuria.

- Alkaptonuria is an autosomal recessive disorder caused by a deficiency in the enzyme homogentisate 1,2-dioxygenase (HGD), also known as homogentisic acid oxidase.
- When HGD is deficient, HGA accumulates and is excreted in the urine, giving it a characteristic dark color upon standing due to oxidation.
- Other features like bluish-black discoloration in the sclerae and ear cartilage strongly suggest ochronosis, with black discoloration with ochre bodies in the dermis of the skin.
- The patient's chronic back pain and stiffness, which have worsened over time, are consistent with the involvement of intervertebral disc cartilage of spine commonly seen in alkaptonuria.

Signs and Symptoms of Alkaptonuria:

- Urine darkens to brown or black upon standing due to HGA oxidation.
- In infants, diaper staining may be the first sign.
- Connective Tissues: Bluish-black discoloration in cartilage and collagen.
- Eyes: Dark pigment deposits in the sclera.
- Ears and Nose: Dark pigmentation in cartilage.
- Joints: HGA pigment causes cartilage degeneration, leading to osteoarthritis-like symptoms.
- Spine: Degeneration and calcification of intervertebral discs, causing back pain and stiffness.
- Arthritis, especially in shoulders and hips, develops due to chronic inflammation and cartilage damage from HGA deposition.

Urine Analysis for Alkaptonuria:

- Colour Change: Darkening of urine upon standing is a key indicator.
- Fehling's or Benedict's Test: Detects reducing substances like HGA; urine from alkaptonuria will turn red/orange with these reagents but does not react with glucose oxidase.
- Gas Chromatography-Mass Spectrometry (GCMS): Provides definitive identification and quantification of HGA in urine.

Treatment:

- Nitisinone: Inhibits the enzyme that produces homogentisic acid (HGA) and may help slow disease progression.
- Dietary Restrictions of tyrosine and phenylalanine and vitamin C are not very efficacious but seem reasonable.

Ehlers-Danlos Syndrome, Marfan Syndrome, Osteogenesis Imperfecta (Option A,C, & D) do not clinically present with the combination of darkening of urine, ochronosis and large joint arthritis.

Solution for Question 2:

Correct Answer: B) Cystathionine β-synthase

Explanation:

- The patient's presentation is consistent with homocystinuria, which is caused by a deficiency of cystathionine β-synthase (CBS).
- The clinical features described- tall stature, elongated limbs, pectus excavatum, blue eyes, developmental delay, and lens dislocation (ectopia lentis) are typical of this disorder.
- The elevated levels of homocysteine and methionine in blood and urine are characteristic biochemical findings in CBS deficiency.

Classical Homocystinuria:

Pathogenesis	Defective metabolism of methionine and homocysteine Accumulation of homocysteine and methionine Decreased cysteine levels Impaired folate metabolism (in MTHFR deficiency)
Enzymes Involved	Cystathionine β-synthase (CBS) - Most common Methylenetetrahydrofolate reductase (MTHFR) Defect in methylcobalamin metabolism
Clinical Features	Ectopia lentis (Inferonasal subluxation of lens) (Superotemporal seen in Marfan disease) Marfanoid habitus (tall, slender, elongated limbs) Developmental delay and intellectual disability Psychiatric and behavioral disorders Skeletal abnormalities (scoliosis, pectus excavatum) Thromboembolism (recurrent episodes of stroke) (due to oxidised homocysteine) Fair complexion, blue eyes Seizures (in ~20% of untreated patients)
Diagnosis	Elevated homocysteine and methionine in blood and urine Low or absent cysteine in plasma
Management	Coenzyme supplementation B9 and B12 supplementation Methionine-restricted diet

	Cysteine supplementation

Phenylalanine hydroxylase: This enzyme is deficient in phenylketonuria (PKU), not homocystinuria (Option A)

Methylenetetrahydrofolate reductase (MTHFR) (Option C):

- While deficiency of this enzyme can cause a form of non-classical homocystinuria, it typically presents with low or low-normal methionine levels, unlike the elevated methionine seen in this case. This is because MTHFR deficiency leads to THF deficiency causing less conversion of homocysteine to methionine.
- Additionally, MTHFR deficiency doesn't cause the characteristic ocular and skeletal features described in this patient.

Methionine synthase (Option D): Deficiency of this enzyme can cause homocystinuria, but it's typically associated with low methionine levels and methylmalonic aciduria if B12 deficiency is present.

Solution for Question 3:

Correct Answer: B) Impaired transport of monoamine monocarboxylic amino acids (neutral amino acids)

Explanation:

- The patient's presentation is consistent with Hartnup disease (Pellagra-like dermatosis), a rare inborn error of metabolism.
- This condition is associated with impaired transport of monoamine monocarboxylic amino acids across the brush border epithelium of the intestine and renal tubules.

Hartnup Disease:

Inheritance	Autosomal recessive
Pathogenesis	Mutation in SLC6A19 gene; Impaired transport of neutral amino acids across intestinal and kidney brush border epithelium

Clinical features	Photosensitivity Pellagra-like rash (Decreased Niacin due to tryptophan deficiency) Neurologic symptoms (ataxia, emotional instability) - Decreased serotonin due to tryptophan deficiency
Biochemical defect	Deficiency of nicotinamide synthesis
Diagnosis	Aminoaciduria (neutral amino acids) in urine and plasma Demonstrate SLC6A19 defect
Treatment	Administration of nicotinamide/nicotinic acid Protection from sunlight

- In Hartnup disorder, the primary issue is impaired transport, not defective metabolism of Tryptophan (Option A)
- Deficiency of the enzyme tyrosine hydroxylase leads to a rare genetic disorder that affects the production of catecholamines, including dopamine, norepinephrine, and epinephrine leading to a range of neurological and motor symptoms (Option C)
- Deficiency of biotinidase enzyme causes biotin deficiency, a different metabolic disorder (Option D)

Solution for Question 4:

Correct Answer: B) Administer intravenous sodium ammonium benzoate and arginine

Explanation:

- A 2-day-old neonate presenting with poor feeding, vomiting, lethargy, severe hyperammonemia (plasma ammonia 800 µmol/L), and respiratory alkalosis suggests a severe urea cycle disorder.
- The most appropriate immediate management is administering intravenous sodium benzoate and arginine.
- Sodium ammonium benzoate is a nitrogen-scavenging agent that helps reduce ammonia levels by providing an alternative pathway for nitrogen excretion.
- Arginine is essential in most UCDs (except arginase deficiency) as it helps drive the urea cycle and promotes ammonia removal.
- This combination is part of the standard emergency protocol for managing acute hyperammonemia in suspected UCDs.

- Valproate is contraindicated as it suppresses urea cycle enzymes.

Clinical Features:

- In newborns, symptoms may include CNS toxicity, seizures, neuropathy, nausea, vomiting, and raised ICP.

Diagnosis:

- The key diagnostic marker for UCDs is hyperammonemia.

Management:

Acute management of hyperammonemia is critical and includes:

1. Providing adequate calories through IV glucose and lipids while temporarily restricting protein intake.

2. Administer medications to remove ammonia, such as sodium benzoate and sodium phenylacetate.

3. Supplementing with arginine or citrulline (except in arginase deficiency).

4. In severe cases, use hemodialysis or peritoneal dialysis to rapidly lower ammonia levels.

Start a protein-based diet (Option A):

- This would be harmful in UCD as it would increase nitrogen load, exacerbating hyperammonemia.
- UCDs require protein restriction, not increased protein intake.

Perform exchange transfusion (Option C):

- While this might temporarily lower ammonia levels, it's not the first-line treatment for hyperammonemia in UCDs.
- More effective and less invasive methods (like medication and dietary management) are preferred initially.

Initiate valproate for seizure prophylaxis (Option D):

- While seizures can occur in UCDs due to hyperammonemia.
- Valproate is contraindicated as it suppresses urea cycle enzymes.

Solution for Question 5:

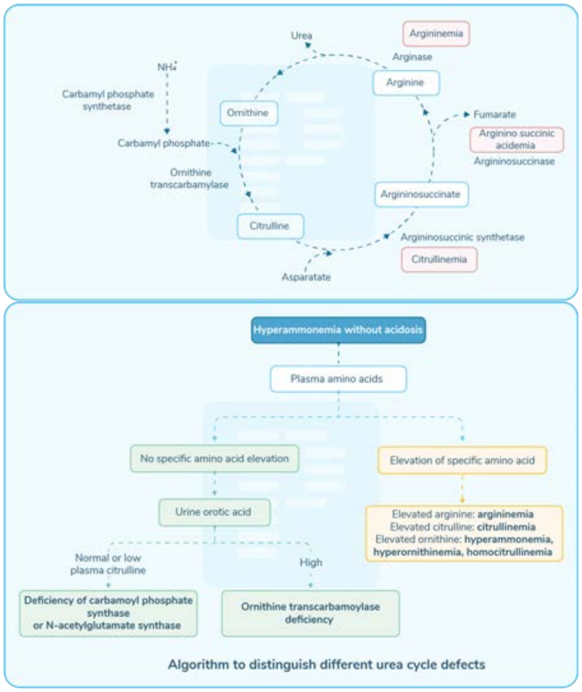
Correct Answer: B) Argininosuccinate synthetase

Explanation:

Argininosuccinate synthetase enzyme deficiency leads to citrullinemia, a urea cycle disorder characterized by elevated citrulline levels.

Urea Cycle Disorders (UCDs):

A group of inborn errors of metabolism affecting the urea cycle.



Algorithm to distinguish different urea cycle defects

Ornithine transcarbamylase (OTC) deficiency (Option A) presents with type 2 hyperammonemia and orotic aciduria. Citrulline levels are characteristically low.

Carbamoyl phosphate synthetase I (CPS1) deficiency (Option C) leads to hyperammonemia with low or absent citrulline levels.

Arginase deficiency (Option D) presents later in life and primarily causes hyperargininemia, but can also lead to hyperammonemia during metabolic crises.

- Citrulline levels are not elevated.

Solution for Question 6:

Correct Answer: D) Start a leucine-restricted diet and provide thiamine supplementation

Explanation:

- The symptoms described for the 5-day-old neonate—poor feeding, lethargy, a sweet, maple syrup-like odor in the urine, metabolic acidosis, and ketonuria—are indicative of Maple Syrup Urine Disease (MSUD).
- The most appropriate next step in management is to start a leucine-restricted diet and provide thiamine supplementation.
- Restricting leucine (and other branched-chain amino acids) helps reduce the accumulation of toxic metabolites.
- Thiamine supplementation is beneficial as it's a cofactor for the deficient enzyme complex and can enhance residual enzyme activity in some patients.

Maple Syrup Urine Disease (MSUD):

- MSUD affects the catabolic pathway of branched-chain amino acids (leucine, isoleucine, and valine).

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- BCKD - Branched-Chain α-Keto Acid Dehydrogenase Complex is deficient in MSUD.
- It has the following components: E1: Decarboxylase (E1α, E1β), E2: Dihydrolipoyl Transacylase, E3: Dihydrolipoyl Dehydrogenase.
- Enzyme complex is responsible for oxidative decarboxylation of BCAA, pyruvate, and alpha-ketoglutarate and hence thiamine is a coenzyme of BCKD enzyme, PDH, and alpha KGD enzymes.

Clinical Features:

Diagnosis:

- Newborn screening: Elevated BCAAs, particularly leucine, on tandem mass spectrometry.
- Plasma amino acid analysis: Elevated leucine, Alloisoleucine, and valine.
- Urine organic acid analysis: Presence of BCKAs and their metabolites.
- Enzyme assay: Reduced BCKD complex activity in cultured fibroblasts or lymphocytes.

Management:

- Acute management: Dietary restriction of BCAA. Careful supplementation of isoleucine and valine. Hemodialysis or hemofiltration in severe cases.

- Dietary restriction of BCAA.
- Careful supplementation of isoleucine and valine.
- Hemodialysis or hemofiltration in severe cases.
- Long-term management: Lifelong dietary restriction of BCAAs, particularly leucine. Regular monitoring of plasma amino acid levels. Thiamine supplementation (responsive in some patients). Liver transplantation may be considered in severe cases.
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- Regular monitoring of plasma amino acid levels.
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- Dietary restriction of BCAA.
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- Hemodialysis or hemofiltration in severe cases.
- Lifelong dietary restriction of BCAAs, particularly leucine.
- Regular monitoring of plasma amino acid levels.
- Thiamine supplementation (responsive in some patients).
- Liver transplantation may be considered in severe cases.

A ketogenic (keto) diet (Option A) involves high-fat and low-carbohydrate intake, which leads to the production of ketone bodies.

- In MSUD, the goal is to limit the intake of BCAAs, which cannot be adequately metabolized.
- A keto diet would not address the underlying metabolic defect and could worsen the metabolic acidosis and ketonuria seen in MSUD.

Administer phenobarbital for seizure prophylaxis (Option B):

- While seizures can occur in MSUD, prophylactic anticonvulsants are not routinely used.

Begin immediate hemodialysis (Option C):

- While hemodialysis may be necessary in severe cases unresponsive to dietary management and medical therapy, it's not the first Tom cat urine line treatment.

Solution for Question 7:

Correct Answer: A) a-4, b-3, c-1, d-2

Explanation:

Inborn Errors of Metabolisms (IEMs) and associated abnormal odors:

Inborn Error of Metabolism (IEM)	Abnormal Odor of urine
Maple Syrup Urine Disease (MSUD)	Maple syrup/ burnt sugar
Phenylketonuria (PKU)	Mousy or musty
Multiple Carboxylase Deficiency	Tomcat urine odor
Tyrosinemia (type I) and Hypermethionemia	Boiled Cabbage-like/rancid butter
Isovaleric acidemia	Sweaty feet/acrid odor
Hawkinsuria	Swimming pool odor
Trimethylaminuria	Rotting fish odor
Cystinuria	Rotten egg/ Sulphur smell

Solution for Question 8:

Correct Answer: C) Respiratory acidosis

Explanation:

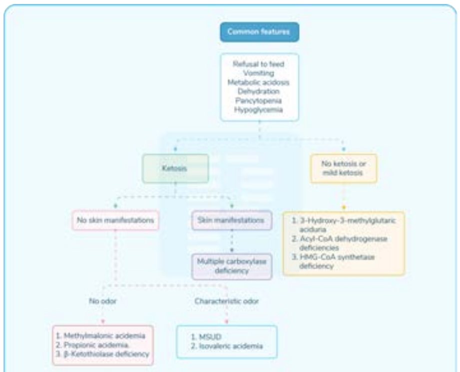
Organic acidemias are severe metabolic disorders that often present in the neonatal period with nonspecific symptoms such as lethargy, poor feeding, vomiting, and hypotonia.

- Due to the accumulation of metabolic organic acids, affected infants develop metabolic acidosis, hypoglycemia, and hyperammonemia.

Clinical Features:

- Lethargy and altered mental status (Option A)
- Vomiting and poor feeding (Option B)
- Respiratory distress (typically with compensatory respiratory alkalosis, not acidosis)
- Seizures (Option D)
- Hypotonia
- Metabolic acidosis (not respiratory acidosis)

- Hyperammonemia
- Ketosis



Clinical approach to infants with organic acidemia

Management:

1. Acute management:

   - Stabilization: Airway, breathing, circulation
   - Fluid resuscitation and correction of electrolyte imbalances
   - Cessation of protein intake for 24-48 hours
   - Glucose infusion to promote anabolism
   - Correction of acidosis with sodium bicarbonate if severe

2. Chronic management:

   - Dietary protein restriction
   - Supplementation with specific amino acids or vitamins based on the specific disorder
   - Carnitine supplementation in some cases
   - Regular monitoring of growth, development, and metabolic parameters.

Solution for Question 9:

Correct Answer: C) Urine succinylacetone test

Explanation:

- The infant’s symptoms including prolonged jaundice, vomiting, failure to thrive, hepatomegaly, a distended abdomen, elevated liver enzymes, a low platelet count, and a distinctive "cabbage-like" odor to the urine suggest a diagnosis of tyrosinemia type I.

- Succinylacetone test: The presence of high succinylacetone in the blood or urine is a hallmark of tyrosinemia type 1 and confirms the diagnosis and it is responsible for hepatorenal failure and HCC.

Types of Tyrosinemia:

Types		
Type I	Type II	Type III
Most common Due to deficiency of the enzyme Fumarylacetoacetate hydroxylase (FAH) FAA accumulates and converted to succinylacetone Cabbage odor urine	Due to deficiency of enzyme tyrosine aminotransferase Clinical features: Palmar or plantar hyperkeratosis, corneal ulcers, and intellectual disability	Least common Due to deficiency of enzyme 4-HPPD (4- Hydroxy Phenyl Pyruvate Dioxygenase) Also known as Hawkinsuria Swimming pool chlorine odor urine

Tyrosinemia refers to a group of inherited metabolic disorders characterized by the body's inability to properly break down the amino acid tyrosine.

Tyrosinemia Type 1:

- Cause: Autosomal recessive disorder due to a deficiency in fumarylacetoacetate hydrolase (FAH), leading to toxic succinylacetone accumulation, which harms the liver and kidneys.
- Symptoms: Infants: Hepatorenal failure and CNS toxicity.
- Infants: Hepatorenal failure and CNS toxicity.
- Diagnosis: Plasma Amino Acid Analysis: Elevated tyrosine, methionine, and phenylalanine. Alpha-Fetoprotein (AFP) Measurement: Elevated levels indicate the condition. Succinylacetone Test: High levels in blood or urine confirm the diagnosis. Genetic Testing: Identifies mutations in the FAH gene for definitive diagnosis.
- Plasma Amino Acid Analysis: Elevated tyrosine, methionine, and phenylalanine.
- Alpha-Fetoprotein (AFP) Measurement: Elevated levels indicate the condition.
- Succinylacetone Test: High levels in blood or urine confirm the diagnosis.
- Genetic Testing: Identifies mutations in the FAH gene for definitive diagnosis.
- Treatment: Nitisinone (NTBC): Blocks tyrosine breakdown to prevent succinylacetone formation. Dietary Restriction: Limits intake of phenylalanine and tyrosine. Liver Transplant: Considered if there is liver failure, cancer, or inadequate response to NTBC.
- Nitisinone (NTBC): Blocks tyrosine breakdown to prevent succinylacetone formation.
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- Infants: Hepatorenal failure and CNS toxicity.
- Plasma Amino Acid Analysis: Elevated tyrosine, methionine, and phenylalanine.
- Alpha-Fetoprotein (AFP) Measurement: Elevated levels indicate the condition.
- Succinylacetone Test: High levels in blood or urine confirm the diagnosis.
- Genetic Testing: Identifies mutations in the FAH gene for definitive diagnosis.
- Nitisinone (NTBC): Blocks tyrosine breakdown to prevent succinylacetone formation.
- Dietary Restriction: Limits intake of phenylalanine and tyrosine.
- Liver Transplant: Considered if there is liver failure, cancer, or inadequate response to NTBC.

Urine culture and sensitivity (Option A): This test is used to identify urinary tract infections, which are not indicated by the infant's symptoms or distinctive odor.

Serum ammonia levels (Option B): Elevated ammonia levels are associated with urea cycle disorders, not Tyrosinemia Type 1.

Urine glucose levels (Option D): While important in evaluating metabolic status, urine glucose levels are not specific for diagnosing Tyrosinemia Type 1.

Solution for Question 10:

Types of Glycogen Storage Disorders: Hepatic and Muscle

Type	Hepatic Glycogen Storage Disorders	Deficient enzyme
I	Von Gierke disease (most common in younger age group)	Glucose 6 Phosphatase
III	Cori disease (C affects D)	Debranching enzyme
IV	Anderson disease (A affects B)	Branching enzyme
VI	Her's disease	Hepatic phosphorylase

Type	Muscle Glycogen Storage Disorders	Deficient enzyme
II	Pompe disease (Pompe affects pump - Dilated Cardiomyopathy)	Acid maltase/α-1,4-glucosidase

V	McArdle disease (No hemolysis) (Most common GSD to present in adolescents)	Muscle phosphorylase
VII	Tarui disease (hemolysis present)	Phosphofructokinase

Solution for Question 11:

Correct Answer: A) Phenylalanine hydroxylase

Explanation:

The infant’s symptoms like failure to meet developmental milestones, eczema, a musty odor, and lighter hair and skin pigmentation are characteristic of Phenylketonuria (PKU).

- PKU is an autosomal recessive disorder caused by a defect in the enzyme phenylalanine hydroxylase (PAH).
- This enzyme deficiency disrupts the body's ability to metabolize the amino acid phenylalanine, leading to its accumulation in the body.

Clinical features:

- Severe, irreversible intellectual disability- due to less dopamine and serotonin in the brain (as excess phenylalanine prevents transport of tyrosine and tryptophan to the brain)
- Microcephaly
- Epilepsy
- Behavioral issues
- Musty odor "mousy" smell in sweat and urine, due to phenylacetate
- Eczema
- Hypopigmentation of skin and hair

Diagnosis of PKU:

- Plasma Phenylalanine Concentration: Raised phenylalanine levels
- Raised phenylalanine
- Other tests: Ferric chloride test, Guthrie test

Treatment of PKU:

- Lifelong Low-Protein Diet

- Palynziq enzyme supplementation
- Phenylalanine-Free Formula for infants and children instead of breastmilk.
- Plasma monitoring: Regular checks of phenylalanine levels to adjust treatment.
- Sapropterin supplementation based on the tetrahydrobiopterin (BH4) responsiveness, where BH4 is the coenzyme of PAH.

Deficiency in tyrosinase (Option B) causes albinism.

- It does not typically present with a musty odor or developmental delays.

Deficiency in tyrosine aminotransferase (Option C) can cause tyrosinemia type 2, which primarily affects the eyes and skin but does not present with a musty odor or developmental delays

Deficiency in 4-HPPD (Option D) is less common and typically does not include the musty odor.

- It causes tyrosinemia type- 3 (Hawkinsuria) with a swimming pool (chlorine) odor.

Solution for Question 12:

Correct Answer: B) Tyrosine aminotransferase

Explanation:

The phenylalanine pathway is crucial for our body, and deficiencies in any of its enzymes can result in serious health disorders like:

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Phenylketonuria	Caused due to the deficiency of phenylalanine hydroxylase causing accumulation of phenylalanine in the body Accumulation of phenylalanine will cause brain damage, and CNS manifestations and gets converted into metabolites like phenylacetate & phenylpyruvate which can be detected in urine and blood.
Albinism	Caused due to the deficiency of the enzyme tyrosinase
Tyrosinemia	Caused due to the deficiency of any of the 3 enzymes: Tyrosine aminotransferase Fumarylacetoacetae hydrolase 4 HPPD (Hydroxy phenyl pyruvate dioxygenase)

Alkaptonuria	Caused due to the deficiency of homogentisic acid oxidase

Solution for Question 13:

Correct Answer: B) Hereditary fructose intolerance

Explanation:

The infant's symptoms began after the introduction of sugars. (fructose or sucrose).

- The sweet odor of the diapers suggests the presence of reducing sugars in the urine, such as fructose.
- Hepatomegaly is consistent with the accumulation of fructose-1-phosphate in the liver, a key feature of hereditary fructose intolerance.

Hereditary fructose intolerance:

- It is a metabolic disorder characterized by the body's inability to properly break down fructose from a deficiency in the enzyme aldolase B.

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- The deficiency of aldolase B leads to the accumulation of fructose-1-phosphate in the liver, which leads to liver toxicity, leading to the clinical features of the disorder.

Clinical features:

- Nausea, Vomiting, Poor feeding, Lethargy, and Jaundice occur following the introduction of weaning foods containing fructose into the diet
- On physical examination: Abdominal pain Hepatomegaly Chronic growth restriction.
- Abdominal pain
- Hepatomegaly
- Chronic growth restriction.
- Abdominal pain
- Hepatomegaly
- Chronic growth restriction.

Initial Evaluation:

- Test for reducing substances in urine(Positive Benedict and Selewinoff test for ketose sugar)
- Dipstick test for glucose is usually negative (Negative glucose oxidase test)

Confirmatory Testing:

- Genetic Testing
- Measurement of Aldolase Activity: Performed via liver biopsy sample
- Performed via liver biopsy sample
- Performed via liver biopsy sample

Management:

- Complete elimination of foods that contain fructose and, sucrose is essential.
- Advisable to use a multivitamin supplement

Phenylketonuria (Option A):

- Caused by a deficiency in phenylalanine hydroxylase, leading to elevated phenylalanine.
- Symptoms include developmental delay and a musty odor in urine, not a sweet odor.
- Typically, symptoms are not triggered specifically by formula feeding.

Galactosemia (Option C):

- Due to a deficiency in galactose-1-phosphate uridyltransferase.
- Symptoms present very early within days of birth when compared to HFI and include poor feeding, jaundice, and hepatomegaly.
- Very early presentation within days of birth as compared to Hereditary Fructose Intolerance.

Glycogen Storage Disease Type I (Option D):

- Caused by a deficiency in glucose-6-phosphatase.
- Symptoms include hypoglycemia, growth retardation, and hepatomegaly.
- Typically does not present with a sweet odor in the diapers and is not specifically triggered by fructose in formula.

Solution for Question 14:

Correct Answer: C) Discontinue all lactose-containing foods (lactose = glucose + galactose)

Explanation: This clinical presentation of poor feeding, vomiting, jaundice, hepatomegaly, cataracts, and positive reducing substance in urine in a child less than 6 months of age is consistent with a diagnosis of galactosemia, an inborn error of galactose metabolism.

- Lactose is a disaccharide made up of glucose and galactose.
- The most critical initial step in management is to remove the source of galactose, which is lactose from milk (including breast milk) and milk products.

Ruling out Hereditary Fructose Intolerance:

- If the clinical scenario describes a child older than 6 months, Hereditary Fructose Intolerance (HFI) can be ruled out.
- HFI presents with symptoms like vomiting, lethargy, seizures, and hypoglycemia after introduction of fructose-containing foods.
- Cataracts are not a feature of HFI but strongly indicate galactosemia, due to accumulation of galactitol.
- Liver findings are common in both HFI and galactosemia, but HFI symptoms arise after fructose intake.
- Renal tubular acidosis or Fanconi syndrome may occur in HFI due to proximal tubular dysfunction following fructose ingestion.

Galactosemia	
Enzyme Deficiency	Galactose-1-phosphate uridylyltransferase (GALT) - Classical type Galactokinase- benign disease with only oil drop cataract. Epimerase- SNHL + symptoms of classical type (Low galactose-based diet can be given which is contraindicated in the classical type)
Pathophysiology	Accumulation of galactose-1-phosphate (toxic causing jaundice, and liver damage) Inability to metabolize galactose from milk Damage to liver, kidney, and brain
Clinical Features	Hepatic: Jaundice, hepatomegaly GI: Vomiting, diarrhea, failure to thrive CNS: Seizures, intellectual disability Eyes: Oil drop cataracts Increased risk of E. coli sepsis
Diagnosis	Reducing substance in urine (positive)(Positive Benedict test but negative glucose oxidase test) Direct enzyme (GALT) assay Genetic testing (Perinatal diagnosis is possible)

Management	Lactose-free diet Breast Milk contraindicated Soy-based or elemental formulas Calcium supplementation

Start phototherapy for jaundice (Option A): Phototherapy may help with jaundice but doesn't address the underlying metabolic problem.

Administer intravenous glucose (Option B): While intravenous glucose might help with hypoglycemia, it doesn't address the primary issue of galactose toxicity.

Begin enzyme replacement therapy (Option D): Enzyme replacement therapy is not currently available for galactosemia.

Solution for Question 15:

The infant’s hypotonia, generalized muscle weakness, feeding difficulties, and cardiomegaly, along with elevated serum creatine kinase and liver enzymes, are characteristic of Pompe disease (Glycogen storage disease type II).

- This condition is caused by a deficiency of acid α-1,4-glucosidase, leading to lysosomal glycogen accumulation.
- Here, glycogen typically accumulates within lysosomes, as opposed to its accumulation in cytoplasm in the other glycogenoses.
- While cytoplasmic glycogen can be present as the disease progresses due to lysosomal rupture, the primary site of accumulation is within lysosomes.

Pompe disease (Glycogen Storage Disease Type II)

Pompe disease (Glycogen Storage Disease Type II)	
Definition	Glycogen storage disorder caused by deficiency of acid α-1,4-glucosidase (acid maltase) (Option A)
Inheritance	Autosomal recessive
Pathophysiology	Accumulation of glycogen in lysosomes, primarily affecting cardiac, skeletal, and smooth muscle cells
Forms	Infantile Pompe disease (IPD) - Severe Late-onset Pompe disease (LOPD) - Less Severe
Clinical Features	IPD: Hypotonia, muscle weakness, feeding difficulties, hepatomegaly, hypertrophic cardiomyopathy (Option D) LOPD: Proximal limb girdle muscle weakness, respiratory muscle involvement, cardiac rhythm disturbances
Laboratory Findings	Elevated serum CK, AST, ALT, LDH Cardiomegaly on chest X-ray (infantile form) ECG changes- shortened P-R interval, tall QRS complexes
Treatment	Enzyme replacement therapy with alglucosidase alfa (Option C) High-protein diet (for LOPD)

Histopathology findings: muscle biopsy/ cardiac biopsy has vacuoles within muscle fibers which stain with PAS strongly (glycogen is pas positive, diastase sensitive)

Solution for Question 16:

On liver histology:

- Type 1 GSD (Von Gierke Disease): Distension of hepatocytes by glycogen and fat deposition
- Type 3 GSD (Cori Disease): Fibrosis (branched glycogen acts as a trigger for fibrosis); Paucity of fat

The fibrosis and paucity of fat distinguish type III glycogenosis from type I.

	Type I GSD/Von Gierke Disease	Type III GSD/Cori Disease
Kidneys	Enlarged (Option A)	Normal

Muscles involvement	Not seen (Option B)	Seen
CPK levels	Normal	Elevated
LFT	Normal or slightly elevated	SGPT, SGOT levels elevated - Transaminitis & fasting ketosis present
Lactate level	Elevated (Option C)	Normal (Usually)
Effect of glucagon	No rise in blood glucose but lactic acid levels may increase	In fed state: Increase in glucose level In fasting state: No increase in blood glucose level
Liver biopsy	Distension of hepatocytes by glycogen and fat deposition	Fibrosis; Paucity of fat

Solution for Question 17:

- The infant's symptoms are indicative of Von Gierke or Type I Glycogen Storage Disease (GSD).
- Here, glucagon administration leads to a negligible increase in blood glucose levels but causes a significant rise in lactate levels.
- This is due to the deficiency in glucose-6-phosphatase or its translocase, which impairs the conversion of glucose-6-phosphate to glucose.

Gargoylic face with hepatosplenomegaly

Von Gierke or Type I Glycogen Storage Disease	
Pathogenesis	Deficiency of glucose-6-phosphatase (Type Ia) or its translocase (Type Ib) (Option A) Impaired conversion of glucose-6-phosphate to glucose Autosomal recessive inheritance (Option B)
Clinical Features	Hepatomegaly Hypoglycemic seizures Doll-like face with fat cheeks (Gargoylic face with HSM (hepatosplenomegaly)) Short stature Protuberant abdomen Thin extremities Kidney enlargement Additionally, Type Ib: Neutropenia, recurrent infections
Biochemical Characteristics	Hypoglycemia Lactic acidosis Hyperuricemia Hyperlipidemia
Diagnosis	Clinical presentation Laboratory findings Minimal glucose response to glucagon (Negative glucagon challenge test) Gene-based variant analysis
Treatment	Maintain normal blood glucose levels by continuous nasogastric glucose infusion or uncooked cornstarch administration (Option D) Dietary restrictions (fructose, galactose) Allopurinol for hyperuricemia Lipid-lowering drugs Type Ib: G-CSF for neutropenia

Lysosomal Storage Disorders

1. A 4-year-old boy presents with recurrent epistaxis, abdominal distension, and splenomegaly. Laboratory investigations reveal anaemia, thrombocytopenia, and reduced glucocerebrosidase enzyme activity. What is the most likely diagnosis?

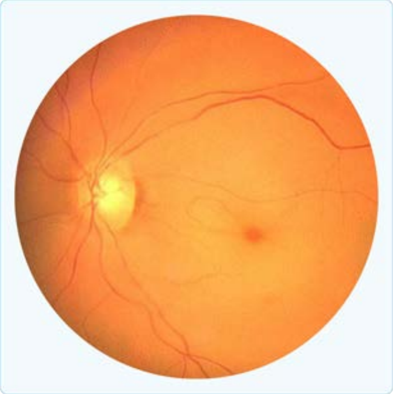
A. Krabbe disease

B. Gaucher disease

C. Fabry disease

D. Neiman pick disease

2. A 6-month-old infant presents with developmental delay, coarse facial features, and hepatosplenomegaly. Ophthalmological examination reveals a cherry-red spot on the macula. X-rays show skeletal dysplasia with vertebral abnormalities. What is the most likely diagnosis? {{caption_text}}



A. GM1 gangliosidosis

B. Niemann-Pick disease

C. Tay-Sachs disease

D. Krabbe disease

3. A 2-year-old child presents with hepatosplenomegaly, progressive neurological deterioration, and lung involvement. Foam cells are found in the bone marrow, and genetic testing reveals a mutation in the SMPD1 gene. What disease is the child suffering from?

A. Niemann-Pick Disease

B. Gaucher Disease

C. Hunter Disease

D. GM1 Gangliosidosis

4. A 7-month-old infant presents with developmental regression, loss of head control, spasticity, and an exaggerated startle response. The infant is diagnosed with Tay-Sachs disease. Which of the following regarding this condition is not correct?

A. Inherited in an autosomal recessive pattern

B. Caused by hexosaminidase A deficiency

C. Cherry-red spot on fundus

D. Associated with splenomegaly

5. Match the following types of mucopolysaccharidosis (MPS) with their corresponding enzyme deficiencies: Types of MPS Enzyme deficiencies 1.MPS I a. N-acetylgalactosamine-6-sulfatase 2.MPS II b. Iduronate-2-sulfatase 3.MPS IV c. Alpha-L-iduronidase 4.MPS VI d. Arylsulfatase B

Types of MPS	Enzyme deficiencies
1.MPS I	a. N-acetylgalactosamine-6-sulfatase

2.MPS II	b. Iduronate-2-sulfatase
3.MPS IV	c. Alpha-L-iduronidase
4.MPS VI	d. Arylsulfatase B

A. 1-c, 2-b, 3-a, 4-d

B. 1-b, 2-c, 3-d, 4-a

C. 1-a, 2-d, 3-b, 4-c

D. 1-d, 2-a, 3-c, 4-b

6. Which of the following is not a clinical feature of Hunter Syndrome (MPS-II)?

A. Coarse facial features

B. Short stature

C. Dysostosis multiplex

D. Corneal clouding

7. A 10-year-old boy presents with recurrent burning pain in his hands and feet, exacerbated by exercise. He has dark red spots on his skin, particularly on his lower back and thighs, and experiences visual disturbances. Laboratory evaluation revealed uremia. Which of the following enzyme deficiency is associated with this condition?

A. Alpha-galactosidase A

B. Iduronate sulfatase

C. Beta-galactosidase

D. N-acetylgalactosamine-4-sulfatase

8. A 3-year-old boy is diagnosed with a deficiency of hypoxanthine-guanine phosphoribosyltransferase (HGPRT) based on genetic testing. Which of the following findings is not typically associated with this condition?

A. Self-mutilation
B. Hyperuricemia
C. Intellectual disability
D. Hepatomegaly

9. A 6-month-old male infant presents with developmental delay, hypotonia, and recurrent myoclonic seizures. On examination, he is found to have hypopigmented skin, kinky hair and a distinctive facial appearance with a depressed nasal bridge. Which of the following is true regarding this condition?

A. Deficiency of the enzyme lysyl oxidase
B. Mutations involving the ATP7B gene
C. Treatment with copper-histidinase
D. Primarily affects females

10. A 3-month-old infant presents with abdominal distension, failure to thrive, and hepatosplenomegaly. Laboratory tests reveal severe adrenal insufficiency and elevated cholesteryl esters. A liver biopsy shows lipid accumulation in lysosomes. Which genetic mutation is associated with this condition?

A. LIPA gene mutation
B. ATP7A gene mutation
C. SLC22A4 gene mutation
D. GBA gene mutation

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	1
Question 4	4
Question 5	1
Question 6	4
Question 7	1
Question 8	4
Question 9	3
Question 10	1

Solution for Question 1:

Correct Answer: B) Gaucher disease

Explanation:

- The given findings are characteristic of Gaucher disease, an autosomal recessive lysosomal storage disorder caused by mutations in the GBA gene, which encodes the enzyme glucocerebrosidase.
- This enzyme deficiency leads to the accumulation of glucocerebroside (glucosylceramide) within lysosomes of macrophages, forming characteristic Gaucher cells.
- Clinical Features: Hematologic: Pancytopenia Skeletal: Bone crisis, severe bone pain. Neurological: Parkinsonism, cognitive impairment, neuropathy .
- Hematologic: Pancytopenia
- Skeletal: Bone crisis, severe bone pain.

- Neurological: Parkinsonism, cognitive impairment, neuropathy
- .
- Investigations: Bone marrow examination: Shows Gaucher cells with a characteristic wrinkled tissue paper appearance. Gaucher cells are macrophages accumulated with glucocerebroside Enzyme assay: Demonstrates reduced activity of glucocerebrosidase in leukocytes or fibroblasts. Long bone X-rays may show the Erlenmeyer flask deformity.
- Bone marrow examination: Shows Gaucher cells with a characteristic wrinkled tissue paper appearance. Gaucher cells are macrophages accumulated with glucocerebroside
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- Enzyme assay: Demonstrates reduced activity of glucocerebrosidase in leukocytes or fibroblasts.
- Long bone X-rays may show the Erlenmeyer flask deformity.
- Hematologic: Pancytopenia
- Skeletal: Bone crisis, severe bone pain.
- Neurological: Parkinsonism, cognitive impairment, neuropathy
- .
- Bone marrow examination: Shows Gaucher cells with a characteristic wrinkled tissue paper appearance. Gaucher cells are macrophages accumulated with glucocerebroside
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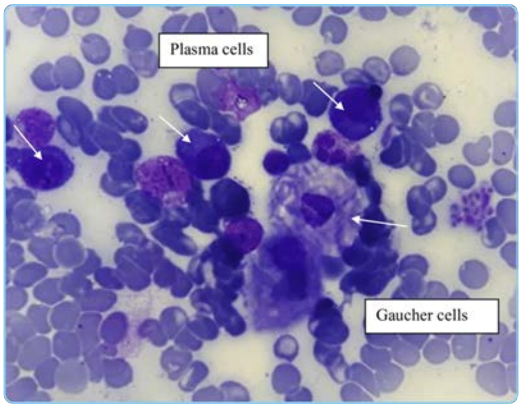
- Enzyme assay: Demonstrates reduced activity of glucocerebrosidase in leukocytes or fibroblasts.
- Long bone X-rays may show the Erlenmeyer flask deformity.
- Gaucher cells are macrophages accumulated with glucocerebroside
- Management: Enzyme replacement therapy (ERT): Imiglucerase, velaglucerase alfa, or taliglucerase alfa Substrate reduction therapy: Miglustat or eliglustat Supportive care: Blood transfusions, pain management, orthopaedic interventions
- Enzyme replacement therapy (ERT): Imiglucerase, velaglucerase alfa, or taliglucerase alfa
- Substrate reduction therapy: Miglustat or eliglustat
- Supportive care: Blood transfusions, pain management, orthopaedic interventions
- Enzyme replacement therapy (ERT): Imiglucerase, velaglucerase alfa, or taliglucerase alfa
- Substrate reduction therapy: Miglustat or eliglustat
- Supportive care: Blood transfusions, pain management, orthopaedic interventions



Krabbe disease(OptionA) is due to the deficiency of galactocerebrosidase enzyme leading to peripheral neuropathy, destruction of oligodendrocytes, developmental delay, and optic atrophy.

Fabry disease (Option C) is due to the deficiency of alpha-galactosidase A leading to angiokeratomas, hyperhidrosis, corneal or lenticular opacities, acroparesthesias, severe and debilitating pain due to involvement of nerves.

Niemann pick disease (Option D) Type A and B results from the deficiency of enzyme acid sphingomyelinase.

- Type A is a rapidly progressive neurodegenerative disorder with the clinical features of hepatosplenomegaly, lymphadenopathy, and death usually by 3 years of age.

Solution for Question 2:

Correct Answer: A) GM1 gangliosidosis

Explanation:

- The given scenario is highly suggestive of GM1 gangliosidosis, a rare inherited lysosomal storage disorder caused by deficiency of the enzyme beta-galactosidase resulting in accumulation of GM1 gangliosides in various tissues, particularly in the central nervous system.
- Types: Three main types based on age of onset: Infantile, Juvenile, and Adult.
- Clinical manifestations:

Progressive neurodegeneration	Coarse facial features
Developmental delay or regression	Cherry-red spot on the macula
Seizures	Skeletal abnormalities
Visceromegaly (enlarged liver and spleen)	Hypotonia

- Diagnosis: Reduced beta-galactosidase activity and GLB1 gene mutations with characteristic clinical features with skeletal abnormalities observed on X-rays.
- Treatment: Primarily supportive care; enzyme replacement and gene therapy are being explored as potential treatments.

Niemann-Pick disease (Option B) presents with hepatosplenomegaly, lung involvement, and neurological deterioration but typically lacks progressive neuropathy and neuro-regression causing intellectual disability seen in GM1 gangliosidosis.

Tay-Sachs disease (Option C) features a cherry-red spot on the macula and developmental regression but lacks significant visceromegaly or skeletal abnormalities.

Krabbe disease(Option D) is due to the deficiency of galactocerebrosidase enzyme leading to peripheral neuropathy, destruction of oligodendrocytes, developmental delay, and optic atrophy.

Solution for Question 3:

Correct Answer: A) Niemann-Pick Disease

Explanation:

- Niemann-Pick Disease is a severe form that typically presents in early childhood with neurological decline, organ enlargement, and lipid-laden foam cells due to a deficiency in sphingomyelinase caused by mutations in the SMPD1 gene.
- It is an autosomal recessive disorder of sphingomyelin and cholesterol in the lysosomes.

Types:

- In the classical form (type A): clinical features begin in early life with feeding difficulties, failure to thrive and developmental delay and later neuroregression.
- There is a protuberant abdomen with hepatosplenomegaly.
- Cherry-red spot on fundus examination is seen in about half the cases.
- Diagnosis is confirmed by measurement of acid sphingomyelinase (ASM) levels.

Type B disease : is a milder form with hepatosplenomegaly but no neurological involvement.

Type C disease : has variable age at onset, organomegaly, developmental delay, vertical supranuclear gaze palsy, extrapyramidal manifestations and psychiatric features depending upon the age of onset.

Gaucher Disease (Option B) :

- This is the commonest autosomal recessively inherited lysosomal storage disease. It occurs due to the deficiency of the tissue enzyme glucocerebrosidase
- It is characterized by visceral (hepatosplenomegaly), and bone marrow involvement leading to anemia, thrombocytopenia, leukopenia, bone pains and fractures.
- There may be associated neurological symptoms like developmental delay, seizures and ocular involvement.

Hunter Disease (Option C) :

- Hunter syndrome is a genetically associated lysosomal storage disorder due to the deficiency of the iduronate 2-sulfatase enzyme (IDS).
- It is an X-linked recessive disorder and occurs predominantly in males. Iduronate 2-sulfatase (IDS) is responsible for the breakdown of large sugar molecules called glycosaminoglycans.
- Decreased activity of IDS results in intracellular and extracellular accumulation of heparan sulfate (HS) and dermatan sulfate (DS) in multiple organs of the body.

GM1 gangliosidosis (Option D) :

- It is a rare inherited lysosomal storage disorder caused by deficiency of the enzyme beta-galactosidase resulting in accumulation of GM1 gangliosides in various tissues, particularly in the central nervous system.

Solution for Question 4:

Correct Answer: D) Associated with splenomegaly.

Explanation:

Organomegaly is not seen in Tay-Sachs disease.

Tay-Sachs Disease:

Features	Description
Inheritance	Autosomal recessive (Option A)
Cause	Deficiency of hexosaminidase A enzyme (Option B)
Accumulation	GM2 ganglioside in ganglion cells and neurons

Clinical Manifestations	Initial symptoms include loss of head control, inability to sit, exaggerated startle response to sound, and a disproportionately large head. Progressive features include spasticity, deafness, and blindness. Not associated with organomegaly.
Fundus Finding	Cherry-red spot (Option C) is caused by GM2 ganglioside accumulation in the ganglion cells of the retina, creating a contrast with the normal red fovea.

Solution for Question 5:

Correct Answer: A) 1-c, 2-b, 3-a, 4-d

Explanation:

MPS Type	MPS I (Hurler syndrome)	MPS II (Hunter syndrome)	MPS IV (Morquio syndrome)	MPS VI (Maroteaux-Lamy syndrome)
Enzyme Deficiency	Alpha-L Iduronidase MPS 1= Alpha LID	Iduronate-2-sulfatase	N-acetylgalactosamine-6-sulfatase	Arylsulfatase B
Gene	IDUA	IDS	GALNS (Type A) GLB1 (Type B)	ARSB
Inheritance	Autosomal recessive	X-linked recessive (Males- Hunters- No corneal clouding (Clear vision for hunting))	Autosomal recessive	Autosomal recessive
Primary GAGs Accumulated	Heparan sulfate, Dermatan sulfate	Heparan sulfate, Dermatan sulfate	Keratan sulfate, Chondroitin-6-sulfate	Dermatan sulfate

Types	Three subtypes: Hurler Syndrome: Severe form, Type Ia. Hurler-Scheie Syndrome: Intermediate form, Type Ia. Scheie Syndrome: Attenuated form, Type Ib.		Two subtypes: Type A (more common) Type B	
Features	Doll like facies Developmental delay Organomegaly Skeletal abnormalities Corneal clouding	Primarily affects males It is the same clinically as Hurler except there is no corneal clouding and peppble-like skin present.	Skeletal dysplasia Short stature Joint hypermobility Corneal clouding Preserved intelligence	Variable progression rate Coarse facial features Skeletal abnormalities Heart valve problems Normal intelligence

Solution for Question 6:

Correct Answer: D) Corneal clouding

Explanation:

Corneal clouding, is not a clinical feature of Hunter Syndrome and is instead a distinguishing feature of Hurler Syndrome.

Feature	Hunter Syndrome (MPS-II)	Hurler Syndrome (MPS-I)
Gene/Enzyme Deficiency	Iduronate sulfatase (IDS)	α-L-iduronidase
Inheritance	X-linked recessive	Autosomal recessive
Corneal Clouding	Absent	Present
Coarse Facial Features	Present	Present
Short Stature		
Dysostosis Multiplex		

Joint Stiffness		
Intellectual Disability	Mild to moderate, usually less severe than Hurler syndrome	Severe
Hepatosplenomegaly	Present	Present
Skeletal Dysplasia		
Hearing Loss		
Life Expectancy	Variable, often longer than Hurler syndrome	Usually fatal before 10 years of age
	 ... (content truncated)	 ... (content truncated)

Solution for Question 7:

Correct Answer: A) Alpha-galactosidase A

Explanation:

The symptoms described in the case scenario (burning pain, angiokeratomas, hypohidrosis, and visual disturbances) are characteristic of Fabry disease, which is due to the deficiency of alpha-galactosidase A.

Table rendering failed

Iduronate sulfatase (Option B) deficiency leads to Hunter syndrome, characterised by developmental delays and skeletal abnormalities but not burning pain or visual disturbances.

Beta-galactosidase(Option C) deficiency leads to GM1 gangliosidosis, a rare inherited lysosomal storage disorder resulting in the accumulation of GM1 gangliosides in various tissues, particularly in the central nervous system leading to progressive neurodegeneration, developmental delay or regression, and seizures.

N-acetylgalactosamine-4-sulfatase (Option D) deficiency is associated with Maroteaux-Lamy syndrome, marked by skeletal abnormalities and organomegaly, but not burning pain or visual disturbances, rather than the symptoms described in the scenario..

Solution for Question 8:

Correct Answer: D) Hepatomegaly.

Explanation:

Self-mutilation, hyperuricemia, and intellectual disability are all typical findings in Lesch-Nyhan syndrome, whereas hepatomegaly is not commonly associated with this condition.

Table rendering failed

Solution for Question 9:

Correct Answer: C) Treatment with copper-histidinase

Explanation:

- The given clinical presentation is characteristic of Menke’s disease, also known as Menke’s kinky hair syndrome, caused by a deficiency of the copper-transporting ATPase enzyme, leading to impaired copper absorption and transport.
- Early treatment with copper-histidinase can improve outcomes if started within the first six weeks of life.

Menkes Disease (Kinky Hair Disease)	
Inheritance	X-linked recessive and primarily affects males (Option D)

Affected Gene	ATP7A (codes for copper-transporting P-type ATPase)
Gene Location	Xq21.1
Pathophysiology	Deficiency in copper absorption and transport, copper accumulation in some tissues and deficiency in others
Clinical Features	Hypothermia Hypotonia Generalised myoclonic seizures Distinctive facies (chubby cheeks, depressed nasal bridge) Kinky, colourless, brittle hair (trichorrhexis nodosa, pili torti) Hypopigmentation Feeding difficulties (failure to thrive) Severe cognitive impairment (optic atrophy) Neuropathologic changes (tortuous degeneration of grey matter, cerebellar changes)
Severity	Progressive and often fatal by age three if untreated.
Treatment	Early initiation of copper-histidinase therapy within the first six weeks of life.

Deficiency of the enzyme lysyl oxidase (Option A) affects collagen cross-linking and leads to connective tissue disorders like Ehlers-Danlos syndrome.

Mutations in the ATP7B gene (Option B) are associated with Wilson's disease, which involves copper accumulation rather than the copper transport deficiency seen in Menke's disease.

Solution for Question 10:

Correct Answer: A) LIPA gene mutation

Explanation:

- The clinical presentation is most likely indicative of Wolman disease, which is caused by a mutation in the LIPA gene.
- This leads to lysosomal acid lipase deficiency, resulting in the accumulation of cholesteryl esters and triglycerides in various tissues, including the liver.

Wolman Disease	
Inheritance	Autosomal recessive

Causing Gene	LIPA gene (located on chromosome 10q23.2-23.3)
Enzyme Deficiency	Lysosomal acid lipase (LAL)
Function of Enzyme	Breakdown of cholesteryl esters and triglycerides in endo-lysosomes
Pathophysiology	Accumulation of cholesteryl esters and triglycerides in histiocytic foam cells, leading to organ failure.
Onset of Symptoms	Typically, within the first few weeks of life
Clinical Features	Failure to thrive Relentless vomiting Abdominal distension Steatorrhea (due to lymphatic obstruction caused by lipid accumulation in the small intestine) Hepatosplenomegaly
Adrenal Insufficiency	 X-Ray showing Adrenal Calcification (Arrow) ... (content truncated)
Prognosis	Death usually occurs within the first year of life
Diagnosis	Imaging showing adrenal calcification and genetic testing for mutations in the LIPA gene.

ATP7A gene mutation (Option B) is associated with Menkes disease, not Wolman disease. ATP7A codes for a copper-transporting enzyme and mutations in this gene lead to copper deficiency and related symptoms.

SLC22A4 gene mutation (Option C) is related to various conditions such as rheumatoid arthritis and Crohn’s disease

GBA gene mutation (Option D) is linked to Gaucher disease, which is also a lysosomal storage disorder caused by glucocerebrosidase deficiency, leading to the accumulation of glucocerebrosides.

Previous Year Questions

1. A 3-year-old child immediately cries after breastfeeding, followed by vomiting and diarrhea. Which enzyme deficiency might be responsible?

A. Aldolase B

B. Phenylalanine Hydroxylase

C. Galactose-1-Phosphate Uridyl Transferase

D. Glucocerebrosidase

2. A 10-year-old boy is brought to the OPD with pain, redness, and swelling around his great toe. His serum uric acid level is 9.4 mg/dL. His mother also mentions that he frequently bites his lips and has symptoms of spasticity and dystonia. Which enzyme is most likely deficient in this condition?

A. Homocysteine oxidase

B. Dihydrothymidine oxidase

C. Hypoxanthine-guanine phosphoribosyltransferase

D. Xanthine oxidase

3. An infant presents with vomiting after feeding. A Benedict's test was performed and returned positive for a non-glucose substance. Which of the following is the most likely diagnosis?

A. Galactosemia due to GAL-1-P uridyl transferase enzyme deficiency

B. Hereditary fructose intolerance due to aldolase B deficiency

C. Glucose-6-Phosphate dehydrogenase (G6PD) deficiency

D. Galactosemia due to galactokinase enzyme deficiency

4. A floppy infant presents with hypotonia, hepatomegaly, and an X-ray showing cardiomegaly. Which of the following is the most likely diagnosis?

A. Pompe disease

B. Transposition of the Great Arteries (TGA)

C. Tetralogy of Fallot (TOF)

D. Ebstein's anomaly

5. Which enzyme deficiency causes Pompe disease?

A. Lysosomal acid alpha glucosidase

B. β-hexosaminidase-A

C. β–galactocerebrosidase

D. β-glucocerebrosidase

6. Menke's disease is a metabolic defect in the process of:

A. Zinc

B. Copper

C. Selenium

D. Iron

7. What is the most likely diagnosis for a 10-year-old child who is brought to the clinic with developmental delay and extrapyramidal symptoms and is found to have

a lens dislocation towards the inferonasal area during further examination?

A. Marfan syndrome
B. Homocystinuria
C. Alkaptonuria
D. Wilson disease

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	1
Question 4	1
Question 5	1
Question 6	2
Question 7	2

Solution for Question 1:

Correct Option C - Galactose-1-Phosphate Uridyl Transferase:

- Galactose-1-phosphate uridyl transferase is responsible for converting galactose-1-phosphate to UDP-galactose in the galactose metabolism pathway. In galactosemia, there is a deficiency of GALPUT; therefore, galactose-1-phosphate accumulates, which is toxic to the liver, kidneys, and brain.

Incorrect Options:

- Options A, B, and D are not deficient in galactosemia.

Solution for Question 2:

Correct Option C - Hypoxanthine-guanine phosphoribosyltransferase:

- Lesch-Nyhan Syndrome is a genetic disorder caused by a deficiency in the enzyme Hypoxanthine-guanine phosphoribosyltransferase. Without sufficient HGPRT, there is an accumulation of purine metabolites, leading to excessive production of uric acid. This causes gout-like symptoms, such as pain and swelling in joints. Additionally, it is associated with neurological symptoms, including spasticity and dystonia.

Incorrect Options:

- Options A, B, and D are not seen in Lesch Nyhan syndrome.

Solution for Question 3:

Correct Option A - Galactosemia due to GAL-1-P uridyl transferase enzyme deficiency:

- This is the most likely diagnosis, given the presentation of vomiting after feeding and a positive Benedict's test for a non-glucose substance, consistent with the presence of galactose in the urine.

Incorrect Options:

Option B - Hereditary fructose intolerance due to aldolase B deficiency:

- Hereditary fructose intolerance results from a deficiency in aldolase B, leading to the accumulation of fructose-1-phosphate after ingestion of fructose, sucrose, or sorbitol. Symptoms include vomiting, hypoglycemia, and liver damage. The Benedict's test would show a positive result for fructose, but this condition typically presents after the introduction of fructose-containing foods, not in the newborn period.

Option C - Glucose-6-Phosphate dehydrogenase (G6PD) deficiency:

- G6PD deficiency leads to hemolytic anemia, particularly after exposure to certain triggers like infections or certain foods/drugs. This condition does not typically cause a positive Benedict's test for a non-glucose substance, nor does it present with vomiting after feeding.

Option D - Galactosemia due to galactokinase enzyme deficiency:

- This milder form of galactosemia results from a deficiency in galactokinase, leading to the accumulation of galactose, which can cause cataracts. However, it usually does not cause severe systemic symptoms like those seen in GAL-1-P uridyl transferase deficiency, and the Benedict's test would still be positive for galactose.

Solution for Question 4:

Correct Option A - Pompe disease:

- Pompe disease is a lysosomal storage disorder characterized by the deficiency of the enzyme acid alpha-glucosidase. This leads to the accumulation of glycogen in various tissues, including the heart, liver, and muscles. Clinical features include hypotonia (floppiness), cardiomegaly (enlarged heart), and hepatomegaly. The presence of a "floppy infant" with these findings strongly suggests Pompe disease as the correct diagnosis.

Incorrect Options:

Option B - Transposition of the Great Arteries (TGA):

- TGA is a congenital heart defect where the two main arteries leaving the heart are switched (transposed). While TGA can cause cardiomegaly and cyanosis, it does not typically present with hypotonia or hepatomegaly, making it less likely in this case.

Option C - Tetralogy of Fallot (TOF):

- TOF is another congenital heart defect involving four anatomical abnormalities. It presents with cyanosis and a heart murmur but does not typically cause hypotonia, hepatomegaly, or significant cardiomegaly on X-ray, making it an unlikely diagnosis here.

Option D - Ebstein's anomaly:

- Ebstein's anomaly is a congenital heart defect that involves the abnormal development of the tricuspid valve. It can lead to cyanosis and heart failure, but it does not typically present with hypotonia, hepatomegaly, or significant cardiomegaly seen in this case.

Solution for Question 5:

Correct Option A - Lysosomal acid alpha glucosidase:

- Pompe disease, also known as Glycogen Storage Disease Type II, is caused by a deficiency of the enzyme lysosomal acid alpha glucosidase (also called acid maltase or acid alpha-glucosidase).

- This enzyme is responsible for breaking down glycogen, a complex sugar molecule, into glucose within the lysosomes of cells.
- In Pompe disease, the deficiency or malfunction of lysosomal acid alpha glucosidase leads to the accumulation of glycogen within various tissues and organs, particularly in skeletal and cardiac muscles.
- This buildup of glycogen impairs the normal function of these tissues, resulting in the signs and symptoms associated with Pompe disease.

Incorrect Options:

Option B - β-hexosaminidase-A: Deficiency of this enzyme is seen in Tay-Sachs disease.

Option C - β-galactocerebrosidase: Deficiency of this enzyme is seen in Krabbe disease.

Option D - β-glucocerebrosidase: Deficiency of this enzyme is seen in Gaucher disease.

Solution for Question 6:

Correct Option B - Copper:

- Menke's disease is a rare X-linked recessive disorder characterized by impaired copper absorption and transport.
- It is caused by mutations in the ATP7A gene that regulates the metabolism of copper in the body.
- As a result, copper metabolism is disrupted, leading to copper deficiency and subsequent neurological and developmental abnormalities.

Incorrect Options:

Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 7:

Correct Option B - Homocystinuria:

- Homocystinuria is an autosomal recessive disorder characterized by impaired metabolism of the amino acid methionine.
- It is caused by deficiencies in enzymes involved in the conversion of methionine to cysteine.
- The dislocation of the lens (ectopia lentis) is a characteristic feature of homocystinuria.
- Additionally, patients may present with developmental delays, intellectual disability, and extrapyramidal symptoms.

Incorrect Options:

Option A - Marfan syndrome:

- Marfan syndrome is a connective tissue disorder caused by mutations in the FBN1 gene.
- While it can also present with ectopia lentis, it is not associated with the developmental delays and extrapyramidal symptoms.

Option C - Alkaptonuria:

- Alkaptonuria is a metabolic disorder characterized by the accumulation of homogentisic acid.
- It does not typically present with dislocated lenses or extrapyramidal symptoms.

Option D - Wilson disease:

- Wilson disease is a disorder of copper metabolism characterized by the accumulation of copper in various organs, particularly the liver and brain.
- It does not typically present with ectopia lentis or developmental delays.

Primary Immunodeficiency and Vasculitis in Children

1. A 6-month-old male infant presents with recurrent bacterial infections, including otitis media and pneumonia. Laboratory tests reveal very low levels of immunoglobulins (IgG, IgA, and IgM) and an absence of B cells (CD19+ cells) in the peripheral blood. What is the most likely diagnosis?

A. Severe Combined Immunodeficiency (SCID)

B. DiGeorge Syndrome

C. Common Variable Immunodeficiency (CVID)

D. Bruton's Disease

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2. A 5 year old girl presents with a history of recurrent sinopulmonary infections, over the past several years. She also reports chronic diarrhea and has had several episodes of unexplained weight loss. On examination her lymph nodes are not palpable and sarcoid like granuloma seen. Laboratory tests reveal low serum immunoglobulin levels (IgG, IgA, and IgM). Which of the following is the most likely diagnosis?

A. Severe Combined Immunodeficiency

B. Common Variable Immunodeficiency

C. IgA Deficiency

D. Hyper IgM Syndrome

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3. Which of the following values of serum IgA is typically indicative of selective IgA deficiency?

A. < 5 mg/dL

B. < 10 mg/dL

C. < 15 mg/dL

D. < 20 mg/dL

4. A 6-year-old boy presents with recurrent infections, including pneumocystis pneumonia, and has a history of chronic diarrhea. Laboratory tests reveal elevated serum IgM levels and low levels of IgG and IgA. Which of the following genetic mutations is most commonly associated with this condition?

A. RAG1

B. CD40L

C. IL-12Rβ1

D. ADA

5. A child presents with recurrent respiratory infections, but his total IgG level is normal. Testing reveals a low IgG2 subclass level. What is the most appropriate management strategy?

A. Immediate initiation of intravenous immunoglobulin (IVIG) therapy

B. Administration of live vaccines

C. Prophylactic antibiotics and tailored vaccination schedule

D. Initiation of corticosteroid therapy

6. A 6-month-old male infant presents with recurrent ear infections, eczema, and petechiae. He has a history of prolonged bleeding from the umbilical stump. His complete blood count reveals thrombocytopenia with small platelets. Immunoglobulin levels show low IgM and elevated IgA and IgE. What is the most

likely diagnosis?

A. Severe Combined Immunodeficiency (SCID)
B. Henoch schonlein purpura
C. Wiskott-Aldrich Syndrome
D. Ataxia-Telangiectasia

7. A 10-year-old child presents with unsteady gait, poor coordination, and visible dilated blood vessels on the sclera. The child's history includes recurrent respiratory infections and a family history of a similar condition. What is the most likely diagnosis?

A. Friedreich's Ataxia
B. Ataxia-Telangiectasia
C. Wiskott-Aldrich Syndrome
D. None of the above

8. A 3-month-old infant presents with chronic diarrhea, failure to thrive and recurrent respiratory infections. The physical examination reveals underdeveloped lymph nodes and absent tonsils. A complete blood count shows profound lymphopenia. What is the next best step in the management of this patient?

A. Start intravenous immunoglobulin (IVIG) therapy
B. Perform genetic testing for IL2RG mutation
C. Initiate broad-spectrum antibiotics
D. Schedule bone marrow biopsy

9. A 5-year-old has a history of chronic eczema and frequent infections. On investigation, he was found to have STAT3 gene mutation. Which of the following is the most likely mode of inheritance?

A. Autosomal dominant

B. Autosomal recessive

C. X linked dominant

D. X linked recessive

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10. An infant is being investigated for DiGeorge syndrome. Which of the following is a feature that is not likely to be seen in DiGeorge syndrome?

A. Thymic aplasia

B. Cleft palate

C. Antimongoloid slant

D. Long philtrum

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11. A 5-year-old male has been suffering from pneumonia and lymphadenitis since early infancy. He has been recently diagnosed with chronic granulomatous disease. Which of the following is the enzyme defect seen in the disease?

A. NADPH oxidase

B. Lactate dehydrogenase

C. Pyruvate dehydrogenase

D. Glucose-6-phosphate dehydrogenase

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12. A 6-month-old infant with recurrent skin infections and pneumonia is diagnosed with Leukocyte Adhesion Deficiency (LAD) due to a defect in selectins.

Which type of LAD does the infant have?

A. LAD I

B. LAD II

C. LAD III

D. LAD IV

13. A 9-year-old boy is evaluated for recurrent severe infections, partial albinism, and progressive neurological decline. Family history reveals consanguinity. Genetic testing identifies a mutation in the LYST gene. Which of the following findings is most consistent with the diagnosis?

A. Elevated serum IgE levels and eczema

B. Presence of large peroxisomes in the blood smear

C. Detection of large granules in neutrophils on Wright-stained blood films

D. Low NADPH oxidase levels with hepatosplenomegaly

14. A 9-year-old child presents with a finding, as in the image. He has joint pain, abdominal pain, and blood in the urine. Laboratory tests reveal normal platelet counts. Which of the following statements about condition is true?

A. Females are most commonly affected

B. Palpable purpura predominantly on the trunk and arms

C. Palpable purpura predominantly on lower limbs

D. Acute liver failure is a complication.

15. A 4-year-old child presents with a high fever, unresponsive to antipyretics,for 6 days. The child also has a polymorphous rash, swelling of the hands and feet, bilateral conjunctivitis without purulent discharge, and strawberry tongue. Laboratory tests show elevated CRP. What is the most appropriate next step in the management of this child?

A. Initiate Ceftriaxone for a suspected bacterial infection.

B. Perform an echocardiogram to assess for coronary artery involvement.

C. Start high-dose intravenous immunoglobulin (IVIG) and high-dose aspirin.

D. Start corticosteroids

16. A 9-year-old boy presents with progressive proximal muscle weakness, difficulty in climbing stairs, Physical findings as below in the image. Which of the following tests is crucial for a definitive diagnosis for the condition?

A. Skin biopsy

B. Muscle Biopsy

C. Antinuclear Antibody (ANA) Test

D. Electromyography (EMG)

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	1
Question 4	2
Question 5	3
Question 6	3
Question 7	2
Question 8	2
Question 9	1
Question 10	4
Question 11	1
Question 12	2
Question 13	3
Question 14	3
Question 15	3
Question 16	2

Solution for Question 1:

Correct Answer: D) Bruton's Disease

Explanation:

Brutons’s Disease

- Also known as X-linked agammaglobulinemia (XLA), caused by mutations in the BTK (Bruton’s tyrosine kinase) gene present on the q22 long arm of the X chromosome.
- Defective B-cell maturation leads to an absence of mature B cells and very low levels of all immunoglobulins (IgG, IgA, IgM).
- Clinical features: Presents after 6-9 months of age. Recurrent bacterial infections (eg, encapsulated bacteria like Streptococcus pneumonia, Haemophilus influenzae) starting around 6 months of age, including otitis media, pneumonia, and sinusitis, Neutropenia and enterovirus infections.
- Diagnosis: Absence of B cells (CD19+ cells) in peripheral blood and Markedly low immunoglobulin levels No germinal centres in lymph nodes
- Absence of B cells (CD19+ cells) in peripheral blood and
- Markedly low immunoglobulin levels
- No germinal centres in lymph nodes
- Absence of B cells (CD19+ cells) in peripheral blood and
- Markedly low immunoglobulin levels
- No germinal centres in lymph nodes

Severe Combined Immunodeficiency (SCID) (Option A): Affects both T-cells and B-cells and causes severe infections early in life but not limited to B-cell deficiency as in Bruton's disease.

DiGeorge Syndrome (Option B): Involves T-cell deficiency due to thymic hypoplasia and presents with congenital heart defects and facial anomalies, but B-cell numbers are typically normal.

Common Variable Immunodeficiency (CVID) (Option C): Usually presents later in life (adolescence or adulthood) with B cells present but dysfunctional and does not involve an absence of B cells like in Bruton's disease.

Solution for Question 2:

Correct Answer: B) Common Variable Immunodeficiency (CVID)

Explanation:

Common Variable Immunodeficiency (CVID)

- CVID is a primary immunodeficiency disorder characterized by hypogammaglobulinemia where IgG must be <2 standard deviation below the age-adjusted norms with low IgA or IgM levels.
- Genes known to produce CVID phenotype when mutated include ICOS deficiency, SH2DIA, CD19, CD20, CD21, BAFF- R, TACI.
- Both B and T cells are affected.
- Can present at any age, but most commonly diagnosed in late childhood or early adulthood (age 20-40).

Clinical Features:

- Sinopulmonary infections (e.g., pneumonia, bronchitis, sinusitis, otitis media) are particularly common due to impaired antibody response.
- Chronic diarrhea, malabsorption, and weight loss are common,
- Increased risk for autoimmune diseases.
- Higher risk of developing lymphoma and Non-infectious granulomas can occur in various organs, including the lungs, liver, and spleen.

Diagnosis:

- Markedly low levels of IgG and IgA, and sometimes IgM.

Severe Combined Immunodeficiency (SCID) (Option A): Severe immunodeficiency that typically presents in infancy with life-threatening infections due to profound T-cell dysfunction, not in adulthood with isolated respiratory symptoms.

IgA Deficiency (Option C): Typically presents with isolated low IgA levels, while IgG and IgM remain normal, unlike CVID, which involves multiple immunoglobulin deficiencies.

Hyper IgM Syndrome (Option D): It presents in early childhood with recurrent infections and is characterized by elevated or normal IgM with low IgG and IgA.

Solution for Question 3:

Correct Answer: A) < 5 mg/dL

Explanation:

Selective IgA Deficiency

- Isolated absence or near absence(< 5 mg/dL) of serum and secretory IgA is the most common well-defined immunodeficiency disorder.
- Patients with Selective IgA deficiency also report IgG2 subclass deficiency

Clinical features:

- Recurrent infections, particularly of the respiratory and gastrointestinal tracts. Intestinal giardiasis is common.
- They may also have increased susceptibility to allergies and autoimmune conditions.
- Non-hemolytic transfusion reaction.

Diagnosis:

- Diagnosis cannot be made until 4 years of age, when IgA levels should be mature to adult levels.
- Based on serum immunoglobulin levels showing low IgA with normal IgG and IgM levels.

Solution for Question 4:

Correct Answer: B) CD40L

Explanation:

The above features are suggestive of Hyper IgM syndrome due to mutation in CD40 Ligand gene.

X-Linked Hyper IgM Syndrome	Autosomal Recessive Hyper IgM Syndrome
Mutation in the CD40 ligand on T cells (CD40L) gene	Mutations in Activation Induced Cytidine Deaminase(AICDA) and CD40 gene
X-linked recessive; predominantly affects males.	Autosomal recessive; affects both males and females equally
Deficiency in CD40L impairs B cell and T cell communication, affecting immunoglobulin class switching.	Defects in class switch recombination due to mutations in AICDA and CD40 gene.
Recurrent bacterial infections, opportunistic infections, autoimmune disorders and increased cancer risk. Decreased antigen-specific T-cell function. Marked susceptibility to P. jiroveci pneumonia. Decreased frequency of CD27+ memory B cells.	Recurrent infections, Polyarthritis, Autoimmune hepatitis, Crohn’s disease and hemolytic anemia. Do not have susceptibility to P. jiroveci pneumonia. Presence of lymphoid hyperplasia.

Elevated IgM, low IgG, IgA, and IgE; genetic testing for CD40L mutations.	Elevated IgM, low IgG, IgA, and IgE; genetic testing for mutations in CD40, AICDA.
Immunoglobulin (IVIG) replacement therapy, prophylactic antibiotics.	Immunoglobulin (IVIG) replacement therapy, prophylactic antibiotics.

RAG1 (Option A): Mutations in the RAG1 gene are associated with Severe Combined Immunodeficiency (SCID) and are crucial for the development of functional B and T lymphocytes.

IL-12Rβ1 (Option C): Mutations in the IL-12 receptor β1 chain are associated with Mendelian Susceptibility to Mycobacterial Disease (MSMD) is involved in the immune response (granuloma formation) to intracellular pathogens like mycobacteria.

ADA (Option D): Adenosine deaminase (ADA) deficiency is a cause of SCID, as the enzyme is essential for the purine salvage pathway and lymphocyte survival.

Solution for Question 5:

Correct Answer: C) Prophylactic antibiotics and tailored vaccination schedule

Explanation:

Prophylactic antibiotics and tailored vaccination schedule is appropriate for managing IgG subclass deficiency to prevent infections and monitor vaccine responses.

Table rendering failed

- Immediate initiation of intravenous immunoglobulin (IVIG) therapy (Option A): IVIG is typically reserved for those with broader antibody deficiencies or severe infections, not for isolated IgG subclass deficiency without evidence of a broader deficiency.
- Administration of live vaccines (Option B): Live vaccines can be risky in individuals with certain immune deficiencies.
- Initiation of corticosteroid therapy (Option D): Corticosteroids are not typically used for managing IgG subclass deficiency.

Solution for Question 6:

Correct Answer: C) Wiskott-Aldrich Syndrome

Explanation:

The triad of eczema, thrombocytopenia with small platelets, and immunodeficiency is characteristic of Wiskott-Aldrich Syndrome.

Wiskott-Aldrich Syndrome	
Definition	Rare X-linked recessive disorder affecting males; characterized by eczema, thrombocytopenia, and immunodeficiency.
Genetic Basis	Caused by mutations in the WASP gene, affecting actin cytoskeleton in blood cells.
Clinical Features	Eczema: Develops within the first year; ranges from mild to severe. Thrombocytopenia: Present at birth; leads to prolonged bleeding from the umbilical stump, easy bruising, petechiae, and purpura. Immunodeficiency: Susceptible to infections (bacterial, viral, opportunistic); increased risk of autoimmune disorders and malignancies (lymphomas, leukemia).
Diagnosis	Clinical Triad: Eczema, thrombocytopenia, and immunodeficiency. Lab Tests: CBC (small, abnormally shaped platelets), immunoglobulin levels (low IgM, high IgA and IgE), lymphocyte subset analysis (low T cells), genetic testing (WASP mutations). Mnemonic: WATER-WASP gene mutation, Thrombocytopenia, Eczema, Recurrent infections
Treatment	Supportive Care: Preventative antibiotics, immunoglobulin replacement, eczema management, use of killed vaccines. Curative Treatment: Hematopoietic stem cell transplantation (HSCT)
Prognosis	Without HSCT: Often fatal in childhood due to infections, bleeding, or malignancies. With HSCT: Can significantly improve life expectancy; risks include graft-versus-host disease (GvHD).

- Severe Combined Immunodeficiency (SCID) (Option A) presents with severe immunodeficiency but typically lacks thrombocytopenia or small platelets.
- Henoch-Schonlein purpura (Option B) is characterised by palpable purpura, joint Involvement, abdominal pain, and renal involvement, but does not usually present with eczema or thrombocytopenia.

- Ataxia-Telangiectasia (Option D) involves cerebellar ataxia, telangiectasia and immunodeficiency but does not present with thrombocytopenia or eczema.

Solution for Question 7:

Correct Answer: B) Ataxia-Telangiectasia

Explanation:

Ataxia-Telangiectasia is characterized by cerebellar ataxia, telangiectasias, and immunodeficiency.

Ataxia-Telangiectasia	
Definition: Rare, multisystem disorder from ATM gene mutations, causing neurological, immune, endocrine, hepatic, and skin abnormalities. Genetic Basis: Autosomal recessive inheritance of ATM gene mutations, affecting DNA repair and cell cycle regulation.	
Clinical Features	Neurological: Progressive cerebellar ataxia (poor coordination, unsteady gait) Oculocutaneous telangiectasias (dilated blood vessels on sclera and ears). Immunodeficiency: Variable with low immunoglobulins and T cells; common IgA deficiency (selective absence of IgA in 50-80 %); impaired T- and B-cell responses; small, disorganized thymus. Other Features: Chronic respiratory infections, increased risk of cancers (lymphomas, leukemia), endocrine issues (diabetes), liver dysfunction and growth retardation.
Diagnosis	Clinical signs, family history, and lab tests including genetic testing for ATM mutations, immunoglobulin levels, lymphocyte counts, raised AFP Chromosomal analysis and radiation sensitivity.
Treatment	Supportive care: physical, occupational, and speech therapy; immunoglobulin replacement; prophylactic antibiotics; cancer screening; genetic counseling. No cure.

- Friedreich's Ataxia (Option A) typically does not present with telangiectasias or immunodeficiency.

- Wiskott-Aldrich Syndrome (Option C) includes thrombocytopenia, eczema, and immunodeficiency but lacks cerebellar ataxia and telangiectasias.

Solution for Question 8:

Correct Answer: B) Perform genetic testing for IL2RG mutation

Explanation:

Genetic testing for IL2RG mutation is crucial as X-linked SCID is the most common form, and confirming the diagnosis guides treatment.

Table rendering failed

- Start intravenous immunoglobulin (IVIG) therapy (Option A): IVIG therapy is supportive but not diagnostic; further testing is needed to confirm SCID.
- Initiate broad-spectrum antibiotics (Option C): Broad-spectrum antibiotics may be necessary but do not address the underlying immune deficiency.
- Schedule bone marrow biopsy (Option D): A bone marrow biopsy is not the first diagnostic step; genetic testing is more specific for SCID.

Solution for Question 9:

Correct Answer: A) Autosomal dominant

Explanation:

The clinical scenario is of Hyper IgE syndrome, aka Job syndrome. Its mode of inheritance is autosomal dominant.

Hyper IgE syndrome:

Inheritance Pattern	Autosomal dominant, typically due to mutations in the STAT3 gene Deficiency of Th17 cells
Presentation	Elevated serum IgE levels (> 2000 IU/mL)

	Eosinophilia Chronic eczema Frequent infections Recurrent cold staphylococcal abscesses Pneumonia leading to the development of pneumatoceles No shedding of primary (baby) teeth
Investigation	Screening: Elevated serum IgE levels (> 2000 IU/mL) Confirmatory: STAT3 mutation

Solution for Question 10:

Correct Answer: D) Long philtrum

Explanation:

In DiGeorge syndrome, the facial feature is short philtrum, not long philtrum

DiGeorge Syndrome: Mnemonic - CATCH-22

Genetic Cause	Microdeletion on chromosome 22q11.2. TBX1 is an important gene affected
Embryonic Defect	Arises from defects in embryogenesis of the third and fourth pharyngeal pouches.
Features	 ... (content truncated)
Definitive diagnosis	Microdeletions detected by: fluorescence in situ hybridization (FISH), multiplex ligation-dependent probe amplification (MLPA), single nucleotide polymorphism (SNP) array, comparative genomic hybridization (CGH) microarray, or quantitative polymerase chain reaction (qPCR)

Solution for Question 11:

Correct Answer: A) NADPH oxidase

Explanation:

Chronic Granulomatous Disease (CGD) is caused by a defect in the NADPH oxidase complex, which is essential for producing reactive oxygen species (ROS) that kill ingested pathogens.

Chronic Granulomatous Disease (CGD)	
Enzyme Defect	Reduced activity of nicotinamide adenine dinucleotide phosphate (NADPH) oxidase, leading to impaired generation of superoxide radical M/c defect is of subunit NOX-2
Genetic Inheritance	X-linked recessive Autosomal recessive
Age of Onset	Early infancy
Clinical Presentations	Persistent smoldering pneumonia Prominent lymphadenitis Multiple liver abscesses Osteomyelitis of small bones of hands and feet Recurrent, life-threatening infections usually by catalase-positive bacteria (e.g., Staphylococcus aureus, Serratia) and fungi (e.g., Aspergillus)
Diagnosis	Screening: Nitroblue tetrazolium (NBT) dye reduction test Definitive test: Dihydrorhodamine assay on flow cytometry

Lactate dehydrogenase (Option B) is released from damaged cells and is elevated in conditions like hemolytic anemia and cancer. LDH deficiency is not typically associated with infections or immune deficiencies.

Pyruvate dehydrogenase (Option C) deficiency, leading to lactic acidosis, neurologic symptoms, and developmental delay. Glucose-6-phosphate dehydrogenase (Option D) deficiency causes hemolytic anemia triggered by oxidative stress (e.g., infections, drugs).

Solution for Question 12:

Correct Answer: B) LAD II

Explanation:

Leukocyte Adhesion Deficiency Type II (LAD II) is a rare autosomal recessive disorder that results from a defect in the selectin ligand, which is crucial for proper leukocyte adhesion to the endothelium.

Leukocyte Adhesion Deficiency (LAD):

Type	Genetic/Pathophysiological Defect	Clinical Features
LAD I	Mutations in the CD18 gene resulting in defective or deficient beta-2 integrin Defective steady adhesion of leukocytes to endothelial surfaces.	Delayed separation of the umbilical cord Recurrent pyogenic infections (onset in the first weeks of life) Infections mainly by Staphylococcus aureus and Pseudomonas aeruginosa Absent pus formation and neutrophilia Periodontitis
LAD II (A/w Bombay blood group)	Defective E-selectin Sialyl Lewis X defect Rolling problem of neutrophils	Recurrent skin infections Pneumonia Bronchiectasis Tuberculosis Denture abnormalities Infections are less severe and fewer than in LAD I
LAD III	Defects in beta integrins 1, 2, and 3 Impaired integrin activation cascade Mutation in kindlin-3, FERMT3 gene.	Omphalitis Osteoporosis-like bone features Bleeding complications Hematological abnormalities (e.g., bone marrow failure)

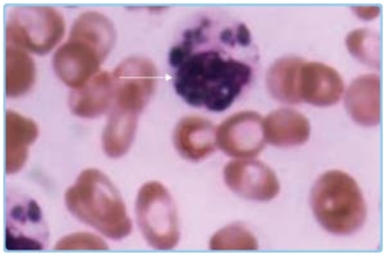
Solution for Question 13:

Correct Answer: C) Detection of large granules in neutrophils on Wright-stained blood films

Explanation:

The clinical scenario of Hypopigmentation, recurrent infection and neurological involvement indicates Chediak-Higashi syndrome. It is marked by the presence of large, abnormally fused granules in various cells, including neutrophils. They are visible on Wright-stained blood films and accentuated by peroxidase staining.

Chediak-Higashi Syndrome

Etiology	Caused by mutations in the LYST gene. Autosomal recessive inheritance. Microtubule dysfunction Disrupts lysosome trafficking and fusion.
Pathophysiology	Enlarged, dysfunctional lysosomes. Impaired phagolysosome fusion. Reduced intracellular killing of pathogens.
Clinical Features	Hypopigmentation: Light skin, silvery hair, partial albinism. Recurrent Infections: Frequent bacterial and viral infections. Neurological Manifestations: Peripheral neuropathy (numbness, tingling, weakness).
Diagnosis	 Laboratory Findings: Giant granules in neutrophils on blood smears. Pancytopenia Genetic Testing: Identifies LYST gene mutations.
Treatment	Ascorbic acid, Controversial yet used. Prophylactic Antibiotics: Long-term antibiotics to prevent infections. Bone Marrow Transplantation: Can be curative for severe cases. Supportive Care: Prompt treatment of infections, antivirals, management of neurological symptoms.

Elevated serum IgE levels and eczema (Option A). These are more indicative of Hyper-IgE Syndrome (Job's syndrome). Hyper-IgE Syndrome is associated with elevated IgE levels and eczema.

Presence of large peroxisomes in the blood smear (Option B). Large peroxisomes are not a feature of Chediak-Higashi Syndrome. CHS is characterized by large granules in neutrophils due to lysosomal dysfunction.

Low NADPH oxidase levels with hepatosplenomegaly (Option D). Low NADPH oxidase levels are not seen in Chediak-Higashi Syndrome.

Solution for Question 14:

Correct Answer: C) Palpable purpura predominantly on lower limbs

Explanation:

The clinical scenario of purpura in legs and buttocks, arthritic pain, abdominal pain and normal platelet level are characteristic of Henoch-Schönlein Purpura (HSP).

Henoch-Schönlein Purpura (HSP)

Etiology	Exact cause unknown. Immune-mediated vasculitis, often following an upper respiratory infection. Genetic predisposition, with HLA-B34 and DRB1 observed in some cases. Most common in Males (Option A).
Pathology and Pathophysiology	Inflammation of small blood vessels affecting skin, joints, intestines, and kidneys. Triggered by immune system response to infections. Leakage of fluid and blood cells into tissues. Immune complex deposition, particularly IgA, in affected vessels.
Clinical Features	Palpable Purpura: Raised, purplish-red rash, mainly on lower limbs (Option C) and buttocks.(Option B) Joint Involvement: Joint swelling, commonly in knees and ankles. Abdominal Pain: Often severe, may include vomiting, diarrhea, gastrointestinal bleeding, or intussusception Renal Involvement: Less common, presents as Chronic renal failure.

Diagnosis	Clinical Criteria: Palpable purpura plus at least one of: Diffuse abdominal pain Arthritis or arthralgia Biopsy-proven IgA deposition Renal involvement Laboratory involvement: Elevated ESR or CRP indicating inflammation. Platelet Counts are normal.
Complications	Intussusception. Chronic Renal Failure due to Long-term kidney damage, in severe cases. Neurological Complications: Rare; may include seizures, headaches, or peripheral neuropathy.
Treatment	Mild Cases: Rest, hydration, and pain relievers. Moderate to Severe Cases: Corticosteroids (e.g., prednisone) to reduce inflammation. Severe or Resistant Cases: Additional immunosuppressive medications may be required.
Prognosis	Full recovery in most cases. Variable based on renal involvement.

Acute liver failure is a complication (Option D). Acute liver failure is not a typical complication. Renal involvement, such as hematuria and proteinuria, is a more common and significant concern in HSP

Solution for Question 15:

Correct answer C) Start high-dose intravenous immunoglobulin (IVIG) and high-dose aspirin.

Explanation:

The clinical scenario is indicative of Kawasaki disease. The primary treatment for KD involves high-dose IVIG along with high-dose aspirin.

Kawasaki Disease (KD)

Etiology and Pathophysiology	Higher incidence in children of Asian descent.. Vasculitis affecting medium-sized arteries, especially coronary arteries.

Pathology	Acute Phase (Days 1-10): Neutrophilic inflammation of blood vessels. Infiltration of inflammatory cells, vessel wall swelling and damage. Subacute Phase (Weeks 2-4): Risk of fusiform aneurysm formation as inflammation subsides. Convalescent Phase (Weeks >4 weeks): Resolution of inflammation; potential scarring and thickening of arterial walls leading to stenosis.
Clinical Features	Fever: For at least 5 days. Extremities: Swelling, redness, peeling skin around nails, fingers, and toes. Mucocutaneous : Bilateral conjunctivitis, red and cracked lips, "strawberry tongue," and pharyngeal congestion. Polymorphous Rash: Various forms (macular, papular, target-like lesions). Cervical Lymphadenopathy: Unilateral, painful swelling of neck lymph nodes.
Diagnosis	Clinical Criteria: Fever lasting at least 5 days. At least 4 of the following: 5-“CREAM” features C - Conjunctivitis, B/Lnon-purulent R - Rash involving trunk E- Erythema & Edema of palms & soles along with desquamation A - Adenopathy (usually unilateral, cervical) M- Mucositis (Strawberry Tongue) Laboratory Findings: Elevated CRP and ESR. Elevated WBC count. Echocardiography coronary artery aneurysms.
Complications	Myocardial infarction. Coronary Artery Aneurysms. Coronary Artery Stenosis:
Differential Diagnosis	Bacterial Infections (e.g., Scarlet Fever, Staphylococcal Scalded Skin Syndrome): Fever, rash, mucosal involvement with additional features like sore throat or blistering.
Treatment	Intravenous Immunoglobulin (IVIG): Reduces inflammation and lowers risk of aneurysms. Aspirin: High doses initially to reduce inflammation and fever; then low-dose regimen for anti-platelet effects. IVIG-Resistant KD: Second dose of IVIG, corticosteroids, or other immunosuppressive medications (e.g., infliximab, cyclosporine).

Initiate Ceftriaxone for a suspected bacterial infection (Option A): Kawasaki Disease is not caused by a bacterial infection, and antibiotics are not effective in treating KD.

Perform an echocardiogram to assess for coronary artery involvement (Option B): Although echocardiogram is crucial in KD for detecting coronary artery aneurysms and monitoring heart involvement, the immediate next step in management, given the clinical presentation, is to start specific treatment for KD.

Start corticosteroids (Option D): Corticosteroids are used only in IVIG-resistant cases. It is not the first step in management.

Solution for Question 16:

Correct Answer: B) Muscle Biopsy

Explanation:

The clinical scenario with Heliotrope rash, Gottrons papule and mechanic hands are indicative of Juvenile Dermatomyositis. A muscle biopsy is crucial for diagnosing juvenile dermatomyositis (JDM)

Juvenile Dermatomyositis

Table rendering failed

Skin Biopsy (Option A) is not done in JDM.

Antinuclear Antibody (ANA) Test (Option C): ANA test can be positive in JDM but is not specific.

Electromyography (EMG) (Option D): EMG can help assess muscle function and detect abnormalities consistent with myopathy but does not provide a definitive diagnosis.

Viral Diseases & COVID-19 in Children

1. An HIV-positive primigravida delivered a baby at 37 weeks of gestation. The baby cried immediately after birth, and samples for HIV testing were collected. The mother has been advised to exclusively breastfeed the baby. What is the recommended prophylactic regimen in this situation?

A. Nevirapine +/- Zidovudine

B. Zidovudine + Lamivudine + Nevirapine

C. Tenofovir + Lamivudine + Efavirenz

D. No prophylaxis needed

2. A 10-year-old child with a history of COVID-19 with significant symptoms is diagnosed with Multisystem Inflammatory System in Children (MIS-C). What is the recommended management to improve outcomes?

A. Intravenous immunoglobulin (IVIG) alone

B. IVIG plus Methylprednisolone

C. Corticosteroids alone for suspected toxic shock phenotype

D. Low-dose aspirin therapy

3. A 7-year-old child presents with a 4-day history of fever, abdominal pain, and rash following a COVID-19 infection three weeks ago. Laboratory tests show elevated ESR, C-reactive protein, and d-dimers. Which of the following is NOT included in the WHO diagnostic criteria for Multisystem Inflammatory Syndrome in Children (MIS-C)?

A. Persistent fever >3 days

B. Elevated D-dimers

C. Evidence of staphylococcal shock syndrome

D. Recent COVID-19 infection

4. A 5-year-old child with COVID-19 presents with mild symptoms, including fever and cough. As the child progresses through the illness, which of the following complications is most commonly associated with COVID-19 in children?

A. Acute respiratory distress syndrome

B. Myocarditis

C. Multisystem inflammatory syndrome in children

D. Chronic kidney disease

5. A 7-year-old child with COVID-19 presents to the hospital with 92% oxygen saturation without any danger signs and experiencing persistent fever and cough. What is the most appropriate management for this patient?

A. Home management with paracetamol and hydration

B. Admit and administer O2, Remdesivir

C. Immediate ICU admission and invasive ventilation

D. Use of antibiotics and convalescent plasma.

6. A 10-year-old boy with COVID-19 presents with fever, cough, and difficulty breathing. His oxygen saturation is 88%, and he exhibits intercostal retractions. How would you classify his COVID-19 severity based on ICMR criteria?

A. Mild

B. Moderate

C. Severe

D. Critical

7. A toddler presents to the emergency department with a 2-day history of cough, high fever, and difficulty breathing. She is irritable, has reduced appetite and fluid intake, and appears lethargic and unconscious. Her respiratory rate is 60 breaths per minute. What is the next step in management?

A. IV antibiotic + Hospital admission

B. Oral antibiotic and 48 hours observation

C. Supportive care at home

D. Reassurance and sending home

8. A female brought her infant to the clinic with a low-grade fever and a rash as depicted in the image below. What is the causative organism of this condition?

A. Parvovirus B19

B. Coxsackie A16 virus

C. Rickettsia prowazekii

D. Paramyxovirus

9. A 7-year-old boy presents to the pediatric clinic with a 3-day history of fever, cough, sore throat, body aches, chills, headaches, and a runny nose, along with fatigue and loss of appetite. Which of the following is the drug of choice for this condition?

A. Doxycycline

B. Tenofovir

C. Azithromycin

D. Oseltamivir

10. An HIV-positive mother brought her 2.5-year-old child to the paediatrician. The child has tested positive for HIV, and the mother is seeking information about precautions and management. What is the first-line treatment regimen for her child?

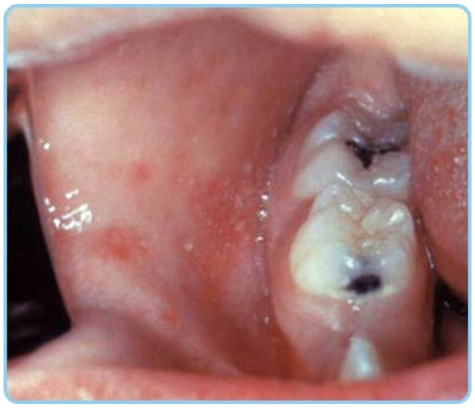
A. Zidovudine + Lamivudine + Nevirapine

B. Zidovudine + Lamivudine + Lopinavir

C. Abacavir + Lamivudine + Efavirenz

D. Tenofovir + Lamivudine + Efavirenz

11. A mother brings her unimmunized toddler to the doctor with a 5-day history of fever, cough, and conjunctivitis, along with a rash that first appeared behind the ears and then spread to the face and chest over the past two days. Based on the observed lesion (as shown in the image), what is the period of infectivity for this condition?



A. From 3 days before to 4 days after the rash onset

B. From 2 days before to 7 days after the rash onset

C. From 5 days before to 6 days after the rash onset

D. Non-communicable after the rash onset

12. A 6-week-old infant born to an HIV-positive mother has a positive HIV antibody test. What is the best method to confirm or exclude HIV infection in this infant?

A. Repeat the antibody test

B. HIV RNA PCR test

C. IgA anti-HIV assay

D. Rapid HIV test

13. A 6-year-old child comes in with sudden weakness in the legs, a recent fever, and muscle pain. Poliomyelitis is suspected. What type of paralysis is this child most likely experiencing?

A. Spastic paralysis with hyperreflexia

B. Flaccid paralysis with absent reflexes

C. Rigid paralysis with limited movement

D. Ataxic paralysis with coordination issues

14. A 7-year-old boy presents with a high-grade fever, headache, and progressively enlarged swelling near the right ear. Physical examination shows tender parotid gland swelling, raising suspicion of mumps. What is the most common neurological complication of mumps?

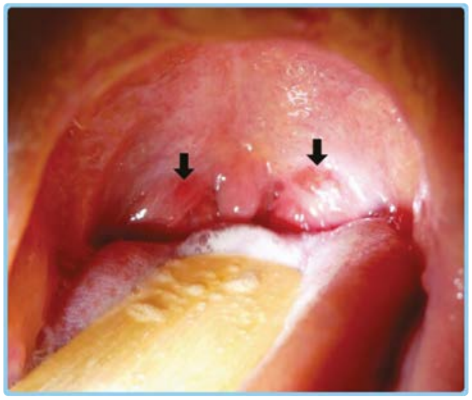
A. Encephalitis

B. Meningitis

C. Transverse myelitis

D. Facial palsy

15. A 10-month-old child is brought to the pediatric clinic with a 3-day history of high fever. On the fourth day, the fever abruptly subsided, and the child developed a rash as given in the image. Oral examination revealed ulcers and small red papules at uvulopalatoglossal junction as shown. What are these called?



A. Koplik spots

B. Nagayama spots

C. Forchheimer spots

D. Café au Lait spots

16. A 5-year-old child with sickle cell disease presents with sudden onset of pallor, fatigue, mild fever, and a rash. Investigations reveal a significant drop in haemoglobin levels. What is the most likely causative agent of this condition?

A. Parvovirus B19

B. Epstein-Barr virus

C. Cytomegalovirus

D. Human Herpesvirus 6

17. A 6-year-old child presents with a history of mild fever and a rash. The rash began with red, flushed cheeks, as shown in the image below and has now developed into a lacy, reticular pattern on the arms and legs. The child has no significant respiratory symptoms or joint pain. What is the most likely diagnosis?

A. First disease

B. Second disease

C. Fifth disease

D. Sixth disease

18. A 6-year-old child presents with a 3-day history of fever and a rash that started on the trunk and spread to the face and extremities, progressing to fluid-filled

vesicles with crusting. The parent reports that the child's rash has become increasingly painful, with signs of severe swelling and redness spreading rapidly from the initial rash sites. Which of the following complication is most likely?

A. Reye syndrome

B. Necrotising fasciitis

C. Purpura fulminans

D. Impetigo

19. A 32-year-old woman who contracted rubella during her first trimester is worried about the potential effects on her baby. Which of the following findings is not typically associated with congenital rubella syndrome?

A. Patent ductus arteriosus

B. Cataract

C. Pleomorphic rash

D. Deafness

20. A 4-year-old child is diagnosed with measles and presents with persistent fever and cough. The child is at risk for complications associated with measles. Which of the following is the most common complication of measles?

A. Pneumonia

B. Otitis media

C. Subacute Sclerosing Panencephalitis

D. Reye syndrome

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	3
Question 4	3
Question 5	2
Question 6	3
Question 7	1
Question 8	2
Question 9	4
Question 10	3
Question 11	1
Question 12	2
Question 13	2
Question 14	2
Question 15	2
Question 16	1
Question 17	3
Question 18	2
Question 19	3
Question 20	2

Solution for Question 1:

Correct Answer: A) Nevirapine +/- Zidovudine

Explanation:

- In the above case, the preferred prophylaxis for the infant born to an HIV-positive mother is Nevirapine +/- Zidovudine for at least 12 weeks.
- Infants born to HIV-infected mothers will test positive for HIV antibodies at birth due to passive transfer from the mother, so IgG testing cannot diagnose HIV. Instead, the following tests can be performed: HIV RNA PCR HIV DNA PCR HIV culture
- HIV RNA PCR
- HIV DNA PCR
- HIV culture
- HIV RNA PCR
- HIV DNA PCR
- HIV culture
- In children >18 months old, HIV diagnosis can be confirmed by detecting IgG antibodies using ELISA or Western blot tests. A positive result requires confirmation with at least two tests.

Risk Category	Maternal Factors
Low Risk	On effective ART with suppressed viral load < 1,000 copies/mL
High Risk	Not on ART or high viral load ≥ 1,000 copies/mL Preterm delivery Rupture of membranes > 4 hours Low maternal CD4 count Maternal drug use HIV acquired during pregnancy

Prophylaxis for infants born to HIV-positive mother

- Baby should receive prophylaxis irrespective of whether the mother has received ART or not. (Option D)
- The risk of HIV transmission through breastfeeding is highest during the early months, and if safe and sustainable alternatives are available, breastfeeding should be avoided.
- Currently, exclusive breastfeeding is recommended during the first months of life, as the benefits of breastfeeding far outweigh the risks.
- The regimen for prophylaxis is, Nevirapine +/- Zidovudine for at least 6 weeks In breastfed babies, Nevirapine +/- Zidovudine for at least 12 weeks

- Nevirapine +/- Zidovudine for at least 6 weeks
- In breastfed babies, Nevirapine +/- Zidovudine for at least 12 weeks
- Nevirapine +/- Zidovudine for at least 6 weeks
- In breastfed babies, Nevirapine +/- Zidovudine for at least 12 weeks

Zidovudine + Lamivudine + Nevirapine (Option B) is the 1st line treatment regimen of HIV positive for infants ■ 2 weeks of age.

Tenofovir + Lamivudine + Efavirenz (Option C) is the 1st line treatment regimen for HIV-positive adolescents.

Solution for Question 2:

Correct Answer: B) IVIG plus Methylprednisolone

Explanation:

The recommended initial treatment for managing MIS-C, with significant symptoms, involves a combination of intravenous immunoglobulin (IVIG) and corticosteroids (such as Methylprednisolone). This combined approach improves outcomes by reducing the hyper-inflammatory response.

Management of MIS-C:

- Lab Diagnosis: Test for antibodies to nucleocapsid protein (avoid anti-spike protein as it may indicate past vaccination)
- Management: Supportive Care First-line treatment: A combination of IVIG and steroids is preferred. Steroids alone may be used in non-Kawasaki phenotypes (Option B) Treatment typically involves 3-5 days of IVIG and steroids, followed by a tapering course of oral steroids over 2-3 weeks. Additional Therapies: Low-Dose Aspirin: Administer 3-5 mg/kg (maximum 75 mg) for 6 weeks (Option D). Anticoagulation: Reserved for severe cases with low ejection fraction, documented thrombosis, or giant coronary aneurysms. Biologics: Consider infliximab, tocilizumab, or anakinra for refractory cases.
- Supportive Care
- First-line treatment: A combination of IVIG and steroids is preferred. Steroids alone may be used in non-Kawasaki phenotypes (Option B) Treatment typically involves 3-5 days of IVIG and steroids, followed by a tapering course of oral steroids over 2-3 weeks.

- A combination of IVIG and steroids is preferred.
- Steroids alone may be used in non-Kawasaki phenotypes (Option B)
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- Low-Dose Aspirin: Administer 3-5 mg/kg (maximum 75 mg) for 6 weeks (Option D).
- Anticoagulation: Reserved for severe cases with low ejection fraction, documented thrombosis, or giant coronary aneurysms.
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- Biologics: Consider infliximab, tocilizumab, or anakinra for refractory cases.
- A combination of IVIG and steroids is preferred.
- Steroids alone may be used in non-Kawasaki phenotypes (Option B)
- Treatment typically involves 3-5 days of IVIG and steroids, followed by a tapering course of oral steroids over 2-3 weeks.
- Low-Dose Aspirin: Administer 3-5 mg/kg (maximum 75 mg) for 6 weeks (Option D).
- Anticoagulation: Reserved for severe cases with low ejection fraction, documented thrombosis, or giant coronary aneurysms.
- Biologics: Consider infliximab, tocilizumab, or anakinra for refractory cases.

- Follow-up: Echocardiography: Recommended at 2 and 6 weeks, then as needed. Prognosis: Short-term mortality between 0-5%.
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- Prognosis: Short-term mortality between 0-5%.
- Echocardiography: Recommended at 2 and 6 weeks, then as needed.
- Prognosis: Short-term mortality between 0-5%.

Solution for Question 3:

Correct Answer: C) Evidence of staphylococcal shock syndrome

Explanation:

Evidence of staphylococcal shock syndrome is NOT part of the MIS-C criteria and suggests a bacterial cause rather than the post-viral inflammatory syndrome seen in MIS-C.

Multisystem Inflammatory Syndrome in Children (MIS-C):

- Common Symptoms: Persistent fever (>3 days) Gastrointestinal issues, Mucocutaneous symptoms Neurocognitive symptoms (headache, lethargy, confusion)
- Persistent fever (>3 days)
- Gastrointestinal issues, Mucocutaneous symptoms
- Neurocognitive symptoms (headache, lethargy, confusion)
- Severe Cases may include encephalopathy, seizures, acute kidney injury, serositis, and various cardiac issues (e.g., ventricular dysfunction, coronary artery dilation).
- Phenotypes: Kawasaki Phenotype: Prominent mucocutaneous manifestations in children under the age of 5. Toxic Shock Phenotype: Shock and myocarditis seen in older children. Febrile Inflammatory Phenotype: Fever without organ involvement; may resolve or evolve into other phenotypes.
- Kawasaki Phenotype: Prominent mucocutaneous manifestations in children under the age of 5.
- Toxic Shock Phenotype: Shock and myocarditis seen in older children.
- Febrile Inflammatory Phenotype: Fever without organ involvement; may resolve or evolve into other phenotypes.
- Persistent fever (>3 days)

- Gastrointestinal issues, Mucocutaneous symptoms
- Neurocognitive symptoms (headache, lethargy, confusion)
- Kawasaki Phenotype: Prominent mucocutaneous manifestations in children under the age of 5.
- Toxic Shock Phenotype: Shock and myocarditis seen in older children.
- Febrile Inflammatory Phenotype: Fever without organ involvement; may resolve or evolve into other phenotypes.

Diagnostic Criteria (WHO) of MIS-C:

Fever >3 days (Option A)
Two or more of the following:
Rash or conjunctivitis (Mucocutaneous Inflammation)
Hypotension or Shock
Myocardial dysfunction or Coronary abnormalities
Evidence of Coagulopathy (Elevated D-dimers) (Option B)
Gastrointestinal symptoms. (abdominal pain, vomiting, diarrhoea)
Elevated inflammatory markers (ESR, CRP, procalcitonin).
No alternative microbial cause. (Option C)
Evidence of COVID-19 exposure or infection. (Option D)

Solution for Question 4:

Correct Answer: C) Multisystem inflammatory syndrome in children

Explanation:

Multisystem inflammatory syndrome in children (MIS-C) is a known and significant complication associated with COVID-19 in children, characterised by severe inflammation affecting multiple organ systems.

Acute Complications of COVID-19:

- Multisystem Inflammatory Syndrome in Children (MIS-C) (Most common)
- Acute Respiratory Distress Syndrome (ARDS)
- Septic Shock
- Acute Kidney Injury (AKI)
- Liver Dysfunction
- Thrombosis

- Disseminated Intravascular Coagulation (DIC)
- Myocarditis

Acute respiratory distress syndrome (Option A) is a serious complication associated with COVID-19; it is more commonly observed in adults than in children.

Myocarditis (Option B) can occur as a complication of COVID-19, but it is less common compared to other complications in children.

Chronic kidney disease (Option D) is not a common immediate complication of COVID-19 in children.

Solution for Question 5:

Correct Answer: B) Admit and administer O2, Remdesivir

Explanation:

The child with 92% oxygen saturation without any danger signs indicates Moderate disease for which hospital admission is required, and remdesivir is administered for saturation < 94% as it can help reduce the severity and duration of the illness.

Management of COVID-19 in Children:

- Supportive Care: Primarily symptomatic and supportive. Avoid polypharmacy. Azithromycin, hydroxychloroquine, doxycycline, ivermectin, antibiotics, favipiravir, and convalescent plasma are not recommended.
- Monitoring: Parents should watch for red flags like persistent high fever, rapid breathing, and low oxygen saturation (<94%).
- Isolation: Mild cases can be deisolated 7 days from symptom onset, while severe cases require 14 days of isolation.

Treatment Protocols:

Mild Disease	Home Management & Hydration Medications: Use paracetamol for fever and nasal saline drops for congestion. Avoid antibiotics and multivitamins (no proven benefit). Monitoring: Watch for red flag signs like rapid breathing, oxygen saturation below 95%, persistent fever, lethargy, or poor feeding.

Moderate Disease	Admit in Covid ward SpO2 > 94: Hydration, paracetamol and antimicrobials SpO2 < 94: Oxygen, IV Steroids, Remdesivir, Antimicrobials
Severe/Critical	Covid - 19 ICU / HDU ward Oxygen, CPAP, Invasive ventilation if required IV Steroids, Remdesivir, Enoxaparin, Antimicrobials Manage Shock, ARDS, AKI, and Myocarditis as per protocol

- Steroids: Administer IV dexamethasone at 0.15 mg/kg (max 6 mg) once daily for 5-14 days.
- Remdesivir should be administered only in a hospital setting.
- Enoxaparin: For <2 months: 1.5 mg/kg/dose every 12 hours. For >2 months: 1 mg/kg/dose every 12 hours.
- For <2 months: 1.5 mg/kg/dose every 12 hours.
- For >2 months: 1 mg/kg/dose every 12 hours.
- For <2 months: 1.5 mg/kg/dose every 12 hours.
- For >2 months: 1 mg/kg/dose every 12 hours.

Home management with paracetamol and hydration (Option A) is done for only mild cases with Spo2>94%

Immediate ICU admission and invasive ventilation (Option C) are not preferred as there are no immediate danger signs or severe symptoms (e.g., severe respiratory distress or shock) that would necessitate immediate ICU admission.

Use of antibiotics and convalescent plasma (Option D): Antibiotics are not recommended for COVID-19 unless there is a confirmed bacterial infection. Convalescent plasma has not shown consistent benefits in managing COVID-19 and is not a standard treatment

Solution for Question 6:

Correct Answer: C) Severe

Explanation:

According to ICMR, COVID-19 is classified as "Severe" when oxygen saturation is below 90% with signs of respiratory distress, such as intercostal retractions, which require immediate medical management.

COVID-19:

- Causative Agent: SARS-CoV-2
- Transmission Modes: Through droplets from breathing, coughing, and sneezing.
- Incubation Period: 2-14 days (mean: 5 days)
- The virus binds to ACE2 receptors, leading to various symptoms and complications.
- Symptoms: Fever, headache, fatigue, sore throat, cough, loss of taste/smell (initially) and severe cases leading to pneumonia, ARDS, thrombosis, and multi-organ dysfunction.
- Variants: Delta variant caused severe cases early, while Omicron resulted in higher transmissibility but milder disease.
- Diagnosis: RT-PCR Tests are the gold standard for detecting viral RNA in respiratory specimens. They have high specificity and a sensitivity of approximately 70-90%. Rapid Antigen Tests (RAT): Lower sensitivity (30-50%) but high specificity. Negative RAT results should be confirmed with RT-PCR. Antibody Tests: Useful for epidemiological studies, not for acute infection.
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- Antibody Tests: Useful for epidemiological studies, not for acute infection.

ICMR Criteria of COVID-19 Disease Severity:

	Features	Saturation
Mild	URTI without lower respiratory tract involvement.	> 94%
Moderate	Fast breathing (age-dependent) or hypoxemia	90-93%
Severe	Retractions, severe dehydration, drowsiness, or seizures. (Danger signs)	<90%
Critical	Severe disease with ARDS, shock, multiorgan dysfunction syndrome, or acute thrombosis.	-

Solution for Question 7:

Correct Answer: A) IV antibiotic + Hospital admission

Explanation:

The given scenario is a case of severe pneumonia having danger signs and fast breathing which requires immediate hospital admission after 1st dose of IV antibiotics, preferably ampicillin and gentamicin.

- Acute lower respiratory tract infection (LRTI) is a leading cause of mortality in children below 5 years of age.
- Common bacteria causing LRTI in preschool children include H. influenzae, S. pneumoniae and. Staphylococci.
- According to the latest IMNCI (Integrated Management of Neonatal & Childhood Illness) guidelines, assessing a child with breathing difficulties or a cough involves counting breaths per minute, checking for chest indrawing, observing for stridor and wheeze and identifying any danger signs.
- Danger signs include convulsions, lethargy, inability to drink or breastfeed, persistent vomiting, unconsciousness, stridor in a calm child.
- Fast breathing is assessed as, < 2 months: > 60/ min 2-12 months: > 50/ min 1-5 years: > 40/ min
- < 2 months: > 60/ min
- 2-12 months: > 50/ min
- 1-5 years: > 40/ min
- < 2 months: > 60/ min
- 2-12 months: > 50/ min
- 1-5 years: > 40/ min

Management:

Category	Signs/symptoms	Treatment

No Pneumonia (cough or cold only) (Option C)	No signs of rapid breathing, chest indrawing, or general danger signs.	Home-based supportive care
Pneumonia (Option B)	Presence of either chest indrawing or rapid breathing, but no danger signs.	Start oral Amoxicillin for 5 days. The child is sent home and will be reassessed after 48 hours. If the child has improved, continue with the same treatment. If the child's condition has worsened, refer to a higher level of care.
Severe Pneumonia (Option A)	Presence of danger signs	Administer the first dose of Inj. Ampicillin and Gentamicin, then urgently refer the patient to the hospital.

Reassurance and sending home (Option D) won’t be enough as this is an emergency situation which requires hospital admission.

Solution for Question 8:

Correct Answer: B) Coxsackie A16 virus

Explanation:

The given clinical presentation is suggestive of Hand Foot Mouth Disease (HFMD) caused by Coxsackie A16 virus.

Hand Foot Mouth Disease (HFMD)	
Causative organism	Coxsackie A16 virus
Age group	<5 years
Clinical features	Low grade fever, feeling of being unwell, sore throat are commonly observed. Characterized by papulovesicular skin rash commonly on buttocks, knees, elbows and genital area. Ulcers/blisters are also seen in the oral cavity, mostly on the posterior aspect.

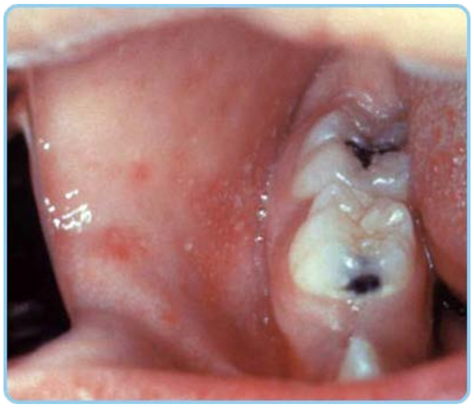
Management	Resolves on its own in 5-7 days, till then symptomatic and conservative treatment is given. Isolation of affected children at home and promotion of hand hygiene to prevent disease spread is important.

Parvovirus B19 (Option A) is the causative agent of erythema infectiosum which is an erythematous illness characterized by slapped cheek appearance.

Rickettsia prowazekii (Option C) is the causative agent of Epidemic typhus, a rickettsial disease characterised by fever and a maculopapular rash. It is transmitted primarily by body lice (Pediculus humanus corporis).

Rickettsia prowazekii" data-author="D. Raoult, V. Roux, J.B. Ndihokubwayo, G. Bise, D. Baudon, G. Martet, and R. Birtles" data-hash="992" data-license="Public Domain" data-source="https://commons.wikimedia.org/wiki/File:Epidemic_typhus_Burundi.jpg" data-tags="March2025" height="621" src="https://image.prepladder.com/notes/KLfVh0jxDgU23Uu3B1bW1742149386.png" width="500" />

Paramyxovirus (Option D) causes measles, which is a disease characterized by fever before rash along with its characteristic Koplik spots (grey-white lesions in buccal mucosa).



Solution for Question 9:

Correct Answer: D) Oseltamivir

Explanation:

The above clinical scenario is diagnosed as a case of Influenza A infection, in which the drug of choice is oseltamivir.

Causative organism	Influenza A virus
Incubation period	1-3 days
Clinical features	Flu like symptoms: Fever, runny nose, sore throat, cough, body ache, headache are very commonly seen. Abdominal pain, diarrhea and vomiting is also seen. Some cases are asymptomatic.
Complications	Pneumonia acute respiratory distress syndrome (most dreaded) Respiratory failure Kidney failure Shock

Treatment,

- Definitive treatment is with a neuraminidase inhibitor (oseltamivir, zanamivir, peramivir).
- Drug of choice is Oseltamivir
- Dose of oseltamivir is given as,

Age/weight	Dose
Infants	3 mg/kg/dose
<15 kg	30 mg/dose
15-23 kg	45 mg/dose
24-40 kg	60 mg/dose
>40 kg	75 mg/dose

- Rest other symptomatic and supportive management is done as per clinical need.

Doxycycline (Option A) is an antibiotic used to treat bacterial infections such as scrub typhus, malaria, and certain sexually transmitted infections, but it is not effective against viral infections like influenza. .

Tenofovir (Option B) is an antiretroviral drug used in ART (Antiretroviral Therapy) regimens for the treatment of HIV-AIDS and is ineffective against influenza viruses.

Azithromycin (Option C) is an antibiotic and drug of choice in traveler’s diarrhea, prophylaxis in pertussis, campylobacter infections and also used in many other infections, not for influenza.

Solution for Question 10:

Correct Answer: C) Abacavir + Lamivudine + Efavirenz

Explanation:

- The first-line antiretroviral regimen for children under three years old typically includes: Zidovudine (AZT): a nucleoside reverse transcriptase inhibitor (NRTI) Lamivudine (3TC): another NRTI Lopinavir/ritonavir (LPV/r): a protease inhibitor (PI)
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- Lamivudine (3TC): another NRTI
- Lopinavir/ritonavir (LPV/r): a protease inhibitor (PI)
- Zidovudine (AZT): a nucleoside reverse transcriptase inhibitor (NRTI)
- Lamivudine (3TC): another NRTI
- Lopinavir/ritonavir (LPV/r): a protease inhibitor (PI)
- The treatment of HIV-infected children includes antiretroviral therapy (ART), prophylaxis, and treatment of common and opportunistic infections, adequate nutrition and immunisation.
- ART should be given to all children with HIV (irrespective of the clinical stage or cell count.).
- High-priority groups (sometimes, ART cannot be given to all children; therefore, these groups are made) Children <2 years of age WHO Stage 3 or 4 Children <5 years of age with CD4 count < 25% or < 750/mm3.
- Children <2 years of age
- WHO Stage 3 or 4
- Children <5 years of age with CD4 count < 25% or < 750/mm3.
- Following is the regimen for preferred 1st line ART therapy,
- Children <2 years of age
- WHO Stage 3 or 4
- Children <5 years of age with CD4 count < 25% or < 750/mm3.

Age	1st Line Treatment
Infants ■ 2 weeks of age (Option A)	Zidovudine + Lamivudine + Nevirapine
Children ■ 3 years of age (Option B)	Abacavir/Zidovudine + Lamivudine + Lopinavir/Ritonavir
3-10 years of age (Option C)	Abacavir + Lamivudine + Efavirenz
Adolescents (Option D)	Tenofovir + Lamivudine + Efavirenz

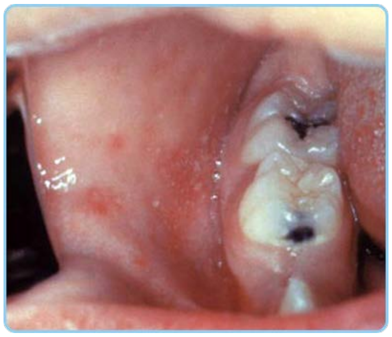
Solution for Question 11:

Correct Answer: A) From 3 days before to 4 days after the rash onset

Explanation:

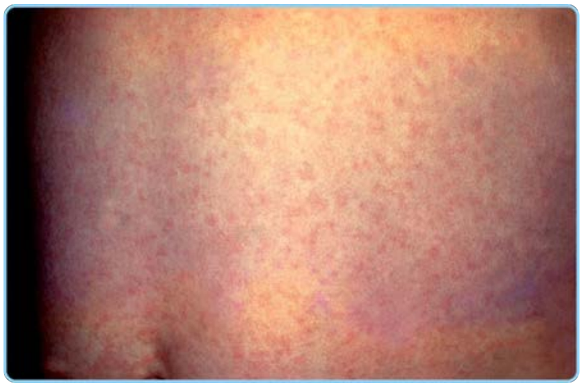
- The given clinical scenario suggests Measles, with the image showing its characteristic, Koplik spots.
- The period of infectivity is 3 days before to 4 days after rash onset.

Aspects	Description
Causative agent	Measles (rubeola) is a highly communicable childhood exanthematous illness caused by the measles virus, an RNA virus from the Paramyxoviridae family.
Transmission	Contact with droplet aerosols (through the respiratory tract or conjunctiva)
Pathogenesis	Primary viremia occurs, resulting in infection of the reticuloendothelial system, followed by secondary viremia, which results in systemic symptoms.
Incubation period	8-12 days
Period of infectivity	From 3 days before to 4 days after the rash onset (Option A).

Clinical manifestations	 Common in preschool children Prodromal phase: Fever (before rash), rhinorrhea, conjunctival congestion and a dry hacking cough. ... (content truncated)
Rash	The erythematous maculopapular rash in measles appears on the 4th day, starting behind the ears, along the hairline, and posterior cheeks, then spreading to the rest of the body over the next 2-3 days.
Management	Symptomatic and conservative (antipyretics, maintenance of hygiene, ensuring adequate fluid and caloric intake, and humidification).

From 2 days before to 7 days after the rash onset (Option B) is a period of infectivity in chicken pox, a mild exanthematous illness caused by the varicella-zoster virus (VZV). Fever, along with a pleomorphic rash, is seen, which begins on the 1st day of the fever.

From 5 days before to 6 days after the rash onset (Option C) is a feature of rubella, a mild exanthematous illness caused by Rubella virus (RNA) belonging to the Togaviridae family. The rash begins on the face and spreads centrifugally.



Non-communicable after the rash onset (Option D): This is consistent with the fifth disease (erythema infectiosum), where the patient is most contagious before the rash appears. Once the rash develops, the risk of transmission decreases. In contrast, measles remains infectious throughout the rash phase.

Solution for Question 12:

Correct Answer: B) HIV RNA PCR test

Explanation:

In infants born to HIV-positive mothers, antibody tests (Option A) are not reliable for diagnosing HIV infection due to the presence of maternal antibodies. Instead, the best method to confirm or exclude HIV infection is through direct detection of the virus through techniques like PCR.

Maternal Antibodies: Infants born to HIV-positive mothers will test positive for HIV antibodies due to passive transfer from the mother. These antibodies typically disappear in uninfected infants by 6-18 months, but some may persist up to 24 months.

Diagnosis of HIV in Infants:

< 24 Months	Antibody Tests: Not reliable for definitive diagnosis due to maternal antibody presence. IgA (Option C) & IgM Tests: Not clinically useful d/t poor sensitivity and specificity. Viral Diagnostic Tests: HIV DNA PCR: Highly sensitive and specific by two weeks of age. HIV RNA PCR: Similar to DNA PCR.

> 24 Months	IgG Antibody Tests: Reliable if repeatedly positive, confirmed with HIV PCR.
Testing Protocols	Rapid HIV Tests (Option D) are useful for immediate diagnosis in newborns, but positive results must be confirmed with a second test or PCR. For breastfed infants, test 12 weeks after breastfeeding ends to detect any potential infection acquired through lactation. High-Risk Infants: Initial viral testing within the first 12-24 hours, followed by additional tests at 2-3 weeks, 4-8 weeks, and 4-6 months if needed.
Definitive Diagnosis	For non-breastfed infants, HIV infection can be excluded with Two negative virologic tests at ≥1 month and ≥4 months Two negative antibody tests at ≥6 months.

Solution for Question 13:

Correct Answer: B) Flaccid paralysis with absent reflexes

Explanation:

- Poliomyelitis often leads to flaccid paralysis because it primarily affects the anterior horn cells in the spinal cord, which are responsible for motor control resulting in muscle weakness and atrophy.
- Reflexes may be diminished or absent due to the disruption of the motor neurons.

Poliovirus (Family: Picornaviridae)

- Transmission: Fecal-oral route
- Incubation Period: 8-12 days from exposure to the onset of symptoms.
- It has three serotypes: PV1, PV2, and PV3
- Pathogenesis: The virus spreads from the intestinal tract to the central nervous system (CNS), potentially causing aseptic meningitis and poliomyelitis.
- Types of Poliovirus Infections:

Inapparent Infection	90-95% of cases; asymptomatic
Abortive Poliomyelitis	Mild, flu-like symptoms with full recovery
Nonparalytic Poliomyelitis	More severe symptoms, including neck stiffness and pain, with a good prognosis

Paralytic Poliomyelitis	Includes spinal, bulbar, and polio encephalitis forms, presenting with various degrees of paralysis

- Diagnosis: Consider polio in any unimmunised child with acute flaccid paralysis, particularly in areas with past or current polio cases. Isolation of poliovirus from stool specimens collected 24-48 hours apart.
- Consider polio in any unimmunised child with acute flaccid paralysis, particularly in areas with past or current polio cases.
- Isolation of poliovirus from stool specimens collected 24-48 hours apart.
- Consider polio in any unimmunised child with acute flaccid paralysis, particularly in areas with past or current polio cases.
- Isolation of poliovirus from stool specimens collected 24-48 hours apart.
- Management: Supportive care is primary; ventilatory support may be necessary for severe respiratory involvement.
- Prevention (Vaccination): Oral Polio Vaccine (OPV): Live attenuated vaccine administered orally that induces both systemic & mucosal immunity with a risk of vaccine-associated polio. Inactivated Polio Vaccine (IPV): Killed virus vaccine. Given via injection that provides systemic immunity without the risk of vaccine-derived polio.
- Oral Polio Vaccine (OPV): Live attenuated vaccine administered orally that induces both systemic & mucosal immunity with a risk of vaccine-associated polio.
- Inactivated Polio Vaccine (IPV): Killed virus vaccine. Given via injection that provides systemic immunity without the risk of vaccine-derived polio.
- Oral Polio Vaccine (OPV): Live attenuated vaccine administered orally that induces both systemic & mucosal immunity with a risk of vaccine-associated polio.
- Inactivated Polio Vaccine (IPV): Killed virus vaccine. Given via injection that provides systemic immunity without the risk of vaccine-derived polio.

Spastic paralysis with hyperreflexia (Option A) is more characteristic of upper motor neuron lesions, which are not typical of poliomyelitis.

Rigid paralysis with limited movement (Option C) is not typical of poliomyelitis and is more associated with other neurological conditions like Parkinsonism and multiple sclerosis.

Ataxic paralysis with coordination issues (Option D) describes a problem with motor coordination, often seen in cerebellar dysfunction, not poliomyelitis.

Solution for Question 14:

Correct Answer: B) Meningitis

Explanation:

The most common neurological complication of mumps is Aseptic meningitis.

Mumps	
Causative Organism	Mumps virus (Paramyxoviridae family)
Age Group	5-9 years
Transmission	Respiratory droplets
Period of communicability	7 days before to 7 days after the onset of symptoms
Symptoms	High-grade fever, Headache, Parotitis (Often Bilateral)
Complications	Most Common complication in child : Aseptic Meningitis Most Common complication in Adult : Orchitis (does not lead to infertility) Oophoritis, pancreatitis, myocarditis, arthritis, thyroiditis & conjunctivitis. Mumps during early pregnancy increases fetal wastage but not malformations.
Neurological Complications	Meningitis (Most Common) - 10-30% of cases. Encephalitis, transverse myelitis, and facial palsy are less common. (Option B, C, D)
Treatment	Self-limiting Management focuses on symptom relief—pain reduction, hydration, and antipyretics.

Solution for Question 15:

Correct Answer: B) Nagayama spots

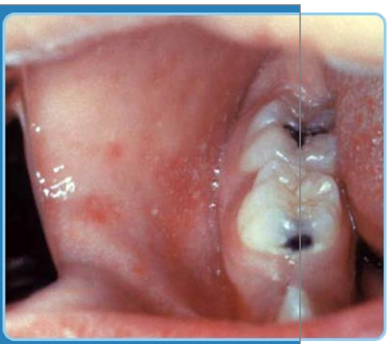
Explanation:

- Nagayama spots are small red papules or ulcers found at the uvulopalatoglossal junction and are characteristic of Roseola infantum (sixth disease).

- They typically appear following a high fever that abruptly resolves, accompanied by the development of a rash.

Roseola Infantum or Exanthem subitum or Sixth disease

- Causative agent: Human Herpesvirus 6 (HHV-6B).
- Nearly all children are infected by age 2, and peak infection occurs between 6 and 9 months of age.
- Spread via saliva or respiratory droplets from asymptomatic individuals.
- Characterised by a sudden high fever lasting 3-5 days, followed by the abrupt onset of a maculopapular rash that begins on the trunk and spreads to the extremities.

Koplik Spots (Option A) are small, bluish-white spots with a red halo located on the buccal mucosa at the level of premolars and are diagnostic for measles (Rubeola).	
Forchheimer spots (Option C) are small reddish spots on the soft palate or uvula, indicative of rubella (German measles).	
Café au Lait Spots (Option D) are light brown skin patches with irregular borders, often associated with neurofibromatosis type 1.	

Solution for Question 16:

Correct Answer: A) Parvovirus B19

Explanation:

- The presentation of pallor, fatigue, mild fever, rash, and a significant decrease in haemoglobin, suggests an aplastic crisis due to parvovirus B19.
- This virus causes cell lysis and depletion of erythroid precursors, leading to transient arrest of erythropoiesis, which exacerbates anemia in children with sickle cell disease.

Clinical Manifestations and Complications of Parvovirus B19 Infection

- Low grade fever, malaise, URTI.
- Erythema infectiosum (Fifth Disease): Characterized by a "slapped-cheek" rash followed by a lacy, reticular rash on the body.
- Arthropathy: Common in adults, presenting as arthritis or arthralgia.
- Transient Aplastic Crisis: Severe anaemia in individuals with chronic hemolytic anaemia.
- Fetal Infection: Risk of nonimmune fetal hydrops and stillbirth typically results in normal deliveries.
- Papular purpural gloves and socks syndrome
- Myocarditis

Epstein-Barr virus (Option B) causes infectious mononucleosis, characterised by fever, sore throat, lymphadenopathy, and splenomegaly. It mainly infects B cells and does not target erythroid progenitor cells like Parvovirus B19.

Cytomegalovirus (Option C) typically causes asymptomatic or mild illness in immunocompetent individuals but leads to severe disease and anemia in immunocompromised patients due to bone marrow suppression, without causing an aplastic crisis.

Human Herpesvirus 6 (Option D) (HHV-6) causes roseola in young children, characterised by high fever and rash, and can cause febrile seizures. It affects T cells without disrupting erythropoiesis.

Solution for Question 17:

Correct Answer: C) Fifth disease

Explanation:

The clinical presentation and image depicting rash, starting with a "slapped cheek" appearance and progressing to a lacy, reticular pattern, is characteristic of the fifth disease, or erythema infectiosum, caused by parvovirus B19.

Erythema Infectiosum (Fifth disease)	
Causative agent	Parvovirus B19
Age group affected	5 to 15 years (70% of cases)

Incubation period	4-28 days
Transmission	Respiratory droplets. Transmission rates among contacts are 15-30%.
Clinical features	 Prodromal phase: Low-grade fever, headache, and mild upper respiratory tract infection symptoms. The characteristic rash in this condition is a "slapped cheek appearance", presenting as erythematous flushing on the face. The erythematous rash spreads to the trunk and proximal extremities as a lacy or reticulated pattern.
Management	Self-limiting disease

First disease (Option A) is another name for Measles / Rubeola, an exanthematous illness caused by an RNA virus belonging to the paramyxovirus family. Here, fever occurs before the onset of rash.

Second disease (Option B) is scarlet fever, which is characterised by fever and an erythematous rash along with strawberry tongue, caused by Group A βStreptococcus.

Sixth disease (Option D) is also known as Roseola infantum or erythema subitum, in which a rash appears once the fever disappears. It is commonly caused by human herpes virus (HHV)-6 and less commonly by HHV-7 and echovirus 16.

Solution for Question 18:

Correct Answer: B) Necrotizing fasciitis

Explanation:

- This question is designed to highlight the urgency of necrotizing fasciitis as a severe and specific complication of chickenpox that requires prompt medical attention.
- A severe, life-threatening infection characterized by rapid tissue destruction and systemic toxicity.

- It often presents with intense pain, swelling, redness, and fever. The infection can spread quickly along the fascial planes, leading to extensive tissue necrosis.

Chickenpox (Varicella)	
Causative agent	Varicella-zoster virus (VZV)
Incubation period	10-21 days.
Clinical manifestations	Fever along with rash.
Rash	 ... (content truncated)
Management	Symptomatic & supportive treatment.- Antipyretics like paracetamol is used (Aspirin avoided due to risk of Reye's syndrome) The child should not attend school until no new lesions appear and all lesions have crusted
Complications	Secondary bacterial infections of the skin lesions like Impetigo (self-limiting) Cellulitis Necrotising fasciitis (Associated with the use of Ibuprofen) Lymphadenitis Subcutaneous abscesses Purpura fulminans (life-threatening condition) Reye syndrome Neurological symptoms like, Stroke Optic neuritis Meningoencephalitis Cerebellar ataxia ... (content truncated)
Prevention	MMRV (measles-mumps-rubella-varicella) vaccine, administered as, 1st dose: 12-15 months of age 2nd dose: 28 days after the 1st dose.

Reye syndrome(Option A) is a rare but serious condition that can occur in children following viral infections, especially if aspirin is used. It involves liver failure and brain inflammation.

Purpura fulminans(Option C) is a severe, acute condition involving widespread skin necrosis and purpura, often due to bacterial infections like meningococcemia.

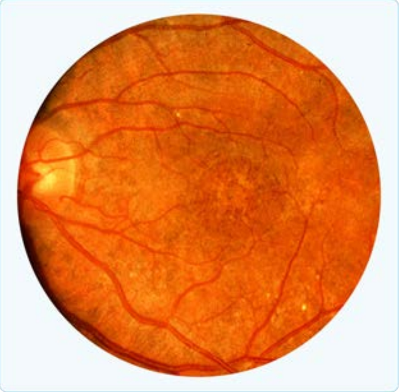
Impetigo (Option D) A secondary bacterial infection which can develop from scratching or secondary infection of the chickenpox lesions, is typically self-limiting with appropriate treatment.

Solution for Question 19:

Correct Answer: C) Pleomorphic rash

Explanation:

- The classical triad of congenital rubella syndrome consists of deafness, cataracts and congenital heart disease (commonly patent ductus arteriosus).
- Transmissibility is highest earliest before 11 weeks of gestation.
- Other clinical features include glaucoma, hepatosplenomegaly, salt and pepper retinopathy, blueberry muffin lesions (Bluish-red nodular lesions), jaundice, etc.
- No treatment exists.
- Diagnosis of rubella in the first trimester of pregnancy is an indication for maternal termination of pregnancy.



Pleomorphic rash (Option C) described as having various kinds of lesions simultaneously (e.g., papules, macules, vesicles, and pustules) is characteristic of chickenpox (varicella). The rash typically begins on the trunk (chest and back) and then spreads outward to the face, scalp, and extremities. The appearance of the vesicles with their clear, fluid-filled content on an erythematous base resembles dew drops sitting on a rose petal. This distinctive appearance helps in differentiating chickenpox from other rashes.

Solution for Question 20:

Correct Answer: B) Otitis media

Explanation

- Otitis media is the most common complication of Measles.
- Complications are more frequent in the very young, the malnourished and the immunocompromised.

Complications of Measles:

- Otitis media (most common complication) (Option B)
- Pneumonia (most common cause of death) (Option A)
- Other respiratory complications - bronchitis, laryngitis, etc.
- Subacute Sclerosing Panencephalitis (SSPE): rare, fatal and long term complication. (Option C)
- Acute encephalitis

Prevention

- Measles is a preventable and potentially eradicable disease.
- Maternal antibodies protect infants until 6-9 months of age, so administering the vaccine at 9 months in endemic countries ensures early protection and seroconversion.
- The measles vaccine, derived from the Edmonston-Zagreb strain, is available as a monovalent preparation or in combination with rubella (MR), mumps (MMR), and varicella (MMR-V).
- The vaccine schedule is as follows:

Vaccine	MR (Measles-Rubella)
Type	Live attenuated
Dose	0.5 ml
Site	Anterolateral thigh or right upper arm (at the insertion of deltoid)
Schedule	1st dose - 9 months; 2nd dose - 16-24 months

Reye syndrome (Option D) is a rare condition associated with aspirin use in children recovering from a viral illness. It is characterised by sudden onset of vomiting, encephalopathy, hepatic dysfunction, and hypoglycemia and can occur as a complication of chickenpox.

Congenital Infections and Bacterial Diseases in Children

1. A 6-day-old newborn presents with difficulty feeding, excessive crying and muscle stiffness. The mother reports that the umbilical cord was cut with an unsterilized blade at home. The baby has spasms triggered by touch. Which of the following is the most appropriate initial management step?

A. Immediate administration of tetanus toxoid (TT)

B. Administration of human tetanus immunoglobulin (TIG)

C. Intravenous administration of broad-spectrum antibiotics

D. Surgical debridement of the wound

2. A 28-year-old pregnant woman at 16 weeks of gestation is diagnosed with varicella. She is concerned about the risk of congenital varicella syndrome (CVS) in her fetus. Which statement regarding CVS in this scenario is most accurate?

A. Risk is 0.4% if varicella is before 16 weeks of gestation.

B. Causes skin scarring and limb hypoplasia.

C. Antiviral treatment prevents further damage in infants.

D. VZV IgM antibodies are highly reliable for diagnosing CVS.

3. A baby is brought to the pediatric department after his mother noticed a yellowish discoloration of his skin and small petechial rashes were noted on examination. Which of the following investigations would be the most reliable and sensitive to diagnose congenital CMV infection?

A. Blood PCR

B. IgM detection

C. Viral culture

D. Urine PCR

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4. A neonate presents with congenital CMV infection. Which of the following clinical features is least commonly associated with this scenario?

A. Hepatosplenomegaly

B. Purpura

C. Jaundice

D. Petechiae

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5. A neonate on examination has bilateral cataracts and, after investigations, is found to have a patent ductus arteriosus. The baby had a low birth weight and is diagnosed with congenital rubella syndrome. On examination, the retina appears as follows. Which of the following retinal changes is seen here?

A. Macular star

B. Retinal tear

C. Salt and pepper retinopathy

D. Macular edema

6. **A 10-year-old child presents with bilateral hearing loss of high tones and has episodes of vertigo. On examination, he has a prominent forehead and linear scars in a spoke-like pattern around his mouth. Which of the following signs is not seen in this stage of the disease?**

A.

B.

C.

D.

7. **An 8-month-old baby is brought to the hospital after her mother noticed rashes all over the baby's body, as shown in the image. The mother gives a history of syphilis during pregnancy. Which of the following is NOT an early onset sign of syphilis?**

A. Mucocutaneous rashes

B. Osteochondritis

C. Hepatosplenomegaly

D. Saddle nose

8. Which of the following maternal treatments for syphilis is most likely to effectively prevent the transmission to the baby?

A. Doxycycline for 2 weeks before delivery

B. Doxycycline for 30 days before delivery

C. Penicillin for 2 weeks before delivery

D. Penicillin for 30 days before delivery

9. A newborn presents with hydrocephalus, chorioretinitis, and intracranial calcifications. Given these findings and a positive Toxoplasma-specific IgM in the CSF, what is the most appropriate treatment regimen for this condition?

A. Pyrimethamine and sulfadiazine with folinic acid

B. Intravenous acyclovir

C. Ganciclovir and valganciclovir

D. Spiramycin

10. An infant presents with the following clinical findings: poor feeding, lethargy, and hydrocephalus. A CT Brain is performed, and the image below shows characteristic intracranial calcifications. Given these findings, which of the following conditions is most likely?

A. Cytomegalovirus (CMV) infection

B. Congenital rubella syndrome

C. Congenital toxoplasmosis

D. Neonatal herpes simplex virus (HSV) infection

11. A 4-year-old child has a history of night sweats, mild fever, and recent weight loss. The child also has a nonproductive cough and mild dyspnea. Chest X-ray is taken and shown below. What is the most likely diagnosis?

A. Primary Tuberculosis

B. Miliary Tuberculosis

C. Bacterial Pneumonia

D. Bronchiectasis

12. A 10-day-old male is brought to the emergency department by his parents, who report that he has been feeding poorly and has had episodes of rigidity. The mother says that she did not receive any antenatal care and delivered the baby at home. On examination, the infant is irritable, has a clenched jaw, and is experiencing frequent, painful muscle spasms. What is the most likely diagnosis?

A. Generalized tetanus

B. Cephalic tetanus

C. Neonatal tetanus

D. Localized tetanus

13. A 3-month-old unimmunized infant presents with a history of severe paroxysms of coughing, which are often followed by a high-pitched inspiratory sound and occasional vomiting. The parents report these symptoms have persisted for 2 weeks. Based on these symptoms, which drug is recommended for management?

A. Azithromycin

B. Erythromycin

C. TMP-SMX

D. Penicillin

14. A 6-year-old child, fully vaccinated against diphtheria but not having received a booster dose in the past 5 years, is identified as an asymptomatic contact of a confirmed diphtheria case. What is the most appropriate management for this child?

A. Administer a single dose of diphtheria antitoxin

B. Administer benzathine penicillin G

C. Administer a booster dose of diphtheria toxoid vaccine

D. No treatment or vaccination needed as the child is asymptomatic

15. An unimmunized 7-year-old child presents to the emergency department with a sore throat, fever, and a thick grayish membrane covering the tonsils. The physician suspects diphtheria and begins treatment. What is the most appropriate initial step in management?

A. Administer erythromycin

B. Perform throat culture and wait for results

C. Administer diphtheria antitoxin immediately

D. Administer DPT vaccine

16. A 13 y/o girl presents with a sore throat, fever, and dysphagia. On examination, her oropharynx appears as shown below. The patient also exhibits symptoms of airway obstruction and has a "bull-neck" appearance. Which of the following is the most likely etiology of these symptoms?

A. Streptococcus pneumoniae

B. Corynebacterium diphtheriae

C. Haemophilus influenzae

D. Adenovirus

17. A 5-year-old girl presents with fever, sore throat, and a red, finely papular rash that started on her neck and spread to her body. The rash is more pronounced in the skin creases, and her examination reveals the findings in the image. What is the most likely diagnosis?

A. Measles

B. Kawasaki disease

C. Scarlet fever

D. Hand, foot and mouth disease

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18. A 3-year-old child is a household contact of a relative with active tuberculosis (TB). The child is asymptomatic and has been evaluated, confirming no active TB disease. The child is also HIV-negative. What is the most appropriate management for this child according to current recommendations?

A. Pyrazinamide daily for 6 months.

B. Isoniazid daily for 6 months.

C. BCG vaccine.

D. Ethambutol daily for 3 months.

19. A 4-year-old child presents with a persistent cough and low-grade fever for 2 weeks. Chest X-ray was taken, and the image is shown below. Based on the X-ray findings, the child was started on a treatment regimen but later developed optic neuritis as a side effect. Which drug from the treatment regimen is most likely responsible for this adverse effect?

A. Isoniazid

B. Rifampin

C. Ethambutol

D. Pyrazinamide

20. A study was conducted on the tuberculin skin test (TST) in a population of children. The researchers aimed to identify groups of children at increased risk of developing tuberculosis. Which of the following groups of children is at an increased risk even with TST induration ≥5 mm?

A. Children in close contact with tuberculosis patient

B. Children born in high tuberculosis prevalence areas

C. Children with Hodgkin's disease

D. Children ≥4 yr old without any risk factors

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	2
Question 5	3
Question 6	2
Question 7	4
Question 8	4
Question 9	1
Question 10	3
Question 11	1
Question 12	3
Question 13	1
Question 14	3
Question 15	3
Question 16	2
Question 17	3
Question 18	2
Question 19	3
Question 20	1

Solution for Question 1:

In a patient presenting with typical symptoms of tetanus, such as difficulty feeding, excessive crying, muscle stiffness and spasms triggered by touch, WHO recommends the immediate administration of human tetanus immunoglobulin (TIG), which is crucial to neutralize the circulating tetanus toxin.

- The diagnosis of Tetanus is mostly clinical, and routine laboratory studies are usually normal. The spatula test is a bedside diagnostic test that is highly sensitive and specific for tetanus, where a spatula or tongue blade is touched to the oropharynx.
- Normally, this elicits a gag reflex, causing the patient to expel the spatula. In tetanus, the patient reflexively bites the spatula due to masseter muscle spasm, indicating a positive test.

A serum antibody titer of ≥0.01 U/mL is considered protective.

Management Step	Comments
Human Tetanus Immunoglobulin (TIG) (Option B)	Neutralizes circulating toxin; Given immediately. If TIG is not available, Human IVIG is given.
Tetanus Toxoid (Option A)	Provides immunity; complete course of tetanus toxoid given once recovered.
Broad-Spectrum Antibiotics (e.g., Metronidazole, Penicillin) (Option C)	Metronidazole is the antibiotic of choice. Parenteral penicillin G is an alternative treatment. Total duration: 7-10 days
Wound Debridement (Option D)	Reduces bacterial load and toxin production

An Arthus-type hypersensitivity reaction or a fever >39.4°C (103°F) after a dose of tetanus toxoid-containing vaccine, often associated with high serum tetanus antitoxin levels, indicates localized vasculitis from immune complex deposition. Such individuals should not receive Td more frequently than every 10 years, even with a significant tetanus-prone injury.

Solution for Question 2:

Correct Answer: B) Causes skin scarring and limb hypoplasia.

Explanation: Congenital varicella syndrome (CVS) is characterized by cicatricial zigzag skin scarring in a dermatomal pattern and limb hypoplasia.

Congenital Varicella	
Transmission	Results from in utero transmission of varicella-zoster virus (VZV).
Incidence	Occurs in 0.4% of infants born to women who have varicella before 13 weeks of gestation. (Option A) 2% of those born to women with varicella between 13 and 20 weeks of gestation.
Clinical Features	Cicatricial zigzag skin scarring in a dermatomal pattern Limb hypoplasia (Option B) Neurological issues: Microcephaly Cortical atrophy Seizures Mental retardation Eye problems: Chorioretinitis Microphthalmia Cataracts Renal issues: Hydroureter ... (content truncated)
Diagnosis	Based on a history of maternal varicella. Presence of characteristic abnormalities in the newborn. Viral DNA can be detected using polymerase chain reaction (PCR) in tissue samples. VZV IgM antibodies may be found in cord blood, though these are not always reliable.(Option D) Ultrasound may help identify limb atrophy.
Treatment	VZV-specific immune globulin (VZIG) and acyclovir may be administered to the mother to reduce disease severity, but their impact on fetal outcomes is uncertain. Antiviral treatment of infants with CVS is not indicated as the damage caused by fetal VZV infection does not progress after birth. (Option C)
Image	congenital varicella" data-author="A. Sauerbrei" data-hash="1041" data-license="CC BY 4.0" data-source="https://commons.m.wikimedia.org/wiki/File:CongenitalVaricellaStill.png" data-tags="March2025" height="779" src="https://image.prepladder.com/notes/73MqvE34ypWZCxF5LMss1742185907.png" width="500" />

Solution for Question 3:

Correct Answer: D) Urine PCR

Explanation:

Urine PCR is the most reliable and sensitive test for diagnosing congenital CMV infection. It is preferred over blood PCR, IgM detection, and viral culture due to its high sensitivity and specificity for detecting CMV in newborns.

- Early Diagnosis: It is essential to identify CMV within the first 2-3 weeks of life for accurate diagnosis.
- Testing Methods: Polymerase Chain Reaction (PCR): The most reliable and sensitive method; Detects viral DNA in bodily fluids and quantifies viral load. Viral Culture (Option C): Culturing the virus combined with immunofluorescence to detect CMV-specific proteins.
- Polymerase Chain Reaction (PCR): The most reliable and sensitive method; Detects viral DNA in bodily fluids and quantifies viral load.
- Viral Culture (Option C): Culturing the virus combined with immunofluorescence to detect CMV-specific proteins.
- Sample Collection: Preferred: Urine and saliva. (Option B) Alternative: Blood, though less sensitive. (Option A)
- Preferred: Urine and saliva. (Option B)
- Alternative: Blood, though less sensitive. (Option A)
- Screening Programs: Some hospitals use saliva samples for newborn screening; positive results should be confirmed with urine tests.
- Challenges with Antibody Tests: IgM Antibodies (Option D): These may suggest recent infection but can persist and are not always reliable for congenital diagnosis. IgG Antibodies: Persistent throughout life; their presence does not confirm congenital infection.
- IgM Antibodies (Option D): These may suggest recent infection but can persist and are not always reliable for congenital diagnosis.
- IgG Antibodies: Persistent throughout life; their presence does not confirm congenital infection.
- Polymerase Chain Reaction (PCR): The most reliable and sensitive method; Detects viral DNA in bodily fluids and quantifies viral load.
- Viral Culture (Option C): Culturing the virus combined with immunofluorescence to detect CMV-specific proteins.
- Preferred: Urine and saliva. (Option B)
- Alternative: Blood, though less sensitive. (Option A)

- IgM Antibodies (Option D): These may suggest recent infection but can persist and are not always reliable for congenital diagnosis.
- IgG Antibodies: Persistent throughout life; their presence does not confirm congenital infection.
- Treatment Guidelines: No universal guidelines; treatment is case-specific based on infection severity and symptoms.
- Antiviral Therapy: Ganciclovir (IV for 6 wks): Used for symptomatic newborns and severe perinatal infections. May improve long-term outcomes but requires more research.
- Ganciclovir (IV for 6 wks): Used for symptomatic newborns and severe perinatal infections. May improve long-term outcomes but requires more research.
- Long-Term Follow-Up: Hearing Tests: Regular monitoring due to the risk of hearing loss. Developmental Assessments: Periodic evaluations to identify delays or disabilities early. Vision Monitoring: Routine eye exams to detect less common vision problems.
- Hearing Tests: Regular monitoring due to the risk of hearing loss.
- Developmental Assessments: Periodic evaluations to identify delays or disabilities early.
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- Ganciclovir (IV for 6 wks): Used for symptomatic newborns and severe perinatal infections. May improve long-term outcomes but requires more research.
- Hearing Tests: Regular monitoring due to the risk of hearing loss.
- Developmental Assessments: Periodic evaluations to identify delays or disabilities early.
- Vision Monitoring: Routine eye exams to detect less common vision problems.

Solution for Question 4:

Correct Answer: B) Purpura

Explanation:

Purpura is the least common symptom in congenital CMV infection.

Most infants who are infected are asymptomatic.

Congenital CMV infection can be described as follows:

Human cytomegalovirus (CMV)	

Mode of Transmission	Bodily fluids: saliva, urine, blood, vaginal fluids, breast milk Community exposure: Daycare centres Perinatal exposure: During birth Breastfeeding
Risk of Transmission	30% of women with primary CMV infection during pregnancy transmit the virus to the fetus. In-utero infections also occur in previously immune women but at a reduced rate
Transmission by Gestational Age	Transmission rates are higher following a primary infection, with the risk increasing as gestational age advances.
Clinical Features of congenital CMV infection	Asymptomatic Infection (90% of infected newborns): No signs at birth; detected through newborn screening. Symptomatic Infection (10% of infected newborns): Jaundice (direct bilirubin >2 mg/dL) (Option C) Petechiae (small, red, or purple spots on the skin) giving a blueberry muffin appearance (Option D) Hepatosplenomegaly (Option A) Microcephaly Intrauterine growth restriction Chorioretinitis Purpura (3% infants) (larger areas of bleeding under the skin) Less Common Findings: Hearing loss (M/c long-term consequence; may develop later). Lab findings align with the clinical presentation: Direct hyperbilirubinemia, ■ ALT, thrombocytopenia, and Head CT abnormalities

Solution for Question 5:

Correct Answer: C) Salt and pepper retinopathy

Explanation:

The retinal change seen in congenital rubella is “salt and pepper” retinopathy.

Congenital rubella	

Rubella (German measles) is a mild disease in infants and children but can cause congenital rubella syndrome (CRS) if contracted during pregnancy. The risk is highest in the first trimester, especially within the first 8 weeks.	
Pathogenesis	Viremia is most intense from 10 to 17 days after infection. Viral shedding is from day 10 of infection up to 2 weeks.
Risk of Defects by Gestational Age	Highest in the first 12 weeks of gestation and decreases significantly thereafter, with defects being rare after 20 weeks.
Clinical features of congenital rubella syndrome	 ... (content truncated)
Chronicity and Complications	Chronic presence of the virus can cause ongoing damage The risk of progressive rubella panencephalitis (PRP) years later is a rare complication. Less common complications: Growth retardation, hepatitis, bone lesions. Risk of diabetes and thyroid dysfunction
Diagnosis	Rubella IgM enzyme immunosorbent assay (detects antibodies) Viral isolation (growing the virus from samples) Polymerase chain reaction (PCR) (detects viral RNA)
Management	No specific treatment for CRS; supportive care is key Early intervention for complications (e.g., hearing loss) is crucial
Prevention	Standard with droplet precaution in hospitalized cases Rubella vaccination RA 27/3 live attenuated vaccine (MMR or MMRV with varicella) effectively prevents rubella and CRS MMR vaccine has a 95% efficacy after one dose and 99% after two doses Vaccination side effects: fever, rash, joint pain
Considerations for Pregnant Women	Immunoglobulin may be considered if pregnancy is not terminated, though it does not guarantee prevention of fetal infection

Macular star, Retinal tear, and Macular edema (Options A, B, and D) are not associated with congenital rubella syndrome.

Solution for Question 6:

Explanation:

The clinical scenario points to the diagnosis of congenital syphilis with late-onset symptoms but Gottron's papules are seen in dermatomyositis.

- Late manifestations of congenital syphilis appear after the first two years of life and are less common in developed countries. These signs are mainly due to hypersensitivity and chronic granulomatous inflammation affecting the bones, teeth, and central nervous system.

Late-onset manifestations are:

Feature	Description
Olympian brow	Bony prominence of the forehead caused by persistent or recurrent periostitis
Clavicular or Higoumenaki's sign	Unilateral or bilateral thickening of the sternoclavicular third of the clavicle
Saber shins (Option C)	 Anterior bowing of the midportion of the tibia
Hutchinson's teeth (Option A)	 Peg-shaped upper central incisors, abnormal enamel, a notch along the biting surface

Mulberry molars	 Abnormal 1st lower molars, small biting surface, and excessive number of cusps
Saddle nose	 ... (content truncated)
Rhagades (Option D)	 ... (content truncated)
Juvenile paresis	Latent meningovascular infection
Juvenile tabes	Rare spinal cord and cardiovascular involvement (aortitis)
Hutchinson triad	Hutchinson teeth, interstitial keratitis, and 8th nerve deafness
Clutton joint	Unilateral or bilateral painless joint swelling from synovitis with sterile synovial fluid

Interstitial keratitis	Intense photophobia, and lacrimation, followed by corneal opacification and complete blindness within weeks or months
8th nerve deafness	Unilateral or bilateral, presents at any age as vertigo and high-tone hearing loss, and later with permanent deafness.

Gottron’s papules (Option B) are seen in dermatomyositis which is an idiopathic inflammatory condition of the skin and skeletal muscle. They are small purple or red flat papules on extensor surfaces, particularly the elbows and joints of the hand.

Solution for Question 7:

Correct Answer: D) Saddle nose

Explanation:

Saddle nose is a late onset sign of congenital syphilis that presents after the age of 2 years.

- Congenital syphilis, caused by Treponema pallidum, can present with early signs within the first two years of life and late signs appearing gradually over the first two decades.

Early signs of congenital syphilis (within first 2 yrs of age)
 ... (content truncated)

Solution for Question 8:

Correct Answer: D) Penicillin for 30 days before delivery

Explanation:

- Penicillin is the recommended treatment for syphilis during pregnancy (Option D), if clinical or serologic findings indicate active syphilis, or if active infection cannot be confidently ruled out.
- It is crucial to initiate treatment at least 30 days before delivery, not 2 weeks (Option C) to effectively prevent congenital syphilis.
- Doxycycline and tetracycline should not be given during pregnancy. (Options A and B)
- Pregnant women allergic to penicillin should undergo desensitization and then be treated with penicillin.
- Macrolides are not effective in preventing fetal infection.
- Women who were previously treated adequately do not require further therapy unless there is a 4-fold increase in titer, indicating possible reinfection.
- Congenital syphilis, caused by Treponema pallidum, occurs through the transplacental transmission of spirochetes or, less commonly, through direct contact with infectious lesions during childbirth.
- Women with primary or secondary syphilis and spirochetemia have a higher likelihood of passing the infection to the fetus compared to those with latent infection.
- Kassowitz law states that “if a woman with untreated syphilis has a series of pregnancies, the likelihood of infection of the fetus in later pregnancies becomes less”.
- Transmission can happen at any stage of pregnancy, leading to outcomes such as early fetal loss, preterm birth, low birth weight, stillbirth, neonatal death, or infants born with congenital syphilis.

Clues Suggesting Diagnosis of Congenital Syphilis	
Epidemiologic Background	Untreated early syphilis in the mother Untreated latent syphilis in the mother Mother has contact with a known syphilitic during pregnancy and remains untreated Mother treated less than 30 days prior to delivery Mother treated with a non-penicillin drug for syphilis during pregnancy Mother treated for syphilis during pregnancy but lacks follow-up demonstrating a 4-fold decrease in titer Mother coinfected with HIV

Solution for Question 9:

Correct Answer: A) Pyrimethamine and sulfadiazine with folinic acid

Explanation:

This presentation with hydrocephalus, chorioretinitis, intracranial calcifications, and a positive Toxoplasma-specific IgM in the CSF is consistent with congenital toxoplasmosis. Pyrimethamine and Sulfadiazine with Folinic Acid (Option A) is the standard treatment for congenital toxoplasmosis. Pyrimethamine and sulfadiazine work synergistically to inhibit the growth of Toxoplasma gondii. Folinic acid is added to prevent bone marrow suppression, a common side effect of pyrimethamine, which inhibits dihydrofolate reductase, leading to reduced folic acid synthesis.

Diagnosis

Diagnosis Method	Details
Fetal Ultrasound	Performed every 2 weeks during gestation after diagnosis of maternal infection.
PCR Analysis of Amniotic Fluid	Used for prenatal diagnosis of fetal infection.
Serologic Tests	Persistent or rising titers in dye/IFA tests, positive IgM-ELISA.
Newborn Evaluation	General, ophthalmologic, neurologic exams; head CT scan for intracranial calcifications.
CSF Analysis	Lumbar puncture with analysis for Toxoplasma-specific IgG and IgM, total IgG, cells, glucose, and protein.
Toxoplasma Isolation	Isolation from placenta, infant’s leukocytes, and CSF using PCR

Management

Treatment	Details
Pyrimethamine and Sulfadiazine, along with Folinic Acid (Calcium Leucovorin)	Synergistic action against Toxoplasma gondii. Standard treatment for congenital toxoplasmosis. Folinic Acid is administered to prevent bone marrow suppression caused by pyrimethamine
Spiramycin	Used in pregnant women during the first trimester to prevent vertical transmission.
Monitoring	Regular leukocyte counts twice weekly to detect neutropenia and other hematologic effects.

Acyclovir (Option B) is the treatment of choice for neonatal herpes simplex virus (HSV) infection.

Ganciclovir and Valganciclovir (Option C) are antiviral agents used to treat congenital cytomegalovirus (CMV) infection. Although CMV can also present with intracranial calcifications, the presence of positive Toxoplasma-specific IgM in the CSF and the characteristic intracranial calcifications are more indicative of congenital toxoplasmosis.

Spiramycin (1 g PO every 8 hr without food) is recommended for the prevention of fetal infection if the mother develops toxoplasmosis during pregnancy. It is used for infection acquired early in gestation when fetal infection is uncertain. (Option D)

Solution for Question 10:

Correct Answer: C) Congenital toxoplasmosis

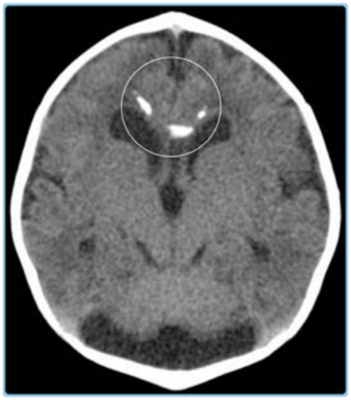
Explanation:

The presence of characteristic intracranial calcifications is a hallmark of Congenital toxoplasmosis. This condition, caused by Toxoplasma gondii, often presents with the classic triad of intracranial calcifications, hydrocephalus, and chorioretinitis. Early diagnosis and treatment are crucial to prevent long-term complications such as mental retardation and deafness.

Category	Details
Etiology	Primary maternal infection with Toxoplasma gondii during pregnancy is the primary cause
Risk Factors	Infection during the first trimester: severe outcomes, lower transmission rate (~17%) Infection during third trimester: milder outcomes, higher transmission rate (~65%)
Clinical Features	CNS: Hydrocephalus, intracranial calcifications, seizures, developmental delays, microcephaly in severe brain damage
Ocular: Chorioretinitis, vision impairment, strabismus, photophobia, epiphora	
Systemic: Hepatosplenomegaly, lymphadenopathy, prematurity	

Other: Jaundice, thrombocytopenia, rash, hearing loss	

Cytomegalovirus (CMV) Infection (Option A) can also cause intracranial calcifications; however, these are more commonly seen in a periventricular pattern. While there is some overlap, the calcifications in congenital toxoplasmosis are often more diffuse.



Congenital Rubella Syndrome (Option B) is typically associated with cataracts, heart defects, and sensorineural deafness rather than intracranial calcifications.

Neonatal Herpes Simplex Virus (HSV) Infection (Option D) may cause encephalitis and lead to brain lesions, but it does not typically present with the diffuse intracranial calcifications seen in congenital toxoplasmosis.

Solution for Question 11:

Correct Answer: A) Primary Tuberculosis

Explanation:

The chest x-ray describes a localized lesion and prominent left hilar lymph nodes, consistent with the Ghon focus and Ghon complex. These features, along with symptoms like cough and weight loss, match the typical presentation of Primary Tuberculosis.

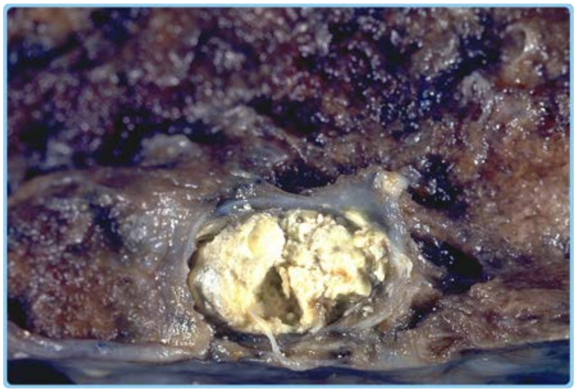
Chest x-ray showing bilateral hilar adenopathy of primary pulmonary TB

Etiological Facts:

- Most common age group in children: <5 years (55%).
- Most frequent presentation of tuberculosis in childhood: Pulmonary tuberculosis.
- Most common extrapulmonary tuberculosis in children: Tubercular lymphadenitis/ Scrofula.

Primary Tuberculosis

- Infection Location: Always begins in the lungs. Typically, the bacilli implant in the lower part of the upper lobe or upper part of lower lobe close to pleura.
- Lung is the portal of entry in >98% of cases.
- Ghon Focus: A 1 - 1.5 cm area of gray-white inflammation with consolidation, often undergoing caseous necrosis.
- Ghon Complex: Combination of the parenchymal lung lesion and caseating regional lymph nodes.
- Dissemination: Initial lymphatic and hematogenous spread to other body parts.
- Immune Response: In 95% of cases, cell-mediated immunity controls the infection, leading to fibrosis and calcification of the Ghon complex.



Clinical Features: Up to 50% of infants and children with moderate to severe pulmonary tuberculosis may have no physical findings. Infants are more likely to exhibit symptoms, including:

- Nonproductive cough and mild dyspnea (M/c symptoms).
- Systemic complaints: Fever, night sweats, anorexia, decreased activity.
- Difficulty in gaining weight or failure-to-thrive syndrome.
- Pulmonary signs are less common but may include Localized wheezing, Decreased breath sounds, Tachypnea, Rarely respiratory distress.
- Localized wheezing,
- Decreased breath sounds,
- Tachypnea,
- Rarely respiratory distress.
- Localized wheezing,
- Decreased breath sounds,
- Tachypnea,
- Rarely respiratory distress.

Miliary Tuberculosis (Option B) is characterized by numerous tiny nodules scattered throughout the lungs. This differs from the scenario's localized pleural lesion and hilar lymphadenopathy, which are more typical of primary pulmonary tuberculosis.

miliary tuberculosis" data-author="James Heilman, MD" data-hash="1010" data-license="CC BY-SA 4.0" data-source="https://en.wikipedia.org/wiki/Miliary_tuberculosis#/media/File:PulmonaryTBCXR.png" data-tags="March2025" height="529" src="https://image.prepladder.com/notes/alk0LxwRTRzS6zwVx0Ep1742181615.png" width="500" />

Bacterial Pneumonia (Option C) can present as either patchy consolidation (bronchopneumonia) or widespread consolidation of a lobe (lobar pneumonia). However, the scenario describes a localized pleural lesion and hilar lymphadenopathy, which are not typical of bacterial pneumonia.

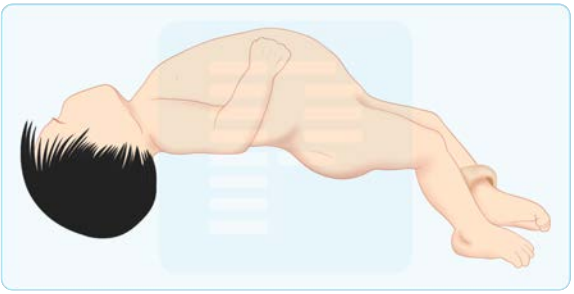
Bronchiectasis (Option D) typically affects the lower lobes and presents with dilated airways filled with mucopurulent secretions. It often involves bilateral lung fields, not the localized pleural lesion and hilar lymphadenopathy described in the scenario.

Solution for Question 12:

Explanation:

Neonatal Tetanus presents with difficulty feeding, rigidity, and spasms typically beginning within the first two weeks of life, often around the third day. It is a spastic paralytic illness caused by tetanus toxin, a neurotoxin produced by Clostridium tetani. The lack of maternal immunization is a critical risk factor.

The baby may also develop opisthotonic posturing.



Type of Tetanus	Clinical Features	Comments
Generalized Tetanus (Option A)	Trismus/locked jaw (classical manifestation) Risus sardonicus (from intractable spasms of facial and buccal muscles) Opisthotonic posturing (arched posture of extreme hyperextension of the body) Headache, restlessness, and irritability are early symptoms.	Common form in older children and adults. Incubation period: 2–14 days
Neonatal Tetanus (Option C)	Presents during 3–12 days of life Progressive difficulty in feeding (sucking and swallowing) Associated hunger and crying Paralysis or reduced movement Stiffness and rigidity to touch May present with opisthotonos	High mortality, linked to lack of maternal immunization. The umbilical stump, often the entry point for the infection, may show signs of dirt, clotted blood, or serum, but can also appear relatively normal.
Cephalic Tetanus (Option B)	Cranial nerve involvement Facial paralysis Difficulty swallowing	Poor prognosis, rare. In association with chronic otitis media, injury to head
Localized Tetanus (Option D)	Rigidity and pain confined to the muscles near the wound.	Less severe, may progress to generalized tetanus.

Solution for Question 13:

Correct Answer: A) Azithromycin

Explanation:

- The infant’s symptoms (severe coughing spells with a "whoop" and vomiting) are classic for pertussis (whooping cough).
- Azithromycin is the recommended first-line antibiotic for all age groups, including infants, for both treatment and post-exposure prophylaxis of pertussis.
- It is preferred over other macrolides due to a lower risk of side effects, such as infantile hypertrophic pyloric stenosis (IHPS), which is more commonly associated with erythromycin (Option B), especially in infants younger than 14 days.
- TMP-SMX (Option C): Alternative for infants >2 months if azithromycin is not suitable.

Pertussis	
Causative agent	Bordetella pertussis, Bordetella parapertussis
Secondary attack rate	100%
Clinical Stages	Catarrhal (1-2 weeks): Cold-like symptoms, worsening cough Paroxysmal (2-6 weeks): Severe cough, "whoop", vomiting Convalescent (1-4 weeks): Recovery
Complications	Apnea, Otitis media, Pneumonia, Pulmonary hypertension, Cardiogenic shock, Hemorrhage, Seizures, hernia and malnutrition.
Diagnosis	Clinical signs, nasopharyngeal swab culture, PCR, serology.
Differential Diagnosis	Adenoviral infection, TB, foreign body inhalation, airway compression.
Management	First-line antibiotics for treating pertussis: Azithromycin is the first-line treatment for pertussis in all age groups. (option A) Erythromycin and clarithromycin are alternative treatments but are not recommended in <1 month old due to risk for IHPS. Alternative antibiotics: For patients allergic to macrolides or there is unavailability of azithromycin, trimethoprim-sulfamethoxazole (TMP-SMX) may be used, but it is C/I in <2 months old (due to the risk for kernicterus) Timing of treatment: Antibiotics are most effective during the catarrhal stage Isolation: Patients diagnosed with pertussis should be isolated from others, especially vulnerable populations like infants and the elderly, for at least 5 days after starting antibiotics.

Prevention	DTaP vaccine (Diphtheria, Tetanus, and Pertussis): Given to children under 7 years of age as part of routine childhood vaccinations. Tdap vaccine (Tetanus, Diphtheria, and Pertussis): Recommended for adolescents, adults, and pregnant women (preferably between 27 and 36 weeks gestation) to provide protection and reduce transmission to infants.

Penicillin (Option D) is not effective against Bordetella pertussis and is not used for either treatment or prophylaxis of pertussis.

Pediatric indications of penicillin administration:

- Congenital syphilis, Diphtheria (post-exposure prophylaxis), Rheumatic Fever, Tetanus (alternative to metronidazole), Bacterial Pneumonia, acute pharyngitis, Acute Otitis Media, Impetigo, Erysipelas.

Solution for Question 14:

Correct Answer: C) Administer a booster dose of diphtheria toxoid vaccine

Explanation: The child is due for a booster (having not received one in the past 5 years), administering the diphtheria toxoid vaccine booster is the most appropriate action to ensure immunity.

(Option A) Diphtheria antitoxin is used to treat individuals with symptomatic diphtheria, not for prophylaxis in asymptomatic contacts. The child does not need antitoxin because they are asymptomatic.

(Option B) Benzathine penicillin G is used for prophylaxis in individuals who have not been fully vaccinated.

(Option D) Even asymptomatic contacts need to have their vaccination status reviewed and updated.

Diphtheria - Management of contacts	

Asymptomatic Case Contacts	Includes household members and close contacts. Monitored for 7 days for symptoms. Prophylaxis: IM benzathine penicillin G (600,000 units for <6 years, 1,200,000 units for ≥6 years) or 10-day oral erythromycin (40-50 mg/kg/day, max 2 g/day). (Option B) Assess and update immunization status. Booster Dose: Administer the diphtheria toxoid vaccine to individuals who have not received a booster within the past 5 years. (Option C) Vaccinate children who have not yet received their 4th dose of diphtheria toxoid. Incomplete or Uncertain Immunization: For those who have received fewer than 3 doses of diphtheria toxoid or have an uncertain immunization status, provide the vaccine according to the primary immunization schedule with an age-appropriate preparation
Asymptomatic Carriers	Receive 10-14 days of antimicrobial prophylaxis. Administer diphtheria toxoid vaccine if no booster has been given in the past year. Droplet precautions for respiratory carriers; contact precautions for cutaneous carriers. Precautions continue until two negative cultures (24 hours apart) post-treatment. Antitoxin is not recommended for asymptomatic carriers.
Follow-up & Repeat Cultures	Repeat cultures for cases and carriers 2 weeks post-treatment. If positive, repeat a 10-day erythromycin course and further cultures. Complete eradication is not guaranteed after one course of antibiotics.
Casual Contacts	Routine investigation is not recommended due to low carriage rates.
Hospital-Acquired Transmission	Rare; only those with direct contact with respiratory/oral secretions need management as contacts.

Solution for Question 15:

Correct Answer: C) Administer diphtheria antitoxin immediately

Explanation : Diphtheria antitoxin is crucial for treating diphtheria because it neutralizes the diphtheria toxin that is causing the disease's symptoms. The toxin can cause severe complications, including damage to the heart and nerves. Early administration of the antitoxin is essential to prevent further systemic effects and improve patient outcomes.

Diphtheria - Diagnosis and management	
Clinical Suspicion	High index of suspicion in exudative pharyngitis with greyish-white pseudomembrane (bleeds on removal). Symptoms: Fever, malaise, sore throat, dysphagia, muffled voice, "bull neck".
Specimen Collection	Collect from nose, throat, mucocutaneous lesions, and membrane exudate (if present).
Laboratory Confirmation	Albert Stain: Rapid visualization of Corynebacterium diphtheriae from oropharynx/larynx swabs. Culture: Definitive identification (8 hours) from affected area swabs nose, throat, lesions). Species-level identification, toxigenicity, antimicrobial susceptibility testing for Coryneform organisms. Send isolates to reference labs.
MANAGEMENT	
Diphtheria Antitoxin	Role: Cornerstone of treatment; administer immediately. Dosage: 20,000-100,000 units based on toxicity and illness duration. Sensitivity testing required; not effective for asymptomatic carriers.
Antibiotic Therapy	Aims: Stop toxin production, treat infection, prevent transmission. Regimens: Erythromycin (40-50 mg/kg/day) or Penicillin G (100,000-150,000 units/kg/day) for 14 days; shorter for cutaneous diphtheria. 2 negative cultures 24 hours apart confirm elimination of organisms. If C.Diphtheria is still present, erythromycin is repeated.
Supportive Care	Precautions: Droplet (pharyngeal), contact (cutaneous) until negative cultures. Wound Cleaning: Soap and water for cutaneous diphtheria. Rest: Bed rest during the acute phase; gradual recovery.
Vaccination	Administer diphtheria toxoid post-infection to complete the primary series or provide booster doses.

Administer erythromycin (Option A): Erythromycin is part of the treatment but should not delay the administration of the antitoxin.

Perform throat culture and wait for results (Option B): Waiting for culture results may delay the necessary treatment.

Administer DPT vaccine (Option D): While vaccination is administered for treating the patient, it can only provide future protection, and does not help in neutralizing the toxins immediately

.

Solution for Question 16:

Correct Answer: B) Corynebacterium diphtheriae

Explanation:

- Corynebacterium diphtheriae causes diphtheria, which is characterized by the formation of a greyish-white pseudomembrane in the throat, airway obstruction, and systemic effects.
- Removing the pseudomembrane is difficult and reveals a bleeding, edematous submucosa.
- May extend to the tonsils, soft palate, uvula and posterior pharyngeal wall.

Diphtheria	
Etiology	Caused by Corynebacterium diphtheriae or, rarely, toxigenic C. ulcerans. Aerobic, nonencapsulated, non-spore-forming, pleomorphic, gram-positive bacilli.
Incubation Period	2-4 days (can range from 1-10 days for respiratory diphtheria)
Transmission	Through respiratory droplets, direct contact with secretions or skin lesions, fomites, or contaminated food/milk.
Asymptomatic Carriers	Carry for up to 6 months.
Clinical Manifestations	Respiratory tract (tonsillar/pharyngeal): Sore throat, fever, dysphagia, pseudomembrane, airway obstruction. Cutaneous: Superficial, ecthyma-like, non-healing ulcer with a grey-brown membrane. (Extremities > Trunk/head) Other sites: Ear (otitis externa), eye (purulent and ulcerative conjunctivitis), and genital tract (purulent and ulcerative vulvovaginitis).

Toxin Effects	Mechanism: Inhibits protein synthesis, causing cell death and local tissue necrosis. Local Effects: Paralysis of the palate and hypopharynx (early local effect of toxin), Pseudomembrane formation, airway obstruction, "bull-neck" appearance. Systemic Effects (Heart, nervous system, kidney): Cardiomyopathy, Demyelination of nerves, Thrombocytopenia, Acute tubular necrosis
Complications of Diphtheria	Risk of complications ∝ Severity of local infection Respiratory Tract Obstruction: Pseudomembrane can block airways, leading to suffocation, especially in laryngeal diphtheria. Toxic Cardiomyopathy: Symptoms include tachycardia (disproportionate to the degree of fever), prolonged PR interval, ST-T changes, and arrhythmias Toxic Neuropathy (Corticosteroids are ineffective) Rarely septicemia (often fatal), endocarditis, and pyogenic arthritis.
Differential diagnosis of membrane over tonsil	Membranous tonsillitis due to pyogenic organisms forming an exudative membrane; Bacteria like streptococcus, viruses like adenovirus. Vincent angina due to fusiform bacilli and spirochetes causing an easily removable membrane usually over one tonsil revealing an ulcer. Infectious mononucleosis often seen in young adults leads to membrane on both tonsils with enlarged, congested tonsils and lymph node enlargement in the posterior triangle of neck. Agranulocytosis has ulcerative necrotic lesions on tonsils and other regions like oropharynx and the patient is ill. Leukaemia Aphthous ulcers present in any part of the oral cavity or oropharynx and is very painful and may involve the tonsils and the pillars. Malignancy tonsil Traumatic ulcer due to any injury heals by forming a membrane. Candidial infection of tonsil

Bull neck appearence

Streptococcal (Option A) membranous tonsillitis patients have high fever but the membrane is confined to the tonsils.

Hemphilus influenza (Option C) causes a group of diseases according to the variant infecting. Most common presentation is pneumonia with high grade fever, chills, productive cough, lethargy, shortness of breath, chest pain and generalized body aches. A membrane is not a feature in this case.

Viral membranous tonsillitis (Option D) caused by adenovirus presents with high fever, sore throat, membranous tonsillitis with normal leukocyte count and is a self limiting disease with a course of 4-8 days.

Solution for Question 17:

Correct Answer: C) Scarlet fever

Explanation: The presentation of fever, sore throat, characterisitc rashes that is more pronounced at the skin creases (Pastia lines) as in the image, and a strawberry tongue all point to the diagnosis of scarlet fever.

Scarlet Fever	Group A Streptococcus (GAS) pharyngitis is associated with a characteristic rash caused due to a pyrogenic exotoxin (erythrogenic toxin) produced by GAS in individuals lacking antitoxin antibodies.
Caused By	Group A Streptococcus (GAS)
Incidence and Virulence	Less common and less virulent now compared to the past, with incidence depending on toxin strains and the population immunity.

Clinical Features	Rash appears within 24-48 hours, starting around the neck and spreading to the trunk and extremities. Rash is a diffuse, finely papular, erythematous eruption causing bright-red discoloration, which blanches on pressure. Rash is accentuated in the creases of the elbow, axilla, and groin (Pastia lines) and gives the skin a goose-pimple appearance. Erythematous cheeks with pallor around the mouth. After 3-4 days, the rash fades, followed by desquamation starting on the face and progressing downward. Desquamation may occur around fingernails, palms, and soles. Tongue is coated with swollen papillae, which later give a "strawberry tongue" appearance after desquamation. Pharyngitis.
Diagnosis	Classic scarlet fever is typically straightforward to diagnose. A history of recent exposure to GAS is helpful. Diagnosis is confirmed by identifying GAS in the pharynx.
Treatment	Antibiotic therapy should be started immediately. Drug of choice: Penicillin or amoxicillin for 10 days. Alternative for Penicillin Allergy: Oral cephalosporins, clindamycin, Macrolides (erythromycin, clarithromycin) for 10 days or azalide (azithromycin) for 5 days. Injection: Benzathine penicillin G single dose intramuscular injection for those unlikely to complete oral therapy.
Complication	Suppurative Complications: Cervical lymphadenitis Peritonsillar abscess Retropharyngeal abscess Otitis media Mastoiditis Sinusitis Group A Streptococcus (GAS) pneumonia Nonsuppurative Complications: Acute rheumatic fever Acute post-streptococcal glomerulonephritis

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Hand, foot, and mouth disease (HFMD) (Option C) presents with low-grade fever, sore throat, followed by oral ulcers or blisters in the oral cavity, and a papulovesicular rash on the palms and soles, which are not mentioned in the scenario. The described rash is sandpaper-like and accentuated in skin creases, characteristic of scarlet fever rather than HFMD.

Vesicles are seen on ventral aspect of 4th toe of right foot

Superficially erroded papules on lips

Solution for Question 18:

Correct Answer: B) Isoniazid daily for 6 months.

Explanation: Isoniazid (INH) daily for 6 months is the appropriate preventive therapy for children in contact with active TB, effectively managing latent TB infection and reducing the risk of progression to active disease.

Isoniazid Prophylaxis:

- Indications: Asymptomatic contacts under 6 years old of a smear-positive TB case, irrespective of their BCG or nutritional status, after ruling out active disease. HIV-infected children who either had a known exposure to an infectious TB case or tuberculin skin test (TST) positive (induration ≥5 mm), with no active TB disease. All TST-positive children receiving immunosuppressive therapy. Children born to mothers with TB during pregnancy, after ruling out congenital TB.
- Asymptomatic contacts under 6 years old of a smear-positive TB case, irrespective of their BCG or nutritional status, after ruling out active disease.
- HIV-infected children who either had a known exposure to an infectious TB case or tuberculin skin test (TST) positive (induration ≥5 mm), with no active TB disease.
- All TST-positive children receiving immunosuppressive therapy.
- Children born to mothers with TB during pregnancy, after ruling out congenital TB.
- Asymptomatic contacts under 6 years old of a smear-positive TB case, irrespective of their BCG or nutritional status, after ruling out active disease.
- HIV-infected children who either had a known exposure to an infectious TB case or tuberculin skin test (TST) positive (induration ≥5 mm), with no active TB disease.
- All TST-positive children receiving immunosuppressive therapy.
- Children born to mothers with TB during pregnancy, after ruling out congenital TB.
- Dosage and Duration:
- Isoniazid Dosage: 10 mg/kg per day, with a range of 7-15 mg/kg and a maximum dose of 300 mg/day.
- Treatment Duration: 6 months.
- Purpose:
- Prevents the development of TB disease in children who have latent TB infection.
- Reduces the likelihood of TB disease developing during childhood.

- Administration: Single isoniazid dispersible tablets of 100 mg are available from the Global Drug Facility (GDF).
- Follow-Up: Regular follow-up should be conducted at least every 2 months until treatment is complete.
- Resistance Risk: No risk of isoniazid resistance developing in children receiving IPT, even if active TB is missed.

Pyrazinamide daily for 6 months (Option A): Pyrazinamide causes hepatitis more often in children than isoniazid. It is used to treat active TB, not for preventive therapy.

BCG vaccine (Option B) is used for preventing severe forms of TB, in highly endemic settings. It is not recommended for use in this preventive therapy. BCG does not prevent latent TB progression, and revaccination offers no additional benefit.

Ethambutol daily for 3 months (Option C): Ethambutol is used to treat active TB, not for preventive therapy. Using it alone for prevention can lead to drug resistance, so it is not recommended for latent TB infection management.

Solution for Question 19:

Correct Answer: C) Ethambutol

Explanation: The child has primary tuberculosis, indicated by right-sided hilar lymphadenopathy and collapse-consolidation lesions on the chest X-ray. The treatment regimen includes multiple antitubercular drugs, which include ethambutol, which causes optic neuritis.

- Standard Regimen for Intrathoracic Tuberculosis: 6-month regimen

Intensive Phase (2 months)	Continuation Phase (4 months)
Isoniazid (H) Rifampin (R) Pyrazinamide (Z) Ethambutol (E)	Isoniazid (H) Rifampin (R) Ethambutol (E)
H, R, Z: Bactericidal drugs E: Bacteriostatic drug	

Medication	Dose (mg/kg/day)	Maximum Dose (per day)	Side Effects

Ethambutol (Option C)	15 - 25	2.5 g	Optic neuritis (usually reversible) Decreased red-green color discrimination Gastrointestinal tract disturbances Hypersensitivity
Isoniazid (Option A)	7 - 15	300 mg	Mild hepatic enzyme elevation Hepatitis Peripheral neuritis Hypersensitivity
Pyrazinamide (Option D)	30 - 40	2 g	Hepatotoxic effects Hyperuricemia Arthralgias Gastrointestinal tract upset
Rifampicin (Option B)	10 - 20	600 mg	Orange discoloration of secretions or urine Staining of contact lenses Vomiting Hepatitis Influenza-like reaction Thrombocytopenia Pruritus

- Corticosteroids:
- Indications: Tuberculous meningitis, endobronchial tuberculosis, tuberculous pericardial effusion, tuberculous pleural effusion, severe miliary tuberculosis.
- Corticosteroid Regimen: Prednisone 1-2 mg/kg/day in 1-2 divided oral doses for 4-6 weeks, followed by a taper.

Solution for Question 20:

Correct Answer: A) Children in close contact with tuberculosis patient

Explanation:

Children in close contact with a tuberculosis patient or an adult with high risk of TB are at an increased risk of developing tuberculosis.

Diagnosis of Childhood Tuberculosis:

- Method: Intradermal injection of 0.1 mL of purified protein derivative.
- Timing: Induration measured 48-72 hours later.
- Reaction of >10mm is considered “positive” and reaction of <6mm is considered “negative”.
- Limitations: False negatives in immunocompromised or recent viral infections False positives with BCG vaccination.
- False negatives in immunocompromised or recent viral infections
- False positives with BCG vaccination.
- False negatives in immunocompromised or recent viral infections
- False positives with BCG vaccination.

Size of Induration	Group
≥ 5 mm (Option A)	Children in close contact with known or suspected contagious people with tuberculosis disease Children suspected to have tuberculosis disease: Findings on chest radiograph consistent with active or previously tuberculosis disease Clinical evidence of tuberculosis disease Children receiving immunosuppressive therapy or with immunosuppressive conditions, including HIV infection
≥10 mm (Option B and C)	Children at increased risk of disseminated tuberculosis disease: Children <4 yr old Children with other medical conditions, including Hodgkin disease, lymphoma, diabetes mellitus, chronic renal failure, or malnutrition Children with increased exposure to tuberculosis disease: Children born in high-prevalence regions of the world Children often exposed to adults with HIV infection, homeless, users of illicit drugs, residents of nursing homes, incarcerated or institutionalized, or migrant farm workers Children who travel to high-prevalence regions of the world
≥15 mm (Option D)	Children ≥4 yr old without any risk factors

- Tests: QuantiFERON-TB and T-SPOT.TB.
- Advantages: Higher specificity, not affected by BCG vaccination.
- Limitations: Cannot differentiate between latent TB infection and active TB.
- Specimens: Sputum, gastric aspirate, induced sputum, bronchoalveolar lavage or pleural fluid.
- Methods: Ziehl-Neelsen staining, special stains, cultures

- Culture: Lowenstein-Jensen (LJ) medium or Mycobacterial growth indicator tube (MGIT) system.
- Method: Real-time polymerase chain reaction (PCR).
- Purpose: Diagnosis of pulmonary tuberculosis (TB) and detection of Rifampicin resistance.
- Results: Available in less than 2 hours.
- Commercial Tests Available: GeneXpert, GeneXpert Ultra, TrueNAT.
- Method: ELISA for detecting antibodies.
- Not widely used in children.
- Chest X-ray: Identifies pulmonary lesions, lymphadenopathy, and miliary disease.
- CT: Useful for detecting lymphadenopathy.
- MRI: For CNS tuberculosis.

Previous Year Questions

1. A 4-year-old unvaccinated child presents with fever, rash, and Bitot spots. What is the appropriate line of management?

A. Measles Vaccine + Vitamin A Supplementation

B. Only Measles Vaccine

C. Only Vitamin A Supplementation

D. Supportive Care

2. A baby is born with microcephaly, intracerebral calcification and chorioretinitis. Which teratogenic infections is the mother suffering from?

A. Toxoplasma

B. CMV

C. Varicella

D. Rubella

3. An unvaccinated 5-year-old child of a migrant worker family was brought to the causality with fever and a characteristic skin rash. Upon eye examination, Bitot's spot was seen. What is the appropriate management?

A. Isolate and give nutritional supply with vitamin A supplement

B. Give measles vaccine and vitamin A supplement

C. Institutional Isolation, nutritional management with vitamin A supplement

D. Only measles vaccine

4. A neonate on examination has bilateral cataracts and, after investigations, is found to have a patent ductus arteriosus and salt & paper retinopathy. What is the most likely congenital infection?

A. Rubella

B. CMV

C. Toxoplasma

D. Varicella

5. A 3-year-old boy presents with a rash, bleeding gums, and a positive tourniquet test. His platelet count is 25,000. Which of the following is the most important prognostic feature in this clinical scenario?

A. Low platelet count

B. Elevated total leukocyte count (TLC)

C. Hematocrit

D. Peripheral smear

6. Congenital rubella syndrome has all of the following except:

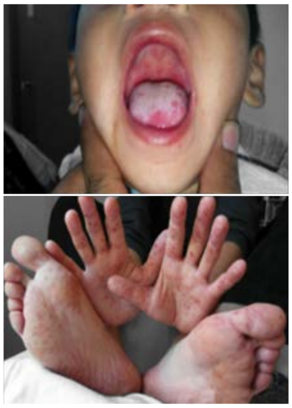
A. Sensorineural deafness

B. Cataract

C. Cardiac defects

D. Hydrocephalus

7. A 5 year old boy had painful sores in the mouth with a rash on hand along with fever. Which of these is the causative agent?



A. Pox virus

B. Coxsackie virus

C. HHV 7 virus

D. HIV virus

8. What is the most frequently observed heart defect in congenital rubella syndrome?

A. ASD

B. VSD

C. PDA

D. PS

9. During infancy, what is the most prevalent causative agent of acute bronchiolitis in a child?

A. Influenza virus

B. Parainfluenza virus

C. Rhinovirus

D. Respiratory syncytial virus

10. A 10-day-old newborn presents with jaundice, hepatosplenomegaly, microcephaly, diffuse petechiae, and periventricular calcifications. Which of the following will be the best sample for diagnosis in this case?

A. Urine examination

B. Liver biopsy

C. Blood examination

D. CSF examination

11. Warthin-Finkeldey cells are seen in?

A. Rubella

B. Rubeola

C. Rabies

D. Typhoid

12. Which of the following is not a manifestation of congenital syphilis?

A. Gumma

B. Hutchinson's teeth

C. Olympian brow

D. Interstitial keratitis

13. What could be the potential diagnosis for a 6-year-old child who exhibits hepatosplenomegaly, generalized lymphadenopathy, fever, and develops a rash following the administration of ampicillin?

A. Infectious mononucleosis
B. Scarlet fever
C. Kawasaki disease
D. HIV infection

14. What is the recommended preventive dosage of Oseltamivir for infants?

A. 1 mg/kg/day
B. 3 mg/kg/day
C. 5 mg/kg/day
D. 7 mg/kg/day

15. 6th day disease is:

A. Erythema infectiosum
B. Exanthema subitum
C. Erythema marginatum
D. Erythema nodosum

16. What is the most effective approach for diagnosing HIV in infants?

A. Western blot

B. ELISA

C. PCR

D. All of the above

17. What is the probable pathogen responsible for the lesions seen on the palm of a 5-year-old child with a 3-day history of fever and changes in consciousness?

A. Staphylococcus aureus

B. Measles virus

C. Rubella virus

D. Picornavirus

18. A 5 year old male child was got to the OPD with complaints of low grade fever, headache and mild upper respiratory tract symptoms since the last 4 days. On examination, an erythematous rash was found on the cheeks. The parents said the rash developed on the cheeks first and then spread out to involve his trunk and limbs. The causative agent of this disease has a single stranded DNA. What is the diagnosis?

A. Hand, foot and mouth disease

B. Measles

C. Erythema infectiosum

D. Rubella

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19. What is the rare and hazardous complication associated with Subacute Sclerosing Panencephalitis?

A. Measles

B. Mumps

C. Rubella

D. Varicella

---

20. What is the diagnosis of a 5-year-old girl presented to the clinic with symptoms of fever, cough, red eyes, running nose, and rash, as depicted in the image of the child's appearance?

A. Mumps

B. Measles

C. Chickenpox

D. Erythema infectiosum

---

21. What is the most probable diagnosis for a 3-month-old baby presenting with complaints of deafness, cataract, and patent ductus arteriosus?

A. Congenital herpes simplex virus infection

B. Congenital toxoplasmosis

C. Congenital cytomegalovirus infection

D. Congenital rubella syndrome

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22. What is the probable diagnosis for a child who has a fever and a rash that starts on the face, spreads behind the cheeks and buccal mucosa, and then extends to other areas of the body, with the presence of Koplik's spots upon examination?

A. Measles

B. Rubella

C. Varicella

D. Mumps

23. For a 3-week-old baby suffering from a cough and sore throat, where the mother states that the baby experiences a sudden bout of coughing followed by a temporary cessation of breathing, with a total leukocyte count exceeding 50,000 cells per microliter, which medication would you recommend for this patient?

A. Azithromycin

B. Amoxicillin

C. Cotrimoxazole

D. Clarithromycin

24. What is the probable diagnosis for an infant who has enlarged liver and spleen, low platelet count, and CT scan revealing calcifications around the brain's ventricles?

A. Congenital rubella syndrome

B. Congenital herpes simplex virus infection

C. Congenital toxoplasmosis

D. Congenital cytomegalovirus infection

25. What antibiotics would you prescribe empirically for a child who is experiencing symptoms of fever, rash, body ache, and throat ache and had a thorn prick injury one week ago?

A. Ceftriaxone

B. Amoxicillin + clavulanate

C. Vancomycin

D. Meropenem

26. Which of the following statements accurately describe cytomegalovirus (CMV) infection? Neonates who are asymptomatic at birth have a lesser risk of later sequelae. 20-40% are symptomatic at birth. In developing countries, the transmission rate of CMV infection to the infant is more common from primary maternal infection than reactivation. Diagnosis by urine specimen at 5 weeks of age.

A. 3, 4

B. 1, 2

C. 2, 4

D. 1, 3

27. Koplik spots are seen in:

A. Mumps

B. Measles

C. Rubella

D. Varicella

28. What should be the next course of action in the case of a 2-month-old baby who was born to a mother infected with HIV and is experiencing repeated episodes of diarrhea?

A. Test stool for giardia and give antibiotics

B. Dried spot sample for HIV DNA PCR

C. Antibody test for HIV

D. Aerobic culture

---

29. What would be your diagnosis for a 3-year-old male who has the given buccal mucosa findings and subsequently develops a rash after three days?

A. Measles

B. Leukoplakia

C. Scarlet fever

D. Kawasaki disease

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30. Which maternal antibodies do not confer substantial immunity in a newborn?

A. Tetanus

B. Polio

C. Diphtheria

D. Measles

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31. What would be the most appropriate course of action for the care of Ajay, the 3-year-old brother of Anuj, who resides in the same household and received diphtheria vaccination approximately 16 months ago, while Anuj is currently undergoing treatment for diphtheria in the ward?

A. One booster dose

B. Nothing as the child is already exposed

C. Erythromycin + diphtheria toxoid

D. Erythromycin only

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32. Which of the statements below accurately describes congenital CMV infection?

A. About 20-30% of the infections are symptomatic

B. Triad of SNHL, periventricular calcification and enamel hypoplasia

C. Most children asymptomatic at birth can develop conductive hearing loss later in life

D. If mother is IgG positive for CMV, the child is unlikely to develop an infection

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33. As a component of the treatment for a child experiencing severe COVID, which of the options below is administered?

A. Corticosteroids

B. Ivermectin

C. Remdesivir

D. All of the above

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34. What is the most probable diagnosis for a child who has a fever and vesicular lesions on both upper and lower limbs, along with neck stiffness? Similar lesions are also found on the palms, soles, and inside the mouth. The analysis of

cerebrospinal fluid (CSF) shows normal glucose levels but elevated lymphocytes and protein.

A. Bacterial meningitis with sepsis

B. Coxsackie viral meningitis and Hand Foot Mouth disease

C. Herpes simplex gingivostomatitis and meningoencephalitis

D. Tuberculous meningitis

35. What is the most frequent symptom of congenital toxoplasmosis?

A. Hydrocephalus

B. Chorioretinitis

C. Hepatosplenomegaly

D. Thrombocytopenia

36. What is the primary reason for HIV infection in a newborn?

A. Perinatal transmission

B. Breast milk

C. Transplacental

D. Exchange transfusion with infected blood

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	1
Question 4	1
Question 5	1
Question 6	4
Question 7	2
Question 8	3
Question 9	4
Question 10	1
Question 11	2
Question 12	1
Question 13	1
Question 14	2
Question 15	2
Question 16	3
Question 17	4
Question 18	3
Question 19	1
Question 20	2
Question 21	4
Question 22	1

Question 23	1
Question 24	4
Question 25	2
Question 26	4
Question 27	2
Question 28	2
Question 29	1
Question 30	2
Question 31	4
Question 32	4
Question 33	1
Question 34	2
Question 35	2
Question 36	1

Solution for Question 1:

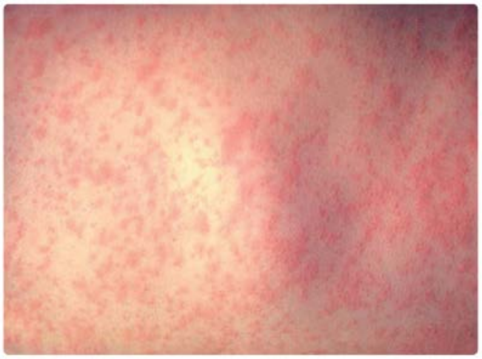
Correct Answer: C) Only Vitamin A Supplementation

Explanation:

- The child already has measles, so the priority is to address the vitamin A deficiency (as evidenced by Bitot spots) to reduce the risk of complications and support recovery.
- The measles vaccine is not required at this stage because the child has already been infected and will develop lifelong immunity.

Measles	
Causative agent	ssRNA paramyxovirus
Incubation period	10 days from exposure to onset of fever and 14 days to the appearance of rash

Source of infection	Person with measles Infective material
Period of communicability	4 days before and 4 days after the appearance of rash
Clinical features	Prodromal stage: Fever Cough Coryza with sneezing Conjunctivitis with lacrimation Koplik's spots- grayish-white spots on the buccal mucosa (Pathognomonic feature) Eruptive stage: Dusky red macular/maculopapular rash, which starts behind the ears and progresses to the face, neck and then to trunks and limbs. In the absence of complications, the lesions and fever disappear after 3-4 days. Post measles stage: After the eruptive stage, it leads to weight loss, weakness, fatigue, diarrhea.
Isolation period	7 days from the onset of rash
Difference between measles and rubella rash	Both rashes are maculopapular The rash in measles is more confluent and botchy, whereas the rash in rubella is discrete or non-confluent.
Diagnosis	Clinically diagnosed based on rash and koplik's spots Gold standard: PCR
Treatment	No specific treatment Symptomatic management only



Measles Rash

Measles Vaccine + Vitamin A Supplementation (Option A): The measles vaccine is not needed after measles infection, as the child will have lifelong immunity. The vaccine is for prevention, not for post-infection management.

Only Measles Vaccine (Option B): The vaccine is unnecessary post-infection, and vitamin A supplementation is critical in this case.

Supportive Care (Option D): While supportive care is part of the management, it does not address the important need for vitamin A supplementation in a measles case.

Solution for Question 2:

Correct Answer: A) Toxoplasma

Explanation:

The presence of characteristic intracranial calcifications, microcephaly, and chorioretinitis are features of congenital toxoplasmosis.

Etiology and Clinical Features of Congenital Toxoplasmosis:

Category	Details
Etiology	Primary maternal infection with Toxoplasma gondii during pregnancy is the primary cause
Risk Factors	Infection during the first trimester: severe outcomes, lower transmission rate (~17%) Infection during third trimester: milder outcomes, higher transmission rate (~65%)
Clinical Features	CNS: Hydrocephalus, intracranial calcifications, seizures, developmental delays, microcephaly in severe brain damage
Ocular: Chorioretinitis, vision impairment, strabismus, photophobia, epiphora	
Systemic: Hepatosplenomegaly, lymphadenopathy, prematurity	
Other: Jaundice, thrombocytopenia, rash, hearing loss	

Diagnosis:

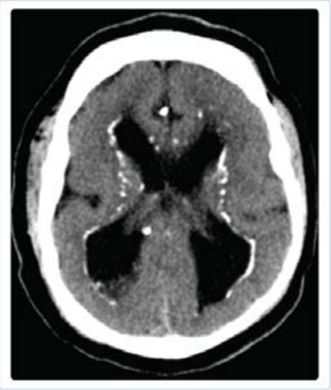
Diagnosis Method	Details
Fetal Ultrasound	Performed every 2 weeks during gestation after diagnosis of maternal infection.

PCR Analysis of Amniotic Fluid	Used for prenatal diagnosis of fetal infection.
Serologic Tests	Persistent or rising titers in dye/IFA tests, positive IgM-ELISA.
Newborn Evaluation	General, ophthalmologic, and neurologic exams; head CT scan for intracranial calcifications.
CSF Analysis	Lumbar puncture with analysis for Toxoplasma-specific IgG and IgM, total IgG, cells, glucose, and protein.
Toxoplasma Isolation	Isolation from placenta, infant's leukocytes, and CSF using PCR

Management:

Treatment	Details
Pyrimethamine and Sulfadiazine, along with Folinic Acid (Calcium Leucovorin)	Synergistic action against Toxoplasma gondii. Standard treatment for congenital toxoplasmosis. Folinic Acid is administered to prevent bone marrow suppression caused by pyrimethamine
Spiramycin	Used in pregnant women during the first trimester to prevent vertical transmission.
Monitoring	Regular leukocyte counts twice weekly to detect neutropenia and other hematologic effects.

Cytomegalovirus (CMV) Infection (Option B) can also cause intracranial calcifications; however, these are more commonly seen in a periventricular pattern. While there is some overlap, the calcifications in congenital toxoplasmosis are often more diffuse.



Varicella (Option C) presents with microcephaly and chorioretinitis. But it is associated with microphthalmia, cataracts, cortical atrophy, seizures, mental retardation, etc. as well.

Rubella (Option D) presents with patent ductus arteriosus, bilateral cataracts, and salt & pepper retinopathy, which are classical signs of congenital rubella syndrome

Solution for Question 3:

Correct Answer: A) Isolate and give nutritional supply with vitamin A supplement

Explanation:

The child has a fever and rash with cough and coryza with a non-immunised status which points to the diagnosis of measles. However, the presence of Bitot's spot also shows vitamin A deficiency. Hence, the management here is isolation of the child, and nutritional management with vitamin A supplementation.

Aspects	Description
Causative agent	Measles (rubeola) is a highly communicable childhood exanthematous illness caused by the measles virus, an RNA virus from the Paramyxoviridae family.
Transmission	Contact with droplet aerosols (through the respiratory tract or conjunctiva)
Pathogenesis	Primary viremia occurs, resulting in infection of the reticuloendothelial system, followed by secondary viremia, which results in systemic symptoms.
Incubation period	8-12 days
Period of infectivity	From 3 days before to 4 days after the rash onset.
Clinical manifestations	 ... (content truncated)
Rash	The erythematous maculopapular rash in measles appears on the 4th day, starting behind the ears, along

	the hairline, and posterior cheeks, then spreading to therest of the body over the next 2-3 days.
Management	Isolation. Symptomatic and conservative (antipyretics, maintenance of hygiene, ensuring adequate fluid and caloric intake, and humidification).

Treatment of vitamin A deficiency:

Age Group	Treatment
Under 6 months	50,000 IU once daily on D1, D2, and 4 weeks later
6 to 12 months	100,000 IU once daily on D1, D2, and 4 weeks later
Over 1 year	200,000 IU once daily on D1, D2, and 4 weeks later

Giving a measles vaccine and vitamin A supplement (Option B) or only a measles vaccine (Option D) is not appropriate since the child is already suffering from measles.

Institutional Isolation, nutritional management and vitamin A supplement (Option C): Institutional isolation is unnecessary in this case since the child does not have any general danger signs, respiratory distress or pneumonia.

Solution for Question 4:

Correct Answer: A) Rubella

Explanation:

The patent ductus arteriosus, bilateral cataracts, and salt & pepper retinopathy are classical signs of congenital rubella syndrome.

Congenital Rubella	
Rubella (German measles) is a mild disease in infants and children but can cause congenital rubella syndrome (CRS) if contracted during pregnancy. The risk is highest in the first trimester, especially within the first 8 weeks.	
Pathogenesis	Viremia is most intense from 10 to 17 days after infection. Viral shedding is from day 10 of infection up to 2 weeks.

Risk of Defects by Gestational Age	Highest in the first 12 weeks of gestation and decreases significantly thereafter, with defects being rare after 20 weeks.
Clinical features of congenital rubella syndrome	 ... (content truncated)
Chronicity and Complications	Chronic presence of the virus can cause ongoing damage The risk of progressive rubella panencephalitis (PRP) years later is a rare complication. Less common complications: Growth retardation, hepatitis, bone lesions. Risk of diabetes and thyroid dysfunction
Diagnosis	Rubella IgM enzyme immunosorbent assay (detects antibodies) Viral isolation (growing the virus from samples) Polymerase chain reaction (PCR) (detects viral RNA)
Management	No specific treatment for CRS; supportive care is key Early intervention for complications (e.g., hearing loss) is crucial
Prevention	Standard with droplet precaution in hospitalized cases Rubella vaccination RA 27/3 live attenuated vaccine (MMR or MMRV with varicella) effectively prevents rubella and CRS MMR vaccine has a 95% efficacy after one dose and 99% after two doses Vaccination side effects: fever, rash, joint pain
Considerations for Pregnant Women	Immunoglobulin may be considered if pregnancy is not terminated, though it does not guarantee prevention of fetal infection

Congenital CMV infection (Option B) is usually asymptomatic at birth. If symptomatic, these neonates present with jaundice, petechiae, hepatosplenomegaly, microcephaly, and chorioretinitis.

Congenital toxoplasmosis (Option C) caused by Toxoplasma gondii often presents with the classic triad of intracranial calcifications, hydrocephalus, and chorioretinitis

Congenital varicella (Option D) presents with cicatricial zigzag skin scarring in a dermatomal pattern, limb hypoplasia, microcephaly, seizures, mental retardation, chorioretinitis, microphthalmia, cataracts, etc.

Solution for Question 5:

Correct Option A - Low platelet count:

- In this clinical scenario, the most important prognostic feature for the 3-year-old boy presenting with a rash, bleeding gums, and a positive tourniquet test, along with a platelet count of 25,000, is the low platelet count.
- This is because a significantly low platelet count is directly associated with a higher risk of bleeding complications and is crucial for determining the management and prognosis of the condition.

Incorrect Options:

Option B - Elevated total leukocyte count (TLC):

- While an elevated TLC may indicate an underlying infection or inflammation, it is less directly related to the prognosis of bleeding risk compared to platelet count. In conditions with bleeding, the platelet count is a more immediate indicator of potential complications.

Option C - Hematocrit:

- Hematocrit measures the proportion of blood volume occupied by red blood cells. While it can provide information on anemia or blood volume status, it is not the primary prognostic factor for bleeding disorders related to low platelet counts.

Option D - Peripheral smear:

- A peripheral smear can reveal the morphology of blood cells and help in diagnosing various hematological conditions. However, in the context of immediate bleeding risk and prognosis, platelet count is more critical than the peripheral smear findings.

Solution for Question 6:

Correct Option D - Hydrocephalus:

- Hydrocephalus is not a typical feature of Congenital Rubella Syndrome. The primary manifestations are sensorineural deafness, cataract, and cardiac defects.

Incorrect Options:

Option A - Sensorineural deafness: Sensorineural deafness is a common feature of Congenital Rubella Syndrome.

Option B - Cataract: Cataract is another manifestation observed in infants with CRS.

Option C - Cardiac defects: Cardiac defects, particularly patent ductus arteriosus (PDA) and pulmonary artery stenosis, are commonly associated with Congenital Rubella Syndrome.

Solution for Question 7:

Correct Option B - Coxsackie virus:

- The patient’s symptom of painful sores in the mouth with a rash on the hand along with a fever is suggestive of Hand Mouth Foot disease.
- The Coxsackie virus is the causative agent of Hand Mouth Foot disease.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 8:

Correct Option C - PDA:

- Among the heart defects commonly associated with congenital rubella syndrome, the most prevalent is Patent Ductus Arteriosus (PDA).
- PDA is a condition where the ductus arteriosus, a blood vessel connecting the pulmonary artery to the aorta in the fetus, remains open after birth.

Incorrect Options:

Options A, B, and D can be associated with congenital rubella syndrome but are not the most frequently observed.

Solution for Question 9:

Correct Option D - Respiratory syncytial virus:

- The most common etiological agent responsible for acute bronchiolitis during infancy is Respiratory Syncytial Virus (RSV).
- While influenza virus and adenovirus can also cause bronchiolitis, RSV is the most common.

Incorrect Options:

Options A, B, and C are incorrect.

Solution for Question 10:

Correct Option A - Urine examination:

- The patient’s presentation with jaundice, hepatosplenomegaly, microcephaly, diffuse petechiae, and periventricular calcifications is suggestive of congenital CMV infection.
- The best sample for diagnosis of the congenital CMV infection is PCR / viral culture of urine sample.

Incorrect Options:

Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 11:

Correct Option B - Rubeola:

- Warthin-Finkeldey cells are multinucleated giant cells that are characteristic of measles (rubeola) infection.
- These cells are formed by the fusion of infected lymphocytes and are found in the respiratory epithelium and lymphoid tissues during the acute phase of measles.

Incorrect Options:

Warthin-Finkeldey cells are not seen in Options A, C, and D.

Solution for Question 12:

Correct Option A - Gumma:

- Congenital syphilis is a condition caused by transmission of the Treponema pallidum from an infected mother to her fetus during pregnancy.
- It can lead to various manifestations, including Hutchinson's teeth, Olympic brow (saddle nose), and interstitial keratitis.
- However, gumma, which is a granulomatous lesion, is seen in the tertiary stage of syphilis.

Incorrect Options:

Option B - Hutchinson's teeth: Hutchinson's teeth y refer to notched and peg-shaped permanent incisors.

Option C - Olympic brow (saddle nose): Olympic brow, also known as saddle nose, is a deformity seen in congenital syphilis where the bridge of the nose collapses.

Option D - Interstitial keratitis: Interstitial keratitis, which is inflammation of the cornea, can occur as a manifestation of congenital syphilis.

Solution for Question 13:

Correct Option A - Infectious mononucleosis:

- The patient’s presentation with hepatosplenomegaly, generalized lymphadenopathy, fever, and a rash following the administration of ampicillin is suggestive of infectious mononucleosis.
- Infectious mononucleosis is caused by the Epstein-Barr virus (EBV).
- The rash, known as an ampicillin rash, is a characteristic reaction seen in patients with EBV infection who are treated with ampicillin or other penicillin-based antibiotics.

Incorrect Options:

- Rash following the administration of ampicillin is characteristic of infectious mononucleosis and not seen in Options B, C, and D.

Solution for Question 14:

Correct Option B - 3 mg/kg/day:

- Oseltamivir is an antiviral medication commonly used for the prevention and treatment of influenza.
- The prophylactic dose of Oseltamivir for infants is 3 mg/kg/day.
- This dose is used to prevent influenza infection in infants at risk.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 15:

Correct Option B - Exanthema subitum:

- Exanthema subitum, also known as Roseola infantum or sixth disease, is a viral infection primarily affecting infants and young children.
- It is typically caused by human herpesvirus 6 (HHV-6) or human herpesvirus 7 (HHV-7).
- It is characterized by a high fever that lasts for a few days, usually followed by the sudden appearance of a rash.
- The fever is often abrupt in onset and may be accompanied by other mild symptoms like irritability, runny nose, and mild cough.
- After the fever subsides, a rash appears, which is typically pink or rose-colored.
- The rash usually starts on the trunk and then spreads to the neck, face, and extremities.
- Exanthema subitum is a self-limiting condition, and the rash usually resolves within a few days.

Incorrect Options:

Option A - Erythema infectiosum:

- Erythema infectiosum, also known as fifth disease or slapped cheek syndrome, is a viral illness caused by parvovirus B19.

Option C - Erythema marginatum:

- Erythema marginatum is a skin manifestation that can occur in rheumatic fever.
- It is characterized by pink or red, non-itchy, and well-defined rings or patches with a clear center.

Option D - Erythema nodosum:

- Erythema nodosum is a type of skin inflammation that presents as painful, red nodules or bumps, usually on the shins.
- It can be associated with various underlying conditions, such as infections, inflammatory bowel disease, or certain medications.

Solution for Question 16:

Correct Option C - PCR:

- PCR is a highly sensitive and specific laboratory test that can detect the genetic material of HIV in a person's blood.

- It can be used to directly detect and quantify the virus, even at low levels.
- PCR-based tests, such as the HIV DNA PCR or HIV RNA PCR, are commonly used to diagnose HIV infection in infants.
- These tests can detect the presence of the virus in an infant's blood as early as a few days after birth, before the development of detectable HIV antibodies.

Incorrect Options:

Options A, B, and D are not the most effective tests for diagnosing HIV in infants.

Option A - Western blot:

- Western blot is a laboratory technique used to confirm the presence of specific antibodies against HIV in a person's blood sample.
- It is commonly used as a supplemental test to confirm the results of an initial screening test, such as ELISA.

Option B - ELISA:

- ELISA detects the presence of HIV-specific antibodies or antigens in a person's blood sample and has limitations in diagnosing HIV infection in infants.
- This is because infants born to HIV-positive mothers can have maternal antibodies present in their blood, which can lead to false-positive results.

Solution for Question 17:

Correct Option D - Picornavirus:

- The patient’s presentation with fever and lesions on the palm is suggestive of hand foot mouth disease.
- Picornaviruses are a family of RNA viruses that include various viruses such as Enterovirus and Coxsackievirus.
- These viruses can cause hand, foot, and mouth disease (HFMD).
- HFMD typically presents with fever, sore throat, and characteristic skin lesions on the palms, soles of the feet, and mouth.

Incorrect Options:

Palm lesions are not seen in options A, B, and C.

Solution for Question 18:

Correct Option C - Erythema infectiosum:

- Erythema infectiosum is a disease with low grade fever, headache, malaise, upper respiratory tract infection, characterized by "Slapped cheek" appearance of rash.
- It is caused by Parvovirus B-19, a virus with a single stranded DNA.

Incorrect Options-

Option A- Hand, foot and mouth disease: Is caused by Coxsackie A virus which is a single stranded RNA virus.

Option B - Measles: Also presents with fever, rash and URTI but the rash starts from the back of the ear and spreads around the body. Measles virus is a single stranded RNA virus

Option D- Rubella: Also presents with fever, rash and URTI but the virus is a single stranded RNA virus.

Solution for Question 19:

Correct Option A - Measles:

- Subacute Sclerosing Panencephalitis is a rare fatal, severe, long-standing complication of measles.
- It is a progressive neurological disorder that affects the central nervous system.
- It typically develops 7-13 years after primary measles infection.

Incorrect Options:

Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 20:

Correct Option B - Measles:

- The patient’s presentation with fever, cough, red eyes, and running nose with the image showing Koplik spots on buccal mucosa is suggestive of measles.
- Koplik spots are considered pathognomonic of measles.
- Measles disease is most common in preschool children and the prodromal phase is characterized by fever, rhinorrhea, conjunctival congestion, and a dry hacking cough.

Incorrect Options:

Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 21:

Correct Option D - Congenital rubella syndrome:

- The presentation of a 3-month-old baby with complaints of deafness, cataract, and patent ductus arteriosus is suggestive of congenital rubella syndrome.
- The characteristic triad of findings in congenital rubella syndrome includes deafness, cataract, and cardiac abnormalities such as patent ductus arteriosus.

Incorrect Options:

Option A - Congenital herpes simplex virus infection:

- Congenital herpes simplex virus infection present with various symptoms, including skin lesions, neurological abnormalities, and systemic involvement.

Option B - Congenital toxoplasmosis:

- While cataracts and other eye abnormalities can occur in congenital toxoplasmosis, the presence of deafness and patent ductus arteriosus are not commonly associated with this condition.

Option C - Congenital cytomegalovirus infection:

- Congenital cytomegalovirus infection can cause hearing loss, vision problems, and cardiac abnormalities.
- Patent ductus arteriosus is not a typical finding associated with congenital CMV infection.

Solution for Question 22:

Correct Option A - Measles:

- The patient's presentation with fever and a rash spreading from the face, behind cheeks, and buccal mucosa to other body parts with Koplik's spot present on examination is suggestive of measles.
- Koplik's spots are small, white spots with a bluish-white center and red halo that appear on the buccal mucosa and are pathognomonic for measles.

Incorrect Options:

Koplick's spots are not seen in Options B, C, and D.

Solution for Question 23:

Correct Option A - Azithromycin:

- The patient's presentation with sudden bout of coughing followed by a temporary cessation of breathing along with a total leukocyte count (>50,000 cells/µL) is suggestive of pertussis, commonly known as whooping cough.
- Azithromycin is the drug of choice for the treatment of pertussis in infants.
- Early treatment with azithromycin can help reduce the severity and duration of symptoms and prevent complications.

Incorrect Options:

Options B, C, and D are not the drug of choice for pertussis.

Solution for Question 24:

Correct Option D - Congenital cytomegalovirus infection:

- The patient’s presentation with hepatosplenomegaly, thrombocytopenia, and periventricular calcifications on CT scan is suggestive of congenital CMV infection.

Incorrect Options:

Periventricular calcifications are not seen in options A, B, and C.

Solution for Question 25:

Correct Option B - Amoxicillin + clavulanate:

- The patient’s presentation with fever, rash, body ache, and throat ache with a history of thorn prick injury one week ago is suggestive of skin or soft tissue infection from the thorn prick injury.
- Empirical antibiotic therapy is initiated to cover the likely causative pathogens until the specific microorganism is identified.
- Amoxicillin + clavulanate combination provides broad-spectrum coverage against various bacteria, including those commonly associated with skin and soft tissue infections..

Incorrect Options:

Options A, C, and D are not the first line choice of empirical antibiotic in skin or soft tissue infections from thorn prick injury

Solution for Question 26:

Correct Option D - 1, 3:

1 - Neonates who are asymptomatic at birth have a lesser risk of later sequelae: Neonates who are asymptomatic at birth, meaning they do not show any symptoms of CMV infection, generally have a lower risk of developing long-term complications or sequelae.

3 - In developing countries, the transmission rate of CMV infection to the infant is more common from primary maternal infection than reactivation:

- In developing countries, the rate of transmission of CMV infection to the infant is higher when the mother acquires a primary CMV infection during pregnancy rather than experiencing a reactivation of a previous infection.
- Primary maternal infection refers to a new CMV infection that occurs during pregnancy, while reactivation refers to the reactivation of a latent CMV infection that the mother had before becoming pregnant.

Incorrect Options:

Options A, B, and C are incorrect.

2 - 20-40% are symptomatic at birth: The actual percentage of neonates who are symptomatic at birth due to CMV infection is around 10-15%.

4- Diagnosis by urine specimen at 5 weeks of age: The recommended time for collecting a urine specimen to diagnose CMV infection in infants is within the first 3-4 weeks of life, not at 5 weeks of age.

Solution for Question 27:

Correct Option B - Measles:

- Koplik's spots are small, white spots with a bluish-white center and red halo that appear on the buccal mucosa and are pathognomonic for measles.

Incorrect Options:

Koplik’s spots are not seen in options A, C, and D.

Solution for Question 28:

Correct Option B - Dried spot sample for HIV DNA PCR:

- The recommended course of action for an infant with recurrent diarrhea who is 2 months old and born to an HIV-positive mother is to take a dried spot sample for HIV DNA PCR.
- Infants with recurrent diarrhea may have HIV infection, and early diagnosis is essential to halt the spread of the virus and its associated implications.
- For infants younger than 18 months of age who might not yet have produced detectable HIV antibodies, the dried spot sample for HIV DNA PCR is a very sensitive and specific test for HIV detection.

Incorrect Options:

Option A - Test stool for giardia and give antibiotics: In cases of infectious diarrhea brought on by parasites or bacteria, testing the stool for giardia and administering antibiotics may be acceptable, but they would not address the potential for an underlying HIV infection.

Option C - Antibody test for HIV: Infants younger than 18 months of age shouldn't get an antibody test for HIV since they might still have circulating maternal antibodies that would affect the test's accuracy. Option D - Aerobic culture: Aerobic culture is typically ineffective in viral infections.

Solution for Question 29:

Correct Option A - Measles:

- The image shows Koplik's spot on the buccal mucosa, which is pathognomonic of measles.
- Koplik's spots are small, white spots with a bluish-white center and red halo that appear on the buccal mucosa.

Incorrect Options:

Koplik's spots are not seen in options B, C, and D.

Solution for Question 30:

Correct Option B - Polio:

- Maternal antibodies play a crucial role in providing passive immunity to newborns during the early stages of life.
- These antibodies are transferred from the mother to the fetus through the placenta or via breast milk.
- They offer protection against various infections until the baby's immune system matures and can produce its own antibodies.
- Polio maternal antibodies against polio do not provide significant immunity to the neonate.
- This is because the transfer of polio-specific antibodies from the mother to the fetus is limited, and the protection offered by these antibodies against polio infection is not as robust.

Incorrect Options:

- Maternal antibodies against options A, C, and D can provide significant immunity to the neonate.

Solution for Question 31:

Correct Option D - Erythromycin only:

- When a case of diphtheria is identified in a household, it is important to take appropriate measures to prevent the spread of the infection to other individuals, especially those who may be at risk, such as close family members.
- In situations where there is a known case of diphtheria in the household, it is recommended to administer erythromycin to individuals who have been in close contact with the infected person.
- Since Ajay, the younger sibling, has not developed symptoms but has been exposed, administering erythromycin alone is appropriate to prevent the development of the infection.

Incorrect Options:

Option A - One booster dose:

- While a booster dose of the diphtheria vaccine is generally recommended as part of routine immunization schedules, it is not the next best step in this case.
- The purpose of a booster dose is to enhance and maintain long-term immunity in individuals who have already received the primary vaccine.
- It is not indicated as a response to a known exposure or case of diphtheria.

Option B - Nothing as the child is already exposed:

- Even though the younger sibling has received vaccination against diphtheria 16 months ago, it does not ensure complete protection or eliminate the risk of infection.
- Vaccination provides immunity, but it may not be 100% effective in preventing infection.
- Therefore, appropriate measures should be taken to minimize the risk of transmission and potential illness.

Option C - Erythromycin + diphtheria toxoid:

- Administering both erythromycin and diphtheria toxoid is not the recommended next step in this case.
- In the case of the younger sibling who has not developed symptoms, administration of erythromycin alone is sufficient as a preventive measure.

Solution for Question 32:

Correct Option D - If mother is IgG positive for CMV, the child is unlikely to develop an infection:

- Ig G antibodies can cross the placenta and provide passive immunity to the infants.
- In case of congenital CMV infection, if the mother was previously infected and is Ig G positive, she would provide passive immunity to the fetus.

Incorrect Options:

Option A - About 20-30% of infections are symptomatic:

- About 5-10% of the fetus are symptomatic with congenital CMV infection

Option B - Triad of SNHL, periventricular calcification and enamel hypoplasia:

- Jaundice, petechiae and hepatosplenomegaly is a classical triad seen in symptomatic congenital CMV infection.

Option C - Most children asymptomatic at birth can develop conductive hearing loss later in life:

- The most important long-term sequelae of congenital CMV infection is sensorineural hearing loss and not conductive hearing loss.

Solution for Question 33:

Correct Option A - Corticosteroids:

- Corticosteroid is an immunosuppressant act by inhibiting the arachidonic acid pathway thereby decreasing the synthesis of inflammatory mediators like prostaglandin, leukotrienes, etc.
- In case of severe covid where SpO2 < 90%, due inflammatory damage to alveoli, the corticosteroid is administered causing a decrease in inflammation and improving the oxygen diffusion through alveoli and preventing the fibrosis of alveoli.

Incorrect Options:

Option B - Ivermectin:

- Ivermectin was earlier thought to have antiviral properties but evidence shows there is no decrease in mortality and hospital stay, hence it is not recommended to be used for COVID-19.

Option C - Remdesivir:

- There is controversy in the use of Remdesivir between guidelines given by IAP and MoHaFW.
- As per IAP Remdesivir can be used at more than 1 month of age, for moderate to severe COVID-19 disease in hospital settings.
- But as per MOHFW, Remdesivir is not recommended in children.
- There is lack of sufficient evidence on safety and efficacy with respect to Remdesivir in children below 18 years of age.

Solution for Question 34:

Correct Option B - Coxsackie viral meningitis and Hand Foot Mouth disease:

- The patient's presentation with vesicular lesions on both upper and lower limbs and also on the palms, soles, and inside the mouth is suggestive of Hand Foot Mouth disease.
- The patient also has fever and neck stiffness with CSF examination showing normal glucose level and elevated protein and lymphocytes is suggestive of viral meningitis.

Incorrect Options:

Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 35:

Correct Option B - Chorioretinitis:

- The manifestations of congenital toxoplasmosis can vary, but chorioretinitis is the most frequently observed and significant manifestation

Incorrect Options:

Options A, C, and D can occur in congenital toxoplasmosis but are not the most frequent symptom of congenital toxoplasmosis.

Solution for Question 36:

Correct Option A - Perinatal transmission:

- The most common cause of HIV infection in a newborn is through perinatal transmission, which occurs during pregnancy, childbirth, or breastfeeding.
- Perinatal transmission accounts for the majority of pediatric HIV cases.

Incorrect Options:

Option B - Breast milk:

- Although breast milk can contain the HIV virus, the risk of transmission through breastfeeding is lower compared to perinatal transmission.
- However, there is still a risk of HIV transmission through breast milk, especially in the absence of appropriate preventive measures.

Options C - Transplacental:

- HIV transmission through the placenta, known as transplacental transmission, is rare.
- However, in rare cases, such as when the mother has a high viral load, transplacental transmission can occur.

Options D - Exchange transfusion with infected blood:

- It is an unlikely mode of transmission, as appropriate screening and testing of blood products significantly reduce the risk of transmitting HIV through transfusions.

Immunization and Vaccines

1. Which of the following statements about the BCG vaccine is false?

A. Provides ~40% protection against primary TB and up to 100% against miliary TB.

B. Leaves a characteristic scar 6-12 months after injection.

C. Can be given until age 5 according to IAP 2021 guidelines.

D. Loses potency rapidly once reconstituted and should be discarded after 4 hours.

2. A nurse at a Primary Health Center (PHC) is preparing vaccines for an outreach program. She needs to prioritize handling the most heat-sensitive vaccines to prevent them from losing potency. Which of the following vaccines is the most sensitive to heat?

A. BCG before reconstitution

B. BCG after reconstitution

C. Hepatitis B

D. Pentavalent (Penta) vaccine

3. A nurse is inspecting vaccines in a clinic before administering them to children. She notices that one of the vials of hepatitis B vaccine appears suspicious. To ensure the vaccine's integrity, she performs a shake test. After vigorous shaking, she observes that the vaccine in the vial has a sediment that settles at the bottom, unlike another vial she shakes which remains uniformly cloudy. Based on the results of the shake test, which of the following actions should the nurse take?

A. Administer the vaccine with the sediment as it is still safe to use.

B. Administer the uniformly cloudy vaccine as it is not damaged by freezing.

C. Discard both vials as they show signs of potential damage.

D. Use the vial with the sediment as a control and proceed with the uniformly cloudy vaccine.

4. Which of the following Vaccine Vial Monitors (VVMs) indicates that the vaccine is safe to use?

A. only 4

B. 1 & 2

C. 3 & 4

D. only 3

5. A 16-month-old child is visiting the clinic for a routine check-up. The child's parents enquire about starting the varicella vaccination series. The child is healthy and has no known allergies or medical conditions. Which of the following vaccination schedules is most appropriate for this child?

A. Administer the varicella vaccine today and schedule the second dose 23 months later.

B. Administer the varicella vaccine today and schedule the second dose 3-6 months later.

C. Administer the varicella vaccine today and schedule the second dose 2 weeks later.

D. Administer the varicella vaccine today and schedule the second dose 1 month later.

6. Which of the following statements about the HPV vaccines is false?

A. Gardasil 9 protects against 9 strains of HPV.

B. The recommended no. of doses for girls under 15 receiving Gardasil 9 is 2.

C. Cervavac (HPV4) is recommended for only girls in the Indian immunization program.

D. Catch-up vaccination for the Gardasil 4 HPV vaccine is available for individuals up to 45 years of age.

7. A 2 y/o child has received only one dose of PCV13 at 6 months. What is the appropriate course of action to complete the pneumococcal vaccination series?

A. Administer three additional PCV13 doses: two 4 weeks apart, the third 8 weeks later.

B. Give one dose of PPSV23 now, no more PCV13 is needed.

C. Administer one additional dose of PCV13 at 24-59 months.

D. No further PCV13 doses are needed since the child is over 2 years old.

8. Which of the following statements about the Rotavirus and Typhoid vaccines is not accurate?

A. The Rotavirus vaccine should be administered at 6, 10, and 14 weeks of age.

B. The Typhoid conjugate vaccine is administered as a single dose at ≥6 months of age.

C. The Typhoid Vi capsular polysaccharide vaccine is recommended every 5 years.

D. The Rotavirus vaccine vial can be used for up to 4 hours after opening.

9. A 1 y/o, born to a Hepatitis B-positive mother, has completed the Hepatitis B series but hasn't received the Hepatitis A vaccine. The family travels frequently to high-risk areas. What is the most appropriate vaccination plan?

A. Give the first Hepatitis A dose now; no more Hepatitis B doses are needed.

B. Give both the first Hepatitis A dose and an additional Hepatitis B dose.

C. Delay Hepatitis A until age 2.

D. Give a second Hepatitis B dose and delay Hepatitis A until travel is planned.

10. A 6-month-old infant is brought to the clinic due to concerns about a local measles outbreak. The parents are worried about the child contracting measles and inquire about vaccination options. The child has not yet received any measles vaccination. What is the most appropriate recommendation regarding measles vaccination for this child?

A. Delay vaccination until the child reaches 9 months of age.

B. Administer the MMR vaccine now and no further doses are needed.

C. Administer the first dose of the Measles vaccine now and give MR vaccine again at 9 months and 16-18 months

D. Administer the first dose of the Measles vaccine now and schedule the next dose at 4-6 years.

11. A 18-month-old, previously vaccinated with the Pentavalent series, is due for the 1st DPT booster. The parents are concerned about side effects. What should the healthcare provider explain about the appropriate vaccine and potential adverse effects?

A. Administer DPT as the booster; local reactions like pain or redness are common.

B. Administer Pentavalent again to maintain immunity; severe side effects are rare.

C. Administer DPT and monitor for severe side effects like encephalopathy.

D. No further vaccines are needed after completing the Pentavalent series.

12. In response to the emergence of circulating vaccine-derived polioviruses (cVDPVs), which of the following strategies has been implemented globally to address this issue?

A. Exclusively using trivalent oral polio vaccine (OPV) to enhance herd immunity.

B. Transitioning from trivalent OPV to bivalent OPV (bOPV) and introducing inactivated polio vaccine (IPV) into routine schedules.

C. Replacing all OPV doses with IPV doses in the primary immunization schedule.

D. Discontinuing the use of any polio vaccines and relying on natural immunity.

13. A 10-week-old infant is brought to the clinic for a routine check-up. The mother reports that the baby received the first dose of Hepatitis B at birth but missed the scheduled vaccinations at 6 weeks. Which of the following should be administered to the infant at this visit to catch up with the National Immunisation Schedule?

A. bOPV, Pentavalent, Rotavirus

B. MMR, Hepatitis A

C. Vitamin A, Tetanus Toxoid

D. Hepatitis B, HPV

14. A preterm infant born at 28 weeks of gestation weighs 1.8 kg at birth. The infant's mother is HBsAg-positive. Which of the following is the correct management for hepatitis B vaccination in this infant?

A. Administer the hepatitis B birth dose within the first 24 hours of life.

B. Defer the hepatitis B birth dose until the infant reaches 1 month of age.

C. Administer hepatitis B immunoglobulin (HBIG) and hepatitis B vaccine within 12 hours of birth.

D. Administer hepatitis B vaccine only after discharge from the hospital.

15. According to the World Health Organization (WHO) classification, which of the following is classified as a severe vaccine reaction?

A. Pain, swelling, or redness at the injection site

B. Mild fever

C. Severe allergic reaction requiring emergency treatment

D. Convulsions occurring in clusters

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16. Which of the following is an example of a vaccine-induced adverse event following immunization?

A. Syncope after vaccination

B. Toxic shock syndrome due to a contaminated vaccine

C. Anaphylaxis after the measles vaccine

D. Gastroenteritis after the MMR injection

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17. Which of the following vaccines is not administered via intramuscular (IM) injection according to the National Immunization Schedule (NIS) of India?

A. Hepatitis B

B. Measles & Rubella (MR)

C. Pentavalent Vaccine

D. Diphtheria, Pertussis & Tetanus (DPT) Booster-1

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18. A 14 y/o girl who missed her HPV vaccination series as a pre-teen is now seeking to catch up. What is the most appropriate action according to the vaccination guidelines?

A. Start the full HPV vaccine series immediately.

B. Administer a single dose of HPV vaccine.

C. Delay vaccination until she is 18 years old.

D. Perform serologic testing before initiating the HPV vaccine series.

19. Which of the following vaccines should be discarded at the end of an immunization session if partially used?

A. Diphtheria-tetanus-pertussis (DTP) vaccine

B. BCG Vaccine

C. Hepatitis B vaccine (monovalent)

D. Pneumococcal conjugate vaccine (PCV)

20. Which of the following correctly describes the purpose of the shake test in vaccine management?

A. To check if the vaccine has been exposed to excessive heat

B. To determine if the vaccine has been frozen and potentially damaged

C. To verify the expiration date of the vaccine

D. To assess the potency of the vaccine by checking its colour

21. A 7 y/o foster child presents to the clinic with an undocumented immunization history. The caretakers are unsure if the child received any vaccines. What is the most appropriate next step in managing this child?

A. Delay vaccination until the immunization history is confirmed.

B. Administer all age-appropriate vaccines immediately.

C. Vaccine-preventable diseases are no longer a concern.

D. Administer only the first dose of each vaccine and wait for serologic confirmation.

22. A 6-month-old infant arrives at the clinic for their scheduled vaccinations. The mother mentions that the baby has a mild cough and runny nose but no fever. She is also concerned because the baby vomited 15 minutes after receiving the last dose of OPV during the previous visit. Which of the following is the most appropriate action for the healthcare provider to take today?

A. Administer all scheduled vaccines, including a repeat dose of OPV.

B. Withhold all vaccines and reschedule for when the child is symptom-free.

C. Restart the entire vaccination series from the beginning.

D. Administer all scheduled vaccines except OPV due to the previous vomiting episode.

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23. A 1 y/o child is assessed for vaccination status. The records show the child has received all vaccines recommended for the first year. Which of the following vaccines has the child not received as part of being fully immunized according to the National Immunisation Schedule?

A. 3 doses of OPV

B. 1 dose of Measles/MR

C. DPT Booster

D. 2 doses of IPV

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	2
Question 5	2
Question 6	3
Question 7	3
Question 8	3
Question 9	1
Question 10	3
Question 11	1
Question 12	2
Question 13	1
Question 14	3
Question 15	3
Question 16	3
Question 17	2
Question 18	1
Question 19	3
Question 20	2
Question 21	2
Question 22	1

Question 23	3

Solution for Question 1:

Correct Answer: B) Leaves a characteristic scar 6-12 months after injection.

Explanation:

The BCG vaccine typically causes a papule at the injection site that ulcerates within 3-6 weeks and heals with a characteristic scar over 6-12 weeks.

Aspect	Details
About BCG	The BCG vaccine contains live attenuated M. bovis Calmette—Guérin strain. In India, the Copenhagen (Danish 1331) strain is used. Available as a lyophilized powder in vacuum-sealed dark multidose vials. When reconstituted with sterile normal saline, each 0.1 mL dose contains 0.1-0.4 million live bacilli. It remains stable in lyophilized form at 2- 8°C for one year but loses potency rapidly once reconstituted and thus discarded after 4 hours. (Option D ruled out)
Dose, Route, Site	Infants under one year of age: 0.05 mL Children over one year and adults: 0.1 mL Intradermal at Left upper arm at insertion of deltoid.
Protective Efficacy	Primary Infection: ~40% protective efficacy. (Option A ruled out) Pulmonary Tuberculosis: 8-79% protective efficacy. All Forms of Tuberculosis: ~50% protective efficacy. Severe Forms (e.g., miliary tuberculosis, tubercular meningitis): 72-100% protective efficacy. (Option A ruled out) Childhood Tuberculosis: Accounts for 15-20% of cases, often disseminated; vaccination prevents serious morbidity. Reduces the risk of mortality from severe tuberculosis. Immunity Duration typically wanes over 10-15 years; Repeating vaccination in adolescence or adulthood does not extend protection.
National Program	At birth; catch up until 1 year (if missed).
IAP 2021	At birth; catch up until 5 years. (Option C ruled out)

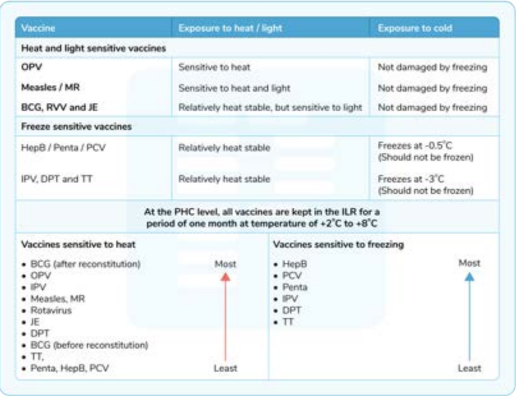
Injection and Reaction	Initial Reaction: A 5-7 mm wheal forms. One Week: Bacilli multiply, forming a 4-8 mm papule. 3-6 Weeks: The papule ulcerates. 6-12 Weeks: The ulcer heals with scarring.
Adverse Reactions	Local ulceration; discharging sinus; axillary lymphadenitis; If immunodeficient, disseminated infection, osteomyelitis; scrofuloderma.
Contraindications	Cellular immunodeficiency; Symptomatic HIV.
Storage	2-8°C; sensitive to heat and light; Discard reconstituted vaccine after 4 hours.

Solution for Question 2:

Correct Answer: B) BCG after reconstitution

Explanation: BCG after reconstitution is the most heat-sensitive and requires careful handling to avoid potency loss.

Amber coloured vials



Vaccine	Exposure to heat / light	Exposure to cold
Heat and light sensitive vaccines		
OPV	Sensitive to heat	Not damaged by freezing
Measles / MR	Sensitive to heat and light	Not damaged by freezing
BCG, RVV and JE	Relatively heat stable, but sensitive to light	Not damaged by freezing
Freeze sensitive vaccines		
HepB / Penta / PCV	Relatively heat stable	Freezes at -0.5°C (Should not be frozen)
IPV, DPT and TT	Relatively heat stable	Freezes at -3°C (Should not be frozen)
At the PHC level, all vaccines are kept in the ILR for a period of one month at temperature of +2°C to +8°C		

Vaccines sensitive to heat

Most

- BCG (after reconstitution)
- OPV
- IPV
- Measles, MR
- Rotavirus
- JE
- DPT
- BCG (before reconstitution)
- TT,
- Penta, HepB, PCV

Least

Vaccines sensitive to freezing

Most

- HepB
- PCV
- Penta
- IPV
- DPT
- TT

Least

BCG vaccine should be packed in amber glass ampoules to protect it from ultraviolet and fluorescent light.

BCG before reconstitution (Option A): BCG before reconstitution is less heat-sensitive compared to after reconstitution.

Hepatitis B (Option C): The Hepatitis B vaccine is relatively heat-stable, though sensitive to freezing.

Pentavalent (Penta) vaccine(Option D): The Pentavalent vaccine is more sensitive to freezing than heat.

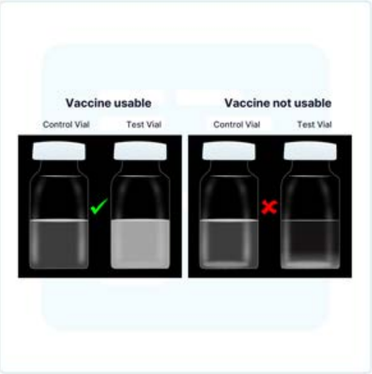
Solution for Question 3:

Correct Answer: B) Administer the uniformly cloudy vaccine as it is not damaged by freezing.

Explanation:

The uniformly cloudy appearance shows that the vaccine has not been damaged by freezing and is safe to use.

Control and Test Vials



Freeze-sensitive vaccines are shaken and assessed.

- Test Vial (Frozen): Sediment settles at the bottom after vigorous shaking; indicating freezing damage.
- Control Vial (Frozen): Used to compare the sedimentation rate with the suspected frozen test vials.
- Test Vial (Non-Frozen): Uniformly cloudy appearance; indicates the vaccine is undamaged by freezing.

Administer the vaccine with the sediment as it is still safe to use (Option A):

Sediment settling at the bottom indicates the vaccine has been damaged by freezing and should not be used.

Discard both vials as they show signs of potential damage (Option C):

Only the vial with sediment should be discarded; the uniformly cloudy vial is safe.

Use the vial with the sediment as a control and proceed with the uniformly cloudy vaccine(Option D):

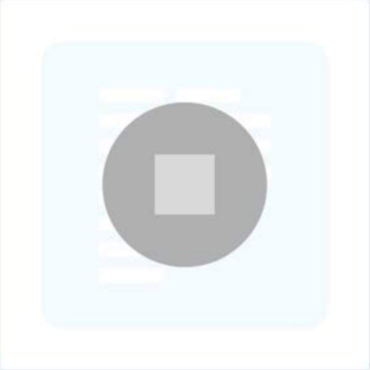
While the sedimented vial can serve as a control for the shake test, it should not be used for vaccination.

Solution for Question 4:

Correct Answer: B)1 & 2

Explanation:

- Both vaccine vials 1 & 2 are safe to use as long as the inner square is lighter than the outer circle, indicating it has not been exposed to excessive heat.
- Vaccine vials 3 and 4: The inner square is darker than the outer circle and hence has to be discarded



- Consists of an Inner lighter square within an outer dark circle.
- Function: The inner square darkens gradually and irreversibly upon heat exposure.
- Usability Indicator: Vaccine usable: Inner square lighter than the outer circle. Vaccine unusable: The inner square matches or is darker than the outer circle.
- Vaccine usable: Inner square lighter than the outer circle.
- Vaccine unusable: The inner square matches or is darker than the outer circle.
- VVM2 (Oral Polio Vaccine): assigned to OPV, the most heat-sensitive vaccine, reaches its end-point in 48 hours at 37°C
- VVM30 (Hepatitis B Vaccine): Reaches end-point in 30 days at 37°C; used for more heat-stable vaccines.
- Vaccine usable: Inner square lighter than the outer circle.
- Vaccine unusable: The inner square matches or is darker than the outer circle.

✓ The inner square is lighter than outer circle. If the expiry date has not passed, **USE** the vaccine.

✓ As time passes the inner square is still lighter than the outer circle. If the expiry date has not passed, **USE** the vaccine.

✗ **Discard point:** the colour of the inner square matches that of the outer circle. **DO NOT** use the vaccine.

✗ Beyond the discard point: inner square is darker than the outer circle. **DO NOT** use the vaccine.

Solution for Question 5:

Correct Answer: B) Administer the varicella vaccine today and schedule the second dose 3-6 months later.

Explanation:

The appropriate vaccination schedule for a 16-month-old child is the second dose correctly scheduled 3-6 months after the first dose.

Table rendering failed

Administer the varicella vaccine today and schedule the second dose for 23 months, 1 month, and 2 weeks later. (Options A, C, and D) do not follow the recommended guidelines for children 15-18 months old, the first dose of the varicella vaccine should be given, and the second dose should be administered 3-6 months later.

Solution for Question 6:

Correct Answer: C) Cervavac (HPV4) is recommended for only girls in the Indian immunization program.

Explanation:

- Cervavac (HPV4) is licensed and recommended for both girls and boys in India, according to the recent guidelines by The Indian Academy of Pediatrics (IAP) as of January 2024.
- Cervarix (HPV2), on the other hand, was used previously and only licensed and recommended for girls in India, and not for boys.

Vaccine Name	Cervavac (HPV4)	Gardasil 4 (HPV4)	Gardasil 9 (HPV9)
Strain Number	4	4	9
L1 protein of Virus-like particles	6, 11, 16, 18	6, 11, 16, 18	6, 11, 16, 18, 31, 35, 45, 52, 58 (Option A ruled out)

Target Age (Girls)	9–26 years	9–45 years	9–45 years
Target Age (Boys)	9-26 years	9–26 years	9–45 years
Dose (Age < 15)	2 doses (Option B ruled out)		
Dose (Age ≥ 15)	3 doses		
Schedule (Age < 15)	0, 6–12 months		
Schedule (Age ≥ 15)	0, 2, 6 months		
Dosage & Route	0.5 mL, intramuscular		
Injection Site	Upper arm (deltoid)		
National Program	In certain states for girls aged 11–13 years.		
IAP 2021	All young girls are recommended. (and boys in Cervavac)		
Catch-up Coverage	Up to 45 years (preferably pre-sexual activity) in Gardasil 4 & 9. (Option D ruled out) Only up to 26 years in Cervavac.		
Side Effects	Local pain, swelling, erythema, fever, syncope (due to injection, not vaccine)		
C/I	Anaphylaxis after the previous dose.		
Storage	2–8°C, protect from light.		

Solution for Question 7:

Correct Answer: C) Administer one additional dose of PCV13 at 24-59 months.

Explanation:

For a 2 y/o child who missed previous doses, the recommended catch-up schedule includes giving one additional dose at 24-59 months.

Aspect	Pneumococcal Conjugate Vaccine (PCV)	Pneumococcal Polysaccharide Vaccine (PPSV23)
Type	Conjugate vaccine	Polysaccharide vaccine
Dose, Route	0.5 mL; Intramuscular	0.5 mL; Intramuscular or Subcutaneous
Site	Anterolateral thigh or deltoid	Deltoid
Schedule in National Program	Three doses at 6 weeks, 14 weeks, and 9 months.	Not included in the national program. Recommended for: All adults 65 years or older. Individuals aged 2 years or older with specific medical conditions that increase their risk for pneumococcal disease should receive the PPSV23 vaccine. Note that PPSV23 is not approved for children under 2 years.
IAP 2021	Three doses at ≥ 6 weeks; given ≥4 weeks apart. One booster at 15-18 months.	One dose ≥8 weeks after the primary course with conjugate vaccine. Repeat once after 5 years, if risk persists.
Catch-Up (IAP)	At 7-11 months of age: Two doses ≥4 weeks apart; one booster at 15-18 months. At 12-23 months of age: Two doses ≥8 weeks apart. At 24-59 months of age: One dose ≥60 months age: One dose, if high-risk	Use only if high-risk and ≥2 years old.
Adverse Reactions	Fever, local pain, soreness, malaise.	Local pain, redness, and soreness (30-50%).
C/I	Anaphylaxis after the previous dose.	Anaphylaxis after the previous dose.
Storage	2-8°C; do not freeze	2-8°C

Administer three additional PCV13 doses: two 4 weeks apart, the third 8 weeks later. (Option A):

- The catch-up schedule does not require three additional doses with specific intervals.
- Instead, it requires one additional dose now and another dose later.

Give one dose of PPSV23 now, no more PCV13 is needed. (Option B):

Pneumococcal Polysaccharide Vaccine (PPSV23) is not recommended for routine use in children under 2 years of age.

No further PCV13 doses are needed since the child is over 2 years old (Option D):

- This option is incorrect because it does not follow the catch-up vaccination guidelines.
- Even though the child is over 2 years of age, additional doses of PCV13 are needed to complete the vaccination series.

Solution for Question 8:

Correct Answer: C) The Typhoid Vi capsular polysaccharide vaccine is recommended every 5 years.

Explanation:

The Typhoid Vi capsular polysaccharide vaccine is recommended every 3 years, not 5.

Aspect	Typhoid Conjugate Vaccine	Typhoid Vi Capsular Polysaccharide Vaccine
Dose	0.5 mL (25 µg)	0.5 mL (25 µg)
Route	Intramuscular (IM)	Subcutaneous (SC) or Intramuscular (IM)
Site	Anterolateral thigh or deltoid	Anterolateral thigh or deltoid
Schedule (IAP)	One dose at ≥6 months of age. (preferred) (Option B ruled out)	One dose at ≥2 years of age; Every 3 years if repeated dose is required.
Booster Doses	None	Required every 3 years. (Option C)
Catch-Up	One dose up to 18 years.	One dose at ≥2 years.
Adverse Reactions	Local pain, swelling, redness; fever	Local pain, swelling, redness; fever
Contraindication	Anaphylaxis after the previous dose.	Anaphylaxis after the previous dose.
Storage	2-8°C	2-8°C; do not freeze

ROTAVIRUS VACCINE	

Schedule	3 doses at 6, 10, and 14 weeks (Option A ruled out)
About Vaccine	Live attenuated oral vaccine Each dose is of 5 drops (0.5ml)-oral. The Rotavirus vaccine vial can be used for up to 4 hours after it has been opened. (Option D ruled out)
Strain Used	Recently, the 116 E strain of rotavirus found in AIIMS NICU has been used to produce the Indian rotavirus vaccine.
Adverse reactions	Fever, diarrhoea, vomiting, cough, rhinitis, rash, irritability; Intussusception is rare.
C/I	Known or Documented Allergic Reaction to the Rotavirus vaccine. History of Intussusception. History of Abdominal Surgery or Intestinal Malformation. Immunodeficiency or severe immunosuppression. Infants with moderate or severe acute illness. Vaccination should be postponed if the infant is experiencing moderate to severe diarrhoea or vomiting requiring rehydration therapy.
Catch-Up on Immunization Schedule	Permissible up to 1 year of age.
	

Solution for Question 9:

Correct Answer: A) Give the first Hepatitis A dose now; no more Hepatitis B doses are needed.

Explanation:

The Hepatitis A vaccine can be given starting at 12 months of age. Since the child has already completed the Hepatitis B vaccination series, no further Hepatitis B doses are needed.

HEPATITIS B VACCINE	
Dose, Route, Site	0.5 mL (10 µg) for infants; 1 mL (20 µg) for adults; Intramuscular. Site: Anterolateral thigh or deltoid; avoid the gluteal region.
Dosage	Birth Dose: As soon as possible after birth to prevent transmission from mother to baby. Follow-up Doses: Administered as part of the Pentavalent vaccine at 6, 10, and 14 weeks.
Catch-Up	Ideally, a gap of ≥4 weeks between the first two doses; ≥8 weeks between doses 2 and 3; and ≥16 weeks between the first and final doses, with the final dose given at ≥6 months of age.
Adverse Reactions	Local soreness, fever, fatigue.
Contraindication	Anaphylaxis after the previous dose.
Storage	2-8°C; do not freeze
Hepatitis B Positive Mother	Initial Dose: Hepatitis B vaccine given immediately after birth. HBIG: Ideally administered within 24 hours of birth.

HEPATITIS A VACCINE	
Disease Overview	Hepatitis A is a liver infection caused by the Hepatitis A virus, which can lead to symptoms such as jaundice, weakness, loss of appetite, fever, nausea, vomiting, abdominal pain, and dark urine. Rarely, this leads to liver failure and death.
Vaccines Available	Inactivated (killed) Vaccine: Administered in two doses, with the second dose given at least 6 months after the first. Live, Weakened Vaccine: Given as a single dose starting at 12 months of age.
Safety and Administration	Both vaccines are generally safe with minor side effects such as local pain, swelling, and low-grade fever. All children over 12 months should receive the vaccine. The vaccine is not recommended for those with severe allergies to vaccine components or those with weakened immune systems.
Prevention	Maintaining good hygiene and sanitation is crucial, but vaccination is the most effective way to prevent Hepatitis A.

Give both the first Hepatitis A dose and an additional Hepatitis B dose. (Option B):

- The child has completed the Hepatitis B vaccine series, so no additional Hepatitis B dose is required.
- However, the Hepatitis A vaccine should be started now.

Delay Hepatitis A until age 2. (Option C):

- The Hepatitis A vaccine should be given at 12 months of age, and delaying it to 2 years is not recommended, especially given the family's travel history.

Give a second Hepatitis B dose and delay Hepatitis A until travel is planned (Option D):

- There is no need for additional Hepatitis B vaccines once the series is completed.
- The Hepatitis A vaccine should be administered now at 12 months, regardless of immediate travel plans.

Solution for Question 10:

Correct Answer: C) Administer the first dose of the Measles vaccine now and give MR vaccine again at 9 months and 16-18 months

Explanation:

In the case of a measles outbreak, vaccination can be administered as early as 6 months of age, followed by completion of immunization as per the national immunization schedule.

MEASLES VACCINE	
Type of vaccine & Strain	Live-attenuated vaccine; Edmonston Zagreb strain; Each dose of the vaccine contains a minimum of 1000 infectious units of the attenuated virus.
Dose, Route	0.5 mL; Subcutaneous
Site	The right upper arm (at the insertion of the deltoid) or anterolateral thigh.
National Program	At 9-12 months and 15-24 months as measles or measles-rubella vaccine.
IAP	At 9-12 months, 15-18 months, and 4-6 years; preferably as MMR; avoid MMR-V <2 years.

In case of measles outbreak	The vaccine can be administered as early as 6 months of age, followed by a booster dose at 12-15 months.
Adverse Reactions	Local pain, tenderness; febrile seizures (especially with MMR-V in <2 years); Measles vaccine: fever or transient macular rash (after 7-12 days); MMR vaccine: Transient rash, arthralgia, aseptic meningitis, lymphadenopathy.
Contraindications	Immunosuppression; Pregnancy; malignancy; immunodeficiency (e.g., symptomatic HIV); recent infusion of immunoglobulin-containing blood product.
Storage	2-8°C; sensitive to heat and light; use within 4-6 hours of reconstitution.

Delay vaccination until the child reaches 9 months of age (Option A):

- During an outbreak, the first dose of the Measles vaccine should be given as early as 6 months to protect the infant.
- Delaying vaccination increases the child's risk of contracting measles.

Administer the MMR vaccine now and no further doses are needed (Option B):

- The MMR vaccine is generally administered after 9 months of age. At 6 months, only the Measles vaccine should be given, and a booster is still required later.

Administer the first dose of the Measles vaccine now and schedule the next dose at 4-6 years (Option D):

- While the first dose should be administered now, the next dose should be given at 12-15 months.
- The third dose is scheduled for 4-6 years.

Solution for Question 11:

Correct Answer: A) Administer DPT as the booster; local reactions like pain or redness are common.

Explanation:

After completing the Pentavalent series at 6, 10, and 14 weeks, the DPT vaccine is administered as the first booster dose between 16-24 months and a second booster between 5-6 years.

- Local reactions such as redness, swelling, and mild fever are common but not severe.

DPT VACCINE	
Protects Against	Diphtheria, Pertussis (Whole cell and Acellular), Tetanus
Dosage	6 weeks, 10 weeks, 14 weeks; 1st Booster: 16-24 months; 2nd Booster: 5-6 years
Route of Administration	Intramuscular, on the anterolateral aspect of the left thigh.
Adverse Effects	Local: Redness and pain at the injection site; Systemic: Fever; Severe: Persistent inconsolable cry (>3 hours), seizures, hypotonic hyporesponsive episodes, encephalopathy/altered sensorium, anaphylaxis.
C/I	Progressive neurological illness; Anaphylaxis to a previous dose; Encephalopathy within 7 days of the previous dose.
Precautions	Previous dose associated with fever >40.5°C within 48 hours; Hypotonic-hyporesponsive episode <48 hours; Persistent inconsolable crying for >3 hours, <48 hours; Seizures <72 hours.
Storage	2-8°C; sensitive to light.
PENTAVALENT VACCINE	Protects a child from 5 life-threatening diseases – Diphtheria, Pertussis, Tetanus, Hepatitis B, and H. Influenza type B. Administer at 6 weeks, 10 weeks, and 14 weeks. 0.5 mL given intramuscularly in the anterolateral mid-thigh. Store and transport at +2 to +8°C.

Administer Pentavalent again to maintain immunity; severe side effects are rare (Option B):

- The Pentavalent vaccine is only given in the primary series at 6, 10, and 14 weeks.
- The first booster at 16-24 months should be DPT, not Pentavalent.

Administer DPT and monitor for severe side effects like encephalopathy (Option C):

- While severe side effects like encephalopathy are possible, they are extremely rare.
- The most common adverse effects are mild, such as fever or pain at the injection site.

No further vaccines are needed after completing the Pentavalent series. (Option D):

- Even though the Pentavalent series has been completed, the DPT booster is still necessary at 16-24 weeks to maintain immunity against diphtheria, pertussis, and tetanus.

Solution for Question 12:

Correct Answer: B) Transitioning from trivalent OPV to bivalent OPV (bOPV) and introducing inactivated polio vaccine (IPV) into routine schedules

Explanation:

The transition to bivalent OPV and incorporating IPV helps mitigate cVDPV risks.

Polio Vaccines Overview:

Types of Polio Vaccines:

Attribute	Oral Polio Vaccine (OPV)	Inactivated Polio Vaccine (IPV)
Composition	Weakened (attenuated) Sabin poliovirus strains grown in monkey kidney cells.	Killed (inactivated) poliovirus - Salk's.
Administration	Oral	Injection
Immunity	Provides mucosal immunity in the gut.	Primarily induces a systemic immune response.
Advantages	Stronger mucosal immunity "Herd immunity" reducing wild poliovirus transmission	No risk of VAPP or cVDPV circulation. Effective in preventing paralytic poliomyelitis.
Disadvantages	Requires multiple doses. Rare risk of vaccine-associated paralytic poliomyelitis (VAPP). Emergence of circulating vaccine-derived polioviruses (cVDPVs).	Weaker mucosal immunity. More complex and expensive to manufacture. (Option C ruled out)

- Historical Focus: Relied on OPV for its ease of use and herd immunity benefits.
- Shift in Strategy: The emergence of cVDPVs, especially type 2 (cVDPV2), led to a switch from trivalent OPV to bivalent OPV (bOPV) in 2016, and the introduction of IPV in routine schedules. (Option A ruled out) (Option B)
- Current Approach: Sequential IPV-OPV schedules are used, transitioning to exclusive IPV use once wild poliovirus is no longer circulating. (Option D ruled out)
- WHO Guidelines: OPV-only schedules are no longer recommended.

- India's Success: The last case of wild poliovirus was reported in 2011; continued use of OPV for immunization days to prevent importation risks.
- Future Vision: Aiming to replace OPV with IPV globally, with interim strategies like fractional IPV dosing to enhance coverage.
- Primary Immunisation: IPV-based schedule with three doses at 6, 10, and 14 weeks of age.
- Booster Doses: Recommended in the second and fifth years of life.

Replacing all OPV doses with IPV doses in the primary immunization schedule (Option C) is incorrect as a complete replacement with IPV is not yet feasible due to IPV supply limitations.

Solution for Question 13:

Correct answer: A) bOPV, Pentavalent, Rotavirus

Explanation: At 10 weeks, the infant should receive the following vaccinations as per the National Immunisation Schedule:

- bOPV: Bivalent oral polio vaccine.
- Pentavalent vaccine-1: Diphtheria, pertussis, tetanus, Hepatitis B and H. influenza type B.
- Rotavirus: First dose
- If they have been missed, they should be given as soon as the child presents (but not later than 1year of age)

Age	National Immunisation Schedule
At birth	BCG, bOPV-0, HBV-0
6 weeks	bOPV-1, Pentavalent-1, Rota-1, fIPV-1, PCV-1
10 weeks	bOPV-2, Pentavalent-2, Rota-2
14 weeks	bOPV-3, Pentavalent-3, Rota-3, fIPV-2, PCV-2
9 months	MR-1, JE-1, PCV-3, fIPV-3
16-18 months	DTwP-B1, bOPV-B, JE-2, MR-2
5-6 years	DTwP-B2
10-12 years	Td
9-14 years	HPV-1, HPV-2

16 years	Td

Abbreviations:

- BCG Bacillus Calmette-Guerin;
- bOPV- Bivalent oral polio vaccine.
- HBV hepatitis B virus
- f-IPV - fractional - Inactivated poliovirus vaccine
- Rota - Rotavirus vaccine
- PCV - Pneumococcal vaccine
- DPT - Diphtheria, pertussis and tetanus vaccine
- MR - Measles & Rubella vaccine
- Td - Tetanus-Diphtheria vaccine
- JE - Japanese encephalitis vaccine

Additional Vaccines added by IAP:

- Hepatitis A
- Varicella
- HPV(Human Papilloma Virus) vaccine
- Typhoid vaccine: Given at 2 years of age

MMR and Hepatitis A (Option B) are administered later (MMR between 12-15 months and Hepatitis A from 12 months), so they are not relevant at 6 weeks.

Vitamin A and Tetanus Toxoid (Option C) are given at different times (Vitamin A at 6 months and TT later in childhood/adolescence).

Hepatitis B (Option D) was given at birth and is also a part of pentavalent vaccine.

Solution for Question 14:

Correct Answer: C) Administer hepatitis B immunoglobulin (HBIG) and hepatitis B vaccine within 12 hours of birth.

Explanation:

For preterm infants born to HBsAg-positive mothers, regardless of birth weight, it is crucial to administer both hepatitis B immunoglobulin (HBIG) and the hepatitis B vaccine within 12 hours of birth to prevent hepatitis B infection.

- This is a critical step in managing potential exposure to the hepatitis B virus.

HIV +ve Individuals:

- Inactivated Vaccines: Most are recommended, but effectiveness may vary based on immune deficiency.
- Live Vaccines: Generally contraindicated, with exceptions: MMR: Administer if asymptomatic or with mild symptoms (CD4+ ≥ 15%). Varicella: Consider if CD4+ ≥ 15%; MMRV is not recommended. Yellow Fever: Contraindicated for severely immunocompromised under 6; specific precautions for others.
- MMR: Administer if asymptomatic or with mild symptoms (CD4+ ≥ 15%).
- Varicella: Consider if CD4+ ≥ 15%; MMRV is not recommended.
- Yellow Fever: Contraindicated for severely immunocompromised under 6; specific precautions for others.
- Other Vaccines: Pneumococcal: PCV13 and PPSV23 are recommended. Meningococcal: MCV4 is recommended for children ≥ 2 months. Hib: Recommended if not given during infancy.
- Pneumococcal: PCV13 and PPSV23 are recommended.
- Meningococcal: MCV4 is recommended for children ≥ 2 months.
- Hib: Recommended if not given during infancy.
- MMR: Administer if asymptomatic or with mild symptoms (CD4+ ≥ 15%).
- Varicella: Consider if CD4+ ≥ 15%; MMRV is not recommended.
- Yellow Fever: Contraindicated for severely immunocompromised under 6; specific precautions for others.
- Pneumococcal: PCV13 and PPSV23 are recommended.
- Meningococcal: MCV4 is recommended for children ≥ 2 months.
- Hib: Recommended if not given during infancy.

Immunocompromised Individuals:

- General Principles: Vaccination varies by condition, severity, risk, and vaccine type.
- Live Vaccines: Typically contraindicated; exceptions based on condition and vaccine.
- Inactivated Vaccines: Effective but may be reduced based on immune deficiency.

Planned Splenectomy:

- Pneumococcal Vaccines: PCV13: Administer first. PPSV23: Administer at least 8 weeks after PCV13.
- PCV13: Administer first.
- PPSV23: Administer at least 8 weeks after PCV13.
- Meningococcal Vaccine: MCV4 recommended.
- Hib Vaccine: Recommended if not previously completed or given after 14 months.
- Influenza Vaccine: Annual vaccination is recommended to protect against seasonal influenza, which can be more severe in post-splenectomy patients.
- PCV13: Administer first.
- PPSV23: Administer at least 8 weeks after PCV13.

Preterm Infants:

- General Recommendations: Vaccinate based on chronological age, similar to full-term infants.
- Hepatitis B Vaccine: For ≥ 2 kg and stable: Birth dose within 24 hours. For < 2 kg: Birth dose deferred until 1 month or hospital discharge (if HBsAg-negative mother). All preterm, low-birthweight infants born to HBsAg-positive mothers should receive HBIG and hepatitis B vaccine within 12 hours of birth, with three additional doses starting at 1 month.
- For ≥ 2 kg and stable: Birth dose within 24 hours.
- For < 2 kg: Birth dose deferred until 1 month or hospital discharge (if HBsAg-negative mother). All preterm, low-birthweight infants born to HBsAg-positive mothers should receive HBIG and hepatitis B vaccine within 12 hours of birth, with three additional doses starting at 1 month.
- For ≥ 2 kg and stable: Birth dose within 24 hours.
- For < 2 kg: Birth dose deferred until 1 month or hospital discharge (if HBsAg-negative mother). All preterm, low-birthweight infants born to HBsAg-positive mothers should receive HBIG and hepatitis B vaccine within 12 hours of birth, with three additional doses starting at 1 month.

Vaccination in Elderly:

- Influenza (flu): Annual vaccination, especially with high-dose or adjuvanted vaccines, helps protect against flu complications such as pneumonia and organ failure.
- Pneumococcal vaccine: A one-time vaccine against bacterial infections that can cause severe pneumonia.
- Shingles: A two-dose series to prevent shingles and its complications, like postherpetic neuralgia.
- Tdap (Tetanus, Diphtheria, Pertussis): Booster shots every 10 years to maintain immunity.

• RSV (Respiratory Syncytial Virus): Newly approved for older adults, especially those with heart or lung conditions.

Solution for Question 15:

Correct Answer: C) Severe allergic reaction requiring emergency treatment.

Explanation:

It is life-threatening and requires immediate medical attention.

WHO Classification of Vaccine Reactions:

Category	Definition	Examples
Minor Reactions (Options A&B ruled out)	Common, self-limiting; resolve within a few days without specialized care.	Pain, swelling, or redness at the injection site Mild fever Malaise or fatigue Muscle pain or headache
Severe Reactions	Usually are enhanced minor reactions, generally due to a heightened immune response of the vaccinated individual.	Severe allergic reactions requiring emergency treatment. Convulsions or serious neurological events. All adverse effects are not categorized as minor or serious.
Serious Reactions (Option D ruled out)	May require hospitalization; can lead to permanent disability or death.	Severe allergic reactions (e.g., anaphylaxis) Convulsions Reactions occurring in clusters (two or more cases at one vaccination centre or in a geographical area)

Solution for Question 16:

Correct Answer: C) Anaphylaxis after the measles vaccine

Explanation:

- This is a vaccine-induced AEFI, where the adverse event is directly caused by the active component of the vaccine.
- Anaphylaxis is a severe allergic reaction triggered by the vaccine.

Adverse Event Following Immunization (AEFI):

Definition:

- An AEFI is any unfavourable medical event occurring after immunization, including signs, symptoms, abnormal lab findings, or diseases.
- Not all AEFIs are causally linked to the vaccine.

Classification of AEFIs:

Classification	Description	Examples
Vaccine-Induced	Caused by an active component of the vaccine.	Anaphylaxis (measles vaccine), vaccine-associated paralytic poliomyelitis, BCG-related adenitis, pertussis encephalopathy.
Vaccine Potentiated	Precipitated by vaccination but could occur even without it.	First febrile seizure.
Injection Reaction (Option A ruled out)	Caused by anxiety or pain from the injection, not the vaccine.	Syncope after vaccination.
Program/Immunization Errors (Option B ruled out)	Result from errors in vaccine preparation, handling, or administration.	Toxic shock syndrome (contaminated measles vaccine), injection site abscess.
Coincidental (Option D ruled out)	Temporally linked to the vaccine by chance or due to an unrelated illness.	Gastroenteritis after MMR injection.

Assessing AEFI Severity:

- An AEFI is considered serious if it: Results in death Leads to hospitalisation Causes persistent or significant disability Occurs in clusters (two or more cases within a geographical area)
- Results in death
- Leads to hospitalisation
- Causes persistent or significant disability
- Occurs in clusters (two or more cases within a geographical area)

- Results in death
- Leads to hospitalisation
- Causes persistent or significant disability
- Occurs in clusters (two or more cases within a geographical area)

Note: Immunisation error-related reactions are the most common cause of serious AEFIs.

Solution for Question 17:

Correct Answer: B) Measles & Rubella (MR)

Explanation:

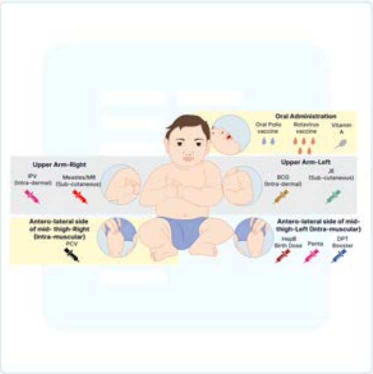
The MR vaccine is administered subcutaneously in the upper arm.

National Immunization Schedule (NIS) of India

Age	Vaccine	Dose	Route	Site
At Birth	BCG	0.1 ml (0.05 ml until 1 month)	Intradermal	Left upper arm
Hepatitis B (Birth Dose) (Option A ruled out)	0.5 ml	Intramuscular	The anterolateral side of the mid-thigh	
Oral Polio Vaccine (OPV-0)	2 drops	Oral	Oral	
6 Weeks	OPV-1	2 drops	Oral	Oral
Pentavalent-1 (Option C ruled out)	0.5 ml	Intramuscular	The anterolateral side of the mid-thigh	
Rotavirus Vaccine (RVV-1)	5 drops	Oral	Oral	
Fractional dose of Inactivated Polio Vaccine (fIPV-1)	0.1 ml	Intra dermal	Right upper arm	
Pneumococcal Conjugate Vaccine (PCV-1)	0.5 ml	Intramuscular	The anterolateral side of the mid-thigh	

10 Weeks	OPV-2	2 drops	Oral	Oral
Pentavalent-2 (Option C ruled out)	0.5 ml	Intramuscular	The anterolateral side of the mid-thigh	
Rotavirus Vaccine (RVV-2)	5 drops	Oral	Oral	
14 Weeks	OPV-3	2 drops	Oral	Oral
Pentavalent-3 (Option C ruled out)	0.5 ml	Intramuscular	The anterolateral side of the mid-thigh	
fIPV-2	0.1 ml	Intra dermal	Right upper arm	
Rotavirus Vaccine (RVV-3)	2 drops	Oral	Oral	
Pneumococcal Conjugate Vaccine (PCV-2)	0.5 ml	Intramuscular	The anterolateral side of the mid-thigh	
9-12 Months	Measles & Rubella (MR-1) (Option B)	0.5 ml	Subcutaneous	Upper arm
Japanese Encephalitis (JE-1)	0.5 ml	Subcutaneous (Live attenuated vaccine) Intramuscular(Killed vaccine)	Left upper Arm (Anterolateral aspect of mid-thigh	
PCV Booster (if applicable) [The only booster dose administered before 1 year]	0.5 ml	Intramuscular	The anterolateral side of the mid-thigh	
16-24 Months	MR-2 (Option B)	0.5 ml	Subcutaneous	Upper arm
JE-2	0.5 ml	Subcutaneous (Live attenuated vaccine) Intramuscular(Killed vaccine)	Left upper Arm Anterolateral aspect of mid-thigh.	
Diphtheria, Pertussis & Tetanus (DPT) Booster-1 (Option D ruled out)	0.5 ml	Intramuscular	Upper arm	
OPV Booster	2 drops	Oral	Oral	

5-6 Years	DPT Booster-2	0.5 ml	Intramuscular	Upper arm
10 Years	Tetanus & adult Diphtheria (Td)	0.5 ml	Intramuscular	Upper arm
16 Years	Td	0.5 ml	Intramuscular	Upper arm



Solution for Question 18:

Correct Answer: A) Start the full HPV vaccine series immediately

Explanation:

- The HPV vaccine series should be started as soon as possible for adolescents who missed the vaccination at the recommended age.
- The full series is needed to ensure complete protection against HPV-related diseases.

Recommended Vaccines for Adolescents (WHO):

Vaccine	Recommended Age	Dosage Schedule
Tetanus-diphtheria-pertussis (Tdap/Td)	Tdap: 10 years Td: Every 10 years	Tdap at age 10. Td boosters every 10 years.
Measles, Mumps, and Rubella (MMR)	12–13 years	One dose if not previously received.
Rubella	12–13 years	One dose specifically for girls if MMR was not given earlier.

Hepatitis B	Birth to 18 years	Series of three doses at 0, 1, and 6 months.
Hepatitis A	12 months to 18 years	Two doses at 0 and 6 months.
Typhoid	2 years to 18 years	A single dose of intramuscular Vi polysaccharide or three doses of oral vaccine. Boosters every 3 years.
Varicella (Chickenpox)	2–12 years and >12 years	One dose for ages 2–12 years. Two doses for ages over 12 years.
Human Papillomavirus (HPV)	10–12 years (Option C ruled out)	Three doses (0, 1, 2 or 0, 1, 6 months depending on the vaccine)

Importance of Adolescent Vaccination:

- Boost Immunity: Helps to enhance immunity that may wane after primary immunization.
- Protect Against Diseases: Addresses increased vulnerability to diseases due to lifestyle changes, such as higher risk of hepatitis B and C infections.

Administer a single dose of HPV vaccine (Option B):

- The HPV vaccine requires a series of doses (typically three) for full efficacy.
- Administering only a single dose is not sufficient for adequate protection.

Perform serologic testing before initiating the HPV vaccine series (Option D):

- Serologic testing is not required before starting the HPV vaccine series.
- The vaccine should be administered based on age and vaccination history, regardless of prior exposure or serologic status.

Solution for Question 19:

Correct Answer: B) BCG Vaccine

Explanation:

BCG Vaccine should be discarded at the end of an immunization session if partially used, as it is not applicable under the open vial policy.

Open Vial Policy:

- The Open Vial Policy allows for the reuse of partially used multi-dose vials of certain vaccines under the Universal Immunization Program (UIP) in subsequent immunization sessions.
- These sessions can be both fixed or outreach and can occur up to four weeks (28 days) after the vial was first opened.
- Vaccines that do not follow the open vial policy include BCG vaccine, MR vaccine and JE vaccine (see table below).

This policy is designed to reduce vaccine wastage.

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Solution for Question 20:

Correct Answer: B) To determine if the vaccine has been frozen and potentially damaged

Explanation:

The shake test is specifically designed to identify if a freeze-sensitive vaccine has been accidentally frozen, which could reduce its efficacy.

Vaccine Vial Monitor (VVM):

- Purpose: A heat-sensitive label on vaccine vials to ensure vaccine potency during transportation and storage.
- Function: The VVM contains a heat-sensitive material that changes colour irreversibly when exposed to heat over time. (Option A ruled out) Healthcare providers use this visual indicator to assess if a vaccine has been exposed to inappropriate temperatures.
- The VVM contains a heat-sensitive material that changes colour irreversibly when exposed to heat over time. (Option A ruled out)
- Healthcare providers use this visual indicator to assess if a vaccine has been exposed to inappropriate temperatures.
- Usage Criteria: The vaccine is usable if the inner square of the VVM is lighter than the outer circle.

- The vaccine is usable if the inner square of the VVM is lighter than the outer circle.
- The VVM contains a heat-sensitive material that changes colour irreversibly when exposed to heat over time. (Option A ruled out)
- Healthcare providers use this visual indicator to assess if a vaccine has been exposed to inappropriate temperatures.
- The vaccine is usable if the inner square of the VVM is lighter than the outer circle.

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- The vaccine should be discarded if the inner square is the same colour as or darker than the outer circle.
- The vaccine should be discarded if the inner square is the same colour as or darker than the outer circle.
- The vaccine should be discarded if the inner square is the same colour as or darker than the outer circle.

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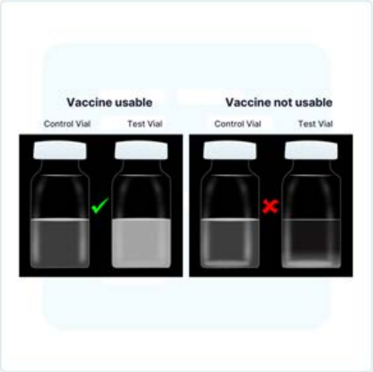
- VVM Color Change Timeline:

Vaccine	VVM Type	Colour Change Time
Oral poliovirus vaccine (OPV)	VVM2	2 days
DTP-hepatitis B vaccine	VVM14	14 days

- Purpose: To determine if a freeze-sensitive vaccine (e.g., DTP, DT, TT, pentavalent, hepatitis B has been accidentally frozen, which can reduce efficacy.
- Procedure: Control Vial: Deliberately freeze a vial from the same batch for a specific time, then thaw at room temperature. Test Vial: The vial suspected of freezing is shaken vigorously along with the control vial. Comparison: Place both vials side by side against a light source to compare sedimentation patterns.
- Control Vial: Deliberately freeze a vial from the same batch for a specific time, then thaw at room temperature.
- Test Vial: The vial suspected of freezing is shaken vigorously along with the control vial.
- Comparison: Place both vials side by side against a light source to compare sedimentation patterns.
- Place both vials side by side against a light source to compare sedimentation patterns.

- Interpretation:
- Control Vial: Deliberately freeze a vial from the same batch for a specific time, then thaw at room temperature.
- Test Vial: The vial suspected of freezing is shaken vigorously along with the control vial.
- Comparison: Place both vials side by side against a light source to compare sedimentation patterns.
- Place both vials side by side against a light source to compare sedimentation patterns.
- Place both vials side by side against a light source to compare sedimentation patterns.

Observation	Interpretation	Action
Sedimentation in the test vial is similar to or faster than the control vial	Likely frozen	Discard the test vial
No sediment or slower sedimentation in the test vial compared to the control vial	Not Frozen	Safe to use



To verify the expiration date of the vaccine (Option C):

- The shake test is unrelated to verifying the expiration date.
- Expiration dates are typically checked visually on the label of the vaccine vial.

To assess the potency of the vaccine by checking its colour (Option D):

- Potency is not assessed by colour in the shake test.
- The shake test evaluates the sedimentation pattern to determine if freezing has occurred.

Solution for Question 21:

Correct Answer: B) Administer all age-appropriate vaccines immediately

Explanation:

- Children with unknown or undocumented immunization histories should be considered at risk and vaccinated without delay based on their current age.
- This ensures they receive protection against vaccine-preventable diseases as soon as possible.

Vaccination of Children with Unknown or No Immunisation History:

Consider Susceptible: Children with unknown or undocumented immunization history should be considered at risk for vaccine-preventable diseases and vaccinated without delay.

- Safety of Vaccination: Administering vaccines to children who are already immune is not harmful.
- Age-Based Immunisation: The immunization schedule should be tailored to the child's current age, ensuring all age-appropriate vaccines are administered promptly.
- Catch-Up Schedule: Refer to the catch-up immunization schedules for children aged 4 months through 18 years. These schedules specify the minimum age for the first dose and the minimum intervals between doses.
- Refer to the catch-up immunization schedules for children aged 4 months through 18 years.
- These schedules specify the minimum age for the first dose and the minimum intervals between doses.
- Refer to the catch-up immunization schedules for children aged 4 months through 18 years.
- These schedules specify the minimum age for the first dose and the minimum intervals between doses.

Options for Addressing Concerns about Previous Vaccinations:

Option	Description
1. Administer or Repeat Missing Doses	Ensures full protection by completing the vaccine series.
2. Perform Serologic Tests	If tests show no antibodies, vaccinate according to the recommended schedule.

Delay vaccination until the immunization history is confirmed (Option A):

- Delaying vaccination is not recommended because it leaves the child susceptible to vaccine-preventable diseases.
- Immediate vaccination should be prioritized when the immunization history is unknown.

Administer only the first dose of each vaccine and wait for serologic confirmation (Option D):

- Administering only the first dose and waiting for serologic results is not the best practice.
- Vaccination should not be delayed, and the child should receive the full age-appropriate schedule as soon as possible to ensure protection.

Solution for Question 22:

Correct Answer: A) Administer all scheduled vaccines, including a repeat dose of OPV.

Explanation:

- This is the correct option because mild illness, such as a cough or cold, is not a contraindication for vaccination.
- Also, since the child vomited within 30 minutes of the last OPV dose, a repeat dose should be given today.

Special case scenarios:

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Withhold all vaccines and reschedule for when the child is symptom-free (Option B):

This is not recommended because mild symptoms like a cough or cold do not justify delaying vaccinations.

Restart the entire vaccination series from the beginning (Option C):

- This is unnecessary. Interrupted or delayed vaccination schedules generally do not require restarting the series.
- The child can continue with the existing schedule, ensuring they receive all due doses.

Administer all scheduled vaccines except OPV due to the previous vomiting episode (Option D):

While OPV might be delayed in cases of uncontrolled vomiting, this child only vomited once, 15 minutes after the previous dose, which warrants a repeat dose rather than skipping it.

Solution for Question 23:

Correct answer: C) DPT Booster

Explanation:

- A Fully Immunized child has received all the vaccines recommended in the National Immunisation Schedule for the first year of life.
- The DPT Booster is recommended in the second year of life, so a fully immunized child (up to 1 year) would not have received this yet.

Fully immunised Child	Completely immunised child
Received all vaccines recommended in the National Immunization Schedule for the first year of life.	Received all vaccines recommended for both the first and second years of life in the National Immunization Schedule.
Specific criteria for monitoring: 1 dose of BCG 1 dose of Measles/MR (Option B ruled out) 3 doses each of OPV (Option A ruled out) and Pentavalent Vaccine 2 doses of IPV (Option D ruled out) Additional vaccines for complete protection (as per the program): 3 doses of PCV 3 doses of RVV 1 dose of JE (where applicable)	First Year: 1 dose of BCG, Measles/MR, JE (where applicable). 3 doses of OPV, Pentavalent vaccine, Rotavirus vaccine (RVV), and PCV (where applicable). 2 doses of IPV. Second Year: 2nd dose of Measles/MR, JE (where applicable). 1 booster dose of OPV and DPT.

Pediatric Cardiology-I

1. Which of the following statements regarding fetal circulation is/are true? The left and right heart in fetal circulation function as a parallel circuit as compared to series in adults. The placenta is not an efficient oxygen exchange organ as compared to lungs. Fetal superior vena cava (SVC) blood preferentially flows across the tricuspid valve, rather than the foramen ovale, into the right ventricle. Because the pulmonary arterial circulation is vasoconstricted, only approximately 5% of right ventricular (RV) outflow enters the lungs.

A. I only

B. I and IV only

C. I, II, and IV

D. All of the above

2. Which of the following statements is true regarding the blood carried by the umbilical vein in fetal circulation, and how many umbilical veins are present?

A. The umbilical vein carries deoxygenated blood and there are two umbilical veins.

B. The umbilical vein carries oxygenated blood and there are two umbilical veins.

C. The umbilical vein carries oxygenated blood and there is one umbilical vein.

D. The umbilical vein carries deoxygenated blood and there is one umbilical vein.

3. Which of the following changes does not occur in fetal circulation following birth?

A. Decrease in pulmonary vascular resistance (PVR)

B. Increase in mass and volume of the left ventricle

C. Closure of the ductus venosus

D. Increase in mass and volume of the right ventricle

4. A 5-year-old child is brought to the pediatric clinic after a heart murmur was detected during a recent febrile illness. On examination, the child has a soft ejection systolic murmur, best heard at the left upper sternal border. The second heart sound (S2) is normal. The child is asymptomatic, with no history of cyanosis, syncope, or respiratory distress. The fever has resolved, and the child is otherwise well. What is the most appropriate next step in managing this patient?

A. Reassure the parents that the murmur is likely innocent, and no further follow-up is necessary.

B. Immediately order an ECG and echocardiogram to rule out congenital heart disease.

C. Reassure the parents and schedule a follow-up assessment in a few weeks to reassess the murmur.

D. Refer to a pediatric cardiologist for further evaluation without additional tests.

5. Which of the following congenital heart defects can exhibit ductus-dependent pulmonary circulation, where the ductus arteriosus is critical for maintaining pulmonary blood flow?

A. Tetralogy of Fallot

B. Transposition of the Great Arteries

C. Coarctation of the Aorta

D. All the above

6. In a large ventricular septal defect (VSD), which factor primarily determines the direction and magnitude of the left-to-right shunt?

A. Size of the left atrium

B. Ratio of pulmonary vascular resistance (PVR) to systemic vascular resistance (SVR)

C. Ratio of left ventricular pressure to right ventricular pressure

D. Ratio of systemic vascular resistance (SVR) to pulmonary arterial pressure

7. A 7-year-old child presents with difficulty in physical activity and a heart murmur. On examination, the child has a widely split and fixed second heart sound (S2), a systolic ejection murmur at the left upper sternal border, and a mid-diastolic murmur at the lower left sternal border. The echocardiogram reveals an atrial septal defect (ASD) located in the fossa ovalis. What is the most likely pathophysiological consequence of this ASD?

A. Decreased right atrial volume due to left-to-right shunting

B. Increased pulmonary vascular resistance leading to cyanosis

C. Left ventricular overload resulting in heart failure

D. Right ventricular volume overload leading to right atrial and ventricular enlargement

8. Determine which syndrome linked to atrial septal defect is accurately paired with its corresponding gene.

A. Cri du chat syndrome- CHD7

B. Noonan Syndrome- MLL2

C. Familial ASD without heart block- NKX2.5

D. Holt-Oram- TBX5

9. A 12-year-old child is evaluated for a heart murmur and an ECG is performed. The ECG shows right axis deviation and minor right ventricular conduction delay.

Which of the following statements is TRUE regarding these ECG findings in the context of atrial septal defect (ASD)?

A. Evidence of left ventricular hypertrophy associated with pulmonary hypertension

B. Evidence of biventricular hypertrophy

C. Minor RV conduction delay

D. Left axis deviation

10. During the physical examination of an infant with a large Patent Ductus Arteriosus (PDA), which of the following statements is true regarding the findings?

A. Continuous machinery-like murmur, radiating to the second intercostal space.

B. High-pitched mitral mid-diastolic murmur at apex

C. Prominent thrill radiating to right clavicle

D. Prominent diastolic component present

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	4
Question 4	3
Question 5	1
Question 6	2
Question 7	4
Question 8	4
Question 9	3
Question 10	1

Solution for Question 1:

Correct Answer: D) All of the above

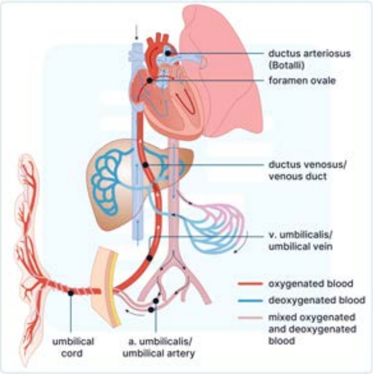
Explanation:

All the above statements regarding fetal circulation are true.

Differences in fetal circulation compared to adults:

	Fetal circulation	Adult circulation
Circulatory Circuit (Statement I)	The right and left ventricles function in a parallel circuit, allowing the mixing of oxygenated and deoxygenated blood.	The right and left ventricles function in a series circuit, with oxygenated blood from the lungs flowing sequentially through the heart and body.
Gas Exchange	Gas exchange occurs via the placenta.	The lungs provide gas exchange, with a much higher oxygen

	The placenta is less efficient than the lungs at oxygen exchange, leading to lower oxygen levels in the blood (umbilical venous PO2 of 30-35 mm Hg). (Statement II)	efficiency.
Pulmonary Circulation	The pulmonary vessels are vasoconstricted, diverting most blood away from the lungs. Only about 5% of right ventricular output reaches the lungs. (Statement IV)	The pulmonary vessels are open, allowing full blood flow through the lungs for oxygenation.
Unique Cardiovascular Structures	Three structures Ductus venosus, Foramen ovale, and Ductus arteriosus allow blood to bypass the liver and lungs, maintaining the parallel circuit.	These structures close or become vestigial after birth, redirecting blood flow through the liver and lungs.
Oxygen Distribution	Blood with higher oxygen content (from the umbilical vein) is preferentially directed to the upper body, including the brain, via the left ventricle. Blood with lower oxygen content (from the superior vena cava) flows through the right ventricle, ductus arteriosus, and aorta supplying the lower body and returning to the placenta. (Statement III)	Oxygenated blood from the lungs is evenly distributed to all parts of the body through systemic circulation.



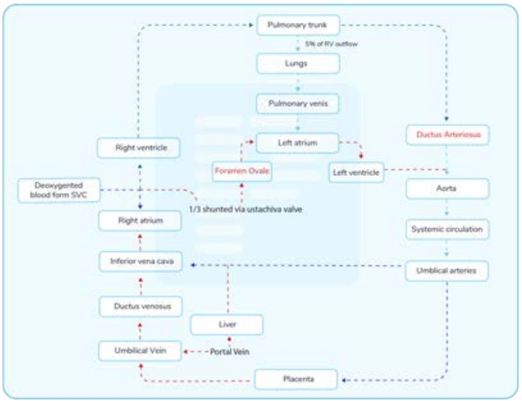
Solution for Question 2:

Correct Answer: C) The umbilical vein carries oxygenated blood and there is one umbilical vein.

Explanation:

In fetal circulation, the single umbilical vein carries oxygenated blood from the placenta to the fetus.

Fetal circulation:



Structure	Number	Type of Blood
Umbilical Artery	2	Deoxygenated blood
Umbilical Vein	1	Oxygenated blood

Solution for Question 3:

Correct Answer: D) Increase in mass and volume of the right ventricle

Explanation:

After birth, the right ventricle's wall thickness and mass decrease due to the shift to low-resistance pulmonary circulation.

Changes in Fetal Circulation Following Birth:

Parameter/Structure	Change	Cause
Pulmonary Vascular Resistance (PVR)	Decreases (Option A ruled out)	Mechanical expansion of the lungs and increased arterial PO2 lead to a rapid decrease in PVR.

Systemic Vascular Resistance (SVR)	Increases	The removal of the low-resistance placental circulation results in an increase in SVR.
Ductus Arteriosus Shunt	Reverses and eventually closes.	As PVR becomes lower than SVR, the shunt through the ductus arteriosus reverses from right-to-left to left-to-right. High arterial PO2 leads to the constriction and eventual closure of the ductus arteriosus, which later becomes the ligamentum arteriosum.
Foramen Ovale	Closes	Increased pulmonary blood flow to the left atrium raises left atrial volume and pressure, leading to the functional closure of the foramen ovale.
Ductus Venosus	Closes (Option B ruled out)	The removal of the placenta from circulation results in the closure of the ductus venosus.
Left ventricular Wall Thickness and Mass	Increases (Option B ruled out)	The left ventricle's wall thickness and mass increase as it adapts to the high-resistance systemic circulation.
Right ventricular Wall Thickness and Mass	Decreases (Option D)	The right ventricle's wall thickness and mass decrease as it adapts to the low-resistance pulmonary circulation.
Left Ventricular Output	Increases	The left ventricle now must deliver the entire systemic cardiac output, which is almost a 200% increase compared to fetal conditions.
Right Ventricular Output	Increases	Opening of pulmonary vasculature results in increased right ventricular output.

Time frame of closure of shunts following birth:

Shunt	Functional Closure	Anatomical closure	Remnant
Ductus Arteriosus	10-15 hrs after birth	2 -3 weeks after birth	Ligamentum arteriosum
Foramen ovale	3rd month	1 year after birth	Fossa ovalis
Ductus venosus	Immediately after birth	3-7 days after birth	Ligamentum venosum.

Solution for Question 4:

Correct Answer: C) Reassure the parents and schedule a follow-up assessment in a few weeks to reassess the murmur.

Explanation:

- In this scenario, the murmur is soft, systolic, and accompanied by a normal second heart sound, which is characteristic of an innocent murmur.
- Since the murmur was detected during a febrile illness, it is likely to disappear as the illness resolves.
- The best approach is to reassure the parents and plan a follow-up assessment to ensure the murmur has resolved, rather than immediately pursuing diagnostic tests or specialist referral.

Nada’s criteria: (used for clinical diagnosis of congenital heart disease)

Major	Minor
Systolic murmur grade III or more	Systolic murmur grade I or II
Diastolic murmur	Abnormal second heart sound
Cyanosis	Abnormal ECG
Congestive cardiac failure	Abnormal X-ray
	Abnormal Blood pressure

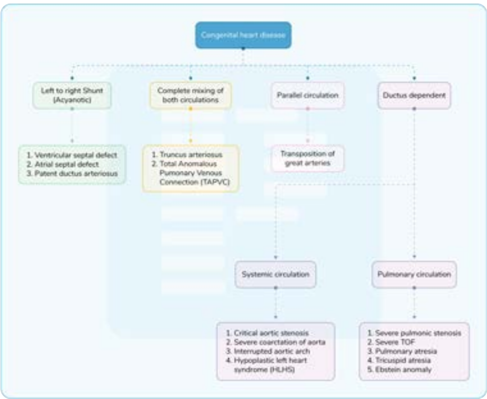
Presence of one major or 2 minor criteria is needed to indicate the presence of CHD.

Solution for Question 5:

Correct Answer: A) Tetralogy of Fallot

Explanation:

Tetralogy of Fallot (ToF) can be associated with ductus-dependent pulmonary circulation, especially in cases with severe right ventricular outflow tract obstruction.



Classification based on the presence of cyanosis:

Acyanotic	Cyanotic
Left to right shunts: Ventricular septal defect (VSD) Atrial septal defect (ASD) Patent ductus arteriosus (PDA) Obstructive lesions: Coarctation of aorta Infundibular stenosis Pulmonary valve stenosis Congenital mitral & tricuspid regurgitation	Right to left shunt: Persistent Truncus arteriosus Transposition of great vessels Tricuspid atresia Tetralogy of Fallot Total anomalous pulmonary venous communication Ebstein anomaly Eisenmenger's syndrome VSD with pulmonic stenosis Pulmonary arteriovenous malformation

Transposition of the Great Arteries (Option B): Results in a parallel circulation.

Coarctation of the Aorta (Option C): Associated with ductus-dependent systemic circulation.

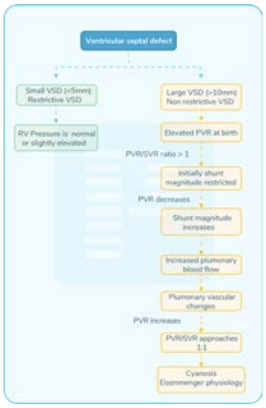
Solution for Question 6:

Correct Answer: B) Ratio of pulmonary vascular resistance (PVR) to systemic vascular resistance (SVR)

Explanation:

In large VSDs, the direction and magnitude of the left-to-right shunt are primarily influenced by the ratio of pulmonary vascular resistance (PVR) to systemic vascular resistance (SVR).

Pathophysiology in VSD:



Size of the Defect and Shunt Magnitude:

- The physical size of the VSD influences the magnitude of the left-to-right shunt.
- In a large VSD, the shunt's magnitude is primarily determined by the ratio of pulmonary vascular resistance (PVR) to systemic vascular resistance (SVR). (Option B)

Solution for Question 7:

Correct Answer: D) Right ventricular volume overload leading to right atrial and ventricular enlargement

Explanation:

The left-to-right shunt through the ASD causes increased blood flow to the right atrium and right ventricle, leading to their enlargement and right ventricular volume overload.

Atrial Septal Defects (ASDs) Features and Pathophysiology

Types of ASDs:

Type	Description
Ostium Secundum Defect	Most common type, located in the fossa ovalis; AV valves are structurally normal.
Ostium Primum Defect	Involves deficiency in the AV septum
Sinus Venosus Defect	Can occur in any part of the atrial septum; may lead to a single functioning atrium.

Pathophysiology:

Feature	Description
Left-to-Right Shunting	Oxygenated blood flows from left to right atrium; degree of shunting varies with defect size and vascular resistance.
Right Heart Volume Overload (Option D) (Option A & C ruled out)	Increased blood flow causes enlargement of the right atrium and ventricle, and dilation of the pulmonary artery.
Pulmonary Vascular Resistance	May increase with time, after remaining low during childhood, and lead to reversal of shunt and cyanosis. (Option B ruled out)

Clinical Manifestations:

Symptoms	Description
Asymptomatic	Many children are asymptomatic.
Failure to Thrive	Poor growth and weight gain.
Exercise Intolerance	Difficulty with physical activity.

Physical Examination Findings:

Finding	Description
Widely Split and Fixed S2	The second heart sound is split and fixed due to increased right ventricular diastolic volume.
Systolic Ejection Murmur	Soft murmur at the left upper sternal border, caused by increased flow across the right ventricular outflow tract.
Mid-Diastolic Murmur	Rumbling murmur at the lower left sternal border due to increased flow across the tricuspid valve.

Indications and timings of surgical intervention:

Indications	Details
Symptomatic ASD	Symptoms such as dyspnea, fatigue, or heart failure due to volume overload.
Large ASD (>10 mm or significant shunt)	Qp ratio ≥ 1.5:1 (significant left-to-right shunt).
Right Heart Enlargement	Right atrial and/or right ventricular dilatation on imaging.
Paradoxical Embolism	History of stroke or embolic events due to right-to-left shunt.
Pulmonary Hypertension	Elevated pulmonary artery pressure without irreversible pulmonary vascular disease.

Asymptomatic with Large ASD	Evidence of right ventricular overload on imaging.
Atrial Arrhythmias	Closure recommended in patients with atrial fibrillation to prevent embolism and heart failure.

Timing	Details
Infants and Children	Elective surgery or device closure before school age (4-5 years). Earlier surgery is symptomatic.
Adults	Surgery once symptomatic or if right heart enlargement or significant shunt is present.
Contraindication	Severe pulmonary hypertension with Eisenmenger syndrome may contraindicate surgery.

Solution for Question 8:

Correct Answer: D) Holt-Oram- TBX5

Explanation:

Genetic Conditions Associated with ASD:

Syndrome	Associated Genes	Cardiac Defects
Holt-Oram (Option D)	TBX5	ASD, VSD, TOF, left isomerism
Down Syndrome	-	ASD, AV septal defects
Patau Syndrome	-	ASD, VSD, PDA, valve abnormalities
Edwards Syndrome	-	ASD, VSD, PDA, valve abnormalities
CHARGE Association	CHD7	ASD, VSD, PDA, TOF
Kabuki Syndrome	MLL2	ASD, VSD, TOF, aortic coarctation, TGA
Ellis–van Creveld syndrome	EVC, EVC2	ASD, common atrium
Noonan Syndrome (Option B ruled out)	PTPN11	Pulmonary stenosis, ASD, cardiomyopathy

Cri du chat syndrome (Option A ruled out)	CTNND2	ASD, VSD, PDA, TOF
Familial ASD	NKX2.5 (with heart block), GATA4 (without heart block) (Option C ruled out)	ASD

Solution for Question 9:

Correct Answer: C) Minor RV conduction delay

Explanation:

Minor right ventricular conduction delay (rSR′ pattern) is a common finding in ASD due to right ventricular volume overload.

ECG findings for ASD and VSD:

Condition	Key ECG Findings
Atrial Septal Defect (ASD)	Right axis deviation (Option D ruled out) Minor RV conduction delay (rSR′ pattern) (Option C) Uncommon RV hypertrophy
Ventricular Septal Defect (VSD)	Left ventricular hypertrophy in small defects- associated with pulmonary hypertension. (Option A ruled out) Biventricular hypertrophy in large defects. (Option B ruled out)

Solution for Question 10:

Correct Answer: A) Continuous machinery-like murmur, radiating to the second intercostal space.

Explanation:

A continuous "machinery-like" murmur is characteristic of a large PDA and is typically heard best at the second left intercostal space.

Features and Pathophysiology of Patent Ductus Arteriosus (PDA):

Definition: A condition where the ductus arteriosus (a fetal blood vessel) fails to close after birth, allowing blood to flow from the aorta to the pulmonary artery.

Small vs Large PDA:

Table rendering failed

Physical Examination Findings:

Feature	Small PDA	Large PDA
Heart Size	Normal	Moderately to grossly enlarged; prominent apical impulse
Thrill	Rarely present	Prominent in second left intercostal space; may radiate to left clavicle, left sternal border, or apex (Option C ruled out)
Murmur	Long-ejection or continuous; best at left upper sternal border; radiates to back	Continuous "machinery-like"; begins after S1; intensifies towards end of systole; can radiate to second left intercostal space, left sternal border, or left clavicle (Option A)
Diastolic Component	Less prominent or absent (Option D ruled out)	Less noticeable or absent if PVR increased
Additional Murmur	None	Low-pitched mitral mid-diastolic murmur at apex (for large left-to-right shunt) (Option B ruled out)

Other Features:

- Premature Infants: Higher susceptibility due to less developed smooth muscle in ductus arteriosus
- Gender: Females are twice as likely to have PDA compared to males
- Maternal Rubella Infection: Associated with higher incidence of PDA

Pediatric Cardiology-II

1. Which of the following is not a component of Tetralogy of Fallot (ToF)?

A. Pulmonary stenosis

B. Ventricular septal defect (VSD)

C. Left ventricular hypertrophy

D. Overriding aorta

2. A 2-month-old infant with Tetralogy of Fallot presents with severe cyanosis from birth and a loud systolic ejection murmur at the left sternal border. The cyanosis worsens with closure of the ductus arteriosus. What is the most likely severity of pulmonary stenosis in this infant?

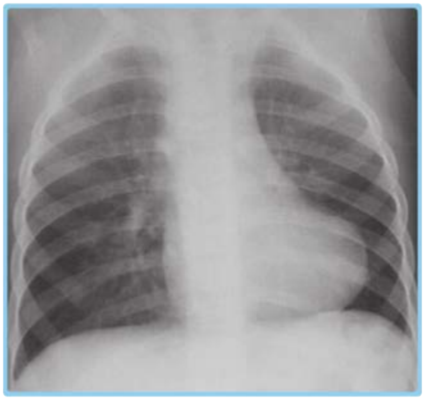
A. Mild

B. Moderate

C. Severe

D. Pulmonary atresia

3. A 6-month-old infant presents with cyanotic spells and a chest x-ray as shown below.. Which of the following investigations is considered the gold standard for diagnosing and characterizing this condition?



A. Electrocardiogram (ECG)

B. Chest X-Ray

C. Cardiac Catheterization

D. Echocardiography

4. A 10-year-old patient presents with symptoms suggestive of rheumatic fever. According to the modified Jones criteria, which of the following is considered a major criterion for diagnosing rheumatic fever for a low risk population ?

A. Arthralgia

B. Elevated ESR

C. Fever

D. Carditis

5. A 10-year-old boy with an ideal weight for his age presents with a history of acute rheumatic fever (ARF) and is being considered for secondary prophylaxis to prevent recurrence. What is the recommended regimen of benzathine penicillin for this patient, assuming he has no carditis?

A. 600,000 units every 4 weeks

B. 1.2 million units every 3 weeks

C. 1.2 million units every 2 weeks

D. 600,000 units every 2 weeks

6. An 8-day-old neonate has severe cyanosis and is in respiratory distress shortly after birth. Chest X-ray is as shown below. An echocardiogram is performed and shows pulmonary veins draining into the right atrium via an abnormal venous connection. Which of the following is the most likely diagnosis?

A. Total Anomalous Pulmonary Venous Connection

B. Truncus Arteriosus

C. Transposition of the Great Arteries

D. Coarctation of the Aorta

7. A neonate has cyanosis and is short of breath shortly after birth. On auscultation grade II ejection systolic murmur is heard. X-ray is taken as shown below. Which of the following congenital heart conditions is most likely associated with this radiographic finding?

A. Hypoplastic Left Heart Syndrome

B. Truncus Arteriosus

C. Transposition of the Great Arteries

D. Ebstein's Anomaly

8. Which congenital cardiac anomaly is associated with CATCH 22 or chromosome 22q11 deletion?

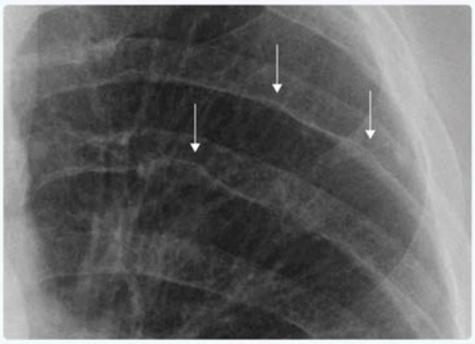
A. Hypoplastic Left Heart Syndrome

B. Truncus Arteriosus

C. Transposition of the Great Arteries

D. Coarctation of the Aorta

9. A 14-year-old boy presents with leg pain during physical activity. He is found to have a radio-femoral delay and murmur is heard on the shoulder region. A part of the chest X-ray is shown below. Which of the following conditions is most likely responsible for these findings?



X-Ray showing Notching of inferior rib border

A. Ventricular Septal Defect

B. Atrial Septal Defect

C. Patent Ductus Arteriosus

D. Coarctation of the Aorta

10. A 6-month-old infant presents with cyanosis, difficulty breathing, and prominent 'a' waves in the jugular pulse. The ECG shows left axis deviation and left ventricular hypertrophy. What is the underlying pathophysiology?

A. No RA-RV communication

B. Obstructed pulmonary outflow with intact septum.

C. Ventricular septal defect and pulmonary stenosis.

D. Underdeveloped left heart.

11. A 3-day-old infant has cyanosis shortly after birth.The mother has a history of Bipolar disorder. On examination, the infant has a prominent pansystolic murmur best heard along the left lower sternal border. The infant's ECG image is shown below. Which of the following is the most likely diagnosis?

A. Tetralogy of Fallot

B. Atrialization of the Right ventricle

C. Transposition of the Great vessels

D. Ventricular Septal Defect

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	4
Question 4	4
Question 5	2
Question 6	1
Question 7	3
Question 8	2
Question 9	4
Question 10	1
Question 11	2

Solution for Question 1:

Correct Answer: C) Left ventricular hypertrophy

Explanation:

- Left ventricular hypertrophy is not a feature of Tetralogy of Fallot.
- The condition is associated with right ventricular hypertrophy, and not left.

Components of Tetralogy of Fallot:

- Pulmonary Stenosis: (Option A ruled out) Narrowing of the pathway from the right ventricle to the pulmonary artery, possibly affecting the infundibulum, valve, or main pulmonary artery.
- Narrowing of the pathway from the right ventricle to the pulmonary artery, possibly affecting the infundibulum, valve, or main pulmonary artery.

- Ventricular Septal Defect (VSD): (Option B ruled out) A large hole between the right and left ventricles, usually misaligned, allowing abnormal blood flow.
- A large hole between the right and left ventricles, usually misaligned, allowing abnormal blood flow.
- Overriding Aorta: (Option D ruled out) The aorta is positioned over the septum, receiving blood from both ventricles.
- The aorta is positioned over the septum, receiving blood from both ventricles.
- Right Ventricular Hypertrophy: (Option C) Thickening of the right ventricle wall due to the extra effort required to pump blood.
- Thickening of the right ventricle wall due to the extra effort required to pump blood.
- Narrowing of the pathway from the right ventricle to the pulmonary artery, possibly affecting the infundibulum, valve, or main pulmonary artery.
- A large hole between the right and left ventricles, usually misaligned, allowing abnormal blood flow.
- The aorta is positioned over the septum, receiving blood from both ventricles.
- Thickening of the right ventricle wall due to the extra effort required to pump blood.

Solution for Question 2:

Correct Answer: C) Severe.

Explanation:

Severe pulmonary stenosis results in severe cyanosis from birth and worsens with the closure of the ductus arteriosus.

General Clinical Features of ToF:

- Cyanosis: Directly linked to the degree of pulmonary stenosis. Ranges from mild to severe based on the extent of RVOT (right ventricular outflow tract) obstruction. Older, untreated children may show dusky skin, gray sclerae, and clubbing.
- Directly linked to the degree of pulmonary stenosis.
- Ranges from mild to severe based on the extent of RVOT (right ventricular outflow tract) obstruction.
- Older, untreated children may show dusky skin, gray sclerae, and clubbing.

- Hypercyanotic Spells (Tet Spells): Sudden increase in cyanosis, often triggered by crying or agitation. Symptoms include irritability, restlessness, rapid breathing, and severe cases like syncope or convulsions. Commonly occurs in the morning or after intense crying.
- Sudden increase in cyanosis, often triggered by crying or agitation.
- Symptoms include irritability, restlessness, rapid breathing, and severe cases like syncope or convulsions.
- Commonly occurs in the morning or after intense crying.
- Murmurs: Loud, harsh systolic ejection murmur at the left sternal border, radiating to the lungs. Murmur intensity may vary, louder with moderate stenosis, and diminishes during tet spells. A continuous murmur may indicate collateral vessels.
- Loud, harsh systolic ejection murmur at the left sternal border, radiating to the lungs.
- Murmur intensity may vary, louder with moderate stenosis, and diminishes during tet spells.
- A continuous murmur may indicate collateral vessels.
- Additional Findings: Palpable systolic thrill along the left sternal border. Single or soft S2 due to a stenotic pulmonary valve. Prominent right ventricular impulse. Bulging of the left anterior chest wall in older children due to RV hypertrophy.
- Palpable systolic thrill along the left sternal border.
- Single or soft S2 due to a stenotic pulmonary valve.
- Prominent right ventricular impulse.
- Bulging of the left anterior chest wall in older children due to RV hypertrophy.
- Directly linked to the degree of pulmonary stenosis.
- Ranges from mild to severe based on the extent of RVOT (right ventricular outflow tract) obstruction.
- Older, untreated children may show dusky skin, gray sclerae, and clubbing.
- Sudden increase in cyanosis, often triggered by crying or agitation.
- Symptoms include irritability, restlessness, rapid breathing, and severe cases like syncope or convulsions.
- Commonly occurs in the morning or after intense crying.
- Loud, harsh systolic ejection murmur at the left sternal border, radiating to the lungs.
- Murmur intensity may vary, louder with moderate stenosis, and diminishes during tet spells.
- A continuous murmur may indicate collateral vessels.
- Palpable systolic thrill along the left sternal border.
- Single or soft S2 due to a stenotic pulmonary valve.
- Prominent right ventricular impulse.
- Bulging of the left anterior chest wall in older children due to RV hypertrophy.

Impact of Pulmonary Stenosis Severity on Clinical Presentation:

Pulmonary Stenosis Severity	Clinical Presentation
Mild (Option A ruled out)	Initial heart failure symptoms due to left-to-right shunt. Cyanosis may appear later as RV hypertrophy worsens. Increased risk of hypercyanotic spells.
Moderate (Option B ruled out)	Inferred presentation: increased cyanosis as left-to-right shunt decreases. Systolic murmur becomes louder, longer, and harsher. Softer pulmonic component of the second heart sound (S2).
Severe (Option C)	Early neonatal cyanosis due to restricted pulmonary blood flow. Right-to-left shunt causing systemic deoxygenation. Potential reliance on ductus arteriosus; closure can lead to severe cyanosis and circulatory collapse.
Pulmonary Atresia (Extreme) (Option D ruled out)	Complete absence of pulmonary valve; entire RV output directed into the aorta. Pulmonary blood flow depends on collateral vessels (MAPCAs) or a persistent ductus arteriosus. Severity of cyanosis and prognosis depends on the development of pulmonary arteries.

Key Points:

- Clinical presentation varies widely based on the combination and severity of the four defining defects of ToF.
- Pulmonary stenosis is the primary determinant of the clinical manifestations in ToF.

Solution for Question 3:

Correct Answer: D) Echocardiography

Explanation:

- Echocardiography is the gold standard for diagnosing and characterizing Tetralogy of Fallot.
- It provides detailed information about the aortic override, RVOT obstruction, and the sizes of related structures such as the pulmonary valve annulus and the pulmonary arteries.

Investigation of ToF

- Physical Examination: Cyanosis, clubbing, heart murmur, intensity/character of murmur, presence of thrill
- Cyanosis, clubbing, heart murmur, intensity/character of murmur, presence of thrill
- Chest X-ray: (Option B ruled out) Enlarged right ventricle, "boot-shaped" heart, decreased pulmonary vascular markings, right-sided aortic arch (20% of cases)
- Enlarged right ventricle, "boot-shaped" heart, decreased pulmonary vascular markings, right-sided aortic arch (20% of cases)
- Electrocardiogram (ECG): (Option A ruled out) Right ventricular hypertrophy (RVH), dominant R wave in V1/V2, RSR' pattern, tall P waves indicating right atrial enlargement
- Right ventricular hypertrophy (RVH), dominant R wave in V1/V2, RSR' pattern, tall P waves indicating right atrial enlargement
- Echocardiography: (Gold standard) (Option D) Assessment of: Aortic override RVOT obstruction Size of pulmonary valve annulus, main pulmonary artery, proximal branch pulmonary arteries Presence of patent ductus arteriosus (PDA)
- Assessment of: Aortic override RVOT obstruction Size of pulmonary valve annulus, main pulmonary artery, proximal branch pulmonary arteries Presence of patent ductus arteriosus (PDA)
- Aortic override
- RVOT obstruction
- Size of pulmonary valve annulus, main pulmonary artery, proximal branch pulmonary arteries
- Presence of patent ductus arteriosus (PDA)
- Cardiac Catheterization (Option C ruled out) Indications: Conclusive anatomy information needed Suspicion of coronary artery anomalies ToF with pulmonary atresia Measure pressures and oxygen saturations Perform interventions (e.g., Blalock-Taussig shunt)
- Indications: Conclusive anatomy information needed Suspicion of coronary artery anomalies ToF with pulmonary atresia Measure pressures and oxygen saturations Perform interventions (e.g., Blalock-Taussig shunt)
- Conclusive anatomy information needed
- Suspicion of coronary artery anomalies
- ToF with pulmonary atresia
- Measure pressures and oxygen saturations
- Perform interventions (e.g., Blalock-Taussig shunt)
- Cyanosis, clubbing, heart murmur, intensity/character of murmur, presence of thrill
- Enlarged right ventricle, "boot-shaped" heart, decreased pulmonary vascular markings, right-sided aortic arch (20% of cases)

- Right ventricular hypertrophy (RVH), dominant R wave in V1/V2, RSR' pattern, tall P waves indicating right atrial enlargement
- Assessment of: Aortic override RVOT obstruction Size of pulmonary valve annulus, main pulmonary artery, proximal branch pulmonary arteries Presence of patent ductus arteriosus (PDA)
- Aortic override
- RVOT obstruction
- Size of pulmonary valve annulus, main pulmonary artery, proximal branch pulmonary arteries
- Presence of patent ductus arteriosus (PDA)
- Aortic override
- RVOT obstruction
- Size of pulmonary valve annulus, main pulmonary artery, proximal branch pulmonary arteries
- Presence of patent ductus arteriosus (PDA)
- Indications: Conclusive anatomy information needed Suspicion of coronary artery anomalies ToF with pulmonary atresia Measure pressures and oxygen saturations Perform interventions (e.g., Blalock-Taussig shunt)
- Conclusive anatomy information needed
- Suspicion of coronary artery anomalies
- ToF with pulmonary atresia
- Measure pressures and oxygen saturations
- Perform interventions (e.g., Blalock-Taussig shunt)
- Conclusive anatomy information needed
- Suspicion of coronary artery anomalies
- ToF with pulmonary atresia
- Measure pressures and oxygen saturations
- Perform interventions (e.g., Blalock-Taussig shunt)

Treatment of ToF:

- Management of Tet Spells Immediate interventions: Knee-chest position Oxygen administration Morphine for sedation Intravenous fluids
- Immediate interventions: Knee-chest position Oxygen administration Morphine for sedation Intravenous fluids
- Knee-chest position
- Oxygen administration

- Morphine for sedation
- Intravenous fluids
- Surgical Repair Complete Repair Procedure: Close VSD, relieve RVOT obstruction Approach: Transatrial-transpulmonary Considerations: Minimizing pulmonary insufficiency with smaller patches Palliative Shunts Blalock-Taussig Shunt: Connects subclavian artery to pulmonary artery Central Shunt: Connects ascending aorta to main pulmonary artery
- Complete Repair Procedure: Close VSD, relieve RVOT obstruction Approach: Transatrial-transpulmonary Considerations: Minimizing pulmonary insufficiency with smaller patches
- Procedure: Close VSD, relieve RVOT obstruction
- Approach: Transatrial-transpulmonary
- Considerations: Minimizing pulmonary insufficiency with smaller patches
- Palliative Shunts Blalock-Taussig Shunt: Connects subclavian artery to pulmonary artery Central Shunt: Connects ascending aorta to main pulmonary artery
- Blalock-Taussig Shunt: Connects subclavian artery to pulmonary artery
- Central Shunt: Connects ascending aorta to main pulmonary artery
- Immediate interventions: Knee-chest position Oxygen administration Morphine for sedation Intravenous fluids
- Knee-chest position
- Oxygen administration
- Morphine for sedation
- Intravenous fluids
- Knee-chest position
- Oxygen administration
- Morphine for sedation
- Intravenous fluids
- Complete Repair Procedure: Close VSD, relieve RVOT obstruction Approach: Transatrial-transpulmonary Considerations: Minimizing pulmonary insufficiency with smaller patches
- Procedure: Close VSD, relieve RVOT obstruction
- Approach: Transatrial-transpulmonary
- Considerations: Minimizing pulmonary insufficiency with smaller patches
- Palliative Shunts Blalock-Taussig Shunt: Connects subclavian artery to pulmonary artery Central Shunt: Connects ascending aorta to main pulmonary artery
- Blalock-Taussig Shunt: Connects subclavian artery to pulmonary artery

- Central Shunt: Connects ascending aorta to main pulmonary artery
- Procedure: Close VSD, relieve RVOT obstruction
- Approach: Transatrial-transpulmonary
- Considerations: Minimizing pulmonary insufficiency with smaller patches
- Blalock-Taussig Shunt: Connects subclavian artery to pulmonary artery
- Central Shunt: Connects ascending aorta to main pulmonary artery

Solution for Question 4:

Correct Answer: D) Carditis

Explanation:

- Carditis is one of the major criteria in the modified Jones criteria for diagnosing rheumatic fever.
- It involves inflammation of the heart and is a significant sign of rheumatic fever.

Diagnosis of Initial ARF:

- 2 Major Manifestations, or
- 1 Major Manifestation plus 2 Minor Manifestations

Modified Jones Criteria for Diagnosing Rheumatic Fever

Historical Background:

- Original Jones criteria established in 1944.
- Significant updates in 1992 and later.

Key Components:

Criteria Type	Components
Major Criteria	Carditis (Option D) Polyarthritis Chorea Erythema marginatum Subcutaneous nodules
Minor Criteria	Fever (Option D ruled out) Arthralgia (Option A ruled out) Elevated acute phase reactants (e.g., ESR, CRP) (Option B ruled out)

	Prolonged PR interval on ECG

Major Criteria

Low-risk Population

- Carditis (clinical or subclinical)
- Arthritis (polyarthritis only)
- Chorea
- Erythema marginatum
- Subcutaneous nodule

Moderate- to High-risk Population

- Carditis (clinical or subclinical)
- Arthritis (polyarthritis, polyarthralgia, and/or monoarthritis)
- Chorea
- Erythema marginatum
- Subcutaneous nodule

Supporting Evidence:

- Importance of clinical history, including: Previous episodes of acute rheumatic fever Presence of streptococcal infection
- Previous episodes of acute rheumatic fever
- Presence of streptococcal infection
- Previous episodes of acute rheumatic fever
- Presence of streptococcal infection

Echocardiographic Findings:

- Recent adaptations include echocardiographic criteria to assess rheumatic heart disease (RHD).
- Focus on valve abnormalities and functional implications.
- World Heart Federation (WHF) guidelines provide specific criteria for diagnosing definite and borderline RHD.

Solution for Question 5:

Correct Answer: B) 1.2 million units every 3 weeks

Explanation:

To determine the recommended regimen of benzathine penicillin for a 10-year-old boy with a history of acute rheumatic fever (ARF) who has no carditis, we need to ist calculate the ideal weight for his age.

According to the formula:

- For an age group of 7-12 years: We calculate weight by (7x–5)/2, where x is the age in years
- For a 10-year-old boy:
- Ideal weight=(7×10–5)/2=(70–5)/2=65/2=32.5 kg.
- The standard regimen for patients over 30 kg with no carditis is 1.2 million units of benzathine penicillin every 3 weeks.

Rheumatic Fever Prevention Strategies

Primordial Prevention:

- Address social determinants to reduce Group A Streptococcus (GAS) transmission.

Primary Prevention:

- Diagnosis & Treatment: Use penicillin antibiotics for GAS infections to prevent acute rheumatic fever (ARF).
- School Programs: Implement early detection and treatment of GAS pharyngitis in schools.

Secondary Prevention:

- Prophylaxis: Administer continuous benzathine penicillin to prevent ARF recurrence.
- Management: Detect and manage rheumatic heart disease (RHD) early to avoid complications.

Role of Benzathine Penicillin:

- Effectiveness: Prevents recurrence of rheumatic fever by eliminating Group A streptococci, reducing the risk of heart damage.
- Administration: Dosage: >30 kg: 1.2 million units every 3 weeks. (Option B) <30 kg (high prevalence areas): 6 lakh units every 2 weeks.
- Dosage: >30 kg: 1.2 million units every 3 weeks. (Option B) <30 kg (high prevalence areas): 6 lakh units every 2 weeks.
- >30 kg: 1.2 million units every 3 weeks. (Option B)
- <30 kg (high prevalence areas): 6 lakh units every 2 weeks.

- Duration: Without Carditis: 5 years after the last episode or until 18 years of age, whichever is longer. With Carditis: 10 years after the last episode or until 25 years of age, whichever is longer. With Established RHD/Valve Surgery: Lifelong or until 40 years of age.
- Without Carditis: 5 years after the last episode or until 18 years of age, whichever is longer.
- With Carditis: 10 years after the last episode or until 25 years of age, whichever is longer.
- With Established RHD/Valve Surgery: Lifelong or until 40 years of age.
- Dosage: >30 kg: 1.2 million units every 3 weeks. (Option B) <30 kg (high prevalence areas): 6 lakh units every 2 weeks.
- >30 kg: 1.2 million units every 3 weeks. (Option B)
- <30 kg (high prevalence areas): 6 lakh units every 2 weeks.
- >30 kg: 1.2 million units every 3 weeks. (Option B)
- <30 kg (high prevalence areas): 6 lakh units every 2 weeks.
- Without Carditis: 5 years after the last episode or until 18 years of age, whichever is longer.
- With Carditis: 10 years after the last episode or until 25 years of age, whichever is longer.
- With Established RHD/Valve Surgery: Lifelong or until 40 years of age.

Tertiary Prevention:

- Medical Treatment: Provide heart surgery and chronic treatment for managing chronic RHD.

Community-Based Programs:

- Combine prevention strategies to reduce the RHD burden.
- More evaluations are needed, especially in low- and middle-income countries, to improve public health strategies and policies.

Solution for Question 6:

Correct Answer: A) Total Anomalous Pulmonary Venous Connection

Explanation:

Onset of symptoms within a few weeks of birth and figure of 8 or snowman appearance on the chest X-ray suggest Total Anomalous Pulmonary Venous Connection.

Total Anomalous Pulmonary Venous Connection:

Pulmonary veins drain into the right atrium directly or indirectly via an abnormal venous connection.

Sites of venous connection:

- Supracardiac- superior vena cava
- Cardiac-coronary sinus, right atrium
- Infracardiac
- Mixed

Clinical presentation:

Cyanosis and congestive cardiac failure

Diagnosis:

- Chest X-ray: Snowman or figure-8 configuration. Cardiomegaly and increased pulmonary vascularity are evident
- Snowman or figure-8 configuration.
- Cardiomegaly and increased pulmonary vascularity are evident
- ECG: Right axis deviation and right ventricular hypertrophy.
- Others: Echocardiography, CT, MRI.
- Snowman or figure-8 configuration.
- Cardiomegaly and increased pulmonary vascularity are evident

Treatment: Surgery

Truncus Arteriosus (Option B):

- A single arterial trunk (truncus arteriosus) originates from the heart, delivering blood to the systemic, pulmonary, and coronary circulations.
- Chest X-ray shows cardiomegaly, increased pulmonary vascular markings, and fullness at the truncal root.

Transposition of the Great Arteries (Option C):

- The main pulmonary artery arises from the left ventricle and the aorta from the right ventricle.
- Chest X-ray shows egg-on-string appearance.

Coarctation of the Aorta (Option D):

- Varying degrees of coarctation occur at any point from the transverse arch to the iliac bifurcation.
- It is an acyanotic heart disease.

- Chest x-ray shows rib notching.
- Other acyanotic congenital heart diseases are atrial septal defect, ventricular septal defect, patent ductus arteriosus, subaortic membrane, valvar aortic stenosis, and supravalvar aortic stenosis.

Solution for Question 7:

Correct Answer: C) Transposition of the Great Arteries

Explanation:

Rapid onset of symptoms immediately after birth and egg-on-string appearance on the chest X-ray suggest transposition of the great arteries.

Transposition of the Great Arteries:

The main pulmonary artery arises from the left ventricle and the aorta arises from the right ventricle.

Pathophysiology:

- Oxygenated pulmonary venous blood recirculates in the lungs and systemic venous blood recirculates in the systemic circulation.

Clinical presentation:

Classified as:

- With intact ventricular septum: Cyanotic at birth Rapid breathing Congestive heart failure secondary to hypoxemia within few days of life Normal S2, single S2. Grade I to II ejection systolic murmur.
- Cyanotic at birth
- Rapid breathing
- Congestive heart failure secondary to hypoxemia within few days of life
- Normal S2, single S2. Grade I to II ejection systolic murmur.
- With VSD: Degree of cyanosis is determined by the severity of VSD. Apical third sound gallop, mid-diastolic rumble. Grade II-IV ejection systolic murmur.
- Degree of cyanosis is determined by the severity of VSD.
- Apical third sound gallop, mid-diastolic rumble. Grade II-IV ejection systolic murmur.

- Cyanotic at birth
- Rapid breathing
- Congestive heart failure secondary to hypoxemia within few days of life
- Normal S2, single S2. Grade I to II ejection systolic murmur.
- Degree of cyanosis is determined by the severity of VSD.
- Apical third sound gallop, mid-diastolic rumble. Grade II-IV ejection systolic murmur.

Investigation:

- Chest X-ray: Mild cardiomegaly + narrow mediastinum = egg-shaped heart (Option C) Plethoric lung fields
- Mild cardiomegaly + narrow mediastinum = egg-shaped heart (Option C)
- Plethoric lung fields
- ECG: Intact septum: Right axis deviation and right ventricular hypertrophy. With VSD: Right axis deviation with biventricular, right ventricular or left ventricular hypertrophy.
- Intact septum: Right axis deviation and right ventricular hypertrophy.
- With VSD: Right axis deviation with biventricular, right ventricular or left ventricular hypertrophy.
- Others: Echocardiography, CT, MRI
- Mild cardiomegaly + narrow mediastinum = egg-shaped heart (Option C)
- Plethoric lung fields
- Intact septum: Right axis deviation and right ventricular hypertrophy.
- With VSD: Right axis deviation with biventricular, right ventricular or left ventricular hypertrophy.

Treatment:

- Infusion of prostaglandin E1: Maintain patency of the ductus arteriosus and improve oxygenation.
- Rashkind balloon atrial septostomy
- Arterial switch (Jatene) procedure: Surgery of choice in neonates with TGA and an intact ventricular septum.

Hypoplastic Left Heart Syndrome (Option A):

- The left side of the heart has different levels of underdevelopment, including stenosis or atresia of the aortic and mitral valves, along with hypoplasia of the left ventricular cavity and ascending aorta.
- Chest x-ray shows an enlarged cardiac silhouette and increased pulmonary vascular markings.

Truncus Arteriosus (Option B):

- A single arterial trunk (truncus arteriosus) originates from the heart, delivering blood to the systemic, pulmonary, and coronary circulations.
- Chest X-ray shows cardiomegaly, increased pulmonary vascular markings, and fullness at the truncal root.

Ebstein's Anomaly (Option D):

- Downward displacement/prolapse of the tricuspid valve and part of the right ventricular volume becomes part of the right atrium (atrialization of the right ventricle).
- Chest x-ray shows a box-shaped heart due to right atrial enlargement and right atrium forming the right heart border.

Solution for Question 8:

Correct Answer: B) Truncus Arteriosus

Explanation:

Truncus arteriosus is a conotruncal heart defect that may be associated with DiGeorge syndrome, which is related to a deletion of a significant portion of chromosome 22q11.

Truncus Arteriosus:

Pathophysiology:

- A single arterial trunk (truncus arteriosus) originates from the heart and supplies the systemic, pulmonary, and coronary circulations.
- VSD is always present.
- Truncus receives blood from both the right and left ventricles
- Associated with DiGeorge syndrome (deletion of a large region of chromosome 22q11 or CATCH 22). (Option B)

CATCH 22:

- 22 signifying the chromosomal abnormality found on the 22nd chromosome. Cardiac abnormality (Truncus arteriosus) Abnormal facies Thymic aplasia or hypoplasia Cleft palate Hypocalcemia/hypoparathyroidism
- Cardiac abnormality (Truncus arteriosus)

- Abnormal facies
- Thymic aplasia or hypoplasia
- Cleft palate
- Hypocalcemia/hypoparathyroidism
- Cardiac abnormality (Truncus arteriosus)
- Abnormal facies
- Thymic aplasia or hypoplasia
- Cleft palate
- Hypocalcemia/hypoparathyroidism

Hypoplastic Left Heart Syndrome (Option A):

- The left side of the heart has different levels of underdevelopment, including stenosis or atresia of the aortic and mitral valves, along with hypoplasia of the left ventricular cavity and ascending aorta.
- It is associated with Jacobsen syndrome.
- The gene implicated is JAM3 on the 11th chromosome

Transposition of the Great Arteries (Option C):

- The main pulmonary artery arises from the left ventricle and the aorta from the right ventricle.
- It is associated with Kabuki syndrome.
- The gene implicated is MLL2 on the 12th chromosome.

Coarctation of the Aorta (Option D):

- Varying degrees of coarctation occur at any point from the transverse arch to the iliac bifurcation.
- It is associated with Kabuki syndrome.
- The gene implicated is MLL2 on the 12th chromosome.

Solution for Question 9:

Correct Answer: D) Coarctation of the Aorta

Explanation:

- Coarctation of the aorta involves narrowing of the aorta leading to decreased blood flow (leg pain during exertion), radio-femoral delay, and murmur in the shoulder region due to collateral formation.
- Chest X-ray shows notching of the inferior border of the ribs due to pressure erosion by enlarged collateral vessels.

Coarctation of the Aorta:

- Constrictions of the aorta of varying degrees at any point from the transverse arch to the iliac bifurcation.
- Juxtaductal coarctation- 98% occurrence

Clinical presentation:

- Differential cyanosis
- Heart failure in severe cases
- Weakness or pain/claudication (or both) in the legs after exercise
- Radio-femoral delay: Disparity in pulsation and BP in the arms and legs
- Systolic or continuous murmurs are heard over the left and right sides of the chest laterally and posteriorly due to collateral blood flow.

Investigation:

- Chest x-ray: The enlarged left subclavian artery produces a prominent shadow in the left superior mediastinum. Notching of the inferior border of the ribs from pressure erosion by enlarged collateral vessels (Common in late childhood after 1st decade of life) Descending aorta has an area of post-stenotic dilation.
- The enlarged left subclavian artery produces a prominent shadow in the left superior mediastinum.
- Notching of the inferior border of the ribs from pressure erosion by enlarged collateral vessels (Common in late childhood after 1st decade of life)
- Descending aorta has an area of post-stenotic dilation.
- ECG, echocardiography, colour Doppler, CT, and MRI.
- The enlarged left subclavian artery produces a prominent shadow in the left superior mediastinum.
- Notching of the inferior border of the ribs from pressure erosion by enlarged collateral vessels (Common in late childhood after 1st decade of life)
- Descending aorta has an area of post-stenotic dilation.

Treatment:

- Infusion of prostaglandin E1 to reopen the ductus and reestablish adequate lower-extremity blood flow.
- Surgical repair

Hypoplastic Left Heart Syndrome (Option A):

- The left side of the heart has different levels of underdevelopment, including stenosis or atresia of the aortic and mitral valves, along with hypoplasia of the left ventricular cavity and ascending aorta.
- Chest x-ray shows an enlarged cardiac silhouette and increased pulmonary vascular markings.

Truncus Arteriosus (Option B):

- A single arterial trunk (truncus arteriosus) originates from the heart, delivering blood to the systemic, pulmonary, and coronary circulations.
- Chest X-ray shows cardiomegaly, increased pulmonary vascular markings, and fullness at the truncal root.

Transposition of the Great Arteries (Option C):

- The main pulmonary artery arises from the left ventricle and the aorta from the right ventricle.
- Chest X-ray shows egg-on-string appearance.

Acyanotic congenital heart defects	Ventricular Septal Defect (Option A)	Atrial Septal Defect (Option B)	Patent Ductus Arteriosus (Option C)
Pathology phy-siology	Abnormal opening/defect in the ventricle septum.	Abnormal opening/defect in the atrial septum.	Failure of closure of ductus areterious (vessel that connects the pulmonary artery to the aorta in a developing fetus).
Chest X-ray	Minimal cardiomegaly and a borderline increase in pulmonary vasculature	Enlarged right ventricle and atrium, depending on the size of the shunt. Increased pulmonary vascularity.	Prominent pulmonary artery with increased pulmonary vascular markings and/or prominent aortic knob.

Solution for Question 10:

Correct Answer: A) No RA-RV communication

Explanation:

- This describes tricuspid valve atresia.
- In this condition, the tricuspid valve is absent or malformed, preventing blood flow from the right atrium (RA) to the right ventricle (RV).
- Blood instead flows through an atrial septal defect (ASD) into the left atrium (LA) and then to the left ventricle (LV).
- The LV becomes hypertrophied due to handling increased blood volume and pressure

Tricuspid Atresia

Definition:

A congenital condition characterized by the absence of the tricuspid valve, leading to a hypoplastic (underdeveloped) right ventricle. (Option D ruled out).

Clinical Presentation:

- Depends on pulmonary blood flow, which may be diminished or increased.
- Diminished pulmonary blood flow: Constitutes 90% of cases. Symptoms and physical signs are similar to Tetralogy of Fallot (TOF).
- Constitutes 90% of cases.
- Symptoms and physical signs are similar to Tetralogy of Fallot (TOF).
- Constitutes 90% of cases.
- Symptoms and physical signs are similar to Tetralogy of Fallot (TOF).

Features Suggesting Tricuspid Atresia:

- Apical Impulse: Left ventricular type of apical impulse.
- Left ventricular type of apical impulse.
- Jugular Venous Pulse: Prominent large 'a' waves.
- Prominent large 'a' waves.
- Liver: Enlarged with presystolic pulsations (a waves).
- Enlarged with presystolic pulsations (a waves).
- Electrocardiogram (ECG): Characterized by left axis deviation. Left ventricular hypertrophy. Mean QRS axis around —45°.
- Characterized by left axis deviation.
- Left ventricular hypertrophy.
- Mean QRS axis around —45°.

- Left ventricular type of apical impulse.
- Prominent large 'a' waves.
- Enlarged with presystolic pulsations (a waves).
- Characterized by left axis deviation.
- Left ventricular hypertrophy.
- Mean QRS axis around —45°.

Immediate Treatment:

- PGE (Prostaglandin E1) Infusion: Medication: Alprostadil infusion. Purpose: Keeps the ductus arteriosus patent to allow blood flow from the aorta into the pulmonary artery.
- Medication: Alprostadil infusion.
- Purpose: Keeps the ductus arteriosus patent to allow blood flow from the aorta into the pulmonary artery.
- Medication: Alprostadil infusion.
- Purpose: Keeps the ductus arteriosus patent to allow blood flow from the aorta into the pulmonary artery.

Definitive Treatment:

Series of Surgical Procedures:

- Modified Blalock-Taussig Shunt: Purpose: Improve pulmonary blood flow.
- Purpose: Improve pulmonary blood flow.
- Glenn Shunt: Procedure: Connects the superior vena cava (SVC) directly to the pulmonary artery. Purpose: Prevents overloading of the left heart chambers and increases pulmonary blood flow.
- Procedure: Connects the superior vena cava (SVC) directly to the pulmonary artery.
- Purpose: Prevents overloading of the left heart chambers and increases pulmonary blood flow.
- Fontan Shunt: Procedure: Connects the inferior vena cava (IVC) directly to the pulmonary artery. Purpose: Diverts blood flow directly to the pulmonary artery, bypassing the right heart.
- Procedure: Connects the inferior vena cava (IVC) directly to the pulmonary artery.
- Purpose: Diverts blood flow directly to the pulmonary artery, bypassing the right heart.
- Purpose: Improve pulmonary blood flow.
- Procedure: Connects the superior vena cava (SVC) directly to the pulmonary artery.
- Purpose: Prevents overloading of the left heart chambers and increases pulmonary blood flow.
- Procedure: Connects the inferior vena cava (IVC) directly to the pulmonary artery.

- Purpose: Diverts blood flow directly to the pulmonary artery, bypassing the right heart.

Obstructed pulmonary outflow with intact septum (Option B):

- This describes pulmonary atresia with an intact ventricular septum.
- The left side of the heart does not typically show left ventricular hypertrophy

Ventricular septal defect and pulmonary stenosis (Option C):

- This describes tetralogy of Fallot.
- The ECG typically shows right axis deviation and right ventricular hypertrophy.

Solution for Question 11:

Correct Answer: B) Atrialization of the Right ventricle

Explanation:

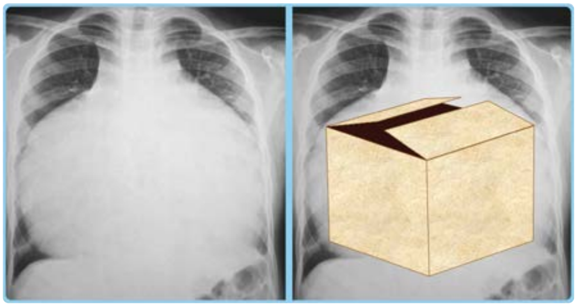
The clinical presentation of cyanosis at birth, pansystolic murmur due to tricuspid regurgitation, mother’s history of Bipolar (Lithium exposed pregnancy), and the characteristic Himalayan P wave on ECG is most consistent with Atrialization of the right ventricle.

- This condition is also known as Ebstein's Anomaly.

Ebstein's Anomaly:

- Associated with Lithium exposure during pregnancy.
- Pathophysiology: Downward displacement/prolapse of the tricuspid valve. Part of the right ventricular volume becomes part of the right atrium (atrialization of the right ventricle).
- Downward displacement/prolapse of the tricuspid valve.
- Part of the right ventricular volume becomes part of the right atrium (atrialization of the right ventricle).
- Consequences: Tricuspid Regurgitation: Prolapsed valve → Blood from the right ventricle regurgitates into the right atrium. RVOT (Right Ventricular Outflow Tract) Obstruction: Prolapsed valve partly occludes blood flow towards the pulmonary artery → Cyanosis.
- Tricuspid Regurgitation: Prolapsed valve → Blood from the right ventricle regurgitates into the right atrium.
- RVOT (Right Ventricular Outflow Tract) Obstruction: Prolapsed valve partly occludes blood flow towards the pulmonary artery → Cyanosis.

- Features: Cyanosis at birth or during the neonatal period. Pansystolic murmur: Due to tricuspid regurgitation. Quadruple Rhythm: Irregular blood flow in the right heart, RVOT obstruction, and tricuspid regurgitation → Multiple added sounds and ejection clicks.
- Cyanosis at birth or during the neonatal period.
- Pansystolic murmur: Due to tricuspid regurgitation.
- Quadruple Rhythm: Irregular blood flow in the right heart, RVOT obstruction, and tricuspid regurgitation → Multiple added sounds and ejection clicks.
- X-Ray Findings: Right atrial enlargement due to regurgitation from the right ventricle and atrialization of the right ventricle. RA forms the right heart border, which displaces laterally and appears on X-ray as a box-shaped heart.
- Right atrial enlargement due to regurgitation from the right ventricle and atrialization of the right ventricle.
- RA forms the right heart border, which displaces laterally and appears on X-ray as a box-shaped heart.
- Downward displacement/prolapse of the tricuspid valve.
- Part of the right ventricular volume becomes part of the right atrium (atrialization of the right ventricle).
- Tricuspid Regurgitation: Prolapsed valve → Blood from the right ventricle regurgitates into the right atrium.
- RVOT (Right Ventricular Outflow Tract) Obstruction: Prolapsed valve partly occludes blood flow towards the pulmonary artery → Cyanosis.
- Cyanosis at birth or during the neonatal period.
- Pansystolic murmur: Due to tricuspid regurgitation.
- Quadruple Rhythm: Irregular blood flow in the right heart, RVOT obstruction, and tricuspid regurgitation → Multiple added sounds and ejection clicks.
- Right atrial enlargement due to regurgitation from the right ventricle and atrialization of the right ventricle.
- RA forms the right heart border, which displaces laterally and appears on X-ray as a box-shaped heart.



- ECG: Himalayan P Waves

Tetralogy of Fallot (Option A):

- Characterized by cyanosis, a systolic ejection murmur, and right ventricular outflow tract obstruction. X-ray typically shows a boot-shaped heart.

Transposition of the Great Arteries (Option C):

- Presents with severe cyanosis, often with a systolic murmur if there is associated outflow obstruction.
- X-ray findings usually show an "egg-on-a-string" appearance.

Ventricular Septal Defect (Option D):

- Presents with a pansystolic murmur and may cause pulmonary overcirculation, but the X-ray findings would generally show an enlarged heart with increased pulmonary markings.

Previous Year Questions

1. In newborns, what is the primary cause of ventriculomegaly?

A. Arnold-Chiari malformation

B. Dandy-Walker syndrome

C. Arachnoid Villi malformation

D. Aqueductal stenosis

2. What is the most common defect in a ventricular septal defect?

A. Muscular defect

B. Membranous defect

C. Non-development of the perimembranous part of the septum

D. Incomplete fusion of the septum

3. A 2-year-old child presents with stable vitals and an ECG showing paroxysmal supraventricular tachycardia (PSVT). Which of the following is the appropriate treatment?

A. Cardioversion

B. Adenosine 0.2 mg/kg

C. Adenosine 0.1 mg/kg

D. IV Adrenaline

4. What could be the possible diagnosis for a newborn exhibiting weak lower limb pulses and strong upper limb pulses?

A. TGA

B. COA

C. TOF

D. Ebstein anomaly

5. Continuous murmur is heard in?

A. Patent ductus arteriosus

B. VSD

C. ASD

D. Tetralogy of Fallot

6. Which of the following congenital heart diseases presents with split-second heart sounds? 1. Atrial Septal Defect (ASD) 2. Ventricular Septal Defect (VSD) 3. Patent Ductus Arteriosus (PDA) 4. Tetralogy of Fallot (TOF)

(or)

Which of the subsequent congenital heart diseases exhibit split-second heart sounds? 1. Atrial Septal Defect (ASD) 2. Ventricular Septal Defect (VSD) 3. Patent Ductus Arteriosus (PDA) 4. Tetralogy of Fallot (TOF)

A. Only 1,2,3 is correct

B. Only 2,3,4 is correct

C. Only 1,2,4 is correct

D. All are correct

7. Which congenital heart disease exhibits equivalent saturation levels across all heart chambers among the options listed below?

A. Tetralogy of Fallot

B. Total anomalous pulmonary venous circulation

C. Tricuspid atresia

D. Transposition of great arteries

8. Continuous murmur is heard in:

A. Patent ductus arteriosus

B. Ventricular septal defect

C. Atrial septal defect

D. Tetralogy of fallot

9. What is the underlying condition in a child who has cold feet and consistently wears socks even in hot weather, with lower limb pulses that are weaker compared to the radial pulse, and a noticeable delay in the radio femoral pulse?

A. There is hypertrophy of ductus arteriosus.

B. Coarctation distal to the origin of the left subclavian artery

C. Supply is from the posterior to the anterior thoracic artery.

D. Collaterals develop to supply the upper limb.

10. What components of Nada’s are minor criteria for congenital heart disease? 1. Diastolic murmur 2. Systolic murmur grade-3 3. Abnormal 2nd heart sound 4 Abnormal BP

A. 1, 2 and 3

B. 3 and 4

C. 2,3 and 4

D. 1, 2 , 3 and 4

11. What is the vessel responsible for returning deoxygenated blood back to the placenta in fetal circulation?

A. Umbilical Vein

B. Umbilical artery

C. Descending aorta

D. Pulmonary Artery

12. Which of the following is the most common association with Tetralogy of Fallot?

A. Klinefelters syndrome

B. Edwards syndrome

C. Downs syndrome

D. Turners’ syndrome

13. A child came to OPD with ■ Fever for 6 days with strawberry tongue, Conjunctival congestion, and Peeling of skin. What should be the ideal treatment?

A. IVIG

B. Aspirin

C. Plenty of fluids

D. Cephalosporins

---

14. What is the most likely diagnosis for a 6-year-old child who was taken to her pediatrician for routine well-child care, has a normal physical examination except for a systolic murmur and a widely split-second heart sound that remains constant during respiration, and has no family history of heart disease?

A. Total anomalous pulmonary venous connection

B. Ventricular septal defect

C. Atrial septal defect

D. Pulmonic stenosis

---

15. What is the most common congenital defect observed in individuals with Down syndrome?

A. PDA

B. ASD

C. Atrioventricular septal defect

D. VSD

---

16. A professor explains the pathophysiology of a “tet spell” and that it is due to a decrease in systemic vascular resistance that leads to more shunting of blood from right to left, causing an overload on the left side of the heart and decreasing the amount of blood reaching the lungs. Which of the following options determines the severity of the cyanotic spells?

A. Size of the ventricular septal defect (VSD)

B. Overriding of the aorta

C. Degree of RVH

D. Degree of pulmonary stenosis

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17. Please match the appropriate answers related to cardiac abnormalities and their corresponding syndromes. 1) Noonan syndrome (A) PDA 2) Downs syndrome (B) Bicuspid aortic valve 3) Turner syndrome (C) Endocardial cushion defects 4) Congenital Rubella Syndrome (D) Pulmonary stenosis

1) Noonan syndrome	(A) PDA
2) Downs syndrome	(B) Bicuspid aortic valve
3) Turner syndrome	(C) Endocardial cushion defects
4) Congenital Rubella Syndrome	(D) Pulmonary stenosis

A. 1-(A), 2-(B), 3-(C), 4-(D)

B. 1-(D), 2-(C), 3-(B), 4-(A)

C. 1-(D), 2-(B), 3-(C), 4-(A)

D. 1-(A), 2-(C), 3-(B), 4-(D)

---

18. Which of the following features can be alleviated by performing surgical correction of VSD in an infant?

A. Respiratory alkalosis

B. Failure to thrive

C. Arrhythmia

D. Heart block

19. Which of the following cyanotic heart diseases causes increased pulmonary blood flow? 1. Ebstein anomaly 2. Tetralogy of Fallot 3. Transposition of the great arteries (TGA) 4. Total anomalous pulmonary venous communication (TAPVC)

A. 1,4

B. 1,2

C. 3,4

D. 2,4

20. A 24-year-old female comes in for a prenatal examination and is presently taking lithium for her bipolar disorder. It is necessary to discontinue this medication to avoid the potential occurrence of which of the subsequent fetal conditions?

A. Ebstein anomaly

B. VACTERL defects

C. Coarctation of aorta

D. Supravalvular aortic stenosis

21. Which of the following is not a characteristic of tetralogy of Fallot?

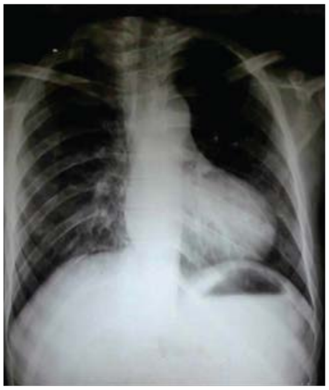
A. Ventricular septal defect

B. Right ventricular hypertrophy

C. Atrial septal defect

D. Pulmonic stenosis

22. Which of the following choices best characterizes the chest X-ray picture?



A. Egg on string appearance

B. Coeur en sabot

C. Scimitar sign

D. Snowman sign

23. What is the primary factor for the closure of the ductus arteriosus after birth?

A. Increase in partial pressure of oxygen (paO2)

B. Increase in systemic vascular resistance

C. Increase in circulating prostaglandin levels

D. Decrease in pulmonary venous resistance

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	3
Question 4	2
Question 5	1
Question 6	1
Question 7	2
Question 8	1
Question 9	2
Question 10	2
Question 11	2
Question 12	3
Question 13	1
Question 14	3
Question 15	3
Question 16	4
Question 17	2
Question 18	2
Question 19	3
Question 20	1
Question 21	3
Question 22	2

Question 23	1

Solution for Question 1:

Correct Option D: Aqueductal stenosis

- Ventriculomegaly refers to the dilation or enlargement of the cerebral ventricles, which are fluid-filled spaces within the brain. It is a common finding in newborns and can be caused by various underlying conditions. Aqueductal stenosis is the most common cause of ventriculomegaly in newborns.
- Aqueductal stenosis is a condition characterized by a narrowing or obstruction of the cerebral aqueduct, a narrow passage that connects the third and fourth ventricles in the brain. When this passage is narrowed or blocked, the flow of cerebrospinal fluid (CSF) is obstructed, leading to the accumulation of fluid in the ventricles and resulting in ventriculomegaly.

Incorrect Choices

Option A: Arnold-Chiari malformation- Arnold-Chiari malformation refers to a group of structural defects in the brain and spinal cord. It involves a downward displacement of the cerebellar tonsils through the opening at the base of the skull. While it can be associated with hydrocephalus, it is not the most common cause of ventriculomegaly in newborns.

Option B: Dandy-Walker syndrome- Dandy-Walker syndrome is a congenital brain malformation characterized by the enlargement of the fourth ventricle and the presence of a cystic structure in the posterior fossa. While it can cause hydrocephalus, it is not the most common cause of ventriculomegaly in newborns.

Option C: Arachnoid Villi malformation- Arachnoid villi are structures that help drain cerebrospinal fluid from the brain into the bloodstream. Malformation or dysfunction of these structures can lead to impaired CSF drainage and hydrocephalus. However, it is not the most common cause of ventriculomegaly in newborns.

Solution for Question 2:

Correct Answer: B) Membranous defect

Explanation

The most common defect in ventricular septal defect (VSD) is membranous defect.

Pathophysiology of VSD:

- The interventricular septum has two parts: membranous and muscular.
- VSD occurs due to incomplete development or fusion of septal components during embryonic cardiac formation.
- A unified classification system is used to describe VSDs.

Types of VSD:

- Infundibular (Outlet) VSD: Location: Below semilunar valves (aortic and pulmonary), within the outlet septum of the right ventricle.
- Location: Below semilunar valves (aortic and pulmonary), within the outlet septum of the right ventricle.
- Perimembranous VSD: Location: Membranous septum, below crista supraventricularis. Characteristics: Most common (80% of cases), may involve muscular septum, potential for spontaneous closure.
- Location: Membranous septum, below crista supraventricularis.
- Characteristics: Most common (80% of cases), may involve muscular septum, potential for spontaneous closure.
- Inlet (Atrioventricular Canal) VSD: Location: Below the inlet valves (tricuspid and mitral), in the inlet portion of the right ventricle septum. Characteristics: Represents 8% of all VSDs, more common in patients with Trisomy 21.
- Location: Below the inlet valves (tricuspid and mitral), in the inlet portion of the right ventricle septum.
- Characteristics: Represents 8% of all VSDs, more common in patients with Trisomy 21.
- Muscular (Trabecular) VSD: Location: Muscular septum (apical, central, outlet regions). Characteristics: Can present as multiple defects ("Swiss cheese"), more common in infants, the tendency to close spontaneously with age.
- Location: Muscular septum (apical, central, outlet regions).
- Characteristics: Can present as multiple defects ("Swiss cheese"), more common in infants, the tendency to close spontaneously with age.
- Location: Below semilunar valves (aortic and pulmonary), within the outlet septum of the right ventricle.
- Location: Membranous septum, below crista supraventricularis.

- Characteristics: Most common (80% of cases), may involve muscular septum, potential for spontaneous closure.
- Location: Below the inlet valves (tricuspid and mitral), in the inlet portion of the right ventricle septum.
- Characteristics: Represents 8% of all VSDs, more common in patients with Trisomy 21.
- Location: Muscular septum (apical, central, outlet regions).
- Characteristics: Can present as multiple defects ("Swiss cheese"), more common in infants, the tendency to close spontaneously with age.

Gerbode Defects:

- A distinct defect with a direct connection between the left ventricle and right atrium.
- Type 1: True Gerbode, congenital deficiency in the membranous septum.
- Type 2: Subvalvular type, ventriculo-atrial shunting due to central VSD and tricuspid valve defect.
- Type 3: Combination of Type 1 and Type 2 characteristics.

Solution for Question 3:

Correct Option C - Adenosine 0.1 mg/kg:

- This is the correct initial treatment for PSVT in a child with stable vitals. Adenosine is a first-line medication for terminating PSVT by temporarily blocking the AV node, often restoring normal sinus rhythm.

Incorrect Options:

Option A - Cardioversion:

- This is typically reserved for patients with PSVT who are hemodynamically unstable or in cases where pharmacologic treatment is ineffective. Given that the child has stable vitals, cardioversion is not the first-line treatment.

Option B - Adenosine 0.2 mg/kg:

- This is the dose used if the initial dose of adenosine (0.1 mg/kg) fails to convert the PSVT. It is not the starting dose, so it is not the appropriate initial treatment.

Option D - IV Adrenaline:

- Adrenaline is not indicated for treating PSVT. It is used in resuscitation protocols, particularly in cases of cardiac arrest or anaphylaxis, not for stable PSVT.

Solution for Question 4:

Correct Option B - COA:

- Coarctation of the Aorta is characterized by weak femoral pulses (lower limb pulses) and strong upper limb pulses.

Incorrect Options:

- Options A, C & D: TGA,TOF and Ebstein anomaly is not characterized by weak lower limb pulses and strong upper limb pulses

Solution for Question 5:

Correct Option A - Patent ductus arteriosus (PDA):

- A continuous (machinery-like) murmur is often heard in PDA. This is due to the persistent connection between the aorta and the pulmonary artery, leading to continuous blood flow during both systole and diastole.

Incorrect Options:

Option B - VSD (Ventricular Septal Defect): VSD typically presents with a holosystolic murmur, not a continuous murmur.

Option C - ASD (Atrial Septal Defect): ASD is associated with a fixed split second heart sound (S2) and a systolic ejection murmur, not a continuous murmur.

Option D - Tetralogy of Fallot: Tetralogy of Fallot is characterized by a harsh systolic ejection murmur, not a continuous murmur.

Solution for Question 6:

Correct Option A: Only 1,2,3 is correct

- Congenital heart diseases that can present with split-second heart sounds include:
- Atrial Septal Defect (ASD): An ASD is a hole in the wall (septum) that separates the two upper chambers of the heart (atria). It can lead to increased blood flow from the left atrium to the right atrium, resulting in more evident splitting of the second heart sound (S2) during auscultation.
- Ventricular Septal Defect (VSD): A VSD is a hole in the septum between the two lower chambers of the heart (ventricles). Like ASD, it can cause increased blood flow from the left ventricle to the right ventricle, leading to accentuated splitting of S2.
- Patent Ductus Arteriosus (PDA): PDA is a condition where a blood vessel called the ductus arteriosus, which connects the pulmonary artery to the aorta during fetal development, remains open after birth. This can result in abnormal blood flow and increased flow across the pulmonary valve, leading to a more pronounced splitting of S2.

Congenital heart diseases that can present with split-second heart sounds include:

Atrial Septal Defect (ASD): An ASD is a hole in the wall (septum) that separates the two upper chambers of the heart (atria). It can lead to increased blood flow from the left atrium to the right atrium, resulting in more evident splitting of the second heart sound (S2) during auscultation.

Ventricular Septal Defect (VSD): A VSD is a hole in the septum between the two lower chambers of the heart (ventricles). Like ASD, it can cause increased blood flow from the left ventricle to the right ventricle, leading to accentuated splitting of S2.

Patent Ductus Arteriosus (PDA): PDA is a condition where a blood vessel called the ductus arteriosus, which connects the pulmonary artery to the aorta during fetal development, remains open after birth. This can result in abnormal blood flow and increased flow across the pulmonary valve, leading to a more pronounced splitting of S2.

Incorrect options:

The congenital heart disease that does not typically present with split-second heart sounds is:

Tetralogy of Fallot (TOF)

Tetralogy of Fallot is a complex congenital heart defect characterized by four specific abnormalities in the heart's structure, including:

Ventricular Septal Defect (VSD): A hole in the septum between the two lower chambers (ventricles) of the heart.

Pulmonary Stenosis: Narrowing of the pulmonary valve or the blood vessel that carries blood from the heart to the lungs.

Right Ventricular Hypertrophy: Thickening of the right ventricular wall due to increased workload.

Overriding Aorta: The aorta, the main artery that carries oxygen-rich blood from the heart to the body, is positioned over both ventricles instead of arising only from the left ventricle.

Solution for Question 7:

Correct Choice: B

- Total anomalous pulmonary venous return is cyanotic heart disease where the pulmonary vein fails to enter into the left atrium and drains abnormally into the systemic venous system resulting in mixed blood into the systemic circulation.
- This results in failure of closure of foramen result into blood flow from right atrium to left follow by blood flow to left ventricle than to systemic circulation.
- This result is equal blood saturation in all four chambers of heart and systemic circulation.

Incorrect Choices:

Option A. Tetralogy of fallot: In Tetralogy of fallot, there is ventricular septal defect, overriding aorta, pulmonary stenosis and right ventricular hypertrophy.

- Thus, the failure in closure of foramen ovale, right to left resulting in mixed blood in left heart and unoxygenated blood in right heart.
- Hence all four chamber don't have same Oxygen concentration.

Option C: Tricuspid atresia: In Tricuspid atresia, there is decreased blood flow from right atrium to ventricule causing right to left shunt and pulmonary oligemia.

Option D: Transposition of great arteries.

- In transposition of great arteries, right ventricle ends at right aorta which carries deoxygenated blood from right ventricle to the tissue while left ventricle drains into pulmonary artery.

- Hence two separate system are created. Deoxygenated blood is circulated in systemic circulation and oxygenated in pulmonary circulation.
- D-looped transposition of the great arteries (TGA). A, Diagram of d-TGA, with the main pulmonary artery (MPA) arising from the left ventricle (LV) and the aorta (Ao) arising from the right ventricle (RV).

Solution for Question 8:

Correct option A - Patent ductus arteriosus:

- A continuous murmur refers to a murmur that is heard throughout systole and diastole without interruption.
- Continuous murmur is heard in patent ductus arteriosus
- Patent ductus arteriosus (PDA) is a congenital heart defect characterized by the persistence of a normal fetal connection between the pulmonary artery and the aorta, known as the ductus arteriosus.

Incorrect options:

Option B: Ventricular septal defect (VSD): VSD produces a holosystolic murmur.

Option C: Atrial septal defect (ASD): ASD typically causes a systolic ejection murmur.

Option D: Tetralogy of Fallot: Tetralogy of Fallot is associated with a systolic ejection murmur, often described as a harsh or crescendo-decrescendo murmur.

Solution for Question 9:

Correct Option B - Coarctation distal to the origin of the left subclavian artery:

- The patient’s presentation with lower limb pulses that are weaker compared to the radial pulse, and a noticeable delay in the radio femoral pulse is suggestive of coarctation of the aorta.

- Coarctation refers to a narrowing or constriction in the aorta.
- The narrowing typically occurs distal to the origin of the left subclavian artery, which is why the lower limb pulses are diminished compared to the radial pulse.

Incorrect Options:

Options A, C, D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 10:

Correct option B - 3 and 4:

Nada's criteria:

- It is used to assess the presence of congenital heart disease.
- 1 major or 2 minor criteria indicates possibility of congenital heart disease.

Major criteria	Minor criteria
Systolic murmur ≥ Grade 3 Any diastolic murmur Central Cyanosis Congestive heart failure	Systolic murmur ≤ Grade 2 Abnormal 2nd heart sound Abnormal BP Abnormal ECG Abnormal Chest X-ray

Incorrect options - Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 11:

Correct Option B - Umbilical artery:

- In fetal circulation, deoxygenated blood is carried back to the placenta through the umbilical artery.

Incorrect Options - Options A, C, and D are incorrect.

Solution for Question 12:

Correct Option C - Down syndrome:

- About 15-20% of cases of TOF are associated with syndromes, with DiGeorge syndrome being the most common.
- Other syndromes associated with TOF include Alagille syndrome and Down syndrome.

Incorrect options:

- Options A, B and D are not associated with Tetralogy of Fallot.

Solution for Question 13:

Correct Option A - IVIG:

- The patient’s presentation with fever for 6 days with strawberry tongue, conjunctival congestion, and peeling of skin is suggestive of Kawasaki disease.
- The treatment of choice of Kawasaki disease is IV immunoglobulins

Incorrect Options - B, C, and D are incorrect.

Option B - Aspirin: Aspirin is given in Kawasaki disease but IVIG remains the mainstay treatment.

Solution for Question 14:

Correct Option C- Atrial septal defect:

- The patient’s presentation with systolic murmur and a widely split-second heart sound that remains constant during respiration is suggestive of atrial septal defect.

Incorrect Options: A widely split-second heart sound that remains constant during respiration is not seen in options A, B, and D.

Solution for Question 15:

Correct Option C - Atrioventricular septal defect:

- The atrioventricular septal defect is a congenital cardiac malformation that is characterized by a variable degree of the atrial and ventricular septal defect along with a common or partially separate atrioventricular orifice..
- AVSD is the most common congenital heart defect associated with Down's syndrome.

Incorrect options - A, B, and D are not the most common congenital defect observed in individuals with Down syndrome.

Solution for Question 16:

Correct Option D - Degree of pulmonary stenosis:

- Tet spells, also called hypercyanotic spells, are characteristic features of Tetralogy of Fallot (TOF).
- During a tet spell, there is an increased right-to-left shunting of blood, leading to decreased oxygenation and cyanosis.
- The severity of these spells is influenced by the degree of pulmonary stenosis, which refers to the narrowing of the pulmonary valve or the pulmonary artery.
- Increased pulmonary stenosis results in more restricted blood flow to the lungs, exacerbating the right-to-left shunting and worsening the severity of the cyanotic spells.

Incorrect options: Options A, B, and C do not determine the severity of the cyanotic spells. Refer to the explanation of the correct answer.

Solution for Question 17:

Correct option B- 1-(D), 2-(C), 3-(B), 4-(A):

Congenital syndromes	Cardiac abnormalities
Noonan syndrome	Pulmonary stenosis
Down syndrome	Endocardial cushion defects
Turner syndrome	Bicuspid aortic valve
Congenital rubella syndrome	PDA

Solution for Question 18:

Correct Option B- Failure to Thrive:

- Failure to thrive refers to inadequate weight gain or growth in infants.
- Infants with a large VSD can develop heart failure and have feeding problems that lead to poor weight gain.
- Surgical correction of VSD aims to repair the structural defect, which reduces the strain on the heart, improves cardiac function, and restores normal blood flow patterns.
- It can enhance the infant's ability to feed, leading to proper weight gain and relieving the feature of failure to thrive.

Incorrect Options- Options A, C, and D are not associated with VSD.

Solution for Question 19:

Correct Option C - 3, 4:

3. Transposition of the great arteries (TGA):

- TGA is a congenital heart defect in which the positions of the pulmonary artery and the aorta are switched.
- This means systemic and pulmonary circulation is completely separate, resulting in deoxygenated blood circulated to the body and oxygenated blood circulated back to the lungs.
- Unless arterial oxygen saturation is only mildly decreased and the atrial communication is adequate, a PGE1 infusion may help by opening and maintaining patency of the ductus arteriosus; this infusion increases pulmonary blood flow, which may promote left-to-right atrial shunting, leading to improved systemic oxygenation.

4. Total anomalous pulmonary venous communication (TAPVC)

- TAPVC is a cyanotic congenital heart disease where all of the pulmonary veins drain directly or indirectly into the right atrium.
- This causes a mixing of oxygenated and deoxygenated blood and leads to increased pulmonary blood flow.

Incorrect Options - Options A, B, and D are incorrect.

1. Ebstein anomaly:

- Ebstein anomaly is a rare congenital heart defect characterized by the abnormal placement of the tricuspid valve, which causes blood to flow backward into the right atrium.
- This condition leads to a decrease in pulmonary blood flow rather than an increase.

2. Tetralogy of Fallot:

- Tetralogy of Fallot (TOF) is a congenital heart defect characterized by four specific abnormalities: a ventricular septal defect (VSD), pulmonary stenosis, an overriding aorta, and right ventricular hypertrophy.
- In TOF, there is a right-to-left shunt of blood due to the VSD and overriding aorta causing deoxygenated blood from the right ventricle to mix with oxygenated blood from the left, leading to cyanosis.
- However, TOF is not associated with increased pulmonary blood flow.

Solution for Question 20:

Correct option A- Ebstein Anomaly:

- Lithium treatment during pregnancy is associated with the Ebstein anomaly.
- Ebstein anomaly is a rare congenital heart disease that involves the apical displacement of the tricuspid valve with adherence of the septal and posterior leaflets to the myocardium and atrialization of the inlet portion of the right ventricle.

Incorrect Options - B, C, and D are not associated with lithium treatment during pregnancy.

Solution for Question 21:

Correct Option C- Atrial septal defect:

Four components of TOF:

- Large non-restrictive ventricular septal defect (VSD)
- Pulmonary infundibular stenosis
- Overriding of aorta
- Right ventricular hypertrophy

Incorrect options - Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 22:

Correct option B - Coeur en sabot:

- Coeur en sabot, or "boot-shaped heart," refers to a characteristic appearance of the heart on a chest X-ray.
- It appears elongated and narrowed at the base, resembling the shape of a boot.

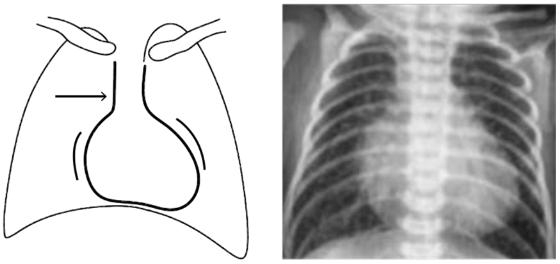
- This is typically associated with congenital heart defects such as Tetralogy of Fallot.

Incorrect Options:

Option A - Egg on string appearance:

- Egg on a string appearance is seen in a condition called Transposition of the Great Arteries.

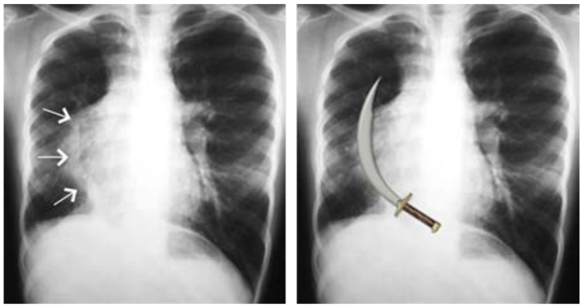
Egg on String Appearance



Also K/A - Egg on side appearance

Option C: Scimitar sign:

- The scimitar sign is seen in a condition called Scimitar syndrome, where a pulmonary vein from the right lung drains into the inferior vena cava instead of the left atrium.
- This creates a characteristic curved appearance on the X-ray resembling a scimitar, a type of curved sword.



Option D - Snowman sign:

- The snowman sign refers to the appearance of the pulmonary artery and its branches on a chest X-ray in certain conditions, such as pulmonary atresia with ventricular septal defect.
- It appears as a "snowman" shape with the main pulmonary artery and its branches resembling the head and body of a snowman.

Solution for Question 23:

Correct option A- Increase in partial pressure of oxygen (paO2).

- The main factor for ductal closure postnatally is an increase in the partial pressure of oxygen in the blood.
- When a newborn transitions from the intrauterine environment to the outside world, there is a sudden increase in the oxygen concentration due to the initiation of lung ventilation.
- This increased oxygen tension acts as a stimulus for the closure of the ductus arteriosus.

Incorrect Options: Options B, C, and D are not the factors for the closure of the ductus arteriosus after birth.

Pediatric Hematology

1. In newborns, phytomenadione is routinely administered to prevent hemorrhagic disease. However, its administration is contraindicated in the presence of a deficiency in which specific condition?

A. Glucose-6-phosphate dehydrogenase (G6PD) deficiency

B. Alpha-1 antitrypsin deficiency

C. Protein C deficiency

D. Cystic fibrosis

2. Which of the following statements regarding Burkitt's lymphoma is incorrect?

A. It is commonly associated with Epstein-Barr virus (EBV) infection, especially in endemic cases

B. The disease predominantly affects children and young adults

C. The standard treatment typically involves low-intensity chemotherapy due to its slow-growing nature

D. The tumor cells typically show translocation t(8;14) involving the MYC gene

3. A 17-year-old male is diagnosed with classical Hodgkin's lymphoma, and staging investigations reveal stage IIA disease, with involvement of lymph nodes only above the diaphragm but no B symptoms. Which of the following treatment regimens is considered the standard approach for initial therapy in this stage of the disease?

A. Radiotherapy alone

B. Single-agent therapy with vinblastine

C. Chemotherapy with ABVD regimen

D. Chemotherapy with vinblastine, etoposide and prednisolone

4. Based on the Ann Arbor staging system, which stage is described by the following features: involvement of two or more lymph node regions on the same side of the diaphragm, without extranodal involvement?

A. IAE

B. X

C. IIIBE

D. IIB

5. A 17-year-old patient undergoing treatment for acute promyelocytic leukemia develops severe adverse reactions, including high fever, difficulty breathing, and peripheral edema. What is the most appropriate treatment to manage this patient?

A. High-dose chemotherapy

B. Corticosteroids

C. Blood transfusion

D. ACE inhibitors

6. In a patient with more than 20% blast cells observed in a smear, which of the following chromosomal abnormalities is NOT useful in differentiating Acute Myeloid Leukemia (AML) from Acute Lymphoblastic Leukemia (ALL)?

A. t(8;21)

B. t(15;17)

C. t(12;21

D. t(9;22)

---

7. Which variant of leukemia is most commonly associated with the development of granulocytic sarcoma?

A. AML with t(8;21)

B. ALL with t(12;21)

C. AML with t(15;17)

D. AML with t(9;22)

---

8. A 2.5-year-old child has been diagnosed with severe Acute Lymphoblastic Leukemia (ALL). Given the high risk of central nervous system (CNS) involvement with this condition, the oncologist is planning appropriate prophylactic measures. What is the most appropriate step for prophylaxis to prevent central nervous system (CNS) involvement in this child?

A. Oral methotrexate

B. Administration of intrathecal methotrexate

C. Use of oral corticosteroids

D. Multiagent chemotherapy

---

9. A 7-year-old child with Down syndrome presents with symptoms of fatigue, pallor, and easy bruising. Laboratory tests reveal anaemia, thrombocytopenia, and an elevated white blood cell count. A bone marrow biopsy shows the presence of lymphoblasts. Cytogenetic analysis confirms the presence of chromosomal abnormalities. Which type of leukaemia is most commonly associated with paediatric patients who have Down syndrome?

A. AML-M7

B. ALL

C. CML

D. CLL

---

10. A 5-year-old child is diagnosed with acute lymphoblastic leukaemia (ALL). The initial diagnostic workup includes a white blood cell count of 15,000/µL, and cytogenetic analysis reveals hyperdiploidy. The child begins treatment with a standard chemotherapy regimen and responds well. Which of the following factors is most indicative of a favourable prognosis for this child?

A. The child’s age at diagnosis >1 and < 10 years of age

B. The elevated initial white blood cell count (>50000/µL)

C. The presence of the Philadelphia chromosome - t(9;22)

D. Presence of t(4;11) translocation

---

11. Which of the following genetic syndromes is not associated with the development of leukemia in a pediatric patient?

A. Down syndrome

B. Fanconi anemia

C. Neurofibromatosis type 1

D. Hereditary spherocytosis

---

12. Which of the following statements are true regarding Heamophilia A? a. It is caused by a deficiency in clotting factor VIII. b. It is inherited in an autosomal dominant pattern. c. It shows increased aPTT and normal PT. d. Patients with severe Hemophilia A often experience spontaneous bleeding into joints and

muscles. Select the correct answer from the codes given below.

A. a,b,c

B. a,b,d

C. a,c,d

D. b,c,d

---

13. A 3-month-old infant is being evaluated for anemia. The pediatrician orders a hemoglobin electrophoresis test, and the results reveal that the majority of the hemoglobin present is fetal hemoglobin (HbF). What is the predominant hemoglobin chain composition of fetal hemoglobin found in this infant?

A. α■β■

B. α■δ■

C. α■γ■

D. α■ε■

---

14. A child is diagnosed with immune thrombocytopenic purpura (ITP). The most recent platelet count is 32,000/µL, and the child does not exhibit any signs of active bleeding. What is the appropriate next step in management for this child?

A. Prednisolone

B. Platelet transfusion

C. Splenectomy

D. Wait and watch

---

15. A newborn presents with an umbilical cord stump that has not separated from the body, even after 1 month. The pediatrician suspects an underlying coagulation

disorder. Which of the following tests would be most appropriate to assess the suspected deficiency in this newborn?

A. Hageman factor (Factor XII) assay

B. Prothrombin Time (PT)

C. Activated Partial Thromboplastin Time (aPTT)

D. Fibrinogen level

---

16. Match the following bleeding and clotting disorders with their corresponding pathology. 1 Glanzmann Thrombasthenia a Defect in (GPIIb/IIIa) 2 Bernard-Soulier Syndrome b Defect in GPIb-IX-V 3 Von Willebrand Disease c Defect in (vWF) 4 Haemophilia A d Defect in Factor 8

1	Glanzmann Thrombasthenia	a	Defect in (GPIIb/IIIa)
2	Bernard-Soulier Syndrome	b	Defect in GPIb-IX-V
3	Von Willebrand Disease	c	Defect in (vWF)
4	Haemophilia A	d	Defect in Factor 8

A. 1-b, 2-a, 3-c, 4-d

B. 1-a, 2-b, 3-c, 4-d

C. 1-a, 2-b, 3-d, 4-c

D. 1-b, 2-a, 3-d, 4-c

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17. A 15-year-old male presents with a history of chronic fatigue, frequent nosebleeds, and recurrent infections over the past few months. His laboratory results show pancytopenia and peripheral blood smear reveals markedly reduced cellularity without evidence of blasts or dysplasia. A bone marrow biopsy demonstrates severe hypocellularity with less than 10% hematopoietic cells and prominent fatty infiltration. Based on these findings, which of the following is the

most likely diagnosis?

A. Acute myeloid leukemia

B. Myelodysplastic syndrome

C. Aplastic anemia

D. Fanconi anemia

18. A 6-year-old child presents with progressive bone marrow failure, short stature, and multiple congenital abnormalities. The child also has distinct skin lesions as shown below. Radiological imaging reveals the following features. Which of the following is the most appropriate management strategy for this condition?

A. High-dose corticosteroids

B. Hematopoietic stem cell transplantation

C. Blood transfusion

D. Targeted gene therapy

19. A 7-year-old boy presents with fatigue, pallor, and recurrent episodes of abdominal pain. His medical history includes multiple hospitalizations for anemia requiring transfusions. Physical examination reveals jaundice and a palpable liver, but no spleen is felt. Laboratory tests show a hemoglobin level of 7 g/dL, elevated reticulocyte count, and indirect hyperbilirubinemia. What is the most likely underlying condition?

A. Beta-thalassemia major

B. Hereditary spherocytosis

C. Sickle cell disease

D. Autoimmune hemolytic anemia

20. A 12-year-old boy presents with complaints of feeling excessively tired. He gives a history of repeated blood transfusions. A skull radiograph revealed the following findings. Which of the following is the most probable cause of this radiographic finding?

X-Ray skull showing hair-on-end appearance

A. Increased osteoclastic activity

B. Increased marrow hyperplasia

C. Decreased bone remodeling

D. Increased osteoblastic activity

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21. A 6-year-old child is diagnosed with folate deficiency based on laboratory tests but shows no signs of Vitamin B12 deficiency. The pediatrician plans to initiate treatment to address the folate deficiency. Which of the following treatment options should be avoided in this child?

A. Oral folic acid supplementation

B. Parenteral folic acid supplementation

C. High-dose folic acid supplementation

D. Dietary modification to include folate-rich food

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22. A 4-year-old child has been diagnosed with iron deficiency anemia and has just started oral iron supplementation. The pediatrician advises the parents that certain changes and observations can be expected during the initial days of treatment. On the 3rd day after initiating oral iron therapy, which of the following is most likely to be observed in the child on evaluation.

A. Immediate normalization of hemoglobin levels

B. Increase in the reticulocyte count

C. Decreased irritability, improved appetite

D. Replenishment of stores

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23. An 8-year-old child is brought to the pediatric clinic with complaints of fatigue and pallor. The pediatrician conducts a complete blood count (CBC) to evaluate the

child’s hemoglobin levels. According to WHO criteria, anemia is classified based on specific hemoglobin thresholds for different age groups. What hemoglobin level would classify this 8-year-old child as anemic according to WHO criteria?

A. <10 g/dl

B. <11g/dl

C. <12 g/dl

D. <13 g/dl

24. At 5 weeks of intrauterine life, which of the following structures is the primary site of hematopoiesis in the developing fetus?

A. Liver

B. Bone Marrow

C. Spleen

D. Yolk Sac

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	3
Question 4	4
Question 5	2
Question 6	4
Question 7	1
Question 8	2
Question 9	2
Question 10	1
Question 11	4
Question 12	3
Question 13	3
Question 14	4
Question 15	1
Question 16	2
Question 17	3
Question 18	2
Question 19	3
Question 20	2
Question 21	3
Question 22	2

Question 23	3
Question 24	4

Solution for Question 1:

Correct Answer: C) Protein C deficiency

Explanation:

Phytomenadione (Vitamin K1) is routinely given to newborns to prevent hemorrhagic disease of the newborn (HDN), which occurs due to Vitamin K deficiency and can cause severe bleeding. However, its administration is contraindicated in newborns with Protein C deficiency, a rare inherited disorder that increases the risk of blood clots.

In these infants, Vitamin K can exacerbate clotting, leading to (Purpura fulminans) a life-threatening condition characterized by widespread blood clots and tissue necrosis. But in most cases , as Protein C deficiency would go undiagnosed at birth, Vitamin K birth dose is almost always given.

Glucose-6-phosphate dehydrogenase (G6PD) deficiency (Option A): This genetic disorder affects red blood cells and can lead to hemolytic anemia when exposed to triggers like certain medications. Vitamin K does not cause oxidative stress or hemolysis in G6PD deficiency.

Alpha-1 antitrypsin deficiency (Option B): This genetic disorder can lead to lung and liver disease but does not affect the safety of Vitamin K administration in newborns.

Cystic fibrosis (Option D): This genetic disorder affects the lungs and digestive system, leading to malabsorption of fat-soluble vitamins like Vitamin K. Newborns with cystic fibrosis often need Vitamin K supplementation, making its administration necessary rather than contraindicated.

Solution for Question 2:

Correct Answer: C) The standard treatment typically involves low-intensity chemotherapy due to its slow-growing nature

Explanation:

Burkitt's lymphoma is a highly aggressive, rapidly growing lymphoma that requires intensive chemotherapy. High-intensity chemotherapy is the standard treatment approach in Burkitt's lymphoma

Burkitt's Lymphoma

- Most common in children >5 years, males > females
- The disease is associated with Epstein-Barr virus (EBV), human immunodeficiency virus (HIV), and chromosomal translocations that cause the overexpression of oncogene C-MYC.
- Presents with fever, jaw swelling, tonsillar enlargement, abdominal mass, CNS and bone marrow involvement
- Histopathology shows starry-sky appearance with intermediate-sized homogenous cells
- CD markers: CD 10+, CD 20+, soluble immunoglobulin (Sig)+
- Cytogenetics: t(2;8), t(8;14), t(8;22)

Solution for Question 3:

Correct Answer: C) Chemotherapy with ABVD regimen

Explanation: For patients with early-stage Hodgkin lymphoma, the standard initial treatment typically involves chemotherapy with ABVD regimen.

Clinical Features:

- Painless cervical or supraclavicular lymphadenopathy
- About 20-30% of children present with systemic (B) symptoms: fever >38°C, night sweats, unexplained weight loss of >10% of body weight

Types: Nodular and Classical HL

Ann Arbor staging of Hodgkin Lymphoma	
I	Single lymphatic or extra lymphatic site involved
II	2 or more regions on the same side of the diaphragm are involved
III	Lymph node (LN) regions on both sides of the diaphragm are involved

IV	Diffuse or disseminated disease +/- LN involvement

Based on staging, HL can be either low risk or high risk:

Low-risk disease	High-risk disease
Stage I, II, IIIA	Stage IIIB, IV
Absence of B symptoms, no evidence of bulky disease	Presence of B symptoms, bulky mediastinal/peripheral lymphadenopathy, extranodal extension
Treatment: 2-4 cycles of chemotherapy (ABVD regimen), low-dose field radiation	Treatment: 4-6 cycles of chemotherapy (ABVD) with/without radiotherapy. Hematopoietic stem cell transplant in relapse or in refractory cases.

ABVD regimen - Adriamycin (Doxorubicin), Bleomycin, Vinblastine, Dacarbazine

Solution for Question 4:

Correct Answer: D) IIB

Explanation:

Involvement of two or more lymph node regions on the same side of the diaphragm, without extranodal involvement and without B symptoms belongs to the IIB stage of the Ann Arbor grading system.

Ann Arbor Grading System

Stage I	Involvement of a single lymph node region or a single extra lymphatic organ or site
Stage II	involvement of two or more lymph node regions on the same side of the diaphragm or localized involvement of an extra lymphatic organ or site
Stage III	involvement of lymph node regions or structures on both sides of the diaphragm
III1	Abdominal disease is limited to the upper abdomen
III2	Abdominal disease involving the lower abdomen
Stage IV	diffuse or disseminated involvement of one or more extra lymphatic organs,

Additional Substaging	
A	No B symptoms
B	presence of B symptoms (including fever, night sweats, and weight loss of ≥10% of body weight over 6 months)
E	involvement of a single, extranodal site, contiguous or proximal to a known nodal site (stages I to III only; additional extranodal involvement is stage IV)
X	bulky nodal disease: nodal mass >1/3 of intrathoracic diameter or 10 cm in dimension

Treatment:

Stage I,II,IIIA with no B symptoms and no bulky disease are treated with 2-4 cycles of chemotherapy (ABVD) and with low-dose field radiation.

Stage IIIB, 4 / B symptoms / bulky disease are treated with 4-6 cycles of ABVD with or without radiotherapy.

Hematopoietic stem cell transplant can be done in people with relapse or in refractory cases.

Solution for Question 5:

Correct Answer: B) Corticosteroids

Explanation:

The given clinical scenario of a patient undergoing treatment for acute promyelocytic leukemia (AML-M3) is most likely receiving ATRA (all-trans-retinoic acid). The symptoms of dyspnea, fever, and peripheral edema are suggestive of differentiation syndrome, a unique side effect of ATRA therapy.

ATRA therapy is used for the management of acute promyelocytic leukemia (APML). It does not kill the malignant cells directly, instead promotes the spontaneous apoptosis of the differentiated malignant cells.

It is sometimes associated with a unique side effect known as the differentiation syndrome which is associated with the development of dyspnea, fever, weight gain, peripheral edema and is treated with dexamethasone.

Solution for Question 6:

Correct Answer: D) t(9;22)

Explanation:

Translocation t(9;22), known as the Philadelphia chromosome, is not useful in differentiating AML from ALL because it can be found in both, although it is more common in CML.

The Philadelphia chromosome is an abnormal version of chromosome 22 that contains a fusion of two genes, the ABL gene and the BCR gene. The ABL gene from chromosome 9 joins with the BCR gene on chromosome 22 to form the BCR::ABL fusion gene. This fusion gene can cause immature white blood cells to grow uncontrollably and build up in the bone marrow and blood. The Philadelphia chromosome is found in the bone marrow cells of almost all people with chronic myelogenous leukemia and some people with acute lymphocytic leukemia or acute myelogenous leukemia.

t(8;21) (Option A) is associated with AML, and has a favorable prognosis.

t(15;17) (Option B) is associated with AML-M3, also known as acute promyelocytic leukemia

t(12;21) (Option C) is associated with ALL, particularly in children, with favorable prognosis.

Solution for Question 7:

Correct Answer: A) AML with t(8;21)(q22;q22)

Explanation:

AML with t(8;21) is most commonly associated with the development of granulocytic sarcoma, also known as chloroma, an extramedullary tumor composed of immature myeloid cells.

FAB subtype	Name	Salient features

M0	Undifferentiated acute myeloblastic leukemia	Myeloperoxidase negative
M1	Acute myeloblastic leukemia with minimal differentiation	-
M2	Acute myeloblastic leukemia with maturation	Chloromas/myeloid sarcoma; t(8;21)
M3	Acute promyelocytic leukemia	Faggot cells (rich in Auer rods), higher risk of DIC; t(15;17)
M4	Acute myelomonocytic leukemia	Organomegaly, gum hypertrophy or leukemia cutis; inv(16)
M5	Acute monocytic leukemia	Organomegaly
M6	Acute erythroid leukemia	-
M7	Acute megakaryoblastic leukemia	Dry tap, more common in Down's syndrome <3 years

Solution for Question 8:

Correct Answer: B) Administration of intrathecal methotrexate

Explanation:

Intrathecal methotrexate is the most appropriate and direct method for CNS prophylaxis in patients with ALL given repeatedly by LP.

The likelihood of later CNS relapse is thereby reduced to <5%. A small percentage of patients with features that predict a high risk of CNS relapse may receive irradiation to the brain in later phases of therapy.

Treatment for ALL is divided into 4 phases:

INDUCTION PHASE: Vincristine Prednisolone L-asparaginase Anthracycline	CNS PROPHYLAXIS: Intrathecal Methotrexate with or without CNS irradiation

CONSOLIDATION/INTENSIFICATION PHASE: Methotrexate Cyclophosphamide Cytosine-Arabinoside	MAINTENANCE PHASE: Maintenance to prevent relapse: Methotrexate 6-Mercaptopurine

Solution for Question 9:

Correct Answer: B) ALL

Explanation:

Pediatric patients with Down syndrome are at an increased risk of developing ALL.

Acute leukemia occurs about 15-20 times more frequently in children with Down syndrome than in the general population. Approximately 10% of neonates with Down syndrome develop a transient leukemia or myeloproliferative disorder characterized by high leukocyte counts, blast cells in the peripheral blood, and associated anemia, thrombocytopenia, and hepatosplenomegaly.

The ratio of ALL to AML in patients with Down syndrome is the same as that in the general population. The exception is during the 1st 3 yr of life when AML is more common.

Most common leukemia overall - ALL

Most common leukemia in age < 3 years - AML-M7 (Option A)

In children with Down syndrome who have ALL, the expected outcome of treatment is slightly inferior to that for other children.

Solution for Question 10:

Correct Answer: A) The child’s age at diagnosis >1 and < 10 years of age

Explanation:

Younger age at diagnosis, typically between 1 and 10 years old, is generally associated with a better prognosis in ALL. Children within this age range tend to respond better to treatment and have higher rates of remission and cure compared to those under 1 year old or over 10 years old.

Prognostic factors in ALL:

Favourable	Unfavourable
Hyperdiploidy	Hypodiploidy
t(12;21)	t(4;11) , t(9;22) (Option C and D)
Age 1-10 years	Age <1 year and >10 years
Leukocyte count t <50,000/µL	Leukocyte count t >50,000/µL (Option B)
Female	Male

Solution for Question 11:

Correct Answer: D) Hereditary spherocytosis

Explanation:

Hereditary spherocytosis is associated with hemolytic anemia, it does not typically predispose individuals to leukemia.

Hereditary spherocytosis is primarily characterized by the destruction of red blood cells leading to anemia, jaundice, and splenomegaly.

Genetic syndromes associated with the development of leukemia in a pediatric patient

Down syndrome	Increased risk of acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML).
Diamond-Blackfan anemia	Increased risk of acute myeloid leukemia (AML).
Kostmann syndrome	Increased risk of acute myeloid leukemia (AML) due to severe congenital neutropenia.
Fanconi anemia	High risk of acute myeloid leukemia (AML) and other hematologic malignancies.
Neurofibromatosis type 1 (NF-1)	Increased risk of juvenile myelomonocytic leukemia (JMML).

Bloom syndrome	Increased risk of acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML).
Shwachman-Diamond syndrome	Increased risk of acute myeloid leukemia (AML).
Li-Fraumeni syndrome	Increased risk of various cancers, including acute leukemias, due to TP53 mutation.
Ataxia-telangiectasia	Increased risk of acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML).
Paroxysmal nocturnal hemoglobinuria (PNH)	PNH primarily predisposes to aplastic anemia and venous thrombosis, not directly to leukemia.

Solution for Question 12:

Correct Answer: C) a,c,d

Explanation:

Hemophilia A is a genetic bleeding disorder caused by a deficiency in clotting factor VIII, which is essential for proper blood coagulation. It primarily affects males due to its X-linked recessive inheritance pattern, while females are typically carriers.

Genetic Pattern: Hemophilia A is inherited in an X-linked recessive pattern. It predominantly affects males.

Symptoms

Bleeding Characteristics: Individuals with Hemophilia A commonly experience spontaneous bleeding, particularly into joints (hemarthrosis) and muscles, which can lead to pain, swelling, and long-term joint damage.

Prolonged Bleeding: They may also have prolonged bleeding following injuries or surgeries.

- Blood Tests: Diagnosis is confirmed through blood tests that measure the activity of factor VIII. Haemophilia A is associated with a prolonged activated partial thromboplastin time (aPTT) and a normal prothrombin time (PT).
- Factor VIII Infusions
- Desmopressin

With proper management, individuals with Hemophilia A can lead relatively normal lives, though they must take precautions to avoid bleeding complications and manage any injuries promptly.

Solution for Question 13:

Correct Answer: C) α■γ■

Explanation:

α■γ■ is the correct composition for fetal hemoglobin (HbF). It consists of two alpha (α) chains and two gamma (γ) chains.

By 6-8 weeks gestation, HbF (α2γ2) is the predominant hemoglobin; at 24 weeks gestation, it constitutes 90% of the total hemoglobin.

HbF declines modestly in the third trimester, such that the HbF comprises 70–80% of the total hemoglobin.

HbF production decreases rapidly postnatally and by 6-12 mo of age reaches adult levels of <2%.

HbF levels may be elevated with hemoglobinopathies, hereditary persistence of fetal hemoglobin, or bone marrow failure syndromes.

Two disorders resulting from mutations in the β-globin gene, β-thalassemia, and sickle cell disease, become symptomatic postnatally as fetal γ-globin expression decreases and adult β-globin increases. In both these disorders, elevated HbF levels persist in childhood and later.

Normal human hemoglobins:

	Haemoglobin	Structural formula
EMBRYONIC	Hb Gower 1	ζ2■2
Hb Gower 2	α2■2 (Option D)	
Hb Portland	ζ2■2	
FETAL	Hb-F	α2γ2 (0.5-1%)
ADULT	Hb-A	α2β2 (97%) (Option A)
Hb-A■	α2δ2 (1.5-3.2%) (Option B)	

- Gene for ■ globin chain is encoded on chromosome 11

Gene for ■ globin chain is encoded on chromosome 16

Solution for Question 14:

Correct Answer: D) Wait and watch

Explanation:

The child doesn’t have severe thrombocytopenia and/or any life-threatening bleeding. Hence in this case the child only requires close observation.

Management of Immune thrombocytopenic purpura in children.

Based on platelet count:

- Platelet Count >30,000 cells/µL + No severe bleeding- Monitor closely
- Platelet Count < 30,000 cells/µL + No severe bleeding + Platelet Count ≥ 5,000 cells/µL: Glucocorticoids (e.g., Prednisone, Dexamethasone) (Option A)
- Platelet Count < 5,000 cells/µL or Severe Bleeding: IVIg + Glucocorticoids

Other modalities,

- Immediate Need to Raise Platelet Count like surgery or trauma- Platelet Transfusion (Option B)
- Persistent or Severe Cases- Monitor with intermittent IVig, If Necessary, Consider Splenectomy (Option C)

Alternative Medications

- Anti-CD20 Antibody: Rituximab
- TPO Receptor Agonists: Romiplostim, Eltrombopag

Solution for Question 15:

Correct Answer: A) Hageman factor (Factor XII) assay

Explanation:

Factor XII deficiency is a rare coagulation disorder that may present with delayed umbilical cord separation in newborns. Although it typically does not lead to bleeding, it can be linked to thrombotic complications.

Since Factor XII plays a key role in the intrinsic pathway of the coagulation cascade, assessing the Hageman factor (Factor XII) is essential for diagnosis.

Causes of delayed umbilical cord separation

Cause	Mechanism
Leukocyte Adhesion Deficiency (LAD)	Neutrophils fail to adhere and migrate to infection sites, impairing cord separation.
Neutrophil Dysfunction	Disorders like chronic granulomatous disease impair neutrophil function, delaying cord detachment.
Prematurity	Underdeveloped immune responses in premature infants delay inflammatory and necrotic processes.
Infections (e.g., Neonatal Sepsis)	Systemic infections impair healing, leading to delayed cord separation.
Hypothyroidism	Congenital hypothyroidism slows metabolic and healing processes, delaying cord detachment.
Umbilical Cord Care Practices	Use of antiseptics (e.g., alcohol, chlorhexidine) on the cord stump slows desiccation and separation.
Congenital Immunodeficiency Syndromes	Conditions like SCID impair immune function, delaying healing and cord separation.
Factor XII Deficiency	Impaired clot formation and tissue remodeling delay wound healing and cord separation.

Prothrombin Time (PT) (Option B): PT assesses the extrinsic and common coagulation pathways. It’s not the primary test for diagnosing Factor XII deficiency.

Activated Partial Thromboplastin Time (aPTT) (Option C): aPTT measures the intrinsic and common pathways, often prolonged in Factor XII deficiency. However, a specific Factor XII assay is required for a definitive diagnosis.

Fibrinogen level (Option D): Fibrinogen is crucial for clot formation. Measuring its levels helps diagnose fibrinogen-related disorders but isn’t directly related to Factor XII deficiency.

Solution for Question 16:

Correct Answer: B) 1-a, 2-b, 3-c, 4-d

Explanation:

1	Glanzmann Thrombasthenia	a	Defect in (GPIIb/IIIa)
2	Bernard-Soulier Syndrome	b	Defect in GPIb-IX-V
3	Von Willebrand Disease	c	Defect in Von Willebrand Factor (vWF)
4	Hemophilia A	d	Defect in Factor 8

Glanzmann Thrombasthenia: Defect in platelet integrin αIIbβ3 (GPIIb/IIIa). It is autosomal recessive in inheritance and a defect of platelet aggregation.

Bernard-Soulier Syndrome: Defect in platelet glycoprotein complex GPIb-IX-V. It is autosomal recessive in inheritance and a defect of platelet adhesion.

Von Willebrand Disease: Defect in von Willebrand factor (vWF). It is an inherited bleeding disorder and involves defective platelet adhesion.

Hemophilia A: Defect in Factor 8. It is an X-linked recessive disorder that presents with easy bruisability, intramuscular hematoma, and haemarthrosis.

Solution for Question 17:

Correct Answer: C) Aplastic anemia

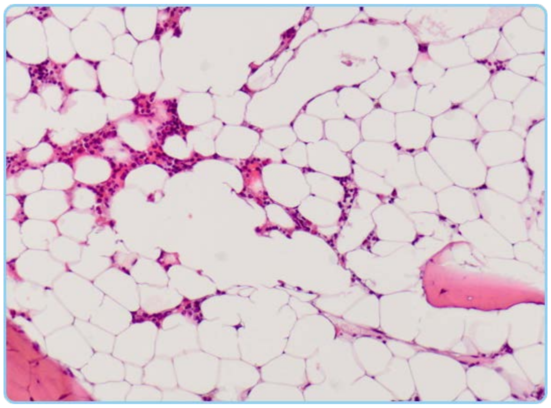
Explanation:

The severe pancytopenia with a hypocellular marrow with fatty infiltration strongly points towards aplastic anemia rather than other marrow failure syndromes.

Aplastic anemia is characterized by the suppression of two or all three hematopoietic cell lines (RBCs, WBCs, platelets).

Clinical features: severe anemia reveals pallor and/or signs of congestive heart failure and bleeding tendencies. Fever, pneumonia or sepsis are due to neutropenia.

Investigations: Peripheral blood smear findings show anemia, the RBCs show macrocytosis (>100fl), thrombocytopenia and neutropenia. Bone marrow aspiration and biopsy are essential for diagnosis and evaluation of severity. The marrow cells are replaced with fat cells and lymphocytes, with very few hematopoietic cells.



Acute myeloid leukemia (Option A): AML typically presents with pancytopenia, but the absence of blasts in this patient’s smear makes AML unlikely. Additionally, the bone marrow in AML would be hypercellular rather than hypocellular.

Myelodysplastic syndrome (Option B): MDS can present with pancytopenia and hypocellular marrow, but it is usually associated with dysplastic changes in the hematopoietic cells and the presence of abnormal cells in the bone marrow.

Fanconi anemia (Option D): Fanconi anemia is a genetic disorder that leads to bone marrow failure, but it usually presents earlier in childhood with additional congenital abnormalities such as absent radius and short stature.

Solution for Question 18:

Correct Answer: B) Hematopoietic stem cell transplantation

Explanation:

The above radiological image of absent thumb and radius along with the history of short stature, progressive bone marrow failure, and café-au-lait macules (hypopigmented skin lesions) are suggestive of Fanconi anemia.

Fanconi anemia is a rare inherited disorder (AR) characterized by progressive bone marrow failure, congenital abnormalities, and a high risk of malignancies, particularly acute myeloid leukemia (AML). The bone marrow failure leads to pancytopenia, resulting in symptoms such as frequent infections, easy bruising, and fatigue.

Hematopoietic stem cell transplantation (HSCT) is the only curative therapy for Fanconi anemia. Criteria for referral for hematopoietic stem cell transplant are: (i) severe aplastic anemia, and (ii) a matched related sibling donor.

High-dose corticosteroids (Option A): While corticosteroids can be used in managing certain types of anemia, they are not a standard treatment for Fanconi anemia.

Blood transfusions (Option C): Blood transfusions may be used temporarily to manage anemia but they do not address the root cause of the bone marrow failure.

Targeted gene therapy (Option D): Although gene therapy is a potential future treatment for Fanconi anemia and is being researched, it is not currently a standard or widely available treatment option.

Vs Fanconi syndrome:

It is an inherited or acquired disorder characterized by generalized dysfunction of the proximal renal tubules, resulting in urinary loss of bicarbonate, phosphate, glucose, and amino acids.

Solution for Question 19:

Answer: C) Sickle cell disease

Explanation:

The above clinical history of fatigue, pallor, and abdominal pain along with the history of multiple blood transfusions along with absence of spleen hints towards the diagnosis of sickle cell disease.

Sickle cell anemia is an autosomal recessive disease that results from the substitution of valine for glutamic acid at position 6 of the beta-globin gene.

Sickled red blood cells lead to the obstruction of the microcirculation, causing ischaemic injury and resulting in progressive damage and eventual loss of splenic function. This leads to an increased risk of infections, particularly from encapsulated organisms such as Streptococcus pneumoniae, Haemophilus influenzae and Neisseira meningitidis.

Beta-thalassemia major (Option A) usually causes severe anemia but doesn’t typically result in spleen absence.

Hereditary spherocytosis (Option B) often presents with splenomegaly.

Autoimmune hemolytic anemia (Option D) can cause anemia, jaundice and splenomegaly.

Learning Objective:

Types of crises in sickle cell disease

Vaso-occlusive crisis	Caused by obstruction of the microcirculation by sickled RBCs
Acute chest syndrome	Caused by vaso-occlusive crisis involving the lungs
Sequestration crisis (spleen or liver)	Obstruction of splenic veins causes sequestration of peripheral blood in an engorged spleen
Infectious crisis	Caused by increased susceptibility to infection by encapsulated organisms
Aplastic crisis	Results from cessation of erythropoiesis in the bone marrow, usually following infections

Solution for Question 20:

Correct Answer: B) Increased marrow hyperplasia

Explanation:

The above radiological image of the “hair on end" appearance along with the history of multiple blood transfusions indicates the diagnosis of thalassemia. The characteristic radiological appearance is due to increased marrow hyperplasia, a response to chronic anemia in thalassemia, which leads to diploic space widening in the skull.

Thalassemias are hemoglobinopathies that occur due to mutations in the globin chain.

Clinical features: severe anemia, intolerance to exercise, irritability, heart murmurs, heart failure, hepatosplenomegaly, frontal bossing, prominent facial bones, and dental malocclusion are present.

Laboratory investigations: Low hemoglobin levels, increased reticulocyte count, and leukocytosis. Peripheral blood film examination reveals microcytosis, nucleated RBCs, and basophilic stippling. HPLC and globin gene sequencing are used for diagnosis.

Treatment: Blood transfusion, hematopoietic stem cell transplant

Complications: “Hair on end" appearance of skull, maxillary bone overgrowth, compression fractures

Increased osteoclastic activity (Option A) leads to bone resorption rather than the formation of new bone.

Decreased bone remodeling (Option C) would result in reduced bone turnover, leading to brittle bones.

Increased osteoblastic activity (Option D): While increased osteoblastic activity could lead to increased bone formation, the "hair-on-end" appearance is specifically due to the hyperplasia of marrow, not just general osteoblastic activity.

Solution for Question 21:

Correct Answer: C) High-dose folic acid supplementation

Explanation:

Doses of folate >0.1 mg can correct the anemia of vitamin B12 deficiency but might aggravate any associated neurological abnormalities therefore high doses of folic acid supplementation should be avoided in a child

Megaloblastic Anemia:

Type of macrocytic anemia with MCV>90 - 100 fL.

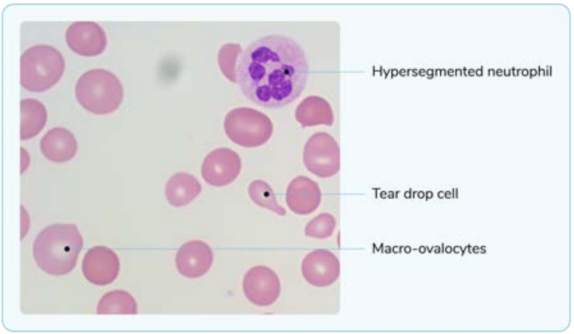
Due to Vit B12 or folate deficiency.

Vit B12 deficiency causes	Folate deficiency causes
Pure vegetarian vegan diet	Exclusive feeding with goat’s milk
Pernicious anemia	Drugs inhibiting folate metabolism Antifolates (e.g., methotrexate), Pyrimethamine; trimethoprim, Sulfasalazine, Valproic acid
Malabsorption: Crohn's disease, chronic pancreatitis	Cerebral folate deficiency (genetic or acquired)
Metabolic: homocystinuria	Defects in absorption: Gastric achlorhydria, Diseases of the upper small intestine- Tropical sprue

Clinical features:

- Pallor
- Smooth, beefy red tongue
- Hepatosplenomegaly
- Hyperpigmented knuckle

On Peripheral smear:



- Macroovalocytes
- Hypersegmented neutrophils (>5 lobes in 5% neutrophils).

Treatment:

When the diagnosis of folate deficiency is established, folic acid may be administered orally or parenterally at 0.5-1.0 mg/day. (Options A and B)

If the specific diagnosis is in doubt, smaller doses of folate (0.1 mg/day) may be used for 1 week as a diagnostic test, because a hematologic response can be expected within 72 hr.

Folic acid therapy (0.5-1.0 mg/day) should be continued for 3-4 weeks until a definite hematologic response has occurred.

Maintenance therapy with a multivitamin (containing 0.2 mg of folate) is adequate.

Encouraging a diet rich in folate is beneficial and supports overall treatment for folate deficiency. It is a complementary approach to oral folic acid supplementation and does not pose a risk. (Option D)

Solution for Question 22:

Correct answer: B) Increase in the reticulocyte count

Explanation:

An increase in the reticulocyte count is observed on the 3rd day of iron therapy.

An increase in reticulocyte count can be an early indicator of effective iron therapy, as the bone marrow responds to the increased availability of iron by producing more red blood cells.

If the anemia is mild, the blood count can be repeated approximately 1 month after initiating therapy. At this point, the hemoglobin has usually risen by at least 1-2 g/dL and has often normalized.

If the anemia is more severe, earlier confirmation of the diagnosis can be made by the appearance of a reticulocytosis, usually within 48-96 hr of instituting treatment. The hemoglobin will then begin to increase by 0.1-0.4 g/dL/day depending on the severity of the anemia.

Iron medication should be continued for 2-3 months after blood values normalize to re-establish iron stores.

Response to treatment in iron deficiency anemia:

Time after iron administration	Response
Within 24 hours	Decreased irritability, improved appetite (Option C)
48-76 hours	Reticulocytosis appears to peak at 5-7 days post-treatment.
1-4 weeks	Hb level becomes normal (Option A ruled out)
2-3 months	Replenishment of stores (Option D)

Iron deficiency anemia

The most common condition causing anemia in children.

Risk factors for iron deficiency anemia include:

- Decreased iron intake-Inadequate diet
- Impaired absorption-Celiac disease
- Increased demands- Infancy, puberty
- Hookworm infestation-causing blood loss and decreased absorption, treated by deworming

Clinical features:

- Pallor, Generalized weakness, lethargy
- Anorexia
- Glossitis, angular stomatitis
- Pica
- Koilonychia-spooning of nails

Laboratory findings:

- Microcytosis, anisocytosis
- Low MCV
- Low serum ferritin
- Increased TIBC
- Mentzer index>14

Treatment:

- Supplement Iron-ferrous sulphate- 3-6 mg/kg/ day of elemental iron.
- Duration: 4-6 months

Solution for Question 23:

Correct Answer: C) <12 g/dl

Explanation:

According to WHO criteria, an 8-year-old child is classified as anemic if their hemoglobin level is below 12.0 g/dL. This threshold helps in the early detection and management of anemia in children.

WHO criteria for anemia in children:

Children 6 months to 5 years	Hb <11g/dl
Children 6 years to 14 years	Hb <12 g/dl

Solution for Question 24:

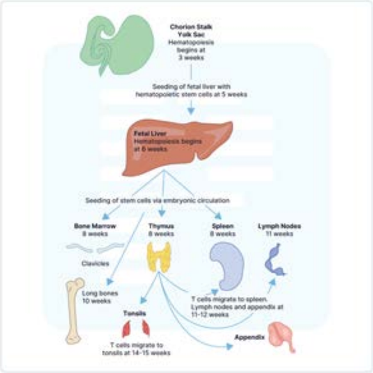
Correct Answer: D) Yolk Sac

Explanation:

Pluripotential hematopoietic stem cells first appear in the yolk sac at 2.5-3 weeks of gestational age, migrate to the fetal liver at 5 weeks gestation, and later reside in the bone marrow, where they remain throughout life.

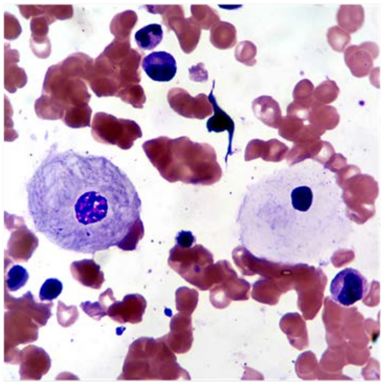
Major sites of hematopoiesis:

Site	Starts at	Continues till
Yolk sac	3rd week	10-12th week
Liver	6-8th week	2nd trimester
Bone marrow	2nd trimester onwards	



Previous Year Questions

1. A child presented with anemia, mental retardation, bone pains, and hepatosplenomegaly. Bone marrow biopsy showed a "crumpled tissue paper" appearance. What is the enzyme deficiency producing this condition?



A. Hexosaminidase-A

B. Iduronate 2-sulphatase

C. Glucose-6-phosphate dehydrogenase

D. Glucocerebrosidase

2. An 18-month-old child presents with hepatosplenomegaly, a hemoglobin level of 7 g/dL, a white blood cell count of 50,000 cells/cu.mm, and petechiae and purpura. What is the recommended drug treatment for this condition?

A. Vincristine + Prednisolone

B. IVIG

C. L-Asparinigase +Prednisolone

D. Daunorubicin + Prednisolone

3. In a child with thalassemia who is receiving frequent blood transfusions, what is the most effective approach to identify iron accumulation?

A. Liver iron concentration

B. MRI T2 myocardium

C. Serum ferritin

D. NTBI

---

4. Which of the following is inherited in an autosomal recessive manner?

A. Hereditary spherocytosis

B. Sickle cell anemia

C. Osteogenesis imperfecta

D. Von Willebrand disease

---

5. Which of the following follows an autosomal dominant inheritance pattern?

A. Von Willebrand disease

B. Wilson disease

C. Cystic fibrosis

D. Friedrichs ataxia

---

6. A patient has bruising, petechiae, and occasional minor epistaxis, which has very little interference with his daily activities. Based on further workup, a diagnosis of a mild case of ITP is made. How will you manage the patient?

A. Short course corticosteroid therapy

B. Intravenous anti-D administration

C. A single dose of intravenous immunoglobulin

D. Education, counseling, and watchful expectancy

7. Which of the following parameters in Acute Lymphoblastic Leukemia (ALL) indicates an unfavorable prognosis?

A. Age > 10 years

B. Leukocyte count < 50,000/mm3

C. Hyperdiploidy

D. Trisomy of chromosomes 4, 10, and 17

8. What is the viral infection that can lead to temporary aplastic anemia?

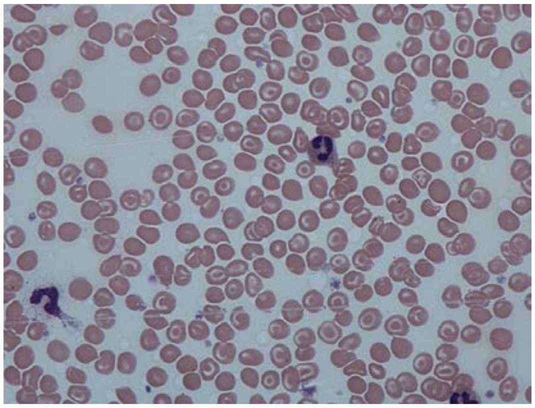
A. HIV

B. Polio

C. Parvovirus B-19

D. HHV-8

9. A 14-month-old child with Hb 6.3 was undergoing repeated blood transfusions. The last transfusion was given in the 9th month. Which of the following investigations is used for a definitive diagnosis?



A. High-performance liquid chromatography

B. Ferritin levels estimation

C. Bone marrow aspiration

D. Globin gene sequencing

---

10. What is the likely diagnosis in a 10-year-old male child who has a history of spontaneous excessive bruising and hemarthrosis, normal platelet count, prolonged PTT, and normal PT?

A. Von Willebrand disease

B. Hemophilia A

C. Bernard-Soulier syndrome

D. Glanzmann thrombasthenia

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11. What of the following malignant tumors commonly found in children that have a high tendency to spread to the bone marrow?

A. Wilms tumor

B. Neuroblastoma

C. Adrenal gland tumors

D. Granulosa cell tumor of ovary

12. A 2-year-old child is experiencing symptoms that indicate anemia, but they are not specific. The examination of a sample of blood under a microscope reveals the presence of target cells. The child also has a low level of hemoglobin (6 gm%) and exhibits a picture of microcytic hypochromic. Additionally, there is a positive family history of this condition. The next step in the diagnostic process is to conduct which investigation?

A. Newborn screening for sickle cell disease

B. HbA1c estimation

C. Hb electrophoresis

D. Osmotic fragility test

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	1
Question 4	2
Question 5	1
Question 6	4
Question 7	1
Question 8	3
Question 9	4
Question 10	2
Question 11	2
Question 12	3

Solution for Question 1:

Correct Option D – Glucocerebrosidase:

- A child presented with anemia, mental r*etardation, bone pains, and hepatosplenomegaly. Bone marrow biopsy showed a "crumpled tissue paper" appearance. The enzyme deficiency producing this condition is Glucocerebrosidase, which is associated with Gaucher disease.

Incorrect Options:

Option A - Hexosaminidase-A:

- This enzyme deficiency is associated with Tay-Sachs disease, which is characterized by progressive neurological degeneration, but it does not typically present with the "crumpled tissue paper" appearance in bone marrow or the specific combination of symptomsdescribed (anemia, mental retardation, bone pains, hepatosplenomegaly).

Option B - Iduronate 2-sulphatase:

- Deficiency of this enzyme is associated with Hunter syndrome, a type of mucopolysaccharidosis (MPS II). This condition can cause developmental delay, skeletal abnormalities, and hepatosplenomegaly but does not typically present with the "crumpled tissue paper" appearance in bone marrow.

Option C - Glucose-6-phosphate dehydrogenase:

- Deficiency in this enzyme leads to G6PD deficiency, which can cause hemolytic anemia but does not typically cause the "crumpled tissue paper" appearance in bone marrow or the combination of symptoms described.

Solution for Question 2:

Correct Option A - Vincristine + Prednisolone:

- This combination is commonly used as part of the treatment regimen for acute lymphoblastic leukemia (ALL). Vincristine is a chemotherapy agent that inhibits cell division, and Prednisolone is a corticosteroid that helps reduce inflammation and has antitumor effects. This option is appropriate if the child's symptoms are indicative of ALL, as the high WBC count and symptoms like hepatosplenomegaly and purpura are consistent with leukemia.

Incorrect Options:

Option B – IVIG:

- Intravenous immunoglobulin (IVIG) is used for treating immune thrombocytopenic purpura (ITP) and other conditions involving immune-mediated platelet destruction. While IVIG can be beneficial in conditions with bleeding tendencies, it is not typically used as the primary treatment for leukemia.

Option C - L-Asparaginase + Prednisolone:

- L-Asparaginase is an enzyme used in the treatment of acute lymphoblastic leukemia (ALL) as part of a chemotherapy regimen. It works by depleting asparagine, an amino acid necessary for leukemia cell survival. Prednisolone is a corticosteroid used to enhance the effects of chemotherapy and reduce inflammation. This combination is appropriate fortreating ALL, but without additional context, it’s less specific than Vincristine + Prednisolonein a classic ALL regimen.

Option D - Daunorubicin + Prednisolone:

- Daunorubicin is a chemotherapy drug used primarily in the treatment of acute myeloid leukemia (AML) and sometimes in acute lymphoblastic leukemia (ALL). Prednisolone is a corticosteroid used to support chemotherapy treatment. This combination is generally used for AML rather than ALL.

Solution for Question 3:

Correct Option A - Liver iron concentration:

- Due to multiple blood transfusions, patients with thalassemia are at risk of developing iron overload.
- Liver iron concentration is the most accurate method for assessing iron overload.
- While invasive, it provides a direct measurement of iron stored in the liver, which is a major site of iron storage.
- It is considered the gold standard for assessing iron overload.

Incorrect Options:

Option B - MRI T2 myocardium: It can be used to assess iron content, but liver iron concentration is a more comprehensive method for overall iron status assessment.

Option C - Serum ferritin:

- Serum ferritin is used to estimate iron levels.
- However, it is not as accurate for detecting iron overload in thalassemia patients, as ferritin levels can be elevated due to other factors (e.g., inflammation).

Option D - NTBI (Non-Transferrin-Bound Iron):

- NTBI refers to iron that is not bound to transferrin.
- NTBI is a form of labile iron that can contribute to iron toxicity.
- While NTBI can be relevant in assessing iron overload, it is not used as a primary method of detection compared to liver iron concentration.

Solution for Question 4:

Correct Option B - Sickle cell anemia: Sickle cell is inherited in an autosomal recessive pattern.

Incorrect Options - A, C, and D are inherited in an autosomal dominant pattern.

Solution for Question 5:

Correct Option A - Von Willebrand disease: Von Willebrand disease is inherited in an autosomal dominant pattern.

Incorrect Options - B, C, and D are inherited in an autosomal recessive pattern.

Solution for Question 6:

Correct Option D - Education, counseling, and watchful expectancy:

- In a patient with mild ITP who has bruising, petechiae, and occasional minor epistaxis but minimal interference with daily activities, the preferred approach is often watchful expectancy.
- This involves providing education and counseling to the patient and closely monitoring their platelet counts and symptoms over time.
- Most cases of mild ITP in adults resolve spontaneously without treatment.
- Regular follow-up visits can help assess the progression of the disease and determine the need for intervention if symptoms worsen or if there is an increase in bleeding risk.

Incorrect Options:

Option A: Short course corticosteroid therapy:

- Corticosteroids, such as prednisone, are commonly used in the management of immune thrombocytopenic purpura (ITP).
- However, in a patient with mild ITP who has minimal symptoms and little interference with daily activities, the use of corticosteroids may not be necessary as the risks and side effects of the medication may outweigh the benefits.

Option B: Intravenous anti-D administration:

- Intravenous anti-D immunoglobulin is typically used in Rh-positive individuals with ITP. It works by inhibiting the destruction of platelets and increasing platelet count. However, it is not the preferred treatment option for mild cases of ITP, and its use is generally reserved for specific situations and more severe cases.

Option C: A single dose of intravenous immunoglobulin

- Intravenous immunoglobulin (IVIG) is commonly used in the treatment of ITP, particularly in cases where there is a significant bleeding risk or severe thrombocytopenia.
- However, for a patient with mild ITP and minimal symptoms, the use of IVIG is not typically indicated as it is associated with potential adverse effects and high cost.

Solution for Question 7:

Correct Option A - Age > 10 years:

- Children older than 10 years of age have a higher risk of relapse and poorer outcomes.

Incorrect Options:

Option B: Leukocyte count < 50,000/mm3:

- A lower leukocyte count is associated with a more favorable prognosis in ALL.
- High leukocyte counts at diagnosis indicate a higher tumor burden and are considered a poor prognostic factor.

Option C: Hyperdiploidy: Hyperdiploidy, which refers to the presence of more than the normal number of chromosomes in leukemic cells, is associated with a favorable prognosis in pediatric ALL.

Option D: Trisomy of chromosomes 4, 10, and 17: Trisomies involving chromosomes 4, 10, and 17 are often observed in pediatric ALL and are associated with a more favorable prognosis.

Solution for Question 8:

Correct Option C - Parvovirus B-19

- Parvovirus B-19 is known to cause a transient aplastic crisis, which is a temporary suppression of red blood cell production in individuals with underlying hemolytic disorders, such as sickle cell disease or hereditary spherocytosis.
- Parvovirus B-19 infects erythroid precursor cells in the bone marrow, leading to their destruction and subsequent reduction in red blood cell production.

Incorrect Options - A, B, and D do not cause temporary aplastic anemia.

Solution for Question 9:

Correct Option D - Globin gene sequencing:

- In a 14-month-old child with a history of repeated blood transfusions and target cells seen on the blood smear, the definitive diagnosis requires investigating the underlying cause of the anemia.
- In this case, the presence of target cells suggests a possible hemoglobinopathy, such as thalassemia or sickle cell disease.
- Globin gene sequencing is a molecular diagnostic test that can identify specific mutations or genetic alterations in the globin genes.
- By sequencing the globin genes, such as the alpha and beta globin genes, it is possible to identify the specific genetic abnormality responsible for the hemoglobinopathy.

Incorrect Options:

Option A: High-performance liquid chromatography:

- High-performance liquid chromatography (HPLC) is a commonly used method to measure the different types of hemoglobin and their variants in the blood.

- While HPLC can help in identifying abnormal hemoglobin variants, it may not provide a definitive diagnosis of the underlying genetic abnormality causing the hemoglobinopathy.

Option B: Ferritin levels estimation: Estimating ferritin levels can provide information about the iron stores in the body but may not directly help in diagnosing the underlying cause of the anemia or the specific genetic abnormality causing the hemoglobinopathy.

Option C: Bone marrow aspiration:

- While bone marrow examination can provide valuable information about the cells in the bone marrow, it may not always be necessary for the definitive diagnosis of hemoglobinopathy.
- In this case, where target cells are seen on the blood smear, bone marrow aspiration may not be required for the diagnosis, and globin gene sequencing would be more specific and informative.

Solution for Question 10:

Correct Option B - Hemophilia A:

- The patient’s presentation with a history of spontaneous excessive bruising, hemarthrosis, normal platelet count, prolonged PTT, and normal PT is suggestive of Hemophilia A.
- Hemophilia A is an X-linked recessive bleeding disorder caused by a deficiency or dysfunction of clotting factor VIII.

Incorrect Options:

Option A - Von Willebrand disease:

- Von Willebrand disease (VWD) is a bleeding disorder caused by a deficiency or dysfunction of von Willebrand factor (vWF), a protein that plays a crucial role in platelet adhesion and clot formation.
- Patients with VWD typically present with mucocutaneous bleeding, such as nosebleeds, gum bleeding, or heavy menstrual bleeding.
- In addition to bleeding symptoms, laboratory tests reveal prolonged bleeding time, decreased levels of vWF antigen and activity, and abnormal platelet function.
- In the case provided, the normal platelet count makes VWD less likely.

Option C - Bernard-Soulier syndrome:

- Bernard-Soulier syndrome (BSS) is a rare inherited bleeding disorder characterized by defective platelet function.
- It is caused by mutations in the genes encoding glycoprotein Ib (GPIb), a receptor complex on platelets involved in platelet adhesion.
- BSS is typically associated with a low platelet count (thrombocytopenia) and giant platelets.
- Given the normal platelet count in the provided case, BSS is less likely.

Option D - Glanzmann thrombasthenia:

- Glanzmann thrombasthenia is characterized by defective platelet function.
- It is caused by mutations in the genes encoding glycoprotein IIb/IIIa (GPIIb/IIIa), an integrin receptor essential for platelet aggregation.
- Patients with Glanzmann thrombasthenia often have normal platelet counts but exhibit a prolonged bleeding time and impaired platelet aggregation.
- Given the normal platelet count in the provided case, Glanzmann thrombasthenia is less likely.

Solution for Question 11:

Correct Option B - Neuroblastoma:

- Neuroblastoma: Neuroblastoma is a cancer that develops from immature nerve cells, primarily affecting children aged 5 years or younger.
- Neuroblastoma has a propensity for metastasizing, and one of the most common sites of metastasis is the bone marrow.

Incorrect Options - A, C, and D are not associated with metastasis to the bone marrow.

Solution for Question 12:

Correct Option C - Hb Electrophoresis:

- Hb electrophoresis is a laboratory test used to identify and quantify the different types of hemoglobin in the blood.
- It is particularly useful in diagnosing various types of hemoglobinopathies, including thalassemias and sickle cell disease.
- In the case of a 2-year-old child presenting with microcytic hypochromic anemia, a positive family history, and target cells on the peripheral smear, Hb electrophoresis is a crucial investigation to assess for the presence of any abnormal hemoglobin variants or thalassemias.

Incorrect Options:

Option A - Newborn screening for sickle cell disease:

- It is a test used to screen newborns for sickle cell disease.
- It is not typically used as a diagnostic test for anemia in a 2-year-old child presenting with nonspecific symptoms.

Option B - HbA1c estimation: HbA1c is a blood test used primarily for the diagnosis and management of diabetes.

Option D - Osmotic fragility test:

- The osmotic fragility test is a laboratory test that measures the susceptibility of red blood cells to burst when exposed to changes in osmotic pressure.
- It is primarily used to diagnose and monitor certain types of hereditary red blood cell disorders, such as hereditary spherocytosis.
- While it can be useful in specific cases of hemolytic anemias, it is not the most appropriate next investigation in this case, as the clinical presentation and peripheral smear findings suggest a microcytic hypochromic anemia.

GIT Disorders in Children

1. An eight-month-old child was brought to paediatrics OPD by his parents, complaining of loose, watery stools along with fever for the last 15 days. On examination, the child is active but malnourished. What is the appropriate management for this condition?

A. Nutritional rehabilitation

B. Elimination diet

C. Empirical antibiotic therapy

D. All of the above

2. A 4-year-old boy is brought to the emergency department by his parents. They report that he swallowed a small object while playing 2 hours ago. A chest X-ray shows a round object in the mid-oesophagus. Which of the following is not an indication of immediate endoscopic removal of the foreign body?

A. Airway compromise

B. Sharp object in the oesophagus

C. Button battery in the oesophagus

D. Retained coin without symptoms

3. A 3-year-old boy presents with intermittent, painless rectal bleeding described as "brick-colored." He has no abdominal pain but appears pale. The pediatrician suspects Meckel's diverticulum. Which of the following is NOT true regarding this condition?

A. Presents with painless rectal bleeding in young children

B. Located about 2 feet proximal to the ileocecal valve

C. Technetium-99m pertechnetate scan has low sensitivity for diagnosis

D. Contains ectopic gastric or pancreatic tissue

4. A 2-day-old newborn presents with bilious vomiting. The pregnancy history reveals polyhydramnios. An abdominal X-ray image has been given. What's the likely diagnosis?

A. Jejunal atresia

B. Duodenal atresia

C. Pyloric stenosis

D. Malrotation with midgut volvulus

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5. A 2-week-old infant presents with persistent constipation since birth, abdominal distension, and failure to thrive. Initial X-rays reveal dilated bowel loops. Hirschsprung disease is suspected. What is the definitive test to confirm the diagnosis in this infant?

A. Contrast enema

B. Anorectal manometry

C. Rectal suction biopsy

D. Colonoscopy

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6. A 2-day-old full-term newborn presents with abdominal distension, failure to pass meconium, and bilious vomiting. Physical examination reveals an empty rectum. A barium enema shows a narrow distal segment with proximal dilation. Which of the following best describes the underlying pathogenesis of this condition?

A. Absence of smooth muscle in the bowel wall

B. Lack of ganglionic cells in the submucosal and myenteric plexuses

C. Hypertrophy of the circular muscle layer in the entire colon

D. Excessive production of acetylcholine in the enteric nervous system

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7. A 3-year-old child was brought to the emergency department as she has been unusually irritable and has had abdominal pain and vomiting over the past 24 hours. On examination, the sausage-shaped mass was palpable over the abdomen. The barium enema radiograph revealed a coiled spring appearance. What is the most likely diagnosis?

A. Hirschsprung disease

B. Duodenal atresia

C. Sigmoid volvulus

D. Intussusception

8. A 2-year-old was taken to the paediatrician with symptoms of abdominal cramps, loose stools, and weight loss over the past few weeks. Her parents observed that the symptoms occurred after she ate bread or pasta. Tests showed elevated anti-tTG IgA levels, though they were less than 10 times the upper limit of normal. What is the next step of diagnosis?

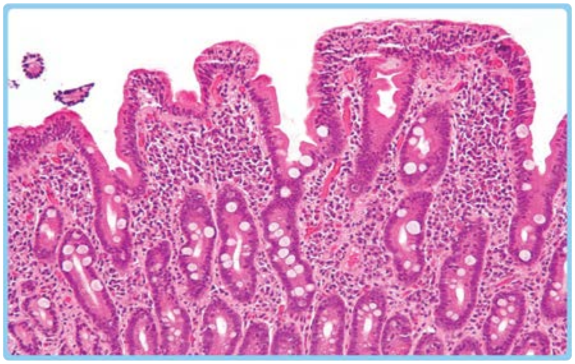
A. Measure total serum IgG levels

B. HLA-DQ2 and HLA-DQ8 genotyping

C. Small intestinal biopsy

D. Test for anti-endomysial antibodies (EMA)

9. A 4-year-old child presents with abdominal pain, diarrhea, weight loss, irritability, and occasional vomiting. The duodenal biopsy is illustrated in the image below. Which of the following does not accurately describe this condition?



A. Associated with HLADQ2 & HLADQ8

B. T-cell-mediated autoimmune disorder

C. Associated with Down’s syndrome

D. Treatment includes a high-fiber wheat diet.

10. An infant with a 2-week history of loose stools now exhibits noisy, loose stools, abdominal bloating, and crying, especially after feeding milk. The stool pH is approximately 6. What is the most probable diagnosis?

A. Irritable bowel syndrome

B. Small intestinal bacterial overgrowth

C. Lactose intolerance

D. Diverticulosis

11. A 4-month-old infant is brought to the pediatrician due to frequent postprandial regurgitation. The parents report that the infant appears irritable during and after feeding, frequently arches her back, and has episodes of choking and gagging. On examination, the infant is mildly irritable with no signs of respiratory distress. What is the next best step in management?

A. Start proton pump inhibitors immediately.

B. Positioning therapy and thickened feedings.

C. Schedule an urgent Esophagogastroduodenoscopy

D. Antireflux surgery (fundoplication).

12. A 6-year-old boy weighing 20 kg is admitted to the pediatric ward with pneumonia. He is hemodynamically stable but requires IV fluids for maintenance hydration. What is the appropriate daily maintenance fluid volume for this patient?

A. 1100 mL/day

B. 1300 mL/day

C. 1500 mL/day

D. 1700 mL/day

13. A 3-year-old girl presents with severe diarrhoea, sunken eyes, and lethargy. She's unable to drink and has cool, mottled extremities. You decide to start IV fluids immediately. Which of the following is the most appropriate initial IV fluid choice?

A. 5% Dextrose

B. Normal Saline

C. Ringer's Lactate with 5% Dextrose

D. Half-Normal Saline

14. A 2-year-old boy presents with a 1-day history of watery diarrhoea and 4 episodes of loose stools in the last 24 hours, but no vomiting. He is alert, drinking normally, and shows no signs of dehydration. What is the most appropriate initial management?

A. Start ORS and withhold solid foods for 24 hours

B. Prescribe an antibiotic and advise clear fluids only

C. Continue normal diet with adequate fluid intake and ORS

D. Administer ondansetron and start a probiotic supplement

15. A 4-week-old male infant presents with projectile vomiting for the past week. The parents report reduced wet diapers, thirst, and irritability. Examination reveals

a slightly sunken fontanelle, dry mucous membranes, normal heart rate, prolonged capillary refill of 3 seconds, and slow skin recoil but of <2 seconds. What is the most likely assessment of this infant's hydration status?

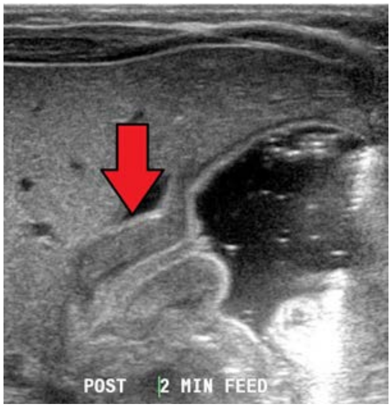
A. No dehydration

B. Mild dehydration

C. Some dehydration

D. Severe dehydration

16. A 4-week-old male infant presents with projectile vomiting after feeds for two weeks. On examination, a palpable "olive-like" mass is noted in the epigastric region. Ultrasound shows a pyloric muscle thickness of 4 mm and a pyloric channel length of 18 mm. The infant is dehydrated but stable. What is the most appropriate next step in management?



A. Proceed immediately with pyloromyotomy

B. Start nasoduodenal feeding

C. Begin IV atropine administration

D. Correct fluid and electrolyte imbalances

17. A 2-week-old infant is admitted with projectile, non-bilious vomiting, visible gastric peristaltic waves, and a palpable olive-shaped mass. An ultrasound confirms congenital hypertrophic pyloric stenosis (CHPS). Which of the following is not a known risk factor for CHPS?

A. Familial history of CHPS

B. Male sex

C. Maternal use of macrolide antibiotics

D. Advanced maternal age

18. A 3-week-old male infant presents with projectile, non-bilious vomiting after feeds and visible peristalsis in the upper abdomen. On examination, a palpable mass is noted in the epigastric region. An abdominal ultrasound is performed. Which of the following radiologic findings would most likely be seen in this condition?

A. Elongated pyloric channel on contrast studies

B. Pyloric muscle thickness is normal on USG

C. Pyloric length of 8 mm on USG

D. Dilated duodenum on contrast studies

19. A 2-week-old infant presents with multiple congenital anomalies, including congenital heart defect (ventricular septal defect), lumbar vertebral defect, and anal atresia. Renal imaging reveals a horseshoe kidney. Which of the following is NOT associated with this condition?

A. Tracheoesophageal fistula

B. Radial limb defects

C. Esophageal atresia

D. Cleft palate

20. A 4-day-old infant is admitted with severe respiratory distress, frothy secretions from the mouth, and worsening cyanosis with feeding. A chest X-ray reveals a gas-filled abdomens, stomach, upper oesophagal pouch, and air in the trachea, but no visible air in the lower oesophagus. What type of tracheoesophageal fistula (TEF) is most likely indicated by this presentation?

A. Type A

B. Type B

C. Type C

D. Type D

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	3
Question 4	2
Question 5	3
Question 6	2
Question 7	4
Question 8	3
Question 9	4
Question 10	3
Question 11	2
Question 12	3
Question 13	3
Question 14	3
Question 15	3
Question 16	4
Question 17	4
Question 18	1
Question 19	4
Question 20	3

Solution for Question 1:

Correct Answer: D) All of the above

Explanation:

The given clinical scenario involves persistent diarrhoea (lasting for >2 weeks), which is managed by dietary modifications, correction of dehydration, and treatment of the cause.

- Persistent diarrhoea is an episode of diarrhoea of presumed infectious aetiology that starts acutely but lasts for more than 14 days.
- It is more common in malnourished children, whose impaired nutritional status hinders gut repair, worsening nutrient absorption and perpetuating a vicious cycle.
- Infections like E. coli or infections of any other site, such as the urinary tract or respiratory tract, can lead to this condition.
- The use of antibiotics in acute diarrhoea can disrupt normal gut flora, leading to bacterial or fungal overgrowth and resulting in persistent diarrhoea and malabsorption.
- Management:
- Correction of dehydration, electrolytes and hypoglycemia.
- Empirical antibiotic therapy (Option C) and antiparasitic agents as per the cause.
- Nutritional rehabilitation (Option A)
- Elimination diet as (Option B) Diet A (Reduced lactose diet) Diet B (Lactose-free diet with reduced starch) Diet C (Monosaccharide-based diet)
- Diet A (Reduced lactose diet)
- Diet B (Lactose-free diet with reduced starch)
- Diet C (Monosaccharide-based diet)
- Some other supplements like Zinc and Vitamin A can also be given
- Diet A (Reduced lactose diet)
- Diet B (Lactose-free diet with reduced starch)
- Diet C (Monosaccharide-based diet)

Solution for Question 2:

Correct Answer: D) Retained coin without symptoms

Explanation:

Retained coin without symptoms does not usually require immediate endoscopic removal, as asymptomatic coins may pass spontaneously through the gastrointestinal tract.

- Foreign body ingestion is most common in children aged 6 months to 6 years, with coins being the most frequently ingested objects.
- Diagnosis: History of ingestion X-rays of the neck, chest, and abdomen
- History of ingestion
- X-rays of the neck, chest, and abdomen
- Management: Small, blunt objects: Observation for 24 hours Sharp objects, batteries, or large items: Early endoscopic removal 95% of objects that reach the stomach pass naturally.
- Small, blunt objects: Observation for 24 hours
- Sharp objects, batteries, or large items: Early endoscopic removal
- 95% of objects that reach the stomach pass naturally.
- Follow-up: Monitor for abdominal pain, vomiting, fever, or bleeding Surgical intervention if obstruction or perforation occurs.
- Monitor for abdominal pain, vomiting, fever, or bleeding
- Surgical intervention if obstruction or perforation occurs.
- History of ingestion
- X-rays of the neck, chest, and abdomen
- Small, blunt objects: Observation for 24 hours
- Sharp objects, batteries, or large items: Early endoscopic removal
- 95% of objects that reach the stomach pass naturally.
- Monitor for abdominal pain, vomiting, fever, or bleeding
- Surgical intervention if obstruction or perforation occurs.

Indications for Immediate Endoscopic Removal of Foreign Bodies in the Esophagus:

Long-Stay Objects: Objects that have been in the oesophagus for an extended period, especially if symptomatic or at risk for causing complications.

Solution for Question 3:

Correct Answer: C) Technetium-99m pertechnetate scan has low sensitivity for diagnosis

Explanation:

The Technetium-99m pertechnetate scan (Meckel scan) is highly sensitive for diagnosing Meckel's diverticulum, with an approximate sensitivity of 85%.

Meckel Diverticulum: Most common congenital GI anomaly (2-3% of infants) with incomplete obliteration of omphalomesenteric duct.

Characteristics ("Rule of 2s")	2% of population 2 feet from the ileocecal valve (Option B) 2 inches long 2 types of ectopic tissue (gastric or pancreatic) (Option D) Often presents before age 2 2 times more common in males
Clinical Manifestations	Painless rectal bleeding (most common) (Option A) Bowel obstruction Diverticulitis
Diagnosis	Meckel scan (technetium-99m pertechnetate) (Option C) Sensitivity: 85% Specificity: 95% Used to diagnose ectopic gastric mucosa Plain radiographs and barium studies are less useful
Complications	Intussusception, Volvulus and Perforation.

Solution for Question 4:

Correct Answer: B) Duodenal atresia

Explanation:

- The ‘double bubble sign’ on the abdominal X-ray is characteristic of duodenal atresia, in which the obstruction occurs distal to the ampulla of Vater, leading to bilious vomiting.
- Polyhydramnios is associated with duodenal atresia in about 50% of cases.

Duodenal Atresia:

Pathophysiology	Failed recanalisation of intestinal lumen during gestation
Associated Conditions	Prematurity (50%)

	Congenital anomalies (heart disease, malrotation, annular pancreas, etc.) Chromosomal abnormalities (e.g., trisomy 21)
Clinical Presentation	Bilious vomiting without abdominal distention. Symptoms typically present within the first few days of life.
Diagnosis	Confirmed by the double bubble sign on the abdominal X-ray and prenatal ultrasonography.
Treatment	Initial: Nasogastric decompression and IV fluids Definitive: Surgical repair (Duodenoduodenostomy) Postoperative care: Gastrostomy tube, nutritional support

Jejunal atresia (Option A) is characterised by multiple dilated loops of the small bowel and a " triple bubble sign" on X-ray, not the double bubble sign. It can cause bilious vomiting, and polyhydramnios association is less common compared to duodenal atresia.

Pyloric Stenosis (Option C) typically presents around 3-6 weeks of age with non-bilious vomiting due to obstruction proximal to the ampulla of Vater. X-ray shows a distended stomach with minimal gas in the small intestine.

Malrotation with midgut volvulus (Option D) presents with bilious vomiting in the newborn period and typically shows a "bird's beak" appearance or paucity of gas in the abdomen on X-ray. It is a surgical emergency, often involving acute symptoms and abdominal distension.

Solution for Question 5:

Correct Answer: C) Rectal suction biopsy

Explanation:

- Rectal suction biopsy is the Gold Standard Test for confirming Hirschsprung disease.
- Procedure: A biopsy must be taken 2 cm above the dentate line and must include adequate submucosa.
- Staining: Acetylcholinesterase: Identifies hypertrophied nerve bundles and the absence of ganglion cells. Calretinin: Used as an alternative when acetylcholinesterase staining is inadequate.
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Contrast enema (Option A): Useful for showing transition zones between dilated and narrow segments but not as reliable as rectal biopsy. Sensitivity ~70%, specificity 50-80%.

Anorectal manometry (Option B): Evaluates rectoanal inhibitory reflex (RAIR) with high sensitivity and specificity (>90%), but not considered the definitive test for young infants.

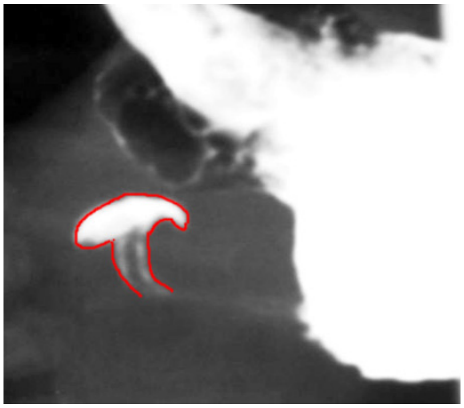
Colonoscopy (Option D) can visualise the bowel but does not provide the definitive histological information needed to confirm Hirschsprung disease.

Solution for Question 6:

Correct Answer: B) Lack of ganglionic cells in the submucosal and myenteric plexuses

Explanation:

The given clinical scenario suggests Hirschsprung disease, characterised by the lack of ganglionic cells in the submucosal (Meissner’s) and myenteric (Auerbach’s), leading to absent peristalsis and functional obstruction in the affected bowel segment.



Hirschsprung Disease:

- It is the most common congenital motility disorder and is characterised by the absence of ganglionic cells in the distal intestine.
- Pathophysiology: The absence of ganglionic cells (aganglionosis) in the distal segment of the intestine leads to tonic contraction and functional obstruction of the affected bowel segment.
- Classification: Short Segment (75%): Limited to the segment distal to the splenic flexure. Long Segment (20%): Extends proximal to the splenic flexure. Total Colonic Aganglionosis (5%): The entire colon is affected.
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- Clinical Presentation Neonates: Delayed passage of meconium, constipation, obstruction Later presentation: Chronic constipation, failure to thrive, enterocolitis
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- Diagnosis: X-ray: Reveals bowel distension and air-fluid levels. Barium enema: Narrow aganglionic segment, transition zone, proximal dilation Anorectal manometry: Absence of rectoanal inhibitory reflex Rectal biopsy (gold standard): Absence of ganglionic cells
- X-ray: Reveals bowel distension and air-fluid levels.
- Barium enema: Narrow aganglionic segment, transition zone, proximal dilation

- Anorectal manometry: Absence of rectoanal inhibitory reflex
- Rectal biopsy (gold standard): Absence of ganglionic cells
- Management: Primarily Surgical: Resection of the aganglionic bowel and pull-through of normal bowel. Medical Management: Stabilization and treatment of enterocolitis prior to surgery. Surgical Approach: Trend toward less invasive techniques, including laparoscopic and single-stage procedures.
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- Surgical Approach: Trend toward less invasive techniques, including laparoscopic and single-stage procedures.

Absence of smooth muscle in the bowel wall (Option A): Hirschsprung disease does not involve a lack of smooth muscle; rather, it affects the neural elements within the bowel wall.

Hypertrophy of the circular muscle layer in the entire colon (Option C): While hypertrophy may occur in the affected segment, the primary pathology is the absence of ganglionic cells, not generalised hypertrophy of muscle layers.

Excessive production of acetylcholine in the enteric nervous system (Option D): The disorder is not due to excessive acetylcholine but to the absence of ganglionic cells, which impairs normal neurotransmission and bowel motility.

Solution for Question 7:

Correct Answer: D) Intussusception.

Explanation:

The clinical features of a sausage-shaped abdominal mass, red currant jelly stools, abdominal pain, and the ‘coiled spring sign’ on barium enema studies suggest intussusception.

Intussusception:

- It refers to the telescoping of a proximal segment of the intestine (intussusceptum) into a distal segment (intussuscipiens).
- Most common cause of intestinal obstruction in 5 months to 6 years of age.
- It may be ileocolic (most common), colocolic or ileoileal.

Clinical features:

- The classic triad of abdominal pain, red currant jelly stools (blood and mucus) and palpable mass are observed.
- Barium enema shows a 'Coiled Spring Sign or Claw Sign'..

Management:

- Spontaneous resolution occurs in a few patients.
- Hydrostatic reduction under Fluoroscopic or USG guidance.
- Surgical reduction may be required if the child has refractory shock, suspected intestinal perforation or necrosis, or in case of multiple recurrences or peritonitis.

Hirschsprung disease (Option A), is characterized by absence of ganglion cells in the distal intestine, can present as delayed passing of meconium or constipation. Barium enema shows the dilated proximal segment and the transition zone.

Duodenal atresia (Option B) is characterised by bilious vomiting without abdominal distention and shows the double bubble sign on the x-ray.

Sigmoid volvulus (Option C) is caused by twisting of the small intestine, leading to symptoms like abdominal pain, bloating, and inability to pass gas and shows a 'coffee bean sign' on the x-ray as shown,

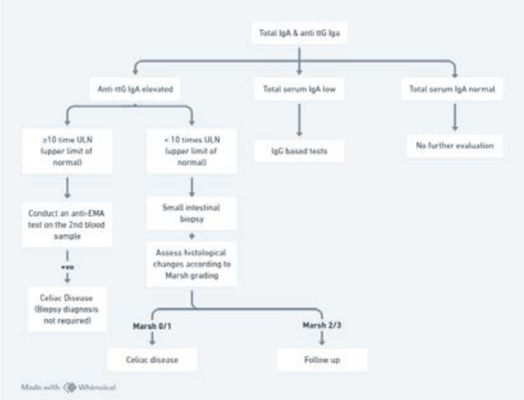
Solution for Question 8:

Correct Answer: C) Small intestinal biopsy

Explanation:

- The symptoms and elevated anti-tTG IgA suggest Celiac disease. According to the 2020 ESPGHAN guidelines, after confirming elevated anti-tTG IgA, a small intestinal biopsy is the next step for a definitive diagnosis.

- If a person is asymptomatic or at risk of developing celiac disease, then HLA-DQ2 and HLA-DQ8 genotyping is done.



Measure total serum IgG levels (Option A): Not needed; total serum IgA levels are checked to ensure accurate anti-tTG IgA results.

Test for anti-endomysial antibodies (EMA) (Option D) are done when anti-ttG IgA levels are elevated to 10 or more times the upper limit of normal.

Solution for Question 9:

Correct Answer: D) Treatment includes a high-fiber wheat diet.

Explanation:

- The given scenario and biopsy findings are consistent with Celiac disease or Gluten-sensitive enteropathy, a T-cell mediated autoimmune disorder associated with HLADQ2 & HLADQ8 and Down’s syndrome.
- Treatment involves a gluten-free diet (avoid wheat, rye, barley and oats).
- High-risk groups include type 1 diabetes mellitus, Down syndrome (Option C), selective IgA deficiency, autoimmune thyroid disease, Turner syndrome, Williams syndrome, Addison’s disease, autoimmune liver disease and first-degree relatives of patients with celiac disease.
- A temporal association may be present with the introduction of wheat products at weaning.
- Clinical manifestations: Diarrhoea, growth failure, and anaemia. Temporal association with the introduction of wheat products at weaning. Abdominal distension, vomiting, anorexia, aphthous stomatitis, and bloating. Rickets, osteoporosis, short stature, muscular atrophy,

peripheral neuropathy, and irritability might be seen in some cases. Dermatitis herpetiformiscan be seen.

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- Temporal association with the introduction of wheat products at weaning.
- Abdominal distension, vomiting, anorexia, aphthous stomatitis, and bloating.
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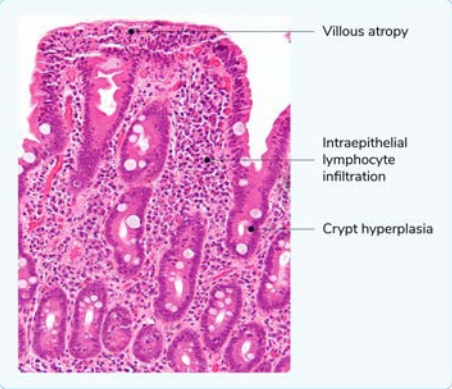
- Rickets, osteoporosis, short stature, muscular atrophy, peripheral neuropathy, and irritability might be seen in some cases.
- Dermatitis herpetiformis can be seen.

Diagnosis:

ESPGHAN (European Society for Pediatric Gastroenterology, Hepatology & Nutrition) guidelines 2020 suggest that in all suspected cases (symptomatic, asymptomatic, and at-risk group), the first test should be total serum IgA and IgAtTG.

Serology	Anti-TTG (Tissue Transglutaminase) Anti EMA (Endomysial antibody) Anti-DGP (Deamidated gliadin peptide)
Upper GI endoscopy	The endoscopy may be normal or show the absence of folds or scalloped duodenal folds.
Histopathology	Villous atrophy/Flat villi Crypt hyperplasia Intraepithelial lymphocyte infiltration.



Solution for Question 10:

Correct Answer: C) Lactose intolerance

Explanation:

The given scenario of explosive stools (mixed with gas and passed with noise) after consuming milk suggests Lactose Intolerance, a condition in which the baby cannot digest milk and its products.

Pathogenesis	Impaired digestion of lactose due to insufficient lactase enzyme leads to gut intolerance.
Types	Congenital or Primary: It is due to a mutation of the lactose gene. Acquired or Secondary (most common): Secondary to some infection, persistent diarrhoea, drugs, radiation, etc.
Clinical features	Diarrhea, abdominal pain and vomiting The characteristic feature is the passing of explosive stools (i.e., stools mixed with gas and passed with noise), especially after consuming milk or its products. Perianal excoriation because of acidic stools.
Diagnosis	Resolution of symptoms by avoiding milk products. Stool pH <6.5
Management	Avoid milk and its products (lactose-free diet) Vitamin A and zinc supplements.

Irritable bowel syndrome (Option A) is characterised by abdominal pain occurring at least 4 days per month for 2 months, with pain related to defecation, and accompanied by changes in stool frequency and consistency.

Small intestinal bacterial overgrowth (Option B) (SIBO) is a condition in which large intestinal bacteria overgrow in the small intestine, causing symptoms like bloating, abdominal pain, and diarrhoea, independent of milk consumption.

Diverticulosis (Option D) is the outpouching of the walls of the intestine (colon or small intestine), which can cause bloating, diarrhoea, abdominal pain, or constipation and is unrelated to milk consumption.

Solution for Question 11:

Correct Answer: B) Positioning therapy and thickened feedings.

Explanation:

- The infant’s symptoms suggest gastroesophageal reflux disease GERD, commonly seen in infants due to the immaturity of the lower esophageal sphincter.
- Positioning therapy and thickened feedings are the preferred initial management for mild to moderate GERD symptoms in infants.
- Positioning therapy involves keeping the infant upright during and after feedings to reduce reflux.
- Thickened feedings, often with rice cereal, reduce regurgitation by preventing stomach contents from flowing back into the esophagus.

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Start proton pump inhibitors (PPIs) immediately (Option A): PPIs are not first-line treatment in infants and are reserved for cases with severe symptoms like esophagitis or when non-pharmacologic interventions fail.

Schedule an urgent esophagogastroduodenoscopy (EGD) (Option C): EGD is not the first step for managing infant GERD symptoms and is reserved for cases with persistent symptoms despite treatment or concerning signs like bleeding or significant weight loss.

Antireflux surgery (fundoplication)(Option D) is reserved for severe GERD cases unresponsive to medical therapy or with complications like significant esophagitis, strictures, or a large hiatal hernia.

Solution for Question 12:

Correct Answer: C) 1500 mL/day

Explanation:

To calculate the daily maintenance fluid requirement for pediatric patients, we use the following Hollidag Segar formula based on weight:

- For the first 10 kg: 100 mL/kg/day
- For the next 10 kg: 50 mL/kg/day
- For each kg above 20 kg: 20 mL/kg/day

For a 6-year-old boy weighing 22 kg admitted with pneumonia and requiring IV fluids for maintenance hydration:

- For the first 10 kg: 10 kg × 100 mL/kg = 1000 mL
- For the next 12 kg: 10 kg × 50 mL/kg = 500 mL
- Total Daily Maintenance: 1000 mL + 500 mL = 1500 mL/day

1500 mL/day (Option C) is the most appropriate answer.

The total fluid requirement for this child in the first 24 hours consists of three components:

- Deficit replacement (rehydration): For severe dehydration, we estimate a 10% fluid deficit
- Maintenance fluids
- Ongoing losses: Estimate 10 mL/kg for each watery stool and 2 mL/kg for each episode of vomiting.

Total fluid requirement = Deficit + Maintenance + Ongoing losses

Solution for Question 13:

Correct Answer: C) Ringer's Lactate with 5% Dextrose

Explanation:

- Ringer's Lactate with 5% Dextrose is commonly used for treating severe dehydration in children. It offers a balanced electrolyte composition similar to plasma, helps correct acidosis, and includes glucose to prevent hypoglycemia.
- Rapid intravenous rehydration is crucial. The total fluid volume is typically 100 mL/kg. For children >12 months: Administer over 3-4 hours. For infants <12 months: Administer over 6-8 hours.
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- For infants <12 months: Administer over 6-8 hours.

- Continual reassessment is essential to adjust treatment based on the child’s response and hydration status.
- For children >12 months: Administer over 3-4 hours.
- For infants <12 months: Administer over 6-8 hours.

Warning signs of severe dehydration:

- Lethargy or altered consciousness
- Sunken eyes
- Unable to drink
- Cool, mottled extremities (indicating poor perfusion)
- Severely decreased skin turgor
- Weak or absent pulse

5% Dextrose solution (Option A) provides glucose but lacks adequate electrolytes and does not effectively correct acidosis or replace lost fluids in cases of severe dehydration.

Normal Saline (Option B) is useful for initial fluid resuscitation but does not correct acidosis and is less balanced in electrolytes than Ringer's Lactate.

Half-normal saline (Option C) contains lower sodium and chloride concentrations and does not correct acidosis, making it unsuitable for severe dehydration.

Solution for Question 14:

Correct Answer: C) Continue normal diet with adequate fluid intake and ORS.

Explanation:

- In cases of acute diarrhoea in young children without signs of dehydration, continue the normal diet and use ORS (Oral Rehydration Solution) to support recovery, prevent nutritional deficiencies, and maintain hydration.
- Acute diarrhoea is defined as the passage of three or more loose or liquid stools per day, lasting < 14 days. It is often caused by gastrointestinal infections from viruses, bacteria, or parasites.

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Management

Aspects	Recommendations
Rehydration	Use Oral Rehydration Solution (ORS) for most cases Start with small volumes if vomiting is present Consider IV for severe cases
Nutrition	Continue breastfeeding Reintroduce age-appropriate diet early Avoid fatty foods and simple sugars
Zinc Supplementation	<6 months: 10 mg/day >6 months: 20 mg/day
Medications	Antibiotics: Use judiciously, mainly for severe bacterial cases Ondansetron (for vomiting) Avoid antimotility agents
Probiotics	It may be beneficial, especially for rotavirus

ORS Composition (WHO standard)

Constituent	mmol/L
Sodium	75
Glucose	75
Potassium	20
Chloride	65
Citrate	10
Total osmolarity	245

Start ORS and withhold solid foods for 24 hours (Option A): While ORS is crucial, withholding food is outdated advice. Early refeeding with an age-appropriate diet is recommended to prevent nutritional deficits and doesn't delay recovery.

Prescribe an antibiotic and advise clear fluids only (Option B): Antibiotics aren't routinely recommended for acute diarrhoea, particularly in viral cases, and clear fluids alone do not meet nutritional needs.

Administer ondansetron and start a probiotic supplement (Option D): Ondansetron is not needed since there is no vomiting, and probiotics are supplementary rather than primary treatment for diarrhoea.

Solution for Question 15:

Correct Answer: C) Some dehydration

Explanation:

The infant's sunken fontanelle, dry mucous membranes, prolonged capillary refill, and slow skin recoil indicate 'some dehydration'.

Assessment of Dehydration in Children:

Classification of Dehydration in children:

Sign/Symptom	Mild/No Dehydration	Some Dehydration	Severe Dehydration
Mental status	Alert	Irritable/restless	Lethargic/unconscious
Thirst	Normal	Increased	Unable to drink
Eyes	Normal	Slightly sunken	Very sunken
Tears	Present	Decreased	Absent
Mouth	Moist	Dry	Parched
Skin turgor	Normal (Immediate skin recoil)	Slow recoil (2-3sec)	Very slow recoil (>3sec)
Capillary refill	Normal (<2sec)	Prolonged (2-3sec)	Prolonged (>3sec)
Urine output	Normal	Decreased	Minimal

Solution for Question 16:

Correct Answer: D) Correct fluid and electrolyte imbalances

Explanation:

- The scenario is consistent with congenital hypertrophic pyloric stenosis (CHPS), where persistent vomiting leads to dehydration and hypochloremic metabolic alkalosis.
- The most appropriate next step is to correct fluid and electrolyte imbalances using intravenous normal saline with added potassium.

- Stabilisation is crucial before any surgical intervention to prevent complications during and after surgery.

Surgical Treatment of Congenital Hypertrophic Pyloric Stenosis (CHPS):

- Pyloromyotomy is the definitive surgical treatment indicated for infants with CHPS after stabilisation of fluid and electrolyte imbalances.
- Technique: The procedure involves incising and splitting the hypertrophied pyloric muscle to relieve the obstruction. It can be performed via an open (Ramstedt pyloromyotomy) or laparoscopic approach.
- Postoperative Care: Initiate feeding 12–24 hours after surgery and progress to full feeds within 36–48 hours.
- Conservative Management: Used for patients at high surgical risk or when surgical expertise is unavailable. They include: Nasoduodenal Feeding: To bypass the obstructed pylorus and ensure adequate nutrition. Atropine Sulfate Administration: Administered intravenously or orally to reduce pyloric muscle contractions.
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Challenges: Conservative management may involve prolonged treatment, which can lead to nutritional deficiencies, growth delays, and potential persistence or worsening of symptoms.

Solution for Question 17:

Correct Answer:D) Advanced maternal age

Explanation:

Familial history of CHPS, Male sex, and Maternal use of macrolide antibiotics are known risk factors for CHPS. In contrast, advanced maternal age is not a risk factor for this condition.

Risk Factors of Congenital hypertrophic pyloric stenosis	

Genetic Factors	White race of northern European descent Male sex Familial History: Higher risk in offspring of mothers with CHPS (20% for sons, 10% for daughters) Twin Studies: Higher concordance in monozygotic twins suggests a genetic component Blood Groups: Infants with B and O blood groups.
Environmental Factors	Erythromycin Use: Especially if administered within the first two weeks of life Higher incidence in infants whose mothers used macrolides during pregnancy or breastfeeding, particularly in females.
Other Associated Factors	Other Congenital Defects Syndromes: Occasionally associated with tracheoesophageal fistula and hypoplasia or agenesis of the inferior labial frenulum. Associated with eosinophilic gastroenteritis, Apert syndrome, Zellweger syndrome, trisomy 18, Smith-Lemli-Opitz syndrome, and Cornelia de Lange syndrome.

Solution for Question 18:

Correct Answer: A) Elongated pyloric channel on contrast studies

Explanation:

- The clinical scenario is most likely Congenital hypertrophic pyloric stenosis (CHPS), a condition involving hypertrophy of the pyloric muscle, leading to gastric outlet obstruction in male infants, typically between 2 to 6 weeks of age.
- Clinical Presentation: Symptoms: Projectile, non-bilious vomiting after feeds, starting around 2 to 6 weeks of age. Physical Examination: Visible gastric peristalsis and a palpable “olive-shaped” mass in the epigastric region.
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- Physical Examination: Visible gastric peristalsis and a palpable “olive-shaped” mass in the epigastric region.

- Diagnosis: Ultrasound (USG) is the Investigation of choice for CHPS, characterised by Pyloric muscle thickness of > 3 mm (Option B ruled out) Pyloric length > 14 mm (Option C ruled out) The pyloric channel appears elongated and thickened (10-14 mm)
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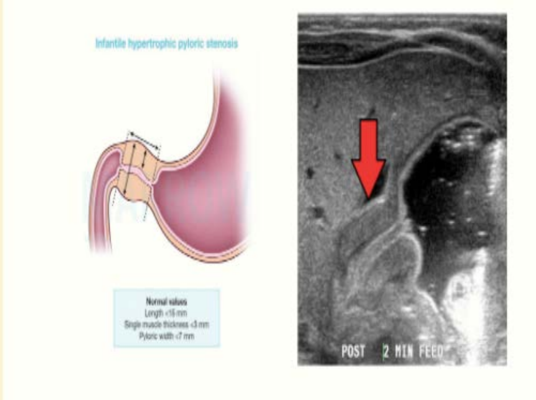
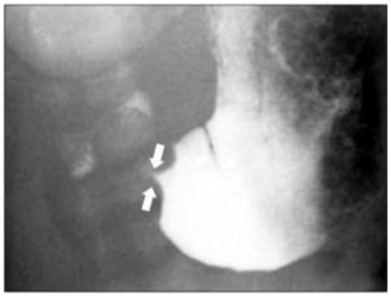
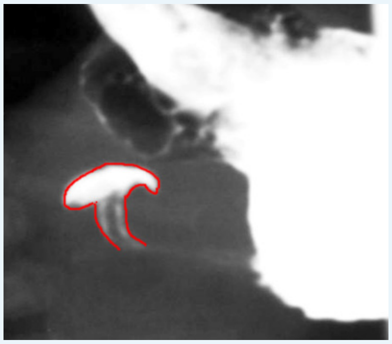


- Contrast Studies (Upper GI Series): Less frequently used, demonstrating the classic "string sign" with a thin line of contrast through the elongated and narrowed pyloric channel and delayed gastric emptying without duodenal dilation. (Option D)

Characteristic findings of Contrast studies in CHPS	
Findings	Description

String Sign	 A thin line of contrast material passes through the narrowed pyloric channel, indicating severe narrowing due to hypertrophied muscle.
Shoulder Sign	A bulge of the hypertrophied pyloric muscle into the gastric antrum gives the appearance of a “shoulder.”
Double Tract Sign	Parallel streaks of contrast material within the narrowed pyloric channel suggest the presence of a narrow passage for contrast flow. hypertrophic pyloric stenosis " data-author="kely mendez" data-hash="954" data-license="NA" data-source="https://radiopaedia.org/articles/double-track-sign-pyloric-stenosis" data-tags="March2025,Radiological Images" src="https://image.prepladder.com/notes/OjRAk7fZQSNVt5raa1ue1742051425.jpg" />
Mushroom sign	 Bulging of the hypertrophic muscle into the 1st part of the duodenum

Solution for Question 19:

Correct Answer: D) Cleft palate

Explanation:

Tracheoesophageal fistula, Radial limb defects, Esophageal atresia, vertebral defect and anal atresia are typical findings seen in VACTERL association, whereas cleft palate is not associated with this condition.

VACTERL Anomalies:

- Vertebral anomalies
- Anorectal malformations
- Cardiac defects
- Tracheoesophageal fistula (TEF)
- Esophageal atresia
- Renal anomalies
- Limb defects (Involve Radial limb defects such as the absence or underdevelopment of the radius bone, thumb anomalies, and shortening of the forearm)

Syndromes associated with Tracheoesophageal fistula

- VACTERL
- Feingold syndrome (N -MYC mutation)
- CHARGE syndrome (CHD7 Mutation): Coloboma of the eyes, CNS anomalies, Atresia of choanae, retardation of growth and development, genital and urinary defects, deafness.
- Anophthalmia-esophageal-genital syndrome (SOX 2 mutation)

Solution for Question 20:

Correct Answer: C)Type C

Explanation:

This scenario emphasises the key radiographic finding of a gas-filled stomach in Type C tracheoesophageal fistula (TEF), aligning with a distal tracheoesophageal fistula with proximal esophageal atresia.

Tracheoesophageal fistula	
Causes	Embryology: Disruption during embryonic development of the trachea and oesophagus. Genetic Factors: Associated with syndromes like Feingold syndrome, CHARGE syndrome, and

	anophthalmia-esophageal-genital syndrome. Unknown Etiology: Factors such as advanced maternal age, ethnicity, obesity, low socioeconomic status, and smoking may be involved.
Types of TEF	Type A: Isolated oesophagal atresia without tracheoesophageal fistula Type B: Esophageal atresia with proximal tracheoesophageal fistula Type C: Proximal oesophagal atresia with distal tracheoesophageal fistula Type D: Esophageal atresia with both proximal and distal tracheoesophageal fistulas Type E: Isolated tracheoesophageal fistula without associated oesophagal atresia
Presentation	Classic Symptoms: Frothing and bubbling at the mouth and nose, coughing, cyanosis, respiratory distress exacerbated by feeding. Aspiration Pneumonia: More severe with distal fistula. H-Type Fistula: Chronic respiratory issues, bronchospasm, recurrent pneumonia in isolated cases.
Diagnosis	Esophagogram Chest X-ray shows a gas-filled or gasless abdomen and a coiled tube in the upper oesophagus. fistula chest xray" data-author="Kinderradiologie Olgahospital Klinikum Stuttgart " data-hash="951" data-license="CC BY-SA 4.0" data-source="https://en.wikipedia.org/wiki/Tracheoesophageal_fistula#/media/File:H-Fistel_neugeb_05032014.jpg" data-tags="March2025" src="https://image.prepladder.com/notes/N2R4zHjhKE3QOhSm7TE21742050022.jpg" />
Management	Immediate Care: Maintain airway patency, prevent aspiration, and use antibiotics. Surgical Repair: Standard involves ligation of TEF and primary end-to-end anastomosis. Alternative Approaches: Delayed primary closure, fistula ligation, gastrostomy tube placement, or using gastric, jejunal, or colonic segments if the gap >3-4 cm.
Complications	Respiratory Complications: Risk of aspiration pneumonia. Esophageal Complications: Feeding difficulties, failure to thrive. Associated Anomalies: Often part of VACTERL association (vertebral, anal, cardiac, tracheal, esophageal, renal, limb).
Prognosis	Improved Outcomes: Better survival rates due to advancements in care. Long-Term Complications: Potential issues include feeding difficulties, esophageal strictures, and respiratory problems.

Type A (Option A): The upper esophageal pouch ends blindly with no connection to the trachea. X-ray shows a gas-filled upper oesophageal pouch but no gas in the stomach or lower esophagus.

Type B (Option B): The fistula is not connected to the distal esophagus, so it does not cause stomach distention. X-ray shows a gas-filled upper and lower esophageal pouch and trachea but no gas in the stomach.

Type D (Option D): Both proximal and distal fistulas connect the upper and lower oesophagal segments to the trachea. X-ray shows air in both upper and lower esophageal pouches and gas in the stomach due to the dual connections.

Tracheoesophageal fistula" data-author="NA" data-hash="950" data-license="NA" data-source="NA" data-tags="Anatomical Illustration,March2025" src="https://image.prepladder.com/notes/Cbm6qSkYGx0Z0RLjc5rx1742049473.png" />

Liver Disorders in Children

1. Which statement about unconjugated hyperbilirubinemia is incorrect?

A. Chronic hemolytic anemias increase unconjugated bilirubin.
B. Crigler-Najjar Type I has a partial UDPGT deficiency.
C. Gilbert Syndrome involves a mild UDPGT deficiency.
D. Phenobarbitone treats Crigler-Najjar Type II.

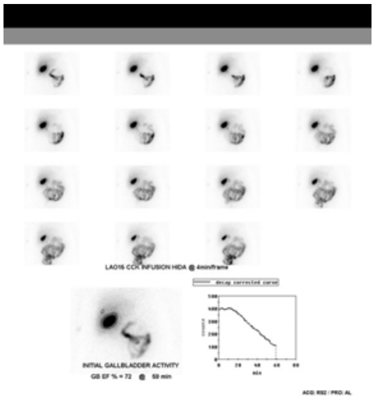
2. Which statement about conjugated hyperbilirubinemia is incorrect?

A. Conjugated bilirubin is >2 mg/dL or >20% of total bilirubin.
B. Dubin-Johnson Syndrome involves a defect in hepatic uptake.
C. Rotor Syndrome affects hepatic uptake, storage, and excretion of bilirubin.
D. PFIC type 3 has elevated GGT levels.

3. A 7-week-old infant presents with jaundice, pale stools, and dark urine. Liver function tests reveal elevated conjugated bilirubin and alkaline phosphatase. An abdominal ultrasound shows a triangular cord sign. Which of the following is considered the best for confirming the diagnosis?

A. Ultrasound of the abdomen
B. HIDA scan
C. Liver biopsy
D. MRCP

4. A neonate presents with conjugated hyperbilirubinemia. A HIDA scan is performed and results are shown below. What is the most probable diagnosis?



A. Extrahepatic biliary atresia (EHBA)

B. Neonatal hepatitis

C. Acute Cholecystitis

D. Bile duct stricture

5. A 7-year-old child, post bone marrow transplantation for leukemia, presents with progressive jaundice, hepatomegaly, and increasing weight due to ascites and fluid retention. The child had undergone total body irradiation prior to the transplant. Which of the following diagnoses should be most strongly considered?

A. Acute hepatic failure

B. Budd-Chiari syndrome

C. Sinusoidal obstruction syndrome

D. Autoimmune hepatitis

6. A 12-year-old boy with a history of chronic liver disease presents with increasing shortness of breath that worsens when he stands up and improves when lying down. His parents also report that he has been more fatigued lately. On physical examination, he has digital clubbing and cyanosis. Contrast echocardiography reveals intrapulmonary vascular dilatations. What is the most likely diagnosis?

A. Congestive cardiac failure

B. Hepatopulmonary syndrome

C. Porto-Pulmonary hypertension

D. Primary ciliary dyskinesia

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7. A 9-year-old girl presents with a 2-week history of abdominal pain, progressive abdominal distension, and leg swelling. On examination, she has hepatomegaly, ascites, and mild jaundice. Doppler ultrasound reveals absent hepatic vein flow with collateral vessels. Which of the following is the most likely diagnosis?

A. Wilson's disease

B. Hepatopulmonary syndrome

C. Budd-Chiari syndrome

D. Extrahepatic portal vein obstruction

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8. A 7-year-old boy is brought to the pediatric clinic with complaints of jaundice, fatigue, and abdominal pain for the past 5 days. The child has a history of poor appetite and dark urine. On examination, he is mildly icteric, with tender hepatomegaly but no splenomegaly. His liver function tests reveal elevated ALT and AST levels, with a significant rise in total and direct bilirubin. Anti-HAV IgM is positive. Which of the following is the most appropriate next step in management?

A. Start antiviral therapy

B. Administer Hepatitis A vaccine

C. Supportive care with hydration and monitoring

D. Admit for immediate liver biopsy

9. A 6-year-old child is brought to the hospital with a 2-week history of fever, joint pain and fatigue. On examination, jaundice with mild hepatic tenderness is noted. The child's mother reports that the child had a blood transfusion a few months back due to massive blood loss in a car accident. Serological testing reveals: Hepatitis B surface antigen (HBsAg): Positive Hepatitis B surface antibody (anti-HBs): Negative Hepatitis B core antibody (anti-HBc): Positive (IgM) Hepatitis B e antigen (HBeAg): Positive Which among the following options best describes the stage of hepatitis B infection?

A. Acute infection

B. Chronic infection

C. Acute resolving infection

D. Early acute infection

10. An 8-hour-old infant was born to a mother who is known to be positive for hepatitis B virus (HBV). The mother's HBV serology shows high levels of hepatitis B surface antigen (HBsAg) and hepatitis B e antigen (HBeAg). What is the most appropriate initial management for this newborn to prevent HBV infection?

A. Administer hepatitis B vaccine only

B. Administer hepatitis B immune globulin (HBIG) only

C. Administer hepatitis B vaccine and hepatitis B immune globulin (HBIG) within 12 hours of birth

D. To complete 2 doses of hepatitis B vaccine universal vaccination series

11. A 3-year-old child's mother is concerned about hepatitis C transmission to her kid as her uncle was recently diagnosed with hepatitis C. The Child does not have any symptoms and examination findings are appropriate for her age. Which of the following is the most common transmission mode of infection in children?

A. Perinatal

B. Blood transfusion

C. Illegal drug use with exposure to blood

D. Breastfeeding

---

12. A 15-year-old patient has a history of progressive dystonia, tremors, and behavioral changes. On physical examination, a Kayser-Fleischer ring is observed in the corneal margin. Laboratory tests reveal elevated liver enzymes. A 24-hour urinary copper excretion test shows markedly increased copper levels. Which of the following statements is NOT true regarding the disorder in this patient?

A. It is an autosomal recessive mode of inheritance

B. Decreased biliary copper excretion leading to copper accumulation in organs

C. Increased serum ceruloplasmin levels

D. D-penicillamine is used for treatment

---

13. A 7-year-old child is brought to the emergency department with a recent history of influenza-like illness which was followed by sudden onset of vomiting, lethargy, and altered mental status. Parents report that the child was treated with aspirin for fever during the influenza episode. On examination, there are no signs of increased intracranial pressure and no focal neurological deficit. Given the diagnosis of Reye syndrome, what is the most appropriate management among the following options?

A. Supportive treatment

B. Mannitol or hypertonic saline

C. Continue aspirin use for fever

D. D-penicillamine administration

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	2
Question 5	3
Question 6	2
Question 7	3
Question 8	3
Question 9	1
Question 10	3
Question 11	1
Question 12	3
Question 13	1

Solution for Question 1:

Answer: B) Crigler-Najjar Type I has a partial UDPGT deficiency.

Explanation:

Crigler-Najjar Type I involves a complete deficiency of UDP-glucuronyltransferase, while Type II involves a partial deficiency.

Unconjugated Hyperbilirubinemia

- Increase in indirect fraction of bilirubin

Increased Production

- Chronic hemolytic anemias (e.g., Thalassemia, Sickle Cell Disease) (Option A ruled out)
- Hereditary spherocytosis
- Autoimmune hemolytic anemia

Decreased Conjugation

- Crigler-Najjar Syndrome
- Gilbert Syndrome

Unconjugated Hyperbilirubinemia					
Condition	Inheritance	Gene Involved	Deficiency	Features	Management
Gilbert Syndrome	Autosomal recessive & dominant	UGT 1A1	Mild deficiency of UDP-GT (Option C ruled out)	Recurrent jaundice during stress, fasting& fatigue	No specific treatment
Crigler-Najjar Syndrome	Autosomal recessive	UGT 1A1	Type1:Complete deficiency of UDP-GT	Severe Jaundice	Phenobarbitone is useful in Type2(Option D ruled out)
Type2:Partial deficiency of UDP-GT	Less Severe Jaundice				

Solution for Question 2:

Answer: B)Dubin-Johnson Syndrome involves a defect in hepatic uptake.

Explanation:

Impaired excretion of conjugated bilirubin occurs in Dubin-Johnson syndrome.

Conjugated Hyperbilirubinemia

- Conjugated (direct) bilirubin >2 mg/dL or >20% of total bilirubin (Option A is ruled out)

Examples:

Dubin-Johnson Syndrome

Rotor Syndrome

Progressive Familial Intrahepatic Cholestasis (PFIC)

Conjugated Hyperbilirubinemia					
Condition	Inheritance	Gene Involved	Deficiency	Features	Management
Dubin-Johnson Syndrome	Autosomal recessive & dominant	ABCC2(MRP2)	Impaired hepatic excretion of conjugated bilirubin	Liver appearance: Dark or pigmented	No specific treatment
Rotor Syndrome	Autosomal recessive	Unknown (deficiency in binding anions)	Defect in hepatic uptake, storage, and biliary excretion of conjugated bilirubin (Option C ruled out)	Liver histology is normal	No specific treatment
Progressive Familial Intrahepatic Cholestasis (PFIC)	Autosomal recessive	ABCB4(MDR3)	Inhepatocellular transport system genes involved in bile formation	Elevated GGT enzyme level only in PFIC type 3 (Option D ruled out) Progressive cholestatic jaundice starting in childhood	–

Solution for Question 3:

Answer: C) Liver biopsy

Explanation:

The presentation is consistent with extra-hepatic biliary atresia (EHBA).Liver biopsy provides definitive histological evidence of extra-hepatic biliary atresia by directly examining liver tissue for bile duct obstruction and damage.

Extra-Hepatic Biliary Atresia (EHBA)

Definition:

- Biliary atresia not associated with cyst formation.
- Also known as Non-Cystic Obliterative Cholangiopathy.
- Characterized by complete obliteration of extrahepatic bile ducts and intrahepatic ducts at the hilum.

Symptoms:

- Typically appear by 2 to 6 weeks of age Jaundice (yellowish coloration of skin and sclera) Abnormally pale stools Dark urine
- Jaundice (yellowish coloration of skin and sclera)
- Abnormally pale stools
- Dark urine
- Jaundice (yellowish coloration of skin and sclera)
- Abnormally pale stools
- Dark urine

Association with other Anomalies:

- The most common is polysplenia or asplenia
- Other associated malformations include absent inferior vena cava, cardiac anomalies, situs inversus, intestinal malrotation, duodenal atresia, anomaly of the portal vein, renal anomalies, and choledochal cyst.

Investigations:

- Increased conjugated bilirubin.
- Markers of cholestasis: 5' nucleotidase Alkaline phosphatase (ALP) GGT (Gamma-glutamyl transferase): Normal: Up to 40 U/L 4x elevation: Intrahepatic causes 10x elevation: Extrahepatic causes Exception: Low in PFIC type I and II
- 5' nucleotidase
- Alkaline phosphatase (ALP)
- GGT (Gamma-glutamyl transferase): Normal: Up to 40 U/L 4x elevation: Intrahepatic causes 10x elevation: Extrahepatic causes Exception: Low in PFIC type I and II
- Normal: Up to 40 U/L
- 4x elevation: Intrahepatic causes
- 10x elevation: Extrahepatic causes

- Exception: Low in PFIC type I and II
- 5' nucleotidase
- Alkaline phosphatase (ALP)
- GGT (Gamma-glutamyl transferase): Normal: Up to 40 U/L 4x elevation: Intrahepatic causes 10x elevation: Extrahepatic causes Exception: Low in PFIC type I and II
- Normal: Up to 40 U/L
- 4x elevation: Intrahepatic causes
- 10x elevation: Extrahepatic causes
- Exception: Low in PFIC type I and II
- Normal: Up to 40 U/L
- 4x elevation: Intrahepatic causes
- 10x elevation: Extrahepatic causes
- Exception: Low in PFIC type I and II
- High-frequency USG (HUS) for neonatal cholestasis.
- Used as initial investigation (Option A ruled out)
- Fasting USG is the gold standard when biliary atresia is suspected.
- Characteristic findings: Triangular cord sign (hyperechoic areas indicating fibrous/atretic bile duct) Non-visualized gallbladder/microgallbladder/gallbladder ghost
- Triangular cord sign (hyperechoic areas indicating fibrous/atretic bile duct)
- Non-visualized gallbladder/microgallbladder/gallbladder ghost
- Triangular cord sign (hyperechoic areas indicating fibrous/atretic bile duct)
- Non-visualized gallbladder/microgallbladder/gallbladder ghost
- HIDA Scan (Hepatobiliary scan with Tc labelled imino-di-acetic acid): Used as a screening test.(Option B ruled out) Decreased tracer excretion due to obstruction
- Used as a screening test.(Option B ruled out)
- Decreased tracer excretion due to obstruction
- Phenobarbitone loading/priming (5 days) to improve tracer uptake
- Used as a screening test, not specific
- Used as a screening test.(Option B ruled out)
- Decreased tracer excretion due to obstruction
- Liver biopsy:IOC for confirming diagnosis.

Surgical Management:

- Exploratory laparotomy
- Per Operative cholangiogram

- Temporary procedure: Kasai procedure/hepaticojejunostomy
- Definitive management: Liver transplantation
- Temporary procedure
- Creates communication between liver and jejunum for bile drainage
- Also known as hepaticojejunostomy
- Should be performed within 8 weeks after birth

MRCP(Option D): It is less definitive than liver biopsy for confirming the diagnosis.

Solution for Question 4:

Answer: B) Neonatal hepatitis

Explanation:

The given image shows normal filling of gallbladder and excretion of radioisotope into the intestine, suggesting neonatal hepatitis.

Neonatal Hepatitis:

It refers to inflammation of the liver in newborns, typically presenting within the first few weeks of life.

Etiology: Causes include viral infections (e.g., cytomegalovirus, rubella, hepatitis B), metabolic disorders, genetic conditions, and idiopathic origins.

Clinical Presentation: Jaundice, hepatomegaly, dark urine, and poor weight gain.The disease can present as either mild or severe, with the potential for liver failure in severe cases.

Diagnosis: Diagnosed through a combination of clinical evaluation, liver function tests (elevated ALT, AST), and specific serologic and metabolic tests.

- HIDA: Shows normal excretion of radioisotope into the intestine.
- Liver biopsy may show giant cell transformation, indicative of neonatal hepatitis.

Differential Diagnosis: Biliary atresia or genetic/metabolic liver diseases.

Treatment: Mainly supportive, including nutritional support and fat-soluble vitamin supplementation.

In acute cholecystitis (Option C): Filling of gallbladder is absent due to obstruction of cystic duct.

In Extrahepatic biliary atresia (EHBA) (Option A) and Bile duct stricture (Option D) there is absence of excretion of radioisotope into the intestines.

Solution for Question 5:

Answer: C) Sinusoidal obstruction syndrome

Explanation:

Sinusoidal obstruction syndrome or veno-occlusive disease is the most frequent cause of portal hypertension in children following bone marrow transplantation after total body irradiation.

Sinusoidal obstruction syndrome (venous occlusive disease):

- It is the most common cause of hepatic vein obstruction in children.
- Occlusion of the centrilobular venules or sublobular hepatic veins occurs.
- Most frequently occurs in bone marrow transplant recipients after total body irradiation with or without cytotoxic drug therapy, but can also be seen in patients on azathioprine, mercaptopurine, thioguanine, and those taking herbal remedies that contain pyrrolizidine alkaloids.

Causes of portal hypertension in children:

Prehepatic	Portal venous thrombosis Extrahepatic portal venous obstruction (cavernous transformation of portal vein) (most common cause) Isolated splenic vein thrombosis
Intrahepatic	Liver cirrhosis (most common intra hepatic cause) Congenital hepatic fibrosis Veno-occlusive disease (Sinusoidal obstruction syndrome) Noncirrhotic portal fibrosis Schistosomiasis Nodular regenerative hyperplasia
Posthepatic	Budd-Chiari syndrome (hepatic vein or inferior vena cava obstruction) Constrictive pericarditis

Acute hepatic failure (Option A), Budd-Chiari syndrome (Option B), Autoimmune hepatitis (Option D) are not associated with portal hypertension following bone marrow transplantations.

Solution for Question 6:

Answer: B) Hepatopulmonary syndrome

Explanation:

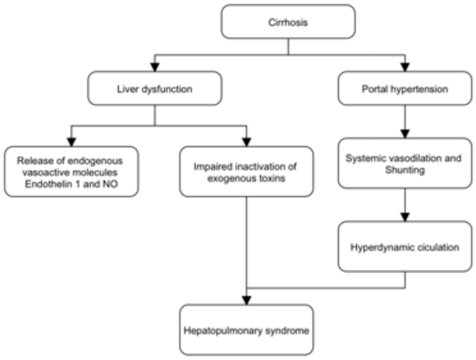
The above scenario describes a pediatric patient with chronic liver disease presenting with platypnea, orthodeoxia, clubbing and cyanosis which are classic features of hepatopulmonary syndrome (HPS)

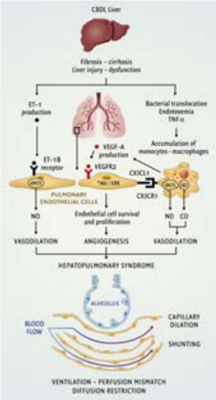
Hepato-pulmonary syndrome:

Triad:

- Chronic liver disease
- Alteration of arterial oxygenation
- Intrapulmonary vascular dilatations

Pathogenesis:





Clinical features:

- Platypnea - dyspnea induced in upright position and relieved by recumbency.
- Orthodeoxia - arterial deoxygenation accentuated in upright position and relieved by recumbency.
- Cyanosis
- Clubbing

Investigations: Contrast echocardiography is the most sensitive test to demonstrate intrapulmonary shunting.

Treatment: The only established effective therapy is liver transplantation.

Clinical manifestations of portal hypertension:

- Bleeding (most common)
- Splenomegaly (second most common)
- Ascites
- Growth failure
- Portal hypertensive biliopathy
- Hepatopulmonary syndrome
- Portopulmonary Hypertension Defined by a pulmonary arterial pressure greater than 25 mm Hg at rest or a left-ventricular end-diastolic pressure of less than 15 mm Hg. Patients with PP-HTN most commonly present with exertional dyspnea. (Option C ruled out) Histologically, these patients have pulmonary arteriopathy with laminar intimal fibrosis.
- Defined by a pulmonary arterial pressure greater than 25 mm Hg at rest or a left-ventricular end-diastolic pressure of less than 15 mm Hg.
- Patients with PP-HTN most commonly present with exertional dyspnea. (Option C ruled out)
- Histologically, these patients have pulmonary arteriopathy with laminar intimal fibrosis.

- Defined by a pulmonary arterial pressure greater than 25 mm Hg at rest or a left-ventricular end-diastolic pressure of less than 15 mm Hg.
- Patients with PP-HTN most commonly present with exertional dyspnea. (Option C ruled out)
- Histologically, these patients have pulmonary arteriopathy with laminar intimal fibrosis.

Congestive cardiac failure (Option A) causes exertional dyspnea and orthopnea.

Primary ciliary dyskinesia (Option D) is associated with other manifestations like dextrocardia, infertility.

Solution for Question 7:

Answer: C) Budd-Chiari syndrome

Explanation:

- In this scenario, the child presents with symptoms and signs consistent with hepatic vein obstruction, including hepatomegaly, ascites, and abdominal pain.
- Doppler ultrasound findings of absent hepatic vein flow with collateral vessel formation further suggest Budd-Chiari syndrome (BCS).

Budd-Chiari syndrome (BCS):

- Post-sinusoidal cause of portal hypertension.

Etiology and Pathogenesis:

Budd-Chiari syndrome occurs with obstruction to hepatic veins anywhere between the efferent hepatic veins and the entry of the inferior vena cava into the right atrium.

In most cases, no specific cause can be found, but thrombosis can occur from

- Inherited and acquired hypercoagulable states Antithrombin III deficiency Protein C or S deficiency Factor V Leiden Prothrombin mutations Paroxysmal nocturnal hemoglobinuria, Pregnancy Oral contraceptives
- Antithrombin III deficiency
- Protein C or S deficiency
- Factor V Leiden
- Prothrombin mutations

- Paroxysmal nocturnal hemoglobinuria,
- Pregnancy
- Oral contraceptives
- Hepatic or metastatic neoplasms
- Collagen vascular disease
- Infection
- Trauma
- Behçet syndrome,
- Inflammatory bowel disease,
- Aspergillosis,
- Dacarbazine therapy,
- Autoinflammatory-recurrent fever syndromes
- Inferior vena cava webs.
- Antithrombin III deficiency
- Protein C or S deficiency
- Factor V Leiden
- Prothrombin mutations
- Paroxysmal nocturnal hemoglobinuria,
- Pregnancy
- Oral contraceptives

Classification

- Primary (Majority): Obstruction due to endoluminal lesion (thrombosis or web).
- Secondary: Obstruction due to lesion outside the venous system (tumor abscess, cysts)(either by invading the lumen or extrinsic compression).

Presentation:

- Chronic course(common presentation): Hepatomegaly, abdominal distension and portal hypertension
- Acute or fulminant forms(rare): Abdominal pain, ascites, hepatomegaly and rapidly progressive hepatic failure.

Investigations:

- Doppler ultrasound
- Venography
- Investigations for hypercoagulable states.

Treatment:

Directed towards restoring the patency of hepatic vein/ inferior vena cava by

- Radiological means (angioplasty, stenting or transjugular intrahepatic portosystemic shunt) or
- Surgery (mesoatrial shunt, mesocaval shunt).

Orthotopic liver transplant is reserved for patients with end-stage liver disease or fulminant failure.

Wilson's disease (Option A) presents with liver dysfunction, neuropsychiatric symptoms, and Kayser-Fleischer rings in the eyes. However, it does not typically cause acute hepatic vein obstruction, ascites, or the specific Doppler findings mentioned in this case.

Hepatopulmonary syndrome (Option B) occurs in chronic liver disease, characterized by liver dysfunction, hypoxemia, and dilated blood vessels in the lungs. It presents primarily with respiratory symptoms like shortness of breath and doesn't typically involve hepatomegaly, ascites, or absent hepatic vein flow.

Extrahepatic portal vein obstruction (Option D) involves obstruction of the portal vein, leading to portal hypertension, splenomegaly, and possibly ascites. The main issue lies in the portal vein, not the hepatic veins.

Solution for Question 8:

Answer: C) Supportive care with hydration and monitoring

Explanation:

Hepatitis A is a self-limiting viral infection, especially in children. Management typically involves supportive care, including maintaining hydration and monitoring liver function.

Hepatitis A:

Etiology: Hepatitis A Virus (HAV) is an RNA virus from the Picornavirus family.

Epidemiology:

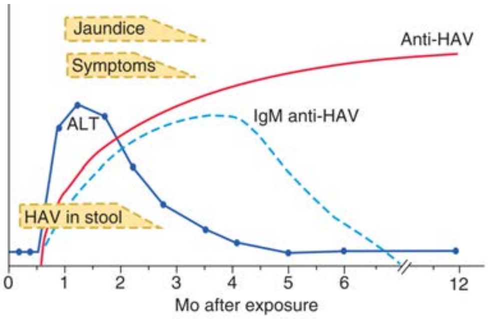
- HAV is highly contagious, transmitted primarily through the fecal-oral route.
- Incubation period: ~3 weeks.

- Contagious (viral shedding) from late in the incubation period until about 2 weeks after the onset of jaundice.

Clinical Manifestations:

- HAV causes acute hepatitis only.
- Often anicteric in young children, but symptomatic in older adolescents, adults, and immunocompromised patients.
- Symptoms: Acute febrile illness with abrupt onset of anorexia, nausea, malaise, vomiting, and jaundice.

Duration of illness: 7-14 days.



Category	Details
Diagnosis	Anti-HAV IgM antibodies: Detectable during symptoms and remains positive for 4-6 months. Anti-HAV IgG antibodies: Provides long-term immunity and appears within 8 weeks of symptom onset. Elevated levels: ALT, AST, bilirubin, alkaline phosphatase, 5′-nucleotidase, γ-glutamyl transpeptidase.
Complications	Full recovery is typical. Acute liver failure (ALF): Rare, particularly in older adolescents, adults, and immunocompromised patients. Prolonged cholestatic syndrome: Leads to pruritus and fat malabsorption, resolves without long-term effects. Acute Relapsing Hepatitis Fulminant hepatitis (Acute Hepatitis E > A)
Treatment	Supportive care: Intravenous hydration, antipruritic agents, and fat-soluble vitamin supplementation for cholestatic disease. Monitor for signs of ALF.

Prevention	Vaccine: Two inactivated vaccines, administered intramuscularly in a 2-dose schedule (second dose 6-12 months after the first). Recommended for children >12 months. Isolation: Infected individuals should be isolated for 2 weeks before and 1 week after jaundice onset. Hand hygiene. Immunoglobulin (Ig) Prophylaxis: For preexposure and postexposure prophylaxis, especially in children <12 months, immunocompromised individuals, those with chronic liver disease, and those allergic to the vaccine.
Prognosis	Excellent prognosis with no long-term sequelae.

Start antiviral therapy (Option A): Hepatitis A virus (HAV) infection typically does not require antiviral therapy. The treatment is generally supportive as the infection is usually self-limiting.

Administer Hepatitis A vaccine (Option B): The vaccine is used for prevention, not for treatment of an acute infection. Since the patient already has a positive Anti-HAV IgM, indicating an acute infection, the vaccine is not appropriate in this context.

Admit for immediate liver biopsy (Option D): A liver biopsy is not indicated in the acute phase of Hepatitis A. The diagnosis is based on clinical presentation and serology, and a biopsy is not necessary for routine management of this infection.

Solution for Question 9:

Answer: A) Acute infection

Explanation: Acute infection with hepatitis B virus is the most likely diagnosis considering the history of recent blood transfusion, short duration of symptoms, and serology suggesting an active infection.

Hepatitis B:

Etiology:

- Partially double-stranded DNA genome
- Hepatitis B surface antigen (HBsAg): Particles present on the surface of the virus.
- Inner portion contains: Hepatitis B core antigen (HBcAg) Viral DNA Nonstructural antigen - hepatitis B e antigen (HBeAg)
- Hepatitis B core antigen (HBcAg)

- Viral DNA
- Nonstructural antigen - hepatitis B e antigen (HBeAg)
- Hepatitis B core antigen (HBcAg)
- Viral DNA
- Nonstructural antigen - hepatitis B e antigen (HBeAg)

Diagnosis:

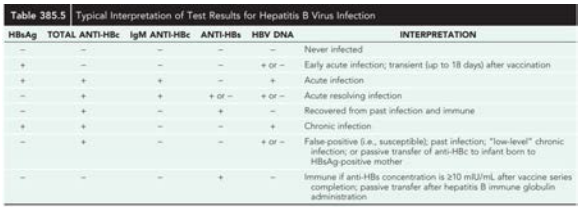
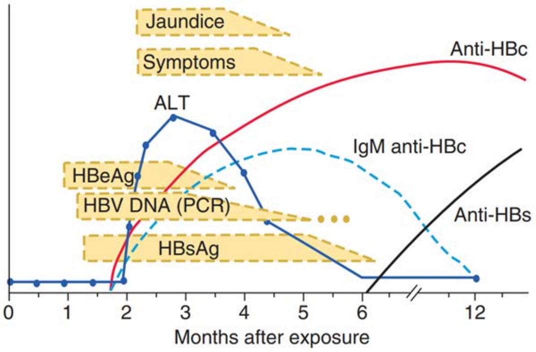


Table 385.5 Typical Interpretation of Test Results for Hepatitis B Virus Infection

HBsAg	TOTAL ANTI-HBc	IgM ANTI-HBc	ANTI-HBs	HBV DNA	INTERPRETATION
–	–	–	–	–	Never infected
+	–	–	–	+ or –	Early acute infection; transient (up to 18 days) after vaccination
+	+	+	–	+	Acute infection
–	+	+	+ or –	+ or –	Acute resolving infection
–	+	–	+	–	Recovered from past infection and immune
+	+	–	–	+	Chronic infection
–	+	–	–	+ or –	False-positive (i.e., susceptible); past infection; "low-level" chronic infection; or passive transfer of anti-HBc to infant born to HBsAg-positive mother
–	–	–	+	–	Immune if anti-HBs concentration is ≥10 mIU/mL after vaccine series completion; passive transfer after hepatitis B immune globulin administration

From the table, Chronic infection (Option B) Acute resolving infection (Option C), and Early acute infection (Option D) are ruled out.



Treatment:

- Acute viral hepatitis: Supportive
- Chronic hepatitis B: Interferon-α2b (IFN-α2b), lamivudine, adefovir, entecavir, tenofovir and peginterferon-α2

Solution for Question 10:

Answer: C) Administer hepatitis B vaccine and hepatitis B immune globulin (HBIG) within 12 hours of birth

Explanation: Babies born to HBV-positive mothers need to be immunized with hepatitis B immunoglobulin (HBIG) and the HBV immunization, given within 12 hr of delivery (protects >95% of neonates).

Management of babies born to HBV-positive mothers:

- Immunoprophylaxis with hepatitis B immunoglobulin (HBIG) and the HBV immunization, given within 12 hr of delivery (protects >95% of neonates)
- Hepatitis universal vaccination series to be completed (3 or 4 doses depending on the weight birth weight of the baby) Birth weight ≥2,000 g = 3 doses Birth weight <2,000 g = 4 doses
- Birth weight ≥2,000 g = 3 doses
- Birth weight <2,000 g = 4 doses
- All infants born to HBsAg-positive mothers should have follow-up HBsAg and anti-HBs tested to determine appropriate follow-up
- Birth weight ≥2,000 g = 3 doses
- Birth weight <2,000 g = 4 doses

Solution for Question 11:

Answer: A) Perinatal

Explanation: In children, perinatal transmission is the most common mode of transmission of the Hepatitis C virus.

Hepatitis C:

Etiology:

HCV is a single-stranded RNA virus

Risk factors:

- Blood transfusion: Before 1992 it was the most common route of infection (Option B ruled out)
- Illegal drug use with exposure to blood or blood products: More than half of adult cases (Option C ruled out)
- Sexual transmission: Second most common cause of infection (Option D ruled out)

- Perinatal transmission: Most common mode of transmission in children (5% of infants born to positive mothers) (Option A)

Clinical Manifestations:

- Acute infection: Mild and Acute liver failure is very rare
- Chronic infection: Liver fibrosis Can progress to cirrhosis, liver failure, and, occasionally, primary HCC.
- Liver fibrosis
- Can progress to cirrhosis, liver failure, and, occasionally, primary HCC.
- Liver fibrosis
- Can progress to cirrhosis, liver failure, and, occasionally, primary HCC.

Diagnosis:

Detection of anti-HCV antigens or viral RNA

Treatment:

Chronic hepatitis C: Peginterferon, IFN-α2b, and ribavirin.

No vaccination: The immune system's selective pressure on HCV quasispecies can lead to the emergence of escape mutants that are resistant to T- and B-cell immune targeting. Thus making it difficult to develop vaccines for the Hepatitis C virus.

Breastfeeding (Option D): There is no evidence of mother-to-infant transmission of the Hepatitis C virus from breastfeeding.

Solution for Question 12:

Answer: C) Increased serum ceruloplasmin levels

Explanation: The most likely diagnosis in this scenario is Wilson's disease. There is decreased serum ceruloplasmin levels (<20 mg/dL) in this disorder.

Wilson disease:

Hepatolenticular degeneration

Pathophysiology:

- Autosomal recessive (positive family history) (Option A ruled out)

- Mutation in ATP7B gene (copper transporting P-type adenosine triphosphatase (ATPase), present in chromosome 13 (13q14).
- Decreased biliary copper excretion and diffuse accumulation of copper in the cytosol of hepatocytes. (Option B ruled out)
- Later, copper is redistributed to other tissues once the liver is saturated with copper.

Clinical features:

- Liver: Asymptomatic hepatomegaly (with or without splenomegaly), subacute or chronic hepatitis, acute hepatic failure, and portal hypertension
- Neurology: Intention tremor, dysarthria, rigid dystonia, Parkinsonism, choreiform movements, lack of motor coordination, deterioration in school performance, psychosis, or behavioral changes
- Psychiatric manifestations: Depression, personality changes, anxiety, obsessive-compulsive behavior, or psychosis
- Eye: KF (Kayser Fleischer) ring in Descement membrane layer of cornea
- Renal: Fanconi syndrome and renal failure

Diagnosis:

- Decreased serum ceruloplasmin levels (<20 mg/dL) (Option C)
- Serum-free copper level >1.6 µmol/L
- Urinary copper excretion (normally <40 µg/day) is increased to >100 µg/day
- Liver biopsy (Gold standard): Hepatic copper content >250 µg/g dry weight of liver

Treatment:

- Dietary intake of copper (Avoid liver, shellfish, nuts, and chocolate) should be restricted to <1 mg/day
- Symptomatic patients: Copper-chelating agents D-penicillamine (Option D ruled out) Triethylenetetramine dihydrochloride
- D-penicillamine (Option D ruled out)
- Triethylenetetramine dihydrochloride
- Zinc impairs the gastrointestinal absorption of copper
- D-penicillamine (Option D ruled out)
- Triethylenetetramine dihydrochloride

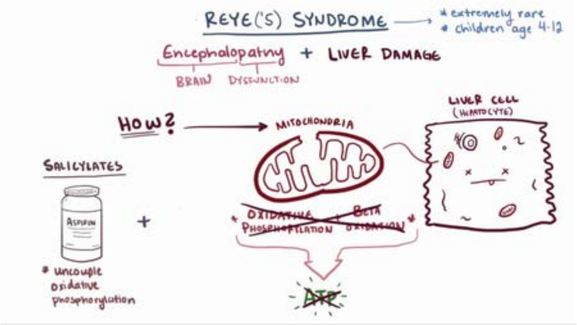
Solution for Question 13:

Answer: A) Supportive treatment

Explanation: Management of Reye syndrome is mainly supportive and needs close monitoring.

Reye syndrome:

- Acute metabolic disorder with generalized mitochondrial dysfunction due to inhibiton of fatty acids oxidation.
- Acute noninflammatory encephalopathy with fatty liver failure



Precipitating factors:

- Preceding viral illness- influenza A and B, varicella, coxsackie, parainfluenza, Epstein-Barr (EBV), cytomegalovirus (CMV), adenovirus.
- Drugs and toxins: Aspirin, valproic acid.

Clinical features:

- Hepatic dysfunction
- Neurological: Altered sensorium, seizures, delirium, comatose, final stage- flaccid paralysis and respiratory arrest
- Increased ICP (intra-cranial pressure)

Investigation:

- Elevated liver function tests (ALT, AST, bilirubin), hyperammonemia, abnormal coagulation studies
- Normal levels of serum bilirubin
- CSF analysis

Treatment:

- Mainstay: Supportive (Option A)
- Stop the precipitating factors if possible (Option C ruled out)

- Treat increased ICP if present

Mannitol or hypertonic saline (Option B): Treatment of increased ICP if present.

D-penicillamine administration (Option D): It is a copper chelating agent, used in treatment of Wilson disease. Hence not used in treatment for Reye syndrome.

Previous Year Questions

1. A 2-week-old neonate was brought to the hospital with a complaint of non-bilious vomiting. While examining the baby, the physician noted a lump in the left epigastric region, which showed movement while feeding. What is the likely diagnosis of this child?

A. Esophageal atresia

B. Duodenal atresia

C. IHPS

D. Intususscception

---

2. What is Celiac Sprue in a child?

A. Anti-tissue transglutaminase (tTG) antibodies are positive and used as a screening test

B. A type of food allergy causing skin rashes

C. A genetic disorder leading to lactose intolerance

D. An infection in the gastrointestinal tract caused by bacteria

---

3. A hospital receives an 18-month-old child with a 3-day history of watery diarrhea, vomiting, and altered sensorium. All of the following differential diagnosis can be considered except:

A. Hyponatremia

B. Hemolytic Uremic Syndrome

C. Severe dehydration

D. Cerebral vein thrombosis

4. What is the most probable diagnosis for a 3-year-old boy who experiences painful bowel movements, has recently started attending school, occasionally complains of fecal incontinence, appears distressed, and shows normal findings?

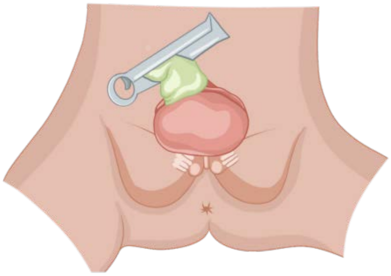
A. Functional constipation

B. Pathologic constipation

C. Toxic megacolon

D. Small bowel obstruction

5. A newborn female presents with continuous dribbling of urine. The image below shows the pathology from which the patient is suffering. The most likely diagnosis is:



A. Gastroschisis

B. Bladder exstrophy

C. Superior vesical fissure

D. Anterior ectopic anus

6. Which of the following is useful in acute diarrhea?

A. Zinc

B. Magnesium

C. Calcium

D. Potassium

---

7. Assertion: Antibiotics should not be given to children having diarrhea Reason: Diarrhea in children is mostly caused by viruses

A. Both assertion and reason are correct and reason is the correct explanation of the assertion

B. Both assertion and reason are correct and reason is not the correct explanation of the assertion

C. Assertion is independently a correct statement but the reason is false

D. Assertion is false but the reason is correct statement

E. Both assertion and reason are false

---

8. The mother of a premature newborn is concerned because the baby has not yet had a bowel movement containing meconium. How long is it safe to wait for the passage of meconium in this infant?

A. 18 hours

B. 48 hours

C. 36 hours

D. 72 hours

---

9. A 2-year-old boy is presented by his parents at a nearby village hospital, reporting loose stools over the past 3 days. The child's mother observes that he has become lethargic and refuses to eat. Upon examination, the child displays lethargy, dry mouth, sunken eyes, and delayed skin pinch recovery. How much fluid should be administered within the initial three hours if the child weighs 10 kg?

A. 1000 ml

B. 100 ml

C. 200 ml

D. 500 ml

10. During the surgery for acute appendicitis in a 2-year-old child, the surgeon discovered a structure measuring 2 inches in length located 2 feet closer to the ileocecal junction. What is the most probable diagnosis in this case?

A. Sarcoma

B. Appendicitis

C. Meckel’s diverticulum

D. Zenker’s diverticulum

11. What is the likely diagnosis for a newborn who has a liver and small intestine protruding through the umbilicus, covered by membranes at birth?

A. Omphalocele

B. Gastroschisis

C. Persistent Vitello intestinal duct

D. Bochdalek hernia

12. What is the diagnosis for a patient who has chronic small bowel diarrhea, claims to have a wheat allergy, and exhibits villous atrophy on duodenal biopsy with tests positive for anti-endomysial antibodies and tissue transglutaminase (tTG) IgA antibodies?

A. Cystic fibrosis

B. Whipple’s disease

C. Tropical sprue

D. Celiac sprue

13. What is the most likely underlying diagnosis in a male child who presents with coarse facial features, macroglossia, thick lips, hepatosplenomegaly, and excessive mucus discharge from the nose?

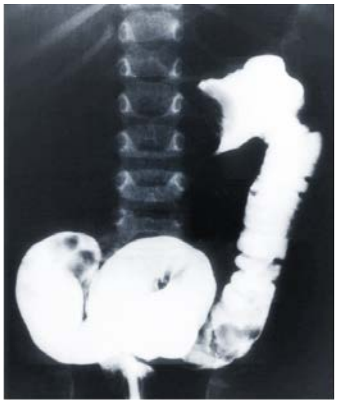
A. Hurler disease

B. Beckwith-Wiedemann syndrome

C. Proteus Syndrome

D. Hypothyroidism

14. A baby presented with abdominal pain. On examination, a mass is palpated in the left lumbar region. A barium enema is done, and the image is given below. What is the likely diagnosis?



A. Intussusception

B. Volvulus

C. Duodenal atresia

D. Intestinal obstruction

15. A 5-year-old child has been presented to you with a history of experiencing 18 episodes of watery bowel movements within the past 24 hours, along with vomiting twice in the last 4 hours. The child appears irritable but is still consuming fluids. What is the most appropriate treatment for this child?

A. Intravenous fluids

B. Oral rehydration therapy with ORS

C. Intravenous fluid initially for 4 hours followed by oral fluids

D. Plain water add lithium

16. What is the most common reason for liver transplantation in children?

A. Alagille syndrome

B. Biliary atresia

C. Caroli disease

D. Hepatocellular carcinoma

17. What is the molecular defect responsible for liver disease in Dubin-Johnson syndrome?

A. ATP7A

B. ATP7B

C. ABCC2

D. SERPINA 1

18. What is the most probable diagnosis for a neonate who continues to have unconjugated hyperbilirubinemia after three weeks of birth, with normal liver enzymes, PT/INR, and albumin levels, no evidence of hemolysis on a peripheral blood smear, and a decrease in bilirubin levels following treatment with phenobarbital?

A. Rotor syndrome

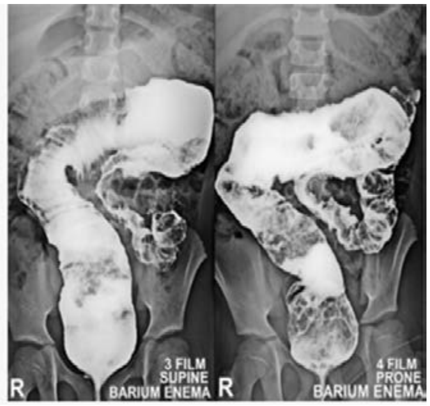
B. Crigler Najjar type 2

C. Dubin Johnson syndrome

D. Crigler Najjar type 1

19. Which of the following is the gold standard for diagnosing the condition in the mentioned case? An abdominal distension was observed in a 3-day-old male

neonate who has not yet passed meconium. There is no history of fever. During rectal examination, forceful expulsion of stool was observed. A barium enema was performed, and the findings are shown in the image below.



A. Ultrasonography

B. MRI abdomen

C. Rectal biopsy

D. Anorectal manometry

20. What is the most likely diagnosis for a newborn who presents with protrusion of bowel loops through a defect in the abdominal wall located to the right side of the umbilicus?

A. Omphalocele

B. Umbilical hernia

C. Gastroschisis

D. Ectopia vesicae

21. What is the cause of the greenish-black color of the initial stool in a newborn?

A. Bilirubin

B. Biliverdin

C. Meconium

D. Gut flora of neonate

22. Which test is most appropriate for the clinical diagnosis of a 3-year-old patient experiencing intermittent loose motions and weight loss?

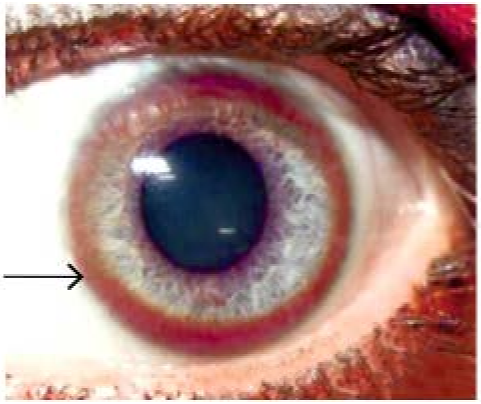
A. Anti – Mitochondrial Antibody

B. Anti-tissue transglutaminase Antibody

C. Anti -SCL 70 antibody

D. Anti-Microsomal antibody

23. What is the likely diagnosis for a case of an 11-year-old boy with intentional tremors, poor scholastic performance, hepatomegaly, and an eye finding shown in the image, when his sister also presents similar complaints?



A. Wilson's disease

B. Huntington's chorea

C. Glutaric aciduria

D. Hepatitis A

24. What is the most probable diagnosis for a 1-month-old baby who has jaundice, clay-colored stools, conjugated hyperbilirubinemia, and periductal proliferation observed in a liver biopsy?

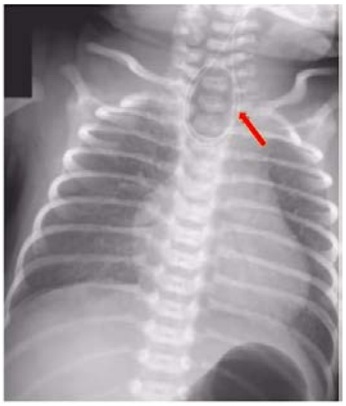
A. Crigler-Najjar syndrome

B. Rotor syndrome

C. Dubin-Johnson syndrome

D. Biliary atresia

25. What is the diagnosis for a neonate who was brought in with a complaint of frothiness from the mouth and respiratory distress, as shown in the provided X-ray image?



A. Transient tachypnea of newborn

B. Congenital diaphragmatic hernia

C. Esophageal atresia with tracheo-esophageal fistula

D. Respiratory distress syndrome

26. A 10-year-old child presents with diarrhea and weight loss. On examination, the height and weight are lesser than expected. Laboratory investigations were positive for class II HLA-DQ2. Which of the following will you advise the child?

A. Fat free diet
B. Lactose free diet
C. Low carbohydrate diet
D. Gluten free diet

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	2
Question 4	1
Question 5	2
Question 6	1
Question 7	1
Question 8	2
Question 9	1
Question 10	3
Question 11	1
Question 12	4
Question 13	1
Question 14	1
Question 15	2
Question 16	2
Question 17	3
Question 18	2
Question 19	3
Question 20	3
Question 21	3
Question 22	2

Question 23	1
Question 24	4
Question 25	3
Question 26	4

Solution for Question 1:

Correct Answer: C) IHPS

Explanation:

The clinical scenario is most likely congenital/ idiopathic hypertrophic pyloric stenosis (CHPS), a condition involving hypertrophy of the pyloric muscle, leading to gastric outlet obstruction in male infants, typically between 2 to 6 weeks of age.

Risk Factors of Congenital Hypertrophic Pyloric Stenosis	
Genetic Factors	White race of northern European descent Male sex Familial History: Higher risk in offspring of mothers with CHPS (20% for sons, 10% for daughters) Twin Studies: Higher concordance in monozygotic twins suggests a genetic component Blood Groups: Infants with B and O blood groups.
Environmental Factors	Erythromycin Use: Especially if administered within the first two weeks of life Higher incidence in infants whose mothers used macrolides during pregnancy or breastfeeding, particularly in females.
Other Associated Factors	Other Congenital Defects Syndromes: Occasionally associated with tracheoesophageal fistula and hypoplasia or agenesis of the inferior labial frenulum. Associated with eosinophilic gastroenteritis, Apert syndrome, Zellweger syndrome, trisomy 18, Smith-Lemli-Opitz syndrome, and Cornelia de Lange syndrome.

- Clinical Presentation: Symptoms: Projectile, non-bilious vomiting after feeds, starting around 2 to 6 weeks of age. Physical Examination: Visible gastric peristalsis and a palpable “olive-shaped” mass in the epigastric region.

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- Physical Examination: Visible gastric peristalsis and a palpable “olive-shaped” mass in the epigastric region.

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- Physical Examination: Visible gastric peristalsis and a palpable “olive-shaped” mass in the epigastric region.

- Diagnosis: Ultrasound (USG) is the Investigation of choice for CHPS, characterised by Pyloric muscle thickness of > 3 mm Pyloric length > 14 mm The pyloric channel appears elongated and thickened (10-14 mm)
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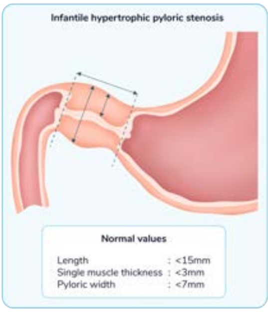
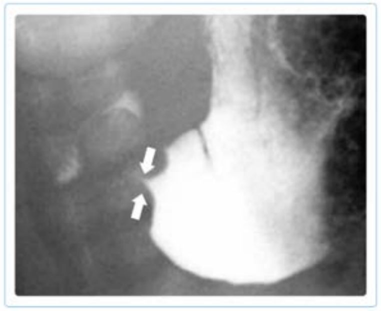
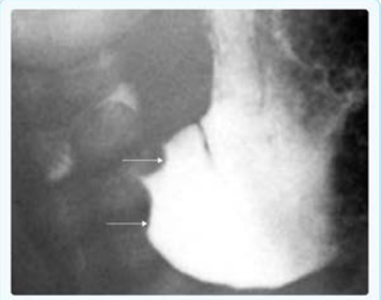
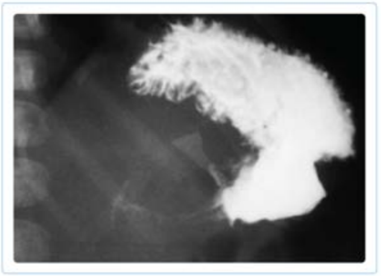
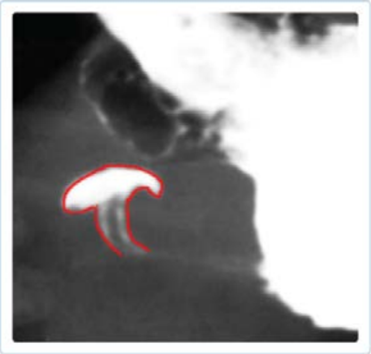


- Contrast Studies (Upper GI Series): Less frequently used, demonstrating the classic "string sign" with a thin line of contrast through the elongated and narrowed pyloric channel and delayed gastric emptying without duodenal dilation.

Characteristic findings of Contrast studies in CHPS	

Findings	Description
String Sign	 A thin line of contrast material passes through the narrowed pyloric channel, indicating severe narrowing due to hypertrophied muscle.
Shoulder Sign	 Shoulder sign A bulge of the hypertrophied pyloric muscle into the gastric antrum gives the appearance of a “shoulder.”
Double Tract Sign	 Parallel streaks of contrast material within the narrowed pyloric channel suggest the presence of a narrow passage for contrast flow.

Mushroom sign	 Bulging of the hypertrophic muscle into the 1st part of the duodenum.

Esophageal atresia (Option A) will have bilious vomiting and does not present with any mass in the epigastrium.

Duodenal atresia (Option B) presents with bilious vomiting immediately after birth. This is not the given scenario.

Intussusception (Option D) presents with a sausage-shaped abdominal mass but is not associated with vomiting. It is more characteristically associated with red currant jelly stools.

Solution for Question 2:

Correct Option A - Anti-tissue transglutaminase (tTG) antibodies are positive and used as a screening test:

- Celiac sprue, also known as celiac disease, is an autoimmune disorder where the ingestion of gluten (a protein found in wheat, barley, and rye) leads to damage in the small intestine.
- The presence of anti-tissue transglutaminase (tTG) antibodies in the blood is used as a screening test for celiac disease. This is the correct answer.

Incorrect Options:

Option B - A type of food allergy causing skin rashes:

- Celiac disease is not a food allergy but an autoimmune disorder. While it may cause skin rashes, such as dermatitis herpetiformis, it primarily affects the small intestine and is not classified as a food allergy.

Option C - A genetic disorder leading to lactose intolerance:

- Celiac disease is a genetic disorder but does not cause lactose intolerance directly. It is characterized by an immune reaction to gluten. However, some individuals with celiac disease may experience lactose intolerance due to damage to the intestinal lining affecting lactase production.

Option D - An infection in the gastrointestinal tract caused by bacteria:

- Celiac disease is not an infection but an autoimmune condition. It is caused by an immune response to gluten, not by bacterial infection.

Solution for Question 3:

Correct Option B: Hemolytic Uremic Syndrome: This can not be a differential diagnosis as there is no thrombocytopenia, bleeding manifestation, anemia, or renal shutdown that are seen in Hemolytic Uremic Syndrome.

Incorrect Options:

Option A: Hyponatremia: Hyponatremia is a possibility due to an to an electrolyte imbalance caused by diarrhea.

Option C: Severe dehydration: Severe dehydration can occur due to the loss of water through diarrhea.

Option D: Cerebral vein thrombosis: Cerebral vein thrombosis can occur as a sequelae in young infants with diarrhea.

Solution for Question 4:

Correct Option:

Option A: Functional constipation Explanation: Functional constipation is the most likely diagnosis in this case. It is a common condition in children and is characterized by infrequent bowel movements, difficulty passing stools, and pain during bowel movements. The boy's recentenrollment in school and complaints of fecal incontinence suggest a functional cause. Thiscondition is usually related to lifestyle factors, such as changes in routine or stress, and does notinvolve any underlying anatomical or pathological abnormalities.

Incorrect Options:

Option B: Pathologic constipation Explanation: Pathologic constipation refers to constipation caused by an underlying medical condition. It can be associated with anatomical abnormalities, such as anorectal malformations or Hirschsprung's disease, or systemic diseases, such as thyroid disorders or metabolic disorders. However, since the findings in this case are normal, pathologic constipation is less likely.

Option C: Toxic megacolon Explanation: Toxic megacolon is a severe condition characterized by massive dilation of the colon and systemic toxicity. It usually presents with symptoms such as abdominal pain, distension, fever, and signs of systemic inflammation. In this case, since all the findings are reported as normal, toxic megacolon is unlikely.

Option D: Small bowel obstruction Explanation: Small bowel obstruction typically presents with symptoms such as abdominal pain, vomiting, distension, and constipation. While small bowel obstruction can cause painful bowel movements, the absence of other signs and symptoms, as well as normal findings, makes it less likely in this case.

Solution for Question 5:

Correct Option B - Bladder exstrophy

- Bladder exstrophy is a congenital malformation of the urinary system characterized by the incomplete closure of the abdominal wall and anterior wall of the bladder during fetal development.
- This results in the exposure of the bladder mucosa on the outside of the body.
- Continuous dribbling of urine is a typical symptom of bladder exstrophy due to the inability to properly contain and control urine.

Incorrect Options:

Options A, C, and D do not present with continuous dribbling of urine.

Solution for Question 6:

Correct Option A - Zinc:

- Zinc supplementation is useful in the management of acute diarrhea.
- It reduces the duration and severity of diarrhea.

Incorrect Options: Options B, C, and D are incorrect.

Solution for Question 7:

Correct Option is A - Both assertion and reason are correct and reason is the correct explanation of the assertion.

- This option is correct because both the assertion and reason are accurate, and the reason provided explains why the assertion is true.
- Antibiotics should not be given to children with diarrhea because most cases of diarrhea in children are caused by viruses, not bacteria.
- Therefore, giving antibiotics to children with viral diarrhea would be ineffective and potentially harmful.

Incorrect Options - B, C, D and E are incorrect. Refer to the explanation of the correct answer.

Solution for Question 8:

Correct Option B - 48 hours:

- Meconium is the thick, sticky, greenish-black substance that makes up the first stool.
- In most cases, newborns pass meconium within the first 24 hours after birth.

- However, for preterm newborns, the passage of meconium may be delayed.
- It is generally considered normal for preterm newborns to pass meconium within 24- 48 hours after birth.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 9:

Correct Option A - 1000 ml:

- The patient’s presentation with lethargy, dry mouth, sunken eyes, and delayed skin pinch recovery is suggestive of severe dehydration.
- In a severely dehydrated child, the initial fluid volume to be given is typically 100 ml/kg.
- Since the child's weight is 10 kg, the total fluid volume would be 1000 ml.

Incorrect Options:

Options B, C, and D are incorrect. Refer to the explanation of the correct answer

Solution for Question 10:

Correct Option C: Meckel’s diverticulum:

- The structure measuring 2 inches in length and located 2 feet closer to the ileocecal junction is suggestive of Meckel’s diverticulum.
- Meckel's diverticulum is a congenital abnormality where a small pouch-like structure forms in the small intestine.
- It is the result of incomplete closure of the vitelline duct, which connects the developing embryo to the yolk sac.

Incorrect Options:

Options A, B, and D are incorrect.

Solution for Question 11:

Correct Option A - Omphalocele:

- Omphalocele is a congenital abdominal wall defect where abdominal organs, such as the liver and intestines, herniate through the umbilicus and are covered by a translucent sac or membrane.
- It occurs due to the failure of the abdominal wall to close properly during fetal development.

Incorrect Options:

Option B - Gastroschisis:

- In gastroschisis, there is a defect in the abdominal wall to the right of the umbilicus, and the abdominal organs, usually the intestines, protrude directly through the defect without a sac or membrane covering.

Option C - Persistent vitellointestinal duct:

- Persistent vitellointestinal duct is a rare condition where the vitellointestinal duct, which connects the developing embryo to the yolk sac during early fetal development, fails to close properly.

Option D - Bochdalek hernia:

- Bochdalek hernia is a type of congenital diaphragmatic hernia.
- It involves a defect in the diaphragm, allowing abdominal organs, such as the intestines, to herniate into the chest cavity.

Solution for Question 12:

Correct Option D - Celiac sprue:

- The above mentioned symptoms are suggestive of celiac sprue.

- Celiac sprue, also known as celiac disease, is an autoimmune disorder triggered by the ingestion of gluten.
- Gluten is a protein found in wheat, barley, and rye.
- When individuals with celiac disease consume gluten, it triggers an immune response that leads to inflammation and damage to the small intestine which is characterized by chronic small bowel diarrhea and malabsorption.

Incorrect Options:

Options A, B, and C are incorrect.

Solution for Question 13:

Correct Option A - Hurler disease:

- The patient’s presentation with coarse facial features, macroglossia, thick lips, hepatosplenomegaly, and excessive mucus discharge from the nose is suggestive of Hurler disease.
- Hurler disease, also known as mucopolysaccharidosis type I (MPS I), is a rare genetic disorder characterized by the deficiency of the enzyme alpha-L-iduronidase.

Incorrect Options:

Option B - Beckwith-Wiedemann syndrome:

- Beckwith-Wiedemann syndrome is a genetic disorder characterized by overgrowth, abdominal wall defects, macroglossia, and an increased risk of certain tumors, such as Wilms tumor and hepatoblastoma.
- Other features can include omphalocele, neonatal hypoglycemia, and ear creases or pits.

Option C - Proteus syndrome:

- Proteus syndrome is an extremely rare genetic disorder characterized by progressive and disproportionate overgrowth of body tissues.
- It typically presents with asymmetric overgrowth of limbs, subcutaneous masses, skin abnormalities (such as thickened skin or prominent veins), and an increased risk of developing tumors.

Option D - Hypothyroidism:

- Common symptoms of hypothyroidism include fatigue, weight gain, cold intolerance, constipation, and dry skin.

Solution for Question 14:

Correct Option A - Intussusception:

- The patient’s presentation with abdominal pain and a mass palpated in the left lumbar region on examination with coiled spring sign or claw sign on barium enema is suggestive of intussusception.
- Intussusception occurs when a section of the intestine folds in on itself, resulting in obstruction and blockage.

Incorrect Options:

Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 15:

Correct Option B - Oral rehydration therapy with ORS:

- Oral rehydration therapy with ORS is intended to rehydrate a moderately dehydrated infant or toddler who is still consuming fluids.
- Its advantages over parenteral therapy include fewer complications, lower cost, lower hospitalization rate, and minimized the risk of iatrogenic hypernatremia or hyponatremia.

Incorrect Options:

Option A, C & D are incorrect.

- Oral rehydration therapy should be offered first to children with dehydration who are drinking orally

Solution for Question 16:

Correct Option B - Biliary atresia:

- Biliary atresia is the most common indication for liver transplantation in children.
- It is a rare, progressive liver disease that affects infants, characterized by the absence or obstruction of the bile ducts, leading to bile flow impairment and subsequent liver damage.
- Without surgical intervention, such as the Kasai procedure (hepatoportoenterostomy), biliary atresia can progress to liver failure, necessitating liver transplantation for long-term survival.

Incorrect Options:

Options A, C, and D are not the most common indication for liver transplantation in children.

Solution for Question 17:

Correct Option C - ABCC2:

- Dubin-Johnson syndrome is a rare autosomal recessive disorder characterized by impaired bilirubin excretion into bile, resulting in a buildup of conjugated bilirubin in the liver and bloodstream.
- This condition is caused by a molecular defect in the ABCC2 gene, which codes for a protein called multidrug resistance-associated protein 2 (MRP2).
- ABCC2 is responsible for the transport of bilirubin glucuronides, conjugated bilirubin, and other compounds into bile.

Incorrect Options:

Options A, B, and D are incorrect.

- ATP7A and ATP7B genes code for copper transporters. Menkes disease is caused by mutations in the ATP7A gene, Wilson's disease arises from mutations in the ATP7B gene.
- The SERPINA1 gene encodes alpha-1 antitrypsin (AAT)

Solution for Question 18:

Correct Option B - Crigler-Najjar type 2:

- Crigler-Najjar type 2 is a condition characterized by a partial deficiency of the enzyme UDP-glucuronosyltransferase (UGT), which leads to unconjugated hyperbilirubinemia.
- In this type, the bilirubin levels can be reduced with the administration of drugs that induce UGT enzyme activity, such as phenobarbital.

Incorrect Options:

Option A - Rotor syndrome:

- Rotor syndrome is a benign condition characterized by conjugated hyperbilirubinemia, where bilirubin is conjugated but not efficiently transported from the liver into the bile.

Option C - Dubin-Johnson syndrome:

- Dubin-Johnson syndrome is a benign condition characterized by impaired bilirubin excretion into the bile.
- It causes a buildup of conjugated bilirubin in the liver, leading to mild conjugated hyperbilirubinemia.

Option D - Crigler-Najjar type 1:

- Crigler-Najjar type 1 is a severe form of the condition characterized by a complete deficiency of UGT enzyme activity.
- It presents early in life and results in severe unconjugated hyperbilirubinemia.
- Unlike Crigler-Najjar type 2, Crigler-Najjar type 1 is not responsive to phenobarbital treatment.
- Management options for Crigler-Najjar type 1 typically involve phototherapy or liver transplantation.

Solution for Question 19:

Correct Option C - Rectal biopsy:

- Infants presenting shortly after birth with constipation and signs of distal obstruction and yet to pass meconium in the first 48 hours of life are symptoms indicating Hirschsprung disease.
- The presence of an empty rectum per rectal examination with a gush of liquid stools on withdrawal of the finger is also suggestive of Hirschsprung disease.
- Barium enema shows an abrupt transition zone between the normal dilated proximal colon and narrow aganglionic segment.
- Rectal biopsy is considered the gold standard for the diagnosis.

Incorrect Options:

Options A, B, and D are not the gold standard for diagnosing Hirschsprung disease.

Solution for Question 20:

Correct Option C - Gastroschisis:

- Gastroschisis is a congenital defect of the abdominal wall, typically to the right of the umbilicus.
- It involves evisceration of abdominal contents, such as bowel loops, without a covering sac.
- The exposed intestines are in direct contact with the amniotic fluid.

Incorrect Options:

Option A - Omphalocele:

- Omphalocele is a congenital abdominal wall defect in which abdominal contents herniate into the base of the umbilical cord and are covered by a sac.
- Omphalocele is typically centrally located.

Option B - Umbilical hernia:

- Umbilical hernia is a protrusion of abdominal contents through a weakness or incomplete closure of the umbilical ring.
- It is not associated with evisceration of bowel loops to the right of the umbilicus.

Option D - Ectopia vesicae:

- Ectopia vesicae refers to the abnormal positioning of the urinary bladder, typically exposed on the abdominal wall due to a defect in the lower abdominal wall.

Solution for Question 21:

Correct Option C - Meconium:

- Meconium is the greenish-black substance that makes up the initial stool of a newborn.
- It is composed of materials that the fetus ingests while in the womb, such as amniotic fluid, mucus, and cells from the intestinal lining.
- Meconium also contains bile, which contributes to its color.
- However, it's important to note that the primary reason for the greenish-black color of meconium is not bilirubin or biliverdin, but rather the various components that make up this substance.

Incorrect Options:

Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 22:

Correct Option B - Anti-tissue transglutaminase Antibody:

- The most-likely provisional diagnosis in a 3-year-old child experiencing intermittent loose motions and weight loss is celiac disease.
- The Anti- tissue transglutaminase Antibody test is a primary screening test for celiac disease.
- It detects the presence of transglutaminase, an enzyme that causes an autoimmune response in celiac disease on gluten ingestion.

Incorrect Options:

Option A - Anti-Mitochondrial Antibody (AMA):AMA is highly specific for primary biliary cirrhosis, which is a chronic liver disease characterized by progressive destruction of the small bile ducts within the liver.

Option C - Anti-SCL 70 Antibody:The presence of anti-SCL 70 antibodies is strongly associated with systemic sclerosis, a chronic connective tissue disease characterized by fibrosis and vascular abnormalities affecting the skin, internal organs, and blood vessels.

Option D - Anti-Microsomal Antibody: It is also known as Anti-Thyroid Microsomal Antibody or Anti-Thyroid Peroxidase Antibody.

Solution for Question 23:

Correct Option A - Wilson's disease:

- The image shows Kayser-Fleischer rings which are associated with Wilson's disease.
- These are golden-brown or greenish-brown rings that appear in the cornea due to the deposition of copper.
- Wilson's disease is a rare genetic disorder characterized by the accumulation of copper in various organs of the body, including the liver, brain, and cornea.

Incorrect Options:

Kayser-Fleischer rings are not seen in options B, C, and D.

Solution for Question 24:

Correct Option D - Biliary atresia:

- The patient’s presentation with jaundice, clay-colored stools, conjugated hyperbilirubinemia, and periductal proliferation observed in a liver biopsy is suggestive of biliary atresia.
- Biliary atresia is a rare condition characterised by the absence or obstruction of the bile ducts, leading to impaired bile flow from the liver to the intestine.

Incorrect Options:

Option A - Crigler-Najjar syndrome:

- Crigler-Najjar syndrome is a rare genetic disorder characterized by a deficiency or absence of the enzyme glucuronyl transferase involved in bilirubin conjugation.
- This leads to unconjugated hyperbilirubinemia.

Option B - Rotor syndrome:

- There is a decreased hepatic uptake, storage and decrease biliary excretion of bilirubin leading to mild conjugated hyperbilirubinemia

Option C - Dubin-Johnson syndrome:

- Impaired excretion of conjugated bilirubin due to mutation in canalicular multidrug resistance protein 2 (MRP-2)
- Dark pigmentation of the liver is seen.

Solution for Question 25:

Correct Option C - Esophageal atresia with tracheo-esophageal fistula:

- The patient’s presentation with frothiness from the mouth, respiratory distress and coiled tube in the upper esophagus shown in the X-ray is suggestive of Esophageal atresia with tracheo-esophageal fistula
- Esophageal atresia is a birth defect in which part of a baby's esophagus does not develop properly.

Incorrect Options:

Option A - Transient tachypnea of the newborn:

- It is a benign, self-limited condition that can present in infants of any gestational age shortly after birth.
- X-Ray shows normally inflated or hyperinflated lungs with streaky perihilar markings, giving the appearance of a shaggy heart border while the periphery of the lungs is clear.

Option B - Congenital diaphragmatic hernia:

- It is a condition resulting from a developmental defect in the diaphragm leading to protrusion of abdominal contents into the thoracic cavity.

- X-Ray shows the left hemithorax is filled with cyst like structures (loops of bowel), the mediastinum is shifted to the right, and the abdomen is relatively devoid of gas.

Option D - Neonatal respiratory distress syndrome:

- It is more common in premature babies.
- It usually develops within the first 24 hours after birth.
- X-Ray shows hypoaeration, bilateral fine granular opacities in the pulmonary parenchyma, and peripherally extending air bronchograms.

Solution for Question 26:

Correct Option D - Gluten free diet:

- The patient's presentation with diarrhea and weight loss with HLA-DQ2 positivity is suggestive of celiac disease.
- The main form of treatment for celiac disease is a gluten-free diet.

Incorrect Options:

Options A, B, and C are not the treatment options for celiac disease.

Respiratory Disorders in Children

1. Which of the following best explains the pathophysiological basis for the chronic lung infections seen in patients with CFTR ΔF508 mutation?

A. Increased bicarbonate secretion leading to decreased mucus viscosity

B. Impaired chloride transport resulting in dehydrated airway surfaces

C. Enhanced ciliary function due to overactivation of ENaC channels

D. Decreased oxidative stress reducing bacterial clearance

2. A 7-year-old child presents with a chronic productive cough, recurrent respiratory infections, and poor weight gain. A high-resolution computed tomography (HRCT) scan of the chest is taken and shown below. What is the most likely diagnosis?

A. Bronchial asthma

B. Bronchiectasis

C. Pneumonia

D. Chronic bronchitis

3. A 4-year-old child presents with multiple episodes of pneumonia over the past year. The pediatrician is concerned about the possibility of an underlying disorder contributing to these recurrent infections. Which of the following best describes the criteria for diagnosing recurrent pneumonia in children?

A. Two or more episodes in six months, with radiographic clearing.

B. Three or more episodes in a year, regardless of radiographic findings.

C. Two or more episodes in a year, with radiographic clearing.

D. Three or more episodes in six months, with or without radiographic clearing.

4. Which of the following radiological findings is most characteristic of Staphylococcal pneumonia in children?

A. Interstitial pneumonia

B. Perihilar and peribronchial infiltrates

C. Pneumatoceles

D. Hazy or fluffy exudates radiating from the hilar region

5. A 3-year-old child is brought to the clinic with a cough and difficulty breathing. The mother reports that the child has been lethargic and unable to drink or breastfeed. On examination, you find that the child's oxygen saturation is 88%, and there is stridor when the child is calm. The AVPU scale indicates a response to painful stimuli only. Based on the signs observed, how should this child's condition be classified?

A. Pneumonia

B. No Pneumonia: Cough or Cold

C. Severe Pneumonia or Very Severe Disease

D. Mild Respiratory Infection

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6. A 6-month-old infant presents with a 4-day history of cough, wheezing, and difficulty breathing. On examination, the infant has a respiratory rate of 65/min and an oxygen saturation of 90% on room air. A chest X-ray shows hyperinflation but no focal infiltrates. What is the most appropriate management?

A. Start oral corticosteroids and bronchodilators

B. Administer antibiotics and nebulized hypertonic saline

C. Provide moist oxygen and ensure proper hydration

D. Initiate ribavirin therapy in all infants with bronchiolitis

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7. A 3-month-old infant is brought with a 2-day history of cough, wheezing, and decreased appetite after an upper respiratory tract infection 1 week ago. On examination, the infant has intercostal retractions, diffuse crackles, and a respiratory rate of 56/min. Oxygen saturation is 95% on room air. What is the most likely diagnosis?

A. Bronchial asthma

B. Bacterial pneumonia

C. Bronchiolitis

D. Foreign body aspiration

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8. A 3-year-old child presents with a 3-day history of a barking cough, hoarseness, and low-grade fever. The child has intermittent stridor, particularly at night, and exhibits mild respiratory distress. An X-ray PA view of the neck shows subglottic narrowing. What is the most likely diagnosis?

A. Epiglottitis

B. Bacterial tracheitis

C. Croup

D. Retropharyngeal abscess

9. A 5-year-old child presents with sudden high fever, severe sore throat, drooling, and labored breathing. The child assumes a tripod position and shows cyanosis. A lateral neck X-ray is performed. Which of the following X-ray findings is most consistent with this clinical presentation?

A. Steeple sign

B. Thumb sign

C. Sail sign

D. Double bubble sign

10. A 10-year-old boy presents with a sore throat, fever, headache, tonsillar exudates, and tender anterior cervical lymph nodes. A classmate was recently diagnosed with streptococcus pharyngitis. What is the best initial management?

A. Symptomatic treatment with antipyretics

B. Throat swab and start penicillin if positive
C. Start oral penicillin immediately
D. Treat with erythromycin

11. A 10-year-old child diagnosed with Cystic Fibrosis presents with recurrent lung infections, poor weight gain, and steatorrhea. Which of the following is the most appropriate combination of therapies for managing this patient's condition?

A. Inhaled bronchodilators, insulin therapy
B. Chest physiotherapy, Ursodeoxycholic acid
C. CFTR modulators, antibiotics, chest physiotherapy
D. Inhaled corticosteroids, bone health management

12. A 6-year-old boy presents with recurrent respiratory infections, failure to thrive, and greasy stools. His older sibling was diagnosed with cystic fibrosis (CF). Which of the following is not a part of the diagnostic criteria of Cystic fibrosis in this patient?

A. Abnormal nasal potential difference measurement
B. Presence of two CFTR mutations
C. Two elevated sweat chloride concentration on separate days
D. Sputum culture positive for Burkholderia cepacia

13. A 3-month-old infant is brought to the pediatric OP for evaluation of failure to thrive, frequent greasy stools, and a persistent cough. The infant was diagnosed with an autosomal recessive disease via a sweat test. Which of the following is most likely to be associated with this patient's condition?

A. Hemoptysis

B. Pancreatic insufficiency

C. Paralytic ileus

D. Cor pulmonale

14. A 5-year-old child is brought to the clinic with a history of chronic productive cough, recurrent respiratory infections, and poor weight gain despite an adequate diet. Additionally, the parents report that the child's sweat is very salty. Which of the following sets of clinical features would most likely indicate the diagnosis?

A. Bronchial asthma, nasal polyps, and aspirin intolerance

B. Chronic sinusitis, meconium ileus in infancy, and rectal prolapse

C. Bronchiectasis, thrombocytopenia, and abdominal pain

D. Frequent diarrhea, poor weight gain, and recurrent otitis media

15. A 5-year-old child is brought to the emergency department following a road traffic injury with a low GCS score. The decision to intubate the child is made during the primary intervention. Which of the following facts should be considered while intubating a child?

A. The subglottis is the narrowest part of the pediatric airway.

B. The cricoid cartilage is situated at the level of vocal cords in children.

C. The pediatric trachea is less prone to collapse.

D. The functional residual capacity is higher than the closing capacity.

16. Which of the following clinical indicators is most characteristic of life-threatening asthma?

A. Mild wheezing

B. Oxygen saturation of 92%

C. Silent chest

D. Peak expiratory flow (PEF) of 60% predicted

17. A 7-year-old boy presents to the emergency department with severe shortness of breath and wheezing unresponsive to his rescue inhaler. His vitals include a respiratory rate of 34/min, heart rate of 125/min, and oxygen saturation of 91% on room air. After giving oxygen and albuterol, what is the next step in management?

A. Intubation and mechanical ventilation

B. Start systemic corticosteroids

C. Administer intravenous magnesium sulfate

D. Provide inhaled anticholinergics like ipratropium bromide

18. Match the following asthma medication devices with its corresponding image. 1. Metered Dose Inhaler (MDI) a. 2. MDI with Spacer b. 3. Dry Powder Inhaler (DPI) c. 4. Nebulizer d.

1. Metered Dose Inhaler (MDI)	 a.

2. MDI with Spacer	 b.
3. Dry Powder Inhaler (DPI)	 ... (content truncated)
4. Nebulizer	 ... (content truncated)

A. 1-b, 2-a, 3-c, 4-d

B. 1-a, 2-b, 3-d, 4-c

C. 1-d, 2-c, 3-a, 4-b

D. 1-c, 2-b, 3-a, 4-d

19. A 10-year-old child with persistent asthma is currently on a low-dose inhaled corticosteroid (ICS) but is still experiencing symptoms. What is the next step in the management of this child according to the stepwise approach for long-term asthma management?

A. Increase to medium-dose ICS

B. Add a long-acting beta-agonist (LABA) to the low-dose ICS

C. Add a leukotriene receptor antagonist (LTRA) or theophylline to the low-dose ICS

D. All of the above

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20. A 10-year-old child presents with symptoms of asthma that occur more than twice a week but not daily. The child has nighttime symptoms 3-4 times a month. Pulmonary function tests reveal an FEV1 of 85% of the predicted value. How would you classify the severity of this child's asthma?

A. Mild Intermittent

B. Mild Persistent

C. Moderate Persistent

D. Severe Persistent

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21. A 10-year-old child presents with a history of recurrent wheezing and cough. The physician suspects asthma and decides to perform some diagnostic tests. Which test result would most strongly support the diagnosis of asthma?

A. FEV1/FVC ratio of 0.9

B. Increase in FEV1 by >10% post-bronchodilator

C. Daily peak expiratory flow <20%

D. FEV1 < 10% after exercise challenge

22. A 5-year-old with recurrent wheezing, coughing, and shortness of breath, worsened by cold air and with a history of eczema and allergic rhinitis, is experiencing persistent symptoms despite antihistamines. Which type of asthma is most likely?

A. Transient nonatopic asthma

B. Seasonal allergic asthma

C. Late-onset asthma

D. Persistent atopy-associated asthma

23. A 10-month-old child is brought with complaints of sudden coughing, wheezing, and difficulty breathing after playing with small toys. On examination, there are decreased breath sounds on one side. A chest X-ray was taken, which is shown below. What is the most appropriate initial management?

A. Administer supplemental oxygen.

B. Administer systemic corticosteroids.

C. Perform rigid bronchoscopy.

D. Perform emergency cricothyrotomy.

24. A 9-month-old infant presented with sudden choking on a small toy part while playing. The infant appears cyanotic and is making weak, ineffective attempts to breathe. What is the most appropriate initial intervention for this infant?

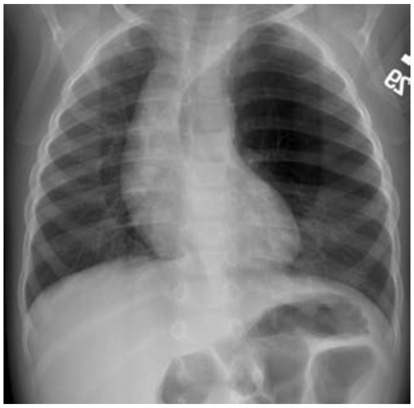
A. Perform heimlich maneuver.

B. Start CPR with chest compressions.

C. Deliver five back blows.

D. Give rescue breaths.

25. A newborn baby is brought to the hospital by her parents with complaints of wheezing. On further examination, she has tachypnea. Radiological examination shows a hyperlucent lobe and a mediastinal shift, as in the image. Which is the most common location of this condition?



A. Right upper lobe

B. Left upper lobe

C. Right lower lobe

D. Left lower lobe

26. An infant on examination has decreased breath sounds over the right lower side of the chest, which is dull on percussion. A chest X-ray shows as follows. Which of the following is true about this condition?

A. The abnormal lung tissue is attached to the bronchial tree.

B. Blood returns to the right side of the heart from this lesion.

C. It usually presents with crackles.

D. Chest X-ray is not a reliable diagnostic test in this case.

27. A 2-week-old infant is brought to a pediatrics OPD due to feeding difficulties. The maternal history is unremarkable. On examination, the baby has a wet inspiratory stridor, that sounds like a low-pitched squeak. The pediatrician suspects laryngomalacia and is concerned about possible dysphagia. Which investigation is most appropriate to confirm the diagnosis?

A. Chest X-ray

B. Airway films

C. Fibre optic endoscopic evaluation

D. Outpatient flexible laryngoscopy

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	3
Question 5	3
Question 6	3
Question 7	3
Question 8	3
Question 9	2
Question 10	3
Question 11	3
Question 12	4
Question 13	3
Question 14	2
Question 15	1
Question 16	3
Question 17	2
Question 18	4
Question 19	4
Question 20	2
Question 21	2
Question 22	4

Question 23	3
Question 24	3
Question 25	2
Question 26	2
Question 27	3

Solution for Question 1:

Correct Answer: B) Impaired chloride transport resulting in dehydrated airway surfaces

Explanation:

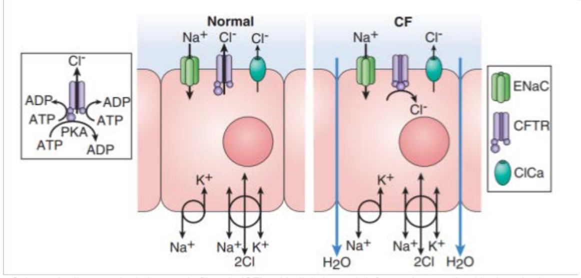
- The CFTR ΔF508 mutation is the most common mutation causing cystic fibrosis.
- This mutation leads to defective CFTR protein, which impairs chloride transport across epithelial cells.
- As a result, water movement is reduced, leading to thick, dehydrated mucus in the respiratory tract.
- This viscous mucus impairs mucociliary clearance, trapping bacteria and leading to chronic infections, a hallmark of CF lung disease.

Genetic Basis of Cystic Fibrosis:

Gene Involved	CFTR (Cystic Fibrosis Transmembrane Conductance Regulator) gene
Chromosome	Located on chromosome 7q31.2
Common Mutations	ΔF508 (deletion of phenylalanine at position 508) is the most common mutation.
Classes of Mutations	Class I: No functional CFTR protein Class II: CFTR trafficking defect Class III: Defective channel regulation Class IV: Decreased channel conductance Class V: Reduced synthesis of CFTR Class VI: Decreased CFTR stability
Inheritance Pattern	Autosomal recessive

Pathophysiology of Cystic Fibrosis:

CFTR Protein Dysfunction	Defective chloride transport across epithelial cells Increased sodium absorption Decreased water movement leading to dehydrated mucous surfaces
Respiratory System	Thick, viscous mucus leads to airway obstruction and bacterial colonization Chronic infection and inflammation Bronchiectasis, atelectasis, and respiratory failure
Gastrointestinal System	Thick secretions block pancreatic ducts leading to enzyme insufficiency Malabsorption of fats and proteins Meconium ileus in neonates Distal Intestinal Obstruction Syndrome (DIOS)
Hepatobiliary System	Bile duct obstruction leading to biliary cirrhosis
Sweat Glands	Impaired chloride reabsorption leading to salty sweat
Reproductive System	Congenital absence of the vas deferens in males leading to azoospermia and infertility. Defective CFTR in the cervix leads to the production of thick, dehydrated cervical mucus that blocks the cervical os, impairing sperm entry and reducing fertility.



Schematic diagram depicting cystic fibrosis (CF) epithelial channel defects, characterized by impaired chloride secretion, massive sodium absorption, and movement of water through the epithelium, leading to a dehydrated airway surface.

Increased bicarbonate secretion leading to decreased mucus viscosity (Option A): In cystic fibrosis, bicarbonate secretion is actually decreased, not increased, leading to a more acidic and viscous mucus, which further complicates mucociliary clearance

Enhanced ciliary function due to overactivation of ENaC channels (Option C): Ciliary function is impaired, not enhanced in cystic fibrosis due to the thickened mucus, due to overactivation of ENaC channels (due to loss of CFTR function).

Decreased oxidative stress reducing bacterial clearance (Option D): Cystic fibrosis is characterized by increased oxidative stress, not decreased. This heightened oxidative stress contributes to inflammation and tissue damage

Solution for Question 2:

Correct Answer: B) Bronchiectasis

Explanation:

The HRCT shows cylindrical bronchiectasis with a "tram line" appearance, and the symptoms of chronic cough, copious purulent sputum, and recurrent infections are consistent with diagnosis of bronchiectasis.

Bronchiectasis	Chronic suppurative disease characterized by destruction of the bronchial and peribronchial tissues, dilatation of the bronchi and accumulation of infected material in the dependent bronchi.
Predisposing Conditions	Proximal Airway Narrowing Airway Injury Altered Pulmonary Host Defenses: Cystic fibrosis Ciliary dyskinesia Altered Immune States Other: Allergic bronchopulmonary aspergillosis Plastic bronchitis Right middle lobe syndrome
Pathogenesis	Three basic mechanisms involved: Obstruction: Due to tumors, foreign bodies, impacted mucus, external compression, bronchial webs, or atresia. Infection: Induce chronic inflammation, progressive bronchial wall damage and dilation. Chronic Inflammation
Progression	Early Stages: Leads to bronchial dilation. Later Stages: Results in pulmonary hypertension.
Pathologic Forms	Cylindrical: Regular bronchial outlines with diffuse dilation and mucous plugging. Varicose: Greater dilation with irregular bronchial outlines and local constriction cause irregularity of outline resembling varicose veins. Saccular (Cystic): Severe dilation with ballooning of bronchi, ending in fluid or mucus-filled sacs.

Symptoms	Cough Copious, purulent sputum Hemoptysis occasionally Fever Anorexia Poor Weight Gain Irritability Recurrent infections
Physical Examination	Crackles: Localized to the affected area Wheezing Digital Clubbing
Investigation	 Chest X-ray: Increased size and loss of definition of bronchovascular markings, crowded bronchi, loss of lung volume. High-Resolution Computed Tomography (HRCT): Gold Standard ... (content truncated)
Treatment	Airway Clearance Techniques Antibiotics: 2-4 weeks of parenteral antibiotics for acute exacerbations Effective Antibiotics: Amoxicillin/clavulanic acid (22.5 mg/kg/dose twice daily) is commonly used Bronchodilators Treat underlying disorders Surgical Options: Resection: Segmental or lobar resection for severe or resistant cases Lung Transplantation: Considered for advanced cases.

Bronchial asthma (Option A) primarily presents with reversible airway obstruction and may show hyperinflation on HRCT, but it is less likely to produce the persistent bronchial dilation patterns depicted in the HRCT image.

Pneumonia (Option C) generally presents with consolidation or infiltrates on imaging rather than the specific bronchial dilation patterns observed in bronchiectasis.

Chronic bronchitis (Option D) is unlikely in children because it primarily affects adults and is linked to smoking or pollution. The child's symptoms and HRCT findings are more consistent with bronchiectasis.

Solution for Question 3:

Correct Answer: C) Two or more episodes in a year, with radiographic clearing.

Explanation:

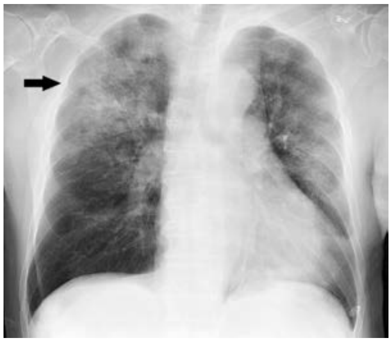
Recurrent Pneumonia	Defined as two or more episodes in a single year or three or more episodes ever, with radiographic clearing between occurrences. (Option C) (Option A, B, and D ruled out)
Persistent Pneumonia	Defined as the persistence of symptoms and radiographic abnormalities for more than one month, despite appropriate antibiotic treatment
Potential Underlying Disorders	
Hereditary Disorders	Cystic fibrosis Sickle cell disease
Disorders of Immunity	HIV/AIDS Bruton agammaglobulinemia Common variable immunodeficiency syndrome Severe combined immunodeficiency syndrome Chronic granulomatous disease Hyper immunoglobulin E syndromes Leukocyte adhesion defect
Disorders of Cilia	Primary ciliary dyskinesia Kartagener syndrome
Anatomic Disorders	Pulmonary sequestration Lobar emphysema Congenital cystic adenomatoid malformation Gastroesophageal reflux Foreign body Tracheoesophageal fistula (H type) Bronchiectasis Aspiration (oropharyngeal incoordination) Aberrant bronchus

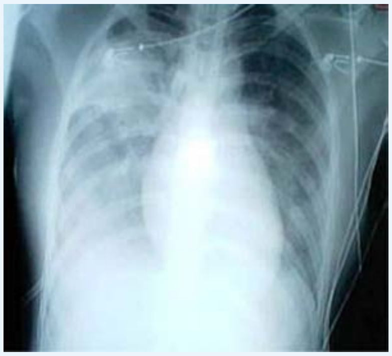
Solution for Question 4:

Correct Answer: C) Pneumatoceles

Explanation:

Pneumatoceles are commonly seen in Staphylococcal pneumonia, resulting from the formation of air-filled cavities within the lung due to the destruction of bronchoalveolar tissue and subsequent abscess formation.

Type of Pneumonia	X-ray Findings
Staphylococcal Pneumonia	 Confluent bronchopneumonia, often unilateral (affecting one lung). Multiple microabscesses leading to pneumatoceles (Option C) ... (content truncated)
Streptococcal Pneumonia	 Confluent lobar consolidation S.pneumoniae produces local edema that aids in the spread of organisms, often resulting in the characteristic focal lobar involvement.

Primary Atypical Pneumonia	Bilateral interstitial infiltrates with bronchial wall thickening Poorly defined hazy or fluffy exudates radiating from the hilar regions. Possible enlarged hilar lymph nodes and pleural effusion.
Viral Pneumonia	 Bilateral interstitial infiltrates ... (content truncated)

Type of Pneumonia Etiology	Clinical Features	Treatment
Staphylococcal Pneumonia Staphylococcus (Primary/Secondary to septicemia)	Predisposing factors: malnutrition, diabetes, macrophage dysfunction. Can complicate measles, influenza, cystic fibrosis, or follow staphylococcal pyoderma.	Hospitalization IV fluids for hydration Oxygen for dyspnea/cyanosis Antibiotics: Penicillin G, Co-amoxiclav, Cloxacillin, or Ceftriaxone. Alternative: Vancomycin, Teicoplanin, Linezolid Drainage for empyema/pyopneumothorax
Streptococcal Pneumonia Group A beta-hemolytic streptococci	May follow measles, varicella, influenza, or pertussis. Interstitial pneumonia, often hemorrhagic. Abrupt onset with fever, chills, cough, dyspnea, rapid respiration, and blood-streaked sputum.	Outpatient Treatment (Mild Cases): High-dose amoxicillin (90 mg/kg/day divided twice daily) Alternative options include cefuroxime or amoxicillin/clavulanate. Inpatient Treatment: Ampicillin or penicillin G Ceftriaxone or cefotaxime in cases with penicillin resistance
Primary Atypical Pneumonia Mycoplasma pneumoniae, Chlamydia, Legionella spp.	Common in winter epidemics among children in overcrowded living conditions. Rare under 4 years old. Incubation: 12-14 days.	Macrolide antibiotics (Erythromycin, Azithromycin, Clarithromycin) Tetracycline for 7-10 days.

Viral Pneumonia Respiratory syncytial virus (RSV), Parainfluenza, Influenza, Adenovirus	RSV is common under 6 months old. Extensive interstitial pneumonia.	Supportive care.

Solution for Question 5:

Correct Answer: C) Severe Pneumonia or Very Severe Disease

Explanation:

- Severe Pneumonia or Very Severe Disease is classified when a child presents with any general danger sign, oxygen saturation below 90%, stridor in a calm child, central cyanosis, or a response of V, P, or U on the AVPU scale.
- In this case, the child shows multiple severe signs, including lethargy, inability to drink, oxygen saturation of 88%, stridor, and a response to painful stimuli only, all of which indicate a very severe disease.

Pneumonia - Etiology :

- Respiratory Syncytial Virus (RSV): The leading viral cause of pneumonia in infants.
- Streptococcus pneumoniae: The most common bacterial cause of pneumonia in children, especially in those aged 1-3 months.
- Pneumocystis jirovecii is a leading cause of pneumonia in HIV-infected infants and young children, particularly in settings with limited access to antiretroviral therapy (ART) and Pneumocystis pneumonia (PCP) prophylaxis.
- Other bacteria: E-coli, other Gram-negative bacilli, Hemophilus influenzae, Group B streptococcus.
- Other viruses: rhinoviruses, parainfluenza viruses, influenza viruses, human metapneumovirus, and adenovirus.
- Other fungal: Histoplasma, aspergillus, blastomyces, cryptococcus, mucormycosis, coccidioides.
- Parasitic: Ascaris, strongyloides.
- Mycobacterium tuberculosis, mycobacterium avium complex.

Assessment and Classification of the Sick Child (Age 2 Months Up to 5 Years)

- Initial Assessment: A thorough history from the mother is essential

- Check for General Danger Signs:

Ask	Look	Action
Able to drink or breastfeed? Does the child vomit everything? H/O convulsions during this illness?	Lethargic or unconscious? Currently convulsing?	If any danger signs are present, the child needs URGENT attention.

- Assess Cough or Difficulty Breathing:
- Ask: How long has the child had a cough or difficulty breathing?
- Look, Listen, Feel: Count the breaths in one minute (use respiratory rate timers if available).
- Fast Breathing Criteria

Age 2 months up to 12 months: Fast breathing is 50 breaths per minute or more.

Age 12 months up to 5 years: Fast breathing is 40 breaths per minute or more.

- Look for chest in-drawing
- Look and listen for stridor, wheeze
- Check for central cyanosis
- Check oxygen saturation
- Check the AVPU scale (Alert, Voice, Pain, Unresponsive)
- Assess for possible asthma/TB if wheezing with chest in-drawing or fast breathing is present.

Classify the cough:

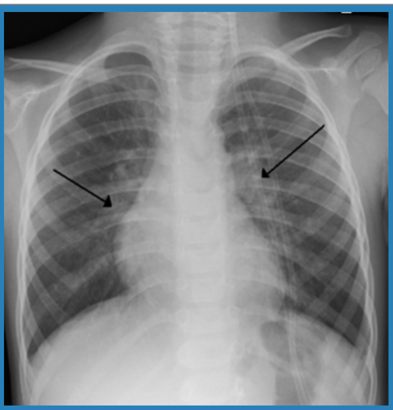
Classification	Signs	Treatment
Severe Pneumonia or Very Severe Disease	Any general danger sign OR Oxygen saturation <90% Stridor in calm child Central cyanosis Convulsions AVPU = V, P, or U	Start oxygen therapy if saturation < 90%. Administer Benzyl Penicillin & Gentamicin. Give diazepam if convulsing. Prevent low blood sugar. Treat wheeze if present. Urgent referral or admission.
Pneumonia	Chest indrawing in calm child OR Fast breathing AND No signs of severe pneumonia	Administer Amoxicillin Dispersible Tablet. Give Vitamin A. Treat wheeze if present. Follow up in 5 days Advise on immediate return signs.
No Pneumonia: Cough or Cold	No signs of pneumonia or very severe disease	Treat wheeze if present - salbutamol for 5 days Advise on immediate return signs.

Solution for Question 6:

Correct Answer: C) Provide oxygen and ensure proper hydration

Explanation:

- Chest X-ray finding of hyperinflation without focal infiltrate and symptoms like cough, wheezing and difficulty breathing in an infant point towards diagnosis of bronchiolitis.
- Supportive care with moist oxygen and hydration is the mainstay treatment of bronchiolitis.

Investigation	 ... (content truncated)
Course and Prognosis	Self-limiting. Resolves in 3-7 days. Mortality in severe cases is around 1%.

Treatment	General: Symptomatic, child should be nursed in a humidified atmosphere in a sitting position. Home Care: For mild cases. Hospital Care: Moist oxygen inhalation (mainstay of treatment) Fluid and electrolyte balance (Option C) Very sick infants: 60% oxygen given through a hood. Pulse oximetry performed regularly to maintain oxygen saturation >92%. Medications: Antibiotics have no role (Option B) Ribavirin: Shorten the course of illness in high-risk infants with underlying congenital heart disease, chronic lung disease and immunodeficiency (Option D) Bronchodilators and steroids are generally not useful. (Option C) Inhaled hypertonic saline is useful (Option B) Severe Cases: May need CPAP (continuous positive airway pressure), mechanical ventilation, or ECMO (extracorporeal membrane oxygenation).
Prevention	Hand Hygiene: Effective for preventing virus transmission. Palivizumab (IM): Monoclonal antibody to RSV F protein (Protects only against RSV); Prophylactically given to reduce hospitalization risk in high-risk infants.

Solution for Question 7:

Correct Answer: C) Bronchiolitis

Explanation:

3 month old infant with cough, wheezing, and crackles, history of an upper respiratory tract infection 1 week ago along with examination findings of intercostal retractions and diffuse crackles point towards diagnosis of Bronchiolitis.

Bronchiolitis	
Age Group	Infants between 1 and 6 months, but can affect children up to 2 years old.
Seasonality	Occurs primarily in winter and spring.
Causative Agent	Respiratory syncytial virus (RSV) (Most common) Others include: Parainfluenza virus

	Adenovirus Influenza virus Rarely Mycoplasma pneumoniae.
Risk factors	M/c in male infants. Infants exposed to tobacco smoke, not breastfed, or living in crowded conditions. Maternal smoking Older family members/siblings (common source of infection)
Protection	IgG3 antibodies (short half-life, limited placental transfer) Breastfeeding provides IgA antibodies.
Pathophysiology	Bronchiolar obstruction with edema, mucus and cellular debris. Increased airway resistance during both inspiration and exhalation. Leads to expiratory wheezing, air trapping, lung hyperinflation and possible atelectasis. Hypoxemia due to ventilation-perfusion mismatch. With severe obstructive disease hypercapnia develops.
Clinical Manifestations	First develop upper respiratory symptoms: Sneezing and clear rhinorrhea. Fever (±) Diminished appetite Persistent Cough Dyspnea Irritability Respiratory distress Cyanosis.
Physical Examination	Prolonged expiratory time Wheezing Increased work of breathing with nasal flaring and retractions Accessory muscles of respiration are working Auscultation: Fine crepitations and rhonchi Faint or inaudible breath sounds in severe disease

Bronchial asthma (Option A) is unlikely because it is uncommon in infants under one year, often has a family history, usually involves multiple episodes, may occur without preceding respiratory infection and responds well to bronchodilators.

Bacterial pneumonia (Option B) often presents with fever and localized lung findings. The diffuse crackles and wheezing here are more characteristic of bronchiolitis.

Foreign body aspiration (Option D) is unlikely because there is no history of aspiration of foreign body. It typically presents with localized wheezing and signs of collapse or localized obstructive emphysema.

Solution for Question 8:

Correct Answer: C) Croup

Explanation:

Young child characterized by a barking cough, hoarseness and stridor with X-ray PA view of the neck showing subglottic narrowing, known as the steeple sign, is consistent with diagnosis of croup.

Laryngotracheobronchitis (Croup): Refers to viral infection of the glottic and subglottic regions.	
Causative Agent	Parainfluenza viruses (types 1,2 and 3) cause about 75% cases. Other Viruses: Influenza A and B, Adenovirus, Respiratory syncytial virus, Measles.
Age Group	Most common in children aged 3 months to 5 years. Peak incidence occurs in the 2nd year of life.
Initial Symptoms	These symptoms last 1-3 days before development of upper airway obstruction signs and symptoms. Upper Respiratory tract infection Rhinorrhea, Pharyngitis, Mild cough, Low-grade fever.
Progressive Symptoms	Barking cough, Hoarseness, Inspiratory stridor, Fever may reach 39-40°C (102.2-104°F) or be absent. Child prefers to sit up in bed or be held upright. Symptoms worsen at night, recur over days and resolve within a week.
Physical Examination Findings	Hoarse voice, Coryza, Normal to moderately inflamed pharynx. Slightly increased respiratory rate. Hypoxia and low oxygen saturation seen only in complete airway obstruction.

Diagnosis	 ... (content truncated)
Treatment	Mainstay of Treatment: Focus on airway management and hypoxia treatment. Mild croup: Single dose dexamethasone. Moderate to severe croup: Nebulized epinephrine in addition to dexamethasone. Nebulized Racemic Epinephrine: Used for moderate to severe stridor at rest, possible need for intubation, respiratory distress and hypoxia. Corticosteroids: Decrease edema in laryngeal mucosa Oral steroids beneficial in mild croup Dexamethasone 0.6 mg/kg single dose; as low as 0.15 mg/kg may be effective. Antibiotics are not indicated.

Epiglottitis (Option A) presents with acute onset of severe sore throat, drooling, difficulty breathing, and may show a thumb sign on X-ray.

- It’s characterized by rapid progression and is not usually associated with the steeple sign or subglottic narrowing seen in croup.

Bacterial tracheitis (Option B) is marked by a brassy cough, biphasic stridor, and an irregular tracheal wall on X-ray.

- Croup, with a barking cough and the steeple sign, differs from this presentation.

Retropharyngeal abscess (Option C) results from a suppurative complication of bacterial pharyngitis or dental infection.

- It presents with high fever, stridor, drooling, and neck stiffness. It is confirmed by increased prevertebral space thickness on X-ray, not the steeple sign.

Solution for Question 9:

Correct Answer: B) Thumb sign

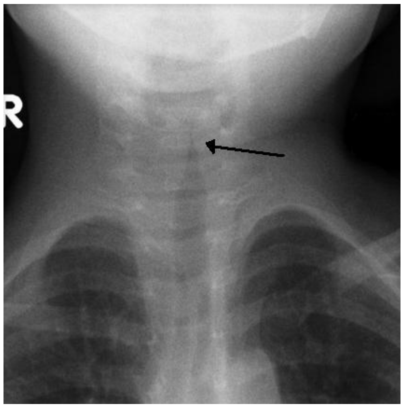
Explanation:

- The given Lateral neck X-ray shows a swollen, rounded epiglottis that resembles a thumb.
- This sign is highly indicative of epiglottitis, along with symptoms like drooling, high fever, and severe sore throat.

Acute Epiglottitis	Life-threatening infection that causes profound swelling of the upper airways, which can lead to asphyxia and respiratory arrest.
Causative Agent	Past Era (Pre-vaccine): Haemophilus influenzae type b (Hib) Post-vaccine Era: Streptococcus pneumoniae (M/c), Streptococcus pyogenes, non-typeable H. influenzae, and Staphylococcus aureus Hib cases have decreased by 99% due to widespread vaccination.
Clinical Features	High fever, Sore throat, Dyspnea, Respiratory obstruction, Respiratory distress, Child appears toxic, Difficulty swallowing, Labored breathing, Drooling, Neck hyperextension to maintain airway, Tripod position (sitting upright, leaning forward) Stridor is a late sign indicating near-complete airway obstruction. Cyanosis, ... (content truncated)
Diagnosis	Laryngoscopy: Large, cherry-red, swollen epiglottis Lateral neck X-ray: Thumb sign
Treatment	Immediate Treatment: Secure the airway with endotracheal or nasotracheal intubation or, less often, with tracheostomy. Intubation is generally for 2-3 days. Intubation decreases mortality from 6% to <1% in epiglottitis cases. Culture Collection: Blood, epiglottic surface and CSF in certain cases after the airway is stabilized. Antibiotics: Ceftriaxone, cefepime or meropenem given parenterally for 10 days.

Recovery	Symptoms improve with a few days of antibiotics, and the patient may be extubated.

Steeple sign (Option A) indicates characteristic narrowing of the subglottic region, seen in acute laryngotracheobronchitis (croup), not related to epiglottis.



Sail sign (Option C) seen in infants as a triangular extension of the normal thymus laterally, not related to epiglottis.

Double bubble sign (Option C) indicates radiograph findings of two adjacent fluid-filled echolucent structures within the abdomen of fetus, is seen in duodenal atresia, not related to epiglottis.

Solution for Question 10:

Correct Answer: C) Start oral penicillin immediately

Explanation:

Symptoms and history strongly suggest Streptococcus pyogenes pharyngitis and early treatment with penicillin is needed to prevent complications like acute rheumatic fever.

Aspect	Bacterial pharyngitis	Viral pharyngitis
Causative Agents	Streptococcus pyogenes (Most common), Arcanobacterium haemolyticum, Fusobacterium necrophorum, Corynebacterium diphtheriae, Neisseria gonorrhoeae, Group C and G streptococci, Francisella tularensis, Chlamydophila pneumoniae, Chlamydia trachomatis, Mycoplasma pneumoniae.	Adenovirus, Coronavirus, Cytomegalovirus, Epstein-Barr virus, Enteroviruses, Herpes simplex virus, Human metapneumovirus, Influenza viruses, Measles virus, Parainfluenza viruses, Respiratory syncytial virus, Rhinoviruses.

Features	Sudden onset of sore throat Age 5-15 yr Fever Headache Nausea, vomiting, abdominal pain Tonsillopharyngeal inflammation Patchy tonsillopharyngeal exudates Palatal petechiae Anterior cervical adenitis (tender nodes) Winter and early spring presentation History of exposure to strep pharyngitis Scarlatiniform rash.	Conjunctivitis Coryza Cough Diarrhea Hoarseness Discrete ulcerative stomatitis Viral exanthema
Diagnosis	Throat swab Throat culture Rapid antigen detection tests (RADTs) Molecular tests: PCR, Nucleic acid amplification tests	Molecular tests: PCR
Treatment	DOC: Oral Penicillin (Option C) Amoxicillin is preferred in children, with once-daily dosing for 10 days A single dose of IM benzathine penicillin or benzathine-procaine penicillin G can ensure compliance. Penicillin Allergy: Use 1st generation cephalosporins (cephalexin or cefadroxil) or macrolides (erythromycin,clarithromycin or clindamycin) for 10 days or azithromycin for 5 days. Main goal of antibiotic therapy is to prevent acute rheumatic fever, if started within 9 days of illness onset.	Symptomatic treatment like oral antipyretics/ analgesics.
Complication	Parapharyngeal abscesses Acute Rheumatic Fever (ARF) Acute Poststreptococcal Glomerulonephritis (APSGN) Post-streptococcal Reactive Arthritis PANDAS (Pediatric Autoimmune Neuropsychiatric Disorders Associated with Streptococci)	Secondary bacterial infections: Middle ear infections (otitis media) and bacterial sinusitis.

Symptomatic treatment with antipyretics (Option A):

- While symptomatic treatment is important for viral pharyngitis, the patient's symptoms and exposure history strongly suggest Streptococcus pyogenes pharyngitis.
- Immediate antibiotic treatment is needed to prevent complications.

Throat swab and start penicillin if positive (Option B):

Although this approach is appropriate in clinical practice, starting penicillin immediately is recommended in this scenario due to the high suspicion of Streptococcus pyogenes pharyngitis and to prevent complications if the antibiotic is started within 9 days of illness onset.

Treat with erythromycin (Option C):

- Erythromycin is generally used for penicillin-allergic patients.
- In this case, penicillin is preferred unless there is a documented penicillin allergy.

Solution for Question 11:

Correct Answer: C) CFTR modulators, antibiotics, chest physiotherapy

Explanation:

- CFTR modulators (e.g., ivacaftor, lumacaftor) directly target the defective CFTR protein, improving its function and thereby helping to alleviate symptoms of CF at a molecular level.
- Trikafta (Elexacaftor-Tezacaftor-Ivacaftor and Ivacaftor): For patients aged 2 years and older with cystic fibrosis, this medication was well tolerated and shown to improve lung function and reduce sweat chloride levels.

- Antibiotics are crucial for managing and preventing chronic pulmonary infections, which are a major cause of morbidity in CF patients.

Management Strategies for Cystic Fibrosis

Management Aspect	Therapies
Airway Clearance	Chest physiotherapy, Percussion, Postural drainage
Infection Control	Antibiotics, CFTR modulators
Nutritional Support	High-calorie diet, Pancreatic Enzyme Replacement Therapy (PERT), Vitamin supplementation (A, D, E, K)
CFTR Modulation	CFTR modulators (e.g., ivacaftor, lumacaftor)
Gastrointestinal Management	Ursodeoxycholic acid, laxatives, enemas
Other Therapies	Insulin therapy (for CFRD), Bone health management (calcium, vitamin D)

Inhaled bronchodilators, insulin therapy (Option A):

- Inhaled bronchodilators can be used for airway obstruction but do not address the underlying CFTR defect or chronic infections
- Insulin therapy is used for managing CF-related diabetes, due to impaired insulin secretion and insulin but is not the first-line treatment for overall CF management. This option lacks the key components like CFTR modulators and infection control.

Chest physiotherapy, Ursodeoxycholic acid (Option B):

- Chest physiotherapy is indeed a cornerstone of CF management for mucus clearance, but without CFTR modulators or antibiotics, it does not fully manage CF.
- Ursodeoxycholic acid is used for managing liver disease in CF but is not relevant for lung infection control or nutritional management, which are more critical in this scenario.

Inhaled corticosteroids, bone health management (Option D):

- Inhaled corticosteroids can reduce inflammation but are not typically first-line therapy for CF and are not generally used unless the patient has severe asthma.
- Bone health management (e.g., calcium and vitamin D supplementation) is essential in the long term but does not address immediate nutritional needs or infection control.

Solution for Question 12:

Correct Answer: D) Sputum culture positive for Burkholderia cepacia

Explanation:

- Though Burkholderia cepacia is pathognomonic for Respiratory tract infections in Cystic Fibrosis, it is not a diagnostic criterion.
- Presence of typical clinical features (such as recurrent respiratory infections and gastrointestinal issues) plus laboratory evidence of CFTR dysfunction is required for a definitive diagnosis of cystic fibrosis.
- Most common microbe causing infection in cystic fibrosis: <18 years: Staphylococcus aureus >18 years: Pseudomonas aeruginosa Burkholderia cepacia: Difficult to detect and treat; despite their low incidence, these infections are associated with a high mortality rate.
- <18 years: Staphylococcus aureus
- >18 years: Pseudomonas aeruginosa
- Burkholderia cepacia: Difficult to detect and treat; despite their low incidence, these infections are associated with a high mortality rate.
- <18 years: Staphylococcus aureus
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- Burkholderia cepacia: Difficult to detect and treat; despite their low incidence, these infections are associated with a high mortality rate.

Diagnostic Criteria for Cystic Fibrosis

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Solution for Question 13:

Correct Answer: B) Pancreatic insufficiency

Explanation:

- Cystic Fibrosis is an autosomal recessive disease that can be detected via sweat chloride test in newborns.

- Pancreatic insufficiency (Option B) is seen in more than 85% of children with CF and leads to malabsorption, resulting in symptoms like failure to thrive, frequent greasy stools, and weight gain difficulties.

Clinical Features of Cystic Fibrosis by Age Group

Age Group	Sinopulmonary	Gastrointestinal	Renal, Endocrine, Other
Infancy	Infection	Fetal echogenic bowel Meconium ileus Pancreatic insufficiency Rectal prolapse	Dehydration Hyponatremic hypochloremic metabolic alkalosis
Childhood	Infection ABPA Polyposis Sinusitis	DIOS (Distal Intestinal Obstruction Syndrome) Intussusception Hepatic steatosis, biliary fibrosis Rectal prolapse	Renal calculi Hyponatremic hypochloremic metabolic alkalosis
Adolescence/Adulthood	Infection ABPA Haemoptysis Pneumothorax Respiratory failure Sinusitis, Polyposis, Anosmia	DIOS Intussusception Biliary fibrosis, cirrhosis Digestive tract cancer (adenocarcinoma)	Delayed puberty, osteoporosis, CFRD (Cystic Fibrosis-Related Diabetes) Renal calculi, renal failure CBAVD (Congenital Bilateral Absence of the Vas Deferens), HPOA (Hypertrophic Pulmonary Osteoarthropathy) Arthritis, vasculitis Hyponatremic hypochloremic metabolic alkalosis

Hemoptysis (Option A) is a complication of advanced lung disease in CF and is more common in older children or adolescents rather than infants.

Paralytic ileus (Option C) refers to a lack of bowel movement due to impaired motility of the intestines

Cor pulmonale (Option D) is a late-stage complication of CF due to chronic lung disease and is not typically seen in infants but rather in older children or adolescents.

Solution for Question 14:

Correct Answer: B) Chronic sinusitis, meconium ileus in infancy, and rectal prolapse

Explanation:

The presentation of chronic productive cough, recurrent respiratory infections, poor weight gain, and salty sweat is consistent with cystic fibrosis.

- Chronic sinusitis is a common manifestation of cystic fibrosis due to persistent inflammation and obstruction in the sinuses.
- Meconium ileus in newborns is often the first sign of cystic fibrosis, caused by thick, sticky meconium obstructing the intestines.
- Rectal prolapse is a common gastrointestinal feature in childhood for patients with cystic fibrosis. It occurs due to increased intra-abdominal pressure from chronic coughing.

Clinical Features of Cystic Fibrosis

System	Clinical Features
Respiratory	Chronic productive cough Recurrent respiratory infections Nasal polyps Bronchiectasis
Gastrointestinal	Greasy stools (steatorrhea) Failure to thrive Rectal prolapse Malabsorption Meconium ileus in newborns
Genitourinary	Infertility in males (due to congenital absence of vas deferens), delayed puberty
Others	Digital clubbing, salt loss syndromes (hypochloremic metabolic alkalosis)

Bronchial asthma, nasal polyps, and aspirin intolerance (Option A) is less indicative of cystic fibrosis as these features are more characteristic of Samter's triad rather than the hallmark symptoms of cystic fibrosis.

Bronchiectasis, thrombocytopenia, and abdominal pain (Option C): Bronchiectasis is a feature of CF due to chronic lung infection and damage.

- However, thrombocytopenia is not commonly associated with CF, and abdominal pain alone is too nonspecific to indicate CF.

Frequent diarrhea, poor weight gain, and recurrent otitis media (Option D): Frequent diarrhea and poor weight gain are features of CF due to malabsorption from pancreatic insufficiency.

- However, recurrent otitis media is not a feature of CF.

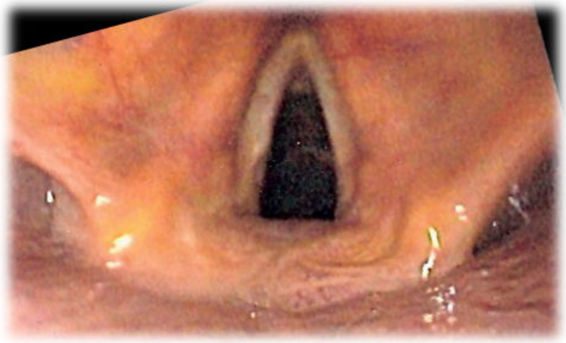
Solution for Question 15:

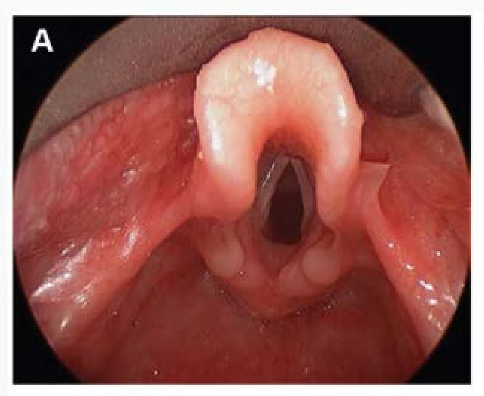
Correct Answer: A) The subglottis is the narrowest part of the pediatric airway

Explanation:

In children, the narrowest part of the airway is the subglottis, which begins below the true vocal cords and ends at the inferior margin of the cricoid cartilage (which is present just below the vocal cords).

Feature	Pediatric airway	Adult airway
Diameter and Resistance	Smaller diameter increases airway resistance significantly	Larger diameter results in lower resistance
Shape	Funnel-shaped	Cylindrical
Carina	At T2	At T4
Epiglottis	Long, floppy, and U-shaped	Flat, spade-shaped
Narrowest point	Subglottis (below vocal cords to the inferior border of cricoid cartilage) (Option B)	At the level of the vocal cords
Tracheal Compliance	Higher tracheal compliance, more prone to collapse (e.g., tracheomalacia) (Option C)	Less compliant trachea, less prone to collapse
Breathing Pattern	Obligate nasal breathers; nasal obstruction can cause significant distress	Can breathe through the mouth if nasal passages are obstructed
Chest Wall Compliance	Higher compliance leads to air trapping and potential hyperinflation	Lower compliance with less air-trapping
Closing Capacity	Higher closing capacity compared to functional residual capacity (FRC), leading to potential ventilation-perfusion mismatch (Option D)	Lower closing capacity relative to FRC, reducing mismatch risk
Susceptibility to Respiratory Distress	More susceptible to respiratory distress from infections, foreign body aspiration, and congenital anomalies	Less susceptible to rapid onset of respiratory distress from the same conditions





Solution for Question 16:

Correct Answer: C) Silent chest

Explanation:

A silent chest, where breath sounds are minimal or absent, is a critical indicator of life-threatening asthma.

Life-Threatening Asthma Management	
Overview	Most severe asthma exacerbation; requires immediate attention. Indicators: Cyanosis, silent chest, poor respiratory effort, exhaustion, altered mental state, PEF <30%, oxygen saturation <90%.
Immediate Management	Oxygen Therapy: Administer for hypoxemia. Systemic Bronchodilators: Terbutaline or adrenaline subcutaneously; inhaled bronchodilators (salbutamol,

	ipratropium). Systemic Corticosteroids: IV hydrocortisone (5 mg/kg). ICU Transfer: For monitoring and advanced care.
ICU Management	Continuous Monitoring: Vital signs, oxygen saturation, PEF. If Improvement Noted: Continue inhaled bronchodilators (salbutamol/terbutaline) every 20-30 minutes. Maintain hydrocortisone at 2-3 mg/kg every 6-8 hours until the patient can take oral medication. If No Improvement or Deterioration: Consider slow intravenous infusion of magnesium sulfate (50 mg/kg). Alternatively, administer a loading dose of theophylline. If these measures fail, prepare for mechanical ventilation. Address Underlying Causes: Check for acidosis, pneumothorax, electrolyte imbalance, and infection.

- Mild wheezing (Option A) is more typical of less severe asthma exacerbations.
- An oxygen saturation of 92% (Option B) does not indicate life-threatening asthma.
- A PEF of 60% predicted (Option D) suggests moderate asthma, not life-threatening.

Solution for Question 17:

Correct Answer: B) Start systemic corticosteroids

Explanation

Systemic corticosteroids should be administered early in the management of an acute asthma attack to reduce inflammation and prevent further deterioration.

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- Intubation (Option A): Considered only if the patient is in respiratory failure or if all other treatments have failed.
- Administer intravenous magnesium sulfate (Option C): Used as an adjunct in severe cases, but corticosteroids are necessary at this stage.
- Inhaled anticholinergics like ipratropium bromide (Option D): Can be used in conjunction with other treatments, but corticosteroids are a priority.

Solution for Question 18:

Correct Answer: D) 1-c, 2-b, 3-a, 4-d

Explanation:

Table of Asthma Medication Devices		
Device	Image	Features
Metered Dose Inhaler (MDI)		Fixed aerosol dose, requires coordination, oropharyngeal deposition possible that may lead to oropharyngeal candidiasis. While using it, at the start of inspiration that is slow and deep, press the canister down and continue to inhale deeply.
MDI with Spacer		Improves lung delivery, reduces oropharyngeal deposition, bulky, costly
MDI with Spacer & Facemask		Facilitates administration for young children
Dry Powder Inhaler (DPI)		Breath-activated, easy for older children, less effective during acute episodes
Nebulizer		Delivers large doses, useful in acute asthma

Solution for Question 19:

Correct Answer: D) All of the above

Explanation:

- Increasing to medium-dose ICS is one possible option in Step 3 for managing persistent asthma in children aged 5-11 years.
- Adding a LABA to the low-dose ICS and adding LTRA or theophylline to the low-dose ICS are also included in Step 3.

Stepwise Approach for Long-Term Management of Asthma in Children						
Age Group	Step 1	Step 2	Step 3 (Option D)	Step 4	Step 5	Step 6
0-4 years	SABA as needed	Low-dose ICS	Medium-dose ICS	Medium-dose ICS + LABA or LTRA	High-dose ICS + LABA or LTRA	High-dose ICS + LABA or LTRA + OCS
5-11 years	SABA as needed	Low-dose ICS	Low-dose ICS +/- LABA, LTRA, or theophylline, or medium-dose ICS	Medium-dose ICS + LABA or LTRA or theophylline	High-dose ICS + LABA or LTRA or theophylline	High-dose ICS + LABA + OCS
≥12 years	SABA as needed	Low-dose ICS	Low-dose ICS + LABA or medium-dose ICS	Medium-dose ICS + LABA	High-dose ICS + LABA + consider omalizumab for allergies	High-dose ICS + LABA + OCS + consider omalizumab for allergies and mepolizumab for eosinophilic asthma

Solution for Question 20:

Correct Answer: B) Mild Persistent

Explanation:

This child’s symptoms align with mild persistent asthma, which includes symptoms more than twice a week but not daily, nighttime symptoms 3-4 times a month, and an FEV1 > 80% predicted.

Asthma Classification by Severity						
Severity	Symptoms	Nighttime Symptoms	Lung Function	Interference with Normal Activity	Frequency of SABA Use	
Age 0-4 yrs	Age ≥5 yrs					
Mild Intermittent	≤2 days/wk	0	≤2 times a month	FEV1 > 80% predicted, normal between episodes	None	During exacerbations
Mild Persistent (Option B)	>2 days/wk but not daily	1-2 times a month	3-4 times a month	FEV1 ≥ 80% predicted, variability of 20-30%	Minor limitation	During exacerbations
Moderate Persistent	Daily	3-4 times a month	>1 time a week but not nightly	FEV1 60-80% predicted, variability > 30%	Some limitation	Daily
Severe Persistent	Throughout the day	>1 time a week	Often 7 times a week	FEV1 < 60% predicted, variability > 30%	Extremely limited	Several times per day

- Mild Intermittent (Option A): Mild intermittent asthma is characterized by symptoms occurring less than twice a week, nighttime symptoms less than twice a month, and normal lung function between episodes.
- Moderate Persistent (Option C): Moderate persistent asthma involves daily symptoms, nighttime symptoms more than once a week, and an FEV1 between 60-80% predicted.
- Severe Persistent (Option D): Severe persistent asthma is characterized by symptoms throughout the day, frequent nighttime symptoms, and an FEV1 < 60% predicted.

Solution for Question 21:

Correct Answer: B) Increase in FEV1 by >10% post-bronchodilator

Explanation:

An increase in FEV1 by > 10% after using a bronchodilator confirms the reversibility of airflow limitation, which is characteristic of asthma.

Common Symptoms of Asthma:

- Recurrent dry cough,
- Expiratory wheezing,
- Shortness of breath,
- Chest congestion and tightness.
- General fatigue due to sleep disturbances.
- Difficulty keeping up with peers in physical activities.
- Severe Symptoms: Exhibit signs of air hunger. Air trapping leads to a hyperresonant chest. As obstruction worsens, it results in feeble breath sounds. Wheezing may disappear as the airflow becomes severely restricted. With improvements, airflow increases, and wheezing may reappear, indicating recovery.
- Exhibit signs of air hunger.
- Air trapping leads to a hyperresonant chest.
- As obstruction worsens, it results in feeble breath sounds.
- Wheezing may disappear as the airflow becomes severely restricted.
- With improvements, airflow increases, and wheezing may reappear, indicating recovery.
- Severe hypoxemia results in cyanosis and cardiac arrhythmias.
- Symptom Timing: Symptoms may worsen at night, associated with sleep and exacerbations from infections or allergens. Daytime symptoms, particularly during physical activities, are frequent in children.
- Symptoms may worsen at night, associated with sleep and exacerbations from infections or allergens.
- Daytime symptoms, particularly during physical activities, are frequent in children.
- Exhibit signs of air hunger.
- Air trapping leads to a hyperresonant chest.

- As obstruction worsens, it results in feeble breath sounds.
- Wheezing may disappear as the airflow becomes severely restricted.
- With improvements, airflow increases, and wheezing may reappear, indicating recovery.
- Symptoms may worsen at night, associated with sleep and exacerbations from infections or allergens.
- Daytime symptoms, particularly during physical activities, are frequent in children.

Investigations of Asthma:

- Spirometry: Airflow limitation: Low FEV1 (relative to the percentage of predicted norms) FEV1/FVC ratio <0.80 (Option A)
- Low FEV1 (relative to the percentage of predicted norms)
- Bronchodilator response (to inhaled ß-agonist) assesses reversibility of airflow limitation. Reversibility is determined by an increase in either FEV1 >12% or Predicted FEV1 >12% after inhalation of a short-acting ß-agonist (SABA) (Option B)
- Reversibility is determined by an increase in either FEV1 >12% or
- Predicted FEV1 >12% after inhalation of a short-acting ß-agonist (SABA) (Option B)
- Exercise challenge: Worsening in FEV1 ≥15% (Option D)
- Daily peak expiratory flow (PEF) or FEV1 monitoring: Day-to-day and/or AM-to-PM variation ≥ 20% (Option C)
- Exhaled nitric oxide (FeNO): A value of >20 ppb supports the clinical diagnosis of asthma in children FeNO can be used to predict response to ICS therapy: <20 ppb: Unlikely to respond to ICS because eosinophilic inflammation is unlikely 20-35 ppb: Intermediate, may respond to ICS >35 ppb: Likely to respond to ICS because eosinophilic inflammation is likely.
- A value of >20 ppb supports the clinical diagnosis of asthma in children
- FeNO can be used to predict response to ICS therapy: <20 ppb: Unlikely to respond to ICS because eosinophilic inflammation is unlikely 20-35 ppb: Intermediate, may respond to ICS >35 ppb: Likely to respond to ICS because eosinophilic inflammation is likely.
- <20 ppb: Unlikely to respond to ICS because eosinophilic inflammation is unlikely
- 20-35 ppb: Intermediate, may respond to ICS
- >35 ppb: Likely to respond to ICS because eosinophilic inflammation is likely.
- Chest Radiographs: Chest radiographs in children with asthma often appear normal. May show subtle signs such as hyperinflation (e.g., flattening of the diaphragms) and peribronchial thickening
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- May show subtle signs such as hyperinflation (e.g., flattening of the diaphragms) and peribronchial thickening

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- Reversibility is determined by an increase in either FEV1 >12% or
- Predicted FEV1 >12% after inhalation of a short-acting ß-agonist (SABA) (Option B)
- A value of >20 ppb supports the clinical diagnosis of asthma in children
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- >35 ppb: Likely to respond to ICS because eosinophilic inflammation is likely.
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- Chest radiographs in children with asthma often appear normal.
- May show subtle signs such as hyperinflation (e.g., flattening of the diaphragms) and peribronchial thickening
- Allergy Testing: Assesses sensitization to inhalant allergens.

Solution for Question 22:

Correct Answer: D) Persistent atopy-associated asthma

Explanation:

Symptoms start in early childhood with a history of atopy (eczema, allergic rhinitis) and are associated with persistent wheezing and other symptoms point towards diagnosis of persistent atopy-associated asthma.

Bronchial Asthma is a disease characterized by episodic airway obstruction and airway hyperresponsiveness, usually accompanied by airway inflammation.

Types of Asthma:

1. Transient non-atopic wheezing (Option A)	2. Persistent atopy-associated asthma (Option D)
Common in early preschool years.	Starting in early preschool years.

Triggered by respiratory viral infections.	Associated with atopy and involves clinical signs (like atopic dermatitis or allergic rhinitis) and biological markers (such as high serum IgE and eosinophils).
Resolves by school age without increasing asthma risk later in life.	Persists into later childhood and adulthood.

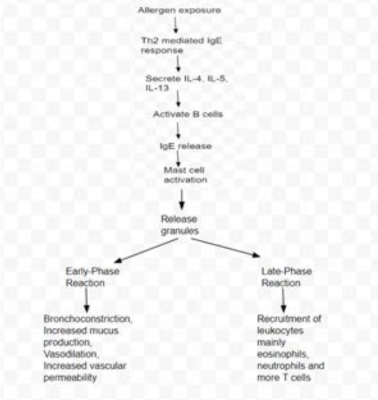
Triggers of Asthma:

Pathophysiology of Asthma:

- Type I hypersensitivity reaction

Seasonal allergic asthma (Option B) is typically triggered by seasonal allergens, which does not fully capture the year-round nature of the child’s symptoms and their history of atopic conditions.

Late-onset asthma (Option C) usually begins in adolescence or adulthood and is inconsistent with the early onset and atopic history of this child.



Solution for Question 23:

Correct Answer: C) Perform rigid bronchoscopy.

Explanation:

The clinical history, reduced breath sounds on the affected side, and chest X-ray showed increased radiolucency (red arrow) and flattening of the diaphragm on the right side (blue arrow) consistent with hyperinflation of the right lung, as well as left mediastinal shift (green arrow), indicating obstruction.

- Immediate rigid bronchoscopy is the best course of action as it allows for the direct removal of the foreign body, resolving the obstruction and alleviating the child’s symptoms.

Lower Airway Foreign Body Aspiration:

Prevalence	Choking is a leading cause of morbidity and mortality in children, especially those under 4 years old. Most victims are older infants and toddlers.
Common Objects	Food items: Nuts, seeds, hard candy, gum, bones, raw fruits and vegetables. Inorganic objects: Coins, latex balloons, pins, jewelry, magnets, pen caps and toys.
Location	Bronchus > Trachea > Larynx
Types	Partial obstruction: This causes a ball valve effect that leads to localized hyperinflation of the affected lung area (a bronchial foreign body blocks the exit of air from the obstructed lung during expiration). Overlying the chest wall shows hyperresonance, diminished vocal resonance and poor air entry. Complete Obstruction: Cause distal atelectasis and suppuration of surrounding parenchyma of lungs result in bronchiectasis.
Clinical Manifestation	Initial Event: Coughing, choking, gagging, potential airway obstruction. Some children may expel the foreign body during this stage. Asymptomatic Interval: Immediate symptoms subside; foreign body remains lodged. Can be misleading and lead to delayed diagnosis. Complications: Fever, cough, hemoptysis, pneumonia, atelectasis.
Diagnosis	Positive history is crucial. Plain Films: Recommended first; may miss radiolucent objects (80-96%). Expiratory Films: Reveal secondary findings like air trapping, asymmetric hyperinflation. CT Scan: High sensitivity and specificity but involves radiation. Bronchoscopy: Should be performed if suspicion remains despite negative or inconclusive imaging.
Treatment	Prompt endoscopic removal with a rigid bronchoscope. Appropriate antibiotics are given for secondary infection.

Administer supplemental oxygen (Option A): While supplemental oxygen can be useful in cases of hypoxia, it does not address the underlying issue of a foreign body obstructing the airway.

Administer systemic corticosteroids (Option B): Systemic corticosteroids are often used to reduce inflammation, but they are ineffective in immediately removing a foreign body from the airway.

Emergency cricothyrotomy (Option C) is a last-resort procedure for severe airway obstruction when other methods fail.

- It is not indicated in this case where the foreign body is in the lower airway and can be accessed with bronchoscopy.

Solution for Question 24:

Correct Answer: C) Deliver five back blows.

Explanation:

- Back blows are the recommended first step for infants under 1 year with signs of airway obstruction.
- This method helps to dislodge the foreign object and restore airflow.

Upper Airway Foreign Body Aspiration:

Causes	Choking on small objects such as toy parts, seeds, nuts, grapes, pebbles, or buttons.
Presentation	Sudden onset of cough, gagging or stridor with or without respiratory distress.
Type of Obstruction	Severe Obstruction: Universal choking sign (clutching the neck), weak or no cough, inability to speak or breathe, cyanosis. Mild Obstruction: Breathing is possible, with possible wheezing, coughing, and stridor.
Immediate Removal Indication	Severe obstruction requires urgent intervention.

Procedure for Infants (<1 year)	 Black blows to relieve foreign body airway obstruction in the infant. ... (content truncated)
Procedure for Children (>1 year) (Option A)	 Abdominal thrusts with the victim standing or sitting (conscious) ... (content truncated)
Complications	Chest Compressions: May cause rib or cardiac damage, though rare. Heimlich Maneuver: If performed incorrectly, the manoeuvre can cause pneumomediastinum, rupture of the spleen or stomach, and injury to the aorta.

Give rescue breaths (Option D): Rescue breaths are part of CPR, which is not the first step in this scenario. The initial focus should be on dislodging the foreign body with back blows.

Solution for Question 25:

Correct Answer: B) Left upper lobe

Explanation:

- Based on the given scenario and the X-ray image, the diagnosis is congenital lobar emphysema.
- This condition is characterized by hyperinflation of a lobe and mediastinal shift.
- The left upper lobe is the most common location affected by congenital lobar emphysema.

Congenital lobar emphysema (Congenital large hyperlucent lobe/ Congenital lobar hyperinflation)	
Causes	Bronchial Abnormalities Congenital deficiency of bronchial cartilage Bronchial stenosis Redundant bronchial mucosal flaps External compression Aberrant vessels Kinking of bronchus due to herniation into mediastinum
Common Location	Left upper lobe > Right middle lobe > Right lower lobe. Involvement of lower lobes is rare.
Symptoms	Mild: Tachypnea (in respiratory distress), Wheezing (due to airway narrowing) Severe: Dyspnea, Cyanosis (Symptoms often appear immediately after birth; can be delayed in some cases)
Diagnosis	 Prenatal Ultrasonography: Often diagnosed before birth Chest X-Ray: radiolucent (hyper lucent) lobe, mediastinal shift (arrow) ... (content truncated)
Treatment	Conservative Management: Watchful waiting for mild cases Surgical Resection: Removal of affected lobe for severe cases Supportive Care: Supplemental oxygen, mechanical ventilation, selective intubation
Differential Diagnoses	Pneumonia Pneumothorax Congenital Cystic Adenomatoid Malformation (CCAM) or Congenital Pulmonary Airway Malformation (CPAM)

Long-Term Outlook	Generally good with early diagnosis and appropriate treatment; Surgical resection often leads to full recovery and normal lung function. Regular follow-up is needed.

Solution for Question 26:

Correct Answer: B) Blood returns to the right side of the heart from this lesion.

Explanation:

The clinical scenario and X-ray point to the diagnosis of pulmonary sequestration from which the blood returns to the right side of the heart.

Pulmonary sequestration	
Definition	Rare congenital lung anomaly where abnormal lung tissue is detached from the normal bronchial tree and receives blood from a systemic artery, usually a branch of the aorta. (Option A)
Types	Intrapulmonary: within the visceral pleura Extrapulmonary: outside the visceral pleura
Pathophysiology	Systemic Arterial Supply: Receives blood from a systemic artery (e.g., aorta) instead of the pulmonary artery. Blood returns to the right side of the heart. (Option B) Extra-pulmonary type drains through IVC and intrapulmonary through the pulmonary veins. Lack of Bronchial Connection: No connection to the bronchial tree; does not participate in gas exchange but may rupture into an airway, potentially leading to infection. Collateral Ventilation: This can occur via the pores of Kohn in intrapulmonary sequestration.
Etiology	There are two theories Embryological Origin Acquired Lesion Theory (Recurrent infection/inflammation ➜ cystic changes and hypertrophy of the feeding artery).
Clinical features	Dullness to Percussion Decreased Breath Sounds over the affected area Crackles: Indicate possible infection (OptionC) Continuous or systolic murmur from turbulent blood flow

Investigations	Chest X-ray: Initial detection of abnormal opacity or mass. (Option D) CECT: Confirms diagnosis, details size, location, and blood supply. MR Angiography: Visualizes abnormal blood vessels. Ultrasonography: Rules out other conditions and visualizes systemic artery
Treatment	Surgical resection: Preferred treatment to remove the abnormal tissue and preserve the systemic artery. Coil sequestration: Minimally invasive alternative, blocks the feeding artery to shrink the sequestration.
Complications of sequestration	Pneumonia Abscess Formation Respiratory Distress Sepsis

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Solution for Question 27:

Correct Answer: C) Fibre optic endoscopic evaluation

Explanation:

- In this scenario a wet stridor indicates dysphagia.
- Fibre optic endoscopic evaluation or a contrast swallow study is the investigation of choice when dysphagia is suspected.

Laryngomalacia	
Definition	M/c congenital laryngeal anomaly in children; characterized by floppy tissues above the vocal cords that collapse during inspiration, causing stridor. Accounts for 45% to 75% of stridor cases in infants.
Onset	Appear within the first 2 weeks of life and can worsen for up to six months, with gradual improvement often occurring thereafter.
Factors deciding severity and clinical course	Gastroesophageal reflux disease Laryngopharyngeal reflux disease Neurological conditions

Investigations	Diagnosis is made mainly by clinical symptoms Outpatient flexible laryngoscopy is done to confirm the diagnosis (Option D) but is not suitable for a baby as well as when dysphagia is suspected. Airway films and chest radiographs are done when the work of breathing is moderate to severe (Options A & B). A contrast swallow study or Flexible endoscopic evaluation of swallowing (FEES) is considered when dysphagia is suspected. (Option C) Complete bronchoscopy is done for those with moderate to severe obstruction.
Treatment	Expectant observation for most infants as the spontaneous resolution of symptoms as the child grows is seen. Laryngoesophageal reflux and GERD are treated with proton pump inhibitors or histamine receptor antagonists, considering the possible iron deficiency anemia. Supraglottoplasty is done in case of severe disease that causes progressive respiratory distress, cyanosis, failure to thrive, or cor pulmonale. Earlier intervention is needed in children with cardiac disease, neurologic disease, pulmonary disorders, or craniofacial anomalies

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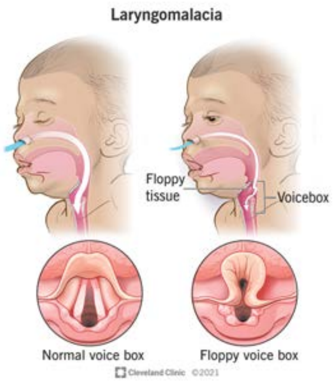


Previous Year Questions

1. A neonate was born to a controlled gestational diabetes mother. At the time of birth, he was fine but 4 days later, he developed dyspnea with Spo2 80% at room air. What is the initial management?

A. Give 21% to 30% O2

B. Give 100% O2

C. Give 50% O2

D. Observation

2. A 3-month-old child has a respiratory rate of 56/min with no chest indrawing and no danger signs. As per IMNCI what is the most appropriate diagnosis?

A. Pneumonia

B. Cold and cough

C. Severe Pneumonia

D. Allergy

3. A newborn child is brought to the emergency department with respiratory difficulty. The doctor evaluated, and the child was found to have a posterolateral defect, as shown in the x-ray given. What is the diagnosis?

A. Bochdalek hernia

B. Morgagni hernia

C. Hiatal hernia

D. Traumatic diaphragmatic hernia

4. An infant presents with recurrent chest infections and tachypnea. Identify the likely clinical condition.

A. Congenital Cystic Adenomatoid Malformation (CCAM)

B. Right-sided Congenital Diaphragmatic Hernia (CDH)

C. Congenital Lobar Emphysema

D. Cystic Fibrosis

5. Which of the following is not a suitable method for free-flow oxygen delivery to a newborn baby?

A. Oxygen tube kept close to the baby's nose and mouth

B. Oxygen mask placed over the baby's mouth and nose

C. Self-inflating bag attached to an oxygen mask placed over baby's mouth and nose

D. A reservoir bag attached to an oxygen mask kept over the baby's nose and mouth

6. INSURE technique is used in which of the following conditions?

A. Respiratory distress syndrome

B. Birth asphyxia

C. Neonatal jaundice

D. Transient tachypnea of the newborn

7. What is the most suitable course of action for managing a one and a half year old child who was brought to the hospital presenting symptoms of breathlessness, fever, cough, and cold, and upon examination has a respiratory rate of 46 breaths per minute, chest retractions, but no danger signs were present?

A. Give IV antibiotics and observe

B. Give IV antibiotics and refer to higher centre

C. Oral antibiotics, warn of danger signs, and send home

D. No need of treatment only home remedies

8. A 9-year-old girl was brought to the emergency room, with fever for 2 days, dyspnea, drooling and inspiratory stridor. Lateral x-ray neck findings were as seen below. Which is the most common organism which causes this condition in children?

A. Haemophilus influenza B

B. Influenza virus

C. Staphylococcus

D. RSV

9. Which of the following medicines can be used in a child with bronchiolitis?

A. Ribavirin

B. Amantadine

C. Oseltamivir

D. Amoxicillin

10. What should be the immediate course of action in managing a 9-month-old infant brought to the hospital presenting with a respiratory rate of 48 breaths per minute but without any signs of chest retractions or other associated symptoms?

A. Home-based supportive care

B. Send back home after starting oral antibiotics

C. Hospitalize and start oral antibiotics

D. Hospitalize and start IV antibiotics

11. What serves as a reliable indicator of perinatal asphyxia?

A. Cord pH < 7.0

B. Hypocalcemia

C. APGAR score of 4-7 beyond 5 min

D. Hypotonia

12. Which of the following PFT findings in a child do not suggest asthma?

A. Increase in FEV1 > 12% after salbutamol inhalation

B. FEV1/FVC< 80%

C. Day-night variation of FEV1 > 15%

D. Decrease in FEV1 > 15% after exercise

13. What is the diagnosis for an infant who is admitted with respiratory distress, prolonged expiration, rhonchi in the chest, and hyperinflation shown on a CXR?

A. Pneumonia

B. Croup

C. Asthma

D. Bronchiolitis

14. Which of the following should be administered in addition to airway management for a 3-year-old child who is experiencing symptoms such as rapid breathing, high fever, noisy breathing during inhalation, and difficulty swallowing with excessive saliva production within 4-6 hours of the onset of fever?

A. Nebulisation with racemic adrenaline

B. IV ceftriaxone

C. Diphtheria antitoxin

D. Start steroids

15. Which of the following is not an indicator of fetal lung maturity?

A. Lecithin-sphingomyelin ratio > 2.5

B. Positive shake test

C. Increased phosphatidyl glycerol

D. Nile blue test showing >50% of blue cells

16. A 2 year old child presented with acute onset breathing difficulty while playing. No previous history of similar complaints. Chest X-ray shows unilateral hyperinflation of right lungs. Which of the following is true?

A. Flexible bronchoscopy is used for removal

B. ■■■In complete obstruction, ball valve mechanism causes hyperinflation

C. Child has laryngotraceobronchitis

D. Focal hyperinflated area with no/decreased breath sounds will be suggestive of foreign body

17. The following image is of the most common cause of stridor in children. Identify it.

A. Laryngomalacia

B. Laryngeal papilloma

C. Chronic epiglottitis

D. Laryngocele

18. Which of the following is absent in a child with cystic fibrosis?

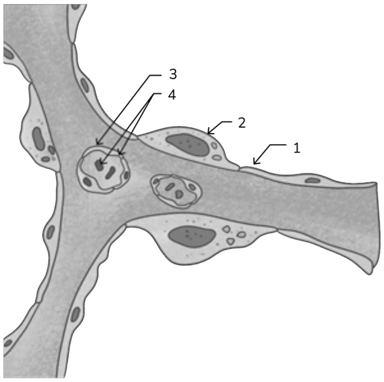
A. Sweat chloride test chloride conc of 70mEq/L

B. Increase immunoreactive trypsinogen level

C. Hyperkalemia

D. Contraction alkalosis

19. Which of the following marked structures is damaged in respiratory distress syndrome in newborn?



A. 1

B. 2

C. 3

D. 4

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20. In a 5-year-old boy with a previously diagnosed asthma, he experienced sudden breathlessness and wheezing. He was immediately taken to the emergency department, where an initial assessment was conducted. The pulse rate is 140 per minute, respiratory rate is 28 per minute, and the forced expiratory volume in one second (FEV1) is 40% of the predicted value. What would be the most appropriate next step in managing this patient's condition?

A. A high concentration of oxygen should be given by a face mask + Salbutamol nebulization

B. High doses of SABA to be given by a nebulizer only

C. Sedatives should be added to alleviate anxiety

D. Nebulized anticholinergics to be added immediately as the mainstay of treatment

21. What is the diagnosis for a 3-month-old baby who has a fever and a respiratory rate of 70/min, but shows no signs of stridor, chest indrawing, or convulsions?

A. Pneumonia

B. Severe pneumonia

C. Very severe pneumonia

D. No Pneumonia

22. What should be the appropriate course of action for a child who is brought to the emergency department after experiencing choking? Upon examination, reduced airflow is observed in the left lung. Bronchoscopy Computed tomography scan ICD insertion Chest X-ray

A. 1, 2, and 4

B. 1 and 2

C. 3 and 4

D. 1 and 4

23. What would you do when a child comes with a sudden worsening of asthma symptoms? Chest X-ray Oxygen administration Salbutamol nebulization three times in 60 minutes Administration of oral corticosteroids

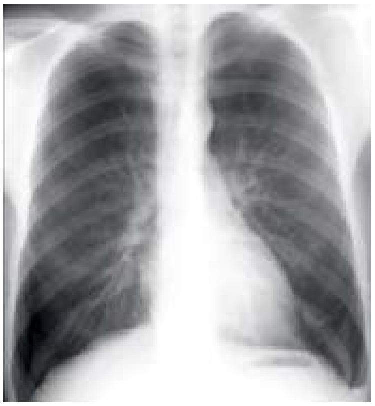
A. 3,4

B. 1,4

C. 1,3,4

D. 2,3,4

24. What is the appropriate order of actions for managing an unresponsive child brought to the emergency department? 1. Assess breathing 2. Assess pulse 3. Assess the response 4. Start compressions 5. Bag and mask ventilation



A. 3-1-2-5-4

B. 1-2-3-4-5

C. 3-1-2-4-5

D. 1-2-4-3-5

25. What is the diagnosis for a child who is brought to the hospital with respiratory distress and biphasic stridor, as indicated by the radiograph provided?

A. Acute epiglottitis

B. Acute laryngotracheobronchitis

C. Foreign body aspiration

D. Laryngomalacia

26. For which disease is the diagnosis made based on the level of chloride in sweat?

A. Phenylketonuria

B. Cystic fibrosis

C. Gaucher's disease

D. Osteogenesis imperfecta

27. The baby had chest retractions, difficulty breathing, and lack of energy. The pediatrician determined that the baby was suffering from respiratory distress syndrome. This condition arises because of the lack of:

A. Dipalmitoyl inositol

B. Lecithin

C. Sphingomyelin

D. Dipalmitoyl Phosphatidylethanolamine

28. An 18-months-old child is brought to the hospital with a history of breathlessness, fever, cough, and cold. The respiratory rate was 46/min on examination, and there were no danger signs. Which is the most appropriate management?

A. Administer IV antibiotics and observe

B. Administer IV antibiotics and refer to higher centre

C. Administer oral antibiotics, warn of danger signs, and send home

D. Administer only home remedies

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	1
Question 4	1
Question 5	3
Question 6	1
Question 7	3
Question 8	1
Question 9	1
Question 10	1
Question 11	1
Question 12	3
Question 13	4
Question 14	2
Question 15	4
Question 16	4
Question 17	1
Question 18	3
Question 19	2
Question 20	1
Question 21	1
Question 22	4

Question 23	4
Question 24	3
Question 25	2
Question 26	2
Question 27	2
Question 28	3

Solution for Question 1:

Correct Answer: B) Give 100% O2

Explanation:

In this scenario, the child developed the symptoms after 4 days of birth, and the aetiology can be cardiac (cyanotic heart disease) or pulmonary. Hence, the best method is to give 100% oxygen to the neonate to distinguish the cause.

If the saturation improves, the cause is respiratory, and if not, the cause is cardiogenic. This is called the hyperoxia test.

Hyperoxia test for distinguishing CCHD from pulmonary disease

- This is the initial method to differentiate congenital critical heart disease (CCHD) from pulmonary disease.
- Procedure: Measure arterial blood gases (ABG) first at room air and then again after administering 100% oxygen for 10 minutes.
- Measure arterial blood gases (ABG) first at room air and then again after administering 100% oxygen for 10 minutes.
- Expected results in CCHD: Neonates with CCHD usually fail to increase PaO2 above 100 mm Hg with 100% oxygen.
- Neonates with CCHD usually fail to increase PaO2 above 100 mm Hg with 100% oxygen.
- Expected Results in Pulmonary Disease: In pulmonary disease, PaO2 typically increases to ≥100 mm Hg when given 100% oxygen, as ventilation-perfusion mismatch is corrected.
- In pulmonary disease, PaO2 typically increases to ≥100 mm Hg when given 100% oxygen, as ventilation-perfusion mismatch is corrected.

- Interpretation: A positive result (failure to increase PaO2) suggests a cardiac origin, warranting further cardiac evaluation to rule out CCHD.
- A positive result (failure to increase PaO2) suggests a cardiac origin, warranting further cardiac evaluation to rule out CCHD.
- Measure arterial blood gases (ABG) first at room air and then again after administering 100% oxygen for 10 minutes.
- Neonates with CCHD usually fail to increase PaO2 above 100 mm Hg with 100% oxygen.
- In pulmonary disease, PaO2 typically increases to ≥100 mm Hg when given 100% oxygen, as ventilation-perfusion mismatch is corrected.
- A positive result (failure to increase PaO2) suggests a cardiac origin, warranting further cardiac evaluation to rule out CCHD.

Solution for Question 2:

Correct Answer: A) Pneumonia

Explanation:

The fast breathing and no other danger signs represent pneumonia as per the IMNCI classification.

Assessment and Classification of the Sick Child (Age 2 Months Up to 5 Years)

- Initial Assessment: A thorough history from the mother is essential
- Check for General Danger Signs:

Ask	Look	Action
Able to drink or breastfeed? Does the child vomit everything? H/O convulsions during this illness?	Lethargic or unconscious? Currently convulsing?	If any danger signs are present, the child needs URGENT attention.

- Assess Cough or Difficulty Breathing: Ask: How long has the child had a cough or difficulty breathing? Look, Listen, Feel: Count the breaths in one minute (use respiratory rate timers if available). Fast Breathing Criteria: Age 2 months up to 12 months: Fast breathing is 50 breaths per minute or more. Age 12 months up to 5 years: Fast breathing is 40 breaths per minute or more.
- Ask: How long has the child had a cough or difficulty breathing?

- Look, Listen, Feel: Count the breaths in one minute (use respiratory rate timers if available).
- Fast Breathing Criteria: Age 2 months up to 12 months: Fast breathing is 50 breaths per minute or more. Age 12 months up to 5 years: Fast breathing is 40 breaths per minute or more.
- Age 2 months up to 12 months: Fast breathing is 50 breaths per minute or more.
- Age 12 months up to 5 years: Fast breathing is 40 breaths per minute or more.
- Look for chest indrawing
- Look and listen for stridor, wheeze
- Check for central cyanosis
- Check oxygen saturation
- Check the AVPU scale (Alert, Voice, Pain, Unresponsive)
- Assess for possible asthma/TB if wheezing with chest in-drawing or fast breathing is present.
- Ask: How long has the child had a cough or difficulty breathing?
- Look, Listen, Feel: Count the breaths in one minute (use respiratory rate timers if available).
- Fast Breathing Criteria: Age 2 months up to 12 months: Fast breathing is 50 breaths per minute or more. Age 12 months up to 5 years: Fast breathing is 40 breaths per minute or more.
- Age 2 months up to 12 months: Fast breathing is 50 breaths per minute or more.
- Age 12 months up to 5 years: Fast breathing is 40 breaths per minute or more.
- Age 2 months up to 12 months: Fast breathing is 50 breaths per minute or more.
- Age 12 months up to 5 years: Fast breathing is 40 breaths per minute or more.

Classify the condition:

Classification	Signs	Treatment
Severe Pneumonia or Very Severe Disease (Option C)	Any general danger sign OR Oxygen saturation <90% Stridor in calm child Central cyanosis Convulsions AVPU = V, P, or U	Start oxygen therapy if saturation < 90%. Administer Benzyl Penicillin & Gentamicin. Give diazepam if convulsing. Prevent low blood sugar. Treat wheeze if present. Urgent referral or admission.
Pneumonia (Option A)	Chest indrawing in calm child OR Fast breathing AND No signs of severe pneumonia	Administer Amoxicillin Dispersible Tablet. Give Vitamin A. Treat wheeze if present. Follow up in 5 days Advise on immediate return signs.

No Pneumonia: Cough or Cold	No signs of pneumonia or very severe disease	Treat wheeze if present - salbutamol for 5 days Advise on immediate return signs.

Options B and D are incorrect.

Solution for Question 3:

Correct Answer: A) Bochdalek hernia

Explanation:

The clinical features and X-ray show a congenital diaphragmatic hernia, and the most common type that appears in the posterolateral part on the left side is the Bochdalek hernia.

Diaphragmatic Hernia:

- It is a communication between the abdominal and thoracic cavities, which may or may not involve abdominal contents in the thorax.
- Aetiology: Rarely traumatic, usually congenital.
- Symptoms and Prognosis: Dependent on the location of the defect and associated anomalies.

Types of diaphragmatic hernia:

- Hiatal hernia (Option C): Occurs at the esophageal hiatus.
- Paraesophageal Hernia: Near the esophageal hiatus.
- Morgagni Hernia (Option B): Retro-sternal, occurring in the foramen of Morgagni (2–6% of CDH cases).
- Bochdalek Hernia: Occurs in the posterolateral portion of the diaphragm; most common, accounting for 90% of congenital diaphragmatic hernias (CDH). It occurs predominantly on the left side (80-90%).

Pathology and etiology:

- CDH involves a diaphragmatic defect and often leads to pulmonary hypoplasia.
- Pulmonary hypoplasia is due to both compression of the lungs and, in some cases, may precede the diaphragmatic defect.

- Pathological features include reduced pulmonary mass, abnormal alveolar septa, thickened alveoli, and biochemical imbalances such as surfactant deficiency.

Clinical Features:

- Symptoms include respiratory distress, tachypnea, cyanosis, and scaphoid abdomen.
- Infants may show delayed symptoms and present with vomiting or mild respiratory distress.

Diagnosis:

- CDH is often diagnosed prenatally via ultrasound (16-24 weeks) and fetal MRI.
- Key findings: polyhydramnios, chest mass, mediastinal shift, abdominal organs in the chest, and fetal hydrops.
- After birth, a chest X-ray confirms the diagnosis.

Treatment:

- Initial Management: Birth in a tertiary care centre with expertise in CDH management is essential. Immediate stabilization includes intubation, gastric decompression, and central venous and arterial access. Goal oxygenation: preductal SpO2 ≥85%. Gentle ventilation and permissive hypercapnia are recommended.
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- Immediate stabilization includes intubation, gastric decompression, and central venous and arterial access.
- Goal oxygenation: preductal SpO2 ≥85%.
- Gentle ventilation and permissive hypercapnia are recommended.
- Ventilation Strategies: Conventional mechanical ventilation, HFOV, and ECMO are primary methods. Permissive hypercapnia and low-pressure ventilation reduce lung injury. Inhaled nitric oxide (iNO) may be used to manage pulmonary hypertension.
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- Permissive hypercapnia and low-pressure ventilation reduce lung injury.
- Inhaled nitric oxide (iNO) may be used to manage pulmonary hypertension.
- Extracorporeal Membrane Oxygenation (ECMO): ECMO is used when conventional ventilation fails. The duration of ECMO may be longer in CDH infants than in other conditions.
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- The duration of ECMO may be longer in CDH infants than in other conditions.
- Surgical Repair: Repair is typically done after stabilization (48 hours post-birth). A subcostal approach is preferred, with a focus on primary repair if possible. If primary repair is not feasible, synthetic patches (e.g., Gore-Tex) may be used. Postoperative monitoring for complications like pulmonary hypertension, bleeding, and bowel obstruction is crucial.

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- Birth in a tertiary care centre with expertise in CDH management is essential.
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- ECMO is used when conventional ventilation fails.
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- Repair is typically done after stabilization (48 hours post-birth).
- A subcostal approach is preferred, with a focus on primary repair if possible.
- If primary repair is not feasible, synthetic patches (e.g., Gore-Tex) may be used.
- Postoperative monitoring for complications like pulmonary hypertension, bleeding, and bowel obstruction is crucial.

Novel Strategies:

- Fetal lung-to-head ratio (LHR) on ultrasound is a reliable predictor of outcomes.
- Fetoscopic tracheal occlusion shows promise for severe cases, but more research is needed.

Traumatic diaphragmatic hernia (Option D) is not the likely diagnosis in the given scenario since there is no significant history of trauma that may indicate this diagnosis.

Solution for Question 4:

Correct Option A - Congenital Cystic Adenomatoid Malformation (CCAM):

- CCAM is a developmental abnormality of the lung characterized by cystic lesions that can cause lung dysfunction and respiratory issues. It often presents with recurrent respiratory infections and tachypnea due to the compromised lung tissue.
- The recurrent chest infections and tachypnea are consistent with CCAM, as the cystic lesions in the lung can lead to chronic respiratory problems and increased susceptibility to infections.

Incorrect Options:

Option B - Right-sided Congenital Diaphragmatic Hernia (CDH):

- CDH occurs when there is a defect in the diaphragm that allows abdominal organs to move into the chest cavity, impairing lung development and function. It often presents with respiratory distress shortly after birth.
- Although CDH can cause tachypnea and respiratory distress, it typically presents more acutely and may be associated with other signs such as mediastinal shift or absence of breath sounds on the affected side. Recurrent chest infections are less characteristic.

Option C - Congenital Lobar Emphysema:

- This condition involves overinflation of a lobe of the lung, which can compress adjacent lung tissue and cause respiratory distress. It usually presents with respiratory symptoms in the newborn period but is less commonly associated with recurrent infections.
- While congenital lobar emphysema can cause tachypnea and respiratory issues, recurrent chest infections are not a typical feature. The clinical presentation is less consistent with this diagnosis compared to CCAM.

Option D - Cystic Fibrosis:

- Cystic fibrosis is a genetic disorder affecting the respiratory, digestive, and reproductive systems. It leads to thick, sticky mucus production that can cause chronic respiratory infections and other complications.
- Although cystic fibrosis can cause recurrent respiratory infections and tachypnea, it usually presents with additional symptoms such as failure to thrive, gastrointestinal issues, and a positive sweat test. The acute presentation in a newborn with recurrent infections is less characteristic.

Solution for Question 5:

Correct Option C - Self-inflating bag attached to an oxygen mask placed over the baby's mouth and nose:

- A self-inflating bag is designed to deliver positive pressure ventilation (PPV) and is not intended for free-flow oxygen delivery. It provides a higher flow rate and pressure suitable for resuscitation rather than the gentle, continuous flow needed for free-flow oxygen.

Incorrect Options:

Option A - Oxygen tube kept close to the baby's nose and mouth:

- This method provides free-flow oxygen by directing the flow from a tube close to the baby's face, ensuring a continuous supply without the need for direct contact with the nose or mouth.

Option B - Oxygen mask placed over the baby's mouth and nose:

- An oxygen mask can be used to deliver free-flow oxygen efficiently, ensuring that the baby inhales the oxygen without requiring active breathing support from the mask itself.

Option D - A reservoir bag attached to an oxygen mask kept over the baby's nose and mouth:

- Like option B, a reservoir bag attached to an oxygen mask can deliver high concentrations of oxygen when placed over the baby's nose and mouth, suitable for providing free-flow oxygen

Solution for Question 6:

Option A - Respiratory Distress Syndrome:

- The INSURE technique (INtubation, SURfactant administration, and Extubation) is specifically designed to treat respiratory distress syndrome (RDS) in preterm infants. This technique aims to administer surfactant therapy effectively while minimizing the duration of mechanical ventilation and associated complications.

Incorrect Options:

Options B, C, and D do not require the INSURE technique.

Solution for Question 7:

Correct Option: C

Fast Breathing

- < 2 months: > 60/ min
- 2-12 months: > 50/ min
- 1-5 years: > 40/ min

General Danger signs

Signs	Classify as	Treatment
Presence of any general danger sign	Severe pneumonia or very severe disease	Give 1 dose of Inj ampicillin & gentamycin & refer urgently to hospital
Presence of either chest indrawing &/or fast breathing but no general danger sign	Pneumonia	Start oral Amoxicillin x 5 days · The child is sent home and assessed after 48 hours. · If the child has improved continue the same treatment. · If the child has worsened, refer to a higher center.
No fast breathing, no chest indrawing, no general danger signs	No pneumonia: cough or cold only	Home-based supportive care

- From the above information the child is classified as having pneumonia. Treatment for Pneumonia is Start oral Amoxicillin x 5 days, the child is sent home and assessed after 48 hours, If the child has improved continue the same treatment. · If the child has worsened, refer to a higher center.
- Therefore, the most appropriate management in this scenario is to give oral antibiotics, warn of danger signs, and send the child home for close monitoring and follow-up.

Solution for Question 8:

Correct options A - Haemophilus influenzae :

- The radiograph finding in the image shows a characteristic thumb sign and is suggestive of acute epiglottitis.

- Most common organism responsible for acute epiglottitis: Previously: H. influenzae Currently: Streptococcus (in vaccinated children) and H. influenzae (in unvaccinated children)
- Previously: H. influenzae
- Currently: Streptococcus (in vaccinated children) and H. influenzae (in unvaccinated children)
- Previously: H. influenzae
- Currently: Streptococcus (in vaccinated children) and H. influenzae (in unvaccinated children)

Incorrect Options - B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 9:

Correct Option A - Ribavirin: Nebulized ribavarin can be used in immunocompromised patients with bronchiolitis.

Incorrect Options: Options B, C, and D are not given in the treatment of bronchiolitis.

Solution for Question 10:

Correct Option A - Home-based supportive care:

Signs	Classify as	Treatment
Presence of any general danger sign	Severe pneumonia	Give 1 dose of injection ampicillin and gentamicin and refer urgently to hospital

Presence of either chest indrawing and/or fast breathing but no general danger sign	Pneumonia	Start oral amoxicillin x 5 days The child is sent home and assessed after 48 hours If the child has improved continue the same treatment If the child has worsened, refer to a higher center
No fast breathing, no chest indrawing, no general danger signs	No pneumonia: Cough or cold only	Home-based supportive care
Fast breathing: < 2 months: > 60/ min 2-12 months: > 50/ min 1-5 years: > 40/ min		
General danger signs: Convulsions Lethargy Inability to drink or breastfeed Persistent vomiting Unconsciousness Stridor in a calm child		

- In the question above, the 9-month-old child has no danger signs, no fast breathing, and no chest indrawing; hence, the child has no pneumonia.

Incorrect Options: Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 11:

Correct Option A - Cord pH < 7.0:

- Perinatal asphyxia refers to a condition in which there is a deprivation of oxygen to a newborn around the time of birth.
- A cord pH less than 7.0 indicates acidosis, which is often associated with perinatal asphyxia.
- Cord pH < 7.0 is a reliable indicator because it directly reflects the acid-base status and the severity of metabolic acidosis associated with perinatal asphyxia.

Incorrect Options:

Option B - Hypocalcemia: While hypocalcemia can be a consequence of perinatal asphyxia, it is not as specific or direct indicator as cord pH.

Option C - APGAR score of 4-7 beyond 5 min: APGAR score of 0-3 for > 5 minutes is a diagnostic criteria for perinatal asphyxia.

Option D - Hypotonia: Hypotonia can be a clinical manifestation of perinatal asphyxia, but it is a subjective observation and not as quantitative or specific as the measurement of cord pH.

Solution for Question 12:

Correct Option C - Day-night variation of FEV1 > 15%:

- In asthma, the lung function typically shows significant variability during the day, but the day-night variation of FEV1 is usually less than 15%.
- A variation greater than 15% may indicate other conditions such as sleep apnea, or vocal cord dysfunction.

Incorrect Options:

Option A: Increase in FEV1 > 12% after salbutamol inhalation: This is a typical finding in asthma, as it indicates a reversible airway obstruction.

Option B: FEV1/FVC < 80%:

- This is a decreased ratio between forced expiratory volume in one second (FEV1) and forced vital capacity (FVC).
- It suggests airflow limitation, which is a characteristic feature of asthma.

Option D: Decrease in FEV1 > 15% after exercise: This is known as exercise-induced bronchoconstriction, which is a common finding in individuals with asthma.

Solution for Question 13:

Correct Option D - Bronchiolitis

- The patient’s presentation with respiratory distress, prolonged expiration, rhonchi in the chest, and hyperinflation shown on a chest x-ray is suggestive of bronchiolitis.
- Bronchiolitis is caused by a viral infection, most commonly respiratory syncytial virus (RSV).

Incorrect Options - A, B, and C do not show hyperinflation on chest x-ray.

Solution for Question 14:

Correct Option B - IV ceftriaxone:

- The patient's presentation with rapid breathing, high fever, noisy breathing during inhalation, and difficulty swallowing with excessive saliva production is suggestive of acute epiglottitis.
- The primary treatment for acute epiglottitis is the prompt administration of intravenous antibiotics, such as ceftriaxone or cefotaxime.

Incorrect Option - A, C, and D are not the treatment for acute epiglottitis.

Solution for Question 15:

Correct Option D - Nile blue test showing >50% of blue cells:

- The Nile blue test is used to assess the presence of mature surfactant in the lungs.
- The presence of more than 50% blue cells suggests the presence of immature surfactant and indicates the need for further lung development.

Incorrect Options:

Option A: Lecithin-sphingomyelin (L/S) ratio > 2.5:

- The L/S ratio measures the ratio of two phospholipids, lecithin, and sphingomyelin, in the amniotic fluid.
- A ratio greater than 2.5 indicates that the fetus has mature lungs and is capable of producing sufficient surfactant.

Option B: Positive shake test:

- The shake test, also known as the foam stability test or bubble stability test, is another method used to assess fetal lung maturity.
- In this test, amniotic fluid is shaken vigorously, and the presence of stable foam indicates the presence of surfactant, suggesting lung maturity.

Option C: Increased phosphatidyl glycerol:

- Phosphatidyl glycerol (PG) is a phospholipid component of surfactant.
- Increased levels of PG in the amniotic fluid are associated with fetal lung maturity.
- The presence of PG is considered a positive indicator of lung maturity.

Solution for Question 16:

Correct Option D - Focal hyperinflated area with no/decreased breath sounds will be suggestive of foreign body: The partial obstruction by a foreign body will lead to air trapping and hyperinflated lung seen on an X-ray with no/decreased breath sounds on auscultation.

Incorrect Options:

Option A: Flexible bronchoscopy is used for removal: Rigid bronchoscopy is used for removal.

Option B: ■■■In complete obstruction, ball valve mechanism causes hyperinflation: In partial obstruction, ball valve mechanism causes hyperinflation.

Option C:Child has laryngotraceobronchitis: Laryngotracheobronchitis is unlikely to cause unilateral hyperinflation of the lung, which is a more specific finding that points towards airway obstruction rather than inflammation.

Solution for Question 17:

Correct option A - Laryngomalacia:

- Laryngomalacia is a congenital softening of the tissues of the larynx above the vocal cords.

- Excess omega-shaped epiglottis is a feature of laryngomalacia.

Incorrect Options - B, C, and D are incorrect.

Solution for Question 18:

Correct option C - Hyperkalemia:

- Hyperkalemia is not seen in a child with cystic fibrosis.
- Cystic fibrosis primarily affects the exocrine glands, leading to abnormal production and secretion of thick, sticky mucus in various organs.
- It is commonly associated with electrolyte imbalances such as hypochloremic metabolic alkalosis due to excessive chloride loss in sweat.

Incorrect Options - A, B, and D are present in a child with cystic fibrosis.

Solution for Question 19:

Correct Option B - 2

- 2 is labeled as type 2 alveolar epithelial cells.
- Respiratory distress syndrome in neonates is caused by surfactant deficiency due to inadequate production or inactivation in immature lungs.
- There is dysfunction of type 2 alveolar epithelial cells,

Incorrect Options:

Option A: 1 is labeled as type I alveolar epithelial cells, which is responsible for gaseous exchange.

Option C: 3 is labeled as blood vessels.

Option D: 4 is labeled as RBC inside the blood vessel.

Solution for Question 20:

Correct Option A - A high concentration of oxygen should be given by a face mask + Salbutamol nebulization

- The patient’s presentation with sudden breathlessness and wheezing, a history of diagnosed asthma, a respiratory rate of 28 per minute, and a forced expiratory volume in one second (FEV1) of 40% of the predicted value suggest an acute attack of severe asthma.
- A high concentration of oxygen should be given by a face mask + Salbutamol nebulization is recommended as the best next step in treatment in this case.

Incorrect Options - B, C, and D are incorrect.

Solution for Question 21:

Correct option A - Pneumonia:

Signs	Classify as	Treatment
Presence of any general danger sign	Severe pneumonia	Give 1 dose of injection ampicillin and gentamicin and refer urgently to hospital
Presence of either chest indrawing and/or fast breathing but no general danger sign	Pneumonia	Start oral amoxicillin x 5 days The child is sent home and assessed after 48 hours If the child has improved continue the same treatment If the child has worsened, refer to a higher center
No fast breathing, no chest indrawing, no general danger signs	No pneumonia: Cough or cold only	Home-based supportive care

Fast breathing: < 2 months: > 60/ min 2-12 months: > 50/ min 1-5 years: > 40/ min		
General danger signs: Convulsions Lethargy Inability to drink or breastfeed Persistent vomiting Unconsciousness Stridor in a calm child		

- In the question above, the 3-month-old child has no danger signs but has fast breathing, i.e., >60/min, hence the child has pneumonia.
- The treatment for children with pneumonia is to administer oral antibiotics, warn of danger signs, and send them home.

Incorrect Options -B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 22:

Correct Option D - 1 and 4:

- When a patient is present with a history of choking and decreased air entry in a lung, it indicates a possible obstruction in the airway.
- In such cases, the primary concern is identifying and removing the foreign body obstruction to restore proper airflow.

1. Bronchoscopy: Bronchoscopy is a procedure that involves inserting a thin, flexible tube with a camera into the airways to visualize the lungs and remove any foreign bodies or obstructions.

4. Chest X-ray: A chest X-ray can help identify the presence of a foreign body, such as inhaled objects, lung collapse, or other abnormalities.

Incorrect options: Options A, B, and C are incorrect.

2. Computed tomography scan: A CT scan may not be the most immediate and appropriate management option for choking with decreased air entry.

3. ICD insertion: An ICD (Implantable Cardioverter-Defibrillator) is not directly related to managing a choking episode or a lung obstruction.

Solution for Question 23:

Correct Option - D - 2, 3, 4:

2. Oxygen administration:

- In acute asthma exacerbations, when there is a significant impairment in oxygenation, supplemental oxygen should be provided to maintain adequate oxygen levels in the blood.
- Oxygen administration can help alleviate hypoxemia and improve the child's respiratory status.

3. Salbutamol nebulization three times in 60 minutes:

- Salbutamol is a bronchodilator that helps relax the airway smooth muscles and improve airflow in asthma.
- In acute exacerbations, frequent administration of salbutamol via nebulization is commonly recommended to provide immediate relief of symptoms and improve lung function.
- The specific frequency and duration of nebulization may vary depending on the severity of the exacerbation and the child's response to treatment.

4. Administration of oral corticosteroids:

- Oral corticosteroids are a cornerstone of therapy for acute asthma exacerbations.
- They help reduce airway inflammation and improve lung function.
- Oral corticosteroids are typically prescribed in a short course, usually for a few days, to control the acute exacerbation and prevent relapses.

Incorrect Options - A, B, and C are incorrect.

1. Chest X-ray:

- Routine chest X-rays are not recommended for the diagnosis or management of acute asthma exacerbations.
- Chest X-rays are typically reserved for cases where alternative diagnoses or complications are suspected.

Solution for Question 24:

Correct Option C - 3-1-2-4-5:

The correct sequence of managing an unconscious child in a casualty follows:

3. Assess response:

- The first step is to assess the child's response to determine their level of consciousness.
- This can be done by calling out to the child, gently shaking them, or tapping on their shoulders.
- The goal is to determine if the child is responsive or unresponsive.

1. Assess breathing:

- Once the child's responsiveness has been assessed, the next step is to assess their breathing.
- Look for chest rise and fall, listen for breath sounds, and feel for air movement.
- If the child is not breathing or only gasping, this indicates a need for immediate intervention.

2. Assess pulse:

- After assessing breathing, check for the presence or absence of a pulse.
- This is typically done by palpating the carotid pulse on the side of the neck.
- If a pulse is absent or weak, it indicates a cardiac arrest and the need for cardiopulmonary resuscitation (CPR).

4. Start compressions:

- If the child is unresponsive, not breathing, and has no palpable pulse, immediate CPR should be initiated.
- This involves performing chest compressions to circulate blood and oxygen throughout the body.

5. Bag and mask ventilation:

- Along with chest compressions, providing ventilation is crucial to deliver oxygen to the child's lungs.
- This can be done using a bag-valve-mask device, providing positive-pressure ventilation.

Incorrect Options - A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 25:

Correct Option B - Acute laryngotracheobronchitis:

- The radiograph finding in the image shows a characteristic steeple sign or narrowing of the subglottic region and is suggestive of acute laryngotracheobronchitis or croup.

Incorrect Options - A, C, and D are incorrect.

Solution for Question 26:

Correct Option B - Cystic fibrosis:

- The chloride level in sweat is used as a diagnostic test for cystic fibrosis.
- Cystic fibrosis is characterised by the production of thick, sticky mucus that can obstruct the airways and impair the function of various organs.
- In individuals with cystic fibrosis, there is an abnormal transport of chloride ions across cell membranes, leading to increased chloride levels in sweat.

Incorrect Options - A, C, and D are incorrect.

Solution for Question 27:

Correct Option B - Lecithin:

- Respiratory distress syndrome occurs due to the deficiency of lecithin, which helps reduce the alveoli’s surface tension in the lungs, preventing their collapse during exhalation.

Incorrect Options - A, C, and D are incorrect.

Solution for Question 28:

Correct Option C- Administer oral antibiotics, warn of danger signs, and send home:

Signs	Classify as	Treatment
Presence of any general danger sign	Severe pneumonia	Give 1 dose of injection ampicillin and gentamicin and refer urgently to hospital
Presence of either chest indrawing and/or fast breathing but no general danger sign	Pneumonia	Start oral amoxicillin x 5 days The child is sent home and assessed after 48 hours If the child has improved continue the same treatment If the child has worsened, refer to a higher center
No fast breathing, no chest indrawing, no general danger signs	No pneumonia: Cough or cold only	Home-based supportive care
Fast breathing: < 2 months: > 60/ min 2-12 months: > 50/ min 1-5 years: > 40/ min		
General danger signs: Convulsions Lethargy Inability to drink or breastfeed Persistent vomiting Unconsciousness Stridor in a calm child		

- In the question above, the 18-month-old child has no danger signs but has fast breathing, i.e., >40/min, hence the child has pneumonia.
- The treatment for children with pneumonia is to administer oral antibiotics, warn of danger signs, and send them home.

Incorrect Options -A, B, and D are incorrect. Refer to the explanation of the correct answer.

Incorrect options:

Option A: Administer IV antibiotics and observe: Intravenous (IV) antibiotics are usually reserved for severe infections or cases where oral antibiotics cannot be administered. In this case, the child's symptoms are not indicative of severe illness, and IV antibiotics would not be necessary at this stage.

Option B: Administer IV antibiotics and refer to a higher center: Referral to a higher center is typically reserved for cases that require specialized care or management beyond the capacity of the current healthcare facility. In this case, the child's symptoms do not suggest the need for immediate referral to a higher center.

Option D: Administer only home remedies: Home remedies alone are not sufficient for managing respiratory tract infections, especially if there is a suspicion of a bacterial infection. While supportive care measures like maintaining hydration and providing symptomatic relief can be recommended, appropriate antibiotic therapy is often required to treat bacterial infections.

Renal Development and Congenital Disorders

1. A 13-year-old boy presents with excessive thirst, increased urination, and growth retardation. His family has a history of kidney disease. Laboratory tests show anemia, and ultrasound imaging reveals small, scarred kidneys with a few cysts. Which of the following genes is most likely involved in this patient's condition?

A. PKD1

B. PKD2

C. NPHP21

D. NPHP1

2. Which of the following is a common electrolyte imbalance associated with Acute Kidney Injury (AKI)?

A. Hypokalemia

B. Hypercalcemia

C. Hyperkalemia

D. Hyponatremia

3. Which type of AKI is most commonly associated with conditions such as urolithiasis or an enlarged prostate?

A. Prerenal AKI

B. Intrarenal AKI

C. Postrenal AKI

D. Hypovolemic AKI

---

4. Which of the following is NOT a criterion for diagnosing AKI according to KDIGO guidelines?

A. Increase in serum creatinine by ≥0.3 mg/dL

B. Anuria for 12 hours

C. Urine output less than 0.5 mL/kg/h for 6 hours

D. Decrease in GFR by 25%

---

5. Which of the following is the most appropriate initial antihypertensive agent for a child with CKD and proteinuria?

A. Metoprolol

B. Amlodipine

C. Enalapril

D. Furosemide

---

6. Which stage of CKD is characterized by a GFR of 60-89 mL/min/1.73 m^2?

A. Stage 1

B. Stage 2

C. Stage 3

D. Stage 4

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7. A 16 year-old patient presents with sudden onset of flank pain and gross hematuria. Imaging studies reveal an enlarged kidney with a blood clot in the renal vein. Which of the following conditions is most strongly associated with renal vein thrombosis?

A. Hypertension

B. Nephrotic syndrome

C. Polycystic kidney disease

D. Diabetes mellitus

8. A 5-year-old child presents with a history of bloody diarrhea followed by decreased urine output, lethargy, and paleness. Laboratory tests reveal hemolytic anemia, thrombocytopenia, and elevated creatinine levels. What is the most likely diagnosis?

A. Immune Thrombocytopenic Purpura (ITP)

B. Disseminated Intravascular Coagulation (DIC)

C. Atypical Hemolytic Uremic Syndrome (aHUS)

D. Typical Hemolytic Uremic Syndrome

9. A 7-year-old female presents with a complaint of continuous urinary incontinence despite normal voiding patterns. Which of the following is the most likely diagnosis?

A. Vesicoureteral reflux

B. Ectopic ureter

C. Urethral stricture

D. Neurogenic bladder

10. What is the most likely diagnosis and it’s best initial management?

A. Omphalocele; surgical repair after stabilization

B. Bladder exstrophy; surgery if required

C. Gastroschisis; urgent surgical closure of the defect

D. Bladder exstrophy; cover the exposed bladder with sterile plastic wrap and plan for surgery within the first 72 hours

11. At what gestational age does glomerular filtration begin, indicating the kidney's initial role in fetal fluid balance?

A. 3-4 weeks

B. 5-9 weeks

C. 10-12 weeks

D. 13-15 weeks

12. Which of the following statements regarding Autosomal Dominant Polycystic Kidney Disease (ADPKD) are true? Kidneys are enlarged Cysts originates from all

regions of a nephron Symptomatic most commonly in the fourth or fifth decade Majority of the cases are PKD1 mutation which is more severe Associated with saccular cerebral aneurysms Cysts seen in other organs as well

A. 1,2,5

B. 2,6

C. 1,2,5,6

D. 1,2,3,4,5,6

13. An 8-year-old boy has a genetic disease involving the PKHD1 gene which encodes for fibrocystin. Which of the following features will not be found in him?

A. Large cystic kidneys

B. Progressive hepatic fibrosis

C. Spontaneous pneumothorax

D. Hypotension

14. Which of the following statements about Horseshoe kidney is false?

A. Lower poles fused in midline

B. Isthmus ascent is prevented by Inferior mesenteric artery.

C. Can be associated with Multicystic Dysplastic Kidney (MCDK) affecting one side.

D. Isthmectomy is done to relieve symptoms

15. Which of the following statements regarding Multicystic Dysplastic Kidney (MCDK) is false?

A. Usually unilateral

B. Increased risk of Wilms tumor

C. Most common cause of abdominal mass in newborn

D. Mostly autosomal dominant inheritance

---

16. A neonate died shortly after birth due to pulmonary hypoplasia. The neonate has characteristic “Potter Facies” as given below. What is the basic etiology of this condition?

A. Bilateral renal agenesis

B. Hepatic agenesis

C. Urinary outlet obstruction

D. Lack of pulmonary circulation

---

17. A child 5 y/o who had recurrent UTI when evaluated was incidentally found to have a simple renal cyst of 1 cm diameter. What is the treatment plan for this child?

A. Nil intervention and follow up

B. Surgical Excision

C. Regular follow up .

D. Medical Management

---

18. A 6-year-old child has a height of 110 cm and a serum creatinine level of 0.8 mg/dL. Using the Schwartz formula with a constant "k" value of 0.42, what is the estimated GFR (mL/min/1.73 m^2)?

A. 50.75 mL/min/1.73 m^2

B. 57.75 mL/min/1.73 m^2

C. 61.75 mL/min/1.73 m²

D. 66.75 mL/min/1.73 m²

19. A 5-year-old child presents with persistent hypertension and growth retardation. Urine examination reveals proteinuria, and blood tests show elevated creatinine levels. Which of the following methods is the most appropriate for estimating the glomerular filtration rate (GFR) in this child?

A. Serum creatinine measurement

B. Blood urea nitrogen (BUN) test

C. Creatinine clearance

D. Clearance of PAH (Paraaminohippuric acid)

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	3
Question 4	4
Question 5	3
Question 6	2
Question 7	2
Question 8	4
Question 9	2
Question 10	4
Question 11	2
Question 12	4
Question 13	4
Question 14	4
Question 15	4
Question 16	1
Question 17	1
Question 18	2
Question 19	3

Solution for Question 1:

Correct Answer: D) NPHP1

Explanation:

Excessive thirst, increased urination, and growth retardation and Laboratory tests showing anemia, ultrasound imaging showing small, scarred kidneys with a few cysts is indicative of Nephronophthisis. NPHP1 is one of the most common genes associated with nephronophthisis..

Nephronophthisis

Definition	Inherited cystic kidney disease Characterized by chronic tubulointerstitial nephritis Leads to kidney failure
Etiology & Pathogenesis	Caused by mutations in over 20 genes NHPH 1-18 (Option C), NHPH 1 being the most common at 13 yrs old.
Disease Process	Tubular atrophy (shrinkage of tubule cells) Interstitial fibrosis (scar tissue formation) Small cyst formation
Clinical Features	Polyuria and polydipsia (excessive thirst & urination) Anemia Growth retardation in children Hypertension
Diagnosis	Detailed medical history & physical exam Blood & urine tests for kidney function & anemia Ultrasound, MRI, CT Genetic testing
Complications	End-stage renal disease (ESRD) Hypertension Anemia (severe fatigue, weakness) Fractures, liver fibrosis, eye abnormalities
Treatment	Adequate hydration, medication for polyuria Iron supplements Blood pressure control medications Management of extra-renal manifestations Renal replacement therapy (dialysis, kidney transplantation)

- PKD1 (Option A) is associated with Autosomal Dominant Polycystic Kidney Disease (ADPKD), and is characterized by multiple large cysts in the kidneys.
- PKD2 (Option B) is also associated with ADPKD, similar to PKD1.

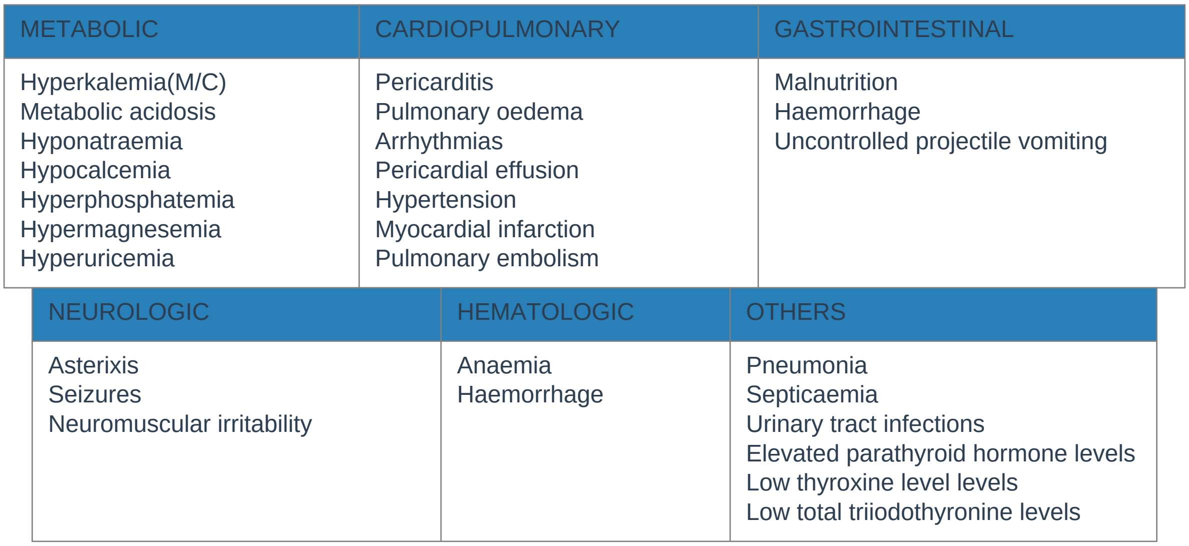
Solution for Question 2:

Correct Answer: C) Hyperkalemia

Explanation:

Hyperkalemia is a common and serious electrolyte imbalance in AKI. The kidneys' impaired function leads to decreased excretion of potassium, causing its accumulation in the blood.

Complication of Acute Kidney Injury(AKI)



METABOLIC	CARDIOPULMONARY	GASTROINTESTINAL
Hyperkalemia(M/C) Metabolic acidosis Hyponatraemia Hypocalcemia Hyperphosphatemia Hypermagnesemia Hyperuricemia	Pericarditis Pulmonary oedema Arrhythmias Pericardial effusion Hypertension Myocardial infarction Pulmonary embolism	Malnutrition Haemorrhage Uncontrolled projectile vomiting

NEUROLOGIC	HEMATOLOGIC	OTHERS
Asterixis Seizures Neuromuscular irritability	Anaemia Haemorrhage	Pneumonia Septicaemia Urinary tract infections Elevated parathyroid hormone levels Low thyroxine level levels Low total triiodothyronine levels

- Hypokalemia (Option A): Not typically associated with AKI. In fact, AKI often leads to hyperkalemia due to the kidneys' reduced ability to excrete potassium.
- Hypercalcemia (Option B): Not a common complication of AKI. AKI can actually lead to hypocalcemia (low calcium) due to reduced conversion of vitamin D to its active form, affecting calcium absorption.
- Hyponatremia (Option D): Not as commonly associated with AKI as hyperkalemia. While fluid overload in AKI can dilute sodium levels, hyperkalemia is a more direct and frequent issue.

Solution for Question 3:

Correct Answer: C) Postrenal AKI

Explanation:

Postrenal Acute Kidney Injury(AKI) is commonly associated with urolithiasis or an enlarged prostate

TYPES OF AKI	CAUSES	CLINICAL FEATURES
Prerenal AKI	Dehydration Gastroenteritis Haemorrhage Burns Sepsis Cirrhosis Anaphylaxis Cardiac failure Capillary leak	Hypotension Tachycardia Dry mucus membrane Oliguria Elevated BUN
Intrinsic / Intrarenal AKI	Acute tubular necrosis Cortical necrosis Renal vein thrombosis Acute interstitial nephritis Glomerulonephritis Haemolytic-uremic syndrome Toxin and drugs Tumor lysis syndrome	Hematuria Proteinuria Edema Hypertension Granular or epithelial cell casts
Postrenal AKI	Urolithiasis Enlarged prostate Ureterocele Posterior urethral valves Ureteropelvic junction obstruction Ureterovesicular junction obstruction Neurogenic bladder Anticholinergic drugs	Flank pain Anuria Intermittent urine flow Incomplete voiding Hydronephrosis

Solution for Question 4:

Correct Answer: D) Decrease in GFR by 25%

Explanation:

- Decrease in GFR by 25% (Option D): GFR is not a criterion for diagnosing AKI according to KDIGO guidelines. KDIGO criteria do not directly involve GFR changes for AKI diagnosis. Instead, the focus is on serum creatinine levels and urine output.

- Acute Kidney Injury (AKI): Defined as abrupt loss of kidney function, leading to rapid decline in the glomerular filtration rate(GFR), accumulation of waste products such as blood urea nitrogen(BUN) and creatinine, and dysregulation of extracellular volume and electrolyte homeostasis.

Staging of AKI based on classification system proposed by Kidney Disease Improving Global Outcomes(KDIGO)

STAGE	SERUM CREATININE	URINE OUTPUT
Stage 1	Increase by ≥0.3 mg/dL OR increase to 1.5–1.9 times baseline	< 0.5 mL/kg/h for 6–12 hours
Stage 2	Increase to 2.0–2.9 times baseline	< 0.5 mL/kg/h for ≥12 hours
Stage 3	Increase to 3.0 times baseline OR increase by ≥4.0 mg/dL	< 0.3 mL/kg/h for ≥24 hours - OR anuria for ≥12 hours

Solution for Question 5:

Correct Answer: C) Enalapril

Explanation:

ACE inhibitors (Enalapril) and Angiotensin II Receptor Blockers (Losartan) are the antihypertensive medications of choice in all children with pediatric CKD.

Treatment of Pediatric CKD: Pediatric Chronic Kidney Disease (CKD) requires a multifaceted approach to treatment, focusing on slowing disease progression, managing complications, and improving quality of life.

Nutrition Management	Dietary Protein restriction is not suggested. Children with CKD stage 2-5 should receive 100% of daily reference intake of vitamins and minerals.
Blood Pressure Control	Antihypertensive drugs, like ACE inhibitors or ARBs, are often prescribed to manage high BP.
Anemia Treatment	Erythropoietin-stimulating agents (ESAs) or iron supplements.
Phosphate Binders	(Calcium carbonate, calcium acetate, Sevelamer, Ferric citrate) to maintain normal serum calcium and

	phosphorus levels.
Ergocalciferol or Cholecalciferol	Typically recommended to treat insufficient 25OH vitamin D
Bicitra or Sodium Bicarbonate tablets	Used to maintain the serum bicarbonate levels > 22mEq/L.
Recombinant human growth hormone(rHuGH)	Given by daily subcutaneous injections and continues until the patient reaches 50th percentile for mid parental height.
Dialysis and Kidney Transplantation	Preferred treatment for children with end-stage renal disease.
Vaccinations	Special considerations for immunocompromised children.

- Metoprolol (Option A): Beta-blocker that can be used for blood pressure control. However, it does not have the added benefit of reducing proteinuria or slowing the progression of CKD, making it less appropriate as a first-line agent.
- Amlodipine (Option B): Calcium channel blocker that is effective in lowering blood pressure but does not significantly reduce proteinuria.
- Furosemide (Option D): Diuretic that can help manage fluid overload and lower blood pressure, but it does not reduce proteinuria or slow CKD progression.

Solution for Question 6:

Correct Answer: B) Stage 2

Explanation:

Chronic Kidney Disease (CKD) is determined by the presence of kidney damage and level of kidney function(GFR).

CKD stages based on NFK KDOQI guidelines

STAGE	DESCRIPTION	GFR(mL/min/1.73 m2)
Stage 1	Kidney damage with normal or increased GFR	≥90
Stage 2	Kidney damage with mild decrease in GFR	60-89

Stage 3	Moderate decrease in GFR	30-59
Stage 4	Severe decrease in GFR	15-29
Stage 5	Kidney failure	<15 or on dialysis

Solution for Question 7:

Correct Answer: B) Nephrotic syndrome

Explanation:

Nephrotic syndrome is strongly associated with the development of renal vein thrombosis (RVT).

Renal Vein Thrombosis (RVT)

Definition	Renal vein thrombosis (RVT) is the formation of a blood clot in the renal vein.
Causes	Dehydration: Risk factor, particularly in infants. Central venous catheters: Disrupt blood flow, increasing clot risk. Inherited thrombophilia: Genetic predisposition to clotting, eg. factor-V leiden mutation, protein-C,S deficiency Nephrotic syndrome: Strongly associated with RVT due to changes in blood clotting factors. Membranous glomerulonephritis Drug induced: Combined oral contraceptives, steroids, asparaginase Immunological disorders: SLE, Rheumatoid arthritis Others: pregnancy, cancer, Hemolytic uremic syndrome
Pathophysiology	Damage to the renal vein's inner lining due to trauma, inflammation, or catheters. Activation of platelets and clotting factors leads to clot formation.The growing clot blocks blood flow, causing kidney swelling and dysfunction.
Clinical Features	Sudden flank pain: Common symptom, especially with rapid clot development. Hematuria: Blood in urine due to obstructed blood flow and kidney damage. Kidney enlargement: Due to obstructed blood flow. Decreased urine output: Impaired kidney function.

Diagnosis	Ultrasound CT Scan: useful if ultrasound is inconclusive. MRI.
Complications	Pulmonary embolism: Clot travels causing obstructing blood flow. Stroke: Clot travels causing a stroke. Kidney failure: Persistent obstruction leads to irreversible damage.
Treatment	Blood thinners (anticoagulants) Thrombolytic therapy (clot busters). Surgery: To remove clot or repair vein.

- Hypertension (Option A): It is not directly associated with an increased risk of RVT.
- Polycystic kidney disease (Option C): This condition causes cyst formation in the kidneys and can lead to chronic kidney disease.
- Diabetes mellitus (Option D) is not specifically linked to an increased risk of RVT compared to nephrotic syndrome.

Solution for Question 8:

Correct Answer: D) Typical Hemolytic Uremic Syndrome

Explanation:

The triad of hemolytic anemia, thrombocytopenia, and acute kidney injury, along with the history of bloody diarrhea, suggests STEC-HUS .

Hemolytic Uremic Syndrome (HUS)

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Solution for Question 9:

Correct Answer: B) Ectopic ureter

Explanation:

An ectopic ureter is a ureter that drains to locations other than the bladder, commonly associated with ureteral duplication.

Ectopic ureter

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Vesicoureteral Reflux (Option A): usually seen in children, presenting with recurrent UTI.

Urethral Stricture (Option C): typically presents with hesitancy, poor flow, prolonged voiding time and rarely retention.

Neurogenic Bladder (Option D): usually presents with increased urinary frequency and nocturia with or without urinary incontinence.

Solution for Question 10:

Correct Answer: D) Bladder exstrophy; cover the exposed bladder with sterile plastic wrap and plan for surgery within the first 72 hours

Explanation:

Bladder exstrophy is a congenital anomaly characterized by the exposure of the bladder mucosa through a defect in the lower abdominal wall.

Bladder exstrophy

Etiology	Disruption in normal development during pregnancy. Exact triggers remain unclear.
Pathology & Pathophysiology	Improper development of the cloacal membrane. Early rupture leads to epispadias (incomplete urethral formation). Later rupture results in bladder exstrophy.
Types	Spectrum from simple epispadias to more complex forms of bladder exstrophy.

Clinical Features	Males: Shortened penis due to pubic bone separation (diastasis of the pubic symphysis). Widely spaced crura (internal, spongy columns within the penis). Underdeveloped corporeal tissue (erectile tissue). Females: Shortened and narrowed vagina, often positioned more anteriorly. Bifid clitoris (split into two parts).
Complications	Urinary function issues. Increased risk of infections. Risk factor for Adenocarcinoma..
Surgical Treatment	Cover the bladder with a wet sterile plastic wrap and plan for surgery. Surgical repair, in multiple stages. Steps Include: Closing the bladder and placing it back inside the body. Reconstructing the urethra. Repairing the abdominal wall defect. Addressing genital abnormalities to ensure normal function and appearance.

Omphalocele (Option A)	Gastroschisis (Option C)
 Herniation of bowel contents that fails to return back	 Evisceration of bowel contents through a defect in anterior abdominal wall just lateral to the umbilicus
Defect through central umbilicus	Defect thru right of umbilicus
Three-layer membrane covering present (peritoneum,wharton's jelly & amnion)	Membrane cover absent
Associated with aneuploidies, genetic syndromes (eg. Beckwith -Wiedemann) and other congenital anomalies	Associated anomalies rare Gut malrotation/bowel atresia may be seen.
Poor prognosis in long term due to associated anomalies	Better prognosis in long term

Bladder exstrophy, immediate surgery to close the bladder and abdominal wall(Option B) immediate surgery is not performed. The priority is to protect the exposed bladder tissue with a sterile covering and ensure the infant's stability before planning surgery within the first few days.

Solution for Question 11:

Correct Answer: B) 5-9 weeks

Explanation:

Development of kidneys	
Structural Development	Pronephros: Emerges at 4 weeks and regresses by the end of the fourth week. Mesonephros: Develops at 4 weeks, temporarily functional, mostly degenerates by the second month. Male mesonephros contributes to genital system. Metanephros Permanent kidney develops from the fifth week. Forms excretory units and collecting system. Kidneys ascend to final position between 6-9 weeks gestation. Size increases from 1.0 cm to 4.0 cm between 12-40 weeks. Ultrasound detects kidneys at 12-13 weeks.
Functional Development	Glomerular filtration begins at 5-9 weeks. (Option B) Tubular function starts at 14 weeks. Renal function matures by two years.

Solution for Question 12:

Correct Answer: D) 1,2,3,4,5,6

Explanation:

All of the statements are correct

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Autosomal Dominant Polycystic Kidney Disease (ADPKD):

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Solution for Question 13:

Correct Answer: D) Hypotension

Explanation:

The scenario is that of Autosomal Recessive Polycystic Kidney Disease (ARPKD)

Hypertension is the clinical feature, not hypotension

Autosomal Recessive Polycystic Kidney Disease (ARPKD):

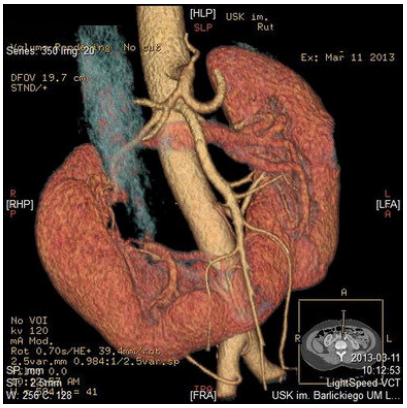
Genetic Basis	Autosomal recessive disorder; gene involved: PKHD1, encoding fibrocystin
Pathology - Kidney	Markedly enlarged kidneys with numerous small cysts in cortex and medulla Dilated, ectatic collecting ducts Progressive interstitial fibrosis and tubular atrophy
Pathology - Liver	Ductal plate abnormality Bile duct proliferation and ectasia Progressive hepatic fibrosis
Clinical Manifestations	Bilateral flank masses Respiratory distress, spontaneous pneumothorax Hypertension Oligohydramnios, Potter syndrome Renal and hepatic disease

Solution for Question 14:

Correct Answer: D) Isthmectomy is done to relieve symptoms

Explanation:

Isthmectomy, which was done before is no more a mode of management. Isthmectomy (surgical separation of connection) should never be done as it contains functional tissue.



Horseshoe kidney:

Description	A condition where the lower poles of the kidneys fuse in the midline. (Option A) More common in males Isthmus lies at level of L3 and its ascent is prevented by the origin of inferior mesenteric artery (IMA). (Option B)
Associations	Turner syndrome Edward syndrome Down syndrome
Risk of Wilms Tumor	Four times more common in children with horseshoe kidneys than in the general population.
Potential Complications	Stone disease. Hydronephrosis secondary to ureteropelvic junction obstruction. Increased incidence of Multicystic Dysplastic Kidney (MCDK) affecting one side. (Option C)
Management	Treatment of complications like PCNL in case of calculus Isthmus is never divided as it contains functional tissue. (Option D)

Solution for Question 15:

Correct Answer: D) Mostly autosomal dominant inheritance

Explanation:

Multicystic Dysplastic Kidney (MCDK) is generally not inherited

Multicystic Dysplastic Kidney (MCDK):

- A congenital condition where the kidney is replaced by cysts and does not function.
- Can result from ureteral atresia.
- Highly variable kidney size
- Usually unilateral (Option A) and generally not inherited. (Option D)
- Bilateral MCDK is incompatible with life.
- Most common cause of an abdominal mass in the newborn (Option C) (though most are nonpalpable at birth).
- Risk of Wilms tumor is increased. (Option B)
- Bunch of grapes appearance seen
- Management: Nephrectomy can be done in unilateral cases

Autosomal recessive polycystic kidney disease (ARPKD):

- Bilateral and severe form of disease
- IVP: Sunburst appearance
- Management: Only palliative care.
- Worst prognosis: Babies die within 2 months of age.

Solution for Question 16:

Correct Answer: A) Bilateral renal agenesis

Explanation:

The scenario is that of Potter syndrome which is caused by bilateral renal agenesis

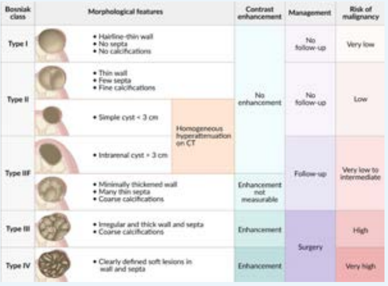
Potter syndrome:

Cause	Bilateral renal agenesis ARPKD Obstructive uropathy (eg. posterior urethral valve) Chronic placental insufficiency
Outcome	Death shortly after birth due to pulmonary hypoplasia
Facial Appearance	Characteristic facial appearance called "Potter facies": Widely separated eyes with epicanthic folds Low set ears Broad and flat nose Receding chin
Other Physical Anomalies	Limb anomalies
Prenatal Indicators	Oligohydramnios Non-visualization of the bladder Absent kidneys in maternal ultrasound

Solution for Question 17:

Correct Answer: A) Nil intervention and follow up

Simple renal cysts	
Overview	Most common renal cystic disease in children Occurs in one or both kidneys Typically benign and asymptomatic Usually discovered incidentally

Diagnosis	Ultrasound, CT Scan, MRI, urine and blood tests (to assess kidney function and rule out serious conditions)
 Treatment Bosniak Classification: Categorizes cystic lesions into four types to guide treatment decisions.	

Bosniak class	Morphological features		Contrast enhancement	Management	Risk of malignancy
Type I	• Hairline-thin wall • No septa • No calcifications		No enhancement	No follow-up	Very low
Type II	• Thin wall • Few septa • Fine calcifications			No follow-up	Low
	• Simple cyst < 3 cm	Homogeneous hyperattenuation on CT			
Type IIF	• Intrarenal cyst > 3 cm			Follow-up	Very low to intermediate
	• Minimally thickened wall • Many thin septa • Coarse calcifications		Enhancement not measurable		
Type III	• Irregular and thick wall and septa • Coarse calcifications		Enhancement	Surgery	High
Type IV	• Clearly defined soft lesions in wall and septa		Enhancement		Very high

Solution for Question 18:

Correct Answer: B) 57.75 mL/min/1.73 m^2

Explanation:

Schwartz formula
GFR (mL/min/1.73 m^2) = (k x height (cm)) / Serum creatinine (mg/dL)

- Substituting the values;

GFR = (0.42 x 110 cm) / 0.8 mg/dL = 57.75 mL/min/1.73 m^2. (Option B)

- Purpose: Quick, bedside estimation of GFR using serum creatinine and height.
- Creatinine-Based: Relies on serum creatinine, influenced by age, muscle mass, and diet.
- Height Inclusion: Reflects correlation between body size and kidney function, especially in children.
- Constant "k": Accounts for age-related variations in creatinine; values range from 0.41 to 0.43.
- Limitations Accuracy issues in populations with altered creatinine production or excretion. Not a gold standard; less accurate compared to inulin or iohexol clearance.
- Accuracy issues in populations with altered creatinine production or excretion.
- Not a gold standard; less accurate compared to inulin or iohexol clearance.

- Clinical use Often used alongside other findings to assess kidney function. Interpret results within the broader clinical picture, recognizing limitations.
- Often used alongside other findings to assess kidney function.
- Interpret results within the broader clinical picture, recognizing limitations.
- Accuracy issues in populations with altered creatinine production or excretion.
- Not a gold standard; less accurate compared to inulin or iohexol clearance.
- Often used alongside other findings to assess kidney function.
- Interpret results within the broader clinical picture, recognizing limitations.

Age Group	GFR (mL/min/1.73 m²)	Inference
Birth (Full-term Infants)	15-20	GFR is low at birth; renal function is still maturing.
Birth (Preterm Infants)	10-15	Lower GFR compared to full-term infants due to underdeveloped kidneys.
2 Weeks	35-45	Rapid increase in GFR as renal function starts maturing.
2 Months	75-80	Continued rapid increase, nearing levels seen in older infants.
Childhood (up to 2 years)	Approaches 80-120 (comparable to adults after correction for body size)	Renal function continues to improve, reaching near-adult GFR levels by age two when corrected for body surface area.

Solution for Question 19:

Correct Answer: C) Creatinine clearance

Explanation:

- Creatinine clearance is considered one of the most accurate for estimating GFR, especially in pediatric patients.
- It accounts for both serum creatinine levels and urine output.

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- Serum creatinine measurement (Option A): While serum creatinine is used to estimate GFR, it is not the most accurate standalone measure, especially in children, where muscle mass can influence creatinine levels.

- Blood urea nitrogen (BUN) test (Option B): BUN levels can be influenced by factors other than kidney function, such as hydration status and protein intake, making it less reliable for estimating GFR.
- Clearance of PAH (Para Aminohippuric acid) (Option D): PAH clearance is used to measure renal plasma flow, not GFR.

Renal Disorders

1. Which of the following features will not be seen in a child with Type I RTA?

A. Urine pH < 5.5

B. Nephrocalcinosis

C. Hypercalciuria

D. Bone diseases

2. Which of the following laboratory findings is not indicative of Minimal Change Disease (MCD) compared to Post-Streptococcal Glomerulonephritis (PSGN)?

A. Normal serum complement C3

B. Hypercholesterolemia

C. Prominent hematuria

D. Heavy proteinuria on dipstick

3. A 7-year-old child presents with symptoms indicative of acute pyelonephritis, including high fever, flank pain and nausea. The child is stable but unable to tolerate oral medications. What is the most appropriate initial treatment for this child?

A. Start oral Nitrofurantoin and follow with a 3-day course of antibiotics.

B. Begin intravenous Ceftriaxone and transition to oral Cefixime after 3 days.

C. Administer oral Amoxicillin and monitor foR 5 days.

D. Initiate treatment with intravenous Fluoroquinolones and assess afteR 7 days.

4. Match the grade of VUR with its description. Grade of VUR Description 1. Grade I A. Reflux into a grossly dilated ureter 2. Grade II B. Reflux into a nondilated ureter 3. Grade III C. Massive reflux with significant ureteral dilation and tortuosity 4. Grade IV D. Reflux into the upper collecting system without dilation 5. Grade V E. Reflux into a dilated ureter and/or blunting of calyceal fornices

Grade of VUR	Description
1. Grade I	A. Reflux into a grossly dilated ureter
2. Grade II	B. Reflux into a nondilated ureter
3. Grade III	C. Massive reflux with significant ureteral dilation and tortuosity
4. Grade IV	D. Reflux into the upper collecting system without dilation
5. Grade V	E. Reflux into a dilated ureter and/or blunting of calyceal fornices

A. 1-B, 2-D, 3-E, 4-A, 5-C

B. 1-A, 2-E, 3-C, 4-D, 5-B

C. 1-D, 2-C, 3-A, 4-E, 5-B

D. 1-E, 2-B, 3-D, 4-C, 5-A

5. Which of the following correctly represents the classic triad of symptoms associated with Prune Belly Syndrome?

A. Deficient abdominal musculature, hydronephrosis and pulmonary hypoplasia.

B. Cryptorchidism, urinary tract abnormalities and pulmonary hypoplasia.

C. Deficient abdominal musculature, cryptorchidism and urinary tract abnormalities.

D. Cryptorchidism, gastrointestinal malformations and hydronephrosis.

6. A 6-year-old child presents with recurrent episodes of flank pain, hematuria and failure to thrive. An abdominal ultrasound reveals bilateral hydronephrosis and a voiding cysto-urethrogram (VCUG) shows no evidence of vesico-ureteric reflux (VUR). Laboratory tests reveal hypercalciuria. What is the most appropriate initial management for this patient?

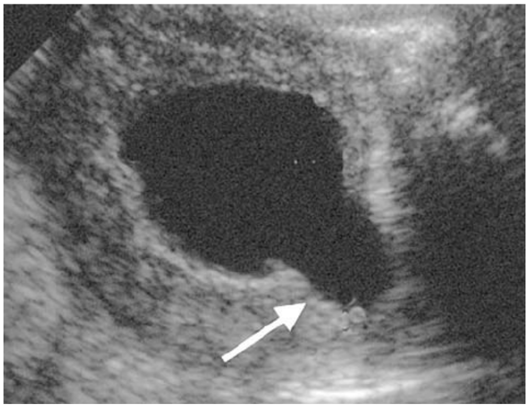
A. Immediate surgical intervention

B. Increased fluid intake and dietary modification

C. Start on antibiotics to prevent UTI

D. Dietary calcium restriction

7. A 2-month-old male infant presents with a history of reduced urine passage and recurrent UTIs. His antenatal scan showed findings as below. Which of the following is the most appropriate next step in confirming the diagnosis ?



A. Renal ultrasound

B. Retrograde Urethrogram

C. Voiding cystourethrogram (VCUG)

D. Intravenous pyelography (IVP)

8. A 5-year-old boy presents with an asymptomatic flank mass. Ultrasonography reveals a dilated renal pelvis without ureteric dilatation. Which of the following is the most appropriate next step in confirming the diagnosis?

A. Intravenous pyelography (IVP)

B. MAG 3 scan

C. MRI

D. CT scan

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9. A 6-year-old boy with muscle spasms and polyuria has been diagnosed with Gitelman syndrome. Which of the following is not likely to be seen in him?

A. Hypochloremia

B. Hypokalemia

C. Hypocalciuria

D. Hypermagnesemia

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10. Which of the following is the cause of Type IV RTA?

A. Hypoaldosteronism

B. Pseudohypoaldosteronism

C. Either A or B

D. Neither A or B

---

11. Which of the following is a characteristic renal biopsy finding in post-streptococcal glomerulonephritis (PSGN)?

A. Immune complex deposition in the mesangial area

B. Subepithelial immune complex deposits with a hump appearance

C. Linear staining of glomerular basement membrane for IgG

D. Diffuse glomerular sclerosis with hyaline deposits

12. A 2-year-old child presents with polyuria, growth failure, fair skin, blond hair, and photophobia. Slit-lamp examination reveals cystine crystals in the cornea. Which genetic mutation is most likely responsible for this patient's condition?

A. CFTR

B. PKD1

C. CTNS

D. HFE

13. Which of the following statements about Fanconi syndrome is false?

A. Rickets is seen

B. Anorexia is a symptom

C. Growth retardation is seen

D. Only tubular dysfunction is bicarbonaturia

14. A 3 month old infant is diagnosed with congenital nephrotic syndrome (CNS) and presents with severe generalised edema, recurrent infections and failure to thrive. Genetic testing reveals a mutation of NPHS1 gene. Which of the following is the definitive treatment for this patient's condition?

A. Albumin infusions and diuretics

B. ACE inhibitors and/or angiotensin II receptor blockers

C. Bilateral nephrectomy followed by chronic dialysis

D. Renal transplantation

15. A 3 month old infant presents with severe generalised edema and poor growth even with adequate feeds and recurrent infections. On further investigations, the infant is found to have hypoalbuminemia and proteinuria. Genetic testing confirms a mutation in the NPHS1 gene. Which of the following conditions is most likely associated with this clinical presentation?

A. Denys-Drash syndrome

B. Finnish type of Congenital Nephrotic Syndrome

C. Pierson syndrome

D. Focal segmental glomerulosclerosis (FSGS)

16. A 2-month-old infant presents with generalised edema, hypoalbuminemia, and oliguria. Prenatal records indicate elevated alpha-fetoprotein (AFP) levels in maternal serum and amniotic fluid. Which of the following genetic mutations is most likely responsible for this condition?

A. WT1

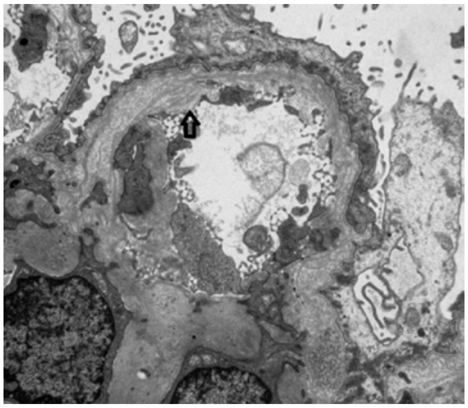
B. NPHS1

C. PLCE1

D. NPHS2

17. A 13 y/o child presents with progressive hearing loss and occasional visual disturbances. His investigation reports show persistent microscopic hematuria and moderate proteinuria. Kidney biopsy shows irregularities in the glomerular basement membrane. Genetic testing reveals a mutation in the COL4A gene. What

is the most likely diagnosis?



A. Alport Syndrome

B. Goodpasture Syndrome

C. IgA Nephropathy

D. Thin Basement Membrane Disease

18. A 13 year old boy who has upper respiratory infections presents with recurrent episodes of blood in his urine and hypertension. A renal biopsy reveals IgA deposition in the mesangium. Which of the following is the most appropriate initial treatment for controlling his condition?

A. Daily use of diuretics

B. ACE inhibitors or angiotensin II receptor blockers

C. Antibiotics to prevent infections

D. High-dose corticosteroids

19. A 13 year old boy presents with hemoptysis. His investigation reports are suggestive of rapidly worsening renal function. Kidney biopsy shows linear IgG deposits along the glomerular basement membrane. Which type of rapidly

progressive glomerulonephritis (RPGN) is most consistent with these findings?

A. Type I RPGN (Anti-GBM Antibody Disease)

B. Type II RPGN (Immune Complex Mediated)

C. Type III RPGN (Pauci-Immune)

D. Membranoproliferative glomerulonephritis

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	2
Question 4	1
Question 5	3
Question 6	2
Question 7	3
Question 8	2
Question 9	4
Question 10	3
Question 11	2
Question 12	3
Question 13	4
Question 14	4
Question 15	2
Question 16	2
Question 17	1
Question 18	2
Question 19	1

Solution for Question 1:

Correct Answer: A) Urine pH < 5.5

Explanation:

- Urine pH in Type I RTA is always > 5.5
- Distal Renal Tubular Acidosis (RTA) = Type I RTA

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Solution for Question 2:

Correct Answer: C) Prominent hematuria

Explanation:

- Prominent hematuria (cola-colored urine) is highly indicative of PSGN and is rarely seen in MCD. This makes it a key differentiator between the two conditions.

Feature	Minimal Change Disease (MCD)	Post-Streptococcal Glomerulonephritis (PSGN)
Clinical Presentation	Nephrotic syndrome: Edema, heavy proteinuria, but normal renal function	Acute nephritic syndrome: Hematuria, oliguria, edema, hypertension
Proteinuria	Heavy (3+ to 4+) on dipstick (Option D)	(1+ to 2+) levels of proteinuria
Hematuria	Rare, microscopic hematuria may occur	Common, with cola-colored urine due to gross hematuria (Option C)
Urinary Casts	Hyaline and granular casts	Red blood cell casts (indicating glomerular inflammation)
Serum Albumin	Low (<1 g/dL)	Usually normal unless severe renal impairment
Blood Urea and Creatinine	Normal unless hypovolemia or acute kidney injury	Elevated due to impaired renal function
Serum Complement C3	Normal (Option A)	Low in 90% of cases, returns to normal in 8-12 weeks
Cholesterol	Hypercholesterolemia (may cause milky plasma) (Option B)	No significant hypercholesterolemia

ASO/Anti-DNase B Titers	Not elevated	Elevated ASO titers (pharyngitis) or anti-DNase B (skin infection)

Solution for Question 3:

Correct Answer: B) Begin intravenous Ceftriaxone and transition to oral Cefixime after 3 days.

Explanation:

Intravenous Ceftriaxone is effective for initial treatment of acute pyelonephritis, then oral Cefixime can be used to complete the treatment course.

Urinary Tract Infections (UTIs)

Etiology and Prevalence:

- Primary Cause: Colonic bacteria ascending the urinary tract.
- Common Pathogen: Escherichia coli (54-67% of cases).
- Other Pathogens: Klebsiella spp., Proteus spp., Enterococcus, Pseudomonas.

Clinical Features and Classification:

- Pyelonephritis (Kidney Infection): Upper renal collecting system involved Symptoms: Abdominal/back/flank pain, fever, malaise, nausea, vomiting, diarrhea. Newborns: Non-specific symptoms like poor feeding, irritability, jaundice, weight loss.
- Symptoms: Abdominal/back/flank pain, fever, malaise, nausea, vomiting, diarrhea.
- Newborns: Non-specific symptoms like poor feeding, irritability, jaundice, weight loss.
- Cystitis (Bladder Infection): Only bladder involved Symptoms: Dysuria, urgency, frequency, suprapubic pain, incontinence, malodorous urine. Typically does not cause high fever or renal injury.
- Symptoms: Dysuria, urgency, frequency, suprapubic pain, incontinence, malodorous urine.
- Typically does not cause high fever or renal injury.
- Symptoms: Abdominal/back/flank pain, fever, malaise, nausea, vomiting, diarrhea.
- Newborns: Non-specific symptoms like poor feeding, irritability, jaundice, weight loss.
- Symptoms: Dysuria, urgency, frequency, suprapubic pain, incontinence, malodorous urine.
- Typically does not cause high fever or renal injury.

Risk Factors for Recurrent UTI:

- Vesicoureteral Reflux (VUR)
- Bladder-Bowel Dysfunction (BBD)
- Obstructive Uropathy
- Urinary Tract Abnormalities: Anatomical issues like labial adhesion.
- Neuropathic Bladder
- Catheterisation: Procedures introducing bacteria.
- Sexual Activity: Increases UTI risk in females.

Diagnosis:

- Urine Culture: Gold standard for UTI diagnosis; identifies and quantifies bacteria.

Interpretation:

A colony count of >50,000 colony-forming units/mL of a single pathogen, along with pyuria or bacteriuria in a symptomatic child, is generally considered diagnostic of a UTI.

- Urinalysis: Leukocyte Esterase and Nitrites positive.
- Imaging Studies:

Age < 1 year	Age > 1 year
USG KUB DMSA scan Micturating cystourethrogram (MCU)	USG KUB (abnormal), DMSA scan (abnormal), MCU

Treatment:

- Antibiotic Therapy: Mainstay of UTI treatment.

Treatment Specific to UTI Presentations	
Acute Cystitis	Common Antibiotics: TMP-SMX, Nitrofurantoin, Amoxicillin (Option C) Duration: 3-5 days
Acute Pyelonephritis	Hospitalized: IV Ceftriaxone, Cefepime, Cefotaxime Outpatient: Oral Cefixime, Cephalexin Avoid Nitrofurantoin (Option A); use Fluoroquinolones for resistant strains (Option D). Duration: 7-14 days
Acute Lobar Nephronia/Abscess	Similar to pyelonephritis, but longer duration (14-21 days) Percutaneous or surgical drainage for abscesses

Solution for Question 4:

Correct answer: A) 1-B, 2-D, 3-E, 4-A, 5-C

Explanation:

Vesico-Ureteric Reflux (VUR)

- Retrograde urine flow from the bladder to the ureters/kidneys due to an incompetent valve at the ureter-bladder junction.

Grading of VUR:

Grade	Description
I	Reflux into a nondilated ureter
II	Reflux into the upper collecting system without dilation
III	Reflux into a dilated ureter and/or blunting of calyceal fornices
IV	Reflux into a grossly dilated ureter
V	Massive reflux with significant ureteral dilation and tortuosity

Solution for Question 5:

Correct Answer: C) Deficient abdominal musculature, cryptorchidism and urinary tract abnormalities

Explanation:

PRUNE BELLY SYNDROME	
Rare congenital disorder mainly in males. Characterized by a triad: (Option C) Deficient abdominal musculature	

Cryptorchidism Urinary tract abnormalities	
Etiology	The exact cause is unknown. Proposed theories: Urethral Obstruction-Malformation Complex: Obstruction during development leads to bladder distension and impacts urinary tract, abdominal wall and testicular development. Yolk Sac Defect: A defect in the yolk sac may contribute to the syndrome. Lateral Plate Mesoderm Defect: Defect in embryological layer forming ureters, bladder, prostate, urethra and gubernaculum.
Clinical Presentation	 ... (content truncated)
Treatment	Medical Management: Prophylactic Antibiotics: To prevent UTIs, especially in patients with vesicoureteral reflux. Management of Constipation: Important to prevent exacerbation of urinary issues. Surgical Interventions: Orchiopexy: Corrects undescended testicles, typically around 6 months of age. Urinary Tract Reconstruction: For recurrent UTIs or progressive renal issues. Abdominal Wall Reconstruction: For significant abdominal wall deficiencies, often with other surgeries.

Solution for Question 6:

Correct Answer: B) Increased fluid intake and dietary modification

Explanation:

The clinical scenario is suggestive of nephrolithiaisis due to hypercalciuria. First-line management for the child includes increased fluid intake to promote urine dilution and dietary modification to prevent stone formation.

Nephrolithiasis

Etiology	Metabolic Causes: Hypercalcemia, hyperphosphatemia, idiopathic hypercalciuria, hyperuricemia, hyperoxaluria, cystinosis. Genetic Disorders: Bartter syndrome, Gitelman syndrome, Dent disease (X-Linked disease). Other Causes: Primary hyperparathyroidism.
Pathophysiology	Idiopathic Hypercalciuria: increased urinary calcium excretion promotes calcium stone formation.
Types of Stones	Common Components: Calcium oxalate, calcium phosphate.
Clinical Features	Neonates/Infants: Weak urinary stream, straining during micturition, distended & palpable bladder. Older Infants/Children: Failure to thrive, recurrent UTIs, sepsis, possible uremia. Common Symptoms: Microscopic or gross hematuria, renal colic (flank/abdomen pain), dysuria, urgency, frequency.
Diagnosis	USG, VCUG or MCU, DMSA scan. Imaging: Abdominal X-ray (radio-opaque calculi), USG, Spiral CT of abdomen & pelvis. Laboratory Tests: Urine calcium (hypercalciuria), uric acid levels, 24-hour urine collection.
Complications	Recurrent stone formation UTI
Treatment	Increased fluid intake, dietary modifications (reduce sodium). Medications: Thiazide diuretics to reduce urinary calcium excretion (in hypercalciuria cases). Surgical Intervention: Required for larger stones (Option A) or if conservative measures fail; procedures used lithotripsy, ureteroscopy, percutaneous nephrolithotomy. Dietary Recommendations: High fluid intake, avoid high-protein diet, dietary calcium usually not restricted (Option D).

Solution for Question 7:

Correct Answer: C) Voiding cystourethrogram (VCUG)

Explanation:

The voiding cystourethrogram (VCUG) is the gold standard imaging test for diagnosing posterior urethral valves (PUVs). It not only confirms the presence of PUVs but also assesses the degree of bladder and ureteral dilation.

Posterior Urethral Valves (PUVs)

Etiology	Male infants Thin, membrane-like structures in the posterior urethra
Pathogenesis	Abnormal fusion of urethral folds forming valves Obstruction leads to possible backflow of urine into ureters and kidneys (Vesicoureteral Reflux - VUR)
Clinical Features	Distended bladder (palpable, firm) Recurrent UTIs Poor urine stream Failure to thrive (difficulty gaining weight, growth delay) Prenatal detection via ultrasound (enlarged bladder, dilated ureters, kidney damage)
Diagnosis	Physical Examination: Palpable enlarged bladder Voiding Cystourethrogram (VCUG): Gold standard test, visualizes urethral valves, assesses bladder and ureteral dilation Ultrasound: Initial test for kidney, bladder and ureter dilation
Complications	Recurrent UTIs Hydronephrosis leading to CKD Bladder dysfunction Electrolyte imbalances due to kidney dysfunction
Treatment	Bladder Drainage: Immediate catheterization in severe cases Cystoscopic Valve Ablation Antibiotics for UTI

- Renal ultrasound (Option A) is often used as an initial imaging test, it is not definitive for diagnosing PUVs.
- Retrograde Urethrogram (Option B): X-ray examination that visualizes the urethra by injecting contrast material directly into it. Primarily used to evaluate urethral strictures or trauma and is not typically the first test for a suspected urinary tract obstruction or reflux in infants.
- X-ray examination that visualizes the urethra by injecting contrast material directly into it.

- Primarily used to evaluate urethral strictures or trauma and is not typically the first test for a suspected urinary tract obstruction or reflux in infants.
- Intravenous pyelography (IVP) (Option D) is an older imaging modality that has largely been replaced by more advanced techniques like VCUG and ultrasound.
- X-ray examination that visualizes the urethra by injecting contrast material directly into it.
- Primarily used to evaluate urethral strictures or trauma and is not typically the first test for a suspected urinary tract obstruction or reflux in infants.

Solution for Question 8:

Correct Answer: B) MAG 3 scan

Explanation:

The diagnosis of PUJ obstruction usually is done by ultrasonography and MAG 3 scan diuretic renography.

Pelviureteric Junction (PUJ) obstruction

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MRI (Option C) and CT scan (Option D) can provide anatomical details but are not the primary investigations for functional assessment of the renal system in PUJ obstruction.

Solution for Question 9:

Correct Answer: D) Hypermagnesemia

Explanation:

Gitelman syndrome presents with hypomagnesemia, not hypermagnesemia.

Gitelman syndrome	
Etiology	Rare autosomal recessive disorder

Pathogenesis	Defects in the gene encoding the sodium chloride co-transporter in the distal convoluted tubule
Clinical Presentation	Presents in late childhood or early adulthood Recurrent muscle cramps Spasms Nocturia Polyuria Occasional hypotension Less prominent growth failure
Biochemical Abnormalities	Hypokalemia (Option B) Hypochloremic metabolic alkalosis (Option A) Hypomagnesemia, elevated urinary magnesium levels Hypocalciuria (Option C)

Solution for Question 10:

Correct Answer: C) Either A or B

Explanation:

Type IV RTA is due to impaired aldosterone production (hypoaldosteronism) or impaired renal responsiveness to aldosterone (pseudohypoaldosteronism)

Hyperkalemic Renal Tubular Acidosis (RTA) = Type IV RTA

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Solution for Question 11:

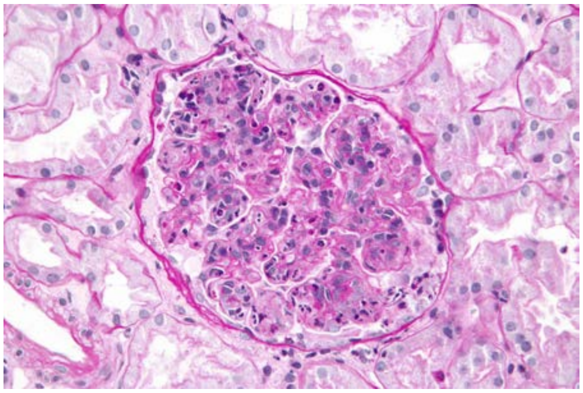
Correct Answer: B) Subepithelial immune complex deposits with a "hump" appearance

Explanation:

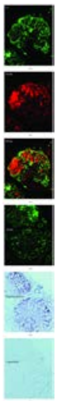
PSGN renal biopsy findings

- Subepithelial immune complex deposits with a characteristic hump appearance is a hallmark of PSGN.
- The hump appearance refers to the subepithelial immune complex deposits that protrude into the glomerular capillary lumen, visible in electron microscopy.
- Diffuse glomerular hypercellularity: Increased number of cells in the glomeruli, often including glomerular endothelial cells and mesangial cells.
- Endocapillary proliferation: Proliferation of cells within the capillary loops.

Light Microscopy:



Immunofluorescence:



Electron Microscopy:

Electron Microscopy" data-author="Kari, Jameela & Bamaga, Ahmed & Jalalah, Sawsan. (2013). Severe acute post-streptococcal glomerulonephritis in an infant. Saudi journal of kidney diseases and transplantation : an official publication of the Saudi Center for Organ Transplantation, Saudi A" data-hash="975" data-license="CC BY-NC-SA 4.0" data-source="https://www.researchgate.net/figure/Transmission-electron-micrograph-showing-numerous-large-hump-like-dense-deposits-located_fig2_236614663" data-tags="March2025" src="https://image.prepladder.com/notes/xwlfkZqlSxyzp6pv45vC1742063914.png" />

Immune complex deposition of IgA and C3 (Option A): This finding is more typical of IgA nephropathy (Berger’s disease).PSGN usually involves subepithelial deposits not mesangial deposits.

Linear IgG staining of the glomerular basement membrane (Option C): This finding is characteristic of anti-glomerular basement membrane (GBM) disease or Goodpasture syndrome not PSGN. PSGN typically shows granular staining rather than linear.

Diffuse glomerular sclerosis with hyaline deposits (Option D): Associated with chronic kidney diseases or diabetic nephropathy, not PSGN, which typically presents with acute changes and immune complex deposits.

Solution for Question 12:

Correct Answer: C) CTNS

Explanation:

- The scenario is that of Cystinosis.
- Cystinosis is due to CTNS gene mutation.

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Solution for Question 13:

Correct Answer: D) Only tubular dysfunction is bicarbonaturia

Explanation:

- Fanconi syndrome causes proteinuria, aminoaciduria, bicarbonaturia, phosphaturia, electrolyte wasting and glycosuria.
- Fanconi syndrome is a type of Proximal Renal Tubular Acidosis (RTA) in which there is reduced proximal reabsorption of bicarbonate.

Table rendering failed

Solution for Question 14:

Correct Answer: D) Renal transplantation

Explanation:

This is the most definitive treatment for primary congenital nephrotic syndrome. It offers the best chance for long-term survival and improved quality of life, even though there is a risk of recurrence post-transplantation.

Treatment of Congenital Nephrotic Syndrome (CNS)	
Primary CNS	Manage symptoms and complications. Slow renal insufficiency progression. Supportive Care: Albumin infusions and diuretics for severe edema and fluid overload. (Option A) Nutritional support: High-protein diet (3-4 g/kg), high caloric intake, lipid supplementation Vitamin and thyroid hormone replacement to compensate for urinary losses Infection prophylaxis: Aggressive prevention, including pneumococcal and annual influenza vaccinations for the child and household contacts Pharmacological Management: ACE inhibitors and/or angiotensin II receptor blockers to reduce proteinuria and slow kidney damage progression. (Option B) Prostaglandin synthesis inhibitors (e.g., indomethacin) to decrease proteinuria and glomerular filtration rate, with close renal function monitoring. Surgical Intervention: Unilateral nephrectomy for severe symptoms to reduce proteinuria. Bilateral nephrectomy for persistent severe edema, recurrent infections, or complications from enlarged kidneys; requires chronic dialysis until kidney transplant is available. (Option C) ... (content truncated)
Secondary CNS	Address the underlying cause (e.g., antibiotics for congenital syphilis) If unresolved: Management similar to primary CNS, including supportive care, medications, potential dialysis, and transplantation

Specific Genetic Syndromes	Denys-Drash Syndrome: Management includes addressing Wilms tumors if present. Pierson Syndrome: Supportive care focusing on CNS and associated eye abnormalities.

Solution for Question 15:

Correct Answer: B) Finnish type of Congenital Nephrotic Syndrome

Explanation:

- The clinical presentation of severe generalised edema, poor growth, recurrent infections, and hypoalbuminemia in a young infant suggests congenital nephrotic syndrome (CNS).
- The identification of a mutation in the NPHS1 gene is strongly indicative of the Finnish type of congenital nephrotic syndrome, which is characterised by massive proteinuria, often present at birth.
- The Finnish type of CNS is caused by mutations in the NPHS1 gene encoding nephrin, a critical protein for glomerular filtration.

Clinical Features	General Symptoms: Severe generalised edema Poor growth, nutrition and hypoalbuminemia. Increased susceptibility to infections Hypothyroidism due to urinary loss of thyroxine-binding globulin. Increased risk of thrombotic events. Progressive renal insufficiency in most infants. Finnish Type: Edema caused by massive proteinuria, often present at birth. Galloway-Mowat syndrome Marked by the presence of microcephaly, hiatal hernia and congenital nephrotic syndrome. Kidney biopsies in affected patients typically reveal distinctive abnormalities, including deficient or poorly formed basement membranes, or basement membranes infiltrated by fibrils.
Investigations	Initial Investigations: Urine Analysis: Confirm nephrotic range proteinuria with a spot urine protein. ratio >2.0. Serum Tests: Assess severity with electrolytes, blood urea nitrogen, creatinine, albumin, and cholesterol levels. Evaluation for Secondary Causes: Especially in infants >10 months, including:

	Complement C3 level Antinuclear antibody Double-stranded DNA Hepatitis B and C testing (high-risk populations) HIV testing (high-risk populations) Additional Investigations: Genetic Testing: Confirm primary CNS. Finnish Type: Mutations in NPHS1 or NPHS2 genes. ... (content truncated)
Indications for Biopsy	Infants >12 months: Less likely to have minimal change nephrotic syndrome (MCNS). Atypical Presentation: Gross hematuria, hypertension, renal insufficiency, or hypocomplementemia. Suspicion of Secondary CNS: Systemic diseases or infections suspected. Steroid-resistant Nephrotic Syndrome: Confirm diagnosis and differentiate between causes like focal segmental glomerulosclerosis (FSGS), MCNS, or membranoproliferative glomerulonephritis.

Solution for Question 16:

Correct Answer: B) NPHS 1

Explanation:

- NPHS1 mutation causes Finnish Type Congenital Nephrotic Syndrome (CNS), which is the most common form of CNS.
- Characterised by severe proteinuria in utero and elevated AFP levels, with symptoms manifesting soon after birth.

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Solution for Question 17:

Correct Answer: A) Alport Syndrome

Explanation:

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- Goodpasture Syndrome (Option B): Presents with hematuria and hemoptysis but is associated with anti-GBM antibodies and not COL4A mutations.
- IgA Nephropathy (Option C): Typically presents with episodic hematuria following respiratory infections and does not involve hearing loss or ocular defects.
- Thin Basement Membrane Disease (Option D): Milder form of GBM disorder, typically without progressive hearing loss or the severe symptoms seen in Alport Syndrome.

Solution for Question 18:

Correct Answer: B) ACE inhibitors or angiotensin II receptor blockers

Explanation

ACE inhibitors or angiotensin II receptor blockers are recommended to control blood pressure and reduce proteinuria in managing IgA nephropathy.

IgA Nephropathy (Berger Disease)	
Overview	Most common chronic glomerular disease in children; immune complex disease causing glomerular inflammation and damage. Can present as a severe form of RPGN in certain cases, though not typically classified as RPGN
Causes	Immune Response: Triggered by environmental factors like infections. Genetic Factors: Higher prevalence in certain ethnic groups.
Pathology & Diagnosis	IgA deposition in the mesangium. Diagnosis: Renal biopsy confirms and assesses histologic severity.
Symptoms	Hematuria: Blood in urine, often following respiratory or gastrointestinal infections. Other Symptoms: Loin pain, proteinuria, hypertension, edema.

Treatment	Blood Pressure Control: ACE inhibitors or angiotensin II receptor blockers. Additional Therapy: Fish oil (anti-inflammatory), corticosteroids for severe cases.
Prognosis	Children: Generally favorable with minimal long-term kidney damage. Adults: 20-30% may develop progressive disease, potentially leading to kidney failure.

Daily use of diuretics (Option A): Not the first line of treatment for IgA nephropathy; primarily used in cases with significant fluid retention or hypertension.

Antibiotics to prevent infections (Option C): While infections can trigger episodes, antibiotics are not a primary treatment for IgA nephropathy.

High-dose corticosteroids (Option D): Considered for severe cases with significant proteinuria or rapid progression but not as an initial treatment.

Solution for Question 19:

Correct Answer: A) Type I RPGN (Anti-GBM Antibody Disease)

Explanation

Type I RPGN (Anti-GBM Antibody Disease); characterized by the presence of anti-GBM antibodies, which cause linear IgG deposits along the glomerular basement membrane, and is associated with Goodpasture syndrome (hemoptysis and glomerulonephritis).

- RPGN (Crescentic glomerulonephritis) is characterized by the formation of crescents in Bowman's space due to the proliferation of parietal epithelial cells.
- Classified into primary and secondary types based on the underlying cause: Primary RPGN Cause is unknown or not linked to systemic disease. Diagnosed primarily through kidney biopsy and exclusion of systemic causes. Secondary RPGN Triggered by systemic diseases that also affect the kidneys. Often associated with systemic autoimmune diseases or infections (Systemic lupus erythematosus, Henoch-Schönlein purpura, membranoproliferative glomerulonephritis, postinfectious GN). Detected by clinical presentation and serological tests.
- Primary RPGN Cause is unknown or not linked to systemic disease. Diagnosed primarily through kidney biopsy and exclusion of systemic causes.

- Cause is unknown or not linked to systemic disease.
- Diagnosed primarily through kidney biopsy and exclusion of systemic causes.
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- Triggered by systemic diseases that also affect the kidneys.
- Often associated with systemic autoimmune diseases or infections (Systemic lupus erythematosus, Henoch-Schönlein purpura, membranoproliferative glomerulonephritis, postinfectious GN).
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- Detected by clinical presentation and serological tests.
- Types of Rapidly Progressive Glomerulonephritis (RPGN)

RPGN Type	Cause	Associated Conditions	Characteristics

Type I RPGN	Autoimmune attack on the glomerular basement membrane (GBM).	Goodpasture syndrome (lung hemorrhage + RPGN).	Presence of anti-GBM antibodies in blood, severe kidney damage, and lung involvement in Goodpasture syndrome.
Type II RPGN	Immune complex deposition in the glomeruli.	Lupus nephritis IgA nephropathy postinfectious GN.	Immune complexes trigger inflammation; can be a severe form of immune-related diseases.
Type III RPGN	Characterized by few or no immune complexes in the glomeruli, associated with ANCA.	Granulomatosis with polyangiitis, microscopic polyangiitis.	Known as pauci-immune; detected by the presence of ANCA in blood, with no significant immune complex deposits on kidney biopsy.

- Type II RPGN (Immune Complex Mediated) (Option B): Involves immune complex deposition but not linear IgG deposits.
- Type III RPGN (Pauci-Immune)(Option C): Characterized by the absence of significant immune complex deposits and presence of ANCAs, not linear IgG deposits.
- Membranoproliferative glomerulonephritis(Option D): Usually presents with immune complex deposition, not linear IgG deposits.

Previous Year Questions

1. A 10-year-old child, treated with antibiotics for follicular tonsillitis, presents with periorbital edema. Blood work reveals eosinophilia and urine microscopy shows WBC casts. What is the most likely diagnosis?

A. Pyelonephritis

B. Allergic interstitial nephritis

C. Glomerulonephritis

D. IgA nephritis

2. A 4-year-old child presents with a 14 cm tender mass in the flank region. The parents report noticing the mass gradually increasing in size over the past few weeks. On physical examination, the child appears well but discomforted due to the mass. The mass is non-fluctuant, firm, and non-mobile, with no signs of inflammation or skin changes. There are no other associated symptoms. What is the most likely diagnosis?

A. Wilm’s tumor

B. Angiomyolipoma

C. Renal cell carcinoma

D. Neuroblastoma

3. A 1-year-old male child with a body weight of 15 kg and height of 100 cm is admitted with renal failure. His blood urea was 100 mg/dL, and serum creatinine was 1 mg/dL. What is the closest calculated eGFR in the patient?

A. 33 ml/min/1.73 m2 BSA

B. 40 ml/min/1.73 m2 BSA

C. 55 ml/min/1.73 m2 BSA

D. 80 ml/min/1.73 m2 BSA

4. What is the appropriate treatment for a 5-year-old child who has been brought to the hospital due to a complaint of loose stools? The child's mother reports that the child is irritable and drinks water quickly when offered. Upon examination, sunken eyes and delayed retraction of the skin after a skin pinch test are observed.

A. Administer the first dose of IV antibiotic and immediately refer to a higher center

B. Give oral fluids and ask the mother to continue the same and visit again the next day

C. Give zinc supplementation and oral rehydration solution only and ask mother to come back if some danger signs develop

D. Consider severe dehydration, start intravenous (IV) fluids, IV antibiotics, and refer to a higher center

5. Identify the condition:

A. Bladder exstrophy

B. Omphalocele

C. Persistent vitellointestinal duct

D. Gastroschisis

6. Steroid-resistant nephrotic syndrome is defined as failure to achieve remission after ___ weeks when on a daily corticosteroid therapy regimen.

A. 4-6

B. 2-4

C. 10-12

D. 8-10

7. What is the preferred initial treatment for a 4-year-old boy who initially had facial swelling but now has experienced total body swelling involving the face, abdomen, scrotum, and feet, and has a urinalysis showing 4+ protein and a urine protein: creatinine ratio of 3?

A. Prednisolone

B. Cyclophosphamide

C. Levamisole

D. Cyclosporine

8. What is the gene associated with Juvenile myoclonic epilepsy?

A. CHYNN-1

B. GABRA-1

C. FMR-1

D. All of the above

9. What is the probable diagnosis for a child (with a history of pharyngitis) who has been experiencing brown-colored urine and reduced urine output for the past three days? The child also exhibits mild swelling in the face and feet, with a blood pressure reading of 126/90. The child has high levels of protein in the urine (2+), more than 100 red blood cells per high power field (HPF), and a few granular casts. Additionally, the child's creatinine level is 0.9 and urea level is 56.

A. PSGN

B. Nephrolithiasis

C. Posterior Urethral Valve

D. Minimal change disease

10. What could be the likely diagnosis for an 8-year-old boy who has sensorineural hearing loss, eye defects, and presents with hematuria in the outpatient department, considering that his renal biopsy showed no abnormalities, and his maternal uncle has end-stage chronic kidney disease (CKD) and is currently undergoing dialysis?

A. Alport syndrome

B. Goodpasture syndrome

C. IgA nephropathy

D. PSGN

11. What is the most probable diagnosis for a young girl who experiences constant urine dribbling, despite voiding normally, and has never been completely dry since birth, according to her parents?

A. Ectopic ureter

B. Ureterocele

C. Congenital megaureter

D. Vesicoureteric reflux

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12. What investigation can be employed to determine the underlying diagnosis in a 3-year-old boy who presents with repetitive episodes of urinary tract infection? The mother reports that he urinates frequently but has a weak urinary stream and dribbling of urine. Upon examination, a palpable bladder is detected.

A. Ultrasonogram

B. Micturating cystourethrography

C. Urine routine examination

D. DMSA scan

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13. What is true about vaccination in a child with steroid-dependent nephrotic syndrome who is brought to the OPD?

A. All live vaccines under NIS can be given as per schedule

B. All killed vaccines under NIS can be given

C. Only pneumococcal vaccine should be given to the child in acute phase

D. Live vaccines are contraindicated in the child's sibling

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14. What is the most probable diagnosis for an 8-year-old boy who has been experiencing dark-colored urine, reduced urine output, high blood pressure, and swollen face for three days? The urine test indicates the presence of albumin. Additionally, the patient had a recent throat infection two weeks ago. Upon

microscopic examination, enlarged and pale glomeruli, along with epithelial humps, are observed.

A. Post-staphylococcal glomerulonephritis

B. Post-streptococcal glomerulonephritis

C. Nephrotic syndrome

D. IgA nephropathy

15. What is the most common presentation of generalised oedema and massive proteinuria?

A. Nephrotic syndrome

B. Acute post-streptococcal glomerulonephritis

C. Renal tubular acidosis

D. Nephrolithiasis

16. Unilateral renal agenesis in an infant is linked with:

A. Single umbilical artery

B. Single umbilical Vein

C. Polyhydramnios

D. Maternal anemia

17. What is the probable diagnosis for a child who exhibits generalized swelling and is diagnosed with high cholesterol levels, as well as protein 3+ and fat bodies observed in urine analysis?

A. Nephrotic syndrome

B. Nephritic syndrome

C. Goodpasture's disease

D. IgA nephropathy

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18. Which of the following is true regarding urinary tract infections (UTIs) in children? Most common cause of UTI is Streptococcus pneumoniae. Bowel and bladder dysfunction increases the risk of recurrence. Cotrimoxazole is not given. Micturating cystourethrogram (MCU) is done in children with a recurrence of UTI.

A. 1, 3, 4

B. 2, 4

C. 2, 3, 4

D. 1, 4

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19. What is the diagnosis of a 5-year-old child who initially had edema of the face that later developed into generalized edema? The urinalysis revealed significant proteinuria, while light microscopy results were normal. However, electron microscopy indicated effacement of podocyte foot processes.

A. Focal segmental glomerulo sclerosis

B. Minimal change disease

C. Membranous glomerulonephritis

D. Post streptococcal glomerulonephritis

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20. A 10-year-old presents with edema and anasarca. A diagnosis of minimal change disease is made. Which of the following is true about this condition?

A. Electron microscopy shows effacement of podocytes

B. No response to steroids

C. Most common in adults

D. Non selective proteinuria

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	2
Question 4	4
Question 5	1
Question 6	1
Question 7	1
Question 8	2
Question 9	1
Question 10	1
Question 11	1
Question 12	2
Question 13	2
Question 14	2
Question 15	1
Question 16	1
Question 17	1
Question 18	2
Question 19	2
Question 20	1

Solution for Question 1:

Correct Option B - Allergic interstitial nephritis:

- The presence of eosinophilia along with WBC casts in the urine of a patient after consuming a course of antibiotics indicates a potential diagnosis of allergic interstitial nephritis (AIN).

Incorrect Options:

Options A, B, and C do not exhibit eosinophilic casts in the urine.

Solution for Question 2:

Correct Option A - Wilm's tumor:

- This is the correct answer. Wilm's tumor, also known as nephroblastoma, is a rare kidney cancer that primarily affects children. The symptoms described, including a large, firm, non-mobile mass in the flank region, are consistent with this diagnosis. Wilm's tumor is the most common kidney cancer in children, and most cases are diagnosed before the age of 5.

Incorrect Options:

Option B - Angiomyolipoma: Angiomyolipomas are benign tumors that typically occur in the kidney. While they can cause a palpable mass, they are much more common in adults and are very rare in children.

Option C - Renal cell carcinoma: Renal cell carcinoma is the most common type of kidney cancer in adults, but it is very rare in children. The child's age and the characteristics of the mass make this an unlikely diagnosis.

Option D - Neuroblastoma: Neuroblastoma is a cancer that often starts in the adrenal glands, which are on top of the kidneys. It can cause a mass in the abdomen, but it is usually associated with other symptoms such as fever, bone pain, and unexplained weight loss.

Solution for Question 3:

Correct Option B - 40 ml/min/1.73 m2 BSA:

- The Schwartz equation is commonly used to calculate a pediatric patient's estimated glomerular filtration rate (eGFR).
- The Schwartz equation estimates renal function based on the child's serum creatinine level, body weight, and height.

Patient details:

- Age: 4 years
- Body weight: 15 kg
- Height: 100 cm
- Blood urea: 100 mg/dL
- Serum creatinine: 1 mg/dL

Schwartz equation for eGFR:

- eGFR (ml/min/1.73 m^2) = (0.413 x Height in cm) / Serum creatinine (mg/dL)
- Calculating eGFR:
- eGFR = (0.413 x 100) / 1
- eGFR = 41.3 ml/min/1.73 m^2

Thus, 40 ml/min/1.73 m^2 BSA: This option is very close to the calculated eGFR of 41.3 ml/min/1.73 m^2. Given the small margin of difference, this is the closest calculated eGFR for the patient. Hence, option b is the correct answer.

Incorrect Options: Option A, C and D are incorrect.

Solution for Question 4:

Correct Option: D

The most appropriate treatment for the child described with signs of dehydration would be to consider severe dehydration, start intravenous (IV) fluids, administer IV antibiotics, and refer to a higher center.

The given history of loose stools, irritability, hasty drinking of water, sunken eyes, and delayed skin pinch test indicates signs of dehydration in the child. In cases of severe dehydration, oral rehydration alone may not be sufficient, and intravenous fluids are necessary to rapidly rehydrate the child.

The options provided can be evaluated as follows:

Option A: Administer the first dose of IV antibiotic and immediately refer to a higher center: While antibiotic therapy may be indicated in some cases of diarrhea, the priority in this scenario is to address the dehydration and provide appropriate fluid replacement. Referral to a higher center is necessary for the management of severe dehydration.

Option B: Give oral fluids and ask the mother to continue the same and visit again the next day: Given the signs of dehydration mentioned, providing oral fluids alone may not be adequate to correct the dehydration rapidly. Prompt intervention with IV fluids is required in severe dehydration cases.

Option C: Give zinc supplementation and oral rehydration solution only and ask the mother to come back if some danger signs develop: Zinc supplementation and oral rehydration solution are essential components of managing diarrheal illnesses, but in cases of severe dehydration, intravenous fluid therapy is necessary.

Option D: Consider severe dehydration, start intravenous (IV) fluids, IV antibiotics, and refer to a higher center: This option is the most appropriate choice based on the description provided. Severe dehydration requires prompt intravenous fluid administration to rehydrate the child quickly. IV antibiotics may be considered if there are signs of systemic infection. Referral to a higher center ensures proper management and monitoring of the child's condition.

It is important to note that the specific management of a dehydrated child may vary depending on clinical guidelines and the healthcare setting. However, in cases of severe dehydration, rapid IV fluid administration and appropriate referral for further management are essential to prevent complications and ensure proper care for the child.

Solution for Question 5:

Correct Option A:

- Bladder exstrophy is a disorder that affects newborns and children from birth and is often identified soon after birth. The bladder forms outside of the body as a result of an improper development of the pelvic bones and lower abdominal muscles that characterizes thedisorder. Therefore, the condition is Bladder exstrophy.

Incorrect Options:

Option B. Omphalocele: The birth abnormality known as omphalocele causes the baby's abdominal organs to protrude through the belly button.

Option C. Persistent vitellointestinal duct: Persistent Vitellointestinal Duct is a congenital abnormality that occurs when a section of the vitelline duct does not completely dissolve during embryonic development, leading to a number of difficulties.

Option D. Gastroschisis: A birth condition known as gastroschisis causes the baby's intestines to protrude through a hole adjacent to the belly button, typically to the right.

Solution for Question 6:

Correct option A- 4-6:

- Steroid-resistant nephrotic syndrome (SRNS) is defined as nephrotic syndrome resistant to steroid therapy, defined by the absence of complete remission after four weeks of daily prednisone therapy at a dose of 60 mg/m2 per day.

Incorrect Options - B, C, and D are incorrect.

Solution for Question 7:

Correct Option A - Prednisolone:

- The above clinical features are suggestive of nephrotic syndrome.
- The initial treatment of choice for nephrotic syndrome is Prednisolone.

Incorrect Options - B, C, and D are not the treatment for nephrotic syndrome.

Solution for Question 8:

Correct Option B - GABRA-1:

- Juvenile myoclonic epilepsy (JME) is a type of epilepsy that typically begins in adolescence and is characterized by myoclonic seizures, generalized tonic-clonic seizures, and sometimes absence seizures.
- GABRA-1, refers to the gamma-aminobutyric acid receptor alpha-1 subunit gene.
- Mutations or variations in this gene have been associated with an increased risk of developing JME.

Incorrect Options - A, C, and D are not associated with juvenile myoclonic epilepsy.

Solution for Question 9:

Correct Option A - PSGN:

- The patient's presentation with gross hematuria (cola colored urine), proteinuria, and edema is suggestive of nephritic syndrome.
- PSGN is an immune-mediated glomerular disease that occurs as a result of a preceding streptococcal infection, such as a throat infection or impetigo. Deposition of immune complexes with streptococcal antigen and complement activation. Antibodies against streptococcus cross react with glomerular antigens due to molecular mimicry.
- Deposition of immune complexes with streptococcal antigen and complement activation.
- Antibodies against streptococcus cross react with glomerular antigens due to molecular mimicry.
- It typically presents with features of nephritic syndrome, including brown-coloured urine, oliguria, facial and pedal edema, and hypertension.
- Dx: proteinuria, RBC and RBC casts, decreased levels of C3, increased ASO titer following pharyngitis, increased Anti DNase B following pyoderma

- Tx: mainly supportive, early antibiotics, diuretics, salt and water restriction
- Deposition of immune complexes with streptococcal antigen and complement activation.
- Antibodies against streptococcus cross react with glomerular antigens due to molecular mimicry.

Incorrect Options - Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 10:

Correct Option A - Alport Syndrome:

- The patient’s presentation with sensorineural hearing loss, eye defects, hematuria, and a history of maternal uncle with end-stage chronic kidney disease (CKD) undergoing dialysis is suggestive of Alport syndrome.
- Alport syndrome is a genetic disorder characterized by kidney disease, hearing loss, and eye abnormalities.
- It is caused by mutations in genes encoding type IV collagen, which is an important component of the basement membrane in the kidney, inner ear, and eyes.
- The classic triad of symptoms includes hematuria, sensorineural hearing loss, and ocular abnormalities.
- In some cases, renal biopsy may appear normal on light microscopy, but electron microscopy or immunohistochemistry may reveal characteristic findings.

Incorrect Options:

Option B: Goodpasture syndrome:

- Goodpasture syndrome is an autoimmune disorder characterized by the presence of antibodies directed against the basement membrane of the kidneys and lungs.
- It typically presents with pulmonary and renal involvement, including pulmonary hemorrhage and glomerulonephritis.

Option C: IgA nephropathy:

- IgA nephropathy, also known as Berger's disease, is a kidney disorder characterized by the deposition of immunoglobulin A (IgA) in the glomeruli.

- It commonly presents with episodes of hematuria and can progress to chronic kidney disease.

Option D: PSGN:

- PSGN is a type of glomerulonephritis that occurs after an infection with certain strains of streptococcus bacteria.
- It is characterized by immune complex deposition in the glomeruli, leading to inflammation and kidney damage.

Solution for Question 11:

Correct Option A - Ectopic ureter:

- The patient’s presentation with constant urine dribbling, despite voiding normally, and has never been completely dry since birth is suggestive of ectopic ureter.
- Ectopic ureter is a condition where the ureter instead of terminating into the urinary bladder terminates at a different site.
- This abnormal connection causes urine to continuously bypass the bladder, resulting in continuous dribbling of urine.
- The continuous leakage occurs because the urine is not being properly stored in the bladder but instead flows directly out through the ectopic ureter.

Incorrect Options:

Option B: Ureterocele:

- Ureterocele is a condition where the lower end of the ureter balloons out and forms a cyst-like structure within the bladder.
- This can cause urinary obstruction and lead to symptoms such as urinary frequency, urgency, or recurrent urinary tract infections.

Option C: Congenital megaureter:

- Congenital megaureter refers to the dilation or enlargement of the ureter.
- It can be caused by an obstruction or a functional abnormality of the ureter.
- Congenital megaureter can lead to urinary reflux and urinary tract infections.

Option D: Vesicoureteric reflux:

- Vesicoureteric reflux is a condition where urine flows backward from the bladder up into the ureters.
- This can increase the risk of urinary tract infections.

Solution for Question 12:

Correct Option B - Micturating Cystourethrography:

- Micturating cystourethrography (MCU) is a specialized imaging study used to evaluate the anatomy and function of the urinary bladder and urethra during urination.
- This procedure involves the use of a contrast agent that is inserted into the bladder via a catheter, and X-ray images are taken as the child urinates.
- MCU helps identify vesicoureteral reflux (VUR), a condition where urine flows backward from the bladder into the ureters and possibly the kidneys.
- VUR is a common cause of recurrent UTIs in children.
- MCU can also detect other abnormalities, such as urethral strictures or bladder abnormalities, which may contribute to poor urinary stream and dribbling.

Incorrect Options - A, C, and D are incorrect. Refer to the explanation of the correct answer.

Option D - DMSA scan: A dimercaptosuccinic acid (DMSA) scan is primarily used to detect and evaluate renal scarring.

Solution for Question 13:

Correct Option B - All killed vaccines under NIS can be given:

- All killed vaccines under NIS (National Immunization Schedule) can be given to a child suffering from steroid-dependent nephrotic syndrome.
- Children with steroid-dependent nephrotic syndrome have compromised immune systems, making them more susceptible to infections.

Incorrect Options:

Option A - All live vaccines under NIS can be given as per schedule:

- Children with compromised immune systems, such as those on long-term steroid therapy, are generally advised to avoid live vaccines.
- Live vaccines contain weakened forms of the virus that can potentially cause the disease in individuals with weakened immune responses.
- Live vaccines under the National Immunization Schedule (NIS) should not be given as per the regular schedule to a child with steroid-dependent nephrotic syndrome.

Option C: Only pneumococcal vaccine should be given to the child in the acute phase:

- While the pneumococcal vaccine is important and can be given to a child with steroid-dependent nephrotic syndrome, it is not the only vaccine that should be administered in the acute phase.
- Other killed vaccines under the NIS that are recommended for children can also be given in accordance with the standard immunization schedule.

Option D: Live vaccines are contraindicated in the child's sibling:

- The vaccination status of a child's sibling does not impact the decision to administer vaccines to a child with this condition.

Solution for Question 14:

Correct Option B - Post-streptococcal glomerulonephritis:

- The patient’s presentation with dark-colored urine (hematuria), reduced urine output, high blood pressure, and swollen face for three days with a history of recent throat infection two weeks ago and microscopic examination showing an enlarged and pale glomeruli, along with epithelial humps is suggestive of post-streptococcal glomerulonephritis.
- Most common cause of nephritic syndrome in children is post-streptococcal glomerulonephritis.)

Hematuria in a child	
Number of days after upper respiratory infection	Probable underlying diagnosis
1 – 2 days	IgA nephropathy

1 – 2 weeks	PSGN

Incorrect Options - A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 15:

Correct Option A - Nephrotic Syndrome:

- Nephrotic syndrome is a condition characterized by the presence of significant proteinuria, hypoalbuminemia, edema, and hyperlipidemia.

Incorrect Options - B, C, and D are incorrect.

Solution for Question 16:

Correct option A - Single umbilical artery:

Unilateral renal agenesis:

- Absent kidney development on one side
- Secondary to defect of wolffian duct / ureteric bud / metanephric blastema
- Increased incidence in neonates with single umbilical artery

Incorrect Options - B, C, and D are not linked to unilateral renal agenesis in an infant.

Solution for Question 17:

Correct Option A - Nephrotic Syndrome

- The patient’s presentation with generalized swelling and high cholesterol levels, protein 3+ and fat bodies observed in urine analysis is suggestive of nephrotic syndrome.
- Nephrotic syndrome is a condition characterized by the presence of significant proteinuria , hypoalbuminemia , edema , and hyperlipidemia.

Incorrect Options: Options B, C, and D are incorrect.

Option B: Nephritic syndrome: Nephritic syndrome is characterized by the presence of hematuria , hypertension, and oliguria .

Option C: Goodpasture's disease:

- Goodpasture's disease is a rare autoimmune disorder that primarily affects the kidneys and lungs.
- It is characterized by the presence of anti-glomerular basement membrane antibodies, leading to glomerulonephritis and pulmonary hemorrhage.

Option D: IgA nephropathy:

- IgA nephropathy, also known as Berger's disease, is a common form of glomerulonephritis characterized by the deposition of immunoglobulin A in the glomeruli of the kidneys.
- It often presents with recurrent episodes of hematuria, particularly following respiratory or gastrointestinal infections.

Solution for Question 18:

Correct Option B - 2, 4:

2. Bowel and bladder dysfunction increases the risk of recurrence:

- Bowel and bladder dysfunction, such as constipation or incomplete bladder emptying, can contribute to the development and recurrence of UTI.
- These dysfunctions may result in incomplete bladder emptying, urinary stasis, or urinary reflux, increasing the risk of bacterial colonization and infection.

4. Micturating cystourethrogram (MCU) is done in children with a recurrence of UTI:

- A micturating cystourethrogram (MCU) is a radiological procedure used to evaluate the structure and function of the urinary bladder and urethra.

- It involves the injection of a contrast dye into the bladder while the child is urinating, allowing visualization of the bladder and identification of any abnormalities or vesicoureteral reflux (VUR).
- In this condition, urine flows backward from the bladder into the kidneys.

Incorrect Options - A, C, and D are incorrect.

1. Most common cause of UTI is Streptococcus pneumoniae: The most common causative organism of UTIs in children is E. coli.

3. Cotrimoxazole is not given: Cotrimoxazole, a combination of sulfamethoxazole and trimethoprim, is one of the antibiotics commonly used for treating urinary tract infections in children.

Solution for Question 19:

Correct Option B - Minimal change disease:

- The patient’s presentation edema of the face which later developed into generalized edema with electron microscopy showing effacement of podocyte foot processes is suggestive of minimal change disease.
- Minimal change disease is a kidney condition that primarily affects young children and is characterized by selective proteinuria.
- Minimal change disease is characterized by the effacement of podocyte foot processes in the renal glomeruli and is seen on electron microscopy.

Incorrect Options - A, B, and D are incorrect. Refer to the explanation of the correct answer

Solution for Question 20:

Correct Option A: Electron microscopy shows effacement of podocytes:

- Minimal change disease is characterized by the effacement of podocyte foot processes in the renal glomeruli and is seen on electron microscopy.

Incorrect options:

Option B: No response to steroids: Minimal change disease is known for its excellent response to corticosteroid therapy.

Option C: Most common in adults: Minimal change disease is actually most commonly seen in children.

Option D: Non selective proteinuria:

- Minimal change disease typically presents with selective proteinuria.
- In minimal change disease, there is a selective loss of albumin in the urine due to damage to the glomerular filtration barrier.

CNS Malformations, Hydrocephalus, and Seizures in Children

1. A 6-month-old infant presents with rapid head growth, a prominent occiput, and developmental delays. An MRI is ordered which shows the finding as shown in the image. What is the most likely diagnosis?

A. Arnold-Chiari malformation Type I

B. Arnold-Chiari malformation Type II

C. Dandy-Walker malformation

D. Hydranencephaly

2. A 6-year-old child with status epilepticus has received two doses of intravenous lorazepam without seizure cessation after initial management. What is the most appropriate next step?

A. IV Fosphenytoin

B. IV Sodium Valproate

C. IV Levetiracetam

D. IV Phenobarbitone

3. A 4-year-old child presents with recurrent, prolonged seizures and developmental delay. Genetic testing reveals a mutation in the SCN1A gene. Based on this genetic finding and the clinical presentation, which of the following is most likely associated?

A. West Syndrome

B. Dravet Syndrome

C. Lennox-Gastaut Syndrome

D. Rett Syndrome

4. A 7-month-old infant presents with recurrent clusters of sudden jerking movements and has recently shown a loss of developmental milestones. An EEG reveals high-voltage, chaotic background activity with multifocal spikes. Given these findings, what is the first-line treatment for the suspected condition?

A. ACTH

B. Vigabatrin

C. Gabapentin

D. Topiramate

5. A 7-year-old girl presents with brief episodes of staring and unresponsiveness lasting a few seconds, occurring multiple times a day. Her teacher reports she often misses instructions. An EEG shows 3-Hz spike-and-slow-wave discharges precipitated by hyperventilation. What is the most likely diagnosis?

A. Myoclonic seizures

B. Typical absence seizures

C. Complex partial seizures

D. Atypical absence seizures

6. A 2-year-old boy has experienced three febrile seizures in the past 6 months, each lasting about 2-3 minutes. He has no neurological deficits and develops normally. His parents are anxious about future episodes and request preventive treatment. What is the most appropriate management approach for this child?

A. Start continuous daily phenobarbital

B. Prescribe intermittent oral clobazam during febrile illnesses

C. Recommend daily valproic acid

D. Advise only antipyretic use during febrile illnesses

7. A 2.5-year-old girl presents to the emergency department with a fever of 39.5°C (103.1°F). She experienced a generalised tonic-clonic seizure lasting 5 minutes, which stopped spontaneously. The child has no history of seizures or neurological issues and appears well after the episode. What is the most appropriate immediate management for this child?

A. Administer high-dose steroids

B. Oral antipyretics and reassurance

C. Start intravenous antiepileptics

D. Perform a lumbar puncture to rule out infection

8. A 14-month-old boy presents to the emergency department with a fever of 39°C (102.2°F) and a generalised- tonic-clonic seizure lasting 5 minutes. The seizure self-resolved, and the child returned to baseline within 10 minutes. There's no history of prior seizures or neurological issues. What is the most likely diagnosis?

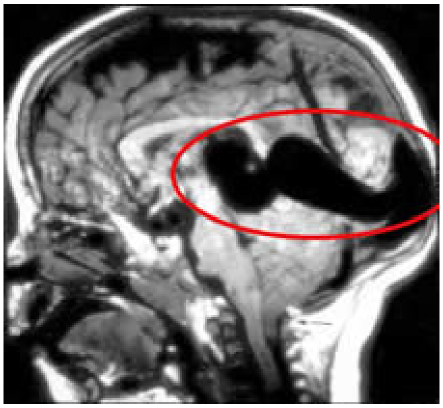
A. Meningitis

B. Simple febrile seizure

C. Complex febrile seizure

D. Febrile status epilepticus

9. A 2-month-old infant presents with signs of heart failure, hydrocephalus, and an enlarging head. Neuroimaging reveals a finding shown in the image. The infant's condition is found to be due to an arteriovenous malformation. Which vessel is primarily involved in this malformation?



A. Internal Jugular Vein

B. Middle meningeal Artery

C. Basal Vein of Rosenthal

D. Median Prosencephalic Vein

10. A 2-day-old male neonate presents with a translucent sac-like swelling in the lumbosacral region. On examination, the infant exhibits normal motor function and bladder control, and the spinal cord remains within the spinal canal. What is the most likely diagnosis?

A. Myelomeningocele

B. Meningocele

C. Spina Bifida Occulta

D. Tethered Cord Syndrome

11. An infant presented with apnea, bradycardia, fever, poor feeding, and irritability for the past 2 days, and has a history of ventriculoperitoneal (VP) shunt placement for hydrocephalus. What is the most common organism responsible for infections related to VP shunt complications?

A. Staphylococcus aureus

B. Staphylococcus hominis

C. Staphylococcus haemolyticus,

D. Staphylococcus epidermidis

12. A 6-month-old infant is diagnosed with hydrocephalus and exhibits a rapidly enlarging head circumference. The clinical team is considering treatment options to manage the condition. Which of the following treatment approaches is generally considered for long-term management of this condition?

A. Acetazolamide

B. Endoscopic third ventriculostomy (ETV)

C. Ventriculoperitoneal (VP) shunt

D. Furosemide

13. A month-old infant presents with an increasing head circumference and signs of raised intracranial pressure. Physical examination reveals an upward gaze palsy.

Cranial ultrasound shows enlarged ventricles. Which of the following conditions is the most likely cause of this infant's condition?

A. Aqueductal stenosis

B. Dandy-Walker malformation

C. Posterior fossa tumor

D. Intraventricular hemorrhage

14. A 32-year-old woman with a history of delivering a child with a neural tube defect (NTD) is planning her next pregnancy. To optimise the prevention of NTDs in this upcoming pregnancy, what is the recommended folic acid supplementation regimen for her?

A. 0.4 mg daily, starting 1 month before conception

B. 1 mg daily, starting 1 week before conception

C. 2 mg daily, starting 1 week before conception

D. 4 mg daily, starting 1 month before conception

15. A 25-year-old pregnant woman undergoes routine prenatal screening during her 16th week of pregnancy. The serum alpha-fetoprotein (AFP) level is found to be elevated. This finding suggests the presence of which of the following conditions?

A. Spina Bifida

B. Down Syndrome

C. Trisomy 18

D. Cystic Fibrosis

16. A 6-month-old female infant presents with developmental delays and frequent seizures. Neuroimaging reveals agenesis of the corpus callosum. The infant also has characteristic eye abnormalities, including distinctive chorioretinal lacunae. Based on these findings, which of the following syndromes is most commonly associated with this presentation?

A. Aicardi Syndrome

B. Tuberous Sclerosis Complex

C. Sturge Weber Syndrome

D. West Syndrome

17. A neonate is found to have a brain imaging result showing a "smooth brain" and an hourglass appearance, with an absence of normal gyri and sulci formation. Which condition does this finding most likely indicate?

A. Lissencephaly

B. Schizencephaly

C. Holoprosencephaly

D. Polymicrogyria

18. A 30-week pregnant woman undergoes an antenatal ultrasound examination. The ultrasound reveals an abnormal fetal posture where the fetus is noted to have a hyperextended neck and head, resembling an upward gaze. Based on this finding, which of the following conditions is most likely?

A. Anencephaly

B. Iniencephaly

C. Porencephaly

D. Encephalocele

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	2
Question 4	1
Question 5	2
Question 6	2
Question 7	2
Question 8	2
Question 9	4
Question 10	2
Question 11	4
Question 12	3
Question 13	1
Question 14	4
Question 15	1
Question 16	1
Question 17	1
Question 18	2

Solution for Question 1:

Correct Answer: C) Dandy-Walker malformation

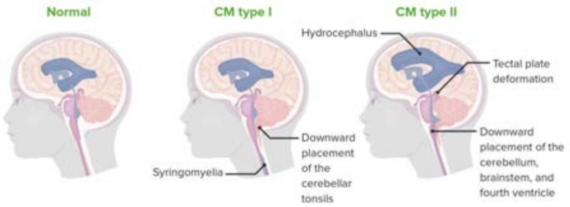
Explanation:

Rapid head growth and a prominent occiput suggest increased intracranial pressure, often associated with conditions affecting CSF flow. Imaging showing the cystic expansion of the fourth ventricle and cerebellar hypoplasia indicates Dandy-Walker malformation.

Dandy-Walker malformation: (Developmental failure (atresia of foramina of Magendie) of the fourth ventricle’s roof)

- Developmental failure of the fourth ventricle’s roof ➜ Cystic expansion of the fourth ventricle and midline cerebellar hypoplasia.
- Associated with Hydrocephalus in 90% of cases and additional anomalies such as agenesis of the cerebellar vermis and corpus callosum.
- Symptoms: Rapid head growth, prominent occiput, cerebellar ataxia, and delayed developmental milestones.
- Transillumination of the skull shows a positive result.
- Treatment: Shunting of cystic cavity.

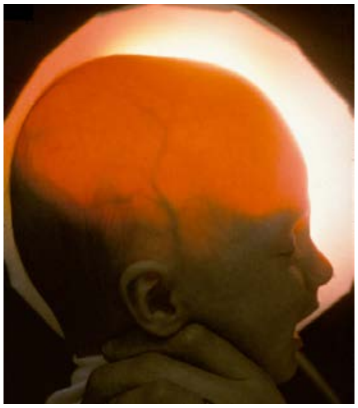
Arnold Chiari Malformation: (Option A, B)



Aspect	Chiari Malformation Type I	Chiari Malformation Type II
Abnormality	Displacement of cerebellar tonsils into the cervical canal.	Elongation of the fourth ventricle with displacement of the hindbrain.
Age	Typically, adolescence or adulthood	Often present at birth or in infancy
Associated Anomaly	Usually not associated with spinal defects	Often associated with myelomeningocele (spinal cord defect)
Symptoms	Headaches, neck pain, urinary frequency, lower extremity spasticity	Stridor, weak cry, apnea, gait abnormalities, spasticity
Imaging Findings	Tonsillar herniation into the cervical canal	Enlarged fourth ventricle, herniation of hindbrain into the cervical canal.
Treatment	Surgical decompression if symptomatic; conservative for mild cases	Surgical decompression, possibly with shunt placement for hydrocephalus.

Hydranencephaly (Option D):

- A severe condition where the cerebral hemispheres are replaced by membranous sacs, with the midbrain and brainstem relatively intact due to bilateral occlusion of the internal carotid arteries early in fetal development.
- Symptoms include irritability, poor feeding, seizures, and spastic quadriparesis.
- Imaging: Transillumination shows the absence of cerebral hemispheres.
- Treatment: Ventriculoperitoneal shunt can help manage head enlargement.



Solution for Question 2:

Correct Answer: A) IV Fosphenytoin

Explanation:

After initial treatment with lorazepam, which is typically effective for managing status epilepticus, the next step involves using additional antiepileptic drugs if seizures persist. IV Fosphenytoin is commonly used as a second-line treatment for continued seizure activity.

Management of Status Epilepticus:

Refractory status epilepsy

- Refractory status epilepticus refers to status epilepticus that has not responded to treatment, typically despite the administration of at least two antiepileptic medications, such as a benzodiazepine and another agent.

Solution for Question 3:

Correct Answer: B) Dravet Syndrome

Explanation:

The SCN1A gene mutation is most commonly associated with Dravet Syndrome, a severe form of epilepsy characterised by frequent, prolonged seizures and developmental delays.

Aspect	Lennox-Gastaut Syndrome (Option C)	West Syndrome (Option A)	Dravet Syndrome
Onset Age	2-10 years	2-12 months	Infancy
Clinical Features	Developmental delay Multiple seizure types (atypical absences, myoclonic, atonic, tonic) EEG: 1-2 Hz spike and slow waves, polyspike bursts in sleep, slow background	Infantile epileptic spasms Developmental regression Hypsarrhythmia on EEG	Initial febrile and afebrile unilateral clonic seizures, occurring every 1-2 months Seizures are prolonged, frequent, focal, and often cluster Progression to seizures with lower fevers and eventually without fever Developmental delay

Genetic Causes	-	ARX gene	SCN1A Mutation
First line	Clobazam Valproate Lamotrigine	ACTH	Benzodiazepines (e.g., Clobazam) Valproate
Alternative treatments	Ketogenic diet Levetiracetam Acetazolamide Corticosteroids IVIG.	Oral Prednisolone Vigabatrin (if associated with Tuberous sclerosis) Ketogenic diet	Ketogenic diet Stiripentol

Rett Syndrome (Option D) is characterised by mutations in the MECP2 gene, which leads to increased NMDA receptor activity and contributes to the symptoms, including neurodevelopmental regression, repetitive hand movements, motor abnormalities, seizures, and cognitive impairment. Trofinetide may be beneficial for treatment.

Solution for Question 4:

Correct Answer: A) ACTH

Explanation:

- The given clinical scenario with infantile seizures, developmental delays, and EEG abnormalities points towards West syndrome.
- ACTH is the first-line treatment for West Syndrome, as it is effective in managing infantile spasms and normalising EEG abnormalities.

West Syndrome	
Age group	Typically occurs between 2 and 12 months of age.
Genetics	ARX Gene Mutations: In males, linked to ambiguous genitalia and cortical migration abnormalities.
Types	Cryptogenic (Unknown Etiology): Normal development before onset. Symptomatic: Associated with developmental delays due to perinatal issues, brain malformations, metabolic disorders, or other causes.

Clinical Features	Infantile Epileptic Spasms: Clusters of spasms occurring mainly during drowsiness or upon waking. Developmental Regression Hypsarrhythmia on EEG: A chaotic, high-voltage background with multifocal spikes.
Management	First-Line Treatment: Hormonal therapy, primarily ACTH (80 mg/mL) Dosage: 150 units/m² daily (75 units/m² twice daily) for 2 weeks, then taper gradually. Monitoring: EEGs at 1, 2, and 4 weeks to assess response. Early intervention is crucial, as a delay beyond 3 weeks can worsen the prognosis.
Alternative Treatments	Oral Prednisolone: Lower-cost alternative (Less effective) Vigabatrin (Option B): Used as a first-line agent in West syndrome associated with tuberous sclerosis complex (TSC)or second-line if hormonal therapy fails. Side effects include retinal toxicity. Ketogenic Diet: Considered a third-line therapy.pyridoxine Other Medications: Valproate, benzodiazepines (nitrazepam, clonazepam), topiramate (Option D), lamotrigine, zonisamide, pyridoxine, and IVIG. These drugs can help reduce seizure frequency and severity but are not uniformly effective and may be used alongside ACTH or vigabatrin in cases with incomplete response.
Side Effects and Precautions	Side Effects of Hormonal Therapy: Hypertension, electrolyte imbalance, infections, hyperglycemia, and gastric ulcers. Vaccination: Live vaccines are contraindicated, and other vaccines may be ineffective during and shortly after hormonal therapy.

Gabapentin (Option D) is an anticonvulsant commonly used for neuropathic pain and partial seizures but is ineffective for infantile spasms.

Solution for Question 5:

Correct Answer: B) Typical absence seizures

Explanation:

The patient's presentation aligns with typical absence seizures, characterised by brief, multiple daily staring spells with 3-Hz spike-and-slow-wave discharges on EEG, often triggered by hyperventilation.

Types of Absent Seizures:

Feature	Typical Absence Seizures	Atypical Absence Seizures	Juvenile Absence Seizures
Age of Onset	5-8 years	Any age often associated with developmental delay	Later onset, typically in adolescence
Duration	Lasts a few seconds	Lasts longer than typical absences	Lasts a few seconds to minutes
Clinical Features	Staring, unresponsiveness, eyelid flutter, upward eye rolling	Staring, myoclonic components, tone changes	Staring may include myoclonic jerks
Frequency	Multiple times per day	Less frequent than typical absences	Occurs less frequently than typical absences
Precipitating Factors	Hyperventilation	Drowsiness	Often occurs spontaneously and may be related to hyperventilation
EEG Findings	3-Hz spike-and-slow-wave discharges	1-2 Hz spike-and-slow-wave discharges	4-6 Hz spike-and-slow-wave and polyspike-and-slow-wave discharges
Postictal State	None	May have a brief postictal period	None
Variants	Jeavons syndrome (eyelid myoclonia), Myoclonic absences (limb jerks)	Myoclonic absence with irregular features	Associated with juvenile myoclonic epilepsy

Myoclonic seizures (Option A) are characterised by sudden, brief, shock-like jerks of muscles, typically without loss of consciousness, with polyspike-and-slow-wave discharges on EEG (faster than 3 Hz).

Complex partial seizures (Option C) involve altered consciousness, longer duration (30 seconds to 2 minutes), often with an aura and postictal confusion. EEG shows focal spikes or sharp waves.

Solution for Question 6:

Correct Answer: B) Prescribe intermittent oral clobazam during febrile illnesses

Explanation:

- Intermittent oral diazepam is appropriate for children with frequent febrile seizures (≥3 in 6 months or ≥4 in one year). It balances seizure prevention with avoiding continuous medication.
- The dose is 0.8-1 mg/kg/day in 2 divided doses and should be started at the first sign of any fever and continued for the first 48 hours.
- For high-risk cases, start intermittent oral diazepam at the first sign of fever and continue for 3 days to manage seizure risk effectively without continuous medication.

Prevention of Recurrent Febrile Seizures:

Category	Intermittent Prophylaxis	Continuous Prophylaxis
Usage	Medications used during periods of fever to prevent seizures and are not used continuously.	Regular, ongoing use of antiepileptic drugs to prevent seizures regardless of fever.
Indications	Frequent recurrences (≥3 in 6 months or ≥4 in one year); High parental anxiety	Complex febrile seizures (focal, prolonged) Recurrences despite intermittent prophylaxis Neurological or developmental issues Family history of epilepsy
Medications Used	Oral Diazepam Rectal Diazepam Oral Clobazam Antipyretics (acetaminophen, ibuprofen)	Valproic Acid (Option C) Phenobarbital (Option A)
Dosage	Diazepam (Oral, rectal) Clobazam Antipyretics (for fever control)	Valproic Acid Phenobarbital

Advise only antipyretic use during febrile illnesses (Option D): Antipyretics manage fever but do not prevent febrile seizures, making them insufficient for seizure prevention in this child's case.

Solution for Question 7:

Correct Answer: B) Oral antipyretics and reassurance

Explanation:

For a child presenting with febrile seizures lasting <5 minutes and no neurological symptoms, the primary management is to administer oral antipyretics (e.g., acetaminophen or ibuprofen) and provide reassurance, as these seizures are typically self-limiting and not indicative of a serious underlying condition.

Management of Febrile Seizures:

- Initial Assessment: Most febrile seizures resolve spontaneously. If the seizure has ended and the child is returning to baseline, active treatment with benzodiazepines is not needed.
- Use antipyretics to manage fever.
- Seizures lasting >5 Minutes: Administer intravenous benzodiazepines (diazepam 0.1 to 0.2 mg/kg or lorazepam 0.05 to 0.1 mg/kg) if the seizure persists, with additional doses as needed.
- Monitoring: Closely monitor respiratory and circulatory status. Prepare for advanced airway management if ventilation is compromised.

Administer high-dose steroids (Option A): Not used for simple febrile seizures. Steroids are reserved for specific conditions like severe inflammatory or autoimmune disorders.

Start intravenous antiepileptics (Option C): Reserved for seizures lasting for >5 minutes or if the seizure is prolonged and does not stop spontaneously.

Perform a lumbar puncture to rule out infection (Option D): It is considered only if there are signs of meningitis or infection.

Solution for Question 8:

Correct Answer: B) Simple febrile seizure

Explanation:

The given scenario is consistent with simple febrile seizures characterised by a short duration (less than 15 minutes), a single generalised tonic-clonic episode, and normal neurological development.

Criteria for Diagnosis of Simple Febrile Seizure:

- Age: 6-60 months (peak incidence at 12-18 months).
- Fever: Associated with a temperature ≥38°C (100.4°F).
- Seizure Characteristics: Generalized, usually tonic-clonic.
- Duration: Seizure lasts ≤15 minutes.
- Recurrence: No recurrence within 24 hours.
- Postictal State: Rapid return to baseline consciousness without neurological deficit.
- Exclusion of Other Causes: No signs of CNS infection, metabolic imbalance, or history of afebrile seizures.

Risk factors for febrile seizures:

- Genetic predisposition (family history)
- Age (peak 12-18 months)
- The rapid rise in body temperature

Meningitis (Option A) often present with signs such as neck stiffness, photophobia, or altered mental status. The child's rapid return to baseline and absence of additional symptoms from the given scenario are not consistent with meningitis.

Simple Febrile Seizure (Option B)	Complex Febrile Seizure (Option C)	Febrile Status Epilepticus (Option C)
Generalised tonic-clonic seizure. Lasts ≤15 minutes. No recurrence within 24 hours. Quick recovery to baseline without deficit.	Lasts >15 minutes. It may be focal in nature. It may recur within 24 hours.	Seizure lasts >30 minutes. Prolonged seizure with significant risk.
Common between 6-60 months, benign, and does not require extensive workup.	Higher risk of recurrence, and it requires further evaluation.	Requires urgent medical management due to the risk of complications.

Solution for Question 9:

Correct Answer: D) Median Prosencephalic Vein

Explanation:

The given clinical scenario with signs of heart failure, hydrocephalus, and radiologic finding of abnormally large venous structure points toward a vein of Galen malformation in which the

Median prosencephalic vein is primarily involved, which is the forerunner of the vein of Galen.

Vein of Galen Malformation:

- Congenital arteriovenous malformation (AVM) in the brain, in which the median prosencephalic vein forms an abnormal connection with multiple feeding arteries, causing direct high-flow shunting of arterial blood into the venous system.
- This results in a significant dilation of the Vein of Galen, leading to hydrocephalus, cardiac failure, and other complications due to excessive volume and pressure overload.
- Management: Medical Management: Symptomatic treatment for heart failure and hydrocephalus. Interventional Treatment: Endovascular Embolization is the primary treatment approach is to occlude abnormal vessels and reduce blood flow to the malformation.
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- Interventional Treatment: Endovascular Embolization is the primary treatment approach is to occlude abnormal vessels and reduce blood flow to the malformation.
- Medical Management: Symptomatic treatment for heart failure and hydrocephalus.
- Interventional Treatment: Endovascular Embolization is the primary treatment approach is to occlude abnormal vessels and reduce blood flow to the malformation.

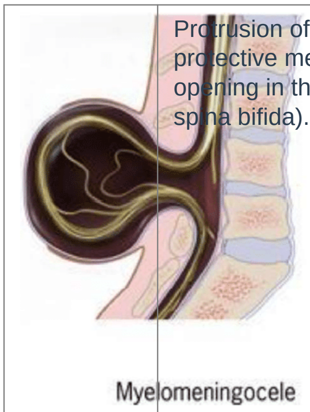
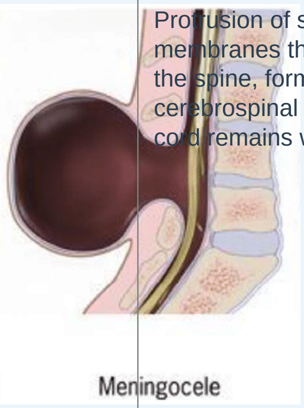
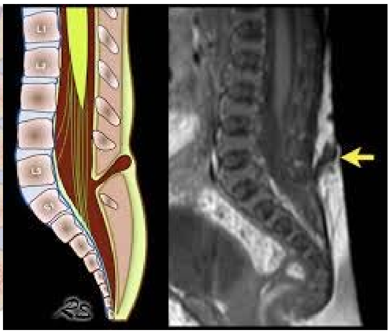
Solution for Question 10:

Correct Answer: B) Meningocele

Explanation:

- The given scenario is most likely Meningocele, a neural tube defect that involves herniation of the meninges through a vertebral defect, forming a fluid-filled sac.
- The spinal cord remains in its normal position, typically preserving motor and bladder control.

Caudal neural tube defects		
Defect	Description	Associations

 ... (content truncated)	Protrusion of the spinal cord and its protective membranes through an opening in the spine (a form of spina bifida).	Hydrocephalus, Chiari malformation Significant neurological impairments, bladder and bowel dysfunction.
 ... (content truncated)	Protrusion of spinal cord membranes through an opening in the spine, forming a sac filled with cerebrospinal fluid, while the spinal cord remains within the spinal canal.	It may have minimal neurological symptoms compared to myelomeningocele Bladder and bowel function is usually preserved.
 ... (content truncated)	Defect in the vertebrae without protrusion of the spinal cord or meninges. Often discovered incidentally. ... (content truncated)	Often asymptomatic, occasionally associated with tethered cord syndrome or skin abnormalities (e.g., dimples, hair tufts).

Tethered cord syndrome (Option D) is characterised by abnormal attachment of the spinal cord, causing progressive neurological symptoms over time, not a congenital sac-like swelling at birth.

Solution for Question 11:

Correct Answer: D) Staphylococcus epidermidis

Explanation:

The presented scenario is indicative of Shunt-related Meningitis, which is most frequently caused by Staphylococcus epidermidis.

- Shunt meningitis, caused by S. epidermidis, is the most common complication of the CSF shunts.
- Diagnosed by “Shunt tapping” (direct aspiration of the shunt).
- Management: Antibiotic therapy and shunt removal.

Complications of CSF Shunt

- Shunt occlusion Manifesting as headache, papilledema, vomiting, altered mental status.
- Manifesting as headache, papilledema, vomiting, altered mental status.
- Bacterial infections, caused by coagulase-negative staphylococci (CoNS), most commonly by S. epidermidis.
- This may result in the following: Meningitis (fever, headache, irritability) Peritonitis (fever, abdominal tenderness, distention, pain,) Wound infection (erythema, swelling, discharge, or dehiscence along the shunt tract and most often occurs within days to weeks of the surgical procedure)
- Meningitis (fever, headache, irritability)
- Peritonitis (fever, abdominal tenderness, distention, pain,)
- Wound infection (erythema, swelling, discharge, or dehiscence along the shunt tract and most often occurs within days to weeks of the surgical procedure)
- Manifesting as headache, papilledema, vomiting, altered mental status.
- Meningitis (fever, headache, irritability)
- Peritonitis (fever, abdominal tenderness, distention, pain,)
- Wound infection (erythema, swelling, discharge, or dehiscence along the shunt tract and most often occurs within days to weeks of the surgical procedure)

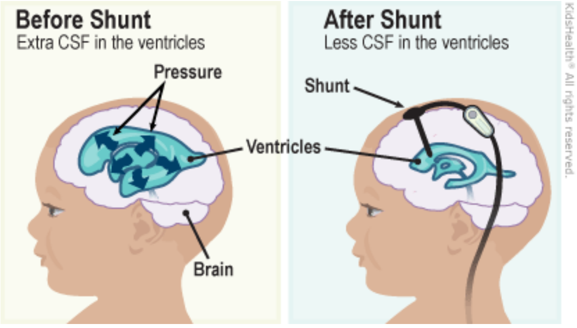
Staphylococcus aureus (Option A), Staphylococcus hominis (Option B), Staphylococcus haemolyticus (Option C) are also widely distributed on the skin, but are less often significant causes of nosocomial infection as compared to S. epidermidis.

Solution for Question 12:

Correct Answer: C) Ventriculoperitoneal (VP) shunt

Explanation:

A Ventriculoperitoneal (VP) shunt is generally considered for long-term management of hydrocephalus. It is implanted surgically to divert excess cerebrospinal fluid (CSF) from the ventricles of the brain to the peritoneal cavity, where it can be absorbed by the body.



Management of Hydrocephalus:

- Medical: Acetazolamide and furosemide can temporarily reduce CSF production but are not effective long-term. (Option A, D)
- Surgical: (CSF Diversion) Ventriculoperitoneal Shunt: Valved tubing from the ventricle to the peritoneal cavity. Endoscopic Third Ventriculostomy: It is a useful treatment for certain cases of hydrocephalus, especially when there is an obstructive pattern that can be bypassed surgically and not always suitable for all patients and is considered an alternative to shunt placement (Option B)
- Ventriculoperitoneal Shunt: Valved tubing from the ventricle to the peritoneal cavity.
- Endoscopic Third Ventriculostomy: It is a useful treatment for certain cases of hydrocephalus, especially when there is an obstructive pattern that can be bypassed surgically and not always suitable for all patients and is considered an alternative to shunt placement (Option B)
- Ventriculoperitoneal Shunt: Valved tubing from the ventricle to the peritoneal cavity.

- Endoscopic Third Ventriculostomy: It is a useful treatment for certain cases of hydrocephalus, especially when there is an obstructive pattern that can be bypassed surgically and not always suitable for all patients and is considered an alternative to shunt placement (Option B)
- Intrauterine Treatment: Results are generally poor except in cases associated with fetal meningomyelocele.

Complications: Shunt occlusion (headache, papilledema, vomiting, altered mental status) and bacterial infection leading to meningitis/peritonitis (fever, headache, meningismus), typically caused by Staphylococcus epidermidis.

Solution for Question 13:

Correct Answer: A) Aqueductal stenosis

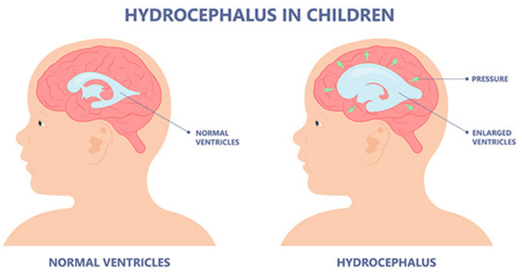
Explanation:

Aqueductal stenosis is a common congenital cause of obstructive hydrocephalus in infants, leading to increased head circumference and elevated intracranial pressure.

The upward gaze palsy, due to pressure on the quadrigeminal plate, and the finding of enlarged ventricles on ultrasound are indicative of this condition.

Hydrocephalus:

- Hydrocephalus is the abnormal accumulation of cerebrospinal fluid (CSF) in the brain, leading to increased intracranial pressure and, in infants before sutural fusion, an enlarged head.



- In normal conditions, the cerebrospinal fluid (CSF) production rate is approximately 20 mL per hour, with a volume of about 50 mL in infants and 150 mL in adults.
- Types of Hydrocephalus:

Obstructive / Non-Communicating	Non-Obstructive / Communicating
Blockage in the ventricular system.	Excess production or defective absorption of CSF due to non-obstructive causes. (Choroid plexus papilloma, meningeal malignancies)

- Causes:

Congenital	Acquired
Aqueductal stenosis (Most Common) Dandy-Walker malformation Arnold-Chiari II malformation Intrauterine infections (e.g., toxoplasmosis, rubella) Malformation of the vein of Galen	Intraventricular hemorrhage Meningitis (bacterial, tuberculous, mumps) Tumours (especially in the posterior fossa) Posterior fossa subdural collection

- Clinical Features: Before Sutural Fusion: Increased head size, bulging anterior fontanel, dilated scalp veins, irritability, lethargy, and poor appetite.
- Before Sutural Fusion: Increased head size, bulging anterior fontanel, dilated scalp veins, irritability, lethargy, and poor appetite.

- Before Sutural Fusion: Increased head size, bulging anterior fontanel, dilated scalp veins, irritability, lethargy, and poor appetite.

- After Sutural Fusion: Symptoms of raised intracranial pressure
- Signs: MacEwen Sign: "Crackpot" sound on skull percussion indicates sutural separation. Setting Sun Sign: Upward gaze palsy due to pressure on the quadrigeminal plate.
- MacEwen Sign: "Crackpot" sound on skull percussion indicates sutural separation.
- Setting Sun Sign: Upward gaze palsy due to pressure on the quadrigeminal plate.

- MacEwen Sign: "Crackpot" sound on skull percussion indicates sutural separation.
- Setting Sun Sign: Upward gaze palsy due to pressure on the quadrigeminal plate.

Solution for Question 14:

Correct Answer: D) 4 mg daily, starting 1 month before conception

Explanation:

For women with a previous child affected by an NTD, the recommended folic acid supplementation is 4 mg daily, beginning at least 1 month before conception. This higher dose is crucial for reducing the risk of recurrence in future pregnancies.

Neural Tube Defects:

- NTDs are the most common congenital CNS anomalies caused by the failure of the neural tube to close between the 3rd and 4th weeks of development.
- Risk Factors: Hyperthermia, drugs like Valproic acid, Trimethoprim and anticonvulsants (carbamazepine, phenytoin, phenobarbital), malnutrition, low folate levels, chemicals, maternal obesity, diabetes, and radiation exposure before conception.

Detection and Screening:

- Biochemical Markers: Elevated α-fetoprotein (AFP) and acetylcholinesterase in amniotic fluid indicate NTDs.

Prevention:

- Ideal Range: Red blood cell folate levels of 900-1,000 nmol/L are associated with a reduced risk of NTDs.

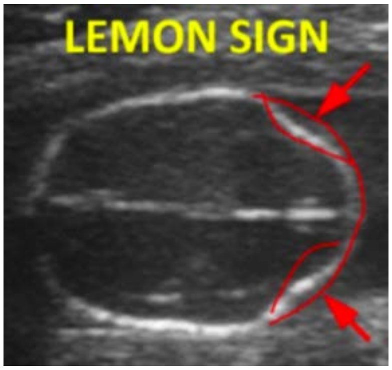
Category	Dose
General Recommendation (All women of reproductive age)	0.4 mg of folic acid daily, starting 1 month before conception.
High-Risk Women (previous child affected by an NTD)	4 mg of folic acid daily, beginning 1 month before conception.
Women on antiepileptics (valproate, carbamazepine)	1-5 mg of folic acid daily in the preconception period.

Solution for Question 15:

Correct Answer: A) Spina Bifida

Explanation:

The elevation of AFP in maternal serum is a significant marker for neural tube defects such as spina bifida. In contrast, conditions like Down syndrome and trisomy 18 are associated with lower AFP levels, and cystic fibrosis is not typically detected through AFP levels.

Markers of Neural tube defects	
Alpha-fetoprotein (AFP)	Elevated levels in maternal serum and amniotic fluid can indicate the presence of an NTD, as fetal substances leak into the amniotic fluid due to neural tube openings. AFP is commonly measured during the 16th-18th weeks of pregnancy.
Acetylcholinesterase	Elevated levels in amniotic fluid can be a marker for NTDs, as it may be released into the amniotic fluid when there is a neural tube defect.
Ultrasound Findings	 ... (content truncated)

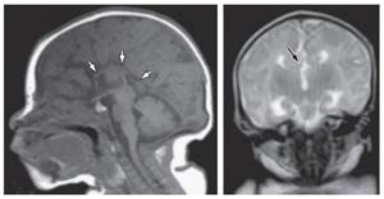
Solution for Question 16:

Correct Answer: A) Aicardi Syndrome

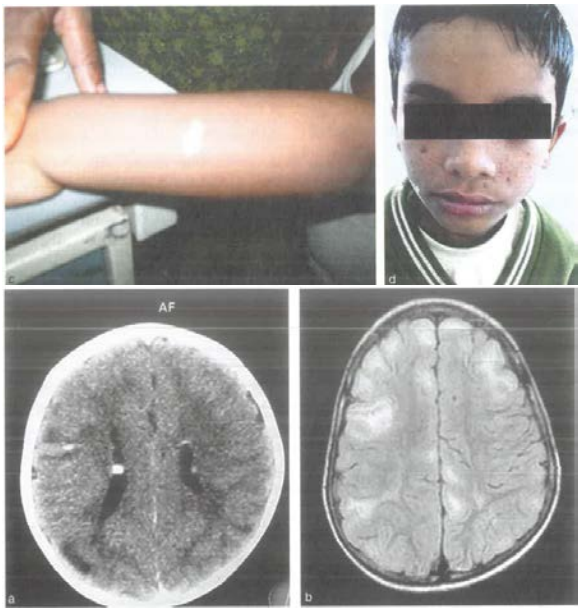
Explanation:

The given scenario is more indicative of Aicardi syndrome characterised by agenesis of the corpus callosum, distinctive retinal lesions (chorioretinal lacunae), and developmental issues, including infantile spasms.

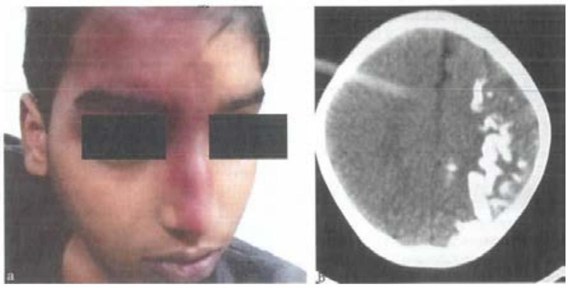
Aicardi Syndrome	

Definition	A complex disorder affecting multiple body systems, often involving the brain, eyes, and other organs.
Genetics	It is an X-linked dominant condition predominantly seen in females since males with a mutation on their single X chromosome are typically non-viable during fetal life.
Characteristic features	 ... (content truncated)
Clinical Presentation	Seizures: Infantile spasms are common and often resistant to anticonvulsant medications. EEG Abnormalities: Independent activity in both hemispheres and a pattern called hemi hypsarrhythmia. Severe Intellectual Disability: Significant cognitive and adaptive limitations. Vertebral Abnormalities: Issues such as fused or hemivertebrae. Retinal Abnormalities: Additional abnormalities like coloboma of the optic disc.

Tuberous Sclerosis Complex (Option B) is an autosomal dominant disorder characterized by a range of symptoms including cns manifestations like seizures (often epileptic spasms), and skin manifestations such as ash leaf macules. Brain imaging may reveal subependymal nodules on CT and cortical tubers on MRI.



Sturge Weber Syndrome (Option C) is characterized by a facial nevus or 'port wine stain' in the distribution of the first branch of the trigeminal nerve, along with ipsilateral 'railroad pattern' intracranial calcifications and glaucoma.



West Syndrome (Option D) is the most common epileptic encephalopathy in infancy, characterized by a triad of symptoms: infantile epileptic spasms (commonly occur in clusters), developmental regression, and a distinctive EEG pattern known as hypsarrhythmia (high voltage chaotic multifocal spike waves).

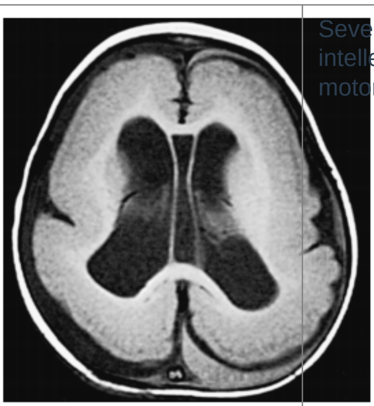
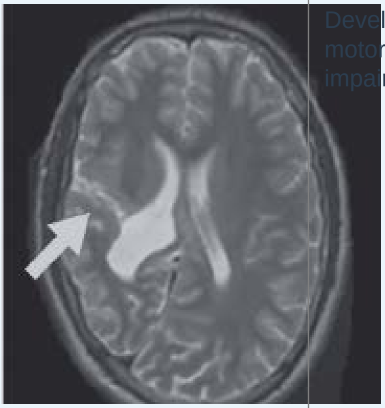
Solution for Question 17:

Correct Answer: A) Lissencephaly

Explanation:

Lissencephaly is characterised by a smooth brain surface caused by the absence of normal gyri and sulci, and hourglass appearance due to narrowing of sylvian fissure, resulting from abnormal neuronal migration during brain development.

Neuronal migration disorders		
Disorder	Radiological features	Clinical Features

Lissencephaly	 ... (content truncated)	Severe developmental delays, intellectual disability, seizures, motor impairments.
Polymicrogyria (Option D)	 ... (content truncated)	Developmental delays, seizures, motor problems, cognitive impairments.
Schizencephaly (Option B)	 ... (content truncated)	Developmental delays, seizures, motor abnormalities, potential hydrocephalus.

Heterotopia	 ... (content truncated)	Seizures, developmental delays, motor impairments.

Holoprosencephaly (Option C) is a condition where the forebrain fails to split into two hemispheres, leading to various brain malformations and facial abnormalities.

Solution for Question 18:

Correct Answer: B) Iniencephaly

Explanation:

- Iniencephaly is a rare and severe neural tube defect characterised by extreme retroflexion of the head (hyperextension) and severe defects in the spine, particularly the cervical vertebrae and occipital bone.
- Clinical Features: Hyperextension of the head ("Stargazer posture"), short and deformed spine, often incompatible with life.
- It is diagnosed prenatally through: Maternal Serum Screening: Elevated alpha-fetoprotein (AFP) levels may indicate a neural tube defect. Ultrasound: Confirms the diagnosis by revealing craniovertebral junction anomalies, including a malformed or absent occipital bone and associated brain and spine abnormalities.
- Maternal Serum Screening: Elevated alpha-fetoprotein (AFP) levels may indicate a neural tube defect.
- Ultrasound: Confirms the diagnosis by revealing craniovertebral junction anomalies, including a malformed or absent occipital bone and associated brain and spine abnormalities.
- Maternal Serum Screening: Elevated alpha-fetoprotein (AFP) levels may indicate a neural tube defect.

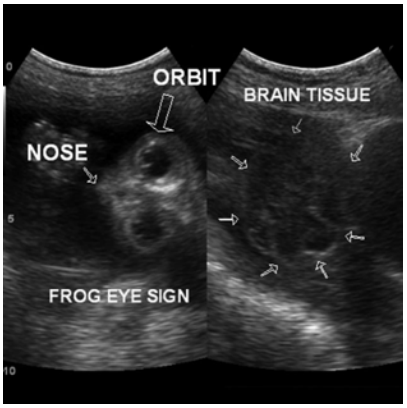
- Ultrasound: Confirms the diagnosis by revealing craniovertebral junction anomalies, including a malformed or absent occipital bone and associated brain and spine abnormalities.

Stargaze posture with meningocele of expelled fetus.

Stargaze posture -abnormal hyperextension of head over cervical spine.

Anencephaly (Option A) is a severe neural tube defect characterized by the partial or complete absence of the brain and cranial vault, resulting from the failure of neural tube closure during early fetal development. During labor, it may present with face presentation or shoulder dystocia.

Prenatal ultrasound shows the absence of the cranial vault, exposed brain tissue, and the "frog eye sign," characterised by a flattened head with prominent, bulging orbits.



Porencephaly (Option C) appears on ultrasound as cystic or cavitary lesions within the cerebral hemispheres but does not affect fetal posture or cause neck hyperextension.

Encephalocele (Option D) is a neural tube defect characterised by the protrusion of brain tissue and meninges through a skull defect, forming a sac-like mass on the fetal head.

CNS Disorders, Brain Death, and Images

1. A 3-year-old child with a history of premature birth presents with spasticity predominantly in the lower limbs. Which of the following pathology is associated with this type of presentation?

A. Parasagittal infarction

B. Periventricular leukomalacia

C. Focal infarct

D. Demyelination

2. A 6-year-old boy presents to the clinic with multiple café-au-lait spots and freckling in the groin area. His mother has a known history of Neurofibromatosis Type 1 (NF1) and similar skin findings. Which of the following additional findings would most strongly support the diagnosis of NF1 in this child? 6 or more café-au-lait spots Sphenoid dysplasia on imaging Multiple hyperpigmented areas, 2-3 mm in diameter. 2 or more hamartomas in iris detected by slit lamp examination

A. 1,2

B. 2,3

C. 2,3,4

D. 1,2,3,4

3. A 15-year-old male child presents with bilateral hearing loss, tinnitus, and unsteadiness. His mother had a history of multiple meningiomas and unilateral vestibular schwannoma. Which of the following genetic mutations is most likely associated with his condition?

A. NF1 gene mutation

B. SPRED1 gene mutation

C. NF2 gene mutation

D. SMARCB1 gene mutation

4. A 5-year-old child presents with multiple hypopigmented macules on the trunk and extremities, facial angiofibromas, and a history of epilepsy. MRI reveals cortical tubers and subependymal nodules. Genetic testing confirms a mutation in the TSC2 gene. Which of the following is most likely true regarding this patient's condition?

A. The condition is inherited in an autosomal recessive pattern.

B. The patient is at increased risk for developing renal cell carcinoma.

C. The mutation in TSC2 gene suggests a more severe clinical presentation.

D. Subependymal giant cell astrocytomas (SEGA) rarely require monitoring.

5. A 12-year-old child presents with multiple facial skin lesions as depicted below. The medical history includes developmental delay and intellectual disability. Based on the clinical presentation and history, which of the following statements is incorrect regarding the diagnosis?

A. Shagreen patches are seen in this condition.

B. Malignant cardiac rhabdomyomas are associated with this condition.

C. Facial angiofibromas may be seen.

D. Seizures and developmental delay are usually seen.

6. Which of the following statements is true about Sturge-Weber syndrome? It is due to a mutation in the GNAQ gene Port-wine stain tends to be unilateral and contralateral to the brain involvement MRI with contrast is the investigation of choice Glaucoma may be seen in these individuals.

A. I and III only

B. I, II, and III only

C. I, III, and IV only

D. All the above

7. A 7-year-old child with a known T-cell defect presents with fever, headache, and neck stiffness. The physician suspects meningitis. Given the child's immunodeficiency, which of the following organisms is the most likely cause of meningitis in this patient?

A. Streptococcus pneumoniae

B. Neisseria meningitidis

C. Haemophilus influenzae

D. Listeria monocytogenes

8. A 6-month-old infant presents with a coma and focal neurologic signs and is diagnosed with acute bacterial meningitis. Despite appropriate antibiotic therapy, the child continues to experience seizures on the fifth day of treatment. The CSF

analysis shows a leukocyte count of 200/mm³. Gram stain of CSF showed a field full of gram positive organisms . Given this clinical scenario, which of the following are poor prognostic indicators in this child?

A. Presence of coma and focal neurologic signs at presentation

B. Gram positive organisms seen in large quantities on CSF microscopy

C. Seizures occurring more than 4 days into therapy

D. All of the above

9. A 10-year-old boy from a rural area presents to the emergency department with high-grade fever, headache, vomiting, and altered sensorium for the past 2 days. On examination, the child is lethargic with neck stiffness and a positive Brudzinski sign. You suspect viral encephalitis and order a lumbar puncture. CSF sent for Japanese Encephalitis Virus (JEV) IgM ELISA, which returned positive, confirming the diagnosis of Japanese Encephalitis. Which of the following serves as an amplifier host for the Japanese Encephalitis Virus (JEV)?

A. Pigs

B. Birds

C. Culex mosquitoes

D. Humans

10. A 12-year-old obese girl presents with persistent headaches, visual disturbances, and dizziness. She has bilateral sixth cranial nerve palsy, no papilledema, normal MRI, and elevated CSF opening pressure of 285 mm CSF. Which neuroimaging finding supports Idiopathic Intracranial Hypertension (IIH) in this patient?

A. Empty sella

B. Flattening of the posterior aspect of the globe

C. Distention of the perioptic subarachnoid space without a tortuous optic nerve

D. All of the above

11. A 5-year-old child is brought to the emergency department after a severe head injury from a motor vehicle accident. Despite aggressive resuscitation efforts, the child remains unresponsive. Which of the following is not true for diagnosing brain death in this child?

A. Performing two separate clinical evaluations, separated by a 24-hour interval.

B. Ensuring the core temperature is above 32°C.

C. Administering ancillary tests.

D. Confirming the absence of all brainstem reflexes.

12. Identify the Image?

A. Anencephaly

B. Inencephaly

C. Craniorachischisis

D. Lissencephaly

13. Which type of neural tube defect is most likely depicted in the picture given below?

A. Spina Bifida Occulta

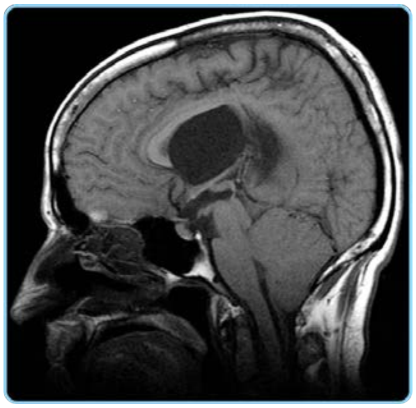
B. Meningocele

C. Myelomeningocele

D. Encephalocele

---

14. A 3-month-old infant is brought with a history of lethargy and poor feeding. The infant has also exhibited some developmental delays and hydrocephalus. On examination, the neurologist notes myelomeningocele. An MRI is shown below. Which of the following is the most likely diagnosis?



A. Arnold-Chiari Malformation Type 1

B. Arnold-Chiari Malformation Type 2

C. Dandy-Walker Malformation

D. Hydranencephaly

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	3
Question 4	3
Question 5	2
Question 6	3
Question 7	4
Question 8	4
Question 9	1
Question 10	4
Question 11	1
Question 12	1
Question 13	1
Question 14	2

Solution for Question 1:

Answer: B) Periventricular leukomalacia

Explanation:

Spasticity predominantly in the lower limbs, often associated with prematurity is suggestive of spastic diplegia, characterized by periventricular leukomalacia, which is a common finding in the MRI of these patients.

Types of Cerebral Palsy (CP) Cerebral Palsy is classified based on motor abnormalities and affected extremities:

Type	Neuropathology	Major Causes
Spastic Diplegia	Periventricular leukomalacia (Option B) Periventricular cysts/scars Enlarged ventricles	Premature birth Ischemia Infection Endocrine/metabolic issues (e.g., thyroid)
Spastic Quadriplegia	Periventricular leukomalacia Multicystic encephalomalacia	Ischemia Infection Endocrine/metabolic issues Genetic/developmental problems
Hemiplegia)	In utero/neonatal stroke Focal infarct (Option C) Cortical/subcortical damage	Thrombophilic causes Infection Genetic/developmental issues
Extrapyramidal (Athetoid, Dyskinetic)	Asphyxia (symmetric scars in putamen/thalamus) Kernicterus (scars in globus pallidus/hippocampus) Mitochondrial scarring	Asphyxia Kernicterus Mitochondrial pathology Genetic/metabolic problems

Parasagittal infarction (Option A) is seen in full term infants with Hypoxic ischemic encephalopathy.

Demyelination (Option D) is not the pathology involved in spastic diplegia.

Solution for Question 2:

Answer: D) 1,2,3,4

Explanation:

Neurofibromatosis Type 1 (NF1)

- Incidence: 1 in 3,000 live births.
- Inheritance: 50% of cases are inherited from a parent. 50% due to spontaneous mutations.
- 50% of cases are inherited from a parent.
- 50% due to spontaneous mutations.

- Genetics: Autosomal dominant loss-of-function mutations in the NF1 gene on chromosome 17q11.2. The gene encodes neurofibromin, which inhibits the oncogene Ras.
- 50% of cases are inherited from a parent.
- 50% due to spontaneous mutations.

Diagnosis Criteria (Two or more required)

Feature	Description
Café-au-lait macules (CALMs)	>5 mm in prepubertal, >15 mm in postpubertal. 6 or more needed for diagnosis (Statement 1)
Axillary/Inguinal Freckling	Multiple hyperpigmented areas, 2-3 mm in diameter. (Statement 3)
Iris Lisch Nodules	Hamartomas in the iris, best seen with a slit-lamp. 2 or more needed for diagnosis (Statement 4)
Neurofibromas	Two or more neurofibromas or one plexiform neurofibroma.
Osseous Lesion	Sphenoid dysplasia or cortical thinning of long bones. (Statement 2) Congenital pseudarthrosis of the tibia.
Optic Gliomas	CNS tumor affecting the optic nerve.
Family History	First-degree relative with an NF1 diagnosis based on these criteria.

Management

- Regular Monitoring:
- Treatment Considerations: Avoid routine brain/optic tract imaging unless symptomatic. Selumetinib: Effective in children with inoperable plexiform neurofibromas.
- Avoid routine brain/optic tract imaging unless symptomatic.
- Selumetinib: Effective in children with inoperable plexiform neurofibromas.
- Neuropsychological and Educational Support.
- Avoid routine brain/optic tract imaging unless symptomatic.
- Selumetinib: Effective in children with inoperable plexiform neurofibromas.

Genetic Counseling

- Inheritance: Autosomal dominant; over 50% of cases are de novo mutations.
- Molecular Testing: Useful in cases with unclear clinical diagnosis or severe disease. Prenatal/preimplantation diagnosis may be considered.
- Useful in cases with unclear clinical diagnosis or severe disease.
- Useful in cases with unclear clinical diagnosis or severe disease.

Solution for Question 3:

Answer: C) NF2 gene mutation

Explanation:

The patient's presentation along with a family history of similar features strongly suggests Neurofibromatosis Type 2 (NF2), which is caused by a mutation in the NF2 gene.

Neurofibromatosis Type 2 (NF2) Less Common: NF2 is rarer than NF1.

- Inheritance: Autosomal dominant; affects Schwann cells, leading to tumors on nerve insulation.
- Incidence: 1 in 25,000 births.
- Genetic Location: NF2 gene on chromosome 22q1.11.

Diagnosis Criteria (Manchester & Baser Criteria)

NF2 is diagnosed with one of the following:

(Mnemonic: MISME- Multiple inherited schwannomas meningioma and ependymoma)

- Bilateral Vestibular Schwannomas.
- Family History + either: Unilateral vestibular schwannoma Any two of the following: meningioma, schwannoma, glioma, neurofibroma, or posterior subcapsular lenticular opacities.
- Unilateral vestibular schwannoma
- Any two of the following: meningioma, schwannoma, glioma, neurofibroma, or posterior subcapsular lenticular opacities.
- Unilateral Vestibular Schwannoma + any two of the features above.
- Multiple Meningiomas (≥2) + either: Unilateral vestibular schwannoma Any two of the above-listed features.
- Unilateral vestibular schwannoma
- Any two of the above-listed features.
- Unilateral vestibular schwannoma
- Any two of the following: meningioma, schwannoma, glioma, neurofibroma, or posterior subcapsular lenticular opacities.

- Unilateral vestibular schwannoma
- Any two of the above-listed features.

Clinical Manifestations

- Common Symptoms: Tinnitus Hearing loss Facial weakness Headache Unsteadiness
- Tinnitus
- Hearing loss
- Facial weakness
- Headache
- Unsteadiness
- Age of Onset: Symptoms usually emerge in childhood; cerebellopontine angle masses are common in 20s-30s.
- Other Signs: Childhood: Café-au-lait macules (CALMs), skin plexiform schwannomas Posterior subcapsular lens opacities in ~50% of cases.
- Childhood: Café-au-lait macules (CALMs), skin plexiform schwannomas
- Posterior subcapsular lens opacities in ~50% of cases.
- Tinnitus
- Hearing loss
- Facial weakness
- Headache
- Unsteadiness
- Childhood: Café-au-lait macules (CALMs), skin plexiform schwannomas
- Posterior subcapsular lens opacities in ~50% of cases.

Genetics

- Mutation: NF2 gene mutation leads to loss of merlin/schwannomin protein.

Management

- Approach: Conservative management focusing on hearing preservation and quality of life.
- Regular Monitoring: Ophthalmologic evaluations Brain and spinal MRI Audiology assessments Brainstem-evoked potentials
- Ophthalmologic evaluations
- Brain and spinal MRI
- Audiology assessments
- Brainstem-evoked potentials
- Ophthalmologic evaluations

- Brain and spinal MRI
- Audiology assessments
- Brainstem-evoked potentials

Comparison Table: NF2 vs. Schwannomatosis vs. Legius Syndrome

Feature	NF2	Schwannomatosis	Legius Syndrome
Tumors	Schwannomas (esp. vestibular)	Multiple schwannomas, No vestibular schwannomas	Multiple CALMs, no neurofibromas, Lisch nodules
Genetic Mutation	NF2 (22q1.11) (Option C)	SMARCB1 (INI1) (Option D ruled out)	SPRED1 (Option B ruled out)
Diagnosis	Based on criteria (bilateral schwannomas, family history)	Multiple schwannomas, family history	CALMs, macrocephaly
Management	Conservative, regular monitoring	Yearly MRI, no established guidelines	Symptomatic management

NF1 (Option A): gene mutation is associated with Neurofibromatosis Type 1 (NF1), which primarily presents with neurofibromas, café-au-lait spots, and Lisch nodules, not vestibular schwannomas.

Solution for Question 4:

Answer: C) The mutation in TSC2 gene suggests a more severe clinical presentation.

Explanation:

Tuberous sclerosis complex (TSC) can be caused by mutations in either the TSC1 or TSC2 gene. Mutations in the TSC2 gene are often associated with more severe forms of TSC.

Tuberous Sclerosis Complex (TSC)

Incidence: Affects 1 in 6,000 to 10,000 newborns. Nature: Multisystem disorder characterized by benign tumors (hamartomas) in various organs.	

Genetics	Inheritance: Autosomal dominant; 65% due to spontaneous mutations. (Option A ruled out) Causing Genes: TSC1 on chromosome 9q34 encodes hamartin. TSC2 on chromosome 16p13 encodes tuberin. Severity: TSC2 mutations often lead to more severe symptoms. (Option C)
Pathophysiology	Key Proteins: Hamartin, Tuberin, and TBC1D7 regulate cell growth via the mTOR pathway. Role of Genes: TSC1 and TSC2 function as tumor suppressor genes.
Clinical Manifestations	Neurologic Symptoms: Epilepsy Cognitive Impairment. Autism. TAND (Tuberous Sclerosis-Associated Neuropsychiatric Disorders). Brain Lesions: Cortical Tubers Subependymal Nodules: Risk of SEGA transformation, causing hydrocephalus. Monitoring: Regular MRI (every 1-3 years). (Option D ruled out) Other Organ Systems: Retina: Hamartomas and white depigmented patches. Heart: Rhabdomyomas ... (content truncated)
Management	Close, continuous monitoring is essential.

Solution for Question 5:

Correct Answer: B) Malignant cardiac rhabdomyomas are associated with this condition.

Explanation:

Based on the clinical presentation of multiple facial skin lesions, developmental delay, and intellectual disability, the condition in question is likely Tuberous Sclerosis Complex (TSC). While Tuberous Sclerosis Complex is associated with cardiac rhabdomyomas, these tumors are typically benign.

Features of Tuberous Sclerosis:

Category	Sign	Description
Brain Lesions	Infantile Spasms	Sudden, brief muscle contractions; common in infancy.
Seizures	Includes focal-onset and generalized myoclonic seizures.(Option D)	
Developmental Delay	Delays in milestones like walking, talking.(Option D)	
Intellectual Disability	Affects ~45% of individuals.	
Autism Spectrum Disorder	Affects up to 50%.	
Behavioral Problems	Includes aggression, anxiety, ADHD.	
Cortical Tubers	Detected via MRI	
Subependymal Nodules	Risk of SEGA transformation, causing hydrocephalus. Monitoring: Regular MRI (every 1-3 years).	
Dermatologic Signs	Hypomelanotic Macules	Flat, light-colored patches; resemble ash leaves. Present at birth or early childhood.
Facial Angiofibromas	Small, reddish or skin-colored bumps on the face.(Option C)	
Shagreen Patches (Option A ruled out)	Thickened, rough patches; texture like orange peel. Typically on lower back or buttocks.	
Forehead Fibrous Plaques	Raised, yellowish-brown or flesh-colored plaques on the forehead.	
Periungual Fibromas	Small, fleshy growths around the nails.	
Ophthalmologic Signs	Retinal Hamartomas	Benign retinal tumors; appear as white or yellowish patches.
Cardiac Signs	Cardiac Rhabdomyomas	Benign heart muscle tumours; often detected prenatally. (Option B)
Renal Signs	Renal Angiomyolipomas	Benign kidney tumours; can cause pain, bleeding, and high blood pressure.

Solution for Question 6:

Answer: C) I, III, and IV only

Explanation:

The port-wine stain in Sturge-Weber syndrome is typically unilateral and ipsilateral (on the same side as the brain involvement). Other statements regarding Sturge Weber syndrome are true.

Sturge-Weber Syndrome:

Sturge-Weber syndrome (SWS) is a rare neurocutaneous disorder characterized by segmental vascular abnormalities affecting the

- Skin - Port-wine birthmarks
- Brain- Leptomeningeal angiomas, and
- Eyes - Glaucoma

Etiology: somatic mutations in the GNAQ gene (Statement I), only sporadic neurocutaneous disorder.

Clinical Manifestations:

- Port-Wine Birthmark (PWB): Present at birth, typically unilateral and ipsilateral to brain involvement. (Statement II ruled out)
- Neurological Symptoms: Seizures, often beginning in infancy, hemiparesis, stroke-like episodes, headaches, and developmental delays.
- Ocular Symptoms: Glaucoma and buphthalmos, particularly in the eye on the same side as the PWB.

Diagnosis:

- MRI Brain with contrast - Investigation of choice (Statement III)
- CT - cerebral calcifications and atrophy (Tram-track calcifications)

Management:

- Anticonvulsants for seizure control.
- Glaucoma Monitoring.
- Laser Therapy: for port wine birthmark.

Solution for Question 7:

Answer: D) Listeria monocytogenes

Explanation:

T-cell defects increase the risk of infections from intracellular pathogens, including Listeria monocytogenes, which is known to cause meningitis in children with T-cell deficiencies.

Bacterial meningitis:

Etiological agents:

	In India	Overall
Neonates	Gram negative organisms (E.Coli > Acinetobacter > Klebsiella)	Group B streptococci
Infants and older children	Streptococcus pneumoniae> Neisseria meningitidis > Haemophilus Influenzae.	Streptococcus pneumoniae
Children T-cell defect	Listeria monocytogenes (Option D)	
Children with CSF shunts	Staphylococci	
Children with Complement System Defects	Meningococcal infections	
Children with Splenic Dysfunction or Asplenia	Streptococcus pneumoniae, Neisseria Meningitidis Haemophilus Influenzae.(Encapsulated organisms)	

Risk Factors for Bacterial Meningitis

Category	Clinical Manifestations
Nonspecific Findings	Fever, anorexia, poor feeding Headache, upper respiratory symptoms Myalgias, arthralgias Tachycardia, hypotension Cutaneous signs: petechiae, purpura, or erythematous macular rash

Meningeal Irritation	Nuchal rigidity Back pain Kernig sign: Pain with extension of the leg after hip flexion Brudzinski sign: Involuntary flexion of the knees and hips upon neck flexion
Increased Intracranial Pressure (ICP) Signs	Headache, emesis Bulging fontanel or diastasis of sutures Cranial nerve involvement: anisocoria, ptosis, abducens nerve paralysis Hypertension with bradycardia, apnea, or hyperventilation Decorticate or decerebrate posturing, stupor, coma, or herniation signs Papilledema
Focal Neurologic Signs	Seizures Altered mental status Photophobia Tâche cérébrale: Raised red streak appearing 30-60 seconds after stroking the skin with a blunt object

Streptococcus pneumoniae, Neisseria meningitidis, Haemophilus influenzae (Options A,Band C) are not the common agents among patients with T-Cell defects.

Solution for Question 8:

Answer: D) All of the above

Explanation:

Presence of coma and focal neurologic signs at presentation, CSF leukocyte count < 250/mm^3 and seizures occurring more than 4 days into therapy are suggestive of poor prognosis in a child with meningitis.

Diagnosis of Acute bacterial meningitis:

Lumbar puncture and CSF analysis : Gold standard.

CSF shows Neutrophilic pleocytosis, increased proteins and decreased glucose.

Treatment:

Antibiotics: Third generation cephalosporins (Ceftriaxone) x 10 to 14 days

Steroids: Inj. Dexamethasone started 1-2 hr before antibiotics and continued for 48 hrs - Prevents cytokine mediated CNS injury.

Poor prognostic factors in Acute bacterial meningitis:

Cerebrospinal fluid (CSF) findings in bacterial, fungal, and viral meningitis:

Parameter	Bacterial Meningitis	Fungal Meningitis	Viral Meningitis
CSF Appearance	Cloudy or purulent	Clear or slightly cloudy	Clear
Opening Pressure	Elevated	Elevated	Normal or slightly elevated
WBC Count	Elevated (1000-5000 cells/µL)	Elevated (100-500 cells/µL)	Elevated (50-500 cells/µL)
Predominant Cells	Neutrophils	Lymphocytes	Lymphocytes
Protein	Elevated (>100 mg/dL)	Elevated (50-200 mg/dL)	Mildly elevated (<100 mg/dL)
Glucose	Low (<40 mg/dL or <40% of blood glucose)	Low (<40 mg/dL)	Normal (>40 mg/dL or >50% of blood glucose)
CSF/Blood Glucose Ratio	Decreased	Decreased	Normal
Gram Stain	Positive for bacteria	Negative	Negative
India Ink (for Cryptococcus)	Negative	Positive (Cryptococcus)	Negative
PCR	Usually not done	Specific to fungi (if needed)	Positive for viral DNA/RNA

Solution for Question 9:

Answer: A) Pigs

Explanation:

Pigs are known as the primary amplifier hosts for JEV, as they can develop high levels of viremia, which is sufficient to infect mosquitoes.

Japanese Encephalitis (JE)

JE is a mosquito-borne viral disease affecting humans and animals.

Vectors & Transmission:

- Culex tritaeniorhynchus (primary vector)
- Culex vishnui
- Pigs act as amplifying hosts. (Option A)

Diagnosis: JE diagnosis is confirmed by detecting virus-specific IgM antibodies or through PCR. JE has B/L thalamic hyperdensities.

Prognosis & Sequelae:

Common sequelae include mental deterioration, personality changes, motor abnormalities, and speech disturbances.

Treatment: There is no specific treatment for JE. Management focuses on intensive supportive care, including seizure control.

Culex mosquitoes (Option C) act as vectors that transmit the virus from animals to humans.

Humans (Option D) are considered dead-end hosts because they do not develop viremia levels sufficient to infect mosquitoes.

Cattle (Option B) do not serve as amplifier hosts for JEV.

Solution for Question 10:

Answer: D) All of the above.

Explanation:

In pediatric patients with suspected IIH without papilledema, the presence of specific neuroimaging features such as an empty sella, flattening of the posterior aspect of the globe, and distention of the perioptic subarachnoid space supports the diagnosis.

Common causes of Idiopathic Intracranial Hypertension (IIH):

- T: Tetracyclines (e.g., minocycline, doxycycline)
- O: Obesity
- A: Vitamin A excess (e.g., hypervitaminosis A, isotretinoin use)
- D: Danazol (a medication used to treat endometriosis)

Diagnostic Criteria for Idiopathic Intracranial Hypertension (IIH):

Table rendering failed

Solution for Question 11:

Answer: A) Performing two separate clinical evaluations, separated by a 24-hour interval.

Explanation:

For a child aged 1-18 years, two separate clinical evaluations are required, but they should be separated by a 12-hour interval, not 24 hours

Clinical Criteria for Brain Death:

- Complete loss of consciousness
- Absence of vocalization
- Absence of voluntary activity

Prerequisites:

- Clinical and/or neuroimaging evidence of acute CNS catastrophe severe enough to explain the condition
- Core temperature more than 32°C (Option B ruled out)
- No drug (sedatives, narcotics) or alcohol intoxication, poisoning, or neuromuscular blockade
- Normal blood pressure
- No severe electrolyte, acid-base, metabolic, or endocrine disturbances

Evaluation Timing:

- <2 months old: Two evaluations separated by 48 hours
- 2-12 months old: Two evaluations separated by 24 hours
- 1-18 years old: Two evaluations separated by 12 hours (Option A)
- 18 years old: Two evaluations at any interval

Evaluation of brain death:

Category	Criteria

Absent Brainstem Reflexes (all absent). (Option D ruled out)	Pupillary responses to light (pupils mid-position, 4-6 mm) Oculocephalic reflex Oculovestibular (caloric) responses Corneal reflex Jaw reflex Facial grimacing to deep pressure on supraorbital ridge or temporomandibular joint Pharyngeal gag reflex Coughing in response to tracheal suctioning Sucking and rooting reflexes
Apnea Test	Absence of respiratory drive at PaCO■ of 60 mm Hg or 20 mm Hg above baseline.
Ancillary Tests (Option C ruled out)	If <2 months old: Two tests required. If 2-12 months old: One test required. If >1 year old: Tests optional Types of tests include Cerebral angiography, Electroencephalography, Transcranial Doppler ultrasonography, and Cerebral scintigraphy.

Solution for Question 12:

Answer: A) Anencephaly

Explanation:

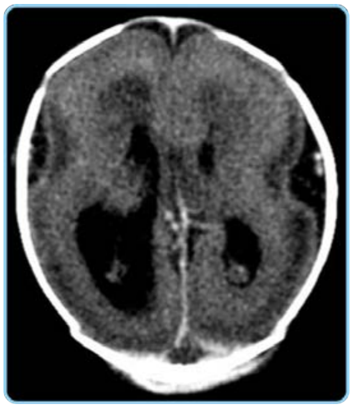
- The given image is of Anencephaly
- Anencephaly is a brain malformation that occurs at birth. It occurs when a newborn’s brain, skull, and scalp do not develop normally during pregnancy.
- Babies with this condition die within a few hours or days of birth.
- Taking the proper dose of folic acid before and throughout pregnancy may reduce the likelihood of having a newborn with anencephaly.

Inencephaly (Option B): is a rare neural tube defect located in the inion, i.e., the nape of the neck

Craniorachischisis (Option C): is the most severe of the NTDs and is characterized by an open defect extending from the skull to the spine and is usually associated with anencephaly.The condition is uniformly fatal

Lissencephaly (Option D):

- Lissencephaly, meaning "smooth brain," is a rare brain malformation caused by abnormal neuronal migration during embryonic development. The cortex in individuals with lissencephaly lacks the normal folds (gyri) and grooves (sulci), leading to a smooth brain surface. This condition is associated with developmental delays, intellectual disabilities, and seizures.
- Cause: Genetic mutations, often affecting genes like LIS1 and DCX

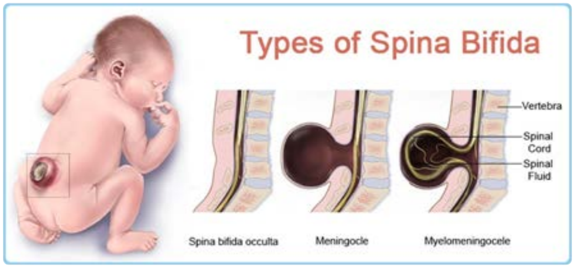


Solution for Question 13:

Answer: A) Spina Bifida Occulta

Explanation:

Based on the given picture, the most likely defect depicted in the image is Spina Bifida Occulta.



Neural Tube Defects (NTDs)

Basic Defect

- Improper closure of the neural tube
- Occurs between the 3rd and 4th week of intrauterine life

Types of Neural Tube Defects (NTDs) and their characteristics:

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Solution for Question 14:

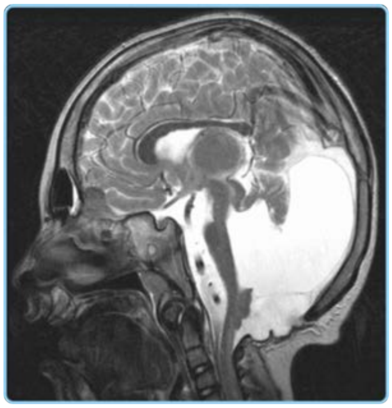
Answer: B) Arnold-Chiari Malformation Type 2

Explanation:

Arnold-Chiari Malformation Type 2 is the most likely diagnosis in this case. This condition is commonly associated with myelomeningocele and can lead to hydrocephalus. It involves the herniation of both the cerebellar tonsils and the brainstem through the foramen magnum and is often seen in conjunction with spina bifida.

Aspect	Type 1	Type 2
Hydrocephalus	No association (Option A ruled out)	Associated with hydrocephalus and myelomeningocele
Herniation	Cerebellar tonsil into the cervical canal	Cerebellar tonsil, vermis, and part of the pons and medulla into the cervical canal
Usual Age of Presentation	Adolescent	Infancy
Clinical Features	Headache Neck pain Urinary problems Lower limb spasticity Infants: Lethargy, poor feeding Older individuals: Gait problems, spasticity	Infants: Lethargy, poor feeding Older individuals: Gait problems, spasticity

Dandy-Walker Malformation (Option C): It involves cystic enlargement of the fourth ventricle and agenesis or hypoplasia of the cerebellar vermis, but not commonly associated with myelomeningocele.



Hydranencephaly (Option D): It involves the replacement of brain tissue with cystic spaces and is not associated with myelomeningocele or the typical Chiari malformations.

Previous Year Questions

1. A child presents to the OPD with intracranial diffuse calcifications, chorioretinitis, and hydrocephalus. Which of the following is the most likely diagnosis?

A. Congenital Toxoplasmosis

B. Dandy-Walker Malformation

C. Budd-Chiari Syndrome

D. Congenital Hydrocephalus

2. Which of the following MRI findings will be seen in a child with spastic quadriplegia?

A. Basal ganglia calcifications

B. Periventricular leukomalacia

C. Cerebellar degeneration

D. Multicystic encephalomalacia

3. What is the probable cause of the patient's condition based on the cerebrospinal fluid (CSF) report, which exhibits increased lymphocyte , increased proteins(63 mg/dL), and normal sugar level and appears clear?

A. Tuberculous meningitis

B. Viral meningitis

C. Bacterial meningitis

D. None of the above

4. In a 1-year-old male child who arrives at the hospital with fever and neck stiffness, the suspected diagnosis is meningitis. Among the options provided, which pathogen is the leading cause of meningitis in this particular age group?

A. Klebsiella

B. Streptococcus pneumoniae

C. Staphylococcus aureus

D. Mycobacterium tuberculosis

5. Which of the following is not part of the criteria for diagnosing neurofibromatosis 1?

A. Two or more iris hamartomas

B. Acoustic neuromas

C. Dysplasia of the sphenoidal and tibial bone

D. Cafe-au-lait spots

6. Anencephaly occurs due to the inability of the neural tube to close at which week of intrauterine life?

A. 3rd week

B. 4th week

C. 5th week

D. 2nd week

7. In which of the following conditions is adrenocorticotropic hormone the preferred medication?

A. Juvenile myoclonic epilepsy

B. Rolandic epilepsy

C. West syndrome

D. Lennox Gastaut syndrome

8. Identify the condition in a neonate with the swelling of the head shown in the image below at a few weeks of life?

A. Caput succedaneum

B. Cephalhematoma

C. Subgaleal bleed

D. Encephalocele

9. What is the most likely reason for an 8-month-old baby being brought to the hospital with issues of inadequate feeding and unusual spasms in the lower limb,

along with symptoms of fever, bulging fontanelle, and lethargy?

A. Febrile seizures

B. Meningitis

C. Pseudotumor cerebri

D. Intracranial haemorrhage

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10. Which vitamin deficiency can result in seizures in newborns?

A. Pyridoxine

B. Thiamine

C. Riboflavin

D. Pantothenic acid

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11. A 7-year-old child from a socioeconomically disadvantaged background has presented with fever, convulsions, and unusual behavior over the past two months, and has now become unresponsive to their parents for the past two days. On examination, the child exhibits neck stiffness, and their cerebrospinal fluid (CSF) analysis reveals an elevated number of lymphocytes, glucose levels of 30 mg%, and protein levels of 300 mg%. What is the diagnosis?

A. Viral meningitis

B. Bacterial meningitis

C. Tubercular meningitis

D. Partially treated bacterial meningitis

---

12. What is the recommended duration for isolating a child after diagnosis of bacterial meningitis in order to prevent further transmission?

A. Till 24 hours after starting antibiotics

B. Till cultures become negative

C. Till antibiotic course is complete

D. Till 12 hours after admission

13. What is the likely diagnosis for an adolescent girl who experiences instances of dropping objects from her hands, particularly in the morning and during exams, without any history of loss of consciousness with EEG showing signs of epileptic spikes?

A. Juvenile myoclonic epilepsy

B. Atypical absence seizures

C. Choreo-athetoid epilepsy

D. Centrotemporal spikes

14. What is the preferred drug for treating the condition characterized by jerking movement of limbs towards the body, regression of developmental milestones, and hypsarrhythmia on electroencephalogram in a 10-month old infant?

A. Phenytoin

B. Adrenocorticotropic hormone

C. Levetiracetam

D. Phenobarbitone

15. What is the probable diagnosis for a newborn who has a patch of hair on the lower back region?

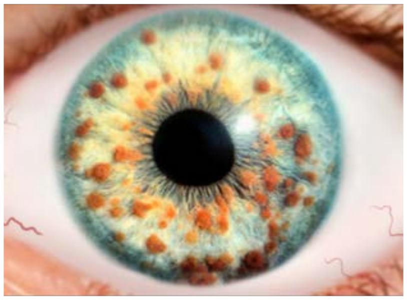
A. Spina bifida occulta

B. Neurofibromatosis type 1

C. Tethered spinal cord

D. Congenital dermal sinus

16. Which of the following statements is incorrect in relation to the provided image?



A. It represents melanocytic hamartoma

B. It causes no visual disturbance

C. It is associated with facial angiofibroma

D. They are diagnostic of a neurocutaneous syndrome

17. Which of the following is not a characteristic of cerebral palsy?

A. Hypotonia

B. Microcephaly

C. Ataxia

D. Flaccid paralysis

18. What is the most probable microorganism responsible for the meningitis in a neonate presenting with heightened irritability, inadequate feeding, and two instances of seizures, as confirmed by lumbar puncture findings?

A. Streptococcus agalactiae

B. Escherichia coli

C. Neisseria meningitidis

D. Listeria monocytogenes

---

19. Which of the following are the causes of neonatal seizures? 1. Hypocalcemia 2. Hyponatremia 3. Hypomagnesemia 4. Hypoglycemia

(or)

What are the factors that can lead to seizures in newborns? 1. Hypocalcemia 2. Hyponatremia 3. Hypomagnesemia 4. Hypoglycemia

A. 1, 2, 3 only

B. 1, 2, 3, and 4

C. 1, 3, 4 only

D. 2, 4 only

---

20. Which drug is typically prescribed as the initial treatment for controlling seizures in a child with febrile seizures?

A. Sodium valproate

B. Phenytoin

C. Midazolam

D. Phenobarbitone

---

21. What guidance should be provided to the students who have been exposed to meningococcal meningitis, considering that two girls in their class have been diagnosed with the disease? Additionally, their 12-year old friend from the same school is concerned about the possibility of contracting the illness.

A. Two doses of polysaccharide vaccine

B. Antibiotic prophylaxis

C. Two doses of conjugate vaccine

D. Single dose of meningococcal vaccine

22. What investigation would be most beneficial in diagnosing the condition of a 10-year-old boy who experienced seizures? His medical history includes a fever accompanied by a rash at 1 year old, which resolved on its own.

A. IgG measles in CSF

B. MRI mesial temporal lobe sclerosis

C. IgM measles in CSF

D. C1Q4 antibodies in the CSF

23. A young child with congenital hydrocephalus was treated with a successful ventriculoperitoneal (VP) shunt. Four months post surgical repair, the child presented with fever, nuchal rigidity, and irritability. You suspect meningitis. What is the next best step?

A. Wait and watch

B. Check shunt patency by nuclear study

C. Blood culture and take CSF sample from shunt tap

D. Blood culture and take CSF sample by lumbar puncture

24. Which of the options listed below are factors that contribute to an increased risk of febrile seizures reoccurring? Age <1 year Temperature of 38-39°C Duration of fever < 24 h Duration of fever > 48 h

A. 1, 2, 4

B. 1, 4

C. 1, 2, 3

D. 1, 3, 4

---

25. What is the most common cause of seizures in a 2-year-old child?

A. Trauma to the head

B. Malnutrition

C. Febrile seizures

D. Neurological deficit

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	2
Question 4	2
Question 5	2
Question 6	2
Question 7	3
Question 8	2
Question 9	2
Question 10	1
Question 11	3
Question 12	1
Question 13	1
Question 14	2
Question 15	1
Question 16	3
Question 17	2
Question 18	1
Question 19	2
Question 20	3
Question 21	2
Question 22	1

Question 23	3
Question 24	3
Question 25	3

Solution for Question 1:

Correct Option A - Congenital Toxoplasmosis:

- Congenital toxoplasmosis is a result of vertical transmission of Toxoplasma gondii from mother to fetus. The classic triad of symptoms includes intracranial calcifications (diffuse), chorioretinitis, and hydrocephalus. This combination strongly suggests congenital toxoplasmosis.

Option B - Dandy-Walker Malformation:

- Dandy-Walker malformation is a congenital brain malformation involving the cerebellum and the fluid-filled spaces around it. It typically presents with hydrocephalus, posterior fossa cyst, and cerebellar hypoplasia, but it does not typically involve intracranial diffuse calcifications or chorioretinitis.

Option C - Budd-Chiari Syndrome:

- Budd-Chiari syndrome involves hepatic venous outflow obstruction, leading to liver-related symptoms like ascites, hepatomegaly, and abdominal pain. It is unrelated to intracranial calcifications, chorioretinitis, or hydrocephalus, making it an unlikely diagnosis in this scenario.

Option D - Congenital Hydrocephalus:

- Congenital hydrocephalus refers to an abnormal buildup of cerebrospinal fluid in the brain's ventricles, leading to increased intracranial pressure. While hydrocephalus is a feature here, the presence of diffuse intracranial calcifications and chorioretinitis is not typical of congenital hydrocephalus alone, making it less likely as the correct diagnosis.

Solution for Question 2:

Correct Option D - Multicystic Encephalomalacia:

- Multicystic encephalomalacia is a condition characterized by the presence of multiple cysts within the brain tissue. It is a result of severe hypoxic-ischemic injury leading to the extensive necrosis of brain tissue, which is replaced by cystic cavities. This condition is often seen in children with severe forms of cerebral palsy, such as spastic quadriplegia, where extensive brain damage has occurred.

Option A - Basal Ganglia Calcifications:

- Basal ganglia calcifications are typically associated with metabolic and genetic disorders, such as Fahr’s disease, rather than with cerebral palsy or spastic quadriplegia.

Option B - Periventricular Leukomalacia (PVL):

- PVL is associated with spastic diplegia, not quadriplegia. It involves damage to the white matter around the ventricles, commonly seen in premature infants, leading to motor control issues predominantly in the legs rather than all four limbs.

Option C - Cerebellar Degeneration:

- Cerebellar degeneration usually presents with ataxia and balance issues rather than spastic quadriplegia. This condition involves the loss of neurons in the cerebellum, which is responsible for coordinating movement.

Solution for Question 3:

Correct Option B-Viral meningitis: This is correct, as protein is raised, glucose is low, and lymphocytes are elevated. Also, a clear appearance on gross examination indicates viral meningitis.

Incorrect Options:

Option A- Tuberculous meningitis: In this, sugar would be depleted, and the gross sample would be straw-colored.

Option C- Bacterial meningitis: In this, sugar would be depleted, and the gross sample would be turbid.

Option D- None of the above: This option is incorrect

Solution for Question 4:

Correct Option B - Streptococcus pneumoniae:

- Streptococcus pneumoniae, also known as pneumococcus, is the most common cause of bacterial meningitis in children.
- Haemophilus influenzae is the second most common cause of bacterial meningitis in children.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 5:

Correct Option B - Acoustic neuromas:

- Acoustic neuromas (also called vestibular schwannomas) are not specific to NF1 and are more commonly associated with neurofibromatosis type 2 (NF2).
- Diagnostic criteria for neurofibromatosis - 1: Presence of 2 or more of the following: Freckling of the axillary or inguinal areas Lisch nodules, 2 or more involving the eyes Optic glioma (most common tumor seen in NF-1) First-degree relative affected with same disorder Presence of 2 or more neurofibromas Presence of 6 or more Caif au lait spots - Significant if the size is >5 mm in prepubertal age and >15 mm in Adolescence A distinctive osseous lesion, such as sphenoid dysplasia or thinning of long bone cortex, also known as dysplasia of the sphenoidal and tibial bone.
- Freckling of the axillary or inguinal areas
- Lisch nodules, 2 or more involving the eyes
- Optic glioma (most common tumor seen in NF-1)
- First-degree relative affected with same disorder
- Presence of 2 or more neurofibromas

- Presence of 6 or more Caif au lait spots - Significant if the size is >5 mm in prepubertal age and >15 mm in Adolescence
- A distinctive osseous lesion, such as sphenoid dysplasia or thinning of long bone cortex, also known as dysplasia of the sphenoidal and tibial bone.
- Freckling of the axillary or inguinal areas
- Lisch nodules, 2 or more involving the eyes
- Optic glioma (most common tumor seen in NF-1)
- First-degree relative affected with same disorder
- Presence of 2 or more neurofibromas
- Presence of 6 or more Caif au lait spots - Significant if the size is >5 mm in prepubertal age and >15 mm in Adolescence
- A distinctive osseous lesion, such as sphenoid dysplasia or thinning of long bone cortex, also known as dysplasia of the sphenoidal and tibial bone.

Incorrect Options:

Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 6:

Correct Option B - 4th week:

- Anencephaly occurs due to the failure of closure of the rostral neuropore.
- It occurs at 4 weeks of gestation.
- Features include absent scalp, cranial bones, and meninges with rudimentary brain exposed to outside.
- This condition is incompatible with life and the baby is not resuscitated due to the poor prognosis.

Incorrect Options:

Options A, C, and D are incorrect.

Solution for Question 7:

Correct Option C - West syndrome:

- The drug of choice for treating West syndrome is adrenocorticotropic hormone (ACTH).
- West syndrome is characterized by a triad of symptoms: Infantile spasms Developmental regression or delay Hypsarrhythmia ACTH has anti-inflammatory and immunomodulatory effects. When used as a medication, synthetic ACTH is administered to control seizures and improve developmental outcomes in children with West syndrome.
- Infantile spasms
- Developmental regression or delay
- Hypsarrhythmia
- ACTH has anti-inflammatory and immunomodulatory effects.
- When used as a medication, synthetic ACTH is administered to control seizures and improve developmental outcomes in children with West syndrome.
- Infantile spasms
- Developmental regression or delay
- Hypsarrhythmia
- ACTH has anti-inflammatory and immunomodulatory effects.
- When used as a medication, synthetic ACTH is administered to control seizures and improve developmental outcomes in children with West syndrome.

Incorrect Options:

ACTH is not the medication used in options A, B, and D.

Option A - Juvenile myoclonic epilepsy:

- Juvenile myoclonic epilepsy is characterized by myoclonic jerks , generalized tonic-clonic seizures , and sometimes absence seizures.
- The drug of choice for treating juvenile myoclonic epilepsy is typically valproic acid or levetiracetam.

Option B - Rolandic epilepsy:

- Rolandic epilepsy is characterized by seizures that usually occur during sleep and involve facial twitching, drooling, or speech difficulties.
- The first-line treatment for rolandic epilepsy is antiepileptic drugs like carbamazepine or oxcarbazepine.

Option D - Lennox Gastaut syndrome:

- Lennox Gastaut syndrome is characterized by multiple seizure types, including tonic seizures , atonic seizures , and atypical absence seizures.
- The treatment for Lennox Gastaut syndrome usually involves a combination of antiepileptic drugs, such as valproic acid, lamotrigine, or clobazam.
- ACTH may sometimes be used but is not considered the drug of choice.

Solution for Question 8:

Correct Option B - Cephalhematoma:

- The above given image is diagnostic of cephalhematoma.
- It occurs between skull and periosteum and usually resolves within 3-6 weeks.
- It does not cross suture lines.
- It is associated with linear skull fracture.

Incorrect Options:

Option A - Caput succedaneum:

- Caput succedaneum presents at birth with maximum size and firmness.

Option C - Subgaleal bleed:

- Subgaleal bleed does not present with a localized swelling, it presents more with a generalized swelling.

Option D - Encephalocele:

- Swelling protrusion of the meninges through a midline cranial defect with or without the presence of the neural elements inside it.

Solution for Question 9:

Correct Option B - Meningitis:

- The patient’s presentation with fever, bulging fontanelle, lethargy, inadequate feeding and unusual spasms in the lower limb are suggestive of meningitis
- In infants and young children, the symptoms of meningitis can be non-specific but may include poor feeding, irritability, abnormal movements, fever, and changes in consciousness.
- The fontanelle, a soft spot on the baby's skull, may appear bulging due to increased intracranial pressure.

Incorrect Options:

Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 10:

Correct Option A - Pyridoxine:

- Deficiency of pyridoxine, also known as vitamin B6, can lead to neonatal seizures.
- Pyridoxine is essential for the proper functioning of the nervous system, and its deficiency can disrupt normal brain development and function, leading to seizures in newborns.

Incorrect Options:

Option B - Thiamine:

- Deficiency of thiamine can result in a condition called beriberi, which primarily affects the cardiovascular and nervous systems but does not typically cause neonatal seizures.

Option C - Riboflavin:

- Riboflavin, also known as vitamin B2, is necessary for energy production and the metabolism of fats, carbohydrates, and proteins.
- While riboflavin deficiency can lead to various symptoms, including skin and eye abnormalities, it is not typically associated with neonatal seizures.

Option D - Pantothenic acid:

- Pantothenic acid, also known as vitamin B5, is involved in the synthesis of coenzyme A, which is essential for energy metabolism.
- Deficiency of pantothenic acid is not associated with neonatal seizures.

Solution for Question 11:

Correct Option C - Tubercular meningitis:

- The patient's presentation with fever, convulsions, and unusual behavior over the past two months, and has now become unresponsive to their parents for the past two days with CSF finding of increased lymphocytes, low glucose levels, and elevated protein levels is suggestive of tubercular meningitis.
- Tubercular meningitis is a form of meningitis caused by Mycobacterium tuberculosis.
- Tubercular meningitis often presents with a subacute or chronic course.
- The symptoms may include fever, headache, neck stiffness, altered mental status, and neurological abnormalities.
- The CSF findings typically show increased lymphocytes, low glucose levels, and elevated protein levels.

Incorrect Options:

Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 12:

Correct Option A - Till 24 hours after starting antibiotics:

- After starting antibiotics for bacterial meningitis, the child should be isolated for at least 24 hours.
- This allows sufficient time for the antibiotics to take effect, reducing the bacterial load and decreasing the risk of transmission to others.
- After this period, the child is generally considered to be less contagious.

Incorrect Options:

Options B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 13:

Correct Option A - Juvenile myoclonic epilepsy:

- JME is a common form of epilepsy that typically begins in adolescence.
- It is characterized by brief, involuntary muscle jerks or myoclonic seizures, which can result in the dropping of objects.
- These seizures often occur in the morning upon waking and can be triggered by fatigue, stress, or sleep deprivation.
- The presence of epileptic spikes on the EEG further supports the diagnosis of JME.

Incorrect Options:

Option B - Atypical absence seizures:

- Atypical absence seizures typically present with a brief loss of awareness, accompanied by subtle motor manifestations such as eye blinking or slight movements.

Option C - Choreo-athetoid epilepsy:

- Choreo-athetoid epilepsy is a rare form of epilepsy and typically presents with a combination of seizures and abnormal movement patterns.
- Choreo-athetoid movements refer to involuntary, irregular, and writhing movements.

Option D - Centrotemporal spikes: Centrotemporal spikes, also known as benign rolandic epilepsy, are characterized by seizures that primarily occur during sleep and involve the face, mouth, and throat muscles.

Solution for Question 14:

Correct Option B - Adrenocorticotropic hormone:

- The patient’s presentation with jerking movement of limbs towards the body, regression of developmental milestones, and hypsarrhythmia on the electroencephalogram are suggestive of West syndrome.

- Adrenocorticotropic hormone (ACTH) is the drug of choice for the treatment of infantile spasms.

Incorrect Options:

Options A, C, and D are not the preferred drugs for treating West syndrome.

Solution for Question 15:

Correct Option A - Spina bifida occulta:

- A tuft of hair over the lumbosacral region in a newborn is a clinical sign that is most commonly associated with spina bifida occulta

Incorrect Options:

Options B and C do not present with a tuft of hair over the lumbosacral region.

Option D - Congenital dermal sinus: It can sometimes occur in the lumbosacral region, the presence of a tuft of hair alone is not a specific sign of this condition.

Solution for Question 16:

Correct Option C - It is associated with facial angiofibroma:

- The given image shows discrete nodules over the iris which are Lisch nodules seen in neurofibromatosis 1.
- It is not associated with facial angiofibroma.
- Facial angiofibromas, also known as adenoma sebaceum, are skin lesions characterized by red papules or nodules that commonly occur in the central face, particularly the cheeks and nose.
- Facial angiofibromas are associated with tuberous sclerosis.

Incorrect Options:

Option A - It represents melanocytic hamartoma: Lisch nodules are the melanocytic hamartomas found in adults with NF1 and are the most consistent features of NF1.

Option B - It causes no visual disturbance: Lisch nodules, although present in the eye, do not typically cause visual disturbance.

Option D - They are diagnostic of a neurocutaneous syndrome: Lisch nodules found in adults with NF1 and are the most consistent features of NF1.

Solution for Question 17:

Correct Option B - Microcephaly:

- Microcephaly is a condition in which the head is smaller than expected for age and sex.
- It is not a characteristic feature of cerebral palsy.

Incorrect Options:

Options A, C, and D are the characteristic features of cerebral palsy.

Option A - Hypotonia:

- Hypotonia refers to decreased muscle tone, resulting in floppy or loose muscles.

Options C - Ataxia:

- Ataxia refers to problems with coordination and voluntary muscle movements, resulting in unsteady or shaky movements.
- While ataxia is not present in all cases of cerebral palsy, it can occur in individuals with cerebral palsy, especially when the cerebellum is affected.

Option D - Flaccid paralysis:

- Flaccid paralysis refers to a condition where there is a complete loss of muscle tone and voluntary movement.
- It occurs when there is damage to the lower motor neurons or peripheral nerves, leading to a loss of muscle control.

Solution for Question 18:

Correct Option A - Streptococcus agalactiae:

- The most common organism responsible for meningitis in neonates is Group B streptococcus or Streptococcus agalactiae.

Incorrect Options:

Options B, C, and D are not the most common organisms responsible for meningitis in neonates.

Solution for Question 19:

Correct Option B - 1, 2, 3, and 4:

The electrolyte abnormalities that can be responsible for neonatal seizures are:

- Hypocalcemia
- Hyponatremia
- Hypomagnesemia
- Hypoglycemia
- Hypokalemia
- Hypernatremia

Incorrect Options:

Option A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 20:

Correct Option C - Midazolam:

- The initial treatment for controlling seizures in a child with febrile seizures: Make the child lie down in the left lateral (recovery position) - Do not anything give orally In cases where seizure lasts > 5 minutes Home Management: Rectal/Buccal/Nasal Midazolam Hospital Management: IV lorazepam or midazolam
- Make the child lie down in the left lateral (recovery position) - Do not anything give orally
- In cases where seizure lasts > 5 minutes Home Management: Rectal/Buccal/Nasal Midazolam Hospital Management: IV lorazepam or midazolam
- Home Management: Rectal/Buccal/Nasal Midazolam
- Hospital Management: IV lorazepam or midazolam
- Make the child lie down in the left lateral (recovery position) - Do not anything give orally
- In cases where seizure lasts > 5 minutes Home Management: Rectal/Buccal/Nasal Midazolam Hospital Management: IV lorazepam or midazolam
- Home Management: Rectal/Buccal/Nasal Midazolam
- Hospital Management: IV lorazepam or midazolam
- Home Management: Rectal/Buccal/Nasal Midazolam
- Hospital Management: IV lorazepam or midazolam

Incorrect Options:

Options A, B, and D are not used in the initial treatment for controlling seizures in a child with febrile seizures.

Solution for Question 21:

Correct Option B - Antibiotic prophylaxis:

- Meningococcal meningitis is a bacterial infection that can spread through close contact with respiratory and throat secretions of an infected person.
- It is important to take precautions to prevent the spread of the disease among other students by taking antibiotic prophylaxis.
- Antibiotics can be given to individuals who have been in close contact with someone diagnosed with meningococcal meningitis to help prevent the spread of the bacteria.

Incorrect Options:

Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 22:

Correct Option A - IgG measles in CSF:

- The patient's history of fever with rash at 1 year of age, which resolved spontaneously, suggests a possible previous measles infection.
- Subacute sclerosing panencephalitis (SSPE) is a rare complication of measles that can occur years after a measles infection.
- In SSPE, there is persistent measles virus infection in the central nervous system, leading to inflammation and progressive neurological deterioration.
- To diagnose SSPE, the presence of elevated measles-specific IgG antibodies in the cerebrospinal fluid (CSF) is considered a helpful investigation.
- IgG antibodies are produced in response to previous infection and can persist in the CSF for an extended period.

Incorrect Options:

Option B - MRI mesial temporal lobe sclerosis:

- Mesial temporal lobe sclerosis (MTS) is a specific finding on MRI associated with temporal lobe epilepsy.
- It is characterized by atrophy and abnormal signal intensity in the mesial temporal structures, particularly the hippocampus.

Option C - IgM measles in CSF:

- The detection of IgM antibodies specific to the measles virus in the cerebrospinal fluid (CSF) may indicate recent or ongoing measles infection.
- However, in the case of SSPE, the initial measles infection usually occurred several years before the onset of symptoms.
- Therefore, the presence of IgM antibodies in CSF is not helpful in diagnosing SSPE.

Option D - C1Q4 antibodies in the CSF:

- C1Q4 antibodies are not associated with SSPE or measles infection.
- These antibodies are typically associated with certain autoimmune diseases, such as systemic lupus erythematosus (SLE).

Solution for Question 23:

Correct Option C - Blood culture and take CSF sample from shunt tap:

- In cases where a child with a ventriculoperitoneal (VP) shunt presents with symptoms suggestive of meningitis, obtaining a CSF sample from the shunt tap is the preferred method to evaluate for infection.
- This involves accessing the shunt reservoir and collecting CSF directly from it.
- The CSF sample can then be sent for analysis, including cell count, protein levels, glucose levels, and culture to identify the causative organism.
- The shunt tap procedure allows for a more accurate assessment of CSF infection, as it is more likely to reflect the current status of the shunt system.
- Additionally, blood cultures should also be obtained to identify any potential bloodstream infection.

Incorrect Options:

Option A - Wait and watch: is not appropriate in this case, as prompt evaluation and treatment are necessary for suspected meningitis.

Option B - Check shunt patency by nuclear study: It may be considered in certain situations but does not address the immediate need to evaluate for suspected meningitis.

Option D - Blood culture and take CSF sample by lumbar puncture: It is not the preferred choice because obtaining CSF from the shunt tap provides a more direct and specific assessment of the shunt system.

Solution for Question 24:

Correct Option C - 1, 2, 3:

- Febrile seizures are seizures with significant fever (≥100. 4°F) without any evidence of CNS infection in the age group of 6 months – 5 years

- Factors increasing the risk of recurrence are: Age <1 year at onset Temperature 100.4■ -102.2■ (38■-39■) Duration of fever <24 hrs Family history of febrile seizures Complex febrile seizures Lower serum Na+ level at the time of presentation
- Age <1 year at onset
- Temperature 100.4■ -102.2■ (38■-39■)
- Duration of fever <24 hrs
- Family history of febrile seizures
- Complex febrile seizures
- Lower serum Na+ level at the time of presentation

- Age <1 year at onset
- Temperature 100.4■ -102.2■ (38■-39■)
- Duration of fever <24 hrs
- Family history of febrile seizures
- Complex febrile seizures
- Lower serum Na+ level at the time of presentation

Incorrect Options:

Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 25:

Correct Option C - Febrile seizures:

- Febrile seizures are the most common cause of seizures in children aged 6 months to 5 years.
- Febrile seizures are seizures with significant fever (≥100. 4°F) without any evidence of CNS infection in the age group of 6 months - 5 years

Incorrect Options:

Option A, B, and D can also cause seizures in children but are less common than febrile seizures.

Musculoskeletal Disorders in Children

1. A 3-year-old child presents with bowing of the legs, delayed growth, and poor weight gain. Lab tests show normal serum calcium, low serum phosphate, high serum alkaline phosphatase, normal parathyroid hormone (PTH), and normal 25-hydroxyvitamin D levels. Which of the following is the most likely diagnosis?

A. Hypophosphatemic rickets

B. Vitamin D deficiency

C. Vitamin D-dependent rickets type 1

D. Vitamin D-dependent rickets type 2

---

2. A 10-year-old girl presents with ptosis, difficulty swallowing, and generalized muscle weakness for the past 2 weeks. The symptoms worsen with activity and improve after rest. On examination, ptosis and mild proximal muscle weakness in the upper limbs are noted. Reflexes and sensation are normal.What is the most likely diagnosis?

A. Juvenile Myasthenia Gravis

B. Guillain-Barré Syndrome

C. Muscular Dystrophy

D. Duchenne Muscular Dystrophy

---

3. A newborn presents with limbs that resemble the flippers of a seal without intermediate segments. The mother reports taking medication for anxiety during early pregnancy. What is the most likely cause of this condition?

A. Thalidomide exposure

B. Vitamin D deficiency

C. Genetic mutation

D. Impaired blood flow to the limbs

4. A 5-year-old child presents with multiple fractures following minimal trauma, blue sclerae, and mild hearing loss. Physical examination reveals joint hypermobility, and radiographs show osteopenia with evidence of previous fractures. What is the most likely underlying pathophysiology?

A. Defective type I collagen synthesis

B. Deficiency of vitamin D

C. Impaired osteoclast function

D. Abnormal calcium metabolism

5. A 4-month-old infant presents with macrocephaly, hepatosplenomegaly, and recurrent fractures. The radiograph shows diffuse bone sclerosis and a “bone within bone” appearance. Which of the following is the most appropriate treatment to prevent the progression of this condition?

A. Vitamin D supplementation

B. Hematopoietic stem cell transplantation (HSCT)

C. Bisphosphonate therapy

D. Physical therapy and fracture management

6. A 5-year-old boy presents with asymmetric arthritis of the lower extremities, dactylitis, and erythematous plaques on his soles 4 weeks after an episode of dysentery. Which of the following is true regarding the diagnosis?

A. HLA-B27 decreases the likelihood of gastrointestinal inflammation

B. Arthritis is symmetric and commonly affects small joints

C. Typically follows a Gastrointestinal or genitourinary infection

D. Broad-spectrum antibiotics are essential for treatment

7. A 10-year-old child has swelling and pain around the wrist joints and fingers. Classic rheumatoid nodules were seen over the spine and bony prominences of the hands. Which of the following about the management of juvenile idiopathic arthritis is false?

A. Methotrexate is used as it acts within a few days of intake

B. Abatacept and tocilizumab target specific immune system components

C. Anakinra is effective for systemic JIA

D. Corticosteroids are used for the control of uveitis

8. An 8-year-old child is brought to the pediatrics department with a low-grade fever and swelling around both the knees, elbows, and shoulders for 5 months. There is pain and swelling around the wrists as well. Numerous nodules are noted on the hands and over the spine, as shown. Which of the following findings is associated with this disease?

A. Asymmetric joint involvement

B. Uveitis

C. Involvement of smaller joints

D. Positive rheumatoid factor

9. A 4-year-old male child is brought to the hospital with swelling around knee joints and both ankles for three months. The mother of the child noticed this after an episode of bacterial pneumonia, which is now cured, and explains that the swelling has stayed in these joints since the beginning. What is the diagnosis here?

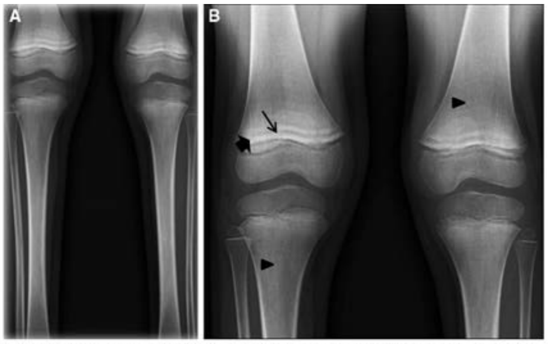
A. Rheumatoid factor-negative polyarthritis

B. Systemic Juvenile idiopathic arthritis

C. Persistent oligoarthritis

D. Rheumatoid factor-positive polyarthritis

10. A 12-year-old child was brought to the pediatric department with complaints of reduced appetite, irritability, and leg swelling. On examination, he has a low grade fever and tenderness in the legs. The following X-ray was taken.



e radiographic findings. (A) Thinning of cortical bones. (B) White line of Frankel (thick

A. Ground glass appearance

B. Trummerfeld zone

C. Pelkan spur

D. Wimberger sign

11. A 14-year-old child presents with swelling of gums and associated bleeding especially while brushing teeth. On examination, perifollicular hemorrhages are noted. Which of the following micronutrients is usually found deficient in association with this condition?

A. Zinc

B. Magnesium

C. Copper

D. Iron

12. A 5-year-old girl presents with short stature, leg bowing and a waddling gait. Her lab results show low serum phosphate, elevated alkaline phosphatase and increased urinary phosphate excretion. A genetic test reveals a mutation on the X chromosome. Which gene is most likely mutated in this patient?

A. SLC34A3

B. FGF23

C. PHEX

D. DMP1

13. A 2-year-old girl presents with delayed walking, bowed legs and frontal bossing. Her lab results show low calcium, low phosphorus, normal 25-hydroxyvitamin D (25-D), low 1,25-dihydroxyvitamin D (1,25-D), and elevated parathyroid hormone (PTH). What is the most likely diagnosis?

A. Vitamin D-Dependent Rickets Type 1A

B. Vitamin D-Dependent Rickets Type 1B

C. Vitamin D-Dependent Rickets Type 2A

D. Vitamin D-Dependent Rickets Type 2B

14. A 4-year-old boy presents with poor growth, bowed legs, muscle weakness, low serum phosphate, low bicarbonate, normal calcium, serum pH of 6.5 and normal creatinine. The child has no response to vitamin D therapy. What is the most likely diagnosis?

A. Renal tubular acidosis associated rickets

B. Vitamin D-dependent rickets

C. Hypophosphatemic rickets

D. Chronic kidney disease associated rickets

15. A 2-year-old child presents with delayed growth and bowed legs. The child has a history of inadequate sun exposure and a diet low in vitamin D. An x-ray image of the forearm is shown below. What is the most appropriate treatment?

A. Daily low-dose vitamin D (2000 IU) with oral calcium

B. Weekly low-dose vitamin D (2000 IU) with oral calcium

C. Single high-dose vitamin D (600,000 IU) with oral calcium

D. Daily high-dose vitamin D (60,000 IU) for 10 days

16. A 10-year-old boy presents with progressive proximal muscle weakness and calf hypertrophy. Genetic testing reveals a mutation in the dystrophin gene. Which of the following statements about the diagnosis is true?

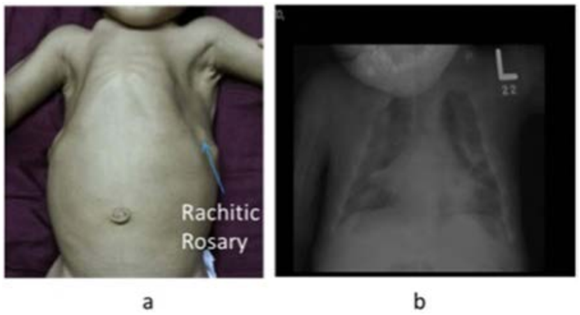
A. Most cases are autosomal dominant.

B. It is caused by the complete loss of dystrophin.

C. It is due to frameshift mutations causing a truncated protein.

D. It is milder with slow progression

17. A 7-year-old child presents with delayed dentition and prominent frontal bossing. During the physical examination, non-tender, round deformity over the chest is noted as shown in the image given below. What is the most likely diagnosis in this case?



- A. Acromegaly
- B. Scurvy
- C. Rickets
- D. Osteomalacia

18. A 5-year-old child presents with bone pain and noticeable deformities, including bowed legs. The child has a diet predominantly consisting of processed foods with minimal dairy products and very limited outdoor activity. What is the most common cause of the child’s condition?

- A. Vitamin D deficiency
- B. Calcium deficiency
- C. Phosphate deficiency
- D. Fanconi syndrome

19. Which of the following statements about the management of spinal muscular atrophy is true?

- A. Nusinersen is administered intravenously for increasing SMN protein production.
- B. Gene therapy focuses on delivering a functional SMN2 gene intrathecally.
- C. AVXS-101 is used for spinal muscular atrophy Type I.
- D. Olesoxime is an oral medication that increases SMN protein production.

20. A 4-month-old baby with severe hypotonia is brought to the hospital and has a weak cough and lie, as shown in the image. Which type of SMA is the child suffering from?

A. Type 0

B. Type II

C. Type I

D. Type IV

21. A baby at birth has profound weakness of the limbs and trunk and severe hypotonia. On examination, there are facial diplegia and joint contractures. The neonate is in respiratory distress. Which type of disease in this scenario is this baby affected by?

A. Werdnig-Hoffmann: Type I

B. Intermediate: Type II

C. Kugelberg-Wulander: Type III

D. Very severe: Type 0

22. A 2-day-old newborn baby developed respiratory insufficiency, difficulty in sucking and swallowing, and generalized hypotonia. There is little spontaneous motor activity and no respiratory infection. The mother has a history of myasthenia gravis. What would be the cause of this disease in the infant?

A. Choline acetyltransferase deficiency

B. Endplate AChE deficiency

C. MuSK deficiency

D. Placental transfer of anti-AChR antibodies

23. A child is being treated for juvenile myasthenia gravis and suddenly develops symptoms like diarrhea, cramping, excessive sweating, excessive urination, and salivary secretion. Which of the following drugs used for the treatment of myasthenia gravis caused this?

A. Prednisone

B. Atropine

C. Rituximab

D. Pyridostigmine

24. A worried pair of parents came to the pediatrician as their child was showing weakness, difficulty in swallowing, and inability to coordinate her eye movements. The child is unable to raise her hands higher, especially towards the end of the day. Her serological evaluation for acetylcholine receptor antibodies came back negative. What would be the next best investigation to confirm the diagnosis?

A. Antinuclear antibody detection

B. Repetitive nerve stimulation (RNS)

C. Electromyography

D. Muscle-specific tyrosine Kinase (MuSK) Antibody detection

25. A 7-year-old unimmunized child presents with sudden onset of asymmetric leg weakness and reduced reflexes. Sensation remains intact. Lumbar puncture reveals lymphocytic pleocytosis with normal or slightly increased protein levels. What is the most likely diagnosis?

A. Guillain-Barré Syndrome

B. Transverse Myelitis

C. Poliomyelitis

D. Traumatic Neuritis

26. A pediatrician is evaluating an infant with generalized muscle weakness and hypotonia with a history of feeding difficulties and delayed motor response. He suspects a congenital myopathy and orders genetic testing along with a muscle biopsy. Which of the following conditions is not classified as a type of congenital myopathy?

A. Nemaline Myopathy

B. Duchenne Muscular Dystrophy

C. Centronuclear Myopathy

D. Core Myopathy

27. A mother brings her 7-year-old son to the clinic because he has difficulty buttoning his shirt. She notes that his eyes remain partially open during sleep. Upon examination, the child exhibits asymmetric wrist and finger weakness, a puckered and rounded mouth, and winging of scapula. What is the most likely diagnosis?

A. Emery-Dreifuss muscular dystrophy

B. Facioscapulohumeral muscular dystrophy

C. Congenital muscular dystrophies

D. Limb-girdle muscular dystrophy

28. A 6-year-old girl presents with decreased muscle tone. Physical examination reveals a high-arched palate, an inverted V-shaped upper lip, and prominent facial muscle wasting. During physical examination, the patient was asked to make tight fists and then quickly open her hands, but a delay in the relaxation of the muscles was noted. What is the most likely diagnosis?

A. Duchenne muscular dystrophy

B. Becker muscular dystrophy

C. Emery-Dreifuss muscular dystrophy

D. Myotonic dystrophy

29. A 6-year-old boy with a history of progressive muscle weakness and elevated serum CK levels is being evaluated for Duchenne Muscular Dystrophy (DMD). Genetic testing did not reveal any deletions or duplications in the dystrophin gene. What would be the next best step in confirming the diagnosis?

A. Initiation of glucocorticoid therapy without further testing

B. Performing a muscle biopsy with dystrophin immunohistochemistry

C. Starting exon-skipping therapy with Eteplirsen based on clinical suspicion

D. Conducting an electromyography (EMG) to assess for myopathic changes

30. A 4-year-old boy presents with a history of delayed motor milestones. He began walking at 18 months and has recently started exhibiting a waddling gait and difficulty climbing stairs. On examination, the boy displays a positive Gowers sign and hypertrophy of the calf muscles. His serum creatine kinase levels are markedly elevated. What is the most probable diagnosis?

A. Duchenne muscular dystrophy

B. Becker muscular dystrophy

C. Emery-Dreifuss muscular dystrophy

D. Myotonic muscular dystrophy

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	1
Question 5	2
Question 6	3
Question 7	1
Question 8	4
Question 9	3
Question 10	4
Question 11	4
Question 12	3
Question 13	1
Question 14	1
Question 15	1
Question 16	4
Question 17	3
Question 18	1
Question 19	3
Question 20	3
Question 21	4
Question 22	4

Question 23	4
Question 24	4
Question 25	3
Question 26	2
Question 27	2
Question 28	4
Question 29	2
Question 30	1

Solution for Question 1:

Correct Answer: A) Hypophosphatemic rickets

Explanation:

The child’s symptoms and lab results of normal serum calcium, low serum phosphate, high alkaline phosphatase and normal 25-hydroxyvitamin D point towards diagnosis of Hypophosphatemic Rickets.

Laboratory Diagnosis:

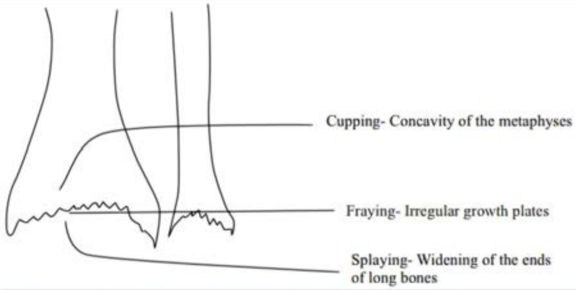
- Serum calcium,
- Serum phosphorus,
- Alkaline phosphatase (ALP),
- Parathyroid hormone (PTH),
- 25-hydroxyvitamin D (25-(OH)D),
- 1,25-dihydroxyvitamin D (1,25-(OH)D),
- Creatinine and electrolytes.

Disorder	Calcium	Pi	PTH	25-(OH)D	1,25-(OH)2D	ALP	Urine Calcium	Urine Pi
Vitamin D deficiency (Option B)	N,■	■	■	■	■,N,■	■	■	■

Vitamin D dependent rickets Type 1 (Option C)	N,■	■	■	N,■	N,■	■	■	■
Vitamin D dependent rickets Type 2 (Option D)	N,■	■	■	N	■■	■	■	■
Chronic kidney disease	N,■	■	■	N	■	■	N,■	■
X- linked hypophosphatemic rickets	N	■	N, ■	N	■	■	■	■
Autosomal dominant hypophosphatemic rickets (Option A)	N	■	N	N	■	■	■	■
Hereditary hypophosphatemic rickets with hypercalciuria	N	■	N, ■	N	■	■	■	■
Autosomal recessive hypophosphatemic rickets	N	■	N	N	■	■	■	■
Tumor-induced rickets	N	■	N	N	■	■	■	■
Fanconi syndrome	N	■	N	N	■ or ■	■	■ or ■	■
Dent’s disease	N	■	N	N	N	■	■	■
Dietary calcium deficiency	N,■	■	■	N	■	■	■	■

Radiological Changes:

- Decreased Calcification: Leads to the thickening of the growth plate.
- Fraying: Initial change is the loss of the normal zone of provisional calcification adjacent to the metaphysis, which appears as a blurring or frayed appearance of the metaphyseal margin.
- Cupping: Edge of the metaphysis changes from a convex or flat surface to a more concave surface, particularly noticeable at the distal ends of the radius, ulna, and fibula.
- Splaying: Cartilage hypertrophy leads to the widening of the growth plate resulting in the widening (splaying) of the metaphyseal ends.
- Generalized reduction in bone density: Osteopenia
- Widening of the Metaphysis



Solution for Question 2:

Correct Answer: A) Juvenile Myasthenia Gravis

Explanation:

Myasthenia Gravis is a chronic autoimmune disease of the neuromuscular junction (postsynaptic endplate) that leads to abnormal neuromuscular transmission or blockade.

- In children it is called Juvenile Myasthenia Gravis.

Clinical features:

- Symptoms can show from the age of 11 months to 17 years
- Characteristic feature: rapid fatigue of muscles.

Ocular symptoms:

- Asymmetric ptosis
- Progressive extraocular muscle weakness
- Diplopia

Bulbar symptoms:

- Dysphagia
- Disarthria
- Facial weakness

Limb weakness:

- Neckflexor weakness: difficulty holding the head up and poor head control.
- Arm weakness: weakness of limb girdle muscle and distal muscles of hands cause difficulty raising arms or performing tasks requiring strength.

Other characteristics:

- Fluctuating weakness
- Normal reflexes
- Absence of sensory symptoms

Complications:

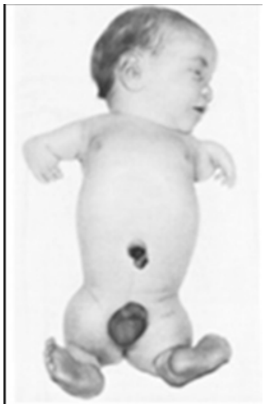
- Respiratory muscle paralysis.
- Myasthenic crisis: acute or subacute increase in weakness caused by an infection, surgery or emotional stress. Treated by IV cholinesterase inhibitors , immunoglobulins , plasma exchange, gavage feeding and if needed transient ventilator support.
- Infants of a mother with MG can have transient neonatal myasthenia characterized by respiratory insufficiency , inability to suck or swallow, generalized hypotonia and weakness.

Solution for Question 3:

Correct Answer: A) Thalidomide exposure

Explanation:

- Phocomelia is a congenital defect in which the limbs lack intermediate segments.
- As a result, the hands or feet attach directly to the trunk, resembling seal flippers.
- Thalidomide, a drug historically used to treat anxiety and morning sickness, is known for its teratogenic effects that lead to this condition, particularly when taken during pregnancy.



Vitamin D deficiency (Option B) leads to conditions like rickets, characterized by bone softening and deformities, but it does not result in the specific limb malformations seen in phocomelia.

While genetic mutations (Option C) can cause various congenital anomalies, the distinct history of taking medication for anxiety (thalidomide use) is a key factor pointing specifically to phocomelia.

Impaired blood flow to the developing limbs (Option D) can cause limb defects, but the unique presentation of phocomelia, along with the maternal history of thalidomide use, makes this less likely than thalidomide exposure.

Solution for Question 4:

Correct Answer: A) Defective type I collagen synthesis

Explanation:

- Osteogenesis Imperfecta (OI) is caused by mutations in the genes responsible for producing type I collagen, which is the primary component of the bone matrix.
- This defective collagen synthesis leads to brittle bones, resulting in frequent fractures from minimal trauma.

Additional clinical features like blue sclerae, hearing loss, and joint hypermobility further support the diagnosis of OI.

Features of Osteogenesis Imperfecta

Pathophysiology	Defective type I collagen synthesis
Genetic Mutations	Type I-IV: COL1A1, COL1A2 Type V: IFITM5 Type VI: SERPINF1
Key Clinical Features	Recurrent fractures from minimal trauma, blue sclerae, hearing loss, joint hypermobility, osteopenia
Complications	Cardiopulmonary issues (pneumonia, cor pulmonale), neurological complications (basilar invagination, brainstem compression)
Differential Diagnosis	Rickets, osteopetrosis, and disorders related to abnormal calcium metabolism
Treatment	Physical rehabilitation, Bisphosphonates, growth hormone, orthopedic management (fracture care, surgery)

Deficiency of vitamin D (Option B) typically leads to conditions like rickets or osteomalacia, characterized by bone pain, muscle weakness, and deformities like bowing of the legs.

- While vitamin D deficiency can cause osteopenia, it is not associated with the other features seen in this child, such as blue sclerae and hearing loss.

Impaired osteoclast function (Option C) is seen in osteopetrosis, a condition characterized by overly dense but brittle bones that lead to fractures.

- Osteopetrosis is not associated with blue sclerae, hearing loss, or joint hypermobility, which distinguishes it from OI.

Abnormal calcium metabolism (Option D) can result in conditions like hypercalcemia or hypocalcemia, which can affect muscle function and potentially lead to bone density changes.

- However, this does not explain the combination of blue sclerae, frequent fractures, and hearing loss seen in OI.

Solution for Question 5:

Correct Answer: B) Hematopoietic stem cell transplantation (HSCT)

Explanation:

- This infant's features of macrocephaly, hepatosplenomegaly, and recurrent fractures, with a radiograph showing diffuse bone sclerosis and a “bone within bone” appearance, are consistent with autosomal recessive osteopetrosis aka marble bone disease.
- Hematopoietic stem cell transplantation (HSCT) is the most effective treatment for this severe disease.
- HSCT addresses the underlying osteoclast dysfunction due to failure of osteoclast to resorb bone, by replacing the defective bone marrow with healthy donor cells capable of normal bone resorption.
- It can prevent or reverse bone abnormalities if performed before irreversible complications, such as vision loss, develop.
- Autosomal Dominant Osteopetrosis, also known as Albers-Schönberg disease or Marble bone disease, is more common and less severe. It is generally asymptomatic and presents in childhood or adolescence with fractures, mild anemia, and sometimes cranial nerve dysfunction.

Features of Osteopetrosis

Aspect	Severe Autosomal Recessive Osteopetrosis	Mild Autosomal Dominant Osteopetrosis
Incidence	~1/250,000 births	~1/20,000 births
Genetic Mutations	TCIRG1, CLCN7, OSTM1, SNX10, TNFSFR11A, TNFSF11, PLEKHM1	CLCN7
Onset	Infancy	Childhood or adolescence
Clinical Features	Macrocephaly, hepatosplenomegaly, vision/hearing loss, severe anemia,	Fractures, mild anemia, possible cranial nerve dysfunction, dental

	fractures	issues, osteomyelitis
Radiographic Findings	Diffuse bone sclerosis, "bone within bone" appearance	Increased bone density, clubbed metaphyses, "sandwich" vertebrae
Prognosis	Often fatal in infancy, survivors may live into the 2nd decade with disabilities	Generally good, may be asymptomatic or have mild symptoms
Treatment	HSCT, RANKL replacement, calcitriol, interferon-γ, supportive care	Symptomatic management

Vitamin D supplementation (Option A) is not sufficient to correct the underlying osteoclast dysfunction in osteopetrosis.

- While Vitamin D can help manage calcium levels, it does not address the genetic cause of the disease.

Bisphosphonates (Option C) inhibit bone resorption and would further worsen the bone density issues in osteopetrosis.

- This treatment is not appropriate for a condition where the problem is already excessive bone density.

Physical therapy and fracture management (Option D) are important for supportive care. They do not prevent the progression of the underlying disease.

- The root cause, osteoclast dysfunction, must be addressed with HSCT.

Solution for Question 6:

Correct Answer: C) Typically follows a Gastrointestinal or genitourinary infection

Explanation:

- Reactive arthritis often follows a gastrointestinal or genitourinary infection, commonly occurring within 3 days to 6 weeks after the initial infection.
- The presentation of asymmetric arthritis of the lower extremities, dactylitis, and erythematous plaques 4 weeks after an episode of dysentery strongly suggests this diagnosis.
- Markers of inflammation such as ESR, CRP, and platelets are elevated in early disease. The classic triad of arthritis, urethritis, and conjunctivitis is not commonly seen in children.

Features of Reactive Arthritis

Feature	Details
Typical Trigger	Recent gastrointestinal (e.g., Salmonella, Shigella, Yersinia, Campylobacter) or genitourinary (e.g., Chlamydia) infection (Option C)
Onset	3 days to 6 weeks post-infection
Joint Involvement	Asymmetric oligoarthritis (Option B), predominantly in the lower extremities
Extra-Articular Manifestations	Dactylitis, enthesitis, erythematous plaques (keratoderma blennorrhagica)
HLA-B27 Association	Increases risk of chronic spondyloarthritis and persistent GI inflammation (Option A)
Systemic Manifestations	Fever, malaise, fatigue; Less common: conjunctivitis, optic neuritis, aortic valve involvement
Lab values	Elevated ESR, CRP, and platelets in early disease
Primary Treatment	NSAIDs; Corticosteroids for severe cases; Antibiotics only if ongoing infection (Option D)
Complications	Chronic spondyloarthritis, IBD, uveitis, carditis

Solution for Question 7:

Correct Answer: A) Methotrexate is used as it acts within a few days

Explanation:

Methotrexate is a disease-modifying anti-rheumatic agent used to treat JIA, but it takes 6-12 weeks to work.

Diagnosis:

- The steps in diagnosing JIA start with the clinical evaluation of the child. The symptoms, medical history, physical examinations, and exclusion of other conditions give a good grasp of the possible diagnosis.
- Investigations are the next step in diagnosis:

Investigations	Observations
Laboratory tests	CBC:

	Microcytic anemia Elevated platelets WBC Low platelets may indicate a macrophage activation system (MAS). ESR & CRP: Indicate inflammation Raised but could also be normal in children Falling ESR in systemic JIA can signal MAS ANA: Elevated in oligoarticular/polyarticular JIA Increases uveitis risk RF: Positive in 5-15% of polyarticular JIA cases ... (content truncated)
Radiological findings	 ... (content truncated)

Treatment of JIA:

Treatment goals	Achieve remission: minimal or no symptoms Prevent joint damage: early, aggressive treatment to minimize long-term damage Support growth: ensure normal bone growth and development

Treatment	NSAIDs: Reduce pain and inflammation Side effects include gastritis and toxicity Disease-modifying anti-rheumatic drugs (DMARDs): Methotrexate is effective but may take 6-12 weeks (Option A) Side effects include nausea and liver toxicity Sulfasalazine & Leflunomide are alternatives for polyarticular JIA Biologics: Target-specific immune system components Includes TNF inhibitors(abatacept & tocilizumab) (Option B ruled out) Side effects are immunosuppression and injection site reactions IL-1 Inhibitors: Effective for systemic JIA ... (content truncated)

Learning Objective:

Differential diagnosis:

- Rheumatic diseases: SLE, juvenile dermatomyositis, sarcoidosis, scleroderma, and vasculitis.
- Infections: Viral (parvovirus B-19, Epstein-Barr), bacterial, Lyme disease, and septic arthritis.
- Other: ERA, psoriatic arthritis, IBD, hypermobility, malignancies (leukemia), and congenital/metabolic disorders.

Solution for Question 8:

Correct Answer: D) Positive rheumatoid factor

Explanation:

The presence of symmetrical (≥5 large) joint involvement and nodules point to the diagnosis of rheumatoid factor-positive polyarticular juvenile idiopathic arthritis/ RF-positive polyarthritis.

Polyarthritis:

- Inflammation of 5 or more joints in both upper and lower limbs.

Prevalence: Affects five or more joints within the first six months.

Feature	Rheumatoid Factor (RF)-Negative	Rheumatoid Factor (RF)-Positive
Prevalence	More common (20-35% of all JIA cases)	Less common (<10%)
Joint pattern	 ... (content truncated)	 ... (content truncated)
Clinical features	 ... (content truncated)	 Rheumatoid nodules (10%) Low-grade fever
Diagnosis	ANA positive in 40% RF negative ESR is slightly or very high CRP is raised or normal Mild anemia	RF positive ESR is very high CRP is raised or normal Mild anemia
Prognosis	Variable Standard therapy with methotrexate (MTX) and NSAIDs Biologics considered if unresponsive to MTX and NSAIDs such as anti-TNF agents and abatacept	Poorer prognosis Early aggressive therapy is needed Long-term remission unlikely Predictors of severe disease include Young age at onset RF seropositivity Rheumatoid nodules Many affected joints

Solution for Question 9:

Correct Answer: C) Persistent oligoarthritis

Explanation:

The involvement of four or fewer joints in the first six months of the onset of the disease and the involvement of the same 4 joints throughout this time indicates the diagnosis of persistent type oligoarthritis.

- Juvenile idiopathic arthritis (JIA) is the most common rheumatic disease in children.
- It is a heterogeneous group of disorders with clinical features similar to arthritis.
- Any subtype of JIA can be diagnosed only if there is arthritis for 6 or more weeks.

Etiology (Causes):

The etiology and pathogenesis of JIA are unknown, but it is believed that an immunogenetic susceptibility and an external trigger are necessary, and genetic factors contribute as well.

- Genetic Factors: Genetic Component: Family and twin studies suggest a genetic predisposition. MHC Genes: Variations in major histocompatibility complex (MHC) genes are linked to different JIA subtypes. Non-MHC Genes: Polymorphisms in genes such as PTPN22, TNF-α, macrophage inhibitory factor, and certain interleukins are implicated.
- Genetic Component: Family and twin studies suggest a genetic predisposition.
- MHC Genes: Variations in major histocompatibility complex (MHC) genes are linked to different JIA subtypes.
- Non-MHC Genes: Polymorphisms in genes such as PTPN22, TNF-α, macrophage inhibitory factor, and certain interleukins are implicated.
- Genetic Component: Family and twin studies suggest a genetic predisposition.
- MHC Genes: Variations in major histocompatibility complex (MHC) genes are linked to different JIA subtypes.
- Non-MHC Genes: Polymorphisms in genes such as PTPN22, TNF-α, macrophage inhibitory factor, and certain interleukins are implicated.
- Environmental Triggers: Infections: Bacterial and viral infections may trigger JIA in genetically susceptible individuals. Heat Shock Proteins: Abnormal immune responses to bacterial or mycobacterial heat shock proteins might be involved. Hormonal Factors: Altered reproductive hormone levels may contribute. Joint Trauma: Injury to a joint may play a role.

- Infections: Bacterial and viral infections may trigger JIA in genetically susceptible individuals.
- Heat Shock Proteins: Abnormal immune responses to bacterial or mycobacterial heat shock proteins might be involved.
- Hormonal Factors: Altered reproductive hormone levels may contribute.
- Joint Trauma: Injury to a joint may play a role.

- Infections: Bacterial and viral infections may trigger JIA in genetically susceptible individuals.
- Heat Shock Proteins: Abnormal immune responses to bacterial or mycobacterial heat shock proteins might be involved.
- Hormonal Factors: Altered reproductive hormone levels may contribute.
- Joint Trauma: Injury to a joint may play a role.

Subtypes of JIA:

Systemic JIA (sJIA) (Option B)	Oligoarthritis	Polyarthritis
Affects 5-15% of JIA cases.	Most common subtype (40-50% of cases).	Affects five or more joints within the first six months. (Option A and D)
 ... (content truncated)	 ... (content truncated)	 ... (content truncated)
RF-Positive Polyarthritis (Option D): Less common (<10% of cases). Resembles adult rheumatoid arthritis with symmetrical joint involvement. Rheumatoid nodules may be present and indicate a more aggressive course. Associated with a poorer prognosis and often requires aggressive treatment.		

Solution for Question 10:

Correct Answer: D) Wimberger sign

Explanation:

The X-ray and clinical symptoms point to the diagnosis of vitamin C deficiency, where the x-ray shows sclerotic rings or Wimberger signs.

- Vitamin C deficiency/ Scurvy is diagnosed based on characteristic clinical findings and detailed history, especially nutritional history.
- The laboratory and radiological examinations help confirm the diagnosis and identify its severity and associated deformities. They are as follows:

Laboratory findings	Radiological findings
Plasma ascorbate levels below 0.2 mg/dL indicate deficiency but are not reliable. Leukocyte vitamin C concentration is a better indicator but more challenging to measure.	X-ray White line of Frankel Subperiosteal hemorrhages are seen as calcified areas during healing Sclerotic rings/Wimberger sign Characteristic changes at long bone ends giving a “ground glass” appearance (Option A) Thin and dense cortex Specific signs: zone of rarefaction/Trummerfeld zone (lucent band) (Option B), Pelkan spur (Option C) Epiphyseal separation MRI: Detects acute and healing subperiosteal hematomas Periostitis Bone marrow changes Metaphyseal changes ... (content truncated)

Management of scurvy:

- Vitamin C supplementation: Dosage: 100-200 mg per day, orally or parenterally. Duration: Typically continue for up to 3 months to ensure complete recovery.
- Dosage: 100-200 mg per day, orally or parenterally.
- Duration: Typically continue for up to 3 months to ensure complete recovery.
- Treat underlying causes: Dietary Modification: Ensure infants receive vitamin C through breast milk, fortified formula, or complementary foods and incorporate vitamin C-rich foods into the diet of older children and adults. Food selectivity: Work with dietitians or behavioral

therapists to expand food choices in restrictive eating habits

- Dietary Modification: Ensure infants receive vitamin C through breast milk, fortified formula, or complementary foods and incorporate vitamin C-rich foods into the diet of older children and adults.
- Food selectivity: Work with dietitians or behavioral therapists to expand food choices in restrictive eating habits
- Monitoring and Follow-up: Regular follow-up to monitor treatment response and ensure adequate long-term vitamin C intake.
- Regular follow-up to monitor treatment response and ensure adequate long-term vitamin C intake.
- Dosage: 100-200 mg per day, orally or parenterally.
- Duration: Typically continue for up to 3 months to ensure complete recovery.
- Dietary Modification: Ensure infants receive vitamin C through breast milk, fortified formula, or complementary foods and incorporate vitamin C-rich foods into the diet of older children and adults.
- Food selectivity: Work with dietitians or behavioral therapists to expand food choices in restrictive eating habits
- Regular follow-up to monitor treatment response and ensure adequate long-term vitamin C intake.

Solution for Question 11:

Correct Answer: D) Iron

Explanation:

The scenario suggests a diagnosis of scurvy due to vitamin C deficiency. Since vitamin C enhances iron absorption, a lack of it can result in iron deficiency.

Role of vitamin C	Collagen synthesis: Vital for forming skin, cartilage, bones, and blood vessels Facilitates hydroxylation of lysine and proline in procollagen Neurotransmitter metabolism Synthesizes norepinephrine and serotonin by conversion of dopamine and tryptophan Cholesterol metabolism Conversion of cholesterol to steroid hormones and bile acids Carnitine biosynthesis Fatty acid transport Energy production Metalloenzyme function Maintains iron and copper (Option C) in their reduced and active state Antioxidant activity ... (content truncated)
Risk groups for deficiency	Infants fed heat-treated milk Destruction of vitamin C by heat Infants fed unfortified formula Individuals on restrictive diets Those avoiding fruits and vegetables Individuals with limited dietary choices Fad and limited diets Infants with delayed feeding Depletion of initial vitamin C stores Infants consuming boiled animal milk Vitamin C degradation Preterm infants Higher requirements for vitamin C
Clinical features	 ... (content truncated)

Zinc (Option A):

- Vitamin C and zinc both help enhance immunity and maintain health.
- These are often given together for the treatment of acne and scars.
- Both of them have been found helpful in treating the common cold.
- However, there is no conclusive evidence that interlinks their absorption.

Magnesium (Option B):

- Magnesium is a cofactor in more than 300 enzyme systems that regulate diverse biochemical reactions in the body, including protein synthesis, muscle and nerve function, blood glucose control, and blood pressure regulation.
- There is no evidence that the absorption of magnesium is affected by vitamin C.

Solution for Question 12:

Correct Answer: C) PHEX

Explanation:

Clinical features like short stature and leg bowing with lab results of low serum phosphate, elevated alkaline phosphatase, increased urinary phosphate and genetic testing showing mutation on the X chromosome point towards a diagnosis of X-linked hypophosphatemic rickets.

- In X-linked hypophosphatemic rickets, the mutated gene is PHEX (a phosphate-regulating gene with homology to endopeptidases on the X chromosome).

X-Linked Hypophosphatemic (XLH) Rickets	Most common among genetic disorders causing rickets
Prevalence	1 in 20,000
Inheritance	X-linked dominant
Gene Mutated	PHEX (phosphate-regulating gene with homology to endopeptidases on X chromosome) Normally, product of this gene appears to have an indirect role in inactivating FGF-23
Pathophysiology	
Clinical Manifestations	Lower limb deformities: Coxa vara Genu valgum Genu varum Poor growth Maxillofacial abnormalities Skull deformities: Due to premature fusion of cranial sutures Delayed dentition Short stature

Laboratory Findings	High renal phosphate excretion Hypophosphatemia Elevated alkaline phosphatase (ALP) Normal PTH and serum calcium Low or inappropriately normal levels of 1,25-D
Treatment	Phosphorus Supplementation: 1-3 g elemental phosphorus daily, divided into 4-5 doses Can be given in the form of Joulie solution (30.4 mg phosphate/mL) or as neutral phosphate effervescent tablets Frequent dosing helps to prevent prolonged decrements in serum phosphorus because there is rapid decline after each dose Calcitriol: 30-70 ng/kg/day in 2 doses Burosumab-twza: Monoclonal antibody targeting FGF-23, an alternative for children >1 year
Prognosis	Generally good response to therapy, though frequent dosing can affect compliance

SLC34A3 (Option A) (Sodium-phosphate cotransporter gene) mutations are associated with hereditary hypophosphatemic rickets, not with hypercalciuria, not with X-linked hypophosphatemic rickets.

FGF23 (Option B): While FGF23 (Fibroblast growth factor-23) levels are elevated in X-linked hypophosphatemic rickets (XLH), the gene responsible for XLH is PHEX, not FGF23 itself. FGF23 mutation is seen in autosomal dominant hypophosphatemic rickets.

DMP1 (Option C) (Dentin matrix protein 1) mutation causes autosomal recessive hypophosphatemic rickets, not X-linked hypophosphatemia.

Solution for Question 13:

Correct Answer: A) Vitamin D-Dependent Rickets Type 1A

Explanation:

Child presents with classic rickets symptoms like bowed legs and frontal bossing and lab results showing low calcium, low phosphorus, normal 25-D, low 1,25-D, and elevated PTH point towards diagnosis of Vitamin D-Dependent Rickets Type 1A.

Feature	Vitamin D Dependent Rickets (VDDR) Type 1	Vitamin D Dependent Rickets (VDDR) Type 2		
Inheritance	Autosomal Recessive	Autosomal Recessive		
Subtypes	VDDR type 1A (Option A)	VDDR type 1B (Option B)	VDDR type 2A (Option C)	VDDR type 2B (Option D)
Genetic Mutation	Mutation in renal 1α-hydroxylase gene	Mutation in 25-hydroxylase gene	Mutation in the vitamin D receptor (VDR) gene	Overexpression of a hormone response element-binding protein
Pathophysiology	Impaired conversion of 25-D to 1,25-D due to defective 1α-hydroxylase	Impaired production of 25-D	Defective vitamin D receptor prevents normal response to 1,25-D	Interference with 1,25-D action due to binding protein
25-Hydroxy vitamin D (25-D)	Normal	Low	Normal	Normal
1,25-Dihydroxy vitamin D (1, 25-D)	Low or lower limit of normal	Normal	Extremely elevated	Extremely elevated
Parathyroid Hormone (PTH)	Elevated	Elevated		
Serum Phosphorus	Low	Low		

Clinical Features	Presents in the first 2 years of life Alopecia absent Presents like vitamin D deficiency rickets Hypotonia Growth failure Motor retardation (poor head control, delayed standing and walking) Convulsions due to hypocalcemia	Presents during infancy or childhood 50-70% have alopecia (ranging from alopecia areata to alopecia totalis) Hypocalcemia Secondary hyperparathyroidism Absence of response to vitamin D or its analogs Ectodermal defects: Oligodontia Milia Epidermal cysts		
Treatment	VDDR1A: Long-term 1,25-D (calcitriol) Initially 0.25-2 µg/day and lower doses used once rickets has healed Adequate calcium (30-50 mg/kg) Calcitriol Radiologically healing occurs within 6 to 8 weeks. Thereafter supplementation with physiological doses of 1α-hydroxy D3 or 1,25-D is continued lifelong. VDDR1B: Pharmacologic doses of vitamin D■ (3,000 IU/day)	Extremely high doses of vitamin D (25-D or 1,25-D) and oral calcium (1,000-3,000 mg/day) All patients should be given a 3-6 months trial of high doses of vitamin D and oral calcium Non-responders may need long-term IV calcium Response to treatment is not satisfactory Clinical, biochemical improvement and radiological healing following long-term administration of large amounts of intravenous or oral calcium		

Solution for Question 14:

Correct Answer: A) Renal tubular acidosis associated rickets

Explanation:

- The child's lab results of low serum phosphate, low bicarbonate, normal calcium, serum pH of 6.5 and normal creatinine suggest renal tubular acidosis-associated rickets.
- This condition is a common cause of refractory rickets in children, characterized by polyuria, rachitic deformities, short stature, and failure to thrive, and investigation reveals normal anion gap metabolic acidosis with normal blood levels of urea and creatinine.

Table rendering failed

Vitamin D-dependent rickets (Option A) usually present with low serum calcium and elevated alkaline phosphatase but not metabolic acidosis or low serum pH.

Hypophosphatemic rickets (Option B) usually present with low serum phosphate but would not typically present with low bicarbonate and normal serum pH.

Chronic kidney disease-associated rickets (Option C) usually present with elevated serum phosphorus and creatinine levels, which are normal in this case. The symptoms can overlap with RTA's, but the lab results do not fit CKD.

Solution for Question 15:

Correct Answer: A) Daily low-dose vitamin D (2000 IU) with oral calcium

Explanation:

Child symptoms, including delayed growth, bowed legs and a history of low vitamin D intake and inadequate sun exposure, combined with X-ray findings of metaphyseal cupping and fraying, point towards a diagnosis of nutritional rickets.

- The recommended treatment for a 2-year-old with nutritional rickets involves daily low-dose vitamin D (2000 IU) for 12 weeks, along with oral calcium supplementation.
- This regimen is effective and minimizes the risk of adverse effects compared to high-dose or infrequent-dosing options.

Treatment of Nutritional Rickets:

1. Vitamin D Supplementation:

- Stoss Therapy: Administer 300,000-600,000 IU of vitamin D orally or intramuscularly over 1 day (2-4 doses). Vitamin D■ is preferred due to its longer half-life. Ideal for patients with questionable adherence.
- Administer 300,000-600,000 IU of vitamin D orally or intramuscularly over 1 day (2-4 doses).
- Vitamin D■ is preferred due to its longer half-life.
- Ideal for patients with questionable adherence.
- Alternative Strategy: Daily Vitamin D minimum dose of 2,000 IU/day for at least 3 months.
- Follow-Up: Both strategies should follow up with 400 IU/day (<1 year old) or 600 IU/day (>1 year old).
- Administer 300,000-600,000 IU of vitamin D orally or intramuscularly over 1 day (2-4 doses).
- Vitamin D■ is preferred due to its longer half-life.
- Ideal for patients with questionable adherence.

2. Calcium Supplementation: Provide oral calcium (30-75 mg/kg/day, maximum 500 mg).

3. Phosphorus Supplementation

4. Monitoring: Radiological improvement (white line of healing in the metaphyses) should be visible within 4 weeks.

Age	Daily dose of vitamin D3 for 12 weeks	Alternative intermittent dose regimen	Daily calcium supplementation	Daily maintenance dose (follow up)
<6 months	2000 IU	Not recommended	30-75 mg/kg/day	400 IU
6-12 months	2000 IU	Equivalent of 2000 IU/ day may be given on a weekly or monthly basis	30-75 mg/kg/day	400 IU
>12 months	3000IU	60,000 IU every 2 weeks for 5 doses	30-75 mg/kg/day (up to 500 mg)	600 IU

Solution for Question 16:

Correct Answer: D) It is milder with slow progression

- The presentation of progressive proximal muscle weakness and calf hypertrophy in a 10-year-old, with Genetic testing revealing a mutation in the dystrophin gene, is consistent with Becker Muscular Dystrophy (BMD).
- Becker Muscular Dystrophy (BMD) has an early onset and shows rapid progression.
- It is an X-linked recessive disease that often results from In-frame mutations of the Dystrophin gene that preserve the reading frame, leading to a partially functional dystrophin protein, which contributes to the milder phenotype.
- It usually presents during later childhood (after 5-7 years) or early adolescence.

DMD vs BMD

Aspect	Duchenne Muscular Dystrophy (DMD)	Becker Muscular Dystrophy (BMD)
Type of Mutation	Frameshift mutations leading to a disrupted reading frame (Option C)	In-frame mutations that preserve the reading frame
Dystrophin Protein	Dystrophin deficient Truncated and unstable (< 3% of normal) (Option C)	Abnormal dystrophin (Option B) Partially functional (20–90% of normal)
Clinical Severity	Severe, early onset, rapid progression	Milder, later onset, slower progression (Option D)
Inheritance Pattern	X-linked recessive	X-linked recessive (Option A)
Age of onset	Early childhood (Before 5 years)	Later childhood (After 5-7 years)

Solution for Question 17:

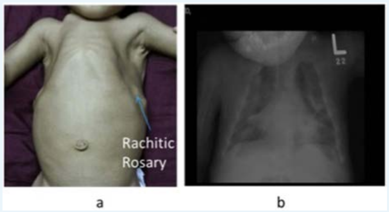
Correct Answer: C) Rickets

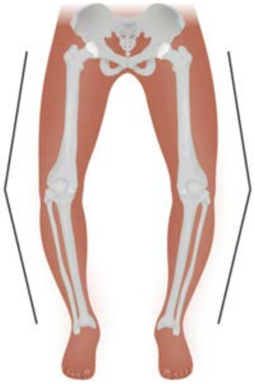
Explanation:

- The given image shows rachitic rosary, which is a characteristic finding of rickets.
- Widening of the costochondral junction results in a rachitic rosary, which feels like a rosary bead as the examiner’s finger moves along the costochondral junctions from rib to rib.

Clinical Features of Rickets:

General: Failure to thrive (malnutrition) Listlessness Protruding abdomen Muscle weakness (especially proximal) Hypocalcemic dilated cardiomyopathy Fractures (pathologic, minimal trauma) Increased intracranial pressure Head: Craniotabes: Softening of cranial bones Frontal bossing Delayed fontanel closure (usually closed by 2 yr) Delayed dentition ... (content truncated)	Chest: Rachitic rosary: Due to widening of the costochondral junction Harrison groove: Horizontal depression along the lower anterior chest due to pulling of softened ribs by the diaphragm during inspiration Respiratory infections and atelectasis Back: Scoliosis Kyphosis Lordosis Extremities: Enlargement of wrists and ankles: Due to growth plate widening Valgus or varus deformities Windswept deformity (valgus deformity of one leg with varus deformity of other leg) Anterior bowing of tibia and femur ... (content truncated)

Aspect	Scorbutic Rosary (Option B)	Rachitic Rosary (Option C)
Underlying Deficiency	Vitamin C deficiency (scurvy) leads to impaired collagen synthesis.	Vitamin D deficiency (rickets) is caused by an unmineralized matrix at the growth plate in children only before the fusion of the epiphysis.
Features	 Beads at costochondral junctions have sharper angulation. Scorbutic rosary is tender.	 Beads have a more rounded appearance Rachitic rosary is non-tender
Additional Clinical Features	Scurvy: Irritability, anorexia, musculoskeletal pain, leg swelling, pseudoparalysis, hemorrhagic manifestations (petechiae, purpura, gum bleeding), and specific radiographic findings in long bones.	Rickets: Delayed closure of fontanelles, craniotabes (soft skull bones), Harrison's groove (indentation along lower ribs), skeletal deformities (bowlegs, knock-knees), and muscle weakness.



Acromegaly (Option A), caused by excess growth hormone, leads to a large lower jaw, prominent forehead, and enlarged hands and feet in adults and occurs after growth plate fusion.

- It does not present the rachitic rosary seen in rickets.

Osteomalacia (Option C) is a condition of bone softening due to prolonged vitamin D deficiency in adults.

- In contrast, rickets results from deficient mineralization at the growth plate in children and is characterized by features like the rachitic rosary appearance, which is not seen in osteomalacia.

Solution for Question 18:

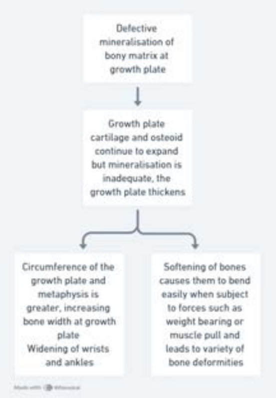
Correct Answer: A) Vitamin D deficiency

Explanation:

- Child with bone pain and noticeable deformities like bowed legs are suggestive of Rickets.
- Vitamin D deficiency is the most common cause of rickets, especially in children with inadequate sun exposure and a diet lacking in vitamin D-rich foods.

Rickets: Disease of growing bone caused by unmineralized matrix at the growth plates in children only before fusion of the epiphysis.

Pathophysiology of Rickets:



Etiology of Rickets:

Vitamin D Disorders:	Phosphorus Deficiency: (Option C) Inadequate intake

Nutritional vitamin D deficiency (Most common cause) (Option A) Congenital vitamin D deficiency Secondary vitamin D deficiency Malabsorption Increased degradation Decreased liver 25-hydroxylase Vitamin D-dependent rickets types 1A and 1B Vitamin D-dependent rickets types 2A and 2B Chronic kidney disease Calcium Deficiency: (Option B) Low intake ... (content truncated)	Premature infants (rickets of prematurity) Aluminum-containing antacids Renal Losses: X-linked hypophosphatemic rickets Autosomal dominant hypophosphatemic rickets Autosomal recessive hypophosphatemic rickets types 1 and 2 Hereditary hypophosphatemic rickets with hypercalciuria Overproduction Or fibroblast growth factor-23 Tumor-induced rickets McCune-Albright syndrome Epidermal nevus syndrome ... (content truncated)

Solution for Question 19:

Correct Answer: C) AVXS-101 is used for spinal muscular atrophy type I

Explanation:

AVXS-101 is used in gene therapy to deliver a functional SMN1 gene intravenously for SMA Type I.

- Proactive and supportive care is necessary to improve the quality of life and maximize functional independence.
- Disease-Modifying Therapies: Nusinersen (Spinraza), an SMN antisense oligonucleotide (Option A), increases SMN protein production from the SMN2 gene through intrathecal administration. Onasemnogene AVXS-101 (Zolgensma): Gene therapy (Option B) delivering a functional SMN1 gene intravenously for SMA Type I. Risdiplam (Evrysdi): Oral medication increasing SMN protein production Neuroprotective agents (Olesoxime) (Option D) to protect motor neurons.
- Nusinersen (Spinraza), an SMN antisense oligonucleotide (Option A), increases SMN protein production from the SMN2 gene through intrathecal administration.
- Onasemnogene AVXS-101 (Zolgensma): Gene therapy (Option B) delivering a functional SMN1 gene intravenously for SMA Type I.
- Risdiplam (Evrysdi): Oral medication increasing SMN protein production
- Neuroprotective agents (Olesoxime) (Option D) to protect motor neurons.

- Nusinersen (Spinraza), an SMN antisense oligonucleotide (Option A), increases SMN protein production from the SMN2 gene through intrathecal administration.
- Onasemnogene AVXS-101 (Zolgensma): Gene therapy (Option B) delivering a functional SMN1 gene intravenously for SMA Type I.
- Risdiplam (Evrysdi): Oral medication increasing SMN protein production
- Neuroprotective agents (Olesoxime) (Option D) to protect motor neurons.
- Supportive Care: Respiratory Support: Non-invasive ventilation, tracheostomies Nutritional Support: Feeding tubes if needed Physical Therapy: Maintain muscle strength and range of motion Orthopedic Care: Manage scoliosis with braces or surgery Gastrointestinal Management: Address constipation, reflux
- Respiratory Support: Non-invasive ventilation, tracheostomies
- Nutritional Support: Feeding tubes if needed
- Physical Therapy: Maintain muscle strength and range of motion
- Orthopedic Care: Manage scoliosis with braces or surgery
- Gastrointestinal Management: Address constipation, reflux
- Respiratory Support: Non-invasive ventilation, tracheostomies
- Nutritional Support: Feeding tubes if needed
- Physical Therapy: Maintain muscle strength and range of motion
- Orthopedic Care: Manage scoliosis with braces or surgery
- Gastrointestinal Management: Address constipation, reflux
- Palliative Care: Symptom management, quality of life optimization.
- Genetic counseling for reproductive planning.
- Prenatal diagnosis for families with an index patient in the family.

Solution for Question 20:

Correct Answer: C) Type I

Explanation:

The above image shows the classical “frog-leg posture” due to severe hypotonia, which is seen in Type I SMA, where the hypotonia causes a “floppy appearance.”

Clinical features of Type1 or the classical type of Spinal Muscular Atrophy (SMA):

- Onset: Symptoms typically appear before six months of age, but some cases may present at birth. Antenatally, decreased fetal movements may be noticed Early motor delays: lack of head control
- Symptoms typically appear before six months of age, but some cases may present at birth.
- Antenatally, decreased fetal movements may be noticed
- Early motor delays: lack of head control
- Characteristic muscle weakness and hypotonia Severe generalized weakness affecting both limbs and trunk Proximal predominance greater weakness in proximal muscles (legs>arms) Hypotonia: leading to a “floppy appearance” Frog-leg posture resting position with hips abducted and knees flexed
- Severe generalized weakness affecting both limbs and trunk
- Proximal predominance greater weakness in proximal muscles (legs>arms)
- Hypotonia: leading to a “floppy appearance”
- Frog-leg posture resting position with hips abducted and knees flexed
- Absent or diminished reflexes (areflexia/hyporeflexia): Deep tendon reflexes are affected
- Tongue fasciculations are fine, rapid muscle twitches in the tongue
- Respiratory distress due to weakness in intercostal muscles leads to paradoxical breathing, a bell-shaped chest, a weak cough, and frequent respiratory infections.
- Feeding difficulties due to challenges in sucking and swallowing carry the risk of aspiration.
- Cognitive functions are preserved as the upper motor neurons are spared.
- Lack of eye muscle/ facial movements: Motor neurons of cranial nerves III, IV, and VI to the extraocular muscles and of the sacral spinal cord to the striated muscle of the urethral and anal sphincters are preserved, but progressive disease might lead to facial weakness later on.
- Symptoms typically appear before six months of age, but some cases may present at birth.
- Antenatally, decreased fetal movements may be noticed
- Early motor delays: lack of head control
- Severe generalized weakness affecting both limbs and trunk
- Proximal predominance greater weakness in proximal muscles (legs>arms)
- Hypotonia: leading to a “floppy appearance”
- Frog-leg posture resting position with hips abducted and knees flexed

SMA Type 0 (Option A) presents with severe hypotonia, facial diplegia, joint contractures, and cardiac abnormalities, but frog leg posture is not seen.

SMA Type II (Option B) has a proximal weakness, scoliosis, hand tremors, and difficulty in walking/standing.

SMA Type IV (Option D) is a mild disease that has slow-progressing muscle weakness and scoliosis and a near-normal lifespan

Solution for Question 21:

Correct Answer: D) Very severe: Type 0

Explanation:

The clinical scenario points to the diagnosis of Type 0 spinal muscular atrophy, which shows profound weakness of the limbs and trunk and severe hypotonia.

Spinal muscular atrophy:

Definition:

- A genetic disorder characterized by degeneration of anterior horn cells of spinal cord and resultant muscle atrophy and weakness. It is the most common genetic cause of infant mortality. Sensory neurons and intelligence are generally unaffected.

Causes:

- Mutations in the survival motor neuron 1 SMN1 gene on chromosome 5
- Most individuals have a homozygous deletion of SMN1
- SMN2 gene influences severity; higher SMN2 copy numbers can lead to milder disease

Spinal muscular atrophy is divided into the following types:

Type	Features	Onset	Clinical features	Prognosis	Genetics
Type 0 (very severe)	Prenatal onset: decreased intrauterine movements Severe hypotonia Respiratory distress Feeding difficulties Cranial nerve, cardiac system involvement involvement Congenital contractures	Prenatal or at-birth	Profound weakness Severe hypotonia Facial diplegia Joint contractures Cardiac abnormalities	Extremely poor; survival typically <1 year	Associated with a single copy of SMN2 gene

	Arthrogryposis Autonomic dysfunction				
Type I (severe Werdnig-Hoffmann) (Option A)	Severe hypotonia Frog-leg posture Absent/ decreased reflexes Tongue fasciculations Respiratory issues Feeding difficulties Preserved cognitive function Lack of head-control	Before 6 months	Profound weakness Frog leg posture Paradoxical breathing Weak cough Normal intelligence	Historically less than 2 years but improved with advancements in care	1-2 copies of SMN2
Type II (Intermediate) (Option B)	Normal early development, followed by plateau/regression Ability to sit independently Proximal muscle weakness Scoliosis	6-18 months	Proximal weakness Difficulty standing/walking Hand tremors Scoliosis	Survival into adulthood is possible with proper management	1-2 copies
Type III (Mild, Intermediate SMA and Kugelberg-Welander) (Option C)	Later onset Progressive proximal muscle weakness Ability to stand/walk independently Possible muscle hypertrophy Tongue fasciculations Hand tremors	After 18 months and some in adulthood	Progressive weakness Muscle hypertrophy Ability to walk independently initially	Generally good near-normal lifespan with possible progression requiring mobility aid	3 copies (intermediate) 3-4 copies(Kugelberg-Welander type)
Type IV (adult-onset)	Adult onset Slowly progressive muscle weakness Mild clinical presentation	After 2nd or 3rd decade	Mild weakness Near-normal lifespan and quality of life Progressive scoliosis High risk of fracturing long bones and the pelvis	Favorable with a good quality of life	≥4 copies

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Solution for Question 22:

Correct Answer: D) Placental transfer of anti-AChR antibodies

Explanation:

The beginning of symptoms within 1-3 days of life and the symptoms all point to the diagnosis of transient neonatal myasthenia gravis, which is caused by the placental transfer of maternal anti-AChR antibodies.

Congenital Myasthenia Gravis (CMS)	Transient Neonatal Myasthenia Gravis (TNMG)
A heterogeneous group of genetic diseases of neuromuscular transmission is collectively called congenital myasthenia gravis syndromes.	Transient symptoms of myasthenia gravis(MG) are caused due to maternal antibodies.
Caused by genetic mutations affecting the neuromuscular junction Examples: Presynaptic: Choline acetyltransferase deficiency (Option A) Synaptic basal lamina-associated Endplate AChE deficiency (Option B) Postsynaptic Primary AChR deficiency=/- minor kinetic abnormality Defects in EP development and maintenance MuSK deficiency (Option C) Rapsyn deficiency Congenital defect of glycosylation Other myasthenic syndromes	Caused by maternal anti-AChR antibodies crossing the placenta
Permanent/lifelong condition inherited in a recessive/autosomal dominant pattern	Due to the passive transfer of antibodies via the placenta
The onset of symptoms at birth or early infancy	Symptoms typically appear within the first 1-3 days of life

Symptoms: Hypotonia External ophthalmoplegia Ptosis Dysphagia Weak cry Facial weakness Easy muscle fatigue Respiratory insufficiency or failure (often triggered by a minor respiratory infection)	Symptoms: Respiratory insufficiency Difficulty in sucking or swallowing Generalized hypotonia Generalized weakness Little spontaneous motor activity
Diagnosis is made by Clinical features Electromyography (EMG) Genetic testing for CMS-associated genes Anti-AChR antibody levels are rarely elevated	Diagnosis is made by Clinical features Detection of anti-AChR antibodies in the infant’s blood (may not always be present)
Treatment: Varies by specific mutations Cholinesterase inhibitors may be helpful or worsen symptoms	Treatment: Supportive care (ventilatory support, gavage feeding) Pyridostigmine for transient management
Prognosis: Static disorder Lifelong symptoms with varying severity Myasthenic crises are rare	Prognosis: Transient: symptoms usually resolve within 2 months as maternal antibodies disappear. Affected infants generally do not develop MG later A small minority may develop arthrogryposis due to fetal akinesia

Solution for Question 23:

Correct Answer: D) Pyridostigmine

Explanation:

Pyridostigmine bromide (Mestinon) is a cholinesterase-inhibiting drug that, if overdosed, causes a “cholinergic crisis” with increased secretions, diarrhea, and cramping.

Management of juvenile myasthenia gravis:

- Mild myasthenia gravis: Some patients may require no treatment.
- Some patients may require no treatment.
- Cholinesterase-Inhibiting Drugs: Pyridostigmine bromide (Mestinon): Dosage: 0.5 to 1 mg/kg per dose every 4 to 6 hours while awake with a maximum daily dose of 7 mg/kg/day.

Overdose causes a cholinergic crisis with increased secretions, diarrhea, and cramping.Atropine (Option B) blocks the muscarinic effects but does not block the nicotinic effects thatproduce additional skeletal muscle weakness.

- Pyridostigmine bromide (Mestinon): Dosage: 0.5 to 1 mg/kg per dose every 4 to 6 hours while awake with a maximum daily dose of 7 mg/kg/day.
- Overdose causes a cholinergic crisis with increased secretions, diarrhea, and cramping. Atropine (Option B) blocks the muscarinic effects but does not block the nicotinic effects that produce additional skeletal muscle weakness.
- Steroids: Long-term treatment with prednisone (Option A) may help as the disease is autoimmune.
- Long-term treatment with prednisone (Option A) may help as the disease is autoimmune.
- Thymectomy: Might provide a cure Most effective in those with high titers of anti-AChR antibodies and symptomatic for less than 2 years.
- Might provide a cure
- Most effective in those with high titers of anti-AChR antibodies and symptomatic for less than 2 years.
- Some patients may require no treatment.
- Pyridostigmine bromide (Mestinon): Dosage: 0.5 to 1 mg/kg per dose every 4 to 6 hours while awake with a maximum daily dose of 7 mg/kg/day.
- Overdose causes a cholinergic crisis with increased secretions, diarrhea, and cramping. Atropine (Option B) blocks the muscarinic effects but does not block the nicotinic effects that produce additional skeletal muscle weakness.
- Long-term treatment with prednisone (Option A) may help as the disease is autoimmune.
- Might provide a cure
- Most effective in those with high titers of anti-AChR antibodies and symptomatic for less than 2 years.
- Treatment of hypothyroidism removes an associated myasthenia.
- IV immunoglobulin Less invasive than plasmapheresis
- Less invasive than plasmapheresis
- Less invasive than plasmapheresis
- Plasmapheresis: For those unresponsive to steroids Provides temporary remission Plasmapheresis with immunoglobulins is effective for those with high circulating anti-AChR antibodies.
- For those unresponsive to steroids
- Provides temporary remission

- Plasmapheresis with immunoglobulins is effective for those with high circulating anti-AChR antibodies.
- For those unresponsive to steroids
- Provides temporary remission
- Plasmapheresis with immunoglobulins is effective for those with high circulating anti-AChR antibodies.
- Monoclonal antibody, rituximab (Option C) for refractory and MuSK-related patients.

Solution for Question 24:

Correct Answer: D) Muscle-Specific tyrosine Kinase (MuSK) Antibodies

Explanation:

- The diagnosis here, according to the clinical features, is myasthenia gravis, which is detected by Acetylcholine Receptor (AChR) antibodies in the serum.
- In case of a negative AChR antibody report, the next step is to evaluate the Muscle-specific tyrosine Kinase (MuSK) antibodies in the serum.

Investigations for Juvenile myasthenia gravis:

- Electrodiagnostic Tests: Assess neuromuscular transmission defects Repetitive Nerve Stimulation (RNS) (Option B) - Decremental response Electromyography (EMG) (Option C)
- Repetitive Nerve Stimulation (RNS) (Option B) - Decremental response
- Electromyography (EMG) (Option C)
- Repetitive Nerve Stimulation (RNS) (Option B) - Decremental response
- Electromyography (EMG) (Option C)
- Serologic Tests: Identify autoantibodies Acetylcholine Receptor (AChR) Antibodies - Confirm autoimmune MG Muscle-Specific Tyrosine Kinase (MuSK) Antibodies (Option C) - Diagnose MG in AChR-negative cases Other Autoimmune Markers - Rule out coexisting autoimmune conditions
- Acetylcholine Receptor (AChR) Antibodies - Confirm autoimmune MG
- Muscle-Specific Tyrosine Kinase (MuSK) Antibodies (Option C) - Diagnose MG in AChR-negative cases
- Other Autoimmune Markers - Rule out coexisting autoimmune conditions

- Acetylcholine Receptor (AChR) Antibodies - Confirm autoimmune MG
- Muscle-Specific Tyrosine Kinase (MuSK) Antibodies (Option C) - Diagnose MG in AChR-negative cases
- Other Autoimmune Markers - Rule out coexisting autoimmune conditions
- Chest Radiograph
- CT/ MRI
- Genetic Testing Differentiate congenital myasthenic syndromes (CMS) from autoimmune MG Clinical genetic panels and targeted sequencing for CMS-specific mutations; specialized electrophysiological studies
- Differentiate congenital myasthenic syndromes (CMS) from autoimmune MG
- Clinical genetic panels and targeted sequencing for CMS-specific mutations; specialized electrophysiological studies
- Differentiate congenital myasthenic syndromes (CMS) from autoimmune MG
- Clinical genetic panels and targeted sequencing for CMS-specific mutations; specialized electrophysiological studies

Solution for Question 25:

Correct Answer: C) Poliomyelitis

Explanation:

- Poliomyelitis typically presents with asymmetric muscle weakness, often affecting different limbs or muscle groups on opposite sides of the body.
- Sensation remains intact and CSF shows lymphocytic pleocytosis with normal or increased protein levels.

Acute Flaccid Paralysis (AFP): Rapid onset of weakness with maximum severity within days to weeks; characterized by lack of spasticity or upper motor neuron signs.

Common Causes:

- Guillain-Barré syndrome
- Poliomyelitis
- Transverse myelitis
- Traumatic neuritis

- Post diphtheritic neuropathy
- Non-polio enteroviral illnesses

Feature	Poliomyelitis	Guillain-Barré Syndrome	Transverse Myelitis	Traumatic Neuritis
Fever	Present; may be biphasic	May have a prodromal illness	May have a prodromal illness	Absent
Symmetry	Asymmetric (Option C)	Symmetrical (Option A ruled out)	Symmetrical (Option B ruled out)	Asymmetric
Sensations	Intact; may have diffuse myalgias	Variable	Impaired below the level of the lesion	Impaired in the distribution of affected nerve
Respiratory Insufficiency	May be present	May be present	May be present	Absent
Cranial Nerves	Affected in bulbar and bulbospinal forms	Usually affected	Absent	Absent
Cerebrospinal Fluid (CSF)	Lymphocytic pleocytosis; normal/increased protein	Albumino-cytologic dissociation	Variable	Normal

Traumatic Neuritis often presents with asymmetric muscle weakness, but it's typically confined to the distribution of the affected nerve rather than being systemic and respiratory insufficiency is absent. (Option D ruled out)

Solution for Question 26:

Correct Answer: B) Duchenne Muscular Dystrophy

Explanation:

Duchenne Muscular Dystrophy (DMD) is a genetic disorder caused by mutations in the dystrophin gene characterized by progressive muscle degeneration and weakness.

- It is classified as muscular dystrophy, not congenital myopathy.

Congenital Myopathies:

- Genetic muscle disorders are characterized by distinctive histochemical or ultrastructural changes on muscle biopsy.

Clinical features:

- The majority present as 'floppy infant' syndrome with hypotonia, static or non-progressive muscle weakness, and normal or decreased deep tendon reflexes.
- Respiratory insufficiency, feeding difficulties, contractures, and skeletal deformities.

Investigations:

- Serum Creatine Kinase: Typically normal or mildly elevated.
- Electromyography: Reveals a myopathic pattern.

Diagnosis:

- Clinically indistinguishable; requires skeletal muscle biopsy with immunohistochemical techniques and electron microscopy for differentiation.

Subtypes:

Condition	Genetics	Key Features
Core Myopathies (Option D ruled out)	AD, AR	Poorly defined fibers, risk of malignant hyperthermia
Nemaline Myopathy (Option A ruled out)	AD, AR, sporadic	Nemaline bodies on muscle biopsy, facial, ocular, and neck muscles affected Chest and skeletal deformities (scoliosis)
Centronuclear Myopathies (Option C ruled out)	AD, sporadic, X-linked	Central nuclei in all muscle fibers, facial weakness, external ophthalmoplegia, weakness of neck flexors, and ptosis
Congenital Fiber Type Disproportion	AD, AR, X-linked	Predominant type 1 fibers, small type 2 fibers

Solution for Question 27:

Correct answer: B) Facioscapulohumeral muscular dystrophy

Explanation:

Based on the given scenario, the likely diagnosis is Facioscapulohumeral muscular dystrophy.

Facioscapulohumeral muscular dystrophy:

- It is an autosomal dominant disease with genetic anticipation.
- The earliest and most severe asymmetrical weakness is in the facial and shoulder muscles (scapular winging shown in the image).
- The mouth is rounded and appears puckered because of protruding lips.
- The inability to close the eye completely in sleep is a common expression of upper facial weakness.
- Flattening or concavity of deltoid contour seen with wasted biceps and triceps. Muscles of the hip and thighs also eventually atrophy, and Gower’s sign is seen.
- Hearing loss, retinal vasculopathy, lumbar lordosis, and kyphoscoliosis are associated complications.

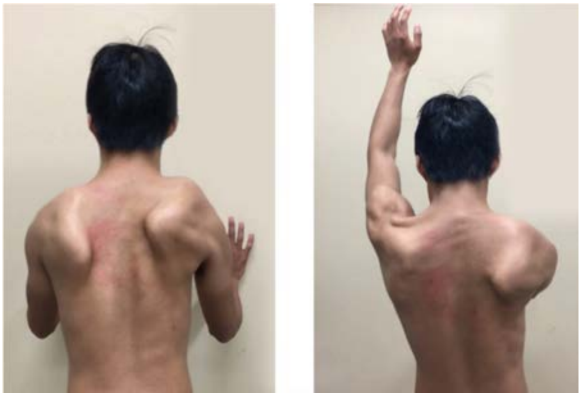


Table:

Feature	Emery-Dreifuss MD (EDMD) (Option A)	Limb-Girdle MD (LGMD) (Option D)	Facioscapulohumeral MD (FSHD) (Option B)	Congenital MD (CMD) (Option C)
Inheritance	XLR (EMD gene) or AD/AR (LMNA gene)	AD (LGMD type1)/ AR (LGMD type 1)	AD with Genetic anticipation	AR
Genetic Mutations	EMD gene (emerin), LMNA gene (lamin A/C)	Over 30 genetic forms	D4Z4 repeat (FSHD1), SMCHD1 mutations (FSHD2)	Multiple genes, associated with brain malformations and severe epilepsy
Onset	Childhood (5-15 yrs)	Variable: childhood (4-5 years) to adulthood	Childhood to adult life	Prenatal, neonatal, or early childhood
Muscle Involvem	Contractures of the elbows, ankles, and neck extensor muscles	Low back pain, Hip and shoulder girdles affected, later distal muscles, weakness	Facial and shoulder girdle, asymmetric weakness, scapular winging, puckered	Diffuse hypotonia, poor head control, contractures, mild facial involvement,

ent	Scapulohumeroperoneal wasting, no facial weakness	of neck flexors andextensors	and rounded mouth	dysphagia
Cardiac Involvement	Dilated cardiomyopathy, ventricular fibrillation	Some subtypes have cardiac involvement	Rare	No specific cardiac involvement
Respiratory Involvement	Early respiratory insufficiency in severe forms	Respiratory compromise in later stages	Rare, some may need respiratory support	Respiratory complications common
Diagnostics	Muscle biopsy, genetic testing, cardiac evaluation	Muscle biopsy, genetic testing, elevated serum CK	Genetic testing, muscle biopsy, EMG	Muscle biopsy, genetic testing, brain imaging
Treatment	Supportive care, cardiac management with ICD, orthopedic management	Supportive care, orthopedic management	Supportive care, light aerobic exercise	Supportive care, physical therapy, feeding support
Prognosis	Slow progression, many survive to late adult life	Variable; Loss of ambulation in early adulthood	Variable; 20% lose independent ambulation	Often benign course despite early onset

Solution for Question 28:

Correct Answer: D) Myotonic dystrophy

Explanation:

The patient’s clinical features of a high-arched palate, an inverted V-shaped upper lip, and prominent facial muscle wasting, along with myotonia, indicate Myotonic dystrophy.

- This autosomal-dominant condition is marked by facial weakness and distal muscle wasting.
- It is caused by defects in CTG (mnemonic: Cannot Terminate Grip (myotonia)) trinucleotide expansion on chromosome 19q13.
- Myotonia is a very slow relaxation of muscle after contraction.

Features and Management of Myotonic Dystrophy

Types	Classic Type 1 (DM1): CTG expansion in the DMPK gene on chromosome 19q13.3 Type 2 (DM2): CCTG expansion in zinc finger protein 9 (ZNF9) gene on chromosome 3q21 Late form (DM3): Linked to locus 15q21-q24
Inheritance	Autosomal dominant
Systemic Involvement	Multisystem disease affecting muscles, cardiac function, endocrine system, GI tract, CNS, and immune system
Clinical Features	Facial Features: Inverted-V upper lip, thin cheeks, scalloped temporalis, high-arched palate Muscle Wasting: Distal > proximal muscles; myotonia present after age 5 Cardiac: Heart block, arrhythmias, sudden death Endocrine: Hypothyroidism, adrenal insufficiency, diabetes, delayed puberty, testicular atrophy Ocular: Cataracts, abnormal visual evoked potentials Intellectual Impairment: Mild to moderate, with increased autism risk Congenital Form: Severe hypotonia, respiratory difficulties, high infant mortality if prolonged ventilation needed
Laboratory Findings	Serum CK: Normal or mildly elevated EMG: Myotonic pattern evident from school age Cardiac Evaluation: Annual ECG; ECHO every 2 years Endocrine and Immune: Regular monitoring of thyroid, adrenal function, glucose tolerance, and immunoglobulins
Diagnosis	Primary: DNA analysis for CTG/CTG repeat expansions Muscle Biopsy: Shows central nuclei, type I fiber atrophy; usually not required for diagnosis

Treatment	Physiotherapy/ Orthopedic: For contractures and to maintain function Myotonia Management: Drugs like mexiletine, phenytoin, carbamazepine Cardiac Care: Annual ECG, consider pacemaker or antiarrhythmics for arrhythmias Respiratory Care: BiPAP, cough assist device, incentive spirometry Anesthesia Precautions: Avoid succinylcholine, use modified induction and extubation

Duchenne muscular dystrophy (Option A):

- The facial features present in this case are not seen.
- It presents with a positive Gowers' sign and hypertrophy of the calf muscles, progresses rapidly, and leads to loss of ambulation in the early teenage years.
- It is associated with significant cognitive impairment and a high incidence of cardiomyopathy and respiratory complications, which typically determine life expectancy.

Becker muscular dystrophy (Option B):

- It presents similar to Duchenne muscular dystrophy but has a later onset (>5 years), slower progression, and allows for ambulation into adulthood.
- Cognitive impairment is less common, and life expectancy is longer.
- The facial features seen in this case are not present.

Emery-Dreifuss muscular dystrophy (Option C):

- Presents with predominant involvement of scapular and numeral muscles.
- It is an X-linked recessive disease.
- Contractures of the elbows, ankles, and neck extensor muscles develop early.
- The muscle becomes wasted in scapulohumeral distribution.

Solution for Question 29:

Correct Answer: B) Performing a muscle biopsy with dystrophin immunohistochemistry

Explanation:

When genetic testing for deletions or duplications in the dystrophin gene is negative and clinical suspicion remains high (e.g., based on elevated serum CK levels and clinical features), the next best step is to perform a muscle biopsy with dystrophin immunohistochemistry.

- This test helps differentiate between DMD and other myopathies by identifying dystrophin abnormalities, providing a definitive diagnosis.

Diagnostic and Therapeutic Approach to Duchenne Muscular Dystrophy (DMD)

Initial Genetic Evaluation	Primary Testing: Deletion/duplication analysis of the dystrophin gene using dosage analysis. Next-generation sequencing (NGS) if initial testing is negative.
Muscle Biopsy	Indications: Used when genetic tests are inconclusive, especially if serum CK levels are elevated. Findings: Endomysial connective tissue proliferation, degenerating/regenerating myofibers, hypercontracted fibers, dystrophin abnormalities.
Steroid Therapy	Purpose: Slows muscle strength decline, prolongs ambulation, and may benefit scoliosis progression. Dosing: Prednisone (0.75 mg/kg/day) or deflazacort (0.9 mg/kg/day); various dosing protocols available.
Advanced Therapeutic Options	Exon Skipping Therapy: Eteplirsen for patients with mutations amenable to exon 51 skipping (~13% of DMD cases). Nonsense Mutation Therapy: Ataluren for patients with nonsense mutations (~10-15% of DMD cases).
Cardiac Management	First-line Agents: ACE inhibitors, ARBs, beta-blockers, aldosterone antagonists. Timing of Intervention: Initiated based on echocardiogram findings (LVEF < 55%) or earlier based on MRI changes.
Pulmonary Management	Infection Prevention: Prompt treatment of pulmonary infections, avoidance of contagious illnesses, routine vaccinations. Sleep-Disordered Breathing: Sleep studies, BiPAP therapy, use of cough assist devices, airway clearance support.
Nutritional Management	Adequate calcium and vitamin D intake, weight management to prevent obesity.
Physiotherapy & Contracture Management	Delays but may not prevent contractures; stretching, bracing, and cautious surgical interventions.

Glucocorticoids (Option A) are essential in managing DMD, but initiating therapy without confirming the diagnosis through further testing (such as a muscle biopsy) is not recommended.

Exon skipping therapy, such as Eteplirsen (Option C), is specific to patients with certain dystrophin gene mutations (e.g., those amenable to exon 51 skipping).

- It should not be started based solely on clinical suspicion without confirming the genetic mutation.

While electromyography (Option D) can show myopathic changes, it is not specific to DMD.

- EMG is less useful than a muscle biopsy for confirming DMD when genetic testing is inconclusive.

Solution for Question 30:

Correct answer: A) Duchenne muscular dystrophy

Explanations:

The findings of positive Gower's sign, hypertrophy of the calf muscle and elevated serum creatine kinase with a history of waddling gait and difficulty climbing stairs in a child <5 years of age are suggestive of Duchenne muscular dystrophy (DMD).

DMD progresses rapidly and leads to loss of ambulation in the early teenage years. It is associated with cognitive impairment and a high incidence of cardiomyopathy and respiratory complications, which typically determine life expectancy.

Clinical Differences Between DMD and BMD

Features	Duchenne Muscular Dystrophy (DMD)	Becker Muscular Dystrophy (BMD)
Age of Onset	Before 5 years	After 5-7 years
Early Symptoms	Hip girdle weakness, delayed walking, Gowers' sign, lordotic	Mild motor delay, later onset of weakness

	posture, frequent falls	
Ambulation Loss	Typically between 10-14 years	Late adolescence or adulthood
Progression Rate	Rapid	Slower
Muscle Involvement	Proximal muscle weakness, calf pseudohypertrophy	Calf pseudohypertrophy, less severe proximal weakness
Cardiac Involvement	Common, usually worsens after loss of ambulation	Common, may appear earlier despite ambulation
Cognitive Impairment	Common (20-30% with IQ < 70), possible autism-like behavior, epilepsy	Less common
Respiratory Involvement	Common, leading to frequent pulmonary infections, sleep apnea, and respiratory failure	Less severe, but present in later stages
Life Expectancy	Late teens to 20s	Into 40s and 50s

Gower sign presents with any other condition associated with weakness of pelvic girdle or proximal muscles of lower extremities, such as:

- Becker muscular dystrophy
- Limb-girdle and other muscular dystrophies
- Proximal ascending pseudomyopathic diseases
- Spinal muscular atrophy
- Sarcoglycanopathy
- Polymyositis
- Discitis
- Juvenile idiopathic arthritis

Becker Muscular Dystrophy (Option B):

- It can also have features of Duchenne muscular dystrophy, including Gower sign and pseudohypertrophy of calf muscles.
- It has a later onset (>5 years), slower progression, and allows for ambulation into adulthood.
- Cognitive impairment is less common, and life expectancy is longer, though cardiac and respiratory complications still contribute to morbidity.

Emery-Dreifuss muscular dystrophy (Option C):

- It is also called Scapuloperoneal or Scapulohumeral muscular dystrophy

- It does not present with hypertrophy of the calf muscles.
- It is an X-linked recessive disease.
- Contractures of the elbows, ankles, and neck extensor muscles develop early, and the muscles become wasted in the scapulohumeral distribution.

Myotonic muscular dystrophy (Option D):

- It does not present with hypertrophy of the calf muscles or elevated CK levels.
- It is the second most common muscular dystrophy.
- It is an Autosomal dominant trait caused by CTG trinucleotide expansion.
- Distal muscles (e.g., hands, forearms, anterior lower legs) show progressive atrophy and very slow relaxation of muscles after contraction (Myotonia) is seen.

Previous Year Questions

1. A 15-year-old boy presents with short stature. On examination, his US/LS ratio is 0.8:1. Which of the following is the likely diagnosis?

A. Spondyloepiphyseal Dysplasia

B. Achondroplasia

C. Growth hormone deficiency

D. Hypothyroidism

2. In a child presenting with a beaded appearance in the chest with the following X-ray, what is the diagnosis?

A. Scurvy

B. Rickets

C. Beri Beri

D. Pellagra

3. What is the most probable diagnosis for a 6-year-old girl, born from parents who are not blood relatives, presenting with rachitic changes in her limbs since the age of 2 despite receiving several rounds of vitamin D treatment? Parameter Lab determined value Calcium 9.5 mg/dL Phosphate 1.6 mg/dL ALP 814 IU PTH Within normal limits Serum electrolytes Within normal limits Serum creatinine Within normal limits ABG Within normal limits

Parameter	Lab determined value
Calcium	9.5 mg/dL
Phosphate	1.6 mg/dL
ALP	814 IU
PTH	Within normal limits
Serum electrolytes	Within normal limits
Serum creatinine	Within normal limits
ABG	Within normal limits

A. Vitamin D dependent rickets

B. Hypoparathyroidism

C. Chronic renal failure

D. Hypophosphatemic rickets

4. What is the diagnosis for a 4-year-old boy who has muscle weakness, difficulty climbing stairs, and getting up from the floor, with muscle biopsy showing small muscle fibrils and absence of dystrophin? Please choose the correct option from the given choices.

A. Becker muscle dystrophy

B. Duchenne muscular dystrophy

C. Myotonic dystrophy

D. Limb-girdle muscular dystrophy

5. In which of the following diseases is the dystrophin protein completely absent?

A. Becker Muscular Dystrophy

B. Duchenne muscular dystrophy

C. Myasthenia gravis

D. Lambert-Eaton myasthenic syndrome

6. Which of the following is accurate regarding the biochemical parameters in vitamin D-dependent rickets? Serum calcium PTH level A Decreased / normal Normal B Decreased / normal Increased C Decreased Increased D Increased Increased

	Serum calcium	PTH level
A	Decreased / normal	Normal
B	Decreased / normal	Increased
C	Decreased	Increased
D	Increased	Increased

A. A

B. B

C. C

D. D

7. A pediatric patient arrives at the hospital complaining of several abnormal fractures, and the X-ray shows the image shown below. The blood tests indicate low levels of phosphorus and calcium, normal levels of Vitamin D, and increased levels of PTH. What is the most likely diagnosis for this patient?

A. Osteopetrosis

B. Osteomalacia

C. Osteosclerosis

D. Histiocytosis X

8. The x-ray image of a child with muscle weakness and failure to thrive is attached below. What is the likely diagnosis?

A. Scurvy

B. Rickets

C. Osteomalacia

D. Osteopetrosis

9. What is the probable diagnosis for a 7-year-old child who complains of leg pain and had a fever and cough three days ago? The calf muscles are tender on palpation, but reflexes and other clinical examinations are normal. Laboratory testing shows CK levels of 2000 IU.

A. Viral Myositis

B. Duchenne muscular dystrophy

C. GBS

D. Dermatomyositis

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	4
Question 4	2
Question 5	2
Question 6	2
Question 7	1
Question 8	2
Question 9	1

Solution for Question 1:

Correct Option A - Spondyloepiphyseal Dysplasia:

- Spondyloepiphysal dysplasia (SED) is an inheritable dysplasia of the bone due to a defect in collagen.
- It can present a significant reduction in the upper segment to a lower segment ratio- short trunk dwarfism.
- The patient's decreased upper segment to lower segment (US/LS) ratio of 0.8:1 points towards the diagnosis of spondyloepiphyseal dysplasia.

Incorrect Options

Option B – Achondroplasia:

- Achondroplasia results in short-limb dwarfism, so US/LS is likely to be increased.

Option C - Growth hormone deficiency:

- Growth hormone deficiency results in proportionate short stature with a normal US/LS ratio.

Option D – Hypothyroidism:

- Hypothyroidism also results in short-limb dwarfism, so US/LS is likely to be increased.

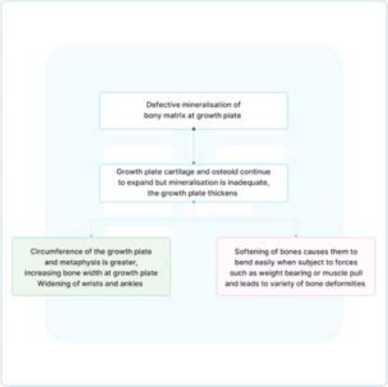
Solution for Question 2:

Correct Answer: B) Rickets

Explanation:

The radiograph showing metaphyseal fraying and cupping and the clinical presentation of a beaded appearance on the chest (rachitic rosary) are characteristic features of rickets caused due to vitamin D deficiency.

Pathophysiology of Rickets:



Clinical Features of Rickets:

General: Failure to thrive (malnutrition) Listlessness Protruding abdomen Muscle weakness (especially proximal) Hypocalcemic dilated cardiomyopathy Fractures (pathologic, minimal trauma) Increased intracranial pressure Head: Craniotabes: Softening of cranial bones Frontal bossing Delayed fontanel closure (usually closed by 2 yr) Delayed dentition ... (content truncated)	Chest: Rachitic rosary: Due to widening of the costochondral junction Harrison groove: Horizontal depression along the lower anterior chest due to pulling of softened ribs by the diaphragm during inspiration Respiratory infections and atelectasis Back: Scoliosis Kyphosis Lordosis Extremities: Enlargement of wrists and ankles: Due to growth plate widening Valgus or varus deformities Windswept deformity (valgus deformity of one leg with varus deformity of other leg) Anterior bowing of tibia and femur ... (content truncated)

Rachitic rosary in a child in rickets

Hand and forearm of a young child with rickets show prominence above the wrist

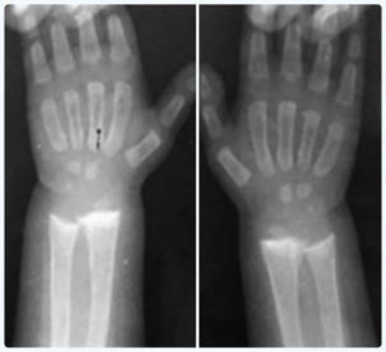
Anterolateral bowing of legs and prominent malleoli in a child with rickets

Windswept deformity of legs in rickets

Treatment of Nutritional Rickets:

1. Vitamin D Supplementation:
   - Stoss Therapy: Administer 300,000-600,000 IU of vitamin D orally or intramuscularly over 1 day (2-4 doses). Vitamin D■ is preferred due to its longer half-life. Ideal for patients with questionable adherence.
   - Administer 300,000-600,000 IU of vitamin D orally or intramuscularly over 1 day (2-4 doses).
   - Vitamin D■ is preferred due to its longer half-life.
   - Ideal for patients with questionable adherence.
   - Alternative Strategy: Daily Vitamin D minimum dose of 2,000 IU/day for at least 3 months.
   - Follow-Up: Both strategies should follow up with 400 IU/day (<1 year old) or 600 IU/day (>1 year old).
   - Administer 300,000-600,000 IU of vitamin D orally or intramuscularly over 1 day (2-4 doses).
   - Vitamin D■ is preferred due to its longer half-life.
   - Ideal for patients with questionable adherence.
2. Calcium Supplementation: Provide oral calcium (30-75 mg/kg/day, maximum 500 mg).
3. Phosphorus Supplementation

4. Monitoring: Radiological improvement (white line of healing in the metaphyses) should be visible within 4 weeks.



Healing of the growth plate following Vitamin D therapy

Age	Daily dose of vitamin D3 for 12 weeks	Alternative intermittent dose regimen	Daily calcium supplementation	Daily maintenance dose (follow-up)
<6 months	2000 IU	Not recommended	30-75 mg/kg/day	400 IU
6-12 months	2000 IU	The equivalent of 2000 IU/ day may be given on a weekly or monthly basis	30-75 mg/kg/day	400 IU
>12 months	3000IU	60,000 IU every 2 weeks for 5 doses	30-75 mg/kg/day (up to 500 mg)	600 IU

Scurvy (Option A) is caused due to vitamin C deficiency and presents with irritability, leg swelling, gum bleeding, and elevated prothrombin time.

Beriberi (Option C) is caused due to the deficiency of vitamin B1 (thiamine) that can lead to symptoms such as lethargy, restlessness, developmental delay, and abdominal distention. Late symptoms like peripheral neuritis, decreased reflexes, loss of vibration sense, muscle tenderness, heart failure, oedema, cardiomegaly, etc.

Pellagra (Option D) is due to niacin (vitamin B3) deficiency leading to dementia, diarrhoea, and dermatitis.

Solution for Question 3:

Correct Option D: Hypophosphatemic rickets:

- The patient’s presentation with rachitic changes in limbs since the age of 2 and receiving multiple courses of vitamin D with a normal calcium level and a low phosphate level is suggestive of hypophosphatemic rickets.
- Hypophosphatemic rickets is a genetic disorder characterized by impaired renal tubular reabsorption of phosphate, resulting in low levels of serum phosphate (hypophosphatemia).
- This leads to defective bone mineralization, resulting in rachitic changes in the bones.

Incorrect Options:

Option A - Vitamin D-dependent rickets:

- Vitamin D dependent rickets can occur due to impaired vitamin D metabolism or receptor defects.
- However, in this case, the serum phosphate level is low, which is not consistent with vitamin D dependent rickets where phosphate levels are typically normal or high.

Option B - Hypoparathyroidism:

- Hypoparathyroidism is characterized by low levels of parathyroid hormone (PTH) leading to low calcium levels and high phosphate levels.
- However, in the given scenario, the PTH level is within normal limits, ruling out hypoparathyroidism.

Option C - Chronic renal failure:

- Chronic renal failure can cause renal tubular dysfunction leading to electrolyte imbalances and bone abnormalities.
- However, in the given scenario, the serum electrolytes and creatinine levels are within normal limits, indicating normal renal function.

Solution for Question 4:

Correct Option B - Duchenne muscular dystrophy:

- Mechanism: Duchenne muscular dystrophy (DMD) is a genetic disorder characterized by progressive muscle degeneration and weakness. It is caused by mutations in the DMD gene that encodes the protein dystrophin.

- Symptoms: Children with DMD typically show signs of muscle weakness, difficulty climbing stairs, and getting up from the floor. The absence of dystrophin in muscle biopsy is a hallmark of DMD.
- Diagnosis: The clinical presentation along with the muscle biopsy showing absence of dystrophin confirms the diagnosis of Duchenne muscular dystrophy.

Incorrect Options

Option A. Becker Muscular Dystrophy

- Mechanism: Becker muscular dystrophy (BMD) is also caused by mutations in the DMD gene but typically results in partially functional dystrophin.
- Symptoms: Symptoms are similar to DMD but milder and with later onset. Muscle biopsy would show reduced but not absent dystrophin.

Option C. Myotonic Dystrophy

- Mechanism: Myotonic dystrophy is a different genetic disorder caused by mutations in the DMPK or CNBP genes.
- Symptoms: It involves myotonia (difficulty relaxing muscles), cataracts, and cardiac abnormalities. Muscle biopsy would not show absence of dystrophin.

Option D. Limb-Girdle Muscular Dystrophy

- Mechanism: Limb-girdle muscular dystrophy (LGMD) involves a group of disorders affecting the voluntary muscles, particularly around the hips and shoulders.
- Symptoms: It typically shows a different pattern of muscle involvement and genetic mutations. Muscle biopsy would not show absence of dystrophin but might show other abnormalities.

Solution for Question 5:

Correct Option B - Duchenne muscular dystrophy: Dystrophin protein completely absent in duchenne muscular dystrophy.

Incorrect Options - A, C, and D are incorrect.

Solution for Question 6:

Correct Option B - In vitamin D-dependent rickets, serum calcium is either decreased or normal and PTH levels are increased.

Incorrect Options - A, C, and D are incorrect.

Solution for Question 7:

Correct option A - Osteopetrosis:

- The patient's presentation with multiple pathological fractures, low phosphorus and calcium levels, normal Vitamin D levels, and elevated PTH along with the X-ray findings is suggestive of osteopetrosis.
- Osteopetrosis is a genetic disorder characterized by defective osteoclast function, leading to increased bone density and susceptibility to fractures.

Incorrect Options:

Option B - Osteomalacia:

- Osteomalacia is characterized by softening of the bones due to vitamin D deficiency or impaired metabolism.
- However, in this case, the vitamin D levels are normal.

Option C - Osteosclerosis: Osteosclerosis refers to abnormal hardening or increased density of the bones, but it does not present with multiple pathological fractures.

Option D - Histiocytosis X:

- Histiocytosis X (also known as Langerhans cell histiocytosis) is a group of disorders characterized by abnormal proliferation of certain immune cells called Langerhans cells.
- It does not present with findings described in the question.

Solution for Question 8:

Correct Option B - Rickets:

- Rickets is a condition that primarily affects growing children and is caused by a deficiency in vitamin D, calcium, or phosphate.
- Rickets primarily affect the growing ends of the long bones, including those in the hand.
- Radiographically, widened or irregular growth plates in the phalanges and metacarpals can be seen.
- The metaphyses near the growth plates may show a cupped or flared appearance.
- This means that the ends of the bones appear wider and more irregular than the shaft of the bone.

Incorrect Options - A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 9:

Correct Option A - Viral Myositis:

- The patient’s presentation with leg pain and a history of fever and cough three days ago with tenderness in calf muscles on palpation and increased creatine kinase is suggestive of viral myositis.
- Viral myositis presents with muscle pain, tenderness, and sometimes swelling.
- The elevation of creatine kinase (CK) levels in the blood is an indication of muscle damage or breakdown.

Incorrect Options:

Option B - Duchenne muscular dystrophy:

- Duchenne muscular dystrophy is a genetic disorder characterized by progressive muscle weakness and wasting.
- It usually manifests in early childhood and is not typically associated with a recent history of fever and cough.

Option C - GBS:

- GBS is an autoimmune disorder affecting the peripheral nervous system.
- It causes muscle weakness and can lead to paralysis.
- It is unlikely in this case as there are no signs of neurological involvement or abnormal reflexes.

Option D - Dermatomyositis:

- Dermatomyositis is an inflammatory muscle disease that primarily affects the skin and muscles.
- It usually presents with characteristic skin rash along with muscle weakness.
- The absence of skin rash in this case makes dermatomyositis less likely.

Pituitary, Thyroid, and Adrenal Disorders

1. A 7-year-old child from a region with known iodine deficiency presents to the clinic with symptoms including developmental delay, short stature, and coarse facial features. The child's parents report a high prevalence of goiter in their community. To prevent similar cases in the future, which of the following preventive measures is the most appropriate?

A. Prescribing high-dose thyroid hormone replacement therapy

B. Recommending the use of iodinated salt or iodized oil

C. Advising on a diet high in vegetables and fruits

D. Performing regular thyroid function tests for all community members

2. In a pediatric patient presenting with hypertension and episodic symptoms such as palpitations, sweating, and headaches, which of the following diagnostic tests is most appropriate?

A. Serum calcium level

B. Plasma fractionated metanephrines

C. Thyroid function test

D. MRI of the brain

3. A pediatric patient presents with resistant hypertension and hypokalemia. Which of the following diagnostic tests is most appropriate to evaluate for primary hyperaldosteronism?

A. Serum cortisol level

B. Plasma aldosterone concentration to plasma renin activity ratio

C. Serum sodium level

D. 24-hour urinary calcium excretion

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4. A 15-year-old girl presents with weight gain, easy bruising, and fatigue. She has a history of hypertension. O/E, she has a rounded face, purple striae on her abdomen, and proximal muscle weakness. Which of the following is the most appropriate initial diagnostic test to confirm the suspected diagnosis?

A. 24-hour urinary free cortisol test

B. Overnight DST

C. Serum cortisol level measurement at 8 am

D. ACTH stimulation test

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5. A 14-year-old girl presents with fatigue, weight loss, and hyperpigmentation of her skin. She also reports experiencing abdominal pain and orthostatic dizziness. Laboratory tests reveal low serum cortisol levels, elevated ACTH levels, and normal aldosterone levels. What is the most likely diagnosis?

A. Primary adrenal insufficiency

B. Secondary adrenal insufficiency

C. Tertiary adrenal insufficiency

D. Cushing’s syndrome

---

6. A 16-year-old girl presents with hypertension, hypokalemia, and amenorrhea. She has a history of reduced secondary sexual characteristics and has not menstruated since puberty. Lab reports reveal elevated serum aldosterone levels, normal cortisol levels, and low serum testosterone levels. What is the most likely diagnosis?

A. 17 ■ hydroxylase deficiency

B. 11 hydroxylase deficiency

C. 3 ■ HSD deficiency

D. None of the above

---

7. A 6-week-old infant is brought to the emergency room with symptoms of vomiting, dehydration, and shock. On physical examination, the infant's genitalia are noted to be ambiguous. Initial laboratory tests reveal severe electrolyte imbalances. Given these clinical findings, which of the following is the most likely diagnosis and the appropriate diagnostic test to confirm it?

A. 21-Hydroxylase deficiency with simple virilizing form; ACTH stimulation test with 17-OHP levels

B. 21-Hydroxylase deficiency with salt-wasting form; elevated 17-OHP levels

C. Non-Classic 21-Hydroxylase deficiency; Measurement of 17-OHP levels before and after ACTH stimulation

D. 11-Hydroxylase deficiency; Hydrocortisone alone

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8. A 5-day-old neonate is admitted to the NICU with tachycardia, irritability, and poor weight gain. The infant's mother has a history of Graves' disease and was treated with propylthiouracil throughout her pregnancy. Initial examination reveals a heart rate of 170 bpm, and thyroid function tests show elevated T3 and T4 levels with suppressed TSH.Which of the following statements about this condition is false?

A. Affects infants born to mothers with Graves' disease

B. Treatment with corticosteroids only

C. Symptoms of neonatal thyrotoxicosis can be delayed due to maternal antithyroid medications

D. Infants are prone to cardiac failure

9. A 10-year-old girl presents with a 3-month history of unexplained weight loss (5 kg) despite increased appetite. She reports feeling constantly warm, sweating excessively, experiencing palpitations, and having decreased concentration in class. On examination, she has visible neck swelling, moist skin, and fine tremors in her outstretched hands.What is the most appropriate initial treatment?

A. Propylthiouracil

B. Methimazole

C. Prednisolone

D. Propranolol

10. A 6-year-old boy is brought to the clinic due to concerns about his short stature. His height is below the 1st percentile for age, and his growth velocity has been consistently below the 25th percentile for the past two years. He was born at term with normal birth weight and length. Physical examination reveals proportionate short stature without dysmorphic features. Laboratory tests show low IGF-1 levels and two provocative growth hormone stimulation tests reveal peak GH levels of 3 ng/mL and 4 ng/mL. Which of the following is the most appropriate next step in management?

A. Observe and reassess in 6 months as normal growth might occur on its own

B. Start recombinant human growth hormone therapy

C. Perform genetic testing for PROP1 mutations

D. Initiate monotherapy with gonadotropin-releasing hormone agonist to delay epiphyseal fusion

11. What is the most effective method for screening newborns for congenital hypothyroidism to ensure early detection and intervention?

A. Heel prick test for TSH

B. Heel prick test for thyroid hormone (T4 and T3)

C. USG of the thyroid gland

D. Clinical assessment of physical signs of hypothyroidism

12. Which of the following is not typically a finding in the shown condition?

A. Poor feeding

B. Constipation

C. Hypertonia

D. Jaundice

13. Which of the following is not typically associated with MEN 1 syndrome?

A. Pituitary adenoma

B. Hyperparathyroidism

C. Pheochromocytoma

D. Medullary thyroid carcinoma

14. A 10-year-old child presents with excessive thirst and frequent urination for the past several months, causing him to wake up multiple times at night to drink water. Lab tests show hypernatremia and dilute urine with low urine osmolality. What is the most appropriate treatment for managing his condition?

A. Insulin therapy

B. Desmopressin

C. Metformin

D. Spironolactone

15. A 10-year-old girl presents to her primary care physician with complaints of excessive thirst and urination for the past 3 months. Laboratory tests reveal: Serum sodium: 147 mEq/L (Normal: 135-145 mEq/L) Serum osmolality: 305 mOsm/kg (Normal: 275-295 mOsm/kg) Urine osmolality: 150 mOsm/kg (Normal: 300-900 mOsm/kg) Urine specific gravity: 1.005 (Normal: 1.005-1.030) A water deprivation test is performed, and after 8 hours, the patient's urine osmolality remains at 200 mOsm/kg. Administration of desmopressin results in a rapid increase in urine osmolality to 600 mOsm/kg within 2 hours. Which of the following is the most characteristic symptom of this condition?

A. Polyuria and polydipsia

B. Hyperglycemia

C. Edema

D. Polyphagia

16. A 12-year-old boy is brought to the endocrinology clinic due to rapid growth over the past year. He has grown 12 cm in the last 12 months and is now 175 cm tall (>97th percentile). Physical examination reveals coarse facial features, large hands and feet, and a deep voice. Laboratory tests show elevated IGF-1 and IGFBP-3 levels. An oral glucose tolerance test fails to suppress GH levels below 5 ng/mL. An MRI of the brain revealed no significant findings. What is the most appropriate initial treatment for this patient?

A. Trans-sphenoidal surgery

B. Pegvisomant

C. Octreotide

D. Cabergoline

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17. A pediatric endocrinology fellow is reviewing hormone profiles of patients with various growth disorders. Match the following conditions with their expected hormone levels: a. GHRH deficiency 1. Low GH, low IGF-1, normal or high GHRH b. GHRH receptor resistance 2. High GH, low IGF-1, high GHRH c. GH deficiency 3. Low GH, low IGF-1, low GHRH d. GH receptor resistance (Laron syndrome) 4. Normal or high GH, low IGF-1, high GHRH

a. GHRH deficiency	1. Low GH, low IGF-1, normal or high GHRH
b. GHRH receptor resistance	2. High GH, low IGF-1, high GHRH
c. GH deficiency	3. Low GH, low IGF-1, low GHRH
d. GH receptor resistance (Laron syndrome)	4. Normal or high GH, low IGF-1, high GHRH

A. a-3, b-1, c-1, d-2

B. a-3, b-4, c-1, d-2

C. a-1, b-2, c-3, d-3

D. a-3, b-1, c-4, d-2

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	1
Question 5	1
Question 6	1
Question 7	2
Question 8	2
Question 9	2
Question 10	2
Question 11	1
Question 12	3
Question 13	4
Question 14	2
Question 15	1
Question 16	3
Question 17	1

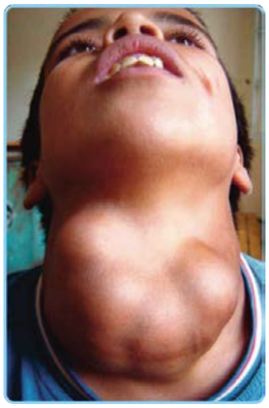
Solution for Question 1:

Correct Answer: B) Recommending the use of iodinated salt or iodized oil

Explanation:

Iodine deficiency disorders, including endemic cretinism, are best prevented through the use of iodinated salt or iodized oil. This approach addresses the root cause of iodine deficiency and is effective in preventing the associated disorders.

Endemic cretinism is associated with endemic goiter and severe iodine deficiency. It manifests in two main forms: neurological cretinism and myxedematous cretinism



A 14-year-old boy with a large nodular goiter - in an area of severe iodine-deficiency disorders in northern Morocco.

- Neurological Cretinism: Characterized by deaf-mutism, squint, proximal spasticity and rigidity, and severe intellectual disability.
- Myxedematous Cretinism: Characterized by less severe intellectual impairment, severe short stature, coarse facial features, and myxedema.
- Prevention: Iodine deficiency disorders are best prevented through iodinated salt or iodized oil, as treatment is largely ineffective.
- Treatment: While signs of hypothyroidism may improve, neuromotor and intellectual deficiencies are irreversible. Surgical removal of large goiters is reserved for airway obstruction or cosmetic reasons.
- Launched in 1962 to control iodine deficiency disorders through salt iodination.
- Recommended daily iodine intake: 40-120 µg for children under 10 years. 150 µg for older children and adults. Additional 25-50 µg during pregnancy and lactation.
- 40-120 µg for children under 10 years.
- 150 µg for older children and adults.
- Additional 25-50 µg during pregnancy and lactation.
- 40-120 µg for children under 10 years.
- 150 µg for older children and adults.
- Additional 25-50 µg during pregnancy and lactation.

Prescribing high-dose thyroid hormone replacement therapy (Option A): This option is not a preventive measure. While thyroid hormone replacement therapy can treat hypothyroidism and its symptoms, it does not prevent iodine deficiency or its associated disorders, such as endemic cretinism.

Advising on a diet high in vegetables and fruits (Option C): While a healthy diet is beneficial for overall health, vegetables and fruits alone do not provide sufficient iodine to prevent iodine deficiency disorders.

Performing regular thyroid function tests for all community members (Option D) focuses on monitoring rather than prevention. Regular thyroid function tests are useful for diagnosing thyroid disorders but do not prevent iodine deficiency.

Solution for Question 2:

Correct Answer: (B) Plasma fractionated metanephrines

Explanation:

Symptoms such as episodic hypertension, palpitations, sweating, and headaches suggest Pheochromocytoma.

- In pheochromocytoma, there is an increase in urinary VMA (Vanillyl mandelic acid) and metanephrines.
- A Plasma fractionated metanephrines test is the most appropriate initial diagnostic test to confirm.

Pheochromocytoma:

- A catecholamine-secreting tumor that arises from the chromaffin cells of the abdominal sympathetic chain/ adrenal medulla
- Co-exist with neurofibromatosis, VHL syndrome, MEN syndrome

Clinical features:

- Classical triad - Headache (most common), profuse sweating, palpitation and tachycardia
- Hypertension (sustained or intermittent)
- Anxiety, panic attacks
- Hyperglycemia and Hypercalcemia

- Pallor- due to alpha-1 mediated vasoconstriction
- Polyuria and polydipsia
- Orthostatic hypotension
- Dilated cardiomyopathy
- Erythrocytosis
- Constipation
- Abdominal pain
- Weakness
- Weight Loss

The rule of 10 in Pheochromocytoma includes:

- 10% are bilateral
- 10% are extra-adrenal
- 10% are malignant
- 25-33% are inherited
- Symptomatic patients do not follow the Rule of 10’s

Diagnosis:

- Measurement of the plasma fractionated metanephrines is the most sensitive screening test
- Increase 24 hour urinary VMA(Vanillyl mandelic acid) and metanephrines
- MRI is the investigation of choice. (Option D)
- Biopsy and CT contrast scans are contraindicated in order to avoid hypertensive crises.

Treatment:

- Adrenalectomy is the treatment of choice
- Size<5cm- Laparoscopic adrenalectomy
- Preop Alpha-blockers should be given
- Mitotane should be started as adjuvant or palliative treatment

Averbuch’s chemotherapy protocol for Malignant pheochromocytoma includes the following:

- Dacarbazine (600mg/msq on days 1 and 2)
- Cyclophosphamide (750mg/msq on day 1)
- Vincristine (1.4mg/msq on day1)
- All are repeated every 21 days for three to six cycles.

(Option A) Serum calcium level is not relevant for diagnosing pheochromocytoma

(Option C) Thyroid function tests are used to evaluate thyroid disorders, not pheochromocytoma.

Solution for Question 3:

Correct Answer: (B) Plasma aldosterone concentration to plasma renin activity ratio

Explanation:

- Plasma aldosterone concentration to plasma renin activity is typically elevated in primary hyperaldosteronism due to high aldosterone levels and suppressed renin activity.
- Primary hyperaldosteronism, also known as Conn's syndrome, is characterized by excess production of aldosterone, leading to hypertension and hypokalemia.

Primary Aldosteronism / Conn Syndrome	Secondary Hyperaldosteronism
Due to adrenal adenoma/ hyperplasia Glucocorticoid remediable hypertension	Due to the stimulation of the renin-angiotensin pathway Renal artery stenosis Renin- secreting tumor Congestive cardiac failure Nephrotic/nephritic syndrome Liver disease

Clinical Presentation:

- Hypertension
- Hypokalemia - Muscle weakness
- Metabolic alkalosis

Serum cortisol level (Option A): Serum cortisol level is used to evaluate for conditions like Cushing's syndrome, not

hyperaldosteronism

Serum sodium level (Option C): Serum sodium level, while related, is not a specific diagnostic hyperaldosteronism

24-hour urinary calcium excretion (Option D): 24-hour urinary calcium excretion is more relevant in conditions like hyperparathyroidism or idiopathic hypercalciuria and is not related to primary hyperaldosteronism

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Solution for Question 4:

Correct Answer: (A) 24-hour urinary free cortisol test

Explanation:

The most appropriate initial diagnostic test is a 24-hour urinary-free cortisol test because it effectively screens for excess cortisol production indicative of Cushing syndrome.

Causes of Cushing's Syndrome:

Exogenous causes(Iatrogenic)	Prolonged administration of glucocorticoids (the most common cause of Cushing's).
ACTH dependent	Cushing's disease: ACTH-secreting pituitary adenoma(most common) Hypothalamic lesions Ectopic source - Neuroblastoma, Wilms tumor, Carcinoid syndrome
ACTH independent	Adrenal adenoma/ carcinoma Pigmented nodular hyperplasia McCune Albright syndrome

Clinical Features:

- Obesity, striae
- Moon facies, and buffalo hump
- Short stature
- Hypertension
- Bone pains
- Muscle weakness
- Behavioral problems
- Hirsutism

Diagnosis:

Screening test

- Loss in diurnal cortisol rhythm
- Overnight Dexamethasone suppression test: Plasma cortisol >1.8microgram at 8 a.m after overnight exposure to dexamethasone (1mg given at 11 p.m)
- 24hr urinary free cortisol excretion: Increased three times the normal value
- High-dose dexamethasone suppression test differentiates between pituitary and ectopic ACTH production; high doses of dexamethasone suppress ACTH levels in pituitary lesions but not in ectopic ACTH cases.
- Low dose dexamethasone suppression test (highly specific test): ACTH levels: a) <5pg/ml: ACTH independent cause b) 5-15pg/ml: ACTH-dependent cause c) 100pg/ml: Ectopic ACTH
- ACTH levels: a) <5pg/ml: ACTH independent cause b) 5-15pg/ml: ACTH-dependent cause c) 100pg/ml: Ectopic ACTH
- a) <5pg/ml: ACTH independent cause
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- c) 100pg/ml: Ectopic ACTH
- a) <5pg/ml: ACTH independent cause
- b) 5-15pg/ml: ACTH-dependent cause
- c) 100pg/ml: Ectopic ACTH

The best test to identify the source of ACTH production - Inferior petrosal sinus sampling

Test to rule out pseudo-Cushing’s syndrome:

- Low-dose DST+CRH stimulation test: Plasma cortisol levels are increased in true Cushing’s but remain unaffected in pseudo-Cushing’s. Dexa-induced negative feedback is overcome by CRH in true Cushing's.

Treatment:

- Surgical resection of pituitary/ adrenal lesions
- Medical: Metyrapone, Ketoconazole, Mitotane

Overnight Dexamethasone suppression test (Option B): Plasma cortisol >1.8 microgram at 8 a.m. after overnight exposure to dexamethasone (1mg given at 11 p.m.). This test helps to differentiate between Cushing syndrome and other causes of elevated cortisol.

Serum cortisol level measurement at 8 AM (Option C) - this is less specific and can be influenced by the body’s natural cortisol rhythm.

ACTH stimulation test (Option D): This test assesses the adrenal gland’s response to ACTH and is typically used to elevate adrenal insufficiency or the cause of Cushing syndrome rather than as an initial diagnostic test.

Solution for Question 5:

Correct Answer: (A) Primary adrenal insufficiency

Explanation:

The most likely diagnosis is Primary adrenal insufficiency, also known as Addison’s disease, which is characterized by damage to the adrenal glands leading to low cortisol and high ACTH levels.

- The increased levels of adrenocorticotropic hormone (ACTH) cause hyperpigmentation of the skin due to the conversion of pro-opiomelanocortin (POMC) to melanocyte-stimulating hormone (MSH).

Etiology:

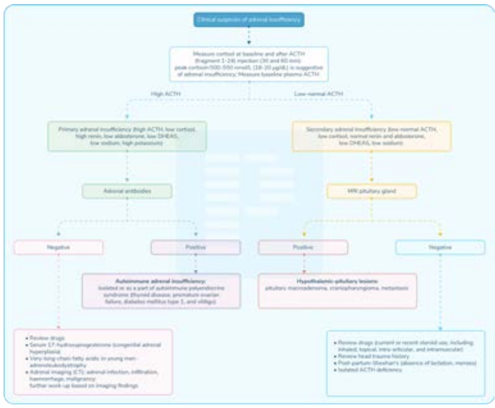
Primary adrenal insufficiency	Secondary adrenal insufficiency

Autoimmune mediated/Addison disease Infections: TB/HIV Adrenal Hemorrhage(Waterhouse- Friderichsen syndrome) CAH due to 3 beta -HSD deficiency Anatomical destruction of the gland Surgical removal Amyloidosis, sarcoidosis, hemochromatosis	Congenital malformations(e.g. Holoprosencephaly) Acquired insults (CNS trauma, surgery, irradiation) Pituitary tumors Sudden discontinuation of steroids after prolonged treatment Genetic defects

Clinical features:

- Hyperpigmentation seen in primary Adrenal insufficiency- seen in palmar creases, dorsal aspect of the foot, nipples, and axillary region
- Dehydration- lethargy
- Shock
- Abdominal pain, nausea, vomiting, salt cravings
- Hypoglycemia
- Dizziness and postural hypotension
- Hyponatremia
- Loss of libido
- Loss of pubic and axillary hair
- Anorexia, weight loss, fatigue, myalgia, arthralgia

Investigations:



Treatment:

Acute adrenal insufficiency requires immediate initiation of the following therapeutic strategies:

- Rehydration- saline infusion at an initial rate of 1 l/hr with continuous cardiac monitoring.
- Glucocorticoid replacement- bolus injection of 100 mg hydrocortisone, followed by the maintenance dose of 100-200mg over 24hr

- Mineralocorticoid replacement- initiated once the daily hydrocortisone dose has been reduced to <50 mg as higher doses of hydrocortisone have sufficient mineralocorticoid action.

Secondary adrenal insufficiency (Option B) is caused by inadequate ACTH secretion from the pituitary gland. This leads to low cortisol levels but normal or low ACTH levels.

Tertiary adrenal insufficiency (Option C) results from inadequate corticotropin-releasing hormone production from the hypothalamus, leading to low ACTH and cortisol levels.

Cushing’s syndrome (Option D) is characterized by excess cortisol levels. Symptoms include weight gain, hypertension, and moon facies, but not typically low cortisol levels or high ACTH levels with skin hyperpigmentation.

Solution for Question 6:

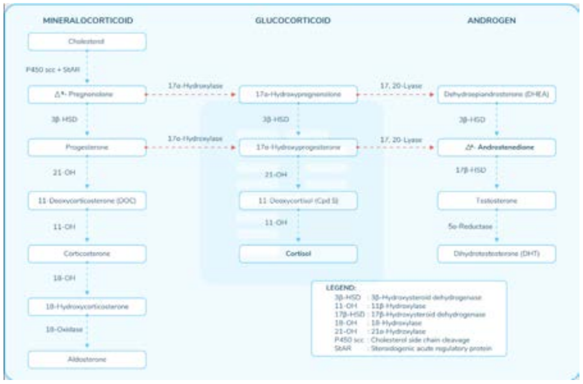
Correct Answer: A. 17 ■ hydroxylase deficiency

Explanation:

The patient’s symptoms of hypertension, hyperkalemia, and amenorrhoea with normal cortisol and low testosterone levels fit with 17 ■ hydroxylase deficiency.

17■ Hydroxylase deficiency:

- Normal cortisol and low sex hormones
- Excess mineralocorticoids: Hypertension, hypokalemia, and salt retention
- No sex steroids production: under virilization (ambiguous genitalia) in males



11 ■ Hydroxylase Deficiency (Option B):

- Deficiency of cortisol

- Increase in deoxycorticosterone (weak mineralocorticoid) and testosterone
- Salt retention and hypertension
- Excess sex steroid - Ambiguous genitalia in females
- Precocious puberty in males

3 ■ HSD deficiency (Option C):

- Deficient glucocorticoids and mineralocorticoids: shock, dehydration and hyperkalemia
- Salt-wasting crisis
- Deficient sex steroids: Normal genitalia in females Ambiguous genitalia in males
- Normal genitalia in females
- Ambiguous genitalia in males
- Normal genitalia in females
- Ambiguous genitalia in males

Solution for Question 7:

Correct Answer: B) 21-Hydroxylase deficiency with salt-wasting form; elevated 17-OHP levels

Explanation:

- The infant's symptoms (vomiting, dehydration, shock, and ambiguous genitalia) are characteristic of the salt-wasting form of 21-hydroxylase deficiency.
- This severe form results in almost absent enzyme activity, leading to critical electrolyte imbalances.
- The definitive diagnostic test for this condition is measuring extremely elevated levels of 17-hydroxyprogesterone (17-OHP), which typically exceeds 10000 ng/dL in salt-wasting forms.

21-Hydroxylase Deficiency	
Most common form of CAH (90%) Enzyme Affected: 21-hydroxylase	
Pathogenesis	21-hydroxylase deficiency impairs the synthesis of cortisol and aldosterone. Low cortisol levels lead to elevated ACTH levels.

	Elevated ACTH causes the accumulation of steroid precursors, including dehydroepiandrosterone (DHEA), androstenedione, and 17-hydroxyprogesterone (17-OHP).
Clinical Manifestations	Salt-Wasting Form: Severe deficiency with nearly absent enzyme activity. Presents in neonates with salt-wasting, virilization, ambiguous genitalia in girls, failure to thrive, polyuria, hyperpigmentation, and shock. Diagnosis may be missed in boys. Simple Virilizing Form: Partial enzyme activity (25%) allows aldosterone production. Features include virilization in girls and peripheral precocious puberty in boys. (Option A) Non-Classic Form: Mild enzyme deficiency with mild hyperandrogenism. Symptoms include hirsutism, acne, and menstrual irregularities in adolescents. (Option C)
Diagnosis	Salt-Wasting Form: Extreme elevations of 17-OHP levels (10000-20000 ng/dL, normal <200 ng/dL) in the context of clinical signs of adrenal insufficiency. Simple Virilizing and Non-Classic Forms: Elevated 17-OHP levels measured before and 60 minutes after an intramuscular ACTH injection (0.25 mg).
Treatment	Salt-wasting and Virilizing Forms treated with Hydrocortisone (10-15 mg/m²/day) and Fludrocortisone (0.1 mg/day). Long-acting glucocorticoid preparations (dexamethasone or prednisolone) may be used after growth and puberty are complete.

11-Hydroxylase Deficiency; Hydrocortisone alone (Option D); 11-Hydroxylase deficiency typically presents with hypertension rather than the salt-wasting and ambiguous genitalia seen in the scenario. This condition is managed with hydrocortisone, but it does not fit the clinical picture described here.

Solution for Question 8:

Correct Answer: B) Treatment with corticosteroids only

Explanation:

The treatment for neonatal thyrotoxicosis involves a combination of antithyroid drugs, propranolol to manage symptoms, and corticosteroids to address inflammation and related issues.

Neonatal Thyrotoxicosis:

- Occurrence: Affects 1% of infants born to mothers with Graves' disease, presenting with fetal thyrotoxicosis and possible cardiac failure. (Options A & D)
- Typically appears within the first week of life but can be delayed if the mother is on antithyroid medications or has TSH receptor-blocking antibodies. (Option C)
- Clinical presentation: Goiter, tachypnea, tachycardia, diarrhea, sweating, low birth weight, irritability, and features of heart failure
- Treatment: Includes a combination of antithyroid drugs, propranolol, and corticosteroids.

Solution for Question 9:

Correct Answer: B) Methimazole

Explanation: The most appropriate initial treatment for this 10-year-old girl with symptoms suggestive of hyperthyroidism (weight loss, increased appetite, heat intolerance, palpitations, tremors, and neck swelling) is Methimazole. The dosage is 0.5-1.0 mg/kg/day in 3-4 divided doses.

Etiology:

- Infancy: Transplacental passage of thyroid antibodies, TSH receptor activating mutation.
- After Infancy: Graves disease (TSH receptor stimulating antibody), toxic thyroid nodule, toxic multinodular goiter, pituitary resistance to T3.
- Transient: Release of preformed thyroid hormone due to viral or bacterial infections of the thyroid (acute or subacute thyroiditis), early Hashimoto thyroiditis (Hashitoxicosis), iatrogenic.

Clinical Features:

- Symptoms: Weight loss with increased appetite, tremors, diarrhea, warm extremities, increased sweating, anxiety, inability to concentrate, personality changes, mood instability, and poor school performance.
- Signs: Firm homogeneous goiter, tachycardia, cardiac arrhythmia, high output cardiac failure.
- Eye Signs: Less common in children compared to adults; may include lid lag, ophthalmoplegia, absence of wrinkling, chemosis, and proptosis.

Diagnosis:

- Elevated serum FT4 and FT3 levels confirm diagnosis.
- Goiter, eye signs, and hyperthyroidism suggest Graves disease.
- Diffusely increased radiotracer uptake suggests Graves disease; reduced uptake indicates AIT.

Management:

- Medication: Antithyroid Drugs: Initially started with Methimazole (0.5-1.0 mg/kg/day in 3-4 divided doses). Propylthiouracil is contraindicated due to hepatotoxicity. (Option A) It is typically reserved for specific situations, such as during the first trimester of pregnancy or in cases of methimazole intolerance. Beta-Blockers: Propranolol (2 mg/kg/day in 2-3 divided doses) to manage sympathetic symptoms. (Option D) It is used to manage the adrenergic symptoms (e.g., tremors, anxiety, palpitations) of hyperthyroidism, but it does not treat the underlying cause. It is often used in conjunction with antithyroid drugs like methimazole. Corticosteroids: Prednisolone (1-2 mg/kg/day) for hyperthyroid storm and to inhibit peripheral conversion of T4 to T3. (Option C) It is not an initial treatment for hyperthyroidism.
- Antithyroid Drugs: Initially started with Methimazole (0.5-1.0 mg/kg/day in 3-4 divided doses).
- Propylthiouracil is contraindicated due to hepatotoxicity. (Option A) It is typically reserved for specific situations, such as during the first trimester of pregnancy or in cases of methimazole intolerance.
- Beta-Blockers: Propranolol (2 mg/kg/day in 2-3 divided doses) to manage sympathetic symptoms. (Option D) It is used to manage the adrenergic symptoms (e.g., tremors, anxiety, palpitations) of hyperthyroidism, but it does not treat the underlying cause. It is often used in conjunction with antithyroid drugs like methimazole.
- Propranolol (2 mg/kg/day in 2-3 divided doses) to manage sympathetic symptoms. (Option D)
- It is used to manage the adrenergic symptoms (e.g., tremors, anxiety, palpitations) of hyperthyroidism, but it does not treat the underlying cause. It is often used in conjunction with antithyroid drugs like methimazole.
- Corticosteroids: Prednisolone (1-2 mg/kg/day) for hyperthyroid storm and to inhibit peripheral conversion of T4 to T3. (Option C) It is not an initial treatment for hyperthyroidism.
- Surgery and Radioiodine: Considered for patients with failure of medical management or relapse. Surgery: Partial or total thyroidectomy for large or toxic nodular goiter. Radioiodine: Increasingly used in childhood Graves disease.
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- Surgery: Partial or total thyroidectomy for large or toxic nodular goiter.

- Radioiodine: Increasingly used in childhood Graves disease.
- Antithyroid Drugs: Initially started with Methimazole (0.5-1.0 mg/kg/day in 3-4 divided doses).
- Propylthiouracil is contraindicated due to hepatotoxicity. (Option A) It is typically reserved for specific situations, such as during the first trimester of pregnancy or in cases of methimazole intolerance.
- Beta-Blockers: Propranolol (2 mg/kg/day in 2-3 divided doses) to manage sympathetic symptoms. (Option D) It is used to manage the adrenergic symptoms (e.g., tremors, anxiety, palpitations) of hyperthyroidism, but it does not treat the underlying cause. It is often used in conjunction with antithyroid drugs like methimazole.
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- Corticosteroids: Prednisolone (1-2 mg/kg/day) for hyperthyroid storm and to inhibit peripheral conversion of T4 to T3. (Option C) It is not an initial treatment for hyperthyroidism.
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- It is used to manage the adrenergic symptoms (e.g., tremors, anxiety, palpitations) of hyperthyroidism, but it does not treat the underlying cause. It is often used in conjunction with antithyroid drugs like methimazole.
- Considered for patients with failure of medical management or relapse.
- Surgery: Partial or total thyroidectomy for large or toxic nodular goiter.
- Radioiodine: Increasingly used in childhood Graves disease.

Solution for Question 10:

Correct Answer: B) Start recombinant human growth hormone therapy

Explanation:

The patient's clinical presentation and laboratory findings are consistent with isolated growth hormone deficiency (IGHD). The most appropriate next step in management is to start recombinant human growth hormone (rhGH) therapy.

IGHD is characterized by impaired linear growth due to insufficient production or secretion of growth hormone (GH) without deficiencies in other pituitary hormones.

Isolated growth hormone deficiency (IGHD):

Definition	Insufficient production or secretion of growth hormone (GH) without deficiencies in other pituitary hormones
Clinical Features	Height below 1st percentile or >2 SD below mid-parental height Growth velocity below 25th percentile Delayed bone age High-pitched voice Normal birth weight and length Proportionate short stature- Upper segment (US): Lower segment (LS) ratio normal
Biochemical Findings	Low IGF-1 levels Peak GH levels <10 ng/mL on two provocative tests (on GH stimulation test)
Diagnosis	Measurements of IGF-1 and IGF-binding protein levels GH stimulation test Done using arginine/clonidine/insulin Peak GH level <10 ng/ml → GH deficiency Bone age < Chronological age
Treatment	Recombinant human growth hormone (rhGH) Initial dose: 0.16-0.24 mg/kg/week Administered subcutaneously once daily GH therapy should be continued until near-final height is achieved.

Option A: Observation is not recommended as the patient has a confirmed diagnosis of IGHD and would benefit from prompt treatment.

Option C: Genetic testing for PROP1 mutations is not necessary at this point, as PROP1 mutations typically cause multiple pituitary hormone deficiencies, which is not the case here.

Option D: Monotherapy with gonadotropin-releasing hormone agonist is not appropriate for a 6-year-old boy and is not the primary treatment for IGHD. However, gonadotropin-releasing hormone agonist is adjuvantly used in the hope that interruption of puberty will delay epiphyseal fusion to reach the target height.

Solution for Question 11:

Correct Answer: (A) Heel prick test for TSH

Explanation:

- The most effective method for screening newborns for congenital hypothyroidism is the heel prick test for TSH.
- TSH is a more sensitive indicator for thyroid disorders compared to T4 or T3 levels.

Diagnosis:

Primary hypothyroidism:

- T4 level: low
- TSH usually>100mIU/L

Central hypothyroidism: Low TSH, T3, and T4 levels

X-ray findings in congenital hypothyroidism:

- Absence of distal femoral and proximal tibial epiphysis at birth
- Punctate epiphyseal dysgenesis
- Deformity or breaking of the 12th thoracic or 1st or 2nd lumbar vertebrae
- X-ray image of skull- large fontanelle and wide sutures

Screening:

- Universal newborn screening for congenital hypothyroidism
- At birth, with umbilical cord blood
- Heel prick: dried blood spots, sample collected beyond 48 hrs or 48-72hrs
- Should not be done in first 1-2 days, to avoid TSH surge
- Most sensitive approach- check for T4 and TSH both

ASH's first strategy:

- Initial TSH screening using dried blood spot
- If TSH>20mIU/L, recheck
- Obtain a venous blood sample to check T4 and TSH levels if TSH remains elevated.

Treatment

- Oral levothyroxine (early morning with empty stomach for older children and for newborns it is to be given at a fixed time)
- The dose is higher in the earlier age group and as the babies grow the dose reduces.
- Newborn- 10-15mg/kg/day

- Children- 2-4mg/kg/day

(Option B) The heel prick test for thyroid hormone (T4 and T3) is less effective as an initial screening test compared to TSH levels because T4 and T3 will be consistently abnormal in the early stages of the condition.

(Option C) USG is not typically done for screening, it is done for diagnosis and assessment after screening.

(Option D) Clinical assessment of physical signs is not reliable for early detection as symptoms of congenital hypothyroidism can be subtle or absent at birth

Solution for Question 12:

Correct Answer: (C) Hypertonia

Explanation:

The given picture is that of a child with untreated congenital hypothyroidism.

In congenital hypothyroidism, hypotonia is seen and not hypertonia

- Congenital hypothyroidism is the most common preventable cause of mental retardation which results from a deficient production of thyroid hormone.
- More common among girls
- The most common cause of congenital hypothyroidism is thyroid dysgenesis.

Etiology of congenital hypothyroidism:

1. Primary hypothyroidism:

- Thyroid dysgenesis
- Thyroid dyshormonogenesis
- Iodine deficiency (endemic goiter)
- Defect in thyroid hormone transporter- Mutation in MCT8 (monocarboxylate transporter 8) gene
- Maternal antibodies- TRBAb (thyrotropin receptor-blocking antibody)
- Maternal medications- Iodides, amiodarone, propylthiouracil, methimazole, radioiodine

2. Central (hypopituitary) hypothyroidism:

- TSH deficiency- mutation in TSH beta subunit gene
- TRH deficiency- mutation in the TRH gene
- TRH unresponsiveness- mutation in TRH receptor gene
- Multiple congenital pituitary hormone deficiencies

Clinical features of congenital hypothyroidism:

- Prolongation of physiological jaundice (Option D) - early sign (it is due to delayed maturation of glucuronide conjugation)
- Constipation - not responsive to treatment (Option B)
- Wide open anterior and posterior fontanelles
- Developmental delay, delayed bone growth, delayed dentition
- A dull-looking face with macroglossia, feeding difficulties (Option A), choking spells
- Hypothermia, hypotonia, slow relaxation of deep tendon reflexes
- Macrocytic anemia refractory to treatment

Solution for Question 13:

Correct Answer: (D) Medullary thyroid carcinoma

Explanation:

Medullary thyroid carcinoma is seen in MEN2A syndrome

MEN1 syndrome Features:

- Chromosome 11q13/MEN1 gene
- Also known as Wermer’s syndrome
- Classical tumors seen are: Parathyroid adenoma (90%) - Most common overall. Enteropancreatic tumors - The most common functioning tumor is gastrinoma(the most common site is duodenum) Pituitary adenoma - Most common in prolactinoma
- Parathyroid adenoma (90%) - Most common overall.
- Enteropancreatic tumors - The most common functioning tumor is gastrinoma(the most common site is duodenum)
- Pituitary adenoma - Most common in prolactinoma

- Associated tumors: Angiofibroma, Collagenoma, Adrenal cortical tumors, lipoma, neuroendocrine tumors, Meningioma, pheochromocytoma, carcinoid tumors

MEN 2 Syndrome Features:

MEN2A

- Known as Sipple's syndrome
- Chromosome 10/RET gene
- Associated conditions are- Medullary thyroid carcinoma Pheochromocytoma Parathyroid adenoma
- Medullary thyroid carcinoma
- Pheochromocytoma
- Parathyroid adenoma

MEN2B

- Known commonly as MEN 3 syndrome
- Chromosome 10/RET gene
- Associated conditions are: 1) Medullary thyroid carcinoma 2) Pheochromocytoma
- 1) Medullary thyroid carcinoma
- 2) Pheochromocytoma
- 1) Medullary thyroid carcinoma
- 2) Pheochromocytoma
- Other associated abnormalities include- Mucosal neuroma, Marfanoid habitus, Megacolon

MEN 4 Syndrome

- Associated with CDKN1B gene on chromosome 12

Associated conditions: parathyroid adenoma, pituitary adenoma, adrenals and renal tumors, and reproductive organ tumors like testicular carcinoma.

Solution for Question 14:

Correct Answer: (B) Desmopressin

Explanation:

The patient's symptoms of excessive thirst, frequent urination, hypernatremia, and dilute urine are consistent with central diabetes insipidus (DI), a condition caused by a deficiency of antidiuretic hormone (ADH).

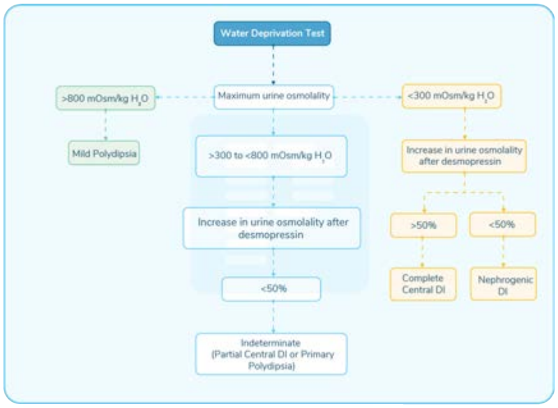
Desmopressin is a synthetic analog of ADH and is the primary treatment for central diabetes insipidus.

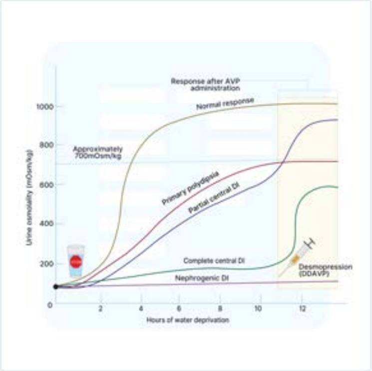
- It helps to reduce the excessive urine output by mimicking the action of ADH, thus improving the body's ability to concentrate urine

Diagnosis:

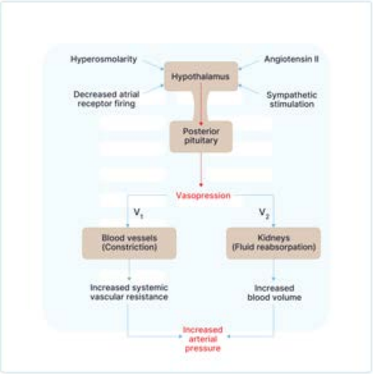
- Low urine osmolality (<600 mosm/L) in association with high plasma osmolality {>300 mosm/L}
- Water deprivation test → differentiates psychogenic polydipsia from DI

Water deprivation test





- The vasopressin response test → to differentiate central DI and nephrogenic DI
- An increase in urine osmolality by >50% of the baseline indicates Central DI



Treatment:

- Central DI: Vasopressin analogue (Desmopressin)
- Nephrogenic DI: Thiazide diuretics, Indomethacin, and salt restriction.

Other Uses of Desmopressin include:

- Central Diabetes Insipidus (CDI)
- Nocturnal Enuresis
- Von Willebrand Disease
- Diagnostic Testing for CDI
- Temporary treatment of Polyuria-Polydipsia Syndrome
- Post-lumbar puncture headache

(Option B) Insulin therapy is used for managing diabetes mellitus

(Option C) Metformin Is used for treating type 2 DM

(Option D) Spironolactone is a potassium-sparing diuretic used to treat conditions like hypertension, and edema.

Solution for Question 15:

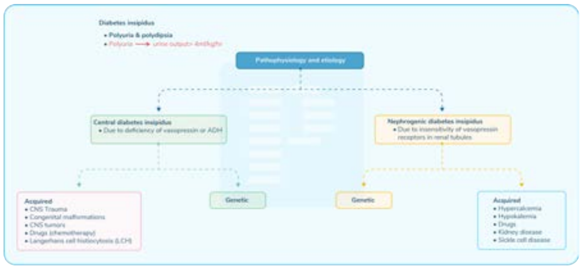
Correct Answer: (A) Polyuria and polydipsia

Explanation:

In diabetes insipidus, the kidney’s inability to concentrate urine leads to large volumes of dilute urine (Polyuria).

Polydipsia means excessive thirst because the body tries to compensate for high fluid loss by increasing thirst to encourage more fluid intake.

- Diabetes insipidus is a condition in which there is the absence of or poor response to anti-diuretic hormone (ADH).
- This results in reduced distal tubule and collecting duct permeability to water.
- This causes the kidneys to excrete large amounts of dilute urine that clinically manifests as polyuria and polydipsia.



(Option B and D) Polyphagia and hyperglycemia - are symptoms of diabetes mellitus

(Option C) Edema is due to fluid retention which is typically not seen in DI

Solution for Question 16:

Correct Answer: C) Octreotide

Explanation:

The patient's clinical features and biochemical tests indicate growth hormone excess disorder, which can occur due to multiple causes.

Since MRI does not reveal the presence of any adenoma, the patient is to be put on medical therapy, which is primarily done using Octreotide (Somatostatin analog) in pediatric patients.

Growth Hormone Excess Disorder:

Clinical Features	Rapid linear growth Tall stature Coarse facial features Enlarged hands and feet Deep voice Possible headaches or visual disturbances
Diagnosis	Elevated IGF-1 and IGFBP-3 levels Failure to suppress GH on oral glucose tolerance test (>5 ng/mL in adolescents) MRI showing pituitary adenoma
Treatment Options	Trans-sphenoidal surgery (first-line for well-circumscribed adenomas) (Option A) Radiation therapy (if surgery fails or is contraindicated) Medical therapy: 1st line- Somatostatin analogs (octreotide, lanreotide) Refractory cases- GH receptor antagonist (pegvisomant) (Option B) Adjuvant- Dopamine agonists (cabergoline, bromocriptine) (Option D)

Solution for Question 17:

Correct Answer: A. a-3, b-1, c-1, d-2

Explanation:

Disease	GHRH	GH	IGF-1
GHRH deficiency	Low	Low	Low

GHRH receptor resistance	Normal / High	Low	Low
GH deficiency	Normal / High	Low	Low
GH receptor resistance (Laron syndrome)	Normal / High	Normal / High	Low

Pediatric Endocrinology and Metabolism

1. Which of the following clinical features is most likely associated with severe Diabetic Ketoacidosis?

A. Venous pH between 7.25 and 7.35, alert patient

B. Venous CO2 between 16 and 20 mEq/L, patient with mild fatigue

C. Venous pH less than 7.15, patient with Kussmaul respirations and depressed sensorium

D. Venous CO2 between 10 and 15 mEq/L, patient who is sleepy but arousable

2. A 17 y/o patient presents with symptoms of polyuria, polydipsia, and fatigue. The patient reports a family history of diabetes and has recently experienced weight loss. Genetic testing identifies a mutation in the GCK gene. Based on this information, what is the most likely diagnosis?

A. Type 1 Diabetes Mellitus

B. Type 2 Diabetes Mellitus

C. Maturity Onset Diabetes of the Young

D. Latent Autoimmune Diabetes of Adults

3. A 13-year-old female presents for a routine check-up. Physical examination reveals terminal pubic hair covering the entire pubic triangle and an elevated areola with a “double scoop” appearance. Which Tanner stage is most appropriate for this patient?

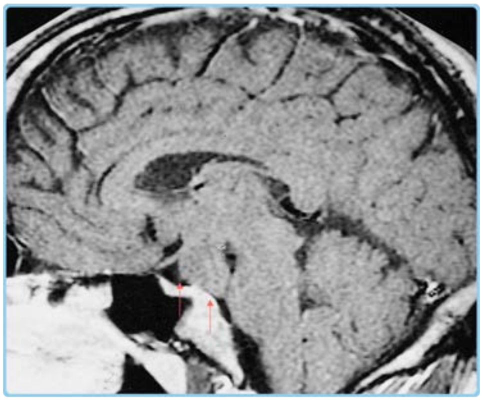
A. Stage 2

B. Stage 3

C. Stage 4

D. Stage 5

4. A 5 y/o girl presents with early-onset and rapid progression of puberty, frequent laughing spells, and seizures. An MRI is performed and the image is given below. What is the most likely diagnosis?



A. Craniopharyngioma

B. Hypothalamic hamartoma

C. Hydrocephalus

D. Head trauma

5. A 10 y/o boy presents with a 2-year history of weight gain. His height is in the 90th percentile, weight is in the 95th percentile, and there are no developmental delays or dysmorphic features. Both parents are obese, and there is no history of neurological issues or chronic medication use. What is the most likely cause of his obesity?

A. Cushing syndrome

B. Hypothyroidism

C. Constitutional obesity

D. Prader-Willi syndrome

6. A 7-year-old child with obesity is enrolled in a weight management program. The family is introduced to the Traffic Light Diet, and they struggle with understanding which foods should be prioritized for their child. Which of the following foods should be categorized as "Yellow" in the Traffic Light Diet?

A. Fried chicken

B. Carrot sticks

C. Whole grain bread

D. Candy

7. A 5 y/o child presents with obesity, feeding difficulties in infancy, developmental delays, and a history of behavioural issues like anxiety and obsessive-compulsive disorder. O/E, you note small hands and dysmorphic facial features. Which genetic syndrome best describes the child’s condition?

A. Prader-Willi Syndrome

B. Laurence-Moon-Bardet-Biedl Syndrome

C. Down Syndrome

D. Turner Syndrome

8. A 6-year-old girl with a BMI in the 98th percentile presents with lethargy, frequent thirst, increased urination, and dark patches on her neck. What is the most likely diagnosis?

A. Type 2 Diabetes Mellitus (T2DM) with associated obesity.

B. Hypothyroidism with weight gain.

C. Cystic Fibrosis-related diabetes (CFRD).

D. Adrenal insufficiency.

9. A 12 y/o girl is undergoing treatment for diabetic ketoacidosis (DKA). After an initial bolus of 0.9% saline, she is started on an insulin infusion at 0.1 U/kg/h after an hour. Several hours later, her blood glucose level decreased to 260 mg/dL. What should be the next step in her management?

A. Increase the insulin infusion rate

B. Stop insulin therapy

C. Add 5% dextrose to the IV fluids

D. Administer an additional bolus of 0.9% saline

10. A 7y/o girl presents with early breast development and rapid growth. Examination shows Tanner's stage 3 breasts and her height is above the 90th percentile. Blood tests reveal elevated LH levels. What is the most likely diagnosis?

A. Peripheral Precocious Puberty

B. Constitutional delayed puberty

C. Premature Thelarche

D. Central Precocious Puberty

11. Which of the following insulin regimes is considered more flexible in terms of meal timing and carbohydrate intake?

A. Basal-Bolus Regimen

B. Three Daily Injection Regimen

C. Once-daily injection of long-acting insulin only

D. Twice-daily injection of short-acting insulin only

12. Which of the following criteria is NOT diagnostic of Type 1 Diabetes Mellitus?

A. A fasting plasma glucose level of ≥ 126 mg/dL.

B. A 2-hour plasma glucose level of ≥ 200 mg/dL on OGTT.

C. A random plasma glucose level of ≥ 180 mg/dL with symptoms of diabetes.

D. Haemoglobin A1c ≥ 6.5%

13. A 16 y/o girl with normal breast development presents with primary amenorrhea. An ultrasound shows an absent uterus and undescended testes. Laboratory results reveal elevated androgen levels, normal estrogen levels, and a 46, XY karyotype. A mutation in the AR gene is confirmed. What is the most likely diagnosis?

A. Complete Androgen Insensitivity Syndrome

B. Partial Androgen Insensitivity Syndrome

C. Mild Androgen Insensitivity Syndrome

D. Klinefelter Syndrome

14. Which of the following conditions is classified as a 46, XY DSD?

A. Placental aromatase deficiency

B. 17β-HSD deficiency

C. Klinefelter syndrome

D. 21-hydroxylase deficiency

15. A 24-week-old fetus is found to have ambiguous genitalia during a routine ultrasound. Karyotyping reveals a 46, XY chromosomal pattern. Ultrasound and examination indicate underdeveloped male internal genitalia and features more typical of female external genitalia. Which of the following is the most likely explanation for this presentation?

A. Deficient Expression of SRY

B. Excessive SOX9 Activity

C. Absent AMH Production

D. Decreased DAX1 Expression

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16. A 14-year-old boy presents with delayed puberty and is significantly below the 3rd percentile for height. His laboratory tests reveal low levels of luteinizing hormone (LH) and follicle-stimulating hormone (FSH). What is the most likely cause of his delayed puberty?

A. Constitutional Delay

B. Hypogonadotropic Hypogonadism

C. Hypergonadotropic Hypogonadism

D. Klinefelter Syndrome

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17. A 7 y/o boy is evaluated due to early signs of puberty, including the development of pubic hair, a noticeable growth spurt, increased testicular volume, and significantly advanced bone age. Blood tests show elevated basal luteinizing hormone (LH) levels. What is the most probable cause of his early pubertal development?

A. Central precocious puberty

B. Peripheral precocious puberty due to congenital adrenal hyperplasia

C. Peripheral precocious puberty due to an hCG-secreting tumour

D. Peripheral precocious puberty due to an androgen-secreting testicular tumour

18. A 5 y/o girl is brought to the clinic with a history of irregular vaginal bleeding, early breast development and an advanced bone age. The examination findings of her skin are given below. What is the underlying cause of her symptoms?

A. McCune-Albright syndrome

B. Central precocious puberty

C. Hypothyroidism

D. Adrenal hyperplasia

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	3
Question 4	2
Question 5	3
Question 6	3
Question 7	1
Question 8	1
Question 9	3
Question 10	4
Question 11	1
Question 12	3
Question 13	1
Question 14	2
Question 15	1
Question 16	2
Question 17	1
Question 18	1

Solution for Question 1:

Correct Answer: C) Venous pH less than 7.15, patient with Kussmaul respirations and depressed sensorium

Explanation:

This is characteristic of severe DKA, where the patient may progress to a coma if not treated promptly.

Kussmaul respirations refer to deep, rapid breathing patterns as a compensatory response to metabolic acidosis. These patients also have a fruity odour in their breath due to acetone.

Diabetic Ketoacidosis (DKA):

Definition:

- An acute, life-threatening complication of T1DM due to severe insulin deficiency.
- Occurs when the body breaks down fat for fuel, leading to ketone production and metabolic acidosis.

Classification of DKA:

Severity	Venous pH	Venous CO2 (mEq/L)	Clinical Presentation
Mild DKA	7.25 - 7.35 (Option A ruled out)	16 - 20 (Option B ruled out)	Alert, possible fatigue.
Moderate DKA	7.15 - 7.25	10 - 15 (Option D ruled out)	Kussmaul respirations, sleepy but arousable.
Severe DKA	< 7.15	< 10	Kussmaul or depressed respirations, sleepy to coma, severe hypernatremia (Na+ > 150 mEq/L).

Solution for Question 2:

Correct Answer: C) Maturity Onset Diabetes of the Young

Explanation:

Maturity Onset Diabetes of the Young (MODY) is characterized by an autosomal dominant inheritance pattern and is often diagnosed in individuals under 25 years of age. GCK mutations are a common cause of MODY, leading to a mild form of diabetes that does not necessarilyrequire insulin treatment. The patient’s age, symptoms, and GCK mutation make MODY themost likely diagnosis.

Aspect	Type 1 diabetes mellitus	Type 2 diabetes mellitus	Maturity onset diabetes of the young (MODY)	Latent autoimmune diabetes of adults (LADA/ T1.5 DM)
Age at diagnosis	6 mo-18 yrs	Puberty; rarely younger than 10 yrs	Younger than 25 yrs	Age greater than 30 years
Causes and genetic factors	Autoimmune; genetic predisposition (HLA and other genes)	Obesity; genetic and ethnic predisposition	Autosomal dominant; HNF1A, HNF4A, GCK, HNF1B (rare)	Positive autoantibodies to islet β cells; glutamic acid decarboxylase antibody is the most predominant.
Associated Features	Lean or weight loss at diagnosis; thyroid autoimmunity; celiac disease	Obesity; acanthosis nigricans; polycystic ovarian syndrome; hypertension; hyperlipidemia; fatty liver disease; family history	Lean or weight loss at diagnosis; GCK mutations are asymptomatic	Polyuria, polydipsia, nocturia, fatigue, visual changes, tingling in the feet, and weight loss; family history; low birth weight history
Diabetic ketoacidosis at presentation	Yes; about 25%	Yes; 5–20%	No	No
Treatment	Insulin	Lifestyle modification; metformin; insulin	Sulfonylurea; no treatment for GCK mutations	Insulin; Dipeptidyl peptidase (DPP) 4 inhibitors

Type 1 Diabetes Mellitus (Option A) typically presents in children or adolescents and is associated with autoimmune destruction of insulin-producing beta cells. It often results in diabetic ketoacidosis at diagnosis and requires insulin treatment. The patient in this scenario is 22 years old and has a GCK mutation, which does not align with Type 1 Diabetes.

Type 2 Diabetes Mellitus (Option B) usually presents in adulthood and is associated with obesity, genetic predisposition, and metabolic syndrome features like hypertension and hyperlipidemia. The patient in this scenario is lean, has weight loss, and genetic testing identifies a GCK mutation, which is more indicative of MODY rather than Type 2 Diabetes.

Latent Autoimmune Diabetes of Adults (Option D) is a form of diabetes that occurs in adults and shares features with both Type 1 and Type 2 diabetes. It is characterized by the presence of autoantibodies to islet beta cells, and patients may initially be misdiagnosed as having Type 2 diabetes. However, the patient's absence of autoimmune markers suggest that LADA is less likely in this case.

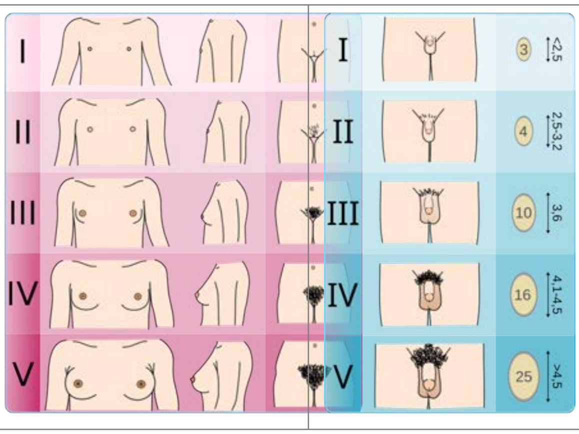
Solution for Question 3:

Correct Answer: C) Stage 4

Explanation:

Stage 4 is characterized by terminal hair filling the entire triangular area over the pubic region and an elevated areola forming a “double scoop” appearance. The patient’s findings match this description.

TANNER STAGING			
Stages	Pubic Hair Scale (both male and female)	Female Breast Development Scale	Male External Genitalia Scale
Stage 1	No hair	No glandular breast tissue palpable	Testicular volume < 4 ml or long axis < 2.5 cm
Stage 2 (Option A ruled out)	Downy hair	Breast bud palpable under the areola (1st pubertal sign in females)	4 ml-8 ml (or 2.5 to 3.3 cm long), 1st pubertal sign in males
Stage 3 (Option B ruled out)	Scant terminal hair	Breast tissue palpable outside areola; no areolar development	9 ml-12 ml (or 3.4 to 4.0 cm long)
Stage 4	Terminal hair that fills the entire triangle overlying the pubic region	Areola elevated above the contour of the breast, forming a “double scoop” appearance	15-20 ml (or 4.1 to 4.5 cm long)
Stage 5 (Option D ruled out)	Terminal hair that extends beyond the inguinal crease onto the thigh	The areolar mound recedes into a single breast contour with areolar hyperpigmentation, papillae development, and nipple protrusion.	> 20 ml (or > 4.5 cm long)

Image			

Solution for Question 4:

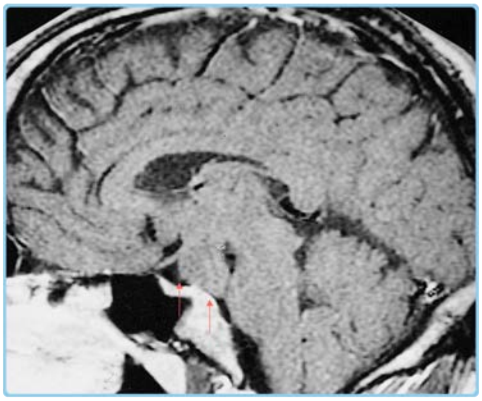
Correct Answer: B) Hypothalamic hamartoma

Explanation:

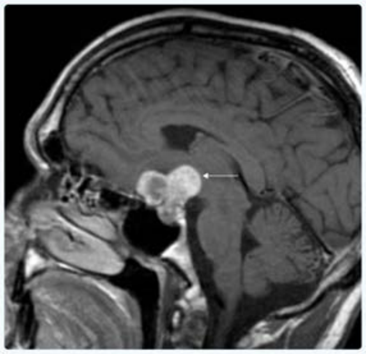
Hypothalamic hamartoma is the most likely diagnosis due to its association with early-onset puberty, gelastic seizures (uncontrolled laughter), and a mass typically in the hypothalamic region. which aligns with the clinical presentation and MRI findings in this case.

Hypothalamic Hamartoma:

- A benign, non-neoplastic tumour located in the hypothalamus. It is the most common cause of central precocious puberty due to its effect on the hypothalamic-pituitary-gonadal axis.
- Symptoms: Characterized by early-onset and rapid progression of puberty, gelastic seizures (uncontrolled laughter), and other seizure types.
- Imaging: MRI typically shows a well-defined mass in the hypothalamus. This is the most consistent with the presentation and imaging described.

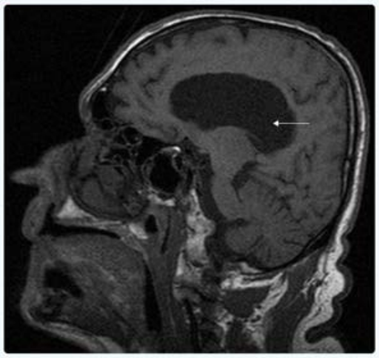


Craniopharyngioma (Option A) is a benign tumour near the pituitary gland that can cause endocrine problems, vision issues, and headaches. It may lead to growth issues and seizures but not early-onset puberty or laughing spells. CT imaging typically shows a cystic mass with calcifications in the suprasellar region.



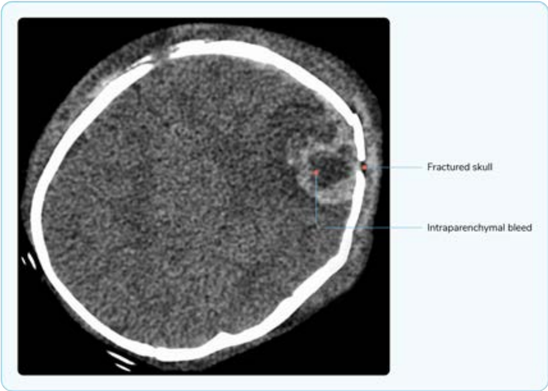
Craniopharyngioma

Hydrocephalus (Option C) is the buildup of cerebrospinal fluid in the brain's ventricles, causing increased intracranial pressure. Symptoms include headaches, vomiting, and possible developmental delays. MRI shows enlarged ventricles, not a hypothalamic mass.



Hydrocephalus

Head trauma (Option D) involves an injury to the head, leading to symptoms like confusion, headache, and possible seizures. MRI would typically reveal signs of trauma or bleeding, not a mass in the hypothalamus.



An intraparenchymal bleed with an overlying skull fracture from abusive head trauma

Solution for Question 5:

Correct Answer: C) Constitutional obesity

Explanation:

The child’s obesity is likely due to environmental factors, as he has normal growth for his age, no developmental delay, and no dysmorphic features. The family history of obesity also supports constitutional obesity, which is more common and accounts for 95% of cases.

Feature	Constitutional (Exogenous) Obesity	Pathological (Endogenous) Obesity
Cause	Imbalance in energy intake and expenditure	Endocrine, genetic, hypothalamic, or drug-related causes
Aetiology	Environmental factors (95% cases)	Endocrine: Cushing syndrome, growth hormone deficiency, hypothyroidism, pseudohypoparathyroidism.
Hypothalamic: Head injury, infection, brain tumour, radiation,		

chemotherapy, followingneurosurgery, ROHHAD(NET).		
Drugs: Antiepileptic drugs, steroids, antidepressants, estrogen.		
Genetic syndromes: Prader-Willi, Bardet-Biedl, Laurence-Moon, Beckwith-Wiedemann, Carpenter syndromes.		
Monogenic disorders: Leptin deficiency, or resistance (leptin receptor defect), abnormalities of melanocortin-4 receptor and proconvertase.		
Psychological: Personal or parental stress, binge eating.		
Growth	Normal or tall for age	Decelerated linear growth, height below genetic potential
Body Fat Distribution	Generalized obesity	Often central obesity (Cushing syndrome)
Appetite	Normal or slightly increased	May show excessive appetite (e.g., hypothalamic lesions)
Family History	Often positive for a family history of obesity	Family history may suggest genetic syndromes (e.g., Prader-Willi)
Developmental Delay/Dysmorphism	Absent specific dysmorphic or syndromic features	May be present (e.g., genetic syndromes like Bardet-Biedl)

Cushing syndrome (Option A) typically presents with central obesity, growth retardation, and features like hirsutism, purple striae, and a buffalo hump. The patient in the scenario has generalized obesity with normal height for age and no signs of Cushingoid features such as striae or hirsutism, which makes this diagnosis less likely.

Hypothyroidism (Option B) is associated with weight gain, growth retardation, coarse skin, and characteristic facial features. This patient has normal growth and no other signs of hypothyroidism, making it an unlikely cause of his obesity.

Prader-Willi syndrome (Option D): Unlike Prader-Willi syndrome, this patient has normal development, no history of hypotonia or feeding difficulties in infancy, and no dysmorphic features, making it the less likely diagnosis.

Solution for Question 6:

Correct answer: C) Whole grain bread

Explanation: Whole grain bread is a moderate energy food that provides essential nutrients and should be categorized as "Yellow," meaning it should be eaten in moderation.

- Pharmacotherapy: Orlistat: It is a gastric lipase inhibitor that reduces fat absorption. GLP-1 Analogs: For pediatric obesity and type 2 diabetes. Metformin: For diabetes and polycystic ovarian disease.
- Orlistat: It is a gastric lipase inhibitor that reduces fat absorption.
- GLP-1 Analogs: For pediatric obesity and type 2 diabetes.
- Metformin: For diabetes and polycystic ovarian disease.
- Specific Management: Treat underlying conditions (e.g., hypothyroidism, Prader-Willi syndrome) Metformin: For insulin resistance, type 2 diabetes, PCOS Statins: For persistent dyslipidemia Antihypertensives: For stage 2 hypertension. PCOS: Lifestyle changes, metformin, contraceptives. Surgery: Bariatric surgery in severe cases after final height
- Treat underlying conditions (e.g., hypothyroidism, Prader-Willi syndrome)
- Metformin: For insulin resistance, type 2 diabetes, PCOS
- Statins: For persistent dyslipidemia
- Antihypertensives: For stage 2 hypertension.
- PCOS: Lifestyle changes, metformin, contraceptives.
- Surgery: Bariatric surgery in severe cases after final height
- Orlistat: It is a gastric lipase inhibitor that reduces fat absorption.
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- Treat underlying conditions (e.g., hypothyroidism, Prader-Willi syndrome)
- Metformin: For insulin resistance, type 2 diabetes, PCOS
- Statins: For persistent dyslipidemia
- Antihypertensives: For stage 2 hypertension.
- PCOS: Lifestyle changes, metformin, contraceptives.
- Surgery: Bariatric surgery in severe cases after final height

Green	Yellow	Red

Low-energy, high-nutrient foods	Moderate energy foods.	High-energy, low-nutrient foods.
Includes fruits and vegetables	Includes grains and pulses.	Includes sweets, sweetened beverages, and fried foods.
Can be eaten often.	Should be eaten in moderation.	Should be eaten sparingly.

Fried chicken (Option A): This is a high-energy, low-nutrient food and should be categorized as " Red." Carrot sticks (Option B) are low-energy, high-nutrient foods and should be categorized as " Green." Candy (Option D): Candy is a high-energy, low-nutrient food and should be categorized as "Red."

Solution for Question 7:

Correct answer: A)Prader-Willi Syndrome

Explanation: The child's symptoms, including obesity, feeding difficulties, developmental delays, and dysmorphic facial features, are classic features of Prader-Willi Syndrome.

Prader-Willi Syndrome	Laurence-Moon-Bardet-Biedl Syndrome(LMBBS)
70% cases- Due to errors in genomic imprinting due to a paternal deletion. 25% cases- Due to maternal uniparental disomy. Infantile Hypotonia: Poor sucking reflex Feeding difficulties Dysmorphic Facial Features: Narrow frontal diameter Almond-shaped palpebral fissures Slender nasal bridge Thin upper vermillion border with downturned mouth corners Small hands and feet (develop later) Developmental Delays: Motor delay (milestones at half the usual pace) ... (content truncated)	Primary Features: Rod and cone dystrophy (night blindness) Polydactyly and polysyndactyly Central obesity Genital abnormalities (hypogonadism) Mental retardation (learning disabilities) Secondary Features: Developmental delay Diabetes mellitus (DM) Congenital heart disease

Laurence-Moon-Bardet-Biedl Syndrome(Option B): While this syndrome also presents with obesity and developmental delays, the absence of features like rod-cone dystrophy and polydactyly makes it less likely.

Down Syndrome(Option C) typically presents with intellectual disability, characteristic facial features (e.g., flat facial profile, upward slanting eyes), and may have congenital heart defects. Obesity and behavioural issues like those described are less typical for Down Syndrome.

Turner Syndrome(Option D) typically presents with short stature, webbed neck, and other distinct features, not consistent with this child's presentation.

Solution for Question 8:

Correct Answer: A) Type 2 Diabetes Mellitus (T2DM) with associated obesity.

Explanation:

The child's BMI at the 98th percentile indicates obesity, a major risk factor for T2DM. The symptoms of lethargy, frequent thirst, and increased urination (polydipsia and polyuria) are classic signs of diabetes. Additionally, the dark patches on the neck, known as acanthosis nigricans, are often associated with insulin resistance, a hallmark of T2DM.

Obesity

	Obesity	Overweight
Adult	BMI ≥ 30	BMI between 25 and 30
Children	BMI ≥ 95th percentile for age and sex	BMI between 85th and 95th percentiles

Aetiology:

- Imbalance between caloric intake and energy expenditure.
- Easy access to energy-dense foods, and large portion sizes.
- Sedentary lifestyle.
- High intake of sugary beverages, processed foods, and large portions.
- Cushing syndrome
- GH deficiency
- Hypothyroidism

Hypothyroidism with weight gain (Option B): Hypothyroidism can cause weight gain and lethargy, but it is less likely to cause the symptoms of increased thirst and urination. Additionally, acanthosis nigricans is not commonly associated with hypothyroidism.

Cystic Fibrosis-related diabetes (CFRD) (Option C) occurs in individuals with cystic fibrosis, which other symptoms, such as recurrent lung infections or gastrointestinal issues would typically accompany. There is no indication of cystic fibrosis in this child's history.

Adrenal insufficiency (Option D) can cause symptoms like fatigue and weight loss rather than weight gain. It does not typically cause polyuria, polydipsia, or acanthosis nigricans.

Solution for Question 9:

Correct answer: C) Add 5% dextrose to the IV fluids

Explanation:

As the blood glucose level approaches ≤ 300 mg/dL, adding 5% dextrose to the IV fluids is essential to prevent hypoglycemia while continuing insulin therapy to resolve the underlying ketoacidosis. This helps maintain glucose levels while the acidosis is treated.

DKA Management in Children	
Immediate Management (First Hour):	
Rapid Assessment:	Evaluate the child's vital signs (heart rate, respiratory rate, blood pressure) and weight. Assess GCS and check for signs of dehydration. Obtain blood samples for urgent measurement of blood glucose, blood beta-hydroxybutyrate (BHB) if available, venous pH, electrolytes (including bicarbonate), serum urea nitrogen, and creatinine. Serum BHB concentration: Most precise approach to measuring the severity of DKA
Establish IV Access:	Place two large-bore peripheral IV lines. Central lines are generally avoided (risk of thrombosis) unless necessary.
Initial Fluid Resuscitation	Administer a bolus of 0.9% saline at a rate of 10-20 ml/kg over 20-30 minutes to restore circulatory volume and address dehydration.
After First Hour	
Initiate Insulin Therapy	Start an intravenous insulin infusion at 0.05-0.1 U/kg/h, typically one hour after initiating fluids.

Continue Fluid Therapy	After the initial bolus, continue fluid replacement with 0.45% to 0.9% saline, to replace the estimated fluid deficit over 24 to 48 hours. Adjust fluids based on clinical status and laboratory values.
Monitoring and Replacing Electrolytes	Begin potassium replacement once urine output is confirmed as insulin causes K+ entry into cells leading to hypokalemia. Monitor vital signs, GCS, fluid balance, and capillary blood glucose every hour. Reassess electrolytes, blood glucose, blood urea nitrogen, calcium, magnesium, phosphate, blood gases, and BOHB every two hours if available.
Transition Phase (6-24 Hours):	
Gradual Reduction of IV Fluids and transition to subcutaneous Insulin	As DKA resolves and the child can tolerate oral intake, gradually reduce IV fluids while maintaining hydration. Transition to subcutaneous insulin once pH > 7.30, bicarbonate > 18 mmol/L, and BOHB < 1 mmol/L. Administer the first subcutaneous insulin dose 15-30 minutes before discontinuing the IV insulin infusion.

Increase the insulin infusion rate (Option A): It is not advisable when the blood glucose level drops, as this could lead to hypoglycemia. The goal of insulin therapy in DKA is to gradually lower blood glucose while resolving ketoacidosis, not to rapidly decrease blood glucose levels.

Stop insulin therapy (Option B): Discontinuing insulin therapy too early could result in a rebound worsening of ketoacidosis. Even if blood glucose levels decrease, insulin therapy must be continued to ensure that ketoacidosis is fully corrected, albeit with adjustments like adding dextrose to prevent hypoglycemia.

Administer an additional bolus of 0.9% saline (Option D): This is generally unnecessary unless there are ongoing signs of significant dehydration or shock. Fluid resuscitation should be carefully managed to avoid complications such as cerebral oedema, and any additional fluid administration should be based on the child's overall clinical status.

Solution for Question 10:

Correct answer: D) Central Precocious Puberty

Explanation:

- Precocious puberty in girls is the development of secondary sexual characteristics before reaching the age of 8.
- Central precocious puberty is characterized by the premature activation of the hypothalamic-pituitary-gonadal axis, leading to increased gonadotropin levels, such as LH.
- This condition is more common in girls and typically presents with early development of secondary sexual characteristics, such as breast development, as well as accelerated growth.

	Central Precocious Puberty (CPP)	Peripheral Precocious Puberty (PPP)
Cause	CPP is due to the early activation of the hypothalamic-pituitary-gonadal axis, leading to the release of gonadotropins (LH and FSH) that stimulate the ovaries to produce estrogen.	PPP results from estrogen production independent of the hypothalamic-pituitary-gonadal axis.
Common Causes	The most frequent cause in girls is idiopathic, with no identifiable underlying condition. Central nervous system abnormalities (e.g., hypothalamic hamartoma, glioma, hydrocephalus) are less common.	The most common cause in girls is estrogen-producing ovarian cysts, leading to fluctuating pubertal development and early vaginal bleeding.
Symptoms	Girls with CPP show advanced growth and bone age	Girls with PPP may have symptoms related to the specific underlying condition, such as delayed bone age in hypothyroidism or skin pigmentation changes in McCune-Albright syndrome.
Diagnostic Tests:	Basal LH and FSH levels, a GnRH stimulation test, and imaging (ultrasound, MRI) help differentiate between CPP and PPP.	
Treatment	GnRH analogues can reverse early-onset CPP and help with height if started before age 6. Stop GnRH analogs at age 11 or at bone age 12.5. MPA: Used to delay postpone menstruation in girls with intellectual disability, but doesn’t improve height. Cyproterone: Another option for suppressing puberty.	Hypothyroidism and Puberty: Thyroxine replacement can reverse puberty changes caused by hypothyroidism. McCune-Albright Syndrome Treatment: Aromatase Inhibitors: Anastrozole, Letrozole (to reduce estrogen production) Estrogen Receptor Antagonists: Tamoxifen, Fulvestrant (to block estrogen action) Ovarian Cysts: Treatment depends on the size and shape of the cysts.

Peripheral Precocious Puberty (Option A): It results from the autonomous production of sex hormones independent of the hypothalamic-pituitary-gonadal axis. In PPP, LH levels are typically low or prepubertal because the central axis is not involved in the process.

Constitutional delayed puberty (Option B): This condition involves a delayed rather than early onset of puberty. Girls with a constitutional delay of growth and puberty typically have low LH levels and delayed development of secondary sexual characteristics, the opposite of what is observed in this case.

Premature Thelarche (Option C) refers to isolated early breast development without other signs of puberty, such as pubic hair or accelerated growth. In this condition, LH levels remain within the normal prepubertal range, unlike in central precocious puberty.

Solution for Question 11:

Correct Answer: A) Basal-Bolus Regimen

Explanation: This regimen provides the most flexibility in terms of meal timing and carbohydrate intake. It involves a long-acting insulin (basal) to cover the body's needs throughout the day and night, and a rapid-acting insulin (bolus) given at mealtimes to manage the rise in blood glucose levels after eating.

This allows patients to adjust their insulin doses based on the amount of carbohydrates they consume, offering greater flexibility.

Type 1 Diabetes Mellitus (T1DM) Management (Goal- HbA1c < 7.5%)

Types of Insulin:

Short- Acting Insulin: Lispro, aspart, and glulisine insulin

Intermediate-acting insulin: Neutral protamine Hagedorm (NPH)

Long-acting insulin: Glargine, detemir, degludec

Insulin Therapy:

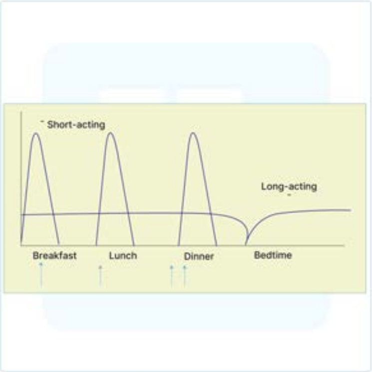
- Initiation: At diagnosis for all T1DM patients
- Starting Dose: 0.4 to 1.2 units/kg/day (based on age, pubertal stage, DKA presence).
- Monitoring: Frequent glucose checks and dosage adjustments by the diabetes team.

There are 2 regimes used for insulin therapy:

1. Basal-Bolus Insulin Regimes: Require multiple daily injections.

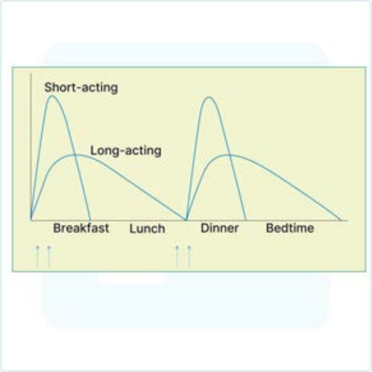
   - Basal: Slow-onset, long-duration insulin for between-meal control (once or twice daily) eg. glargine, detemir, degludec.

Bolus: Rapid-onset insulin is given with meals for carbohydrate coverage and hyperglycemia correction eg. lispro, aspart, and glulisine insulin.



2. Mixed Split Insulin Regimes:

   - Three daily injections regimen: 1st injection with breakfast: A short-acting insulin + Intermediate-acting insulin 2nd injection with dinner: A short-acting insulin only. 3rd injection at bedtime: A long-acting insulin only.
   - 1st injection with breakfast: A short-acting insulin + Intermediate-acting insulin
   - 2nd injection with dinner: A short-acting insulin only.
   - 3rd injection at bedtime: A long-acting insulin only.
   - 1st injection with breakfast: A short-acting insulin + Intermediate-acting insulin
   - 2nd injection with dinner: A short-acting insulin only.
   - 3rd injection at bedtime: A long-acting insulin only.
   - Twice daily injection Regimen:
   - 1st injection at breakfast: Pre-mixed injection (containing short-acting and intermediate-acting insulin)
   - 2nd injection at dinner: same as 1st.



Insulin Pump Therapy:

- Mechanism: Continuous subcutaneous insulin infusion via battery-powered pumps.
- Advantages: Closer approximation of normal insulin profiles, flexible meal/snack timing.
- Transition Timing: 6-12 months post-diagnosis, based on patient preference.

Three Daily Injection Regimen (Option B): Typically includes a mixture of short- and intermediate-acting insulins. While it provides good glucose control, it offers less flexibility compared to the basal-bolus regimen because meal timing needs to align with the insulin action peaks.

Once-daily injection of long-acting insulin only (Option C): This regimen is the least flexible because it does not account for meals, providing only basal coverage without bolus insulin.

Twice-daily injection of short-acting insulin only (Option D): This regimen is also less flexible since it requires strict meal timing to avoid hypoglycemia, with less ability to adjust for variations in carbohydrate intake.

Solution for Question 12:

Correct Answer: C) A random plasma glucose level of ≥ 180 mg/dL with symptoms of diabetes.

Explanation: A random plasma glucose level of ≥ 200 mg/dL (11.1 mmol/L) with symptoms of diabetes is diagnostic of T1DM. A glucose level of 180 mg/dL (10.0 mmol/L) does not meet the diagnostic criteria for T1DM.

Diagnostic Criteria of Type 1 Diabetes (T1DM):

- Fasting Plasma Glucose: ≥ 126 mg/dL. (Option A ruled out)
- 2-Hour Plasma Glucose during OGTT: ≥ 200 mg/dL. (Option B ruled out)
- Random/Casual Plasma Glucose: ≥ 200 mg/dL plus symptoms (polyuria, polydipsia, unexplained weight loss) and glucosuria/ketonuria.
- Haemoglobin A1c ≥ 6.5% (48 mmol/mol). (Option D ruled out)

Conditions Associated with T1DM:

Autoimmune Diseases	
Condition	Description
Chronic Lymphocytic Thyroiditis	~20% of T1DM patients have thyroid antibodies.
Celiac Disease	7–15% of children with T1DM develop celiac disease within 6 years of diagnosis.
Autoimmune Adrenal Insufficiency	Indicated by decreased insulin needs, skin pigmentation, salt craving, weakness, asthenia, postural hypotension, or adrenal crisis.
Atrophic Gastritis	Correlated with antibodies to gastric parietal cells.
Genetic Disorders	
IPEX syndrome	Immune dysfunction, polyendocrinopathy, enteropathy, X-linked; FOXP3 gene mutation.
Wolfram syndrome	DIDMOD: diabetes insipidus, diabetes mellitus, optic atrophy, deafness; Autosomal recessive disease; WFS1 gene mutation.
APS-1	Autoimmune polyendocrinopathy syndrome type 1; AIRE gene mutation.

Cystic Fibrosis-Related Diabetes (CFRD):

- Increasingly common as CF patients live longer.
- Higher prevalence in females, particularly after age 40.
- Associated with pancreatic insufficiency.
- Features overlap with both T1DM and T2DM.

Solution for Question 13:

Correct Answer: A) Complete Androgen Insensitivity Syndrome

Explanation:

In Complete Androgen Insensitivity Syndrome (CAIS):

- Breast development is normal because the body converts androgens to estrogen, allowing for feminization.
- Primary amenorrhea occurs due to the absence of a uterus and female internal genitalia.
- Undescended testes are present because the patient is genetically male (46, XY karyotype) but the body is unresponsive to androgens.
- Elevated androgen levels are typical as the body produces them, but they have little effect due to the androgen receptor mutation.

AR gene mutation confirms androgen insensitivity, and in CAIS, there is complete resistance to androgen action, leading to female external genitalia despite a male genotype.

	Complete Androgen Insensitivity Syndrome (CAIS)	Partial Androgen Insensitivity Syndrome (PAIS)	Mild Androgen Insensitivity Syndrome (MAIS)
Clinical Features	Female external appearance with normal breast development. Short vagina Absent uterus and fallopian tubes Undescended testes Primary amenorrhea Infertility.	Variable genitalia ranging from predominantly female with clitoromegaly to predominantly male with micropenis. Gynecomastia Bifid scrotum Undescended testes Virilization.	Male genitalia with micropenis Gynecomastia, Impaired fertility.
Diagnosis	Suspected in a phenotypically female individual with a 46, XY karyotype, inguinal hernia with testes, elevated testosterone and LH, normal estradiol, confirmed by AR gene mutation.	Suspected with ambiguous genitalia or undervirilization, elevated testosterone, and LH, confirmed by AR gene mutation.	Clinical features and AR gene mutation testing.
Treatment	Gonadectomy to prevent malignancy risk, estrogen replacement, vaginal dilation, and psychological support.	Based on undervirilization, fertility potential, and preference, may include gonadectomy, hormone therapy, genital reconstructive surgery, and psychological support.	Supportive care, psychological counselling, and fertility evaluation.

Partial Androgen Insensitivity Syndrome (Option B): In PAIS, the body has partial responsiveness to androgens, leading to a wide range of phenotypes, from ambiguous genitalia to predominantly male or female external genitalia with varying degrees of virilization. Theytypically have ambiguous genitalia, not the normal female external genitalia seen in this case.They do not have normal breast development as eitherl.

Mild Androgen Insensitivity Syndrome (Option C): Patients with MAIS usually have normal male genitalia and only minor defects such as infertility or gynecomastia. They do not present with female external genitalia or primary amenorrhea,

Klinefelter Syndrome (Option D): This option is incorrect because Klinefelter Syndrome is characterized by a 47, XXY karyotype, not a 46, XY karyotype, and typically presents with male genitalia and gynecomastia, which does not fit the patient’s presentation of female external features and absent uterus.

Solution for Question 14:

Correct Answer: B) 17β-HSD deficiency

Explanation:

17β-HSD (Hydroxy steroid dehydrogenase) deficiency is classified as a 46, XY DSD because it disrupts androgen synthesis, affecting male sexual development.

DSD is defined as a discrepancy in chromosomal/gonadal/genital sex. It can be classified as follows:

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Solution for Question 15:

Correct Answer: A) Deficient Expression of SRY

Explanation:

The SRY (Sex-determining Region Y) gene is crucial for the initiation of testis development. If SRY is not expressed properly, testes may not develop, leading to ambiguous genitalia or underdeveloped male genitalia in a 46, XY fetus. This explanation is consistent with theobserved fetal presentation.

1. Gonadal Differentiation:

   - Early Development: Germ cells migrate to the gonadal ridge and form a bipotential gonad.
   - Sexual differentiation starts with the karyotype of the fetus (XX or XY).
   - Testis Formation: The presence of SRY on the Y chromosome leads to testis development. Key genes involved include: WT1 and SF1: Required for initial gonadal development. SOX9: Works with SRY to promote testis formation. DAX1: Can inhibit testis formation if overexpressed.
   - WT1 and SF1: Required for initial gonadal development.
   - SOX9: Works with SRY to promote testis formation.
   - DAX1: Can inhibit testis formation if overexpressed.
   - Ovary Formation: In the absence of SRY, the gonad develops into an ovary.
   - WT1 and SF1: Required for initial gonadal development.
   - SOX9: Works with SRY to promote testis formation.
   - DAX1: Can inhibit testis formation if overexpressed.

2. Genital Differentiation:

   - In Males: Testis produces the Antimullerian hormone -AMH (from Sertoli cells) and testosterone (from Leydig cells). AMH regresses the Mullerian duct. Testosterone differentiates the wolfian duct into Vas deferens and seminiferous tubules. Testosterone gets converted into Dihydrotestosterone (DHT) which causes male external genitalia development.
   - Testis produces the Antimullerian hormone -AMH (from Sertoli cells) and testosterone (from Leydig cells). AMH regresses the Mullerian duct. Testosterone differentiates the wolfian duct into Vas deferens and seminiferous tubules. Testosterone gets converted into Dihydrotestosterone (DHT) which causes male external genitalia development.
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   - Testosterone differentiates the wolfian duct into Vas deferens and seminiferous tubules.
   - Testosterone gets converted into Dihydrotestosterone (DHT) which causes male external genitalia development.
   - In Females: Without AMH and testosterone, Müllerian ducts form female reproductive organs (fallopian tubes, uterus, upper vagina). Absence of testosterone leads to undifferentiated external genitalia (labia majora and minora, clitoris, and lower vagina).
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- Absence of testosterone leads to undifferentiated external genitalia (labia majora and minora, clitoris, and lower vagina).
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- Testosterone gets converted into Dihydrotestosterone (DHT) which causes male external genitalia development.
- Without AMH and testosterone, Müllerian ducts form female reproductive organs (fallopian tubes, uterus, upper vagina).
- Absence of testosterone leads to undifferentiated external genitalia (labia majora and minora, clitoris, and lower vagina).
- Effects of Androgens: Prenatal Exposure: May cause labioscrotal fusion in females, leading to ambiguous genitalia. Postnatal Exposure: Typically results in clitoromegaly without labial fusion.
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- Postnatal Exposure: Typically results in clitoromegaly without labial fusion.
- Prenatal Exposure: May cause labioscrotal fusion in females, leading to ambiguous genitalia.
- Postnatal Exposure: Typically results in clitoromegaly without labial fusion.

Differential Diagnosis of ambiguous genitalia:

- Transverse testicular ectopia
- True hermaphrodite
- Male pseudo hermaphrodite
- Complete androgen insensitivity syndrome
- Partial androgen insensitivity syndrome
- 5 alpha reductase deficiency
- Persistent Mullerian duct syndrome
- Mixed gonadal dysgenesis
- Female pseudo hermaphrodite
- Congenital adrenal hyperplasia
- Dysgenetic male pseudo hermaphrodite
- Pure gonadal dysgenesis

Excessive SOX9 Activity (Option B): SOX9 is another gene involved in male sexual development. Increased activity of SOX9 typically promotes male differentiation, not ambiguous genitalia, making it an unlikely explanation in this case.

Absent AMH Production(Option C): AMH is produced by the developing testes to prevent the formation of female internal structures (Müllerian ducts). While a lack of AMH could lead to the presence of female internal genitalia, it would not directly cause ambiguous external genitalia in a 46, XY fetus.

Decreased DAX1 Expression (Option D): DAX1 normally acts as an antagonist to SRY in sex determination. Decreased DAX1 expression would typically favour male development, not result in ambiguous genitalia.

Solution for Question 16:

Correct answer: B) Hypogonadotropic Hypogonadism

Explanation:

- Hypogonadotropic hypogonadism is a condition where delayed puberty results from insufficient stimulation of the gonads due to low levels of gonadotropins.
- This can be caused by central nervous system issues affecting the release of gonadotropin-releasing hormone (GnRH).
- Factors contributing to this condition include stress, chronic illness, malnutrition, or congenital abnormalities affecting the hypothalamus. The boy’s low LH and FSH levels align with this diagnosis.

Causes of delayed puberty:

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Constitutional Delay (Option A): This is characterized by a normal hormonal profile and is often a variation in the timing of puberty, with growth eventually catching up. However, constitutional delay usually does not present with low levels of LH and FSH, making it less likely in this case.

Hypergonadotropic Hypogonadism (Option C): This condition involves high levels of LH and FSH resulting from the primary gonadal failure. The boy’s low levels of these hormones suggest that hypergonadotropic hypogonadism is not the cause.

Klinefelter Syndrome (Option D): This genetic disorder often leads to elevated LH and FSH levels due to testicular failure. Since the boy’s tests show low levels of LH and FSH, Klinefelter syndrome is unlikely to be the cause of his delayed puberty.

Solution for Question 17:

Correct answer: A) Central precocious puberty

Explanation:

- Precocious puberty is when puberty begins before 9.5 years of age in boys.
- In this scenario, elevated LH levels and increased testicular volume strongly indicate central precocious puberty, where the body’s normal pubertal mechanisms are activated prematurely.
- Organic neurological cases are more common in boys which include hypothalamic hamartoma, glioma, hydrocephalus, and tubercular meningitis.

Peripheral precocious puberty due to congenital adrenal hyperplasia (Option B): It results from the autonomous production of sex hormones without the involvement of the central axis. CAH leads to the overproduction of androgens, presenting with signs of puberty and advanced bone age. However, elevated LH levels suggest a central rather than peripheral cause.

Peripheral precocious puberty due to an hCG-secreting tumour (Option C): hCG-secreting tumours, such as those found in the liver or brain, can stimulate testicular production of testosterone without involving the hypothalamic-pituitary axis. These tumours do not have elevated LH.

Peripheral precocious puberty due to an androgen-secreting testicular tumour (Option D): This condition involves a tumour within the testes that produces androgens independent of central hormonal control. It presents with signs of early puberty, such as unilateral testicular enlargement, and typically maintains prepubertal LH levels.

Solution for Question 18:

Correct answer: A) McCune-Albright syndrome

Explanation:

McCune-Albright syndrome is caused by a somatic mutation in the GNAS1 gene(codes for G-protein), leading to the autonomous production of sex hormones and symptoms such as irregular vaginal bleeding, early breast development, and characteristic café-au-lait spots. The condition also often presents with advanced bone age due to its rapid progression.

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Endocrine abnormalities associated:

- Hyperthyroidism
- GH excess
- Peripheral Precocious puberty

Diseases associated with Cafe-au-lait macules:

- Neurofibromatosis type 1 (NF1)/ von Recklinghausen’s disease
- Neurofibromatosis type 2 (NF2)
- McCune Albright Syndrome
- Legius syndrome (NF1-like syndrome)
- Noonan syndrome with multiple lentigines (LEOPARD syndrome)

Differential Diagnosis of Cafe-au-lait macules:

- Congenital melanocytic nevus
- Becker nevus
- Lentigo
- Urticaria pigmentosa
- Postinflammatory hyperpigmentation
- Plexiform neurofibroma
- Segmental pigmentation disorder
- Phytophotodermatitis

Central precocious puberty (Option B): It is associated with early activation of the hypothalamic-pituitary-gonadal axis. Unlike McCune-Albright syndrome, it does not typically present with café-au-lait spots.

Hypothyroidism (Option C): While hypothyroidism can cause delayed or even sometimes irregular pubertal development, it does not account for the rapid progression, advanced bone age, or café-au-lait spots seen in McCune-Albright syndrome.

Adrenal hyperplasia (Option D): Congenital adrenal hyperplasia usually presents with virilization or ambiguous genitalia due to excess androgen production, not the estrogen-dominated signs like those seen in McCune-Albright syndrome.

Previous Year Questions

1. What condition is likely in a newborn presenting with dry, rough skin, a big tongue, and rough hair?

A. Congenital hypothyroidism
B. Prader Willisyndrome
C. Edward syndrome
D. Galactosemia

2. When is a blood sample taken to diagnose neonatal hypothyroidism?

A. Immediately from the umbilical cord
B. 24 hours after delivery
C. 23-48 hours after delivery
D. 72 hours after delivery

3. What potential issues might be linked to the newborn of a 31-year-old woman with diabetes mellitus who delivers at full term?

A. Hyperglycemia
B. Tetralogy of Fallot
C. Hypercalcemia
D. Hypoglycemia

4. A 22-year-old mother presented with her 3-week-old newborn who had a large tongue and an umbilical hernia. Which of the following investigations has to be done in view of congenital hypothyroidism?

A. A fresh blood sample and screen first for TSH levels

B. A fresh blood sample and screen first for T4 levels

C. A dried blood sample and screen first for TSH levels

D. A dried blood sample and screen first for T4 levels

---

5. Which of the following is produced in the highest amounts by the fetal adrenal gland?

A. DHEA

B. Cortisol

C. Corticosterone

D. Progesterone

---

6. Wide open posterior fontanelle, large tongue, rough dry skin with constipation is seen in:

A. Pellagra

B. Down syndrome

C. Congenital Hypothyroidism

D. Nutritional rickets

---

7. What specific actions are taken for the management of a 3-week-old neonate with ambiguous genitalia, a sodium level of 127, a potassium level of 6 meq, and a blood pressure of 50/22 mm Hg, in addition to administering intravenous fluids?

A. Spironolactone

B. Potassium binding resins

C. Hydrocortisone

D. Broad spectrum antibiotics

8. The mother brought her 3-year-old daughter to seek medical attention due to the girl's enlarged clitoris. The results of the karyotype test showed a 46XX pattern. What is the probable underlying cause for this condition?

A. 21 hydroxylase deficiency

B. 11 hydroxylase deficiency

C. 3 beta dehydrogenase deficiency

D. 17α-OH deficiency

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	4
Question 4	3
Question 5	1
Question 6	3
Question 7	3
Question 8	1

Solution for Question 1:

Correct Option A - Congenital hypothyroidism:

- Congenital hypothyroidism (also known as cretinism) presents with: Dry, rough skin Macroglossia (an enlarged, protruding tongue) Rough hair
- Dry, rough skin
- Macroglossia (an enlarged, protruding tongue)
- Rough hair
- Dry, rough skin
- Macroglossia (an enlarged, protruding tongue)
- Rough hair

Incorrect Options:

Option B - Prader Willi syndrome:

- Prader-Willi syndrome is characterized by Feeding problems during infancy Excessive eating and obesity Intellectual disability Hypogonadism

- Feeding problems during infancy
- Excessive eating and obesity
- Intellectual disability
- Hypogonadism

- Feeding problems during infancy
- Excessive eating and obesity
- Intellectual disability
- Hypogonadism

Option C - Edward syndrome:

- Edward syndrome is characterized by:

R	• Rocker bottom foot i.e., foot is convex from below like a rocking chair
O	• Overlapping fingers or toes
C	• Cardiac anomalies
K	• Kidney malformations
Y	• Microcephaly
M	• Mental retardation/Intellectual disability

Option D - Galactosemia

- Galactosemia is characterized by jaundice, hepatomegaly, diarrhea, and vomiting.

Solution for Question 2:

Correct Answer: D) 72 hours after delivery

Explanation:

The blood sample taken for diagnosing neonatal hypothyroidism is 48-72 hours after delivery.

Diagnosis:

- Primary hypothyroidism: T4 level: low TSH usually>100mIU/L
- T4 level: low
- TSH usually>100mIU/L
- Central hypothyroidism: Low TSH, T3, and T4 levels
- T4 level: low
- TSH usually>100mIU/L

X-ray findings in congenital hypothyroidism:

- Absence of distal femoral and proximal tibial epiphysis at birth
- Punctate epiphyseal dysgenesis
- Deformity or breaking of the 12th thoracic or 1st or 2nd lumbar vertebrae
- X-ray image of skull- large fontanelle and wide sutures

Screening:

- Universal newborn screening for congenital hypothyroidism
- At birth, with umbilical cord blood (Option A)

Diagnostic:

- Heel prick: dried blood spots, sample collected beyond 48 hrs or 48-72 hrs
- Should not be done in first 1-2 days, to avoid TSH surge (vs Option B and C)
- Most sensitive approach- check for T4 and TSH both

ASH's first strategy:

- Initial TSH screening using dried blood spot
- If TSH>20mIU/L, recheck
- Obtain a venous blood sample to check T4 and TSH levels if TSH remains elevated.

Treatment:

- Oral levothyroxine (early morning with empty stomach for older children and for newborns; it is to be given at a fixed time)
- The dose is higher in the earlier age group, and as the babies grow, the dose reduces.
- Newborn- 10-15mg/kg/day
- Children- 2-4mg/kg/day

Solution for Question 3:

Correct Option D - Hypoglycemia:

- Hypoglycemia is the condition associated with an infant born to a mother with diabetes mellitus.

Maternal Hyperglycemia

↓

Fetal Hyperglycemia

Hyperplasia and hypertrophy of fetal pancreatic beta cells

Fetal hyperinsulinemia → Neonatal Hypoglycemia

Incorrect Options:

Options A, B, and C are not associated with an infant born to a mother with diabetes mellitus

Solution for Question 4:

Correct Option C - A dried blood sample and screen first for TSH levels:

- Screening for congenital hypothyroidism is typically done using a dried blood spot sample taken from the baby's heel.
- Elevated TSH levels indicate hypothyroidism, while low or normal TSH levels suggest normal thyroid function.
- Measuring T4 levels alone is not sufficient for the initial screening of congenital hypothyroidism because T4 levels can be affected by various factors, including stress, illness, and medications.
- TSH is a more sensitive and specific marker for detecting hypothyroidism in newborns.
- Dried blood spots enable minimally invasive sampling of a small blood volume.

Incorrect Options:

Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 5:

Correct Option A - DHEA:

- During fetal development, the adrenal glands play a crucial role in producing various hormones that are important for growth and development.
- Among these hormones, DHEA is produced in the highest amounts by the fetal adrenal glands.
- DHEA (dehydroepiandrosterone) is an androgen hormone and a precursor to other sex hormones, such as testosterone and estrogen.
- It is primarily produced by the zona reticularis layer of the adrenal cortex. In the fetus, DHEA is synthesized from cholesterol, and its production increases as the pregnancy progresses.

Incorrect Options:

Options B, C, and D are produced in relatively lower as compared to the DHEA

Solution for Question 6:

Correct Option C - Congenital Hypothyroidism:

- The clinical features of wide open posterior fontanelle, large tongue, rough dry skin, and constipation is suggestive of congenital hypothyroidism.

Incorrect Options:

Option A - Pellagra:

- Pellagra is a nutritional deficiency disease caused by a lack of niacin (vitamin B3).
- It typically presents with symptoms such as dermatitis, diarrhea, and dementia.

Option B - Down syndrome:

- Down syndrome is a genetic disorder caused by the presence of an extra copy of chromosome 21.

Option D - Nutritional rickets:

- Nutritional rickets is a condition caused by vitamin D deficiency, leading to impaired bone mineralization.
- It is characterized by skeletal deformities, such as bowed legs.

Solution for Question 7:

Correct Option C - Hydrocortisone:

- The patient's presentation with ambiguous genitalia, a low serum sodium level, a high serum potassium level, and a low BP is suggestive of congenital adrenal hyperplasia.
- CAH is a group of autosomal recessive disorders characterized by impaired cortisol synthesis, leading to an accumulation of precursors and subsequent adrenal gland hyperplasia.
- The most common enzyme deficiency causing CAH is 21-hydroxylase deficiency.
- In a neonate with CAH, the low cortisol levels lead to an overproduction of adrenocorticotropic hormone (ACTH), which stimulates the adrenal glands to produce excess androgens.
- This can result in ambiguous genitalia in females and precocious pseudopuberty in males.
- The specific management in this case involves administration of hydrocortisone, a glucocorticoid, to replace the deficient cortisol and suppress the excessive ACTH production.
- Hydrocortisone helps to normalize electrolyte imbalances, such as hyponatremia and hyperkalemia, and improve blood pressure.

Incorrect Options:

Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 8:

Correct Option A - 21 hydroxylase deficiency:

- 21 hydroxylase deficiency is a common form of congenital adrenal hyperplasia (CAH) and is characterized by impaired production of cortisol and aldosterone.
- The lack of cortisol leads to an increase in adrenocorticotropic hormone (ACTH) secretion, which stimulates the adrenal glands to produce excess androgens (male sex hormones).

- In individuals with XX chromosomes (typically female), the excess androgens can cause virilization of the external genitalia, resulting in an enlarged clitoris.

Incorrect Options:

Option B - 11 hydroxylase deficiency:

- 11 hydroxylase deficiency is another form of congenital adrenal hyperplasia (CAH), but it typically leads to hypertension and may cause ambiguous genitalia.
- It is less commonly associated with an enlarged clitoris.

Option C - 3 beta dehydrogenase deficiency:

- 3 beta dehydrogenase deficiency is a rare form of congenital adrenal hyperplasia (CAH) characterized by impaired conversion of pregnenolone to progesterone.
- It can lead to impaired production of cortisol and aldosterone, but it is not typically associated with virilization and an enlarged clitoris.

Option D - 17α-OH deficiency:

- 17α-OH deficiency is generally present with complaints of delayed puberty, primary amenorrhea, and a lack of secondary sexual characteristics in a female patient.
- The presentation in male patients is typically under masculinization and can range from phenotypic females to ambiguous or small male genitalia.